TRACK P (PLANETARY): BLUEPRINT

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Abstract. A proof plan, step by step, for the three frozen statements of track P: the properly degenerate KAM theorem of Chierchia–Pinzari (specialized as frozen), the analytic planar two-planet theorem with a moving Sun, and the fixed-Sun n-planet variant. Part 0 fixes conventions, restates every frozen definition and the three frozen theorems; the remaining parts are listed in `PLAN.md`. Results of track K's blueprint are cited as [K:`label`].

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1. Conventions, definitions and the frozen statements

This part fixes notation and restates, in mathematical form, the three frozen theorems of track P (owner approval D20, 27 October 2026) together with every frozen definition they use: the 33 definitions of `KAM/Frozen/Planetary/Defs/` and the three track-K definitions `KAM.IsDiophantine`, `KAM.hamiltonianVectorField`, `KAM.symplecticForm`. The Lean text is authoritative and the statements below transcribe it, with the same quantifier order and the same order in which constants are chosen. The D22 additions (P-EXP-SAFE, P-AMD-CONFINEMENT, P-SECULAR-BOX) are not frozen and are deliberately absent; a later part is reserved for them (`PLAN.md`, Part 14).

1.1. Conventions.

- *Ambient spaces.* All coordinate spaces are $\mathbb{R}^d = (\mathtt{Fin}\, d \to \mathbb{R})$ with the max norm $|v|_\infty = \max_i |v_i|$ and Lebesgue measure; $\mathrm{Ph}(d) = \mathbb{R}^d \times \mathbb{R}^d$ carries the max norm of the two components and the product Lebesgue measure `volume`. Fourier indices are $k \in \mathbb{Z}^d$. The Lean block conventions are: for $v \in \mathbb{R}^{a+b}$, the *fast* part is $v^{\mathrm{f}} = (v_i)_{i<a}$ and the *slow* part is $v^{\mathrm{s}} = (v_{a+j})_{j<b}$.
- *Phase order.* A phase point is $z = (z_1, z_2)$ with positions (angles) z_1 and momenta (actions) z_2. In the properly degenerate setting $z = (\varphi, q; I, p)$, that is $z_1 = (\varphi, q)$ and $z_2 = (I, p)$, with $\varphi \in \mathbb{R}^a$, $q \in \mathbb{R}^b$ and $I \in \mathbb{R}^a$, $p \in \mathbb{R}^b$ (`assemble`).
- *Physical distance.* On $\mathbb{R}^2$ the physical length is $|x|_2 = \sqrt{x_1^2 + x_2^2}$, never the max norm; $x \cdot y = x_1 y_1 + x_2 y_2$. Neighbourhoods in phase space and the norms $|\cdot|$ of tori data use the max norm; physical formulas use $|\cdot|_2$.
- *Bodies.* Positions of n planets are $q \in \mathbb{R}^{2n}$; planet i has position $q^{(i)} = (q_{(i,1)}, q_{(i,2)}) \in \mathbb{R}^2$, where the index of the pair $(i, k) \in \mathtt{Fin}\, n \times \mathtt{Fin}\, 2$ is $k + 2i$ (Lean `finProdFinEquiv`).
- *Torus parameters have period one* for every torus predicate below. For the cylinder tori of P-G (Definition 1.15) the *canonical fast angles keep period* 2π (CP10 footnote 1, p. 546), so the parameterization winds by 2π in the fast directions. Consequently the frequency vector w of a torus is the physical angular frequency divided by 2π (it changes frequency constants, not the dynamics or the measure ratio; `statement.md`, "Conventions"; owner question Q-P2). For the physical tori the parameter lattice is $\mathbb{Z}^d$ and the angles $2\pi\phi$ enter only through `circularPoint`.
- *Constants and quantifier order.* In each theorem, all data precede the small parameters; the proofs must keep the order of Theorems 1.28–1.30. In particular the thresholds r_*, C, c of P-G depend on the whole fixed family $(V, W, \dots, P, \dots)$, and μ_0, d, D (resp. ε_0, d, D) are chosen before μ (resp. ε).

1.2. Track-K definitions used by track P.

The three frozen track-P theorems import exactly these frozen track-K definitions; no definition of track I or N is used.

Definition 1.1 (Diophantine vectors)**.** For $\gamma, \tau \in \mathbb{R}$ and $\omega \in \mathbb{R}^d$, ω is (γ, τ)-*Diophantine* if $\gamma / |k|_1^\tau \le |k \cdot \omega|$ for every $k \in \mathbb{Z}^d \setminus \{0\}$, where $|k|_1 = \sum_i |k_i|$ and $k \cdot \omega = \sum_i k_i \omega_i$.
Frozen Lean declaration: `KAM.IsDiophantine`
Source: HL19 (3.1), p. 14; Pos01 (1), p. 5; same as [K:`bp:p0:diophantine`].

Definition 1.2 (Hamiltonian vector field and symplectic form)**.** For $H : \mathrm{Ph}(d) \to \mathbb{R}$ and $z = (z_1, z_2)$,

$$X_H(z) = \big(\partial_{z_2} H(z),\ -\partial_{z_1} H(z)\big),$$

that is $X_H(z)_{1,i} = DH(z)(0, e_i)$ and $X_H(z)_{2,i} = -DH(z)(e_i, 0)$, with D the Fréchet derivative (`fderiv`). For $v = (v_1, v_2)$, $w = (w_1, w_2) \in \mathrm{Ph}(d)$,

$$\Omega(v, w) = \sum_i \big(v_{2,i} w_{1,i} - v_{1,i} w_{2,i}\big).$$

With z_1 the position and z_2 the momentum, $\dot{z}_1 = \partial_{z_2} H$, $\dot{z}_2 = -\partial_{z_1} H$ are the usual Hamilton equations, and $\Omega = \sum dz_{2,i} \wedge dz_{1,i}$. The condition $\Omega(Dv, Dw) = 0$ defining a Lagrangian torus is insensitive to the sign of Ω.
Frozen Lean declaration: `KAM.hamiltonianVectorField, KAM.symplecticForm`
Source: HL19 Rem. 2.1, p. 6; same as [K:`bp:p0:hamiltonian`].

1.3. **Elementary planar and index definitions.**

Definition 1.3 (Phase space)**.** $\mathrm{Ph}(d) = \mathbb{R}^d \times \mathbb{R}^d$ (canonical coordinates *(position, momentum)*).
Frozen Lean declaration: `KAM.Planetary.Phase`

Definition 1.4 (Planar length and scalar product)**.** For $v, w \in \mathbb{R}^2$: $\mathrm{sq}(v) = \sum_k v_k^2$, $|v|_2 = \sqrt{\mathrm{sq}(v)}$ and $v \cdot w = \sum_k v_k w_k$.
Frozen Lean declaration: `KAM.Planetary.planeSq, KAM.Planetary.planeLength, KAM.Planetary.planeDot`
Source: Fej04r (28), pp. 46–47 (physical Euclidean length; independent of the ambient max norm).

Definition 1.5 (Body vectors)**.** For $q \in \mathbb{R}^{2n}$ and $i < n$, $\mathrm{body}(q, i) = q^{(i)} = (q_{(i,1)}, q_{(i,2)}) \in \mathbb{R}^2$.
Frozen Lean declaration: `KAM.Planetary.bodyVector`

Definition 1.6 (Fast and slow blocks, assembly)**.** For $v \in \mathbb{R}^{a+b}$, $v^{\mathrm{f}} \in \mathbb{R}^a$ and $v^{\mathrm{s}} \in \mathbb{R}^b$ are the first a and last b coordinates. For $I, \varphi \in \mathbb{R}^a$ and $p, q \in \mathbb{R}^b$, $\mathrm{assemble}(I, \varphi, p, q) = \big((\varphi, q), (I, p)\big) \in \mathrm{Ph}(a+b)$, with the concatenated coordinate vectors.
Frozen Lean declaration: `KAM.Planetary.fast, KAM.Planetary.slow, KAM.Planetary.assemble`
Source: CP10 (1), p. 546; canonical order $(\varphi, q; I, p)$.

Definition 1.7 (Oscillator actions)**.** For $p, q \in \mathbb{R}^b$ and $j \le b$: $R_j(p, q) = (p_j^2 + q_j^2)/2$.
Frozen Lean declaration: `KAM.Planetary.oscillatorAction`
Source: CP10 (A2") and CP11 (A_2), pp. 547 and 51.

Definition 1.8 (Fast shift)**.** For $k \in \mathbb{Z}^a$, $\mathrm{shift}(k) = \big(((2\pi k_i)_{i<a}, 0_b),\, 0\big) \in \mathrm{Ph}(a+b)$: the fast angles are translated by $2\pi k$, the slow position coordinates and all momenta are unchanged.
Frozen Lean declaration: `KAM.Planetary.fastShift`

Definition 1.9 (Fast frequency map)**.** For $H_0 : \mathbb{R}^a \to \mathbb{R}$ and $I \in \mathbb{R}^a$, $\omega_{H_0}(I)_i = DH_0(I)(e_i)$ (the Fréchet derivative is 0 where H_0 is not differentiable).
Frozen Lean declaration: `KAM.Planetary.fastFrequency`
Source: CP10 (A1), p. 546.

1.4. **Definitions for the properly degenerate theorem.**

Definition 1.10 (Cylinder and its fundamental cell)**.** For $V \subset \mathbb{R}^a$ and $r \in \mathbb{R}$,

$$\mathcal{C}(V, r) = \Big\{ z \in \mathrm{Ph}(a+b) : \ z_2^{\mathrm{f}} \in V, \ \ (z_2^{\mathrm{s}})_j^2 + (z_1^{\mathrm{s}})_j^2 < r^2 \ \ \forall j \le b \Big\},$$

that is $I \in V$ and $p_j^2 + q_j^2 < r^2$ for every j: a *product of discs*, not a box. Its fundamental cell for the fast angles of period 2π is

$$\mathcal{F}(V, r) = \big\{ z \in \mathcal{C}(V, r) : \ 0 \le z_{1,i}^{\mathrm{f}} < 2\pi \ \ \forall i \le a \big\}.$$

Uses require $r > 0$.
Frozen Lean declaration: `KAM.Planetary.cylinder, KAM.Planetary.cylinderCell`
Source: CP10 p. 561, below (55) (disc region; audit P1); CP10 footnote 1 (period 2π).

Definition 1.11 (The family $H_0 + \mu P$)**.** For $H_0 : \mathbb{R}^a \to \mathbb{R}$ and $P : \mathbb{R} \times \mathrm{Ph}(a+b) \to \mathbb{R}$, $H_\mu(z) = H_0(z_2^{\mathrm{f}}) + \mu P(\mu, z)$.
Frozen Lean declaration: `KAM.Planetary.degenerateHamiltonian`
Source: CP10 (1), p. 546.

Definition 1.12 (Averaged perturbation)**.** For $\mu \in \mathbb{R}$, $I \in \mathbb{R}^a$, $p, q \in \mathbb{R}^b$,

$$P_{\mathrm{av}}(\mu, I; p, q) = (2\pi)^{-a} \int_{[0,2\pi]^a} P\big(\mu, \mathrm{assemble}(I, \varphi, p, q)\big)\, d\varphi$$

(Bochner integral; $= 0$ for a non-integrable integrand, so every use asserts integrability).
Frozen Lean declaration: `KAM.Planetary.secularAverage`
Source: CP10 (3), p. 547.

Definition 1.13 (Fast diffeomorphism)**.** For H_0 and $W \subset \mathbb{R}^a$, HasFastDiffeo(H_0, W) holds if (i) H_0 is real analytic at every point of W (`AnalyticOnNhd`); (ii) $\omega_{H_0}(W)$ is open; (iii) there is $g : \mathbb{R}^a \to \mathbb{R}^a$, of class C^1 on $\omega_{H_0}(W)$, with $g(\omega_{H_0}(I)) = I$ for $I \in W$, and $g(w) \in W$ and $\omega_{H_0}(g(w)) = w$ for $w \in \omega_{H_0}(W)$. Thus ω_{H_0} is a bijection of W onto its open image with C^1 inverse g (explicit, not a totalized inverse).
Frozen Lean declaration: `KAM.Planetary.HasFastDiffeomorphism`
Source: CP10 (A1), p. 547; neighbourhood version by audit P2.

Definition 1.14 (Secular Birkhoff form (A2"))**.** Let $V \subset \mathbb{R}^a$, $r_0, \mu_0 \in \mathbb{R}$, and functions $P_0 : \mathbb{R} \times \mathbb{R}^a \to \mathbb{R}$, $\Omega : \mathbb{R} \times \mathbb{R}^a \to \mathbb{R}^b$, $\beta : \mathbb{R} \times \mathbb{R}^a \to \mathbb{R}^{b \times b}$. HasBirkhoff$(P, V, r_0, \mu_0, P_0, \Omega, \beta)$ holds if for every $\mu \in [0, \mu_0]$ and $I \in V$:

(i) for all $p, q \in \mathbb{R}^b$ with $p_j^2 + q_j^2 < r_0^2$ for all j, $\varphi \mapsto P(\mu, \mathrm{assemble}(I, \varphi, p, q))$ is integrable on $[0, 2\pi]^a$;

(ii) for every $\epsilon > 0$ there is δ with $0 < \delta \le r_0$ such that for all p, q with $p_j^2 + q_j^2 < \delta^2$ for all j,

$$\Big| P_{\mathrm{av}}(\mu, I; p, q) - \Big(P_0(\mu, I) + \sum_j \Omega_j(\mu, I) R_j + \tfrac{1}{2} \sum_{i,j} \beta_{ij}(\mu, I) R_i R_j \Big) \Big| \le \epsilon\, |(p, q)|_\infty^4,$$

where $R_j = R_j(p, q)$ and $|(p, q)|_\infty = \max(|p|_\infty, |q|_\infty)$. (The case $p = q = 0$ is included and fixes $P_{\mathrm{av}}(\mu, I; 0, 0) = P_0(\mu, I)$.)

Note that δ may depend on (μ, I, ϵ) in this definition; uniformity must be derived from the joint analyticity hypothesis of Theorem 1.28 (see `PLAN.md`, Part 5).
Frozen Lean declaration: `KAM.Planetary.HasSecularBirkhoffForm`
Source: CP10 (A2"), p. 547 (CP11 (A_2), p. 51), with the oscillator-disc guards of audit P1.

Definition 1.15 (Cylinder torus)**.** Let $H : \mathrm{Ph}(a + b) \to \mathbb{R}$, $U \subset \mathrm{Ph}(a + b)$, $w \in \mathbb{R}^{a+b}$, $K : \mathbb{R}^{a+b} \to \mathrm{Ph}(a + b)$. IsCylTorus$(H, U, w, K)$ holds if:

(i) K is real analytic on $\mathbb{R}^{a+b}$;
(ii) $K(\theta) \in U$ and H is differentiable at $K(\theta)$, for all θ;
(iii) $K(\theta + k) = K(\theta) + \mathrm{shift}\big((k_i)_{i<a}\big)$ for all $\theta \in \mathbb{R}^{a+b}$, $k \in \mathbb{Z}^{a+b}$;
(iv) if $K(\theta') = K(\theta) + \mathrm{shift}(k)$ for some $k \in \mathbb{Z}^a$, then $\theta' - \theta \in \mathbb{Z}^{a+b}$;
(v) $DK(\theta)$ is injective for all θ;
(vi) $\Omega(DK(\theta)u, DK(\theta)v) = 0$ for all θ, u, v;
(vii) for all $\theta, t \in \mathbb{R}$, $s \mapsto K(\theta + sw)$ has derivative $X_H(K(\theta + tw))$ at $s = t$.

No Diophantine condition is part of this predicate (see Definition 1.16).
Frozen Lean declaration: `KAM.Planetary.IsCylinderTorus`
Source: CP11 Thm. 11.1, pp. 50–51 (analytic conjugacy to a linear flow, Lagrangian); CP10 Thm. 1.3, p. 548.

Definition 1.16 (Cylinder Kolmogorov set)**.** For $H : \mathrm{Ph}(a + b) \to \mathbb{R}$, $V \subset \mathbb{R}^a$, $r, \mu, C \in \mathbb{R}$ and $S \subset \mathrm{Ph}(a + b)$, IsCylKol$(H, V, r, \mu, C, S)$ holds if S is Lebesgue measurable, $S \subset \mathcal{C}(V, r)$, and for every $z \in S$ there are $w \in \mathbb{R}^{a+b}$, $K : \mathbb{R}^{a+b} \to \mathrm{Ph}(a + b)$ and θ with: $K(\theta) = z$; $K(\mathbb{R}^{a+b}) \subset S$ (a whole torus); IsCylTorus$(H, \mathcal{C}(V, r), w, K)$; there are $\gamma > 0$ and $\tau > a + b$ with w (γ, τ)-Diophantine (the constant may depend on the torus); $|w^{\mathrm{f}}|_\infty \le C$ and $|w^{\mathrm{s}}|_\infty \le C\mu$.
Frozen Lean declaration: `KAM.Planetary.IsCylinderKolmogorovSet`
Source: CP11 Thm. 11.1, pp. 50–51: $\omega_1 = O(1)$, $\omega_2 = O(\mu)$, Diophantine.

1.5. **Definitions for the physical theorems.** Throughout, $G, m_0 > 0$ and the planet weights $m_i > 0$.

Definition 1.17 (Reference circular configurations and the tube)**.** For $m, a \in \mathbb{R}^n$ and $\phi \in \mathbb{R}^n$ (phases in $\mathbb{R}/\mathbb{Z}$),

$$\mathrm{circ}(\phi) = \Big(\big(a_i(\cos 2\pi\phi_i, \sin 2\pi\phi_i)\big)_i,\ \big(m_i\sqrt{Gm_0/a_i}\,(-\sin 2\pi\phi_i, \cos 2\pi\phi_i)\big)_i \Big) \in \mathrm{Ph}(2n)$$

(circular prograde Kepler configurations at coupling 0; $a_i > 0$, $G, m_0 > 0$ guard the square root and the division). For $\eta \in \mathbb{R}$ the *circular tube* is the open η-neighbourhood in the phase max norm, $U_\eta = \{z \in \mathrm{Ph}(2n) : \exists \phi \in \mathbb{R}^n,\ |z - \mathrm{circ}(\phi)|_\infty < \eta\}$.
Frozen Lean declaration: `KAM.Planetary.circularPoint, KAM.Planetary.circularTube`
Source: Fej04r Thm. 63, p. 49 (near circular coplanar Kepler motions); ours for the explicit reference point.

Definition 1.18 (No collision, pair bounds)**.** For $q \in \mathbb{R}^{2n}$: NoColl(q) means $|q^{(i)}|_2 > 0$ for all i and $|q^{(i)} - q^{(j)}|_2 > 0$ for all $i \neq j$ (heliocentric configurations, the Sun at the origin). For N points $Q_0, \dots, Q_{N-1} \in \mathbb{R}^2$, PairBounds$(d, D; Q)$ means $d \le |Q_j - Q_i|_2 \le D$ for all $i \neq j$.
Frozen Lean declaration: `KAM.Planetary.NoCollision, KAM.Planetary.PairBounds`
Source: Fej04r §6.1, pp. 45–47.

Definition 1.19 (Actual masses, barycentric reconstruction)**.** For $\mu \in \mathbb{R}$ let $\mathrm{mass}_\mu = (m_0, \mu m_1, \dots, \mu m_n) \in \mathbb{R}^{n+1}$ (index 0 is the Sun). For heliocentric positions $q \in \mathbb{R}^{2n}$ and scaled momenta $p \in \mathbb{R}^{2n}$,

$$x_0(q) = -\frac{\mu}{m_0 + \mu\sum_i m_i}\sum_i m_i q^{(i)}, \qquad x_i(q) = x_0(q) + q^{(i)},$$

$$v_0(p) = -\frac{\mu}{m_0}\sum_i p^{(i)}, \qquad v_i(p) = \frac{1}{m_i}p^{(i)} \qquad (1 \le i \le n),$$

with the Sun at index 0 and planet i at index i in the order of `Fin.cons`. The denominators are positive for $m_i > 0$, $\mu > 0$.
Frozen Lean declaration: `KAM.Planetary.actualMasses, KAM.Planetary.barycentricPositions, KAM.Planetary.barycentricVelocities`
Source: Fej04r §6.1, (28), pp. 46–47; zero total momentum and barycentre; Fej07e p. 3.

Definition 1.20 (Newton solutions)**.** Let $N \in \mathbb{N}$, $M \in \mathbb{R}^N$ and $Q, V : \mathbb{R} \to (\mathbb{R}^2)^N$. IsNewton$(G, M, Q, V)$ holds if for all $t \in \mathbb{R}$ and $i \neq j$ (all indices $< N$): (i) $|Q_j(t) - Q_i(t)|_2 > 0$; (ii) Q_i has derivative $V_i(t)$ at t; (iii) V_i has derivative

$$\sum_{j \neq i} \frac{GM_j}{|Q_j(t) - Q_i(t)|_2^3}\,(Q_j(t) - Q_i(t)) \quad \text{at } t$$

(the all-time Newton equations; `HasDerivAt`).
Frozen Lean declaration: `KAM.Planetary.IsNewtonSolution`
Source: Fej04r (1), §6.1, with G restored.

Definition 1.21 (Fixed-Sun solutions)**.** For $Q, V : \mathbb{R} \to (\mathbb{R}^2)^n$ and $\varepsilon \in \mathbb{R}$, the predicate IsFixedSol$(G, m_0, m, \varepsilon, Q, V)$ holds if for all t and $i \neq j$: $|Q_i(t)|_2 > 0$, $|Q_j(t) - Q_i(t)|_2 > 0$, Q_i has derivative $V_i(t)$ and V_i has derivative

$$-\frac{Gm_0}{|Q_i(t)|_2^3}\,Q_i(t) + \varepsilon\sum_{j \neq i}\frac{Gm_j}{|Q_j(t) - Q_i(t)|_2^3}(Q_j(t) - Q_i(t)).$$

The Sun is the constant origin.
Frozen Lean declaration: `KAM.Planetary.IsFixedSolution`
Source: Ours (fixed-center law obtained by differentiating H_{fix}), after Fej04r (28).

Definition 1.22 (Moving-Sun Hamiltonian)**.** For $z = (q, p) \in \mathrm{Ph}(2n)$ and $\mu \in \mathbb{R}$,

$$H_{\mathrm{mov}}(z) = \sum_i \Big[\big(\tfrac{1}{m_i} + \tfrac{\mu}{m_0}\big) \tfrac{|p^{(i)}|_2^2}{2} - \frac{G m_0 m_i}{|q^{(i)}|_2} \Big] + \mu \sum_{i<j} \Big[\frac{p^{(i)} \cdot p^{(j)}}{m_0} - \frac{G m_i m_j}{|q^{(i)} - q^{(j)}|_2} \Big].$$

It is the reduced Hamiltonian divided by μ, with G restored, and every term is kept: the Sun moves, and neither the self kinetic term $\mu|p^{(i)}|_2^2/(2m_0)$ nor the cross term is dropped. Equivalently $H_{\mathrm{mov}} = H_{\mathrm{fix}}\big|_{\varepsilon=\mu} + \mu|\sum_i p^{(i)}|_2^2/(2m_0)$ (a calculation for a later part).

Frozen Lean declaration: `KAM.Planetary.movingHamiltonian`

Source: Fej04r (28), pp. 46–47, with Fej07e p. 3 (indices $m_j m_k$ corrected) and G restored.

Definition 1.23 (Fixed-Sun Hamiltonian)**.** For $z = (q, p) \in \mathrm{Ph}(2n)$ and $\varepsilon \in \mathbb{R}$,

$$H_{\mathrm{fix}}(z) = \sum_i \Big[\frac{|p^{(i)}|_2^2}{2m_i} - \frac{G m_0 m_i}{|q^{(i)}|_2} \Big] - \varepsilon \sum_{i<j} \frac{G m_i m_j}{|q^{(i)} - q^{(j)}|_2}.$$

The Sun is the origin; only the direct planet–planet interaction is kept, with coefficient ε.

Frozen Lean declaration: `KAM.Planetary.fixedHamiltonian`

Source: Ours (Newtonian fixed-center model; "our variant of Fej04r (28)"); `statement.md`, P-FIXED.

Definition 1.24 (Physical torus)**.** For $H : \mathrm{Ph}(d) \to \mathbb{R}$, $w \in \mathbb{R}^d$ and $K : \mathbb{R}^d \to \mathrm{Ph}(d)$, $\mathrm{IsPhysTorus}(H, w, K)$ holds if:

(i) K is real analytic on $\mathbb{R}^d$;
(ii) H is differentiable at $K(\theta)$ for all θ;
(iii) $K(\theta + k) = K(\theta)$ for $k \in \mathbb{Z}^d$;
(iv) $K(\theta) = K(\theta')$ implies $\theta' - \theta \in \mathbb{Z}^d$;
(v) $DK(\theta)$ is injective;
(vi) $\Omega(DK(\theta)u, DK(\theta)v) = 0$;
(vii) there are $\gamma > 0$ and $\tau > d - 1$ with w (γ, τ)-Diophantine;
(viii) for all θ, t, $s \mapsto K(\theta + sw)$ has derivative $X_H(K(\theta + tw))$ at $s = t$.

(For such K the image is a compact embedded analytic Lagrangian torus. Track K's primary-torus predicate has affine winding and is not reused. Note $\tau > d - 1$ here against $\tau > a + b$ in Definition 1.16.)

Frozen Lean declaration: `KAM.Planetary.IsPhysicalTorus`

Source: HL19 (2.1), (2.3), pp. 6–7 (modified to period-one embeddings); CP09 Thm. 4, p. 8; ours.

Definition 1.25 (Moving-Sun Kolmogorov set)**.** For $G, m_0 > 0$, $m, a \in \mathbb{R}^n$, $\mu, \eta, d, D \in \mathbb{R}$, $S \subset \mathrm{Ph}(2n)$: $\mathrm{MovKol}(G, m_0, m, a, \mu, \eta, d, D, S)$ holds if S is measurable, $S \subset U_\eta$, and every $z \in S$ has w, K, θ with $K(\theta) = z$, $K(\mathbb{R}^{2n}) \subset S$, $\mathrm{IsPhysTorus}(H_{\mathrm{mov}}, w, K)$ and, for every $\phi \in \mathbb{R}^{2n}$:

(a) $\mathrm{NoColl}\big(K(\phi)_1\big)$;
(b) $\mathrm{IsNewton}\big(G, \mathrm{mass}_\mu,\ t \mapsto x(K(\phi + tw)_1),\ t \mapsto v(K(\phi + tw)_2)\big)$, where x, v are the barycentric reconstructions of Definition 1.19;
(c) $\mathrm{PairBounds}\big(d, D;\ x(K(\phi)_1)\big)$ (all pairs including the Sun; as ϕ is arbitrary this covers every point $K(\phi + tw)$ of every orbit, hence all times).

Frozen Lean declaration: `KAM.Planetary.MovingKolmogorovSet`

Source: Fej04r Prop. 77 and Fej07e §7, pp. 3–4, with CP09 Thm. 4; the physical conclusions are ours (`statement.md`).

Definition 1.26 (Fixed-Sun Kolmogorov set)**.** For ε, η, d, D and $S \subset \mathrm{Ph}(2n)$ the predicate $\mathrm{FixKol}(G, m_0, m, a, \varepsilon, \eta, d, D, S)$ holds

if S is measurable, $S \subset U_\eta$, and every $z \in S$ has w, K, θ with $K(\theta) = z$, $K(\mathbb{R}^{2n}) \subset S$, $\mathrm{IsPhysTorus}(H_{\mathrm{fix}}, w, K)$ and for every ϕ: $\mathrm{NoColl}(K(\phi)_1)$; $\mathrm{IsFixedSol}\big(G, m_0, m, \varepsilon,\ t \mapsto (K(\phi + tw)_1^{(i)})_i,\ t \mapsto (\frac{1}{m_i} K(\phi + tw)_2^{(i)})_i\big)$; $\mathrm{PairBounds}\big(d, D; (0, K(\phi)_1^{(1)}, \dots, K(\phi)_1^{(n)})\big)$, with the Sun at the origin as point 0.

Frozen Lean declaration: `KAM.Planetary.FixedKolmogorovSet`

Source: Ours (new deduction; Fej04r Lemma 64 and Prop. 77, CP09 Thm. 4).

Remark 1.27 (Lean totality). The frozen definitions use Lean total functions, and the frozen theorems guard each of them as follows; proofs must establish these guarded facts before using the defining formulas.

- *Fréchet derivatives* (ω_{H_0}, X_H, DK) are 0 at points of non-differentiability. ω_{H_0} is used only through `HasFastDiffeomorphism`, which asserts analyticity of H_0 on $W \supset \overline{V}$; X_H appears only in the clauses (vii)/(viii) of the torus predicates, which also assert differentiability of H on the range of K (for the cylinder tori at points of $\mathcal{C}(V, r)$, where H_μ is analytic because $r < r_0$ and $\mu \in [0, \mu_0]$, by the hypotheses on P and H_0); DK is guarded by analyticity of K. Injectivity of DK and $\Omega(DK\cdot, DK\cdot) = 0$ are asserted for the genuine derivative.
- *Bochner integrals* (P_{av}) are 0 for non-integrable integrands; Definition 1.14 (i) asserts integrability on the disc $p_j^2 + q_j^2 < r_0^2$, and every point used in (ii) has $\delta \le r_0$.
- *Square roots and divisions*: $\sqrt{\cdot}$ in $|\cdot|_2$ has nonnegative radicand; $1/|\cdot|_2$ and $1/|\cdot|_2^3$ occur only after the collision-free clauses (a), IsNewton (i), IsFixedSol (the first two conjuncts) in the same conjunction; in circ, $\sqrt{Gm_0/a_i}$ and $1/a_i$ are guarded by $G, m_0, a_i > 0$; in x_0 and v_0 the denominators $m_0 + \mu \sum m_i$ and m_0 are positive since $\mu > 0$; the factor $1/m_i$ needs $m_i > 0$.
- *Real powers and logarithms*: $r^6/(\log(1/r))^c$ with $0 < r < r_* < 1$ has positive base $\log(1/r) > 0$; $\sqrt{r}$ has $r > 0$; $|k|_1^\tau$ in the Diophantine condition has $|k|_1 \ge 1$ for $k \ne 0$.
- *Truncated subtraction*: the lower bound is written with `ENNReal.ofReal`, so it is the real number $1 - C\sqrt{r}$ when that is positive and 0 otherwise; the strict inequality $\mathrm{ofReal}(1 - C\sqrt{r})\ \mathrm{meas}\,\mathcal{F} < \mathrm{meas}(S \cap \mathcal{F})$ is then nontrivial only when $1 - C\sqrt{r} > 0$, but must be proved for every $r < r_*$ as stated (it follows from $0 < \mathrm{meas}(S \cap \mathcal{F})$ when $1 - C\sqrt{r} \le 0$).
- *Measure*: meas is Lebesgue (outer) measure; the sets $\mathcal{F}(V, r)$ are Borel and S is measurable by the conclusion; $\mathrm{meas}\,\mathcal{F}(V, r) < \infty$ since V is bounded.

1.6. **The frozen theorems.**

Theorem 1.28 (P-G: corrected properly degenerate KAM)**.** ***Data (all fixed first).*** *Integers $a, b \ge 1$. A set $V \subset \mathbb{R}^a$ that is open, connected and bounded; an open set $W \subset \mathbb{R}^a$ with $\overline{V} \subset W$; a boundary collar condition: there are $A, d > 0$ such that for all $0 < t < d$*

$$\mathrm{meas}\{I \in \mathbb{R}^a :\ \exists J \in \partial V,\ |I - J|_\infty < t\} \le A\, t.$$

Reals $r_0, \mu_0, B > 0$. Functions $H_0 : \mathbb{R}^a \to \mathbb{R}$, $P : \mathbb{R} \times \mathrm{Ph}(a + b) \to \mathbb{R}$, $P_0 : \mathbb{R} \times \mathbb{R}^a \to \mathbb{R}$, $\Omega : \mathbb{R} \times \mathbb{R}^a \to \mathbb{R}^b$, $\beta : \mathbb{R} \times \mathbb{R}^a \to \mathbb{R}^{b \times b}$.

Hypotheses.

(H1) *$(\mu, z) \mapsto P(\mu, z)$ is real analytic at every point of $[0, \mu_0] \times \overline{\mathcal{C}(V, r_0)}$ (jointly, as a function on $\mathbb{R} \times \mathrm{Ph}(a + b)$);*

(H2) *for $\mu \in [0, \mu_0]$, $z \in \mathcal{C}(V, r_0)$ and $k \in \mathbb{Z}^a$: $P(\mu, z + \mathrm{shift}(k)) = P(\mu, z)$;*

(H3) $\mathrm{HasFastDiffeo}(H_0, W)$ *(Definition 1.13);*

(H4) $\mathrm{HasBirkhoff}(P, V, r_0, \mu_0, P_0, \Omega, \beta)$ *(Definition 1.14);*

(H5) *$|\det \beta(\mu, I)| \ge B$ for all $\mu \in [0, \mu_0]$, $I \in V$;*

(H6) *$\beta(\mu, I)$ is symmetric for all $\mu \in [0, \mu_0]$, $I \in V$.*

Conclusion. *There exist r_*, C, c with $0 < r_* < r_0$, $r_* < 1$, $C > 0$, $c > 0$ such that for all r, μ with*

$$0 < r < r_*, \qquad 0 < \mu < \mu_0, \qquad \mu < \frac{r^6}{(\log(1/r))^c},$$

there is $S \subset \mathrm{Ph}(a+b)$ *with* $\mathrm{IsCylKol}(H_\mu, V, r, \mu, C, S)$ *(with* $H_\mu = H_0 + \mu P(\mu, \cdot)$ *of Definition 1.11) and*

$$0 < \mathrm{meas}(S \cap \mathcal{F}(V,r)), \qquad \mathrm{ofReal}(1 - C\sqrt{r})\ \mathrm{meas}\,\mathcal{F}(V,r) < \mathrm{meas}(S \cap \mathcal{F}(V,r)) < \mathrm{meas}\,\mathcal{F}(V,r).$$

Here r_*, C, c *depend on all the data and hypotheses above and not on* r *or* μ*;* C *is simultaneously the frequency bound of Definition 1.16 and the measure constant.*

Frozen Lean declaration: `KAM.Planetary.properlyDegenerateKAM`

Source: CP11 Thm. 11.1, pp. 50–51, whose proof is CP10 Thm. 1.3, p. 548 (with $c_* = 2b$*,* $\rho = r$*). It is* not *the printed theorem: the neighbourhood* W *(CP10 p. 556, (58) p. 562), the collar (CP10 p. 561, below (56)), the product of discs (CP10 p. 561, below (55)), joint analyticity of the family (CP10 footnote 3, p. 546 asks only* C^1 *uniform dependence) and the* 2π *period are corrected-contract premises (audit P1–P4; D-PG-1 to D-PG-5).*

Source gap: CP11 Thm. 11.1 states $\mathrm{meas}\,P_\epsilon > \mathrm{meas}\,K$ *without proof; the strict inequality* $\mathrm{meas}(S \cap \mathcal{F}) < \mathrm{meas}\,\mathcal{F}$ *is obtained in the plan by deleting a small positive-measure subfamily of tori (Part 5), which is possible because the lower bound has slack. CP10 Thm. 1.3 also needs* $P_{\mathrm{av}} - N = O(\epsilon^5)$ *uniformly (CP10 (60), p. 562), which follows from (H1), (H2), (H4) and compactness but is not proved in the source.*

Theorem 1.29 (P-PLANAR: two planar planets with a moving Sun)**.** *Let* $G, m_0 > 0$*; let* $m, a : \mathbb{F}in\,2 \to \mathbb{R}$ *(planets* $0, 1$*;* m *the planet weights,* a *the axes) with* $m_i > 0$ *and* $a_i > 0$ *for* $i = 0, 1$ *and* $a_0 < a_1$ *(strictly ordered axes, no numerical separation); let* $\eta > 0$*. Then there exist* μ_0, d, D *with* $0 < \mu_0$ *and* $0 < d < D$ *such that for every* μ *with* $0 < \mu < \mu_0$ *there is* $S \subset \mathrm{Ph}(2 \cdot 2)$ *with*

$$0 < \mathrm{meas}\,S < \infty \quad \textit{and} \quad \mathrm{MovKol}(G, m_0, m, a, \mu, \eta, d, D, S) \qquad (n = 2).$$

The order is: data (G, m_0, m, a, η)*, then* (μ_0, d, D)*, then* μ*, then* S*. The tori are* 4*-dimensional in the* 8*-dimensional* $\mathrm{Ph}(4)$*.*

Frozen Lean declaration: `KAM.Planetary.planarTwoPlanetKAM`

Source: CP09 Thm. 4, p. 8 with Fej04r Prop. 77, pp. 56–58, using the recorded source-proof repairs (`statement.md`*, P5–P7); Fej07e p. 3–4 supplies the smooth counterpart. No threshold from CP11 §8.1 (audit P4).*

Source gap: The route is not signed off as proved: the analytic weak-nondegeneracy gate (CP09 Thm. 4 resting on Rüssmann's theorem, CP09 Thm. 3, whose source is not pinned), the continuation of eigenvalue branches (Fej04r Lemma 76 is false as printed) and the closeness of the tori to the circular torus (CP09 Thm. 4 says only "close to"); see `PLAN.md`*.*

Theorem 1.30 (P-FIXED: n planar planets around a fixed Sun)**.** *Let* $n \geq 2$*,* $G, m_0 > 0$*,* $m, a \in \mathbb{R}^n$ *with all* $m_i > 0$ *and* $a_i > 0$*,* $a_i < a_j$ *whenever* $i < j$*, and* $\eta > 0$*. Then there exist* ε_0, d, D *with* $0 < \varepsilon_0$ *and* $0 < d < D$ *such that for every* ε *with* $0 < \varepsilon < \varepsilon_0$ *there is* $S \subset \mathrm{Ph}(n \cdot 2)$ *with*

$$0 < \mathrm{meas}\,S < \infty \quad \textit{and} \quad \mathrm{FixKol}(G, m_0, m, a, \varepsilon, \eta, d, D, S).$$

The order is data (n, G, m_0, m, a, η)*, then* (ε_0, d, D)*, then* ε*, then* S*.*

Frozen Lean declaration: `KAM.Planetary.fixedSunPlanetaryKAM`

Source: Ours: "our proposed Newtonian fixed-center deduction, not a published fixed-Sun statement"; Fej04r Lemma 64 (p. 49–50) and Prop. 77, CP09 Thm. 4.

Source gap: The secular identification for the fixed Sun (equal secular quadratic forms and frequency map up to functions of the actions) and the general-n induction of Fej04r Prop. 77 ("par un raisonnement analogue", p. 58) are proof obligations; no secular nondegeneracy is a premise.

Remark 1.31 (Scope of this part). The three statements are the only frozen theorems of track P. P-SPATIAL is a sketch only and has no Lean statement. The D22 additions are under statement audit and are treated in the reserved last part.

1.7. **D25 additions: low-AMD confinement and exponential safety.** Owner approval D25 (27 October 2026) froze two further theorems, P-AMD-CONFINEMENT and P-EXP-SAFE, together with their definitions. The Lean text is authoritative (`KAM/Frozen/Planetary/Theorems/{AmdConfinement,` and `KAM/Frozen/Planetary/Defs/`). Throughout this subsection $G = m_0 = 1$ (the D25 normalisation of the Hamiltonian), $n \geq 2$ is the number of planets, $m \in \mathbb{R}^n$ the planet weights, $\mu > 0$, $z = (q, p) \in \mathrm{Ph}(2n)$ and $q_i = \mathrm{body}(q, i)$, $p_i = \mathrm{body}(p, i)$. H_{mov} is the moving-Sun Hamiltonian of Definition 1.22 with $G = m_0 = 1$, parameter μ and weights m (Lean `movingHamiltonian 1 1 m μ`).

Definition 1.32 (Planar cross product)**.** For $u, v \in \mathbb{R}^2$: $\mathrm{cr}(u, v) = u_0 v_1 - u_1 v_0$ (the planar angular momentum of the canonical pair (u, v)).
Frozen Lean declaration: `KAM.Planetary.planeCross`
Source: LP17 (9), p. 2 (angular momentum $\sum u_i \wedge \tilde{u}_i$).

Definition 1.33 (Osculating Kepler quantities)**.** For $z = (q, p) \in \mathrm{Ph}(2n)$ and $i < n$ (all with $G = m_0 = 1$):

$$E_i(z) = \frac{|p_i|_2^2}{2m_i} - \frac{m_i}{|q_i|_2} \qquad (\texttt{keplerEnergy}),$$

$$a_i(z) = -\frac{m_i}{2E_i(z)} \qquad (\texttt{keplerAxis}),$$

$$\Lambda_i(z) = m_i \sqrt{a_i(z)} \qquad (\texttt{keplerAction}),$$

$$\mathbf{e}_i(z) = \Big(\frac{|p_i|_2^2}{m_i^2} - \frac{1}{|q_i|_2}\Big) q_i - \frac{q_i \cdot p_i}{m_i^2}\, p_i \in \mathbb{R}^2 \qquad (\texttt{keplerEccVector}),$$

as total Lean functions: a division by 0 is 0 and $\sqrt{x} = 0$ for $x < 0$. They are used only where the guard ProgradeElliptic (Definition 1.35) holds, where $|q_i|_2 > 0$, $E_i < 0$, $a_i > 0$ (see Remark 1.43).
Frozen Lean declaration: `KAM.Planetary.keplerEnergy, KAM.Planetary.keplerAxis, KAM.Planetary.keplerAction, KAM.Planetary.keplerEccVector`
Source: LP17 (7), (10), p. 2 ($\Lambda_k = m_k \sqrt{\mu a_k}$, $\mu = G m_0$), with $\mu = 1$; the eccentricity vector is the Laplace–Runge–Lenz vector.

Definition 1.34 (Total momentum and AMD)**.** $J(z) = \sum_{i<n} \mathrm{cr}(q_i, p_i)$ (`totalMomentum`) and $\mathcal{A}(z) = \sum_{i<n} \Lambda_i(z) - J(z)$ (`physicalAMD`).
Frozen Lean declaration: `KAM.Planetary.totalMomentum, KAM.Planetary.physicalAMD`
Source: LP17 (9)–(11), p. 2 (the angular momentum deficit; $\mathcal{A}$ is not a first integral of the full system, J is).

Definition 1.35 (Prograde elliptic points)**.** For $m \in \mathbb{R}^n$ and $z \in \mathrm{Ph}(2n)$, ProgradeElliptic(m, z) holds iff for every $i < n$: $0 < m_i$, $0 < |q_i|_2$, $E_i(z) < 0$ and $0 < \mathrm{cr}(q_i, p_i)$.
Frozen Lean declaration: `KAM.Planetary.ProgradeElliptic`

Definition 1.36 (Low-AMD region)**.** For $m, \ell, u \in \mathbb{R}^n$ and $A \in \mathbb{R}$,

$$\mathrm{LowAMD}(m, \ell, u, A) = \big\{ z \in \mathrm{Ph}(2n) : \ \mathrm{ProgradeElliptic}(m, z),\ \ell_i < \Lambda_i(z) < u_i \ \forall i,\ \mathcal{A}(z) < A \big\}.$$

Frozen Lean declaration: `KAM.Planetary.LowAMDRegion`
Source: Round 2 §2 (`docs/explorations/almost-every-round2.md`); LP17 (10)–(11).

Definition 1.37 (Geometric data $\mathcal{B}$ and admissibility)**.** A *box* $\mathcal{B}$ for n planets is the record of m, ℓ, ℓ^0, u^0, $u \in \mathbb{R}^n$ (fields `mass, lower, innerLower, innerUpper, upper`) and $e_*, A_0, A_1, d, R \in \mathbb{R}$ (fields `ecc, initialAMD, outerAMD, distance, radius`). It is *admissible* if

(B1) for all i: $0 < m_i$, $0 < \ell_i < \ell_i^0 < u_i^0 < u_i$;
(B2) $0 < e_* < 1$, $0 < A_0 < A_1$, and $A_1 < \ell_i\big(1 - \sqrt{1 - e_*^2}\big)$ for all i;
(B3) $0 < d$, $0 < R$, and for all i: $d \leq (\ell_i/m_i)^2(1 - e_*)$ and $(u_i/m_i)^2(1 + e_*) \leq R$;
(B4) for all $i < j$: $(u_i/m_i)^2(1 + e_*) + d \leq (\ell_j/m_j)^2(1 - e_*)$.

No condition ties $\mathcal{B}$ to μ. (The m_i are positive by (B1), so the divisions are guarded; $\sqrt{1-e_*^2}$ has radicand in $(0,1)$ by (B2).)
Frozen Lean declaration: `KAM.Planetary.AMDBox, KAM.Planetary.AMDBox.Admissible`
Source: Nek77 §12.2, p. 55 (domains of planetary motions), specialised to separated annuli and low AMD by the D25 decision; LP17 (10)–(11). The all-pairs radial margin (B4) is an explicit sufficient separation condition (docstring of the Lean definition).

Definition 1.38 (Moving-Sun solutions on a time set)**.** For $J \subset \mathbb{R}$ and $z : \mathbb{R} \to \mathrm{Ph}(2n)$, $\mathrm{IsMovSolOn}(m, \mu, J, z)$ holds iff for every $t \in J$: H_{mov} is differentiable at $z(t)$, z has the derivative $X_{H_{\mathrm{mov}}}(z(t))$ at t (two-sided, also at endpoints of J, z being defined on $\mathbb{R}$), and $\mathrm{NoColl}(z(t)_1)$. (The momentum-domain slot of `KAM.Iso.IsHamiltonianSolutionOn` is `Set.univ`.)
Frozen Lean declaration: `KAM.Planetary.IsMovingSolutionOn, KAM.Iso.IsHamiltonianSolutionOn`
Source: Ours; the frozen differentiability-guarded ODE predicate of track I reused.

Definition 1.39 (Newton solutions on a time set)**.** For $N \in \mathbb{N}$, $M \in \mathbb{R}^N$, $J \subset \mathbb{R}$ and $Q, V : \mathbb{R} \to (\mathbb{R}^2)^N$: $\mathrm{IsNewtonOn}(M, J, Q, V)$ holds iff for all $t \in J$ and all indices: (i) $|Q_j(t) - Q_i(t)|_2 > 0$ for $i \neq j$; (ii) Q_i has derivative $V_i(t)$ at t; (iii) V_i has derivative $\sum_{j \neq i} \dfrac{M_j}{|Q_j(t) - Q_i(t)|_2^3}(Q_j(t) - Q_i(t))$ at t. It is Definition 1.20 with $G = 1$ and the quantifier over all $t \in \mathbb{R}$ replaced by $t \in J$.
Frozen Lean declaration: `KAM.Planetary.IsNewtonSolutionOn`
Source: Fej04r (1), §6.1, p. 46, with $G = 1$.

Definition 1.40 (Uniqueness among local solutions)**.** For $J \subset \mathbb{R}$ and $z : \mathbb{R} \to \mathrm{Ph}(2n)$, $\mathrm{MovUniqueOn}(m, \mu, J, z)$ holds iff for every open convex $U \subset \mathbb{R}$ with $0 \in U$ and every $w : \mathbb{R} \to \mathrm{Ph}(2n)$ with $w(0) = z(0)$ and $\mathrm{IsMovSolOn}(m, \mu, U, w)$: $w(t) = z(t)$ for all $t \in U \cap J$. (No global flow is presupposed.)
Frozen Lean declaration: `KAM.Planetary.MovingUniqueOn`

Definition 1.41 (Stopped action bound)**.** For a box $\mathcal{B}$, $\mu, T, \delta \in \mathbb{R}$ and $z_0 \in \mathrm{Ph}(2n)$, $\mathrm{StoppedBound}(\mathcal{B}, \mu, T, \delta, z_0)$ holds iff for all $l, r \in \mathbb{R}$ and $z : \mathbb{R} \to \mathrm{Ph}(2n)$ with $l \leq 0 \leq r$, $l < r$, $z(0) = z_0$, $\mathrm{IsMovSolOn}(m, \mu, [l, r], z)$ and $z(t) \in \mathrm{LowAMD}(m, \ell, u, A_1)$ for all $t \in [l, r]$ (the outer region), one has

$$\left|\Lambda_i(z(t)) - \Lambda_i(z_0)\right| \leq \delta \qquad \text{for all } t \in [l, r] \cap [-T, T] \text{ and all } i.$$

Both one-sided intervals are allowed; T, δ are arbitrary reals here.
Frozen Lean declaration: `KAM.Planetary.StoppedActionBound`
Source: Nek77 Thm. 4.4, p. 30 (stopped form: arbitrary C, time $\min(C, T)$) and Lemma 12.5.E, p. 62; round 3 (`almost-every-round3-refutation.md`, item 1c).

Definition 1.42 (Exponents)**.** For $n \in \mathbb{N}$: $\alpha_n = \dfrac{2}{6n^2 - 3n + 14}$ (`planetaryTimeExponent`) and $\beta_n = \frac{3}{2}\alpha_n$ (`planetaryDriftExponent`); n is the number of planets.
Frozen Lean declaration: `KAM.Planetary.planetaryTimeExponent, KAM.Planetary.planetaryDriftExponent`
Source: Nek77 (12.4)–(12.5), p. 56 (page image read: $a = 2/(6s^2 - 3s + 14)$, $b = \frac{3}{2}a$, s the number of planets); equal to [N:`bpN:p0:partial-exponents`] with n fast actions.

Remark 1.43 (Lean totality for the D25 objects). $E_i, \Lambda_i, a_i, \mathbf{e}_i$ divide by $|q_i|_2$, m_i, E_i and take a square root; each occurrence in the frozen theorems is under ProgradeElliptic (through $z \in \mathrm{LowAMD}$): in `StoppedActionBound` for all $t \in [l, r]$ (hence for $t = 0$, $z_0 = z(0)$); in the conclusions of both theorems for $z(t) \in \mathrm{LowAMD}(\ell, u, A_1)$, $t \in [-T, T]$, and for $z_0 \in \mathrm{LowAMD}(\ell^0, u^0, A_0)$. There $a_i > 0$ and $\Lambda_i = m_i\sqrt{a_i} > 0$. IsNewtonOn divides by $|Q_j - Q_i|_2^3$ only after clause (i). The exponents $\mu^{-\alpha_n}$, μ^{β_n} are real powers of $\mu > 0$; $6n^2 - 3n + 14 > 0$.

Theorem 1.44 (P-AMD-CONFINEMENT: deterministic stopped-input helper)**.** *Let $n \geq 2$, $\mathcal{B}$ an admissible box, $\mu, T, \delta > 0$ with $\ell_i + 2\delta < \ell_i^0$ and $u_i^0 + 2\delta < u_i$ for all i, and $A_0 + 2n\delta < A_1$. Let $z_0 \in \mathrm{LowAMD}(m, \ell^0, u^0, A_0)$ with $\mathrm{StoppedBound}(\mathcal{B}, \mu, T, \delta, z_0)$. Then there is $z : \mathbb{R} \to \mathrm{Ph}(2n)$ with $z(0) = z_0$ and*

(a) $\mathrm{IsMovSolOn}(m, \mu, [-T, T], z)$ *and* $\mathrm{MovUniqueOn}(m, \mu, [-T, T], z)$*;*
(b) $\mathrm{IsNewtonOn}(\mathrm{mass}_\mu, [-T, T], t \mapsto x(z(t)_1), t \mapsto v(z(t)_2))$ *with the barycentric reconstructions of Definition 1.19 for $G = m_0 = 1$;*
(c) for every $t \in [-T, T]$: $z(t) \in \mathrm{LowAMD}(m, \ell, u, A_1)$; $J(z(t)) = J(z(0))$; for all i: $|\mathbf{e}_i(z(t))|_2 < e_*$ and $d \leq |q_i(t)|_2 \leq R$; and $\mathrm{PairBounds}(d, 2R; x(z(t)_1))$;
(d) for every $t \in [-T, T]$: $\mathcal{A}(z(t)) \leq A_0 + n\delta$ and $|\Lambda_i(z(t)) - \Lambda_i(z_0)| \leq \delta$ for all i.

The order of choice: $\mathcal{B}, \mu, T, \delta$, then z_0 with the stopped bound; there is no dynamical premise except the stopped bound.
Frozen Lean declaration: `KAM.Planetary.amdConfinement`
Source: Round 2 §2, Lemmas 1, 2, 5, 6, with the stopped-source correction of round 3 (`almost-every-round2.md`*, *`almost-every-round3-refutation.md`*); LP17 (9)–(11), p. 2;* `docs/tracks/P/statement.md`*, "P-AMD-CONFINEMENT".*

Theorem 1.45 (P-EXP-SAFE: planar moving-Sun exponential safety for every low-AMD datum)**.** *Let $n \geq 2$ and $\mathcal{B}$ an admissible box. There are $\mu_*, c, C > 0$ such that for every $\mu \in (0, \mu_*)$ and every $z_0 \in \mathrm{LowAMD}(m, \ell^0, u^0, A_0)$ there is $z : \mathbb{R} \to \mathrm{Ph}(2n)$ with $z(0) = z_0$, such that, with $T = \exp(c\mu^{-\alpha_n})$,*

(a) the clauses (a), (b), (c) of Theorem 1.44 hold on $[-T, T]$ (existence, uniqueness, Newton equations on $[-T, T]$, outer region, conservation of J, eccentricity, radius, pair bounds); and
(b) $|a_i(z(t)) - a_i(z_0)| \leq C\mu^{\beta_n}$ for all $|t| \leq T$ and all i.

The order of choice is $n, \mathcal{B}$; then μ_*, c, C; then μ and z_0 (the exponents are α_n, β_n of Definition 1.42).
Frozen Lean declaration: `KAM.Planetary.planetaryExponentialSafety`
Source: Nek77 §12, Thm. 12.3, pp. 55–56, (12.3)–(12.5); `docs/tracks/P/statement.md`*, "P-EXP-SAFE". The symmetric horizon, the heliocentric canonical elements and the explicit geometric bounds are documented deductions (Lean docstring).*

2. Hamiltonian tools for tori: charts, period conversion, Diophantine conversion, Liouville

Track P produces tori in coordinates that are not the frozen Cartesian ones: CP's action–angle variables with angles of period 2π, oscillator action–angle discs, the Poincaré chart. This part supplies, once, the four tools that carry a torus found in such a chart to a frozen predicate ([P:`bpP:p0:cyltorus`], [P:`bpP:p0:phystorus`]):

- `torus-transport` (Theorem 2.9): an analytic embedding Ψ with $\Psi^*\Omega = c\,\Omega$ that is compatible with the angle lattice sends a lattice torus of $(H \circ \Psi)/c$ to a torus of H, clause by clause of the two frozen predicates;
- `period-convert` (Lemma 2.11 and Corollaries 2.12, 2.13): the passage between 2π-angles and the frozen period-one parameter is the chart $S_{2\pi}$, so the frozen frequency is $w = \tilde{\omega}/2\pi$ and the dictionary of [P:`bpP:p3:period-dictionary`] is a special case;
- `diophantine-convert` (Lemmas 2.15–2.16): CP10 (19) and CP09 (6) imply the frozen `KAM.IsDiophantine` with the exponents the two torus predicates require;
- `measure-symplectic` (Lemmas 2.18–2.21): conformally symplectic charts scale Lebesgue measure by c^d, and a union of whole tori that is a Lipschitz image is measurable with controlled measure.

The `ode` item is cited, not repeated (§2.7). Everything here is standard; the point of writing it is to fix signs, lattices and Lean totality exactly as frozen.

2.1. **Conventions and external facts.** Phase space is $\mathrm{Ph}(d) = \mathbb{R}^d \times \mathbb{R}^d$, points $z = (z_1, z_2)$ (position, momentum), max norms, Lebesgue measure ([P:bpP:p0:conventions]). Let $M_0 = \left(\begin{smallmatrix} 0 & -\mathbf{1} \\ \mathbf{1} & 0 \end{smallmatrix}\right)$ and $J_0 = -M_0$. The frozen form ([P:bpP:p0:hamsymp]) is $\Omega(v, w) = \sum_i (v_{2,i} w_{1,i} - v_{1,i} w_{2,i}) = v^\top M_0 w$. For a linear form $\ell : \mathrm{Ph}(d) \to \mathbb{R}$ put $X_\ell := \big((\ell(0, e_i))_i, (-\ell(e_i, 0))_i\big)$; thus $X_H(z) = X_{DH(z)}$, and $X_H(z) = 0$ if H is not differentiable at z (Lean totality, [P:bpP:p0:totality]). Write $\iota_k := \big((k_i)_{i<d}, 0\big) \in \mathrm{Ph}(d)$ for $k \in \mathbb{Z}^d$ (integers coerced to reals: an integer translation of the positions; the nested form $((k_f, 0_b), 0)$ is reserved for the split $d = a + b$ and for `fastShift`) and, for $c > 0$, $S_c(z_1, z_2) := (c z_1, z_2)$.

(X1) Chain rule for `fderiv` and `HasDerivAt`; compositions of real analytic maps are real analytic (`AnalyticAt.comp` and `AnalyticOnNhd.comp`, `Analysis/Analytic/Composition.lean`; the latter needs the range of the inner map inside the set where the outer map is analytic, here $K(\mathbb{R}^d) \subset O$); sums and products of analytic maps are analytic.

(X2) Inverse function theorem: a C^1 map Ψ with $D\Psi(z_0)$ invertible is a homeomorphism from a neighbourhood of z_0 onto an open set, with C^1 inverse (Mathlib: `HasStrictFDerivAt.toOpenPartialHomeomorph` and `HasStrictFDerivAt.to_localInverse` (`Analysis/Calculus/InverseFunctionTheorem/FDeriv.lean`); for C^1 regularity of the inverse `ContDiffAt.to_localInverse` (`.../ContDiff.lean`). A C^1 map has a strict derivative, and a bijective linear map of $\mathbb{R}^{2d}$ is a continuous linear equivalence).

(X3) Area formula (`MeasureTheory/Function/Jacobian.lean`; the derivative field is a family of continuous linear maps with `HasFDerivWithinAt` on the set, supplied by analyticity of Ψ on O): `MeasureTheory.addHaar_image_le_lintegral_abs_det_fderiv` (A *Borel*, no injectivity) gives $\mathrm{meas}\,\Psi(A) \le \int_A |\det D\Psi|$; `MeasureTheory.lintegral_abs_det_fderiv_eq_addHaar_image` ($A$ Borel, Ψ injective on A) and its subscript-0 version (name with the suffix `image`$_0$, i.e. `addHaar_image` followed by the subscript-zero character; A `NullMeasurableSet`) give equality. Also `addHaar_image_eq_zero_of_differentiableOn_of_addHaar_eq_zero`: differentiable images of null sets are null.

(X4) A Lebesgue measurable subset of $\mathbb{R}^N$ is the union of an F_σ set and a null set (inner regularity, `MeasureTheory/Measure/Regular.lean`); a closed set is σ-compact; continuous images of compact sets are compact. *Null sets need not be Borel*: "measurable" below means `NullMeasurableSet` (Lebesgue) unless the word Borel is used, and the frozen predicates need `MeasurableSet`, i.e. Borel. Injective continuous images of Borel sets are Borel: `MeasurableSet.image_of_continuousOn_injOn` (`MeasureTheory/Constructions/Polish/Basic.lean`).

(X5) $\det(AB) = \det A \det B$, $\det A^\top = \det A$, and $\det M_0 = 1$.

Track K: [K:bp:p1:holomorphic-analytic], [K:bp:p1:real-restriction], [K:bp:p1:hamiltonian-real], [K:bp:p1:lip-image], [K:bp:p1:mcshane].

Lemma 2.1 (Algebra of Ω and X_H)**.** *(a) $M_0^\top = -M_0$, $M_0^2 = -\mathbf{1}$, Ω is nondegenerate, and $\Omega(w, X_\ell) = \ell(w)$ for all w and all linear ℓ. Hence X_ℓ is the unique vector with this property, and $\Omega(w, X_H(z)) = DH(z)w$ for every z (both sides vanish if H is not differentiable at z). (b) If A is a real $2d \times 2d$ matrix with $\Omega(Au, Av) = c\,\Omega(u, v)$ for all u, v ($c \neq 0$), then A is invertible, $|\det A| = |c|^d$, and $X_{\ell \circ A} = c\,A^{-1} X_\ell$ for every linear form ℓ. (c) The relation $\Omega(Au, Av) = c\Omega(u, v)$ is equivalent to $A^\top(-M_0)A = c(-M_0)$: it does not depend on the sign convention for the symplectic form (CP10's $dI \wedge d\varphi$ is Ω; CP09 Thm. 3 prints $dx \wedge dy$ with x the angles, which is $-\Omega$; both define the same class of maps, and the vector field is the frozen X_H, not a function of the printed sign).*

Proof. **Step S2.1: Matrices.** Direct from the block form; $X_\ell = J_0 \nabla \ell$ so $M_0 X_\ell = -M_0^2 \nabla \ell = \nabla \ell$, i.e. $\Omega(w, X_\ell) = w^\top M_0 X_\ell = \ell(w)$. Uniqueness: M_0 invertible.

Step S2.2: (b). $A^\top M_0 A = cM_0$ and (X5) give $(\det A)^2 = c^{2d}$. For all w: $\Omega(w, cA^{-1}X_\ell) = \Omega(Aw, AA^{-1}X_\ell) = \Omega(Aw, X_\ell) = \ell(Aw) = (\ell \circ A)(w)$, so $cA^{-1}X_\ell = X_{\ell\circ A}$ by uniqueness. (c) is the displayed matrix identity. □

Remark 2.2 (Reordering of coordinates). Part 3 writes points as (J, θ) (action, angle). The relabelling $\pi(J,\theta) = (\theta, J)$ to the frozen order (position, momentum) = (angle, action) turns $\dot\theta = \partial_J H$, $\dot J = -\partial_\theta H$ into X_H of [P:bpP:p0:hamsymp] and the form $dJ \wedge d\theta$ into Ω; no sign changes. All statements of Part 3 are read in the frozen order after applying π.

2.2. **Conformally symplectic charts.**

Definition 2.3 (Chart)**.** Let $O \subset \mathrm{Ph}(d)$ be open, $c > 0$. A *c-chart on O* is a map $\Psi : \mathrm{Ph}(d) \to \mathrm{Ph}(d)$ (total) that is real analytic at every point of O and satisfies $\Omega(D\Psi(z)u, D\Psi(z)v) = c\,\Omega(u,v)$ for $z \in O$, $u, v \in \mathrm{Ph}(d)$. A 1-chart is a *symplectic* chart. For $H : \mathrm{Ph}(d) \to \mathbb{R}$ the *pulled-back Hamiltonian* is $H^\Psi := c^{-1}\,(H \circ \Psi)$.

By Lemma 2.1(b), $D\Psi(z)$ is invertible on O, so Ψ is a local analytic diffeomorphism there ((X2); analyticity of the local inverse is not needed below).

Lemma 2.4 (Charts and Hamiltonian fields)**.** *Let Ψ be a c-chart on O, $H : \mathrm{Ph}(d) \to \mathbb{R}$, $z \in O$. Then H^Ψ is differentiable at z iff H is differentiable at $\Psi(z)$, and then*

$$D\Psi(z)\, X_{H^\Psi}(z) = X_H(\Psi(z)).$$

If H is not differentiable at $\Psi(z)$ then both fields are 0 (totality) and the identity still holds.

Proof. **Step S2.3: Differentiability.** ($\Leftarrow$) chain rule (X1). ($\Rightarrow$) by (X2) there is a C^1 local inverse Ψ_{loc}^{-1} near $\Psi(z)$ and $H = c\,H^\Psi \circ \Psi_{\mathrm{loc}}^{-1}$ near $\Psi(z)$, a composition of differentiable maps.

Step S2.4: The field. With $\ell = DH(\Psi z)$ and $A = D\Psi(z)$, $D(H^\Psi)(z) = c^{-1}\ell \circ A$, so $X_{H^\Psi}(z) = c^{-1}X_{\ell\circ A} = A^{-1}X_\ell$ by Lemma 2.1(b). Hence $AX_{H^\Psi}(z) = X_\ell = X_H(\Psi z)$. In the non-differentiable case both $D(H^\Psi)(z)$ and $DH(\Psi z)$ are 0 by the Lean convention. □

Corollary 2.5 (Orbits transfer)**.** *Let $y : J \to O$ ($J \subset \mathbb{R}$ an interval) satisfy: H^Ψ is differentiable at $y(t)$ and y has derivative $X_{H^\Psi}(y(t))$ at t, for all $t \in J$. Then $x = \Psi \circ y$ has derivative $X_H(x(t))$ at every $t \in J$ and H is differentiable at $x(t)$.*

Proof. Chain rule (X1) and Lemma 2.4. □

Remark 2.6 (What is new compared with track N). [N:bpN:p5:symplectic] and [N:bpN:p5:canonical](b),(c) give $D\Phi X_{H\circ\Phi} = X_H \circ \Phi$ and the orbit transfer for the time-one map $\Phi = X_\phi^1$ of a *holomorphic* generating function on a complex domain, with symplecticity *derived* from the flow. Here the chart is an arbitrary real analytic map (Poincaré chart, polar chart of oscillators, scalings) with symplecticity *assumed*, in the frozen real setting, with the factor c, with the frozen sign of Ω (Lemma 2.1) and with the Lean-totality guard (differentiability of H is transported, not assumed). The algebra is that of [N:bpN:p5:canonical](b); it is repeated only because the hypotheses differ. Time-one maps of Part N are charts in the sense of Definition 2.3 with $c = 1$ (Lemma 2.7).

Lemma 2.7 (Holomorphic charts, real restriction)**.** *Let $c > 0$, $V \subset \mathbb{C}^{2d}$ be open, $\Psi : V \to \mathbb{C}^{2d}$ holomorphic, real at real points ($\Psi(V \cap \mathbb{R}^{2d}) \subset \mathbb{R}^{2d}$), and assume $D\Psi(z)^\top M_0 D\Psi(z) = cM_0$ (complex matrices) for $z \in V \cap \mathbb{R}^{2d}$. Identify $\mathbb{R}^{2d} \subset \mathbb{C}^{2d}$ through the real embedding; $O = V \cap \mathbb{R}^{2d}$ is open in* $\mathrm{Ph}(d)$*. Then any total map $\mathrm{Ph}(d) \to \mathrm{Ph}(d)$ that agrees with Ψ on O (choose any extension off O; no property off O is ever used) is a c-chart on O, with real derivative $D\Psi(z)|_{\mathbb{R}^{2d}}$.*

Proof. [K:bp:p1:real-restriction] (via [K:bp:p1:holomorphic-analytic]) gives analyticity and that the real derivative is the restriction of the complex one; the identity of matrices is the defining condition of Definition 2.3 (Lemma 2.1(c)). □

2.3. **Transport of tori.**

Definition 2.8 (Lattice torus in a chart)**.** Let $O \subset \mathrm{Ph}(d)$ be open, $G : \mathrm{Ph}(d) \to \mathbb{R}$, $w \in \mathbb{R}^d$, $K : \mathbb{R}^d \to \mathrm{Ph}(d)$. $\mathrm{LT}(G, O, w, K)$ holds if

(L1) K is real analytic on $\mathbb{R}^d$, $K(\theta) \in O$ and G is differentiable at $K(\theta)$ for all θ;
(L2) $K(\theta + k) = K(\theta) + \iota_k$ for $\theta \in \mathbb{R}^d$, $k \in \mathbb{Z}^d$;
(L3) $K(\theta) = K(\theta')$ implies $\theta' - \theta \in \mathbb{Z}^d$;
(L4) $DK(\theta)$ is injective;
(L5) $\Omega(DK(\theta)u, DK(\theta)v) = 0$ for all u, v;
(L6) $s \mapsto K(\theta + sw)$ has derivative $X_G(K(\theta + tw))$ at $s = t$, for all θ, t.

Note that (L2) is the period-one form of "angles are shifted by integers" in the chart; the frozen cylinder predicate differs from it only by the image shift in clause (iii), and the frozen physical predicate by periodicity.

Theorem 2.9 (Torus transport)**.** *Let $c > 0$, $O \subset \mathrm{Ph}(d)$ open, Ψ a c-chart on O, $H : \mathrm{Ph}(d) \to \mathbb{R}$, $G = H^\Psi = c^{-1} H \circ \Psi$, $w \in \mathbb{R}^d$, and K with $\mathrm{LT}(G, O, w, K)$. Assume the lattice compatibility*

(Q1) $O + \iota_k = O$ and $\Psi(z + \iota_k) = \Psi(z) + \sigma_k$ for $z \in O$, $k \in \mathbb{Z}^d$, for vectors $\sigma_k \in \mathrm{Ph}(d)$;
(Q2) (injectivity modulo the lattice) if $z, z' \in O$, $k \in \mathbb{Z}^d$ and $\Psi(z') = \Psi(z) + \sigma_k$, then $z' = z + \iota_m$ for some $m \in \mathbb{Z}^d$.

Put $\mathcal{K} := \Psi \circ K$. Then:

(a) (cylinder) *Let $d = a + b$ and $\sigma_k = \mathrm{shift}((k_i)_{i<a})$ (Definition 1.8). If $U \supset \Psi(K(\mathbb{R}^d))$, then* $\mathrm{IsCylTorus}(H, U, w, \mathcal{K})$ *(Definition 1.15).*
(b) (physical) *Let $\sigma_k = 0$ and assume w is (γ, τ)-Diophantine for some $\gamma > 0$, $\tau > d - 1$. Then* $\mathrm{IsPhysTorus}(H, w, \mathcal{K})$ *(Definition 1.24).*

In both cases $\mathcal{K}(\theta + tw)$ is the transport of the chart orbit, $\mathcal{K}(\mathbb{R}^d) = \Psi(K(\mathbb{R}^d))$, and $D\mathcal{K}(\theta) = D\Psi(K\theta)DK(\theta)$.

Frozen Lean declaration: `KAM.Planetary.IsCylinderTorus, KAM.Planetary.IsPhysicalTorus`

Source: Standard (change of canonical coordinates). Ours as a statement for the frozen predicates; no source page.

Proof. Write $\mathcal{K} = \Psi \circ K$.

Step S2.5: (i) Analyticity, range, differentiability. $K(\mathbb{R}^d) \subset O$ and Ψ is analytic on O, so $\mathcal{K}$ is analytic on $\mathbb{R}^d$ by (X1). $G = H^\Psi$ is differentiable at $K\theta$ by (L1), hence H is differentiable at $\mathcal{K}(\theta)$ by Lemma 2.4. Range: $\mathcal{K}(\theta) \in \Psi(K(\mathbb{R}^d)) \subset U$ in (a).

Step S2.6: (ii) Lattice behaviour. By (L2) and (Q1): $\mathcal{K}(\theta + k) = \Psi(K\theta + \iota_k) = \mathcal{K}(\theta) + \sigma_k$. In (a) this is clause (iii) of Def. 1.15 (with $k \in \mathbb{Z}^{a+b}$, shift by $k_f = (k_i)_{i<a}$); in (b) it is periodicity, clause (iii) of Def. 1.24.

Step S2.7: (iii) Injectivity modulo the lattice. (a): let $k \in \mathbb{Z}^a$ and $\mathcal{K}(\theta') = \mathcal{K}(\theta) + \mathrm{shift}(k)$. Then $\Psi(K\theta') = \Psi(K\theta) + \sigma_{(k,0)}$, so by (Q2) $K\theta' = K\theta + \iota_m = K(\theta + m)$ for some $m \in \mathbb{Z}^d$ (by (L2)), and (L3) gives $\theta' - (\theta + m) \in \mathbb{Z}^d$, i.e. $\theta' - \theta \in \mathbb{Z}^d$. (b): $\mathcal{K}(\theta') = \mathcal{K}(\theta)$ is the case $k = 0$ ($\sigma_0 = 0$) of the same argument.

Step S2.8: (iv) Derivative and Lagrangian property. $D\mathcal{K}(\theta) = D\Psi(K\theta)DK(\theta)$; $D\Psi(K\theta)$ is invertible (Lemma 2.1(b)), so $D\mathcal{K}(\theta)$ is injective by (L4). $\Omega(D\mathcal{K}u, D\mathcal{K}v) = c\,\Omega(DKu, DKv) = 0$ by Definition 2.3 and (L5).

Step S2.9: (v) The flow clause. Fix θ, t and let $y(s) = K(\theta + sw) \in O$. By (L6) y has derivative $X_G(y(t))$ at t and G is differentiable at $y(t)$; Corollary 2.5 (pointwise) gives that $s \mapsto \Psi(y(s)) = \mathcal{K}(\theta + sw)$ has derivative $X_H(\mathcal{K}(\theta + tw))$ at t.

Step S2.10: (vi) Diophantine clause. In (b) it is the hypothesis; in (a) none is part of IsCylTorus (see Def. 1.16 for where it is imposed; Lemmas 2.15–2.16). □

Remark 2.10 (Typical charts). (1) The polar chart of one oscillator $(\psi, R) \mapsto (q,p) = \sqrt{2R}(\sin\psi, \cos\psi)$ on $\{R>0\}$ is a 1-chart ($dp\wedge dq = dR\wedge d\psi$, which is Ω in the frozen order), invariant under $\psi \mapsto \psi + 2\pi$; Part 5 composes it with $S_{2\pi}$ below. (2) S_c is a c-chart (Lemma 2.11). (3) Composition: if Ψ_1 is a c_1-chart on O_1, Ψ_2 a c_2-chart on $O_2 \supset \Psi_1(O_1)$, then $\Psi_2\circ\Psi_1$ is a c_1c_2-chart on O_1 (chain rule), and $(H\circ\Psi_2)^{\Psi_1}$ with the factor c_1^{-1} followed by c_2^{-1} equals $H^{\Psi_2\circ\Psi_1}$.

2.4. **Period conversion.** CP10 (footnote 1, p. 546: "$\mathbb{T}^n$ denotes the standard n-dimensional flat torus $\mathbb{R}^n/(2\pi\mathbb{Z}^n)$") and CP09 (Thm. 3, p. 6: "$\mathbb{T}^n$ the usual n-dimensional torus $\mathbb{R}^n/2\pi\mathbb{Z}^n$") use angles of period 2π. The frozen predicates use a parameter $\theta\in\mathbb{R}^d$ of period one. The relation is one explicit chart.

Lemma 2.11 (Period conversion)**.** *Let $c>0$ and $S_c(z_1,z_2)=(cz_1,z_2)$.*

- *(a) S_c is a c-chart on all of* $\mathrm{Ph}(d)$ *(globally analytic, $\Omega(S_cu,S_cv)=c\,\Omega(u,v)$), $S_c(z+\iota_k)=S_c(z)+(ck,0)$, and $|\det DS_c|=c^d$.*
- *(b)* (Hamiltonian dictionary) *For $\tilde H:\mathrm{Ph}(d)\to\mathbb{R}$ the pulled-back Hamiltonian is $\tilde H^{S_c}(z_1,z_2)=c^{-1}\tilde H(cz_1,z_2)$, and for every z: $S_c\,X_{\tilde H^{S_c}}(z)=X_{\tilde H}(S_cz)$. With $c=2\pi$ and $(I,\phi)=(z_2,2\pi z_1)$ this is $H(I,\theta)=\tilde H(I,2\pi\theta)/2\pi$ of Lemma 4.5: orbits of H (angle θ) and of $\tilde H$ (angle $\phi=2\pi\theta$) correspond with the* same *time; $\omega=\tilde\omega/2\pi$; Lagrangian for Ω iff after S_c (factor c).*
- *(c)* (Parameter conversion) *Let $\tilde\kappa:\mathbb{R}^d\to\mathrm{Ph}(d)$ (parameter of period 2π), $\tilde\omega\in\mathbb{R}^d$, $K(\theta):=\tilde\kappa(2\pi\theta)$, $w:=\tilde\omega/2\pi$. Then K analytic iff $\tilde\kappa$ is; $DK(\theta)=2\pi D\tilde\kappa(2\pi\theta)$, so injectivity of DK and the Lagrangian property (factor $4\pi^2$) hold for K iff for $\tilde\kappa$; $\tilde\kappa(\vartheta+2\pi k)=\tilde\kappa(\vartheta)+\tau_k$ for all ϑ,k iff $K(\theta+k)=K(\theta)+\tau_k$; $\tilde\kappa(\vartheta)=\tilde\kappa(\vartheta')\Rightarrow\vartheta'-\vartheta\in2\pi\mathbb{Z}^d$ iff $K(\theta)=K(\theta')\Rightarrow\theta'-\theta\in\mathbb{Z}^d$; and $s\mapsto K(\theta+sw)$ has derivative v at t iff $s\mapsto\tilde\kappa(2\pi\theta+s\tilde\omega)$ has derivative v at t. In particular, for every Hamiltonian F, the frozen flow clause for (K,w) is CP's flow clause for $(\tilde\kappa,\tilde\omega)$ and the frozen frequency is $w=\tilde\omega/2\pi$.*
- *(d)* (Frozen shift) *The frozen fast shift* $\mathrm{shift}(k_f)=((2\pi k_f,0_b),0)$ *(Def. 1.8) equals $S_{2\pi}$ applied to $((k_f,0_b),0)$.*

Proof. **Step S2.11: (a).** $DS_c=\mathrm{diag}(c\mathbf{1},\mathbf{1})$, so $\Omega(S_cu,S_cv)=\sum_i(u_{2,i}\,cv_{1,i}-cu_{1,i}v_{2,i})=c\Omega(u,v)$; $\det=c^d$; the lattice formula is linearity.

Step S2.12: (b). Lemma 2.4 gives $DS_c\,X_{\tilde H^{S_c}}=X_{\tilde H}\circ S_c$. Explicitly $\partial_{z_2}\tilde H^{S_c}=c^{-1}\partial_2\tilde H$, $\partial_{z_1}\tilde H^{S_c}=\partial_1\tilde H$, which are Part 3's equations $\dot\theta=\partial_I H=\dot\phi/2\pi$, $\dot I=-\partial_\theta H=-\partial_\phi\tilde H$.

Step S2.13: (c). All claims are composition with the linear bijection $\theta\mapsto2\pi\theta$ of $\mathbb{R}^d$ (the chain rule is not even needed for the flow clause: $K(\theta+sw)=\tilde\kappa(2\pi\theta+s\tilde\omega)$ exactly). (d) is (a). □

Corollary 2.12 (Cylinder pipeline (consumed by Part 5))**.** *Let $d=a+b$, $H:\mathrm{Ph}(d)\to\mathbb{R}$, Φ a 1-chart on an open $O_0\subset\mathrm{Ph}(d)$ with $O_0+(2\pi k,0)=O_0$ and $\Phi(z+(2\pi k,0))=\Phi(z)+\mathrm{shift}(k_f)$ for $z\in O_0$, $k\in\mathbb{Z}^d$, and $(\Phi(z')=\Phi(z)+\mathrm{shift}(k_f)\Rightarrow z'=z+(2\pi m,0)$ for some $m\in\mathbb{Z}^d)$; put $O=S_{2\pi}^{-1}(O_0)$, $\Psi=\Phi\circ S_{2\pi}$, $G_3=H^\Psi=(2\pi)^{-1}(H\circ\Phi)(2\pi z_1,z_2)$. If $\mathrm{LT}(G_3,O,w,K_3)$ (for instance, K_3 the torus of Prop. 4.3(a)–(d) with π of Remark 2.2 applied, and the injectivity modulo translations of that proposition for (L3)) and $U\supset\Psi(K_3(\mathbb{R}^d))$, then* $\mathrm{IsCylTorus}(H,U,w,\Psi\circ K_3)$, *with the same w: the frozen frequency equals Part 3's frequency $\omega\in\mathcal{D}_{\gamma_1,\gamma_2,\tau}$, and CP's own angular frequency is $2\pi w$.*

Proof. Ψ is a 2π-chart on O (Remark 2.10(3)); (Q1) holds because $\Psi(z+\iota_k)=\Phi(S_{2\pi}z+(2\pi k,0))=\Psi(z)+\mathrm{shift}(k_f)$; (Q2) because $\Psi(z')=\Psi(z)+\mathrm{shift}(k_f)$ gives $S_{2\pi}z'=S_{2\pi}z+(2\pi m,0)$, i.e. $z'=z+\iota_m$. Apply Theorem 2.9(a). □

Corollary 2.13 (Physical pipeline (consumed by Parts 12, 13)). *Let $H : \mathrm{Ph}(d) \to \mathbb{R}$, Φ a 1-chart on an open O_0 with $O_0 + (2\pi k, 0) = O_0$, $\Phi(z + (2\pi k, 0)) = \Phi(z)$ and $(\Phi(z') = \Phi(z) \Rightarrow z' = z + (2\pi m, 0)$, some $m \in \mathbb{Z}^d$). Let $\tilde\kappa : \mathbb{R}^d \to O_0$ be a torus in the 2π convention of CP09 Thm. 3 for $\tilde G = H \circ \Phi$: i.e. the clauses (L1)–(L5) hold for $\tilde\kappa$ with period 2π as in Lemma 2.11(c) ($\tau_k = (2\pi k, 0)$) and $s \mapsto \tilde\kappa(\vartheta + s\tilde\omega)$ has derivative $X_{\tilde G}(\tilde\kappa(\vartheta + t\tilde\omega))$ at t, and $\tilde\omega$ is $(\tilde\gamma, \tau)$-Diophantine with $\tilde\gamma > 0$ and $\tau > d-1$ (the positivity of $\tilde\gamma$ is a hypothesis: `KAM.IsDiophantine` itself has no positivity guard, and $\tilde\omega = 0$ is $(0, \tau)$-Diophantine). Then* $\mathrm{IsPhysTorus}(H, \tilde\omega/2\pi,\ \theta \mapsto \Phi(\tilde\kappa(2\pi\theta)))$, *and* $K(\mathbb{R}^d) = \Phi(\tilde\kappa(\mathbb{R}^d))$.

Proof. By Lemma 2.11(c), $K_0(\theta) = \tilde\kappa(2\pi\theta)$ satisfies (L1)–(L6) for $G = \tilde G$ and $w = \tilde\omega/2\pi$ with ι_k replaced by τ_k; the argument of Theorem 2.9(b) with $\Psi = \Phi$ (a 1-chart, $\sigma = 0$, lattice τ_k in place of ι_k) applies verbatim. The frequency w is $(\tilde\gamma/2\pi, \tau)$-Diophantine with $\tilde\gamma/2\pi > 0$ (Lemma 2.15(c)). □

2.5. **Diophantine conversion.**

Definition 2.14 (Two-scale Diophantine set). For $n = n_1 + n_2$, $\gamma_1, \gamma_2, \tau > 0$: $\mathcal{D}_{\gamma_1,\gamma_2,\tau}$ is the set of $\omega = (\omega_1, \omega_2) \in \mathbb{R}^{n_1} \times \mathbb{R}^{n_2}$ with $|k \cdot \omega| \geq \gamma_1 |k|_1^{-\tau}$ whenever $k_1 \neq 0$ and $|k \cdot \omega| \geq \gamma_2 |k|_1^{-\tau}$ whenever $k_1 = 0$, $k \neq 0$ ($k = (k_1, k_2) \in \mathbb{Z}^{n_1} \times \mathbb{Z}^{n_2}$). This is the set of Part 3 (§4.1); for $k_1 = 0$, $|k|_1 = |k_2|_1$.
Source: CP10 (19), p. 551 (with $\gamma_2 = \mu\bar\gamma_2$; norm $|k| = \sum |k_j|$, CP10 footnote 5, p. 547), and the definition of $\mathcal{D}_{\gamma_1,\gamma_2,\tau}$, p. 551.

Lemma 2.15 (Elementary properties of `KAM.IsDiophantine`). *Let $\omega \in \mathbb{R}^d$, $\gamma, \tau > 0$ (the frozen definition is $\gamma/|k|_1^\tau \leq |k \cdot \omega|$ for $k \neq 0$, Def. 1.1).*

- *(a) (monotone) (γ, τ)-Diophantine implies (γ', τ')-Diophantine for $0 < \gamma' \leq \gamma$, $\tau' \geq \tau$. In particular $\tau > d - 1$ and $\tau > a + b$ can be arranged by raising τ.*
- *(b) (norms) If $|k \cdot \omega| \geq \gamma\, \nu(k)^{-\tau}$ for all $k \neq 0$ with $0 < \nu(k) \leq |k|_1$ (e.g. $\nu = |\cdot|_2$ or $|\cdot|_\infty$, which are positive on $k \neq 0$), then ω is (γ, τ)-Diophantine.*
- *(c) (scaling) For $s > 0$: ω is (γ, τ)-Diophantine iff ω/s is $(\gamma/s, \tau)$-Diophantine.*
- *(d) (two scales) $\mathcal{D}_{\gamma_1,\gamma_2,\tau} \subset \{\omega :\ \omega$ is $(\min\{\gamma_1, \gamma_2\}, \tau)$-Diophantine$\}$; in particular, if $\gamma_1 \geq \gamma_2$, the set of (γ_2, τ)-Diophantine vectors.*

Proof. (a) For $k \neq 0$, $|k|_1 \geq 1$, so $|k|_1^\tau \leq |k|_1^{\tau'}$ (real powers of a base ≥ 1 are nondecreasing in the exponent; the Lean base is ≥ 1, so no totality issue). (b) On the positive bases $0 < \nu(k) \leq |k|_1$ one has $\nu(k)^{-\tau} \geq |k|_1^{-\tau}$ ($\tau > 0$). The positivity guard is necessary: Lean has $0^{-\tau} = 0$, and $\nu \equiv 0$ would satisfy the hypothesis for $\omega = 0$. (c) $|k \cdot \omega/s| = |k \cdot \omega|/s \geq (\gamma/s)|k|_1^{-\tau}$. (d) For each $k \neq 0$ the bound in the definition is one of the two lines, each $\geq \min\{\gamma_1, \gamma_2\}|k|_1^{-\tau}$. □

Lemma 2.16 (CP10 and CP09 give the frozen Diophantine clauses). *(a)* (CP10, cylinder tori) *Let $\omega \in \mathcal{D}_{\gamma_1,\gamma_2,\tau}$ with $\gamma_1, \gamma_2 > 0$ and $\tau > n = a + b$ (CP10: $\tau_* > n$, footnote 12, p. 550; Prop. 4.3 assumes $\tau > n$). Then ω is $(\min\{\gamma_1, \gamma_2\}, \tau)$-Diophantine, so the clause "$\exists \gamma > 0, \tau > a + b$, w (γ, τ)-Diophantine" of Def. 1.16 holds for $w = \omega$. If $w = \tilde\omega/2\pi$ with $\tilde\omega \in \mathcal{D}_{\tilde\gamma_1,\tilde\gamma_2,\tau}$ then $w \in \mathcal{D}_{\tilde\gamma_1/2\pi,\tilde\gamma_2/2\pi,\tau}$.*

(b) (CP09, physical tori) *Let $\tilde\omega = \omega^\star(b)$ satisfy $|\langle k, \tilde\omega\rangle| \geq \gamma |k|_2^{-\tau}$ for all $k \in \mathbb{Z}^n \setminus \{0\}$ (CP09 Thm. 3(iv), (6), p. 7: $|k|_2$ the Euclidean norm), with $\gamma > 0$ and $\tau > n\mu \geq n$ ($\mu \in \mathbb{Z}_{\geq 1}$ by CP09 Def. 2, p. 6). Then $\tilde\omega$ is (γ, τ)-Diophantine, $w = \tilde\omega/2\pi$ is $(\gamma/2\pi, \tau)$-Diophantine, and $\tau > n - 1$: the clause (vii) of Def. 1.24 holds with $d = n$.*

Proof. (a) Lemma 2.15(d) and (c). (b) Lemma 2.15(b) with $\nu = |\cdot|_2 \leq |\cdot|_1$, then (c); $\tau > n\mu \geq n > n - 1$. □

Remark 2.17 (Dimension). For the planar theorems the torus has $d = 2n$ frequencies; CP09 Thm. 3 is applied with its n equal to this d, so its $\tau > n\mu \geq d$ exceeds $d - 1$ as the frozen clause requires.

2.6. **Liouville measure and unions of tori.**

Lemma 2.18 (Charts scale Lebesgue measure)**.** *Let Ψ be a c-chart on O and $A \subset O$ Lebesgue measurable. Then $\Psi(A)$ is Lebesgue measurable (null-measurable) and* $\operatorname{meas}\Psi(A) \le c^d \operatorname{meas} A$*, with equality if Ψ is injective on A. An* injective *symplectic chart ($c = 1$) preserves the measure of A; without injectivity only the upper bound is asserted (the polar chart of Remark 2.10(1) maps $[0, 4\pi) \times [1, 2]$, of area 4π, onto an annulus of area 2π). If A is Borel then $\Psi(A)$ is Borel (because Ψ is a local homeomorphism, Step S2.16; on a set A where Ψ is* injective *this is also the Lusin–Souslin theorem, (X4), and injectivity on A is what the consumers in Part 12 use).*

Proof. **Step S2.14: Determinant.** $|\det D\Psi| = c^d$ on O (Lemma 2.1(b)).

Step S2.15: Measurability. By (X2) cover O by countably many open O_j on which Ψ is a homeomorphism onto an open set with continuous inverse Ψ_j^{-1} (Lindelöf). Write $A \cap O_j = B \cup N$, B F_σ, N null ((X4)); $\Psi(B)$ is σ-compact, $\Psi(N)$ is null because Ψ is locally Lipschitz ([K:`bp:p1:lip-image`] on small cubes, countable union). So $\Psi(A) = \bigcup_j \Psi(A \cap O_j)$ is Lebesgue measurable.

Step S2.16: Borel sets. If A is Borel, $\Psi(A \cap O_j) = (\Psi_j^{-1})^{-1}(A \cap O_j)$ is the preimage of a Borel set under the continuous map Ψ_j^{-1} on the open set $\Psi(O_j)$, hence Borel; the countable union is Borel. (This uses only the local inverses; for a merely continuous map one would need injectivity on A and Lusin–Souslin, (X4).)

Step S2.17: Measure. For Borel A, (X3) gives $\operatorname{meas}\Psi(A) \le \int_A |\det D\Psi| = c^d \operatorname{meas} A$, with equality if Ψ is injective on A. For Lebesgue measurable A write $A = B \cup N$ with B Borel, N null: $\operatorname{meas}\Psi(A) \le \operatorname{meas}\Psi(B) + 0$, and for equality under injectivity use the subscript-0 version of (X3). □

Lemma 2.19 (Quotient version for whole tori)**.** *Assume the hypotheses of Theorem 2.9 with $\sigma_k = 0$. Let $A \subset O$ be measurable with $A + \iota_k = A$ for all $k \in \mathbb{Z}^d$, and $Q = \{z : z_{1,i} \in [0, 1)\ \forall i\}$. Then $\Psi(A)$ is Lebesgue measurable, $\Psi(A) = \Psi(A \cap Q)$ and* $\operatorname{meas}\Psi(A) = c^d \operatorname{meas}(A \cap Q)$*. Consequently $0 < \operatorname{meas}\Psi(A)$ iff $0 < \operatorname{meas} A$, and $\operatorname{meas}\Psi(A) < \infty$ if $A \cap Q$ is bounded (e.g. the momenta of A are bounded).*

Proof. $A = \bigcup_m (A \cap Q) + \iota_m$, and $\Psi((A \cap Q) + \iota_m) = \Psi(A \cap Q)$ by (Q1) with $\sigma = 0$. If $z, z' \in A \cap Q$ and $\Psi(z) = \Psi(z')$, (Q2) gives $z' = z + \iota_m$ and $z, z' \in Q$ forces $m = 0$: Ψ is injective on $A \cap Q$; Lemma 2.18 gives $\operatorname{meas}\Psi(A \cap Q) = c^d \operatorname{meas}(A \cap Q)$. *Borel version:* if A is Borel, so is $A \cap Q$, and Ψ is continuous and injective on it, so $\Psi(A) = \Psi(A \cap Q)$ is Borel by (X4) (`MeasurableSet.image_of_continuousOn_injOn`); this is the form needed for the frozen `MeasurableSet` clauses. Finally $\operatorname{meas} A = \sum_m \operatorname{meas}((A \cap Q) + \iota_m)$ is 0 iff $\operatorname{meas}(A \cap Q) = 0$. □

Remark 2.20 (Scope of (Q1), (Q2), (L2) and Lemma 2.19). Theorem 2.9(a) and Lemma 2.19 assume lattice compatibility in *all* position coordinates: (L2) and (Q1) shift the whole lattice ι_k, $k \in \mathbb{Z}^d$. This does not fit the cylinder tori of P-G, whose slow coordinates are periodic Cartesian ones (clause (iii) of `IsCylinderTorus` asks the shift only in the fast coordinates), nor the chart compositions whose slow Cartesian coordinates are not lattice-shifted. For those, Part 12 proves the cylinder transport and the measure statement directly: Proposition 18.8 and Proposition 18.9; they need only the fast-angle lattice, injectivity of the chart modulo it, and Lemma 2.18, applied on sets where the chart is injective. Theorem 2.9 and Lemma 2.19 remain valid as stated, in their scope (polar oscillator coordinates, Lemma 9.15).

Lemma 2.21 (Unions of whole tori as Lipschitz images)**.** *Let $E \subset \mathbb{R}^d$ be Lebesgue measurable, $A = E \times [0, 1)^d \subset \mathbb{R}^{2d}$, $\mathcal{F} : A \to \mathbb{R}^{2d}$ with $|\mathcal{F}(x) - \mathcal{F}(y)| \le \Lambda |x - y|$ (max norms). Then $\mathcal{F}(A)$ is Lebesgue measurable (`NullMeasurableSet`;* not *asserted Borel) and* $\operatorname{meas}\mathcal{F}(A) \le \Lambda^{2d} \operatorname{meas} E$*.* ***Borel consumers (the frozen predicates need `MeasurableSet`):*** *(B1) if E is σ-compact (e.g.*

compact, as $\mathcal{K}_\star$ in CP09 Thm. 3), then $\mathcal{F}(A)$ is σ-compact, hence Borel, with no injectivity needed; (B2) if E is Borel and $\mathcal{F}$ is continuous and injective on A, then $\mathcal{F}(A)$ is Borel ((X4)). In both cases the saturated union $\bigcup_k(\mathcal{F}(A)+\iota_k)$ below is a countable union of Borel sets. If moreover $|\mathcal{F}(x)-\mathcal{F}(y)|\geq\lambda|x-y|$ with $\lambda>0$, then $\operatorname{meas}\mathcal{F}(A)\geq\lambda^{2d}\operatorname{meas}E$; in particular $0<\operatorname{meas}\mathcal{F}(A)$ iff $0<\operatorname{meas}E$, and $\operatorname{meas}\mathcal{F}(A)<\infty$ iff $\operatorname{meas}E<\infty$. If $\mathcal{F}(\omega,\vartheta)=\varphi_\omega(\vartheta)$ with $\varphi_\omega(\vartheta+k)=\varphi_\omega(\vartheta)+\iota_k$, then $\bigcup_{\omega\in E}\varphi_\omega(\mathbb{R}^d)=\bigcup_{k\in\mathbb{Z}^d}(\mathcal{F}(A)+\iota_k)$ is measurable.

Proof. **Step S2.18: Extension.** McShane ([K:`bp:p1:mcshane`], componentwise in the max norm) gives a Λ-Lipschitz $\hat{\mathcal{F}}:\mathbb{R}^{2d}\to\mathbb{R}^{2d}$ extending $\mathcal{F}$.

Step S2.19: Measurability. A is measurable; $A=B\cup N$ with B F_σ, N null ((X4)); $\hat{\mathcal{F}}(B)$ is σ-compact and $\operatorname{meas}\hat{\mathcal{F}}(N)=0$ by [K:`bp:p1:lip-image`]; so $\mathcal{F}(A)=\hat{\mathcal{F}}(B)\cup\hat{\mathcal{F}}(N)$ is Lebesgue measurable (completeness of Lebesgue measure; this does not give a Borel set).

Step S2.20: Borel cases. (B1): $E\times[0,1)^d$ is a countable union of compact sets and continuous images of compact sets are compact. (B2): (X4).

Step S2.21: Bounds. [K:`bp:p1:lip-image`]: $\operatorname{meas}\mathcal{F}(A)\leq\Lambda^{2d}\operatorname{meas}A=\Lambda^{2d}\operatorname{meas}E$. Under the lower bound $\mathcal{F}$ is injective with λ^{-1}-Lipschitz inverse on $\mathcal{F}(A)$, so $\operatorname{meas}E=\operatorname{meas}\mathcal{F}^{-1}(\mathcal{F}(A))\leq\lambda^{-2d}\operatorname{meas}\mathcal{F}(A)$ (same lemma). The last claim is the periodicity and countability of $\mathbb{Z}^d$. □

Remark 2.22 (Use). *P-PLANAR/P-FIXED.* The chart-level family is $X_0(b,\xi)$ of CP09 Thm. 3(ii), bi-Lipschitz on $\mathcal{K}_\star\times\mathbb{T}^d$ with $\mathcal{K}_\star$ compact, so the chart-level union is compact (B1), hence Borel, and of positive finite measure (Lemma 2.21, applied *at the chart level*). The Cartesian union is its image under the symplectic chart Φ of Corollary 2.13; Lemma 2.19 gives meas = meas$(A\cap Q_{2\pi})$ with $c=1$ and the cell $Q_{2\pi}=\{z_{1,i}\in[0,2\pi)\}$ (the lattice of the chart is $2\pi\mathbb{Z}^d$ there). In the period-one formulation $\Psi=\Phi\circ S_{2\pi}$ one has $c=2\pi$ and the cell Q_1, and the two formulas agree because $S_{2\pi}$ multiplies cell volume by $(2\pi)^d$. The lower Lipschitz bound of Lemma 2.21 must *not* be applied to a wrapped (periodic) Cartesian embedding, which cannot satisfy it on a whole cell. Borelness of the Cartesian union follows from Lemma 2.19 (Borel version). *P-G.* The family is that of Prop. 4.3; its parameter set $E=\omega_0(D)\cap\mathcal{D}$ must be Borel (or σ-compact) for (B2)/(B1), and injectivity of the family modulo translations is part of that proposition; Part 5 must retain whole tori and Borelness when deleting parameters; the measure bounds of the cell are Part 4.

2.7. **The ODE layer (cited).** Track N's Part 3 supplies, for the three tracks, existence, uniqueness, continuation, energy and the identification of the frozen solution predicates. Track P uses only the following, always by these labels.

[N:`bpN:p3:gen-uniqueness`]	uniqueness on every interval for C^1 fields (orbits of a torus are *the* solutions; Lemma 2.23)
[N:`bpN:p3:ham-smooth`], [N:`bpN:p3:ham-energy`], [N:`bpN:p3:ham-periodic`]	X_H is C^1 where H is C^2; energy conservation; angle periodicity
[N:`bpN:p3:newton-reduction`], [N:`bpN:p3:newton-continuation`]	`IsNewtonSolution`, `IsFixedSolution` as `IsSolIn` (Part 6)
[N:`bpN:p3:gen-continuation`], [N:`bpN:p3:gen-confinement`]	only if a later part needs continuation (not used in Part 1)
[N:`bpN:p5:symplectic`], [N:`bpN:p5:canonical`]	time-one maps are symplectic; $D\Phi X_{H\circ\Phi}=X_H\circ\Phi$ (Remark 2.6)

Lemma 2.23 (Orbits on a torus)**.** *Let $H:\mathrm{Ph}(d)\to\mathbb{R}$ be C^2 on an open $W\supset\mathcal{K}(\mathbb{R}^d)$ and let $\mathcal{K},w$ satisfy the flow clause of Defs. 1.15 (vii) or 1.24 (viii). Then $t\mapsto\mathcal{K}(\theta+tw)$ is a solution of $\dot z=X_H(z)$ in W on $\mathbb{R}$, and every solution $y:\mathbb{R}\to W$ with $y(0)=\mathcal{K}(\theta)$ equals it. $H\circ\mathcal{K}(\cdot+tw)$ is constant in t.*

Proof. The flow clause is the solution property (with values in W by hypothesis). X_H is C^1 on W ([N:bpN:p3:ham-smooth]); uniqueness is [N:bpN:p3:gen-uniqueness] on $J = \mathbb{R}$; energy is [N:bpN:p3:ham-energy]. □

Remark 2.24 (Orbit transfer to Newton). That the frozen clause (b) of MovKol and the fixed-Sun clause hold along torus orbits is Part 6: the barycentric reconstruction is a linear map composed with the orbit, the chain rule is Corollary 2.5 or elementary, and the translation of the Newton predicates is [N:bpN:p3:newton-reduction]; nothing of the existence theory is used.

2.8. **Sources, deviations, forward references.**
Sources. CP10 footnote 1 (p. 546), (19) (p. 551), footnote 5 (p. 547) and footnote 12 (p. 550), all read; CP09 Def. 2 (p. 6), Thm. 3 and (6) (pp. 6–7), read. The rest is standard (the algebra of symplectic matrices, the inverse function theorem, the area formula).
Remarks on the sources. (1) CP09 Thm. 3 prints the symplectic form as $dx \wedge dy$ with $x \in \mathbb{T}^n$ the angles; with the frozen $\Omega = \sum dz_2 \wedge dz_1$ this is $-\Omega$; the class of symplectic maps and the Lagrangian property are unaffected (Lemma 2.1(c)), and the printed Hamiltonian field in CP09 Remark 3(i) ($\xi \mapsto \xi + \omega^\star t$) is the frozen one. (2) The exponent in CP09 (6) is typeset as $|k|_2^\tau$ with a Euclidean norm; we only use $|k|_2 \le |k|_1$. (3) CP09 Thm. 3 gives no statement that $X(b,\cdot,\cdot)$ is defined on all of $\mathbb{R}^{2n}$ or injective modulo 2π translations of ξ; (L3) for $X_0(b,\cdot)$ follows from (ii) (a homeomorphism of $\mathcal{K}_\star \times \mathbb{T}^n$ onto its image).
Forward references. [P:bpP:p5:kam-application] (cylinder pipeline), Parts 6, 12, 13 (physical pipeline, orbit transfer) cite Theorem 2.9 and Corollaries 2.12, 2.13. Part 3 cites [P:bpP:p1:period-convert] (Lemma 2.11; its Lemma 4.5 is the case (b) with $c = 2\pi$). Every other cited label exists in tracks K and N.
Label index. `bpP:p1:omega-algebra`, `chart`, `chart-field`, `chart-orbit`, `hol-chart`, `lattice-torus`, `torus-transport`, `period-convert`, `cyl-pipeline`, `phys-pipeline`, `two-scale`, `dio-basic`, `dio-sources`, `liouville`, `liouville-lattice`, `cyl-scope`, `torus-union`, `torus-orbit`.

3. Averaging with two frequency scales

This part proves the averaging statement that Part 3 consumes, Lemma 4.8 (CP10 Proposition 1 with $m = 0$ and two non-resonance scales $\alpha_1 \ge \alpha_2$), and the averaging step of CP10 §4, Step 1 (p. 563) that Part 5 consumes. CP10 proves its Lemma A.1 only by reference to the unpinned [4, Lemma A.5]; here everything is derived from track N's Parts 4–6.
What track N gives, and what it does not. The one-scale Normal Form Lemma [N:bpN:p6:normal-form] cannot simply be applied with $\alpha = \alpha_2$, for four reasons, each of which is repaired below. (1) *Exponential averaging needs an iteration*: one homological solve and one flow leave an angle-dependent quadratic remainder; the exponential bound $e^{-Ks/6}E$ comes from the finite iteration of [N:bpN:p6:iterative] (as in [N:bpN:p6:normal-form], CP10 p. 565), not from one step. (2) *The strip.* [N:bpN:p6:normal-form] shrinks the strip from S to $S/6$; the interface needs $\bar{s} + s \to \bar{s} + s/6$, that is, a loss budget s that is unrelated to the initial width. The iteration is therefore re-run with N's step lemma as a black box (Proposition 3.4). (3) *Two scales.* The first-action displacement with α_1 is not a consequence of the scalar statement (since α_1/α_2 is unbounded); it comes from the splitting of the generating function (Lemma 3.3, CP10 (70)–(71)). (4) *Hypotheses.* [N:bpN:p6:iterative] asks for non-resonance on the base set plus a Lipschitz bound L with $KLr \le \alpha/p$; Part 3 provides non-resonance on the whole complex neighbourhood, and no Lipschitz constant enters the statement. Lemma 3.2 records that N's proof only uses the divisor bound on the neighbourhood.

3.1. **Conventions: norm, widths, dictionary.**

Norm ruling. Track P uses track N's norm ([N:bpN:p4:norm], ruling of the coordinator after review P3-03): for u holomorphic on $\mathcal{D}_{r,s} = T_s \times V_rD$ and 1-periodic in θ, $|u|_{r,s} = \sup_{I\in V_rD}\sum_k|\hat{u}_k(I)|e^{2\pi|k|_1 s}$, where s is the *geometric* half-width of the strip $T_s = \{|\mathrm{Im}\,\theta|_\infty < s\}$ and $V_rD = \{I \in \mathbb{C}^n : \mathrm{dist}_\infty(I, D) < r\}$ ([N:bpN:p4:domains]). This is not the norm $\sum_k \sup_I$ used in Part 3 as printed; only $|\cdot|, \le \sum_k \sup_I$ holds.
Dictionary to Part 3. (Lemma 4.5 and [N:bpN:p4:dictionary]). Part 3 writes widths as *coefficient radii* σ (weight $e^{\sigma|k|_1}$, strip $\{|\mathrm{Im}\,\theta| < \sigma/2\pi\}$, notation $\mathbb{T}_\sigma$). Hence $\mathbb{T}^{\mathrm{P3}}_\sigma = T_{\sigma/2\pi}$, and Part 3's $(\bar{s}, s)$ correspond to $(\bar{s}/2\pi, s/2\pi)$ here. Frequencies, divisors, the Hamilton equations $\dot{\theta} = \partial_I H$, $\dot{I} = -\partial_\theta H$ and the period-one angles are the same in both; N's $\bar{\alpha} = 2\pi\alpha$ and $\kappa = 2\pi Ks$ are the quantities $2\pi\alpha$ and "Ks of Part 3". Throughout this part

$$s \text{ (geometric)}, \qquad \kappa := 2\pi Ks \quad (\text{so } \kappa = Ks_{\mathrm{P3}}), \qquad \bar{\alpha}_j := 2\pi\alpha_j, \qquad Z_K = \{k \in \mathbb{Z}^n : |k|_1 \le K\}.$$

The statement for Part 3 in its own notation is Corollary 3.6.
Splitting of indices. $n = n_1 + n_2$, $k = (k_1, k_2)$, $\theta = (\theta_1, \theta_2)$, $I = (I_1, I_2)$. A function is *independent of* θ_1 if all its Fourier modes have $k_1 = 0$.
Cauchy estimates used. After the repair of N Part 4 ([N:bpN:p4:cauchy](a)) the angular estimate holds at each fixed parameter point for the *sum* over j and, globally, for each single j; the sum of global norms over j is not available. This part never uses it: Lemma 3.3 and the field bounds use one coordinate at a time ([N:bpN:p5:vf-bound]), and the brackets are taken at a fixed parameter point (Lemma 3.8).
External facts (all inherited): those of [N:bpN:p4:s-ext], [N:bpN:p6:s-ext]; $12\ln 2 > 8$ and $\sum_{i\ge1} 4^{1-i} = \frac{4}{3}$ ([N:bpN:p6:ext-numerics]); $|k\cdot v| \le |k|_1|v|_\infty$.

3.2. N's step lemma under a direct divisor hypothesis.

Definition 3.1 (Direct divisor setting)**.** Let $D \subset \mathbb{C}^n$, $r > 0$, $K \ge 0$, $\alpha > 0$, $q \ge 1$, $\Lambda \subset \mathbb{Z}^n$, and h holomorphic on V_rD, $\omega = \partial_I h$. The data satisfy (Div_q) if $|k\cdot\omega(I)| \ge \alpha/q$ for all $I \in V_rD$ and $k \in Z_K \setminus \Lambda$.

Lemma 3.2 (Iterative Lemma, direct form)**.** [N:bpN:p6:homological] *and* [N:bpN:p6:iterative] *(with its Remark* [N:bpN:p6:iterative-special]*) hold verbatim, with the same* $\phi = \sum_{k\in Z_K\setminus\Lambda} \frac{\hat{f}_k}{2\pi i k\cdot\omega}\chi_k$*, the same constants and the same estimates, if the hypotheses* [N:bpN:p6:cond-nr], [N:bpN:p6:cond-L]*,* $KLr \le \alpha/p$ *are replaced by* (Div_q)*,* $q \ge 1$ *arbitrary. In particular* $q = 1$ *is allowed.*
Source: ours; N Part 6 as written.
Source gap: This is a variant of the written N statement. N's own proof is unchanged (check below); N may adopt (Div_q) *as its primary hypothesis.*

Proof. **Step S3.1: Where the hypotheses are used.** In [N:bpN:p6:homological], [N:bpN:p6:cond-nr], [N:bpN:p6:cond-L] and $KLr \le \alpha/p$ enter only through [N:bpN:p6:divisor], whose conclusion is exactly (Div_q); and [N:bpN:p6:divisor] is used only in Steps [N:stepN:p6:hom-def] (holomorphy, nonvanishing denominators) and [N:stepN:p6:hom-norm] (the bound $|\hat{\phi}_k| \le q|\hat{f}_k|/\bar{\alpha}$). The proof of [N:bpN:p6:iterative] uses the hypotheses only through [N:bpN:p6:homological]; the identity $p^{-1} + q^{-1} = 1$ occurs only in [N:bpN:p6:divisor]. The rest (Lemmas [N:bpN:p5:flow], [N:bpN:p5:composition], [N:bpN:p5:canonical-map], [N:bpN:p5:transform], [N:bpN:p4:bracket]) never mentions ω. □

Lemma 3.3 (Two-scale step: splitting of ϕ and componentwise displacement)**.** *Let* $D \subset \mathbb{C}^n$*,* $r, s > 0$*,* $0 < \alpha_2 \le \alpha_1$*,* $q \ge 1$*,* $\Lambda = \{0\}$*,* $K \ge 0$*, and assume*

$$(\mathrm{Div}^{(2)}_q) \qquad |k\cdot\omega(I)| \ge \alpha_1/q \ \ (k_1 \ne 0), \qquad |k\cdot\omega(I)| \ge \alpha_2/q \ \ (k_1 = 0,\ k \ne 0), \qquad I \in V_rD,\ k \in Z_K.$$

Let $f \in \mathcal{O}(r,s)$ with $\delta = |f|_{r,s}$, $0 < \rho < r/2$, $0 < \sigma < s/2$, $u = q\delta/(\bar\alpha_2 \rho\sigma) \le \frac14$ (the hypothesis of Lemma 3.2 with $\alpha = \alpha_2$), ϕ and $\Phi = X^1_\phi$ as there. Put $\phi^{(1)} = \sum_{k\in Z_K, k_1\neq 0} \hat\phi_k \chi_k$, $\phi^{(2)} = \phi - \phi^{(1)}$. Then

(a) $\phi^{(2)}$ is independent of θ_1; $|\phi^{(1)}|_{r,s} \le q\delta/\bar\alpha_1$ and $|\phi|_{r,s} \le q\delta/\bar\alpha_2$;
(b) on $\mathcal{D}_{r-\frac32\rho, s-\frac32\sigma}$:

$$|\Phi_{I_1} - I_1| \le \frac{q\delta}{e\,\bar\alpha_1\sigma}, \qquad |\Phi_{I_2} - I_2| \le \frac{q\delta}{e\,\bar\alpha_2\sigma}, \qquad |\Phi_\theta - \theta| \le \frac{q\delta}{\bar\alpha_2\rho}.$$

Source: CP10 (70)–(71), p. 565 (splitting of H_ϕ, bound for $|I_1 - I_1'|$); the displacement bookkeeping is ours.

Proof. **Step S3.2: Coefficients.** By the formula for ϕ ([N:bpN:p6:homological]), $\hat\phi_k = \hat f_k/(2\pi i k\cdot\omega)$, a finite family of holomorphic functions. For $k_1 \neq 0$, $(\mathrm{Div}_q^{(2)})$ gives $|\hat\phi_k| \le q|\hat f_k|/\bar\alpha_1$; for every $k \in Z_K \setminus\{0\}$, $\alpha_1 \ge \alpha_2$ gives $|\hat\phi_k| \le q|\hat f_k|/\bar\alpha_2$. Multiply by $e^{2\pi|k|_1 s}$, sum, take the supremum: (a) except independence, which holds because $\phi^{(2)}$ has only modes with $k_1 = 0$. Both $\phi^{(i)}$ lie in $\mathcal{O}(r,s)$ with finite norm ([N:bpN:p4:coefficients](c)).

Step S3.3: Field. Put $\varepsilon = |\phi|_{r,s}/(\rho\sigma) \le u \le \frac14$, $\Omega = \mathcal{D}_{r-\rho,s-\sigma}$, $\nu = \nu_{\rho,\sigma}$ ([N:bpN:p5:setting]). By [N:bpN:p5:flow](a) and Step [N:stepN:p5:cm-ab] of [N:bpN:p5:canonical-map], for $z \in \mathcal{D}_{r-\frac32\rho, s-\frac32\sigma}$ the trajectory $\gamma(t) = X^t_\phi(z)$, $0 \le t \le 1$, stays in $\bar B_\nu(z,\varepsilon) \subset \Omega$ and solves $\dot\gamma = X_\phi(\gamma) = (\partial_I\phi, -\partial_\theta\phi)$. The first n_1 components of $\dot\gamma_I$ are $-\partial_{\theta_1}\phi = -\partial_{\theta_1}\phi^{(1)}$ because $\partial_{\theta_1}\phi^{(2)} = 0$. By [N:bpN:p5:vf-bound] applied to $\phi^{(1)}$, $|\partial_{\theta_1}\phi^{(1)}| \le |\phi^{(1)}|_{r,s}/(e\sigma) \le q\delta/(e\bar\alpha_1\sigma)$ on Ω. Likewise [N:bpN:p5:vf-bound] for ϕ bounds $|\partial_{\theta_2}\phi| \le q\delta/(e\bar\alpha_2\sigma)$ and $|\partial_I\phi| \le q\delta/(\bar\alpha_2\rho)$ on Ω.

Step S3.4: Integrate. The mean value inequality along γ ([K:bp:p1:ext-mvt]) gives (b) with $t = 1$. □

3.3. The averaging proposition in N's conventions.

Proposition 3.4 (Averaging with a loss budget and two scales)**.** *Let $n = n_1 + n_2$, $D \subset \mathbb{R}^n$, $r > 0$, $\bar s \ge 0$, $s > 0$, $K > 0$ with $\kappa = 2\pi K s \ge 6$, $0 < \alpha_2 \le \alpha_1$. Let h be holomorphic on V_rD and real on $V_rD \cap \mathbb{R}^n$, $\omega = \partial_I h$, and assume*

(A1) $|k\cdot\omega(I)| \ge \alpha_1$ if $k_1 \neq 0$, $\ge \alpha_2$ if $k_1 = 0$, for all $I \in V_rD$ and $k \in Z_K \setminus\{0\}$;
(A2) $f \in \mathcal{O}(r, \bar s + s)$, real on real points, with $E := |f|_{r,\bar s+s} \le \dfrac{\alpha_2 r}{128K}$.

Then there is a map $\Psi : \mathcal{D}_{r/2,\bar s+s/6} \to \mathcal{D}_{r,\bar s+s}$, $\Psi(\theta', I') = (\theta, I)$, and functions g, f_* on $\mathcal{D}_{r/2,\bar s+s/6}$ such that

(a) Ψ is holomorphic, injective, real on real points, $\Psi(\theta'+k, I') = \Psi(\theta', I') + (k, 0)$ ($k \in \mathbb{Z}^n$), and $(D\Psi)^\top J\, D\Psi = J$ (Ψ preserves $d\theta \wedge dI$, hence $dI \wedge d\theta$);
(b) $H \circ \Psi = h + g + f_*$, $H = h + f$, with $g = g(I')$ independent of θ', real on real points, and $\sup_{V_{r/2}D}|g - \hat f_0| \le |g - \hat f_0|_{r/2,\bar s+s/6} \le E/4$ (indeed $\le E/\kappa$);
(c) f_* is holomorphic, periodic, real on real points and $|f_*|_{r/2,\bar s+s/6} \le e^{-\kappa/6}E$;
(d) $\bar\alpha_1 s\,|I_1 - I_1'| \le 3E$, $\bar\alpha_2 s\,|I_2 - I_2'| \le 3E$, $\bar\alpha_2 r\,|\theta - \theta'| \le 9E$ on $\mathcal{D}_{r/2,\bar s+s/6}$.

Constants depend on nothing: no n, τ, q or L enter.
Source: CP10 Prop. 1 (p. 552) with $m = 0$, Lemma A.1 (p. 564), proof pp. 564–565; Pöschel 1993 pp. 193–196 via [N:bpN:p6:normal-form]. *Constants recomputed (see Remark 3.5).*

Proof. Throughout $q = 1$: by (A1), $(\mathrm{Div}_1^{(2)})$ holds on V_rD with $\Lambda = \{0\}$, and for every $r' \le r$ and $D' \subset D$-neighbourhoods inside V_rD. Write $\bar\alpha = \bar\alpha_2$, $\varepsilon := E$, $S_i := \bar s + s_i$ for the strip widths, where s_i (the *budget* still to be spent) is chosen below. From (A2): $\varepsilon \le \alpha_2 r/(128K) = \bar\alpha r s/(128\kappa)$. Every application of Lemma 3.2 below is on $\mathcal{D}_{r_i,S_i}$ with $r_i \le r$, so its non-resonance hypothesis is (A1); its requirements $2\rho < r_i$, $2\sigma < S_i = \bar s + s_i$ are checked directly at each step (below; at the last step

$s_N = s/6 + 2\sigma > 2\sigma$ and $r_N > 2\rho$). Because the weights $|g|_*$ of [N:bpN:p6:setting] contain the widths only through the *differences* $s_j - s_i$, the strip offset $\bar{s}$ plays no role in any estimate.

Step S3.5: Step 0. Take $\rho_0 = r/8$, $\sigma_0 = s/6$ and $g = 0$ in Lemma 3.2. Then $y_0 := \varepsilon/(\bar{\alpha}\rho_0\sigma_0) = 48\varepsilon/(\bar{\alpha}rs) \le \frac{3}{8\kappa} \le \frac{1}{16}$, so $u = y_0 \le \frac{1}{4}$. Lemma 3.2(d) with $|g|_* = 0$ gives

$$|f_1|_{\frac{3}{4}r,\bar{s}+\frac{2}{3}s} \le (1-2y_0)^{-1}\big(\tfrac{2}{e}y_0 + e^{-\kappa/6}\big)\varepsilon, \qquad (1-2y_0)^{-1} \le \tfrac{8}{7}$$

($2\pi K\sigma_0 = \kappa/6 \ge 1$). *Case A:* $e^{-\kappa/6} \le y_0$. Then

$$\varepsilon_1 := |f_1|, \le \tfrac{8}{7}\big(1+\tfrac{2}{e}\big)y_0\varepsilon \le 1.984 \cdot \tfrac{3}{8\kappa}\,\varepsilon \le \frac{3\varepsilon}{4\kappa} \le \frac{\varepsilon}{8}. \tag{2.1}$$

Case B: $e^{-\kappa/6} > y_0$. Repeat Step 0 with the doubled losses $\rho' = r/4$, $\sigma' = s/3$ ($u' = y_0/4$): $|f_1|_{r/2,\bar{s}+s/3} \le (1-y_0/2)^{-1}(\frac{2}{e}\frac{y_0}{4} + e^{-\kappa/3})\varepsilon \le \frac{32}{31}(\frac{1}{2e} + \frac{1}{e})e^{-\kappa/6}\varepsilon \le e^{-\kappa/6}\varepsilon$, using $e^{-\kappa/3} \le e^{-1}e^{-\kappa/6}$. We stop with $\Psi = \Phi_0'$ on $\mathcal{D}_{r/2,\bar{s}+s/3} \supset \mathcal{D}_{r/2,\bar{s}+s/6}$, $g = \hat{f}_0$, $f_* = f_1$. Clauses (a)–(c) follow from Lemma 3.2; (d): Lemma 3.3(b) gives $|\Delta I_j| \le \varepsilon/(e\bar{\alpha}_j\sigma') = 3\varepsilon/(e\bar{\alpha}_j s) < 3\varepsilon/(\bar{\alpha}_j s)$ and $|\Delta\theta| \le \varepsilon/(\bar{\alpha}\rho') = 4\varepsilon/(\bar{\alpha}r)$. From now on Case A.

Step S3.6: Number of steps. $N := \lfloor \kappa/(12\ln 2)\rfloor$, so $N < \kappa/8$. If $N = 0$: $\kappa < 12\ln 2$, hence $e^{-\kappa/6} > \frac{1}{4}$, and $\varepsilon_1 \le \varepsilon/8 \le e^{-\kappa/6}\varepsilon$ by (2.1); set $\Psi = \Phi_0$, $g = \hat{f}_0$, $f_* = f_1$ on $\mathcal{D}_{3r/4,\bar{s}+2s/3}$. Displacements: $|\Delta I_j| \le \varepsilon/(e\bar{\alpha}_j\sigma_0) = 6\varepsilon/(e\bar{\alpha}_j s) \le 2.21\,\varepsilon/(\bar{\alpha}_j s)$, $|\Delta\theta| \le 8\varepsilon/(\bar{\alpha}r)$. (a)–(d) follow. Let $N \ge 1$ and for $1 \le i \le N$

$$\rho_i = \rho = \frac{r}{8N}, \quad \sigma_i = \sigma = \frac{s}{4N}, \qquad r_i = \tfrac{3}{4}r - 2(i-1)\rho, \quad s_i = \tfrac{2}{3}s - 2(i-1)\sigma, \quad S_i = \bar{s} + s_i;$$

then $r_{N+1} = r/2$, $s_{N+1} = s/6$. $2\pi K\sigma = \kappa/(4N) \ge 3\ln 2$, so $e^{-2\pi K\sigma} \le \frac{1}{8}$; $\rho\sigma = rs/(32N^2)$. From $\kappa > 8N$, (A2) and (2.1):

$$\varepsilon \le \frac{\bar{\alpha}rs}{2^{10}N}, \qquad \varepsilon_1 \le \frac{3}{4\kappa}\varepsilon < \frac{3\varepsilon}{32N} \le \frac{3\,\bar{\alpha}rs}{2^{15}N^2} < \frac{\bar{\alpha}\rho\sigma}{2^8}. \tag{2.2}$$

Step S3.7: Induction. *Claim:* for $1 \le i \le N$ step i (Lemma 3.2 on $\mathcal{D}_{r_i,S_i}$, losses (ρ,σ)) is admissible, and $\varepsilon_{i+1} := |f_{i+1}|_{r_{i+1},S_{i+1}} \le \varepsilon_i/4$; hence $\varepsilon_i \le 4^{1-i}\varepsilon_1$ ($1 \le i \le N+1$). Here f_{i+1}, $g_i = g_{i-1} + (\hat{f}_i)_0$ are produced by the step, $g_0 := \hat{f}_0$, $|g_i - g_{i-1}| \le \varepsilon_i$ ([N:bpN:p4:projections]). *Admissibility:* $u_i = \varepsilon_i/(\bar{\alpha}\rho\sigma) \le 2^{-8}$ by (2.2). *Weights:* decompose $g_{i-1} = g_0 + \sum_{j=1}^{i-1}(g_j - g_{j-1})$; the term $j \ge 1$ lives on $\mathcal{D}_{r_j,S_j}$ with $r_j - r_i \ge 2\rho$, $s_j - s_i \ge 2\sigma$, weight $\le \frac{1}{2}(\frac{1}{3}+\frac{1}{3}) \le \frac{1}{2}$; the term g_0 lives on $\mathcal{D}_{r,\bar{s}+s}$ with $r - r_i \ge r/4$, $s - s_i \ge s/3$, weight $\le \frac{1}{2}(\frac{1}{2N} + \frac{3}{4N}) \le \frac{1}{N}$. Thus $|g_{i-1}|_* \le \frac{1}{2}\cdot\frac{4}{3}\varepsilon_1 + \varepsilon/N \le \varepsilon_1 + \varepsilon/N \le \bar{\alpha}\rho\sigma\,(2^{-8} + 2^{-5}) \le \bar{\alpha}\rho\sigma/2^4$ (by (2.2) and $\varepsilon/N \le \bar{\alpha}rs/(2^{10}N^2) = 2^{-5}\bar{\alpha}\rho\sigma$). *The step:* Lemma 3.2(d) with $\delta = \varepsilon_i$ gives $\varepsilon_{i+1} \le \frac{128}{127}\big(\frac{2}{e}(\frac{1}{16} + \frac{1}{2^8}) + \frac{1}{8}\big)\varepsilon_i \le \frac{128}{127}\cdot 0.1915\,\varepsilon_i < \frac{1}{4}\varepsilon_i$. (This is the computation of [N:bpN:p6:normal-form], Step [N:stepN:p6:nf-induct], with $q = 1$.)

Step S3.8: Conclusion of the construction. Put $\Psi = \Phi_0 \circ \cdots \circ \Phi_N$ on $\mathcal{D}_{r/2,\bar{s}+s/6}$, $g = g_N$, $f_* = f_{N+1}$. Since $H_{i+1} = H_i \circ \Phi_i$, $H \circ \Psi = h + g + f_*$. g is a finite sum of θ-independent functions (each $(\hat{f}_i)_0$), so $g = g(I')$; $\hat{f}_0 = g_0$. [N:bpN:p6:telescope] with $U_i = \mathcal{D}_{r_i,S_i}$ gives (a) (holomorphic, canonical, injective; periodicity and reality are inherited from each Φ_i, [N:bpN:p6:iterative](a),(e)), and reality of g, f_* is [N:bpN:p6:iterative](e). (c): $\varepsilon_{N+1} \le 4^{-N}\varepsilon_1 \le 4^{-N-1}\varepsilon \le e^{-\kappa/6}\varepsilon$, since $N + 1 > \kappa/(12\ln 2)$ gives $4^{-(N+1)} < e^{-\kappa/6}$. (b): $|g - g_0| \le \sum_{i=1}^{N}\varepsilon_i \le \frac{4}{3}\varepsilon_1 \le \varepsilon/\kappa \le \varepsilon/6$ by (2.1); this is the bound of CP10 (25) in which no K appears in the numerator (review P3-04). The sup bound follows from [N:bpN:p4:sup-weighted](a).

Step S3.9: (d) Displacement. By [N:bpN:p6:telescope] (seminorms $|\Pi_{I_1}\cdot|$, $|\Pi_{I_2}\cdot|$, $|\Pi_\theta\cdot|$) and Lemma 3.3(b) for each step (valid on $V_{i+1}\subset\mathcal{D}_{r_i-\frac{3}{2}\rho_i,S_i-\frac{3}{2}\sigma_i}$, [N:bpN:p6:iterative](a)):

$$|\Delta I_j|\le\frac{\varepsilon}{e\bar\alpha_j\sigma_0}+\sum_{i=1}^{N}\frac{\varepsilon_i}{e\bar\alpha_j\sigma}\le\frac{1}{e\bar\alpha_j s}\Big(6\varepsilon+4N\cdot\tfrac{4}{3}\varepsilon_1\Big),\qquad|\Delta\theta|\le\frac{\varepsilon}{\bar\alpha\rho_0}+\sum_{i=1}^{N}\frac{\varepsilon_i}{\bar\alpha\rho}\le\frac{1}{\bar\alpha r}\Big(8\varepsilon+8N\cdot\tfrac{4}{3}\varepsilon_1\Big).$$

By (2.1) and $N<\kappa/8$: $N\varepsilon_1\le\frac{3N}{4\kappa}\varepsilon<\frac{3}{32}\varepsilon$. Hence $\frac{1}{e}(6+\frac{16}{3}\cdot\frac{3}{32})=\frac{6.5}{e}<2.4\le3$ and $8+\frac{32}{3}\cdot\frac{3}{32}=9$ (strict inequality, since $N<\kappa/8$ strictly). Both bounds are $\le$ the claimed ones. □

Remark 3.5 (Where the constants come from). Smallness: (A2) is the one-scale hypothesis (i) of [N:bpN:p6:normal-form] with $q=1$ ($2^7qK\varepsilon\le\alpha r$). The weaker requirement of CP10, $64e\,KE<\alpha_2 r$ (CP10 (23) with $m=0$, $c_0=e/2$, $d=rs$), implies it since $64e>128$. The number of steps N, the losses, 4^{1-i} and $e^{-\kappa/6}$ are those of [N:bpN:p6:normal-form]; the improvements (E/κ instead of $16c\varepsilon^2$, $3E$ instead of $9E$ for the actions) come from keeping the exact expressions $6/e$ and $y_0\le3/(8\kappa)$ instead of the cruder $\kappa\ge6\Rightarrow1/(\bar\alpha s)\le K/(6\alpha)$ of [N:bpN:p6:normal-form](c), which is a bound in K and is useless for the interface (the interface displacement is independent of K). The deviation of [N:bpN:p6:deviations](1) is respected: all steps satisfy $|\phi|\le\rho\sigma/4$.

3.4. **The interface of Lemma 4.8.**

Corollary 3.6 (Averaging with two scales, in Part 3's notation)**.** *Let $0<\alpha_2\le\alpha_1$, $D\subset\mathbb{R}^n$, $r>0$, $\bar s,s>0$, $K>0$ with $Ks\ge6$. Let h be real analytic on D_r (holomorphic on $D_r=V_rD$, real on real points) with $\omega=\partial h$, and let f be holomorphic on $D_r\times\mathbb{T}_{\bar s+s}$, real analytic, 1-periodic in θ, with the* N-norm $E:=\sup_{I\in D_r}\sum_k|\hat f_k(I)|e^{(\bar s+s)|k|_1}<\infty$. *Assume (A1) (that is, (N1) of Lemma 4.8) and*

$$64e\,KE<\alpha_2 r.$$

Then there is a holomorphic, injective, real, symplectic ($\Psi^*(dI\wedge d\theta)=dI'\wedge d\theta'$) map $\Psi:D_{r/2}\times\mathbb{T}_{\bar s+s/6}\to D_r\times\mathbb{T}_{\bar s+s}$ with $\Psi-\mathrm{id}$ $\mathbb{Z}^n$-periodic in θ', and $g=g(I')$, f_* with $H\circ\Psi=h+g+f_*$ on $D_{r/2}\times\mathbb{T}_{\bar s+s/6}$ and, in the N-norm with weight $e^{(\bar s+s/6)|k|_1}$ on $D_{r/2}$,

$$\sup_{D_{r/2}}|g-f_0|\le E/4,\qquad\|f_*\|_{r/2,\bar s+s/6}\le e^{-Ks/6}E,$$
$$\alpha_1 s|I_1-I_1'|\le3E\ (\le9E),\quad\alpha_2 s|I_2-I_2'|\le3E\ (\le9E),\quad2\pi\alpha_2 r|\theta-\theta'|\le9E.$$

Source: CP10 Prop. 1 (p. 552) with $m=0$, $\Lambda=\{0\}$; proof: Proposition 3.4.

Proof. **Step S3.10: Dictionary.** Put $\bar s_N=\bar s/2\pi$, $s_N=s/2\pi$ (geometric). Then $\mathbb{T}_{\bar s+s}=T_{\bar s_N+s_N}$, the weight $e^{(\bar s+s)|k|_1}=e^{2\pi|k|_1(\bar s_N+s_N)}$, so $E=|f|_{r,\bar s_N+s_N}$ in the notation of Proposition 3.4; $\kappa=2\pi Ks_N=Ks\ge6$; $\bar\alpha_j s_N=\alpha_j s$ and $\bar\alpha_2 r=2\pi\alpha_2 r$.

Step S3.11: Hypotheses. $64e>128$, so $64eKE<\alpha_2 r$ gives $E\le\alpha_2 r/(128K)$. The remaining hypotheses are identical. Proposition 3.4(a)–(d) are the claims; $e^{-\kappa/6}=e^{-Ks/6}$. □

Remark 3.7 (What Part 3 must change (the interface report)). Lemma 4.8 is *provable as stated, with the single change of norm* (made in revised Part 3, which now uses N's norm; the list below records what the consumer relies on): the former Part 3 norm $\|\cdot\|_{r,\sigma}=\sum_k\sup_I|f_k|e^{\sigma|k|}$ by N's $\sup_I\sum_k|\hat f_k(I)|e^{\sigma|k|_1}$ (this is Corollary 3.6). Precisely:

(1) Every occurrence of the norm (done in the revision; hypothesis $E\ge\|f\|_{\rho,\bar s+s}$ of [P:bpP:p3:two-scale-kam], E_j in the iteration, outputs $\|f_*\|$) is read in N's norm. Used properties: monotone under restriction of r, σ and of the base set; $\sup|f|\le\|f\|$ ([N:bpN:p4:sup-weighted](a)); Cauchy estimates for g use only sup. Nothing in Part 3 uses the finer $\sum_k\sup_I$ structure, except the printed definition.

(2) The numerical constants: hypothesis (N2) $64eKE < \alpha_2 r$ and the conclusions $E/4$, $e^{-Ks/6}E$, $\alpha_2 r 2\pi|\Delta\theta| \le 9E$ are unchanged; $\alpha_i s|\Delta I_i| \le 9E$ may be replaced by $3E$ (so the constant $\frac{3}{4}$ of `st-avg` may become $\frac{1}{4}$); no other constant of Part 3 changes.
(3) Hypothesis (N1) is used on the full complex neighbourhood D_r, exactly as in Lemma 4.8; *no Lipschitz hypothesis is added* (Lemma 3.2). Reality of h and f on real points is used, as in Part 3's setting.
(4) The one-flow proof plan of Lemma 4.8 is replaced by Proposition 3.4 (a finite iteration; done in the revision); the Fourier-tail remark is not the source of the exponential.

3.5. **Averaging with an oscillator pair (CP10 §4, Step 1).** CP10 §4, Step 1 (p. 563) averages over the n_1 fast angles while the oscillator variables $(p,q) \in \mathbb{C}^{n_2} \times \mathbb{C}^{n_2}$ are dynamical: the transformation moves them, symplectically for $dI \wedge d\theta + dp \wedge dq$. Track N's Parts 4–6 are written for (θ, I) only. The extension is mechanical but must be recorded; we do it with its own constants.
Setting. $n, m \ge 1$, $D \subset \mathbb{R}^n$, $v = (r, r_p, r_q)$, $s > 0$, $\bar{s} \ge 0$. $\mathcal{D}_{v,S} := T_S \times V_r D \times B_{r_p} \times B_{r_q} \ni (\theta, I, p, q)$ with $B_a = \{x \in \mathbb{C}^m : |x|_\infty < a\}$ (the complex neighbourhood of $\{0\}$ in the max norm; the Euclidean ball of CP10 contains it up to $\sqrt{m}$ in radius), $\mathcal{O}(v,S)$ and $|\cdot|_{v,S}$ as in [N:`bpN:p4:norm`] with the sup over (I,p,q) (I,p,q together play the rôle of N's parameter I; only θ is Fourier expanded). The field of u is $X_u = (\partial_I u, -\partial_\theta u;\ \partial_p u, -\partial_q u)$ in the order $(\dot\theta, \dot I; \dot q, \dot p)$ and $\{u,w\} = \sum_i (\partial_{\theta_i} u\, \partial_{I_i} w - \partial_{I_i} u\, \partial_{\theta_i} w) + \sum_j (\partial_{q_j} u\, \partial_{p_j} w - \partial_{p_j} u\, \partial_{q_j} w) = Du \cdot X_w$. Put $d := \min\{rs, r_p r_q\}$ and $b := 1+m$. For losses $\nu = (\rho, \rho_p, \rho_q, \sigma)$ put $\delta_\nu := \min\{\rho\sigma, \rho_p\rho_q\}$ and the scaled norm $\nu(a, b_I, c_q, c_p) = \max(|a|/\sigma, |b_I|/\rho, |c_q|/\rho_q, |c_p|/\rho_p)$, whose balls are the polydiscs of the four domain losses.

Lemma 3.8 (Extended bracket and field estimates)**.** *Let $u \in \mathcal{O}(v_0, S_0)$, $w \in \mathcal{O}(v, S)$ with finite norms, $0 < \rho < r$, $0 < \rho_p < r_p$, $0 < \rho_q < r_q$, $0 < \sigma < S$, $v - \nu < v_0$ componentwise, $S - \sigma < S_0$. Then $\{u,w\} \in \mathcal{O}(v - \nu, S - \sigma)$ and*

$$|\{u,w\}|_{v-\nu,S-\sigma} \le \frac{W}{\delta_\nu} |u|_{v_0,S_0} |w|_{v,S}, \quad W = \tfrac{1}{e}\Big(\tfrac{\rho}{r_0 - r + \rho} + \tfrac{\sigma}{S_0 - S + \sigma}\Big) + m\Big(\tfrac{\rho_q}{r_{q0} - r_q + \rho_q} + \tfrac{\rho_p}{r_{p0} - r_p + \rho_p}\Big) \le 2b\,R,$$

where R is the largest of the four ratios. Moreover for $\phi \in \mathcal{O}(v,S)$: on $\mathcal{D}_{v-\nu,S-\sigma}$, $\nu(X_\phi) \le |\phi|_{v,S}/\delta_\nu$.

Proof. **Step S3.12: Derivative bounds.** Write $x = (I,p,q)$ for the parameter. [N:`bpN:p4:cauchy`](a) gives, at every fixed x, $\sum_i |\partial_{\theta_i} u|_{x,S} \le |u|_{x,S_0}/(e(S_0 - S))$, and individually $|\partial_{\theta_i} u|_{,S} \le |u|_{/,(}e(S_0 - S))$ globally; [N:`bpN:p4:cauchy`](b) is a one-variable Cauchy integral in one coordinate of x and holds for each coordinate with its own loss: $|\partial_{I_i} u|_{,} \le |u|_{/,(}r_0 - r + \rho)$, $|\partial_{p_j} u|_{,} \le |u|_{/,(}r_{p0} - r_p + \rho_p)$, $|\partial_{q_j} u|_{,} \le |u|_{/,(}r_{q0} - r_q + \rho_q)$ (global norms, each coordinate separately), and likewise for w with losses ρ, ρ_p, ρ_q.

Step S3.13: Products. Fix x. By [N:`bpN:p4:algebra`] at this x and the triangle inequality, $|\{u,w\}|_{x,S-\sigma} \le \big(\sum_i |\partial_{\theta_i} u|_{x,}\big) \max_i |\partial_{I_i} w|_{,} + \max_i |\partial_{I_i} u|_{,} \big(\sum_i |\partial_{\theta_i} w|_{x,}\big) + \sum_{j \le m} \big(|\partial_{q_j} u|_{,} |\partial_{p_j} w|_{,} + |\partial_{p_j} u|_{,} |\partial_{q_j} w|_{,}\big)$, and the last sum is at most m times the product of the maxima (this is the factor m). Insert the bounds of the previous step and take the supremum over x last. The (θ, I) part is $\frac{1}{e}\big(\frac{1}{(r_0 - r + \rho)\sigma} + \frac{1}{(S_0 - S + \sigma)\rho}\big)$, the (q,p) part $m\big(\frac{1}{(r_{q0} - r_q + \rho_q)\rho_p} + \frac{1}{(r_{p0} - r_p + \rho_p)\rho_q}\big)$. Since $\frac{1}{(r_0 - r + \rho)\sigma} = \frac{1}{\rho\sigma} \cdot \frac{\rho}{r_0 - r + \rho}$ etc. and $\rho\sigma, \rho_p\rho_q \ge \delta_\nu$, the bound follows; $\frac{1}{e} + \frac{1}{e} \le 2$ gives $W \le 2bR$.

Step S3.14: Field. As in [N:`bpN:p5:vf-bound`]: $|\partial_I \phi|/\sigma \le |\phi|_{,}/(\rho\sigma)$, $|\partial_\theta \phi|/\rho \le |\phi|_{,}/(e\sigma\rho)$, $|\partial_p \phi|/\rho_q \le |\phi|_{,}/(\rho_p\rho_q)$, $|\partial_q \phi|/\rho_p \le |\phi|_{,}/(\rho_q\rho_p)$. □

Lemma 3.9 (Iterative Lemma with an oscillator pair)**.** [N:`bpN:p6:iterative`] *(in direct form, Lemma 3.2, $\Lambda = \{0\}$, $q = 1$) holds on $\mathcal{D}_{v,S}$ with all losses ν, with $\rho\sigma$ replaced by δ_ν, the factor $\frac{2}{e}$ in the bracket estimate replaced by $2b$ (so $|g|_*$ is replaced by $|g|_R := \sum_j R_j |g_j|_{,}$), and $\Phi = X^1_\phi$ symplectic for $d\theta \wedge dI + dq \wedge dp$:*

(a) if $|f|_{v,S} = \delta \le \bar\alpha\delta_\nu/(16b)$ *then* $|\Phi_\theta - \theta| \le \delta/(\bar\alpha\rho)$, $|\Phi_I - I| \le \delta/(e\bar\alpha\sigma)$, $|\Phi_q - q| \le \delta/(\bar\alpha\rho_p)$, $|\Phi_p - p| \le \delta/(\bar\alpha\rho_q)$;
(b) $|f_+|_{v-2\nu,S-2\sigma} \le (1 - 2eb\,\delta/(\bar\alpha\delta_\nu))^{-1}\big(\frac{2b}{\bar\alpha\delta_\nu}(|g|_R + \delta) + e^{-2\pi K\sigma}\big)\delta$.

Proof. **Step S3.15: Transfer.** [N:`bpN:p6:homological`] involves θ and the parameters only through coefficients and is unchanged ($|\phi|_, \le \delta/\bar\alpha$). [N:`bpN:p5:flow`] is for an arbitrary complex norm and open set; [N:`bpN:p5:symplectic`] uses only $DF = JS$ with S symmetric and $J^\top = -J$, true for the enlarged J; Corollary [N:`bpN:p5:canonical`] follows. Lemma [N:`bpN:p5:adjoint`], Theorem [N:`bpN:p5:composition`], Propositions [N:`bpN:p5:transform`], [N:`bpN:p5:canonical-map`] use the geometry only through the bracket bound and the field bound: replace [N:`bpN:p4:bracket`] and [N:`bpN:p5:vf-bound`] by Lemma 3.8, and $\rho\sigma$ by δ_ν.

Step S3.16: The adjoint ratio. N's iterated-bracket estimate is $|\mathrm{ad}_\phi^h u|_,/h! \le (e\eta/4)^h|u|_,$ $\eta = 2|\phi|_,/(\rho\sigma)$, obtained by h bracket estimates with losses divided by h, each costing the bracket constant $2/(e\rho'\sigma')$. With the constant $2b/\delta'$ each bracket costs a factor eb more, so the ratio becomes $eb\cdot e\eta/4 = e^2b\eta/4 \le eb\eta$ (as $e/4 < 1$), now with $\eta := 2|\phi|_,/\delta_\nu$ (we use this cruder bound; $e^2/4 > 1$, so the ratio $b\eta$ is *not* claimed). Put $\theta_\phi := eb\eta = 2eb|\phi|_,/\delta_\nu$. Then the Lie series converges with $|\mathrm{ad}_\phi^h u|_,/h! \le \theta_\phi^h|u|_,$ and $|u\circ X_\phi^1|_, \le (1-\theta_\phi)^{-1}|u|_,$, and the same for the integral remainders ([N:`bpN:p5:transform`], [N:`bpN:p5:composition-integral`]).

Step S3.17: Smallness range. The equality of the Lie series with the composition needs the ratio, evaluated at the shrunken losses $(1-\varepsilon_\phi)\nu$ with $\varepsilon_\phi = |\phi|_,/\delta_\nu \le \frac14$ ([N:`bpN:p5:composition`], remainder step: factor $\le \frac{16}{9}$), to be < 1. We *assume* $\delta \le \bar\alpha\delta_\nu/(16b)$ (this replaces $u \le \frac14$). Then $\theta_\phi \le 2eb\delta/(\bar\alpha\delta_\nu) \le e/8 < 0.34$ and $\frac{16}{9}\theta_\phi < 0.61 < 1$. The assumption holds at every step of Proposition 3.10 (there $\delta/(\bar\alpha\delta_\nu) \le y_0 \le 1/(64b\kappa)$ or is smaller by (2.3)). Then (a) and (b) follow as in [N:`bpN:p6:iterative`] with $\frac{2}{e} \mapsto 2b$ and $1 - \frac{2q\delta}{\bar\alpha\rho\sigma} \mapsto 1-\theta_\phi$, $\theta_\phi \le 2eb\delta/(\bar\alpha\delta_\nu)$.

Step S3.18: Displacement. As in Step S3.3: components of X_ϕ and Lemma 3.8: $|\partial_I\phi| \le |\phi|_,/\rho$, $|\partial_\theta\phi| \le |\phi|_,/(e\sigma)$, $|\partial_p\phi| \le |\phi|_,/\rho_p$, $|\partial_q\phi| \le |\phi|_,/\rho_q$. □

Source gap: Lemma 3.9 extends N Parts 5–6 by an oscillator pair; the extension is stated by its (short) list of changes, not rewritten in N. It is the oscillator part of CP10 Lemma A.1 / [4, Lemma A.5]. If track N prefers, it can be absorbed into N Part 5 ($\{\cdot,\cdot\}$ on $\mathbb{C}^n\times\mathbb{C}^n\times\mathbb{C}^m\times\mathbb{C}^m$). Constants differ from CP10's $c_m = e(1+em)/2$.

Proposition 3.10 (Averaging with oscillators, one scale)**.** *In the setting above, let h be holomorphic on V_rD, real on real points, $\omega = \partial h$ (h independent of p,q), $\alpha > 0$, $K > 0$, $\kappa = 2\pi Ks \ge 6$,*

(O1) $|k\cdot\omega(I)| \ge \alpha$ *for* $I \in V_rD$, $0 < |k|_1 \le K$;
(O2) $f \in \mathcal{O}(v, \bar s + s)$ *real on real points,* $E := |f|_{v,\bar s+s}$ *with* $2^{12}\,b\,KE\,s \le \alpha d$.

Then there is $\Psi : \mathcal{D}_{v/2,\bar s+s/6} \to \mathcal{D}_{v,\bar s+s}$ holomorphic, injective, real, $\Psi - \mathrm{id}$ periodic in θ, symplectic for $d\theta\wedge dI + dq\wedge dp$, with $H\circ\Psi = h + g + f_*$ on $\mathcal{D}_{v/2,\bar s+s/6}$, $H = h + f$, $g = g(I,p,q)$ independent of θ, real, and with $\bar\alpha = 2\pi\alpha$

$$|g - \hat f_0|_{v/2,\bar s+s/6} \le \frac{400\,b\,E^2}{\bar\alpha d}, \qquad |f_*|_{v/2,\bar s+s/6} \le e^{-\kappa/6}E,$$
$$\bar\alpha s|I - I'| \le 3E, \quad \bar\alpha r|\theta-\theta'| \le 9E, \quad \bar\alpha r_q|p-p'| \le 9E, \quad \bar\alpha r_p|q-q'| \le 9E.$$

Source: CP10 Prop. 1 (p. 552), Lemma A.1 (p. 564) with $\ell_2 = 0$, $\Lambda = \{0\}$; (25), (26). Constants ours.

Proof. Use the notation of Proposition 3.4 ($q = 1$, $\varepsilon = E$, $\bar\alpha = 2\pi\alpha$); (O2) reads $\varepsilon \le \bar\alpha d/(2^{12}b\kappa)$. Step 0 losses $\rho_0 = r/8$, $\rho_{p0} = r_p/8$, $\rho_{q0} = r_q/8$, $\sigma_0 = s/6$; then $\delta_{\nu_0} = \min\{rs/48, r_pr_q/64\} \ge d/64$ and $y_0 := \varepsilon/(\bar\alpha\delta_{\nu_0}) \le 64\varepsilon/(\bar\alpha d) \le \frac{1}{64b\kappa} \le \frac{1}{16}$.

Step S3.19: Step 0. Lemma 3.9(b) with $|g|_R = 0$: $\varepsilon_1 \le (1-2eby_0)^{-1}(2by_0 + e^{-\kappa/6})\varepsilon$. Here $by_0 \le \frac{1}{64\kappa} \le \frac{1}{384}$ (the factor $b = 1+m \ge 2$ is kept; it is absorbed by the b^{-1} in the smallness), so $(1-2eby_0)^{-1} \le (1-\frac{e}{192})^{-1} < 1.015 \le \frac{8}{7}$. *Case A:* $e^{-\kappa/6} \le 2by_0$. $\varepsilon_1 \le \frac{32}{7} by_0\varepsilon \le \frac{32}{7}\cdot\frac{1}{64\kappa}\varepsilon \le 0.08\,\varepsilon/\kappa$, and $\varepsilon_1 \le \frac{32}{7}\cdot 64\, b\varepsilon^2/(\bar\alpha d) \le 294\, b\varepsilon^2/(\bar\alpha d)$. *Case B:* $e^{-\kappa/6} > 2by_0$. Doubled losses ($\delta_{\nu'} = 4\delta_{\nu_0}$): $|f_1|, \le (1-eby_0/2)^{-1}(2b\frac{y_0}{4} + e^{-\kappa/3})\varepsilon$, with $(1-eby_0/2)^{-1} \le (1-\frac{e}{768})^{-1} < \frac{32}{31}$ (this is the denominator $1-2eb\,y_0/4$ with the doubled losses), and $\frac{by_0}{2} < \frac{1}{4}e^{-\kappa/6}$, so the bound is $\le \frac{32}{31}(\frac{1}{4}+\frac{1}{e})e^{-\kappa/6}\varepsilon \le e^{-\kappa/6}\varepsilon$; stop, $g = \hat f_0$, displacements $\le \varepsilon/(e\bar\alpha\sigma') < 3\varepsilon/(\bar\alpha s)$, $\le 4\varepsilon/(\bar\alpha r)$, $\le 4\varepsilon/(\bar\alpha r_q), 4\varepsilon/(\bar\alpha r_p)$.

Step S3.20: Iteration (Case A). $N = \lfloor \kappa/(12\ln 2)\rfloor < \kappa/8$; $N = 0$: stop, since then $e^{-\kappa/6} > \frac{1}{4} > \varepsilon_1/\varepsilon$. $N \ge 1$: losses $\rho = r/8N$, $\rho_p = r_p/8N$, $\rho_q = r_q/8N$, $\sigma = s/4N$, all four radii and the budget following the schedule of Step S3.6 (so every ratio R_j is $\le \frac{1}{3}$ for $j \ge 1$ and $\le \frac{3}{4N}$ for g_0). Then $\delta_\nu \ge d/(64N^2)$ and by $\kappa > 8N$

$$\frac{\varepsilon}{N} \le \frac{\bar\alpha d}{2^{15} bN^2} \le \frac{\bar\alpha\delta_\nu}{2^9 b} \quad (\text{since } d \le 64N^2\delta_\nu), \qquad \varepsilon_1 \le \frac{0.08\,\varepsilon}{8N}. \tag{2.3}$$

Then $|g_{i-1}|_R + \varepsilon_i \le 2\varepsilon_1 + \varepsilon/N \le 1.02\,\varepsilon/N$ and $x := \frac{2b}{\bar\alpha\delta_\nu}(|g_{i-1}|_R + \varepsilon_i) \le 0.004$; Moreover $u_i := \varepsilon_i/(\bar\alpha\delta_\nu) \le 0.01/(2^9 b)$ by (2.3), so $2ebu_i < 1.1\cdot 10^{-4}$ and $(1-2ebu_i)^{-1} < \frac{128}{127}$. Lemma 3.9(b) gives $\varepsilon_{i+1} \le \frac{128}{127}(0.004 + \frac{1}{8})\varepsilon_i < \frac{1}{4}\varepsilon_i$ (using $e^{-2\pi K\sigma} \le \frac{1}{8}$). Hence $\varepsilon_i \le 4^{1-i}\varepsilon_1$.

Step S3.21: Conclusion. As Step S3.8: $|f_*|, \le 4^{-(N+1)}\varepsilon \le e^{-\kappa/6}\varepsilon$; $|g - \hat f_0|, \le \frac{4}{3}\varepsilon_1 \le \frac{4}{3}\cdot 294\, b\varepsilon^2/(\bar\alpha d) \le 400\, b\varepsilon^2/(\bar\alpha d)$. Displacements by Lemma 3.9(a) and [N:`bpN:p6:telescope`]: I: $\frac{1}{e\bar\alpha s}(6\varepsilon + 4N\frac{4}{3}\varepsilon_1) \le 2.3\,\varepsilon/(\bar\alpha s)$; θ: $\frac{1}{\bar\alpha r}(8\varepsilon + \frac{32}{3}N\varepsilon_1) \le 8.2\,\varepsilon/(\bar\alpha r)$ since $N\varepsilon_1 \le 0.01\varepsilon$; the same for p, q with r_q, r_p. □

Proposition 3.11 (Averaged remainder for [P:`bpP:p0:pg`]: CP10 §4, Step 1)**.** *Let $n = n_1 \ge 1$, $m = n_2 \ge 1$, $\bar D \subset \mathbb{R}^n$, $\bar\rho, \epsilon_0 > 0$, $s_0 > 0$ (geometric), $0 < \epsilon < e^{-1/5}$, $\mu > 0$, $v_0 = (\bar\rho, \epsilon_0, \epsilon_0)$. Let $H = H_0(I) + \mu P(\theta, I, p, q)$, H_0 holomorphic on $V_{\bar\rho}\bar D$ and real, $P \in \mathcal{O}(v_0, s_0)$ real with $|P|_{v_0,s_0} \le 1$, and $\langle P\rangle(I,p,q) := \int_{\mathbb{T}^n} P\,d\theta = \hat P_0$. Choose K by $e^{-2\pi K s_0/6} = \epsilon^5$, i.e. $K = \frac{15}{\pi s_0}\log\frac{1}{\epsilon}$ (so $\kappa = 30\log\frac{1}{\epsilon} \ge 6$; CP10 (36) with 2π-conversion $K_{\mathrm{CP}} = K$, $s_{\mathrm{CP}} = 2\pi s_0$). Assume, with $\bar\alpha = 2\pi\alpha$, $d = \min\{\bar\rho s_0, \epsilon_0^2\}$,*

- *(R1) $|k\cdot\partial H_0(I)| \ge \alpha$ for $I \in V_{\bar\rho}\bar D$, $0 < |k|_1 \le K$ (non-resonance on the complex neighbourhood, CP10 (39));*
- *(R2) $2^{12}(1+m)\,K s_0\,\mu \le \alpha d$ (CP10 (65), implied by (61)).*

Then there is a map $\bar\varphi : \mathcal{D}_{v_0/2, s_0/6} \to \mathcal{D}_{v_0,s_0}$ as in Proposition 3.10 (with $\bar s = 0$, $s = s_0$) such that on $\mathcal{D}_{v_0/2,s_0/6}$

$$H\circ\bar\varphi = H_0 + \mu\langle P\rangle + \widetilde P, \qquad |\widetilde P|_{v_0/2,s_0/6} \le \frac{400(1+m)\,\mu^2}{\bar\alpha d} + \mu\,\epsilon^5,$$

where $\widetilde P = (g - \mu\langle P\rangle) + f_*$, $g = g(I,p,q)$, $|f_*|, \le \mu\epsilon^5$ (so $f_* = \mu\epsilon^5\bar P$ with $|\bar P|, \le 1$, in CP10's (40) notation), and

$$\bar\alpha s_0|I - \bar I| \le 3\mu, \quad \bar\alpha\bar\rho|\theta - \bar\theta| \le 9\mu, \quad \bar\alpha\epsilon_0|p - \bar p|,\ \bar\alpha\epsilon_0|q - \bar q| \le 9\mu.$$

Source: CP10 §4, Step 1, p. 563 (36)–(40), (62)–(63); the estimates (41)–(42) of Step 1 of §3 (p. 557). Perturbation size: $E = \mu$ before averaging and $\mu\epsilon^5$ after (CP10 p. 560: $f = \mu\epsilon^5\check P$, $E = C\mu\epsilon^5$; not μ^5).

Proof. **Step S3.22: Apply.** $f = \mu P$, $E = \mu$, $\bar s = 0$, $s = s_0$, $r = \bar\rho$, $r_p = r_q = \epsilon_0$, $h = H_0$; (R1), (R2) are (O1), (O2). Proposition 3.10 gives $H\circ\bar\varphi = H_0 + g + f_*$, and $\hat f_0 = \mu\langle P\rangle$.

Step S3.23: Collect. $\widetilde P := (g - \mu\langle P\rangle) + f_*$; its norm is bounded by the sum of the two bounds of Proposition 3.10, and $e^{-\kappa/6}\mu = \epsilon^5\mu$ by the choice of K. □

Remark 3.12 (How Part 5 uses this). (1) The two terms of CP10 (63) correspond to $\mu^2/(\bar{\alpha}d)$ (the average-correction) and $\mu\epsilon^5$ (the tail). Part 5 must verify $\mu/(\bar{\alpha}d) \le C\epsilon^5$ from its choice of $\bar{\gamma}$ (CP10 (61), with $\mu < \epsilon^6/(\log\frac{1}{\epsilon})^c$ of [P:bpP:p0:pg]); this is a different inequality from CP10's (since the proof here has no $K^{2\tau+1}/\bar{\gamma}^2$), and it is implied by (61) together with $\mu \le \epsilon^5$. (2) Distinguish three radii: the physical oscillator radius ϵ (only through K and the later domain), the ambient radius ϵ_0 of P, and Part 3's enlargement radius $\hat{\rho}$. (3) *Period dictionary with Cartesian oscillators remaining.* The pure all-angle conversion of Lemma 4.5 ($\tilde{H} \mapsto \tilde{H}(I, 2\pi\theta)/2\pi$ with I fixed) must not be applied to (ϕ, I, p, q): it pulls the form back to $2\pi\, dI \wedge d\theta + dp \wedge dq$, which is not a single conformal multiple of the standard form, and dividing the Hamiltonian by 2π would also rescale the oscillator speeds. Use instead the *canonical* chart (a c-chart of [P:bpP:p1:period-convert] with $c = 1$, acting on the angle–action block only)

$$(\theta, I, p, q) = \big(\phi_{\mathrm{CP}}/2\pi,\ 2\pi I_{\mathrm{CP}},\ p_{\mathrm{CP}},\ q_{\mathrm{CP}}\big), \qquad dI \wedge d\theta = dI_{\mathrm{CP}} \wedge d\phi_{\mathrm{CP}}.$$

The Hamiltonian is the same function, μ is unchanged, the frequency is $\omega = \omega_{\mathrm{CP}}/2\pi$ and $\alpha = \alpha_{\mathrm{CP}}/2\pi$; $\bar{\rho} = 2\pi\bar{\rho}_{\mathrm{CP}}$, $s_0 = s_{\mathrm{CP}}/2\pi$, ϵ_0 unchanged. The Fourier coefficients and the norm are unchanged (weight $e^{|k|_1 s_{\mathrm{CP}}}$). Invariants: $\bar{\alpha} = 2\pi\alpha = \alpha_{\mathrm{CP}}$, $\bar{\alpha}s_0 = \alpha_{\mathrm{CP}}s_{\mathrm{CP}}/2\pi$, $d = \min\{\bar{\rho}s_0, \epsilon_0^2\} = \min\{\bar{\rho}_{\mathrm{CP}}s_{\mathrm{CP}}, \epsilon_0^2\}$, $Ks_0/\alpha = Ks_{\mathrm{CP}}/\alpha_{\mathrm{CP}}$, so (R2) is the same inequality in both conventions; and the displacement bounds become exactly CP10's shape: $\alpha_{\mathrm{CP}}s_{\mathrm{CP}}|\Delta I_{\mathrm{CP}}| \le 6\mu$ (from $\bar{\alpha}s_0|\Delta I| \le 3\mu$, $\Delta I = 2\pi\Delta I_{\mathrm{CP}}$), $\alpha_{\mathrm{CP}}\bar{\rho}_{\mathrm{CP}}|\Delta\phi| \le 9\mu$, $\alpha_{\mathrm{CP}}\epsilon_0|\Delta p|, \alpha_{\mathrm{CP}}\epsilon_0|\Delta q| \le 9\mu$. After the oscillators have been put in action–angle form (Part 5), all angles have period 2π again and Lemma 4.5 applies to the whole system. The max-norm polydisc B_{ϵ_0} replaces CP10's Euclidean ball.

3.6. **Suspected errata in CP10 and open points.**

(1) CP10 p. 564, Lemma A.1, (66): the target is printed as $W_{v,\sigma}$; it must be $W_{v,s}$ (the map lands in the original domain).
(2) CP10 p. 563, Step 1 (rendered page checked): the text prints $r_p = r_q = \epsilon_0$, as on p. 556; an earlier version of this part misquoted "$r_p = r_q = 0$" from the text extraction (the subscript is lost) and that claim is *withdrawn*. The genuine inconsistency on that page is the action loss: it takes "$\rho = \bar{\rho}$" and "$r = \bar{\rho}$" while Lemma A.1 (p. 564) requires $\rho < r/2$ and the output action radius claimed is $\bar{\rho}/2$. The loss $\rho = \bar{\rho}/4$ would fit the claimed output radius; this is our inferred repair, not a printed statement. The finite loss schedule of Proposition 3.4 (resp. 3.10) does not depend on it.
(3) CP10 p. 565 proves the exponential bound by "applying A.1 once and then proceeding as in (A21)–(A26) of [4]": the iteration is not in the pinned sources; it is Proposition 3.4, Steps S3.6–S3.8.
(4) The original one-flow proof plan of Lemma 4.8 did not give $e^{-Ks/6}E$ (review P3-01); the finite iteration above replaces it.

Forward and cross-track references used: [N:bpN:p4:norm], [N:bpN:p4:domains], [N:bpN:p4:cauchy], [N:bpN:p4:bracket], [N:bpN:p4:projections], [N:bpN:p4:sup-weighted], [N:bpN:p4:coefficients], [N:bpN:p4:algebra], [N:bpN:p4:dictionary], [N:bpN:p5:setting], [N:bpN:p5:vf-bound], [N:bpN:p5:flow], [N:bpN:p5:symplectic], [N:bpN:p5:canonical], [N:bpN:p5:adjoint], [N:bpN:p5:composition], [N:bpN:p5:transform], [N:bpN:p5:canonical-map], [N:bpN:p6:setting], [N:bpN:p6:homological], [N:bpN:p6:divisor], [N:bpN:p6:iterative], [N:bpN:p6:iterative-special], [N:bpN:p6:telescope], [N:bpN:p6:normal-form], [N:bpN:p6:deviations], [K:bp:p1:ext-mvt], [P:bpP:p0:pg].

4. Two-scale KAM theorem I: statement and the iterative lemma

This part (3a) states the two-scale KAM theorem of Chierchia–Pinzari (CP10 Proposition 3, pp. 554–555) in the frozen conventions and proves the iterative lemma (CP10 Lemma B.1 with

Lemma B.2, pp. 565–570). Part 3b (limit, Lipschitz bounds, measure) completes the proof. *Every constant is recomputed here*: the numerical constants differ from those of CP10 (max norms, track N's weighted norm, no use of the unpinned reference [4]), the shape of $K, \hat\rho, L, \hat E$ and of the KAM condition is the same. Results of track K are cited as [K:`label`]; the averaging step is proved in Part 2 ([P:`bpP:p2:avg`], [P:`bpP:p2:nfb`]), which rests on track N, Parts 4–6.

4.1. **Conventions, norms, constants.**

- *Period one.* $\mathbb{T}^n = \mathbb{R}^n/\mathbb{Z}^n$ with coordinates θ; Fourier characters $e^{2\pi i k\cdot\theta}$, $k \in \mathbb{Z}^n$, $|k|_1 = \sum_i |k_i|$. The symplectic form is $dJ \wedge d\theta$, the Hamilton equations are $\dot\theta = \partial_J H$, $\dot J = -\partial_\theta H$ (this is the order (action, angle) used by CP10; the frozen order $(z_1, z_2) =$ (position, momentum) is restored in Part 5). A frequency $\omega \in \mathbb{R}^n$ is a rotation number per unit time. Part 0 keeps period 2π for the canonical fast angles of P-G; the passage is the one-line dictionary of Lemma 4.5, under which *all constants of this part are invariant.*
- *Norms.* $|\cdot|$ is the max norm on $\mathbb{R}^n, \mathbb{C}^n$; $\|A\| = \max_i \sum_j |A_{ij}|$ the induced operator norm (also for rectangular matrices). Hence for a bilinear form entries and operator norm are comparable ($|A_{ij}| \le \|A\|$).
- *Complex neighbourhoods.* For $D \subset \mathbb{R}^n$ and $r > 0$: $D_r = \{I \in \mathbb{C}^n : \exists I_0 \in D,\ |I - I_0| < r\}$ and $\mathrm{Re}(D_r) = D_r \cap \mathbb{R}^n = \{x \in \mathbb{R}^n : \mathrm{dist}_\infty(x, D) < r\}$; $(D_r)_{r'} \subset D_{r+r'}$ and $D_r \subset D_{r'}$ for $r \le r'$. For $\sigma > 0$, $\mathbb{T}_\sigma = \{\theta \in \mathbb{C}^n/\mathbb{Z}^n : |\mathrm{Im}\,\theta| < \sigma/2\pi\}$ (weight σ, strip width $\sigma/2\pi$).
- *Weighted norm (track N's norm, ruling P3-03).* For $f = \sum_k \hat f_k(I) e^{2\pi i k\cdot\theta}$ analytic on $D_r \times \mathbb{T}_\sigma$: $\|f\|_{r,\sigma} := \sup_{I\in D_r} \sum_{k\in\mathbb{Z}^n} |\hat f_k(I)|\, e^{\sigma|k|_1}$ ([N:`bpN:p4:norm`]; `docs/fourier-conventions.md`). *Width dictionary:* σ here is the coefficient radius, $\sigma = 2\pi\sigma_N$ with σ_N the geometric half-width ($\mathbb{T}_\sigma = T_{\sigma/2\pi}$ in the notation of Part 2). The printed CP10 norm is $\sum_k \sup_I$, which dominates this one; the two are not equivalent. Properties used in Part 3: monotone ($\|f\|_{r',\sigma'} \le \|f\|_{r,\sigma}$ for smaller base set, $r' \le r$, $\sigma' \le \sigma$), $\sup_{D_r\times\mathbb{T}_\sigma}|f| \le \|f\|_{r,\sigma}$, $\sup_{D_r}|f_0| \le \|f\|_{r,\sigma}$, and Cauchy estimates, applied to sup only; the averaging output (Corollary 3.6) is stated in this norm. Every constant below was rechecked for this change (none used $\sum_k \sup_I$; Part 2, Remark 3.7).
- *Two-scale Diophantine set.* For $n = n_1 + n_2$, $k = (k_1, k_2) \in \mathbb{Z}^{n_1} \times \mathbb{Z}^{n_2}$, $\gamma_1 \ge \gamma_2 > 0$, $\tau > 0$: $\mathcal{D}_{\gamma_1,\gamma_2,\tau} = \{\omega \in \mathbb{R}^n :\ |k\cdot\omega| \ge \gamma_1 |k|_1^{-\tau}$ if $k_1 \ne 0$, $\ |k\cdot\omega| \ge \gamma_2 |k|_1^{-\tau}$ if $k_1 = 0, k \ne 0\}$. For $\gamma_1 = \gamma_2$ this is the set of Definition 1.1; $\mathcal{D}_{\gamma_1,\gamma_2,\tau} \subset \mathcal{D}_{\gamma_2,\gamma_2,\tau}$. (CP10 (19) with $\bar\gamma_2\mu = \gamma_2$; for $k_1 = 0$ CP10 writes $|k_2|^\tau$, the same as $|k|_1^\tau$.) For $D \subset \mathbb{R}^n$ and $\omega_0 : D_\rho \to \mathbb{C}^n$ put $D^{\mathrm{dio}} := \omega_0^{-1}(\mathcal{D}_{\gamma_1,\gamma_2,\tau}) \cap D$ (this is $\mathcal{D}_{\gamma_1,\gamma_2,\tau_*}$ of CP10, "the ω_0-preimage in D").
- *Blocks.* $U = \partial^2 h$ is the $n \times n$ Hessian; $\hat U$ is the $n_2 \times n$ matrix of its last n_2 rows (the derivatives of $\omega_2 = \partial_{I_2} h$ with respect to all actions); T_1, T_2 are the first n_1 and last n_2 rows of U^{-1}.

Remark 4.1 (Misprint in the definition of $\hat U$). CP10 p. 554 defines $\hat U$ as "the $n \times n_1$ submatrix of U, i.e. the matrix with entries $\hat U_{ij} = U_{ij}$, for $n_1 + 1 \le i \le n$, $1 \le j \le n$". The size and the index ranges disagree. Only the reading above ($n_2 \times n$, rows $n_1 + 1, \dots, n$, all columns) is used in the proof: $\hat M$ bounds the variation of ω_2 (CP10 (82), via the choice of $\hat\rho$). We adopt it.

Definition 4.2 (Constants of the two-scale theorem)**.** Let $n = n_1 + n_2$, $\tau > 0$, $\gamma_1 \ge \gamma_2 > 0$, $0 < s$, $\rho > 0$, and $E, M, \hat M, \bar M > 0$. Put

$$L := \max\{\bar M,\ M^{-1},\ \hat M^{-1}\}, \qquad x := \frac{EM^2L}{\gamma_1^2}, \qquad K := \frac{6}{s}\log_+\!\big(x^{-1}\big), \quad \log_+ a := \max\{1, \log a\},$$

$$\hat\rho := \min\Big\{\frac{\gamma_1}{3MK^{\tau+1}},\ \frac{\gamma_2}{3\hat MK^{\tau+1}},\ \rho\Big\}, \qquad \hat E := \frac{EL}{\hat\rho^{\,2}}, \qquad \hat c := \max\Big\{2^{13}n,\ \frac{24^{\tau+1}}{3}\Big\}.$$

The *KAM condition* is $\hat{c}\hat{E} < 1$. (CP10 (31): $\hat{c} = \max\{2^8 n, 24^{\tau+1}/6\}$; the larger constants come from the max-norm Cauchy estimates and the Lipschitz bookkeeping of Parts 3a–3b. Only $\hat{c} = \hat{c}(n,\tau)$, an absolute function of n and τ, matters downstream.) *Totality.* $E > 0$ guards $\log(1/x)$, and $\log_+ \geq 1$ gives $Ks \geq 6$; $K, \hat{\rho} > 0$.

Source: CP10 Prop. 3, p. 555, equations (31), (32), with the changes stated.

4.2. **The theorem.**

Proposition 4.3 (Two-scale KAM theorem)**.** ***Data.*** *Integers $n_1, n_2 \geq 1$, $n = n_1 + n_2$; $\tau > n$; $\gamma_1 \geq \gamma_2 > 0$; $s, \bar{s}$ with $0 < 4s \leq \bar{s} < 1$; $\rho > 0$; an arbitrary set $D \subset \mathbb{R}^n$; $A := D_\rho$. A function $H(J,\theta) = h(J) + f(J,\theta)$,* real analytic *on $A \times \mathbb{T}_{\bar{s}+s}$, 1-periodic in θ (real on $\mathbb{R}^n \times \mathbb{R}^n$). Let $\omega_0 := \partial h$; assume ω_0 is an injective holomorphic map $A \to \mathbb{C}^n$ with $U = \partial^2 h$ nonsingular on A (hence a diffeomorphism of A onto an open set). Constants $M, \hat{M}, \bar{M}, \bar{M}_1, \bar{M}_2, E > 0$ with*

$$M \geq \sup_A \|U\|, \quad \hat{M} \geq \sup_A \|\hat{U}\|, \quad \bar{M} \geq \sup_A \|U^{-1}\|, \quad \bar{M}_i \geq \sup_A \|T_i\| \ (i = 1,2), \quad E \geq \|f\|_{\rho,\bar{s}+s},$$

and (normalizations, always satisfiable by lowering the constants; see Remark 4.4) $\hat{M} \leq M$, $\bar{M}_i \leq \bar{M}$. Let $L, K, \hat{\rho}, \hat{E}, \hat{c}$ be as in Definition 4.2 and assume the ***KAM condition*** *$\hat{c}\hat{E} < 1$.*

Conclusions. *Put $r := 32\,\hat{E}\hat{\rho}$, $\Omega_* := \omega_0(D) \cap \mathcal{D}_{\gamma_1,\gamma_2,\tau}$ and $D^{\mathrm{dio}} := \omega_0^{-1}(\Omega_*) \cap D$. There are maps φ_ω, $\omega \in \Omega_*$,*

$$\varphi_\omega : \mathbb{R}^n \to \mathrm{Re}(D_r) \times \mathbb{R}^n, \quad \varphi_\omega(\vartheta) = \big(v(\vartheta;\omega),\ \vartheta + u(\vartheta;\omega)\big),$$

with v, u 1-periodic in ϑ, such that for every $\omega \in \Omega_*$:

- *(a) (analyticity) φ_ω is real analytic and extends holomorphically to $\{|\mathrm{Im}\,\vartheta| < \bar{s}/2\pi\}$, with $|u| \leq \frac{3}{4\pi}\hat{E}s$ there;*
- *(b) (conjugacy) $\frac{d}{dt}\varphi_\omega(\vartheta + t\omega) = X_H(\varphi_\omega(\vartheta + t\omega))$ for all $\vartheta \in \mathbb{R}^n, t \in \mathbb{R}$, where $X_H = (-\partial_\theta H, \partial_J H)$ (Hamilton equations for (J,θ));*
- *(c) (embedded Lagrangian torus) $D_\vartheta \varphi_\omega$ is injective (indeed $\|D_\vartheta u\| \leq \frac{3}{4} n\hat{E}$) and $\sum_i \big(\partial_{\vartheta_a} v_i\, \partial_{\vartheta_b}(\vartheta_i + u_i) - \partial_{\vartheta_b} v_i\, \partial_{\vartheta_a}(\vartheta_i + u_i)\big) = 0$ for all a, b;*
- *(d) (estimates) with $I^0(\omega) := \omega_0^{-1}(\omega) \in D^{\mathrm{dio}}$ and $v = (v_1, v_2) \in \mathbb{R}^{n_1} \times \mathbb{R}^{n_2}$,*

$$|v_1 - I_1^0(\omega)| \leq 20\Big(\frac{\bar{M}_1}{\bar{M}} + \frac{\hat{M}}{M}\Big)\hat{E}\hat{\rho}, \quad |v_2 - I_2^0(\omega)| \leq 20\Big(\frac{\bar{M}_2}{\bar{M}} + 1\Big)\hat{E}\hat{\rho}, \quad |u| \leq \frac{3}{4\pi}\hat{E}s \quad (\text{on } \mathbb{R}^n).$$

Moreover the map $(\vartheta, \omega) \mapsto \varphi_\omega(\vartheta)$, $\mathbb{R}^n \times \Omega_* \to \mathbb{R}^{2n}$, is injective modulo the translations $\vartheta \mapsto \vartheta + k$, $k \in \mathbb{Z}^n$ (precisely: $\varphi_\omega(\vartheta) - \varphi_{\omega'}(\vartheta') \in \{0\} \times \mathbb{Z}^n$ implies $\omega = \omega'$ and $\vartheta - \vartheta' \in \mathbb{Z}^n$), and is Lipschitz in the following sense: it factors as $\Phi \circ (\check{\imath} \circ \omega_0^{-1}, \mathrm{id})$ with $\check{\imath}$ of Lipschitz constant $\leq 1 + 2560 n\hat{E}$ and Φ bi-Lipschitz in the rescaled norm of §5.1; ω_0^{-1} is $\bar{M}$-Lipschitz on Ω_* for pairs at distance $< \rho/\bar{M}$ (globally $\bar{M}\max\{1, \mathrm{diam}\, D/\rho\}$-Lipschitz if D is bounded). Finally, with $K_\Omega := \bigcup_{\omega \in \Omega_*} \varphi_\omega(\mathbb{R}^n)$ (a union of tori), $c_n := 1 + (1+2^n)(1+(4/3)^n)$ and meas *the Lebesgue measure on $\mathbb{R}^n \times \mathbb{T}^n$,*

$$\mathrm{meas}\big((\mathrm{Re}(D_r) \times \mathbb{T}^n) \setminus K_\Omega\big) \leq c_n\Big(\mathrm{meas}\big((D \setminus D^{\mathrm{dio}}) \times \mathbb{T}^n\big) + \mathrm{meas}\big((\mathrm{Re}(D_r) \setminus D) \times \mathbb{T}^n\big)\Big). \tag{3.1}$$

Constant dependence. *$\hat{c}, c_n$ depend only on n, τ; $K, \hat{\rho}, \hat{E}, L$ depend only on $(n_1, n_2, \tau, \gamma_1, \gamma_2, s, \rho, E, M, \hat{M}, \bar{M})$, in the order of Definition 4.2. The sets D, h, f enter only through those constants and the hypotheses; $\bar{s}$ enters only through $4s \leq \bar{s} < 1$ and through the width of analyticity in (a).*

Not frozen itself: Part 5 consumes this statement (§5.6) to prove the frozen theorem [P:bpP:p0:pg].

Source: *CP10 Prop. 3, pp. 554–555, proved in Appendix B, pp. 565–573. Constants recomputed; (35) of CP10 is corrected in (d).*

Source gap: *Prop. 3 has no other source ("not available in literature", CP10 p. 550); its printed proof relies on Prop. 1/Lemma A.1 through the unpinned [4, Lemma A.5]; that input is supplied by Part 2 (Lemma 4.8 is proved there). The scheme (Lemma B.1) is proved inductively on CP10 pp. 567–570; the pointer to Arnold [3, p. 144] in Remark 3 is historical, not a dependency. Misprints and repairs: §5.7.*

Remark 4.4 (Normalizations $\hat{M} \le M$, $\bar{M}_i \le \bar{M}$). $\hat{U}$ and T_i are sub-blocks of U, U^{-1}, so $\|\hat{U}\| \le \|U\|$ and $\|T_i\| \le \|U^{-1}\|$ on A: the hypotheses may be stated with $\hat{M} \le M$ and $\bar{M}_i \le \bar{M}$ *if one replaces* $\hat{M}$, $\bar{M}_i$ *by* $\min\{\hat{M}, M\}$, $\min\{\bar{M}_i, \bar{M}\}$; this only decreases $\hat{M}$ (which can only increase $\hat{\rho}$, hence is not harmless for $\hat{E}$: one should lower after computing $\hat{E}$). We state the proposition with the normalization as a hypothesis, as CP10 uses it implicitly in (74), (77) and (35) ("$\le 5n\hat{E}\hat{\rho}$" needs $\bar{M}_i \le \bar{M}$). In P-G, $\hat{M} = C\mu \le M = C$ and $\bar{M}_i \le \bar{M} = C\mu^{-1}$.

Lemma 4.5 (Period dictionary)**.** *This is* [P:`bpP:p1:period-convert`] *with* $c = 2\pi$ *(chart* $S_{2\pi}(z_1, z_2) = (2\pi z_1, z_2)$*; in Part 3's order,* $Q(I, \theta) = (I, 2\pi\theta)$*). Let* $\tilde{H}(I, \phi) = \tilde{h}(I) + \tilde{f}(I, \phi)$ *be analytic on* $A \times \{|\operatorname{Im}\phi| < \sigma\}$, $\phi \in (\mathbb{R}/2\pi\mathbb{Z})^n$ *(CP10 convention: form* $dI \wedge d\phi$*, characters* $e^{ik\cdot\phi}$*, norm* $\sup_I \sum_k |\tilde{f}_k| e^{\sigma|k|_1}$*,* $\tilde{\omega} = \partial\tilde{h}$*; constants* $\tilde{M}, \hat{\tilde{M}}, \bar{\tilde{M}}, \tilde{E}, \tilde{\gamma}_i$ *and* $\tilde{L}, \tilde{K}, \hat{\tilde{\rho}}, \hat{\tilde{E}}$ *computed by Definition 4.2 with* $\tilde{\gamma}_i$ *the Diophantine constants of* $\tilde{\omega}$*). Put* $H(I, \theta) := \tilde{H}(I, 2\pi\theta)/2\pi$*. Then* $h = \tilde{h}/2\pi$, $f = \tilde{f}(\cdot, 2\pi\theta)/2\pi$, $\omega = \tilde{\omega}/2\pi$, $\|f\|_{\rho,\sigma} = \|\tilde{f}\|_{\rho,\sigma}/2\pi$ *and, with* $M = \tilde{M}/2\pi$, $\hat{M} = \hat{\tilde{M}}/2\pi$, $\bar{M} = 2\pi\bar{\tilde{M}}$, $\bar{M}_i = 2\pi\bar{\tilde{M}}_i$, $E = \tilde{E}/2\pi$, $\gamma_i = \tilde{\gamma}_i/2\pi$*:* $L = 2\pi\tilde{L}$ *and* $K = \tilde{K}$, $\hat{\rho} = \hat{\tilde{\rho}}$, $\hat{E} = \hat{\tilde{E}}$, $\hat{c} = \hat{\tilde{c}}$ identically. Tori. *If* $\varphi_\omega = (v, \mathrm{id} + u)$ *is a torus of Proposition 4.3 for* H *(coordinates* (I, θ)*, frequency* ω*), then* $\tilde{\varphi}(\phi) := Q\big(\varphi_\omega(\phi/2\pi)\big) = (v(\phi/2\pi),\ \phi + 2\pi u(\phi/2\pi))$ *is a torus for* $\tilde{H}$ *in CP's coordinates with flow* $\phi \mapsto \phi + t\tilde{\omega}$*,* $\tilde{\omega} = 2\pi\omega$*, and* $\tilde{u} := 2\pi u$ *satisfies* $|\tilde{u}| \le \frac{3}{2}\hat{E}s$ *(the* output angle is scaled*). The frozen parameterization is* $K = \Psi \circ (\pi \circ \varphi_\omega)$ *with the pipeline chart* $\Psi = \Phi \circ S_{2\pi}$ *of* [P:`bpP:p1:cyl-pipeline`] *(the* unscaled *torus* $\pi \circ \varphi_\omega$ *is the input;* Q *is only the CP-coordinate description of the same torus, never composed again), with frozen frequency* $w = \omega = \tilde{\omega}/2\pi$*:* one *division by* 2π*, already contained in the passage* $\tilde{H} \mapsto H$*;* w *is not divided again (test:* $\tilde{h}(I) = 2\pi I$ *has* $\tilde{\omega} = 2\pi$ *and* $\omega = w = 1$*).*

Proof. **Step S4.1: Substitution.** $\tilde{f}(I, \phi) = \sum_k \tilde{f}_k(I) e^{ik\cdot\phi}$ and $\phi = 2\pi\theta$ give $f(I, \theta) = \frac{1}{2\pi}\sum_k \tilde{f}_k(I) e^{2\pi ik\cdot\theta}$: same weights, coefficients divided by 2π. $U = \tilde{U}/2\pi$, $U^{-1} = 2\pi\tilde{U}^{-1}$, sub-blocks likewise.

Step S4.2: Invariants. $EM^2L/\gamma_1^2 = \frac{\tilde{E}}{2\pi}\frac{\tilde{M}^2}{4\pi^2} 2\pi\tilde{L}\frac{4\pi^2}{\tilde{\gamma}_1^2} = \tilde{E}\tilde{M}^2\tilde{L}/\tilde{\gamma}_1^2$, hence $K = \tilde{K}$; $\gamma_1/(3MK^{\tau+1}) = \tilde{\gamma}_1/(3\tilde{M}K^{\tau+1})$ (same for $\gamma_2, \hat{M}$), so $\hat{\rho} = \hat{\tilde{\rho}}$; $EL/\hat{\rho}^2 = \tilde{E}\tilde{L}/\hat{\rho}^2$.

Step S4.3: Dynamics and tori. $\dot{\phi} = \partial_I\tilde{H} = 2\pi\partial_I H = 2\pi\dot{\theta}$, $\dot{I} = -\partial_\phi\tilde{H} = -\partial_\theta H$ ([P:`bpP:p1:period-convert`](b)); composing the conjugacy with Q gives $\tilde{\varphi}$, and [P:`bpP:p1:period-convert`](c) gives $w = \tilde{\omega}/2\pi = \omega$. □

4.3. **External facts and imported tools.**

(X1) *Holomorphic inverse function theorem.* An injective holomorphic $F : \Omega \to \mathbb{C}^n$ (Ω open) with $\det DF \neq 0$ is open, and $F^{-1} : F(\Omega) \to \Omega$ is holomorphic with $D(F^{-1}) = (DF)^{-1} \circ F^{-1}$.

(X2) *Fourier coefficients of periodic continuous functions* ([K:`bp:p1:ext-fourier`]): $\hat{F}_k = \int_{[0,1]^n} F(\vartheta) e^{-2\pi ik\cdot\vartheta} d\vartheta$, used in Lemma 5.4 (which replaces a several-variable Vitali–Montel theorem; review P3-11).

(X3) *Weierstrass.* Locally uniform limits of holomorphic functions are holomorphic and derivatives converge locally uniformly ([K:`bp:p1:uniform-limits`]).

(X4) Banach fixed point ([K:`bp:p1:ext-banach`]); mean value inequality on convex sets ([K:`bp:p1:ext-mvt`]); Cauchy estimates on polydiscs ([K:`bp:p1:cauchy-open`]).

(X5) Neumann series: $\|B\| \le \frac{1}{2}$ implies $1 + B$ invertible with $\|(1 + B)^{-1}\| \le 2$.

(X6) Lipschitz measure lemmas [K:`bp:p1:lip-image`], McShane extension [K:`bp:p1:mcshane`] (max norm).

4.4. **Two elementary lemmas.**

Lemma 4.6 (Balls in the image of a diffeomorphism)**.** *Let $\Omega \subset \mathbb{C}^n$ be open, $F : \Omega \to \mathbb{C}^n$ injective holomorphic with $\|(DF)^{-1}\| \le m$ on Ω, and $B(x_0, \delta) \subset \Omega$ (max-norm ball). Then $F(B(x_0, \delta)) \supset B(F(x_0), \delta/m)$. Moreover, if $w_1, w_2 \in B(F(x_0), \delta/m)$ then $|F^{-1}(w_1) - F^{-1}(w_2)| \le m|w_1 - w_2|$, with F^{-1} evaluated in $B(x_0, \delta)$.*

Proof. **Step S4.4: Inverse.** By (X1), $G := F^{-1} : F(\Omega) \to \Omega$ is holomorphic with $\|DG\| \le m$ and $F(B(x_0, \delta))$ is open.

Step S4.5: Path lifting. Let $w \in B(F(x_0), \delta/m)$, $\gamma(t) = F(x_0) + t(w - F(x_0))$, $\delta' = m|w - F(x_0)| < \delta$ and $T = \{t \in [0,1] : \gamma([0,t]) \subset F(B(x_0, \delta))\}$. $0 \in T$ and T is relatively open. For $t \in T$ the path $G \circ \gamma$ on $[0,t]$ is C^1 with $|(G \circ \gamma)'| \le m|w - F(x_0)|$, so $|G(\gamma(t)) - x_0| \le t\delta'$ ((X4), mean value inequality along the path).

Step S4.6: Closedness. If $t_k \uparrow t^*$, $t_k \in T$, then $G(\gamma(t_k))$ is Cauchy (Lipschitz in t), with limit x^*, $|x^* - x_0| \le \delta' < \delta$, $F(x^*) = \gamma(t^*)$ by continuity of F, so $t^* \in T$. Hence $T = [0,1]$ and $w = F(G(w)) \in F(B(x_0, \delta))$.

Step S4.7: Lipschitz bound. The segment $[w_1, w_2]$ lies in the convex ball $B(F(x_0), \delta/m) \subset F(B(x_0, \delta))$, so the mean value inequality for G gives the second claim. □

Lemma 4.7 (Lipschitz constants from sup bounds, non-convex sets)**.** *Let $S \subset \mathbb{R}^N$ be arbitrary, $r > 0$, $S_r \subset \mathbb{C}^N$ its complex r-neighbourhood in the max norm, and $\Delta : S_r \to \mathbb{C}^m$ holomorphic with $\sup_{S_r} |\Delta| \le A$. Then $|\Delta(x) - \Delta(y)| \le \frac{4NA}{r}|x - y|$ for all $x, y \in S$.*

Proof. **Step S4.8: Close pairs.** If $|x - y| < r/2$, every point z of the real segment $[x, y]$ has $|z - x| < r/2$, so the closed polydisc $\bar{P}(z, r/2) \subset S_r$ (z is within $r/2$ of $x \in S$, hence $\bar{P}(z, r/2) \subset S_r$ up to the open/closed boundary; use radius $r/2 - \epsilon$ and let $\epsilon \to 0$). By [K:`bp:p1:cauchy-open`], $|\partial_\ell \Delta_i(z)| \le 2A/r$, so $\|D\Delta(z)\| \le 2NA/r$, and the mean value inequality on the segment gives $|\Delta(x) - \Delta(y)| \le (2NA/r)|x - y|$.

Step S4.9: Distant pairs. If $|x - y| \ge r/2$, then $|\Delta(x) - \Delta(y)| \le 2A \le (4A/r)|x - y|$. Since $N \ge 1$, both are $\le 4NA/r$. □

4.5. **Averaging with two frequency scales (the case $m = 0$, $\Lambda = \{0\}$ of CP10 Proposition 1).**

Lemma 4.8 (One averaging step with two scales)**.** *Let $0 < \alpha_2 \le \alpha_1$, $D \subset \mathbb{R}^n$, $r > 0$, $\bar{s}, s > 0$, $K > 0$ with $Ks \ge 6$. Let h be analytic on D_r, real on real points, $\omega = \partial h$, f analytic on $D_r \times \mathbb{T}_{\bar{s}+s}$, real analytic, 1-periodic in θ, and $E := \|f\|_{r,\bar{s}+s}$ (N-norm, §4.1). Assume*

- *(N1) (non-resonance on the complex neighbourhood) for all $I \in D_r$ and $k \in \mathbb{Z}^n$ with $0 < |k|_1 \le K$: $|k \cdot \omega(I)| \ge \alpha_1$ if $k_1 \ne 0$ and $\ge \alpha_2$ if $k_1 = 0$;*
- *(N2) (smallness) $64e\,KE < \alpha_2 r$; equivalently $E < \alpha_2 d/(2^7 c_0 Ks)$ with $d = rs$, $c_0 = e/2$ (CP10 (23), $m = 0$).*

Then there is $\Psi : D_{r/2} \times \mathbb{T}_{\bar{s}+s/6} \to D_r \times \mathbb{T}_{\bar{s}+s}$ ($\Psi(I', \theta') = (I, \theta)$), holomorphic, injective, real on real points, $\Psi - \mathrm{id}$ $\mathbb{Z}^n$-periodic in θ', symplectic ($\Psi^*(dI \wedge d\theta) = dI' \wedge d\theta'$), with $H \circ \Psi = h + g + f_*$ on $D_{r/2} \times \mathbb{T}_{\bar{s}+s/6}$, $g = g(I')$ independent of θ', and

$$\sup_{D_{r/2}} |g - f_0| \le E/4 \quad (\text{indeed} \le E/(Ks)), \qquad \|f_*\|_{r/2,\bar{s}+s/6} \le e^{-Ks/6}E,$$

$$\alpha_1 s|I_1 - I_1'| \le 9E, \quad \alpha_2 s|I_2 - I_2'| \le 9E, \quad 2\pi\alpha_2 r|\theta - \theta'| \le 9E \qquad \text{on } D_{r/2} \times \mathbb{T}_{\bar{s}+s/6}.$$

Here $f_0(I) = \int_{\mathbb{T}^n} f(I, \theta) d\theta$ and $I = (I_1, I_2) \in \mathbb{R}^{n_1} \times \mathbb{R}^{n_2}$.
Source: CP10 Prop. 1 (p. 552), $m = 0$, $\Lambda = \{0\}$, (23)–(26); Lemma A.1 (p. 564).

Proof. This is Corollary 3.6 of Part 2 verbatim (there the first two displacement constants are even $3E$; $9E$ is kept because Lemma 4.9 was computed with it). Its proof is Proposition 3.4: a *finite iteration* of track N's Iterative Lemma (the exponential $e^{-Ks/6}$ comes from the iteration, not from one flow), with the loss schedule $\rho_i = r/8N$, $\sigma_i = s/4N$, $N = \lfloor Ks/12\ln 2\rfloor$, and the generating-function splitting of Lemma 3.3 that gives the α_1-bound for I_1 at every stage. The bound on $g - f_0$ has denominator $\alpha_2 rs$ and no K in the numerator ($|g - f_0| \le E/\kappa$, $\kappa = Ks$; CP10 (25); review P3-04). Only (N1), (N2) are assumed: no Lipschitz hypothesis (Part 2, Remark 3.7). □

4.6. **One step of the scheme.**

Lemma 4.9 (One KAM step)**.** *Let the data of Proposition 4.3 be given with $H = h+f$ on $D_\rho\times\mathbb{T}^n_{\bar s+s}$, constants $M, \hat M, \bar M, \bar M_i, L, E, \dots, K, \hat\rho, \hat E$ as in Definition 4.2, and in addition*

(S1) $\hat c\hat E < 1$ *(so $\hat E < 2^{-13}/n$);*

(S2) $L \ge \max\{\bar M, M^{-1}, \hat M^{-1}\}$ *(this is the definition; it is stated to allow L larger, used in the iteration: replace L by any larger number, and then use the same formulas);*

(S3) $\omega_0(D) \subset \mathcal{D}_{\gamma_1,\gamma_2,\tau}$.

Put $\rho_+ := \hat\rho/16$, $s_+ := s/12$, $M_+ := 2M$, $\hat M_+ := 2\hat M$, $\bar M_+ := 2\bar M$, $\bar M_{i+} := 2\bar M_i$, $L_+ := 2L$, $E_+ := E^2LM^2/\gamma_1^2$ and let $K_+, \hat\rho_+, \hat E_+$ be given by Definition 4.2 with the + data. Then there are a map Ψ_+, a function h_+ and a set $D_+ \subset D_{\hat\rho/16}$, with

(i) $\Psi_+ : (D_+)_{\rho_+}\times\mathbb{T}^n_{\bar s+s_+} \to D_{\hat\rho}\times\mathbb{T}^n_{\bar s+s}$ *holomorphic, real on reals, $\Psi_+ - \mathrm{id}$ periodic, symplectic, and $H_+ := H\circ\Psi_+ = h_+ + f_+$ with $h_+ = h+g$, $g = g(I)$ analytic on $D_{\hat\rho/2}$, $\sup_{D_{\hat\rho/2}}|g| \le \frac54 E$, and $\|f_+\|_{\rho_+,\bar s+s_+} \le E_+$;*

(ii) $\omega_+ = \partial h_+$ *is injective on $D_{\hat\rho/8}$ with $U_+ = \partial^2 h_+$ nonsingular there; $(D_+)_{\rho_+} \subset D_{\hat\rho/8}$, $\omega_+(D_+) = \omega_0(D)$ (so (S3) holds again);*

(iii) $M_+ \ge \sup_{D_{\hat\rho/4}}\|U_+\|$, $\hat M_+ \ge \sup\|\hat U_+\|$, $\bar M_+ \ge \sup\|U_+^{-1}\|$, $\bar M_{i+} \ge \sup\|T_{i+}\|$ *(sup over $D_{\hat\rho/4}$), and $L_+ \ge \max\{\bar M_+, M_+^{-1}, \hat M_+^{-1}\}$;*

(iv) the map $\hat\imath_+ := (\omega_+|_{D_{\hat\rho/8}})^{-1}\circ\omega_0$ is defined and holomorphic on $D_{\hat\rho/16}$, $D_+ = \hat\imath_+(D)$, and

$$\sup_{D_{\hat\rho/16}}|\hat\imath_{+,i} - \mathrm{id}_i| \le 10\frac{\bar M_i}{\bar M}\hat E\hat\rho \ (\le 10\hat E\hat\rho,\ i=1,2), \qquad L_D(\hat\imath_+ - \mathrm{id}) \le 640\, n\hat E;$$

(where ω_0 here means the frequency map of the current h, and L_D is the Lipschitz constant on the set D);

(v) the displacement bounds, for $(I',\theta') \in D_{\hat\rho/2}\times\mathbb{T}^n_{\bar s+s/6}$ and $\Psi_+(I',\theta') = (I,\theta)$:

$$|I_1 - I_1'| \le \tfrac34\tfrac{\hat M}{\bar M}\hat E\hat\rho, \quad |I_2 - I_2'| \le \tfrac34\hat E\hat\rho, \quad 2\pi|\theta - \theta'| \le \tfrac34\hat E s;$$

(vi) $\hat\rho_+ \ge \hat\rho/(2\cdot 24^{\tau+1})$, $K_+ \le 24K$, $\hat E_+ \le \frac89(4s)^{2(\tau+1)}\hat E^2 < \hat E^2$, *and* $\hat c\hat E_+ < 1$.

Proof. **Step S4.10: Non-resonance on $D_{\hat\rho}$.** Let $I \in D_{\hat\rho}$, pick $I_0 \in D$ with $|I - I_0| < \hat\rho \le \rho$; the segment $[I_0, I]$ lies in D_ρ, where $\|U\| \le M$, hence $|\omega_i(I) - \omega_i(I_0)| \le M\hat\rho$ for all i, and for $i > n_1$ (rows of $\hat U$) $\le \hat M\hat\rho$ ([K:`bp:p1:ext-mvt`]). For $0 < |k|_1 \le K$ and $k_1 \ne 0$: $|k\cdot\omega(I)| \ge \gamma_1|k|_1^{-\tau} - |k|_1 M\hat\rho \ge \gamma_1 K^{-\tau} - KM\hat\rho \ge \frac23\gamma_1K^{-\tau} =: \alpha_1$, since $\hat\rho \le \gamma_1/(3MK^{\tau+1})$ and $\omega(I_0) \in \mathcal{D}_{\gamma_1,\gamma_2,\tau}$ by (S3). For $k_1 = 0$ only ω_2 enters: $|k\cdot\omega(I)| \ge \gamma_2K^{-\tau} - K\hat M\hat\rho \ge \frac23\gamma_2K^{-\tau} =: \alpha_2$. Also $\alpha_2 \le \alpha_1$ and (from $\hat\rho \le \gamma_i/(3\cdot K^{\tau+1}\cdot(\mathrm{M}\text{ or }\hat M))$)

$$\alpha_1 \ge 2MK\hat\rho, \qquad \alpha_2 \ge 2\hat MK\hat\rho. \tag{3.2}$$

Step S4.11: Smallness and $Ks \ge 6$. $Ks = 6\log_+(x^{-1}) \ge 6$. By (3.2) and $L \ge \hat M^{-1}$: $64e\,KE/(\alpha_2\hat\rho) \le 64e\,E/(2\hat M\hat\rho^2) \le 32e\,\hat E < 87\hat E < \hat c\hat E < 1$. So Lemma 4.8 applies with $r = \hat\rho$.

Step S4.12: Averaging. Lemma 4.8 gives Ψ, g, f_* on $D_{\hat\rho/2}\times\mathbb{T}_{\bar s+s/6}$. Since $e^{-Ks/6}=\min\{e^{-1},x\}\le x$ ($Ks/6=\log_+(1/x)$), $\|f_*\|\le xE=E^2LM^2/\gamma_1^2=E_+$. Define Ψ_+ as the restriction of Ψ to $(D_+)_{\rho_+}\times\mathbb{T}_{\bar s+s_+}$ once D_+ is defined below ($(D_+)_{\rho_+}\subset D_{\hat\rho/8}\subset D_{\hat\rho/2}$ and $s_+\le s/6$; the norm is monotone). Also $\sup_{D_{\hat\rho/2}}|g|\le\sup|g-f_0|+\sup|f_0|\le E/4+E$. This proves the first part of (i).

Step S4.13: Displacements. By the last line of Lemma 4.8, (3.2), $Ks\ge 6$ and $M\hat M^{-1}$-bounds: $|I_1-I_1'|\le\frac{9E}{\alpha_1 s}\le\frac{9EK}{6\alpha_1}\le\frac{9E}{12M\hat\rho}=\frac34\frac{E}{M\hat\rho}\le\frac34\frac{\hat M}{M}\frac{EL}{\hat\rho}$, using $\hat ML\ge1$; and $\frac{EL}{\hat\rho}=\hat E\hat\rho$. Likewise $|I_2-I_2'|\le\frac{9E}{\alpha_2 s}\le\frac34\frac{E}{\hat M\hat\rho}\le\frac34\hat E\hat\rho$, $2\pi|\theta-\theta'|\le\frac{9E}{\alpha_2\hat\rho}\le\frac{9E}{2\hat MK\hat\rho^2}\le\frac92\frac{\hat E}{K}\le\frac34\hat Es$ (as $K\ge6/s$). This is (v).

Step S4.14: Derivatives of g. For $I\in D_{\hat\rho/4}$ the polydisc of radius $\hat\rho/4$ around I lies in $D_{\hat\rho/2}$. [K:bp:p1:cauchy-open] with $A=\frac54E$, $\delta=\hat\rho/4$ gives $|\partial_i g(I)|\le5E/\hat\rho$ and $|\partial_i\partial_j g(I)|\le4A/\delta^2=80E/\hat\rho^2$, so

$$\sup_{D_{\hat\rho/4}}\|\partial^2 g\|\le\frac{80nE}{\hat\rho^2}=\frac{80n\hat E}{L},\qquad\sup_{D_{\hat\rho/4}}|\partial g|\le\frac{5E}{\hat\rho}. \tag{3.3}$$

Since $L^{-1}\le M$, $L^{-1}\le\hat M$ and $\bar M\le L$: $\|\partial^2 g\|\le80n\hat E\min\{M,\hat M\}\le\frac{1}{100}\min\{M,\hat M\}$ and $\bar M\|\partial^2 g\|\le80n\hat E\le\frac{1}{100}$.

Step S4.15: Bounds on U_+. $U_+=U+\partial^2 g$ on $D_{\hat\rho/4}$ and $\hat U_+=\hat U+(\partial^2 g)_{\text{rows}>n_1}$ with $\|\text{rows}\|\le\|\partial^2 g\|$. Hence $\|U_+\|\le M+\frac{M}{100}\le2M$ and $\|\hat U_+\|\le\hat M+\frac{\hat M}{100}\le2\hat M$. Further $U_+=(1+\partial^2 g\,U^{-1})U$, so $U_+^{-1}=U^{-1}(1+\partial^2 g\,U^{-1})^{-1}$ ($\|\partial^2 g\,U^{-1}\|\le1/100$, (X5)), hence $\|U_+^{-1}\|\le2\bar M$ and, the rows of a product T_iB being T_i times B, $\|T_{i+}\|\le2\|T_i\|\le2\bar M_i$ (the left factor of $U^{-1}\cdot(\dots)$ keeps row blocks). Nonsingular. And $L_+=2L\ge\max\{2\bar M,(2M)^{-1},(2\hat M)^{-1}\}$. This is (iii) on $D_{\hat\rho/4}$.

Step S4.16: Injectivity of ω_+ on $D_{\hat\rho/8}$. Let $I,I'\in D_{\hat\rho/8}$ with $\omega_+(I)=\omega_+(I')$ and put $w=\omega_0(I)$, $w'=\omega_0(I')$. Then $w-w'=\partial g(I')-\partial g(I)$ and $|w-w'|\le10E/\hat\rho$ by (3.3). The ball $B(I,\hat\rho/8)\subset D_{\hat\rho/4}\subset D_\rho$, and ω_0 is injective on D_ρ with $\|U^{-1}\|\le\bar M$; Lemma 4.6 gives $\omega_0(B(I,\hat\rho/8))\supset B(w,\hat\rho/(8\bar M))$. Since $10E/\hat\rho<\hat\rho/(8\bar M)$ (equivalent to $80E\bar M<\hat\rho^2$, implied by $80\hat E<1$ and $\bar M\le L$), $w'=\omega_0(I'')$ with $I''\in B(I,\hat\rho/8)$ and, by injectivity of ω_0 on D_ρ, $I''=I'$. Thus the segment $[I,I']\subset D_{\hat\rho/4}$ and, by Lemma 4.6 (second part) and the mean value inequality for ∂g along $[I,I']$, $|I-I'|\le\bar M|w-w'|\le\bar M\sup\|\partial^2 g\||I-I'|\le80n\hat E|I-I'|$, whence $I=I'$. With (iii), ω_+ is a holomorphic injection with nonsingular derivative on $D_{\hat\rho/8}$; by (X1) an open map with holomorphic inverse, $\|D(\omega_+^{-1})\|\le2\bar M$ (and $\|(D\omega_+^{-1})_i\|\le2\bar M_i$ for the row blocks) on $\omega_+(D_{\hat\rho/8})$.

Step S4.17: The map $\hat\imath_+$ and D_+. For $I\in D_{\hat\rho/16}$, $B(I,\hat\rho/16)\subset D_{\hat\rho/8}$ and Lemma 4.6 for $F=\omega_+$, $m=2\bar M$ gives $\omega_+(B(I,\hat\rho/16))\supset B(\omega_+(I),\hat\rho/(32\bar M))$. But $\omega_0(I)=\omega_+(I)-\partial g(I)$ and $|\partial g(I)|\le5E/\hat\rho<\hat\rho/(32\bar M)$ (as $160E\bar M<\hat\rho^2$, because $160\hat E<1$). So $\hat\imath_+:=\omega_+^{-1}\circ\omega_0$ is defined and holomorphic on $D_{\hat\rho/16}$, with values in $B(I,\hat\rho/16)$, $\omega_+\circ\hat\imath_+=\omega_0$. By Lemma 4.6 (second part, with $w_1=\omega_0(I)$, $w_2=\omega_+(I)$): $|\hat\imath_+(I)-I|\le2\bar M|\partial g(I)|\le10\bar ME/\hat\rho$, and the same estimate holds for the first n_1 and last n_2 components with $\bar M_i$ instead of $\bar M$ (use the row blocks of $D\omega_+^{-1}$). By $\bar M\le L$: $10\bar M_iE/\hat\rho\le10\frac{\bar M_i}{\bar M}\frac{EL}{\hat\rho}=10\frac{\bar M_i}{\bar M}\hat E\hat\rho\le10\hat E\hat\rho$. This is the sup bound in (iv). Then $\sup|\hat\imath_+-\mathrm{id}|\le10\hat E\hat\rho<\hat\rho/16$ since $\hat E<2^{-13}$. Lipschitz bound: Lemma 4.7 with $S=D$, $r=\hat\rho/16$, $N=n$, $A=10\hat E\hat\rho$ gives $L_D(\hat\imath_+-\mathrm{id})\le4n\cdot10\hat E\hat\rho\cdot16/\hat\rho=640n\hat E$. Since h is real analytic, ω_0 and ω_+ are real on $\mathbb{R}^n$; if $\omega_+(I'')=\omega_0(I)$ with $I\in D\subset\mathbb{R}^n$, then $\overline{I''}$ is a second preimage in the same ball, so $I''=\overline{I''}\in\mathbb{R}^n$ by injectivity. Hence $\hat\imath_+(D)\subset\mathbb{R}^n$. Put $D_+:=\hat\imath_+(D)\subset D_{\hat\rho/16}\cap\mathbb{R}^n$. Since $\omega_+\circ\hat\imath_+=\omega_0$ on D and ω_0 is injective, $\hat\imath_+$ is injective on D and $\omega_+(D_+)=\omega_0(D)$. Then $(D_+)_{\rho_+}\subset D_{\hat\rho/16+\hat\rho/16}=D_{\hat\rho/8}$. This is (ii) and the rest of (iv); in particular $\omega_+(D_+)=\omega_0(D)\subset\mathcal{D}_{\gamma_1,\gamma_2,\tau}$.

Step S4.18: The new constants. Put $x = EM^2L/\gamma_1^2$; then $E_+ = Ex$ and $x_+ := E_+L_+M_+^2/\gamma_1^2 = Ex \cdot 2L \cdot 4M^2/\gamma_1^2 = 8x^2$. Hence $\log(1/x_+) = 2\log(1/x) - 3\log 2 \le 2\log_+(1/x)$ and $\log_+(1/x_+) \le 2\log_+(1/x)$ (as $1 \le \log_+$), so

$$K_+ = \frac{6}{s_+}\log_+\frac{1}{x_+} \le \frac{72}{s}\cdot 2\log_+\frac{1}{x} = 24K.$$

Therefore $\gamma_1/(3M_+K_+^{\tau+1}) \ge \frac{1}{2}24^{-(\tau+1)}\gamma_1/(3MK^{\tau+1})$, likewise for $\gamma_2, \hat{M}$, and $\rho_+ = \hat{\rho}/16$; as $\hat{\rho} \le \min\{\gamma_1/(3MK^{\tau+1}), \gamma_2/(3\hat{M}K^{\tau+1})\}$, $\hat{\rho}_+ \ge \hat{\rho}\min\{\frac{1}{2\cdot 24^{\tau+1}}, \frac{1}{16}\} = \hat{\rho}/(2\cdot 24^{\tau+1})$.

Step S4.19: The quadratic bound. $E_+L_+ = 2E^2L^2M^2/\gamma_1^2$ and $\hat{\rho}_+^{-2} \le 4\cdot 24^{2(\tau+1)}\hat{\rho}^{-2}$, so $\hat{E}_+ \le 8\cdot 24^{2(\tau+1)}\,\hat{E}^2\,\hat{\rho}^2M^2/\gamma_1^2$ (using $E^2L^2/\hat{\rho}^2 = \hat{E}^2\hat{\rho}^2$). From $\hat{\rho} \le \gamma_1/(3MK^{\tau+1})$ and $K \ge 6/s$: $\hat{\rho}^2M^2/\gamma_1^2 \le \frac{1}{9}(s/6)^{2(\tau+1)}$. Hence $\hat{E}_+ \le \frac{8}{9}(24s/6)^{2(\tau+1)}\hat{E}^2 = \frac{8}{9}(4s)^{2(\tau+1)}\hat{E}^2 < \hat{E}^2$ since $4s \le \bar{s} < 1$. Finally $\hat{c}\hat{E}_+ < \hat{c}\hat{E}\cdot\hat{E} < 1$. This is (vi). □

Remark 4.10 (Differences from CP10 in the step). CP10 (74)–(75), (86)–(87) have $5n$, 2^6n and 2^7 from a mixture of 1-, ∞- and "sup" norms; the above constants are those of max norms. CP10 injectivity argument (p. 569–570) uses "$\sup\|\partial(\omega^{-1})\|\sup|\omega(I) - \omega(I')|$", a mean value inequality that silently needs $[\omega(I), \omega(I')] \subset \omega(D_\rho)$; Lemma 4.6 supplies it. The Lipschitz bound of $\hat{\imath}_+ - \mathrm{id}$ on the (non-convex) set D is repaired by Lemma 4.7. CP10 (91) prints $(4s)^{\tau+1}$ where the computation gives $(4s)^{2(\tau+1)}$ (a sharpening: the printed bound is true, being weaker).

4.7. **The iterative lemma.**

Definition 4.11 (The sequences)**.** Given the data and constants of Proposition 4.3, put $D_0 = D^{\mathrm{dio}} = \omega_0^{-1}(\mathcal{D}_{\gamma_1,\gamma_2,\tau}) \cap D$, $\rho_0 = \rho$, $s_0 = s$, $M_0 = M$, $\hat{M}_0 = \hat{M}$, $\bar{M}_0 = \bar{M}$, $\bar{M}_{i,0} = \bar{M}_i$, $L_0 = L$, $E_0 = E$, $h_0 = h$, $f_0 = f$ and, recursively for $j \ge 1$: $s_j = s_{j-1}/12$, $M_j = 2M_{j-1}$, $\hat{M}_j = 2\hat{M}_{j-1}$, $\bar{M}_j = 2\bar{M}_{j-1}$, $\bar{M}_{i,j} = 2\bar{M}_{i,j-1}$, $L_j = 2L_{j-1}$, $\rho_j = \hat{\rho}_{j-1}/16$, $E_j = E_{j-1}^2L_{j-1}M_{j-1}^2/\gamma_1^2$, and $K_j, \hat{\rho}_j, \hat{E}_j$ by Definition 4.2 with the data $(s_j, \rho_j, E_j, M_j, \hat{M}_j, \bar{M}_j)$ (so L_j there is $\max\{\bar{M}_j, M_j^{-1}, \hat{M}_j^{-1}\} \le 2^jL$, and in the formulas the larger $L_j = 2^jL$ is used, as allowed by (S2) of Lemma 4.9).

Lemma 4.12 (Iterative lemma)**.** *Under the hypotheses of Proposition 4.3 (in particular $\hat{c}\hat{E}_0 < 1$), for every $j \ge 0$ there are a set $D_j \subset \mathbb{R}^n$, an analytic symplectic map $\Phi_j = \Psi_1 \circ \cdots \circ \Psi_j$ on $W_j := (D_j)_{\rho_j} \times \mathbb{T}_{\bar{s}+s_j}$ (with $\Phi_0 = \mathrm{id}$, D_0 as above) and $H_j = H \circ \Phi_j = h_j + f_j$, such that*

(i) $D_j \subset (D_{j-1})_{\hat{\rho}_{j-1}/16}$, $\omega_j = \partial h_j$ is a diffeomorphism of $(D_j)_{\rho_j}$ onto an open set, $\omega_j(D_j) = \omega_0(D_0) \subset \mathcal{D}_{\gamma_1,\gamma_2,\tau}$; the map $\hat{\imath}_j := \omega_j^{-1} \circ \omega_{j-1} : D_{j-1} \to D_j$ (defined and holomorphic on $(D_{j-1})_{\hat{\rho}_{j-1}/16}$) satisfies $\sup|\hat{\imath}_{j,i} - \mathrm{id}_i| \le 10\frac{\bar{M}_i}{M}\hat{E}_{j-1}\hat{\rho}_{j-1}$ and $L_{D_{j-1}}(\hat{\imath}_j - \mathrm{id}) \le 640n\hat{E}_{j-1}$;

(ii) $\|f_j\|_{\rho_j,\bar{s}+s_j} \le E_j$;

(iii) $\Psi_j : W_j \to W'_{j-1}$ with $W'_{j-1} := (D_{j-1})_{\hat{\rho}_{j-1}} \times \mathbb{T}_{\bar{s}+s_{j-1}} \subset W_{j-1}$ is holomorphic (so $\Psi_j(W_j) \subset W_{j-1}$), real on reals, $\Psi_j - \mathrm{id}$ periodic, and on W_j

$$|\Delta I_1| \le \tfrac{3}{4}\tfrac{\hat{M}}{M}\hat{E}_{j-1}\hat{\rho}_{j-1}, \quad |\Delta I_2| \le \tfrac{3}{4}\hat{E}_{j-1}\hat{\rho}_{j-1}, \quad 2\pi|\Delta\theta| \le \tfrac{3}{4}\hat{E}_{j-1}s_{j-1};$$

(iv) $\hat{E}_j < \hat{E}_{j-1}^2$ for $j \ge 1$, whence $\hat{E}_j \le \hat{E}_0^{2^j}$, and $\hat{c}\hat{E}_j < 1$;

(v) $\hat{\rho}_j \le \hat{\rho}_{j-1}/16$ and $\hat{\rho}_j \ge \hat{\rho}_{j-1}/(2\cdot 24^{\tau+1})$, $s_j = s/12^j$, $K_j \le 24^jK$.

Proof. **Step S4.20: Induction hypothesis.** We prove by induction on $j \ge 0$ the statement $\mathcal{P}_j$: the data $(D_j, h_j, f_j, \rho_j, s_j, M_j, \hat{M}_j, \bar{M}_j, \bar{M}_{i,j}, L_j, E_j)$ satisfy the hypotheses of Lemma 4.9 (on the domain $(D_j)_{\rho_j} \times \mathbb{T}_{\bar{s}+s_j}$): $\hat{c}\hat{E}_j < 1$, $L_j \ge \max\{\bar{M}_j, M_j^{-1}, \hat{M}_j^{-1}\}$, $\omega_j(D_j) \subset \mathcal{D}_{\gamma_1,\gamma_2,\tau}$, bounds $M_j, \hat{M}_j, \bar{M}_j, \bar{M}_{i,j}$ on $(D_j)_{\rho_j}$, ω_j injective on $(D_j)_{\rho_j}$ with nonsingular U_j, $\|f_j\| \le E_j$, $\hat{M}_j \le M_j$, $\bar{M}_{i,j} \le \bar{M}_j$, and $4s_j \le \bar{s} < 1$.

Step S4.21: Base. $\mathcal{P}_0$ is the hypothesis of Proposition 4.3 and $\omega_0(D_0) \subset \mathcal{D}_{\gamma_1,\gamma_2,\tau}$ by definition of D_0 (note $D_0 \subset D$, so the sup bounds on D_ρ give those on $(D_0)_\rho$; injectivity is inherited).

Step S4.22: Induction. Apply Lemma 4.9 to $\mathcal{P}_{j-1}$: it produces $D_j := D_+$, $\Psi_j := \Psi_+$, $h_j := h_+$, $f_j := f_+$, and (i)–(vi) of that lemma are (i)–(iii) here and the recursion of Definition 4.11 (the quantities named $\hat\rho_+, K_+, \hat E_+$ there are $\hat\rho_j, K_j, \hat E_j$). Items (iii) and (v) of the step give $\mathcal{P}_j$ ($\hat M_j \le M_j$, $\bar M_{i,j} \le \bar M_j$ propagate by doubling).

Step S4.23: Composition. $\Phi_j = \Phi_{j-1} \circ \Psi_j$ is defined on W_j because $\Psi_j(W_j) \subset W'_{j-1} \subset W_{j-1}$: $W_j \subset D_{\hat\rho_{j-1}/8} \times \mathbb{T}_{\bar s + s_j}$ (more precisely $(D_j)_{\rho_j}$ with $D_j \subset (D_{j-1})_{\hat\rho_{j-1}/16}$), the displacement is $\le \frac{3}{4}\hat E_{j-1}\hat\rho_{j-1} < \hat\rho_{j-1}/16$, so the action of $\Psi_j(y)$ lies in $(D_{j-1})_{\hat\rho_{j-1}/8+\hat\rho_{j-1}/16} \subset (D_{j-1})_{\hat\rho_{j-1}}$, and $\hat\rho_{j-1} \le \rho_{j-1}$ (definition of $\hat\rho$). The identity $H_j = H_{j-1}\circ\Psi_j = H\circ\Phi_j$ is then immediate, and Φ_j is symplectic, real, with $\Phi_j - \mathrm{id}$ periodic.

Step S4.24: (iv)–(v) of the Lemma. By Step (vi) of Lemma 4.9: $\hat E_j < \hat E_{j-1}^2$, $\hat\rho_j \ge \hat\rho_{j-1}/(2\cdot 24^{\tau+1})$, $\hat\rho_j \le \rho_j = \hat\rho_{j-1}/16$, $K_j \le 24K_{j-1}$. □

5. Two-scale KAM theorem II: the limit, the tori and the measure

We continue with the notation of Definition 4.11 and Lemma 4.12; $c := 24^{\tau+1}/6$ and $q := 2\cdot 24^{\tau+1} = 12c$. Since $\hat c \ge 2c$, $\hat c \ge 2^{13}n$, the KAM condition gives

$$\hat E_0 < \frac{1}{8192\,n}, \qquad c\hat E_0 < \frac{1}{2}. \tag{3.4}$$

Lemma 5.1 (Sums)**.** *For $m \ge 0$: $\hat E_k \le \hat E_m^{2^{k-m}}$ for $k \ge m$, $\sum_{k\ge m}\hat E_k \le 2\hat E_m$, and $\sum_{k\ge 0} c^k \hat E_k \le 2\hat E_0$. Also $\hat\rho_k \le \hat\rho_0\, 16^{-k}$, $s_k = s\,12^{-k}$.*

Proof. **Step S5.1: Doubling.** By Lemma 4.12(iv), $\hat E_k < \hat E_{k-1}^2$, so $\hat E_k \le \hat E_m^{2^{k-m}} \le \hat E_m^{k-m+1}$ (as $2^i \ge i+1$ and $\hat E_m < 1$). Hence $\sum_{k\ge m}\hat E_k \le \hat E_m/(1-\hat E_m) \le 2\hat E_m$.

Step S5.2: Weighted sum. $c^k\hat E_k \le c^k \hat E_0^{k+1} = \hat E_0 (c\hat E_0)^k \le \hat E_0 2^{-k}$ by (3.4); sum $\le 2\hat E_0$.

Step S5.3: Radii. $\hat\rho_k \le \rho_k = \hat\rho_{k-1}/16$ (Definition of $\hat\rho_k$), iterate. □

5.1. The limit actions (CP10 Step 1).

Lemma 5.2 (Limit actions)**.** *Put $\check\imath_0 = \mathrm{id}$, $\check\imath_j := \hat\imath_j \circ \cdots \circ \hat\imath_1 : D_0 \to D_j$. Then $\check\imath_j$ converges uniformly on D_0 to $\check\imath = (\check\imath_1, \check\imath_2)$ and*

- *(i)* $\sup_{D_0}|\check\imath_i - \mathrm{id}_i| \le 20\frac{\bar M_i}{\bar M}\hat E_0\hat\rho_0 \le 20\hat E_0\hat\rho_0$; $\sup_{D_0}|\check\imath - \check\imath_j| \le 20\hat\rho_j\hat E_j$;
- *(ii)* $D_* := \check\imath(D_0) \subset (D_j)_{\rho_j/2}$ *for all j, in particular* $D_* \subset \mathrm{Re}(D_0)_{20\hat E_0\hat\rho_0}$;
- *(iii)* $\lambda := L_{D_0}(\check\imath - \mathrm{id}) \le 2560\,n\hat E_0 < \frac{1}{3}$; *so $\check\imath$ is injective,* $(1-\lambda)|x-y| \le |\check\imath x - \check\imath y| \le (1+\lambda)|x-y|$;
- *(iv)* $\omega_j(\check\imath_j(I_0)) = \omega_0(I_0)$ *for all $I_0 \in D_0$, $j \ge 0$.*

Source: CP10 Step 1, pp. 570–571, (97)–(104); constants recomputed (CP10: $10n$, $2^8 n\hat E$).

Proof. **Step S5.4: Well-posedness.** $\hat\imath_j : D_{j-1} \to D_j$ (Lemma 4.12(i)); so $\check\imath_j(D_0) \subset D_j$ and the composition is defined. (iv) follows from $\omega_j \circ \hat\imath_j = \omega_{j-1}$ by induction.

Step S5.5: Telescoping. $|\check\imath_{j,i}(x) - \check\imath_{j-1,i}(x)| = |\hat\imath_{j,i}(y) - y_i|$ with $y = \check\imath_{j-1}(x) \in D_{j-1}$, so $\le 10\frac{\bar M_i}{\bar M}\hat E_{j-1}\hat\rho_{j-1}$ (the ratio $\bar M_{i,j}/\bar M_j = \bar M_i/\bar M$ is constant). Since $\hat\rho_{j-1} \le \hat\rho_0$, Lemma 5.1 gives the Cauchy property and (i): $\sum_{j\ge1} 10\hat E_{j-1}\hat\rho_{j-1} \le 20\hat E_0\hat\rho_0$; the tail is $\le 10\hat\rho_j\sum_{m\ge j}\hat E_m \le 20\hat\rho_j\hat E_j$.

Step S5.6: Location. $\check\imath_j(D_0) \subset D_j \subset \mathbb{R}^n$ and $20\hat\rho_j\hat E_j \le \rho_j/2$ ($\hat\rho_j \le \rho_j$, $\hat E_j < 1/40$), so $D_* \subset \mathrm{Re}(D_j)_{\rho_j/2}$; for $j=0$ and $D_0 \subset D$ this gives the last claim of (ii).

Step S5.7: Lipschitz constant. Put $i_j = \check\imath_j - \mathrm{id}$. From $\check\imath_{j+1} = \hat\imath_{j+1} \circ \check\imath_j$: $i_{j+1} = (\hat\imath_{j+1} - \mathrm{id}) \circ \check\imath_j + i_j$, so $1 + L(i_{j+1}) \le (1 + L(\hat\imath_{j+1} - \mathrm{id}))(1 + L(i_j))$ and, by Lemma 4.12(i), $1 + L(i_j) \le \prod_{k \le j}(1 + 640 n \hat E_{k-1}) \le \exp(640 n \sum_k \hat E_{k-1}) \le \exp(1280 n \hat E_0)$. As $x := 1280 n \hat E_0 < \frac{1}{6}$, $e^x \le 1 + 2x$: $L(i_j) \le 2560 n \hat E_0$ uniformly in j; the pointwise limit of Lipschitz maps keeps the constant. Injectivity and the two-sided bound follow from $|\check\imath x - \check\imath y| \ge |x - y| - \lambda |x - y|$. □

5.2. **The limit symplectic map (CP10 Step 2).** *Rescaling.* Let $S(I, \theta) := (I/\hat\rho_0,\ 2\pi\theta/s_0)$ and $|(I, \theta)|_* := |S(I, \theta)|$ (max norm); for a map G put $\check G = SGS^{-1}$ and L_* for Lipschitz constants in $|\cdot|_*$. Put $\mathcal{D}_j := \mathrm{Re}(D_j)_{\rho_j/2}$. All maps below are real on real points.

Lemma 5.3 (Domains and rescaled Lipschitz bounds for Ψ_k)**.** *(i) $\Psi_k(\mathcal{D}_k \times \mathbb{R}^n) \subset \mathcal{D}_{k-1} \times \mathbb{R}^n$ $(k \ge 1)$; hence $\Phi_j = \Psi_1 \circ \cdots \circ \Psi_j$ is defined on $\mathcal{D}_j \times \mathbb{R}^n$ with values in $\mathcal{D}_0 \times \mathbb{R}^n$.*
(ii) $L_*(\check\Psi_k - \mathrm{id}; \mathcal{D}_k \times \mathbb{R}^n) \le 192\, n\, c^{k-1} \hat E_{k-1} =: \varsigma_k$, and $\sum_{k \ge 1} \varsigma_k \le 384 n \hat E_0 < \frac{1}{20}$.
(iii) $|\Psi_k(y) - y|_* \le \frac{3}{4} \hat E_{k-1} 12^{-(k-1)} \le \frac{3}{4} \hat E_{k-1}$ for $y \in \mathcal{D}_k \times \mathbb{R}^n$.

Proof. **Step S5.8: Domains.** Let $y = (I, \theta)$, $I \in \mathcal{D}_k$: $\mathrm{dist}(I, D_k) < \rho_k/2 = \hat\rho_{k-1}/32$ and $D_k \subset \mathrm{Re}(D_{k-1})_{\hat\rho_{k-1}/16}$. By Lemma 4.12(iii) the real displacement is $\le \frac{3}{4} \hat E_{k-1} \hat\rho_{k-1}$. Hence $\Psi_k(y)_I$ is within $\hat\rho_{k-1}(\frac{1}{32} + \frac{1}{16} + \frac{3}{4} \hat E_{k-1}) < \hat\rho_{k-1}/8$ of D_{k-1}; and $\hat\rho_{k-1}/8 < \rho_{k-1}/2$ (for $k = 1$ because $\hat\rho_0 \le \rho$; for $k \ge 2$ because $\hat\rho_{k-1} \le \hat\rho_{k-2}/16$).

Step S5.9: Bound on displacement. Scaled: $|\Delta I|/\hat\rho_0 \le \frac{3}{4} \hat E_{k-1} \hat\rho_{k-1}/\hat\rho_0 \le \frac{3}{4} \hat E_{k-1} 16^{-(k-1)}$ (Lemma 5.1; $\hat M \le M$) and $2\pi |\Delta\theta|/s_0 \le \frac{3}{4} \hat E_{k-1} s_{k-1}/s_0 = \frac{3}{4} \hat E_{k-1} 12^{-(k-1)}$. Thus $A_k := \sup_{W_k} |\check\Psi_k - \mathrm{id}| \le \frac{3}{4} \hat E_{k-1} 12^{-(k-1)}$; this is (iii).

Step S5.10: Lipschitz. Apply Lemma 4.7 in the scaled coordinates, $N = 2n$, $S = S(\mathcal{D}_k \times \mathbb{R}^n)$, radius $\varepsilon_k = \min\{\rho_k/(2\hat\rho_0),\ s_k/s_0\}$: the scaled complex ε_k-neighbourhood of a real point of S lies in $(D_k)_{\rho_k} \times \mathbb{T}_{s_k} \subset W_k$ ($\varepsilon_k \hat\rho_0 \le \rho_k/2$ and $\varepsilon_k s_0 \le s_k \le \bar s + s_k$), where the bound A_k of (iii) holds (Lemma 4.12(iii) holds on the complex W_k). Now $\rho_k/(2\hat\rho_0) = \hat\rho_{k-1}/(32\hat\rho_0) \ge q^{-(k-1)}/32$ (Lemma 4.12(v)) and $s_k/s_0 = 12^{-k} \ge q^{-(k-1)}/32$, so $\varepsilon_k \ge q^{-(k-1)}/32$ and

$$L_* \le \frac{4 \cdot 2n\, A_k}{\varepsilon_k} \le 8n \cdot \tfrac{3}{4} \hat E_{k-1} 12^{-(k-1)} \cdot 32 q^{k-1} = 192 n\, (q/12)^{k-1} \hat E_{k-1} = 192 n\, c^{k-1} \hat E_{k-1}.$$

Step S5.11: Sum. Lemma 5.1: $\sum_k \varsigma_k = 192 n \sum_{m \ge 0} c^m \hat E_m \le 384 n \hat E_0 < 384/8192 < \frac{1}{20}$. □

Lemma 5.4 (Compactness of periodic holomorphic functions on a strip)**.** *Let $F_j : \{\vartheta \in \mathbb{C}^n : |\mathrm{Im}\,\vartheta| < w\} \to \mathbb{C}$ be holomorphic, $\mathbb{Z}^n$-periodic, $|F_j| \le B$, and suppose F_j converges uniformly on $\mathbb{R}^n$ to F. Then for* every *$0 < v < w$, F_j converges uniformly, together with all derivatives, on $\{|\mathrm{Im}\,\vartheta| \le v\}$; the limit is a holomorphic periodic function on the whole open strip extending F, with $|F| \le B$ there.*

Proof. **Step S5.12: Coefficient decay.** For $0 < w' < w$ and $k \ne 0$ shift the integration contour to $\mathrm{Im}\,\vartheta = -w' \,\mathrm{sign}(k)$ (componentwise; Cauchy's theorem for the periodic integrand, [K:bp:p1:ext-rectangle]): $|\hat F_{j,k}| \le B e^{-2\pi w' |k|_1}$, uniformly in j.

Step S5.13: Convergence of coefficients. Uniform convergence on $\mathbb{R}^n$ gives $\hat F_{j,k} \to \hat F_k$ for every k ((X2)), and $|\hat F_k| \le B e^{-2\pi w' |k|_1}$ for every $w' < w$.

Step S5.14: Tail and finite part on $|\mathrm{Im}\,\vartheta| \le v$. Fix $0 < v < w$ and put $w' = (v + w)/2 > v$. There $|e^{2\pi i k \cdot \vartheta}| \le e^{2\pi v |k|_1}$, so for every $m \ge 0$ $\sum_{|k|_1 > N} |(2\pi k)^m \hat F_{j,k} e^{2\pi i k \cdot \vartheta}| \le B \sum_{|k|_1 > N} (2\pi |k|_1)^m e^{-2\pi (w' - v) |k|_1} =: \varepsilon_{N,m} \to 0$ $(N \to \infty)$, uniformly in j, and the same for F; the finitely many modes $|k|_1 \le N$ converge uniformly. Hence $F_j \to F^\sharp := \sum_k \hat F_k e^{2\pi i k \cdot \vartheta}$ with all derivatives, uniformly on $|\mathrm{Im}\,\vartheta| \le v$. The limits for different v coincide (same coefficients), so $F^\sharp$ is

defined and holomorphic on the whole open strip ((X3)), equals F on $\mathbb{R}^n$ and, as a pointwise limit of the F_j, satisfies $|F^\sharp| \le B$. □

Lemma 5.5 (The limit map Φ)**.** *On $W_* := D_* \times \mathbb{R}^n$ the maps Φ_j converge uniformly to a map Φ with $\Phi - \mathrm{id}$ $\mathbb{Z}^n$-periodic in θ, real analytic in θ, and*

(i) $\sup_{W_*}|\Pi_{I_1}\Phi - \mathrm{id}| \le \frac{3}{2}\frac{\hat M}{M}\hat E_0\hat\rho_0$, $\sup|\Pi_{I_2}\Phi - \mathrm{id}| \le \frac{3}{2}\hat E_0\hat\rho_0$, $\sup 2\pi|\Pi_\theta\Phi - \mathrm{id}| \le \frac{3}{2}\hat E_0 s_0$, *and the same bounds on complex θ with $|\mathrm{Im}\,\theta| < \bar s/2\pi$;*

(ii) $L_*(\check\Phi - \mathrm{id}; W_*) \le 768\, n\hat E_0 < \frac{1}{10}$; *hence Φ is injective on W_* and $|x-y|_*\cdot\frac{9}{10} \le |\Phi x - \Phi y|_* \le \frac{11}{10}|x-y|_*$;*

(iii) $\Phi(I,\theta+k) = \Phi(I,\theta) + (0,k)$ *for $k \in \mathbb{Z}^n$.*

Source: CP10 Step 2, pp. 571–572, (105)–(108).

Proof. **Step S5.15: Telescoping of the displacements.** Write $(I^{(k)},\theta^{(k)}) = \Psi_{k+1}\circ\cdots\circ\Psi_j(x)$ for $x \in W_* \subset W_j$ (Lemma 5.2(ii)); then $\Pi_I\Phi_j(x) - \Pi_I x = \sum_{k=1}^{j}(I^{(k-1)} - I^{(k)})$ and every summand is bounded by Lemma 4.12(iii). Summing with Lemma 5.1 ($\hat\rho_{k-1} \le \hat\rho_0$, $s_{k-1} \le s_0$, $\sum \hat E_{k-1} \le 2\hat E_0$) gives (i), uniformly in j and also for complex θ (Ψ_k is defined on the complex strip).

Step S5.16: Cauchy sequence. For $x \in W_*$: $\Phi_j(x) - \Phi_{j-1}(x) = \Phi_{j-1}(\Psi_j x) - \Phi_{j-1}(x)$, with $x \in \mathcal{D}_j \times \mathbb{R}^n$ and $\Psi_j x \in \mathcal{D}_{j-1}\times\mathbb{R}^n$ (Lemma 5.3(i)). By the composition rule $L_*(\check\Phi_{j-1} - \mathrm{id}) \le \prod_{k<j}(1+\varsigma_k) - 1 \le e^{\sum\varsigma_k} - 1 \le 2\sum\varsigma_k \le \frac{1}{10}$ (using $e^x \le 1+2x$, $x \le 1$; the composition formula is that of Lemma 5.2, proof step S5.7, valid because each Ψ_k maps its domain into the next one). Therefore $|\Phi_j(x) - \Phi_{j-1}(x)|_* \le \frac{11}{10}|\Psi_j x - x|_* \le \frac{11}{10}\cdot\frac{3}{4}\hat E_{j-1}$, which is summable: uniform convergence on W_*. The limit satisfies (ii) with $L_* \le 2\sum\varsigma_k \le 768 n\hat E_0$ (the constant passes to the pointwise limit).

Step S5.17: Periodicity and injectivity. Each $\Psi_k - \mathrm{id}$ is $\mathbb{Z}^n$-periodic in θ (Lemma 4.12), so (iii) holds for Φ_j and in the limit. From (ii), $|\Phi x - \Phi y|_* \ge \frac{9}{10}|x-y|_*$.

Step S5.18: Analyticity in θ. Fix $I_* \in D_*$; $\vartheta \mapsto \Phi_j(I_*,\vartheta) - (0,\vartheta)$ is holomorphic on $\{|\mathrm{Im}\,\vartheta| < \bar s/2\pi\}$ (as $I_* \in (D_j)_{\rho_j}$), $\mathbb{Z}^n$-periodic, bounded uniformly in j by (i), and converges uniformly on $\mathbb{R}^n$. By Lemma 5.4 (componentwise, $w = \bar s/2\pi$) it converges, with all derivatives, uniformly on $\{|\mathrm{Im}\,\vartheta| \le v\}$ for every $v < \bar s/2\pi$, to a function holomorphic on the *whole* strip $\{|\mathrm{Im}\,\vartheta| < \bar s/2\pi\}$ whose real restriction is $\Phi(I_*,\cdot) - (0,\mathrm{id})$; the bounds (i), being pointwise, pass to the limit. This is claim (a) of Proposition 4.3. □

5.3. **The tori (CP10 Steps 2–3 and the estimates (35)).** Define, for $\omega \in \Omega_* = \omega_0(D)\cap\mathcal{D}_{\gamma_1,\gamma_2,\tau}$, $I_0(\omega) := \omega_0^{-1}(\omega) \in D_0$ (injectivity of ω_0; $\omega \in \omega_0(D)\cap\mathcal{D}$ means $I_0 \in D^{\mathrm{dio}} = D_0$), $I_*(\omega) := \check\imath(I_0(\omega)) \in D_*$ and

$$\varphi_\omega(\vartheta) := \Phi(I_*(\omega),\vartheta) = (v(\vartheta;\omega),\ \vartheta + u(\vartheta;\omega)).$$

Lemma 5.6 (Estimates and location)**.** *For $\omega \in \Omega_*$: $|u| \le \frac{3}{4\pi}\hat E_0 s$ on $\{|\mathrm{Im}\,\vartheta| < \bar s/2\pi\}$; $|v_1 - I_{0,1}| \le 20(\frac{\bar M_1}{M} + \frac{\hat M}{M})\hat E_0\hat\rho_0$, $|v_2 - I_{0,2}| \le 20(\frac{\bar M_2}{M}+1)\hat E_0\hat\rho_0$ on $\mathbb{R}^n$, and $v \in \mathrm{Re}(D_r)$ with $r = 32\hat E_0\hat\rho_0$, so $\varphi_\omega(\mathbb{R}^n) \subset \mathrm{Re}(D_r)\times\mathbb{R}^n$. This is Proposition 4.3(a), (d), and the location claim.*
Source: CP10 (35), (105)–(107), Step 2 end, p. 572. Misprint: *CP10 (35) bounds $|v_2 - I_2^0|$ by $10n(\bar M_2/\bar M + \hat M/M)\hat E\hat\rho$ although (107) only gives $2\hat E\hat\rho$ for I_2 without the factor $\hat M/M$.*

Proof. **Step S5.19: Sup bounds.** $|v_i - I_{0,i}| \le |\Pi_{I_i}\Phi(I_*,\vartheta) - I_{*,i}| + |I_{*,i} - I_{0,i}|$, the two terms bounded by Lemma 5.5(i) and Lemma 5.2(i): $\frac{3}{2}\frac{\hat M}{M}\hat E\hat\rho + 20\frac{\bar M_1}{M}\hat E\hat\rho$ for $i=1$ (and $\frac{3}{2} + 20\frac{\bar M_2}{M}$ for $i=2$), both $\le$ the stated constants. $|u| \le \frac{3}{2}\hat E s/2\pi$ is the angle bound.

Step S5.20: Location. $I_* \in \mathrm{Re}(D_0)_{20\hat E\hat\rho}$ and $|v - I_*| \le \frac{3}{2}\hat E\hat\rho$, so $\mathrm{dist}(v,D) < 22\hat E\hat\rho < r$. □

Lemma 5.7 (Immersion and Lagrangian property)**.** *For $\omega \in \Omega_*$, $\|D_\vartheta u\| \le \frac{3}{4} n\hat{E}_0$ (so $D_\vartheta \varphi_\omega = \binom{D_\vartheta v}{1+D_\vartheta u}$ has injective lower block $1 + D_\vartheta u$, since $\frac{3}{4} n\hat{E}_0 < \frac{1}{2}$), and $\varphi_\omega^*(\sum_i dJ_i \wedge d\theta_i) = 0$.*

Proof. **Step S5.21: Cauchy in the strip.** u is holomorphic on $|\mathrm{Im}\,\vartheta| < \bar{s}/2\pi$ with $|u| \le 3\hat{E}s/4\pi$. At real ϑ the polydisc of radius $\bar{s}/4\pi$ lies in the strip, so $|\partial_a u_i| \le \frac{3\hat{E}s}{4\pi} \cdot \frac{4\pi}{\bar{s}} = 3\hat{E}\frac{s}{\bar{s}} \le \frac{3}{4}\hat{E}$ ($4s \le \bar{s}$), and the row sum is $\le \frac{3}{4} n\hat{E}$.

Step S5.22: Each Φ_j-torus is Lagrangian. Φ_j is symplectic and $\vartheta \mapsto (I_*, \vartheta)$ has pullback $\sum dI_i \wedge d\theta_i = 0$ (the I-coordinates are constant), so $\varphi^{(j)} := \Phi_j(I_*, \cdot)$ has $\Omega_j(a,b) := \sum_i (\partial_a v_i^{(j)} \partial_b \Theta_i^{(j)} - \partial_b v_i^{(j)} \partial_a \Theta_i^{(j)}) = 0$.

Step S5.23: Limit. By Lemma 5.5 (Step S5.18) the first derivatives of $\varphi^{(j)}$ converge to those of φ_ω (uniformly on $\mathbb{R}^n$, Lemma 5.4); Ω_j is bilinear in them, so $\Omega = \lim \Omega_j = 0$. □

Lemma 5.8 (Invariance)**.** *For $\omega \in \Omega_*$, $\vartheta \in \mathbb{R}^n$, $t \in \mathbb{R}$: $\frac{d}{dt}\varphi_\omega(\vartheta + t\omega) = X_H(\varphi_\omega(\vartheta + t\omega))$, $X_H = (-\partial_\theta H, \partial_J H)$.*
Source: CP10 Step 3, pp. 572–573.

Proof. Let $I_0 = I_0(\omega)$, $I_* = \check{\imath}(I_0)$, $\theta_t = \vartheta + t\omega$, $\gamma(t) = \Phi(I_*, \theta_t)$, $\gamma_j(t) = \Phi_j(I_*, \theta_t)$.

Step S5.24: Symplectic conjugation. Since Φ_j is a symplectic diffeomorphism onto its image and $H_j = H \circ \Phi_j$: $X_H \circ \Phi_j = D\Phi_j\, X_{H_j}$ ([N:`bpN:p5:symplectic`]). With $X_{H_j}(I_*, \theta) = (-\partial_\theta f_j,\ \omega_j(I_*) + \partial_I f_j)$ we get $\gamma_j'(t) - X_H(\gamma_j(t)) = D\Phi_j(I_*, \theta_t)\big(\partial_\theta f_j,\ \omega - \omega_j(I_*) - \partial_I f_j\big)$ (with $\gamma_j' = D\Phi_j(0, \omega)$).

Step S5.25: $D\Phi_j$ bounded. $\mathcal{D}_j \times \mathbb{R}^n$ is open and $L_*(\check{\Phi}_j) \le \frac{11}{10}$ (Lemma 5.5(ii) argument), so $|S\, D\Phi_j S^{-1}| \le \frac{11}{10}$: $\|D\Phi_j\| \le \kappa := \frac{11}{10}\max\{1, \frac{2\pi\hat{\rho}_0}{s_0}, \frac{s_0}{2\pi\hat{\rho}_0}\}$, independent of j.

Step S5.26: The three error terms. (1) $\omega = \omega_j(\check{\imath}_j(I_0))$ (Lemma 5.2(iv)); $|I_* - \check{\imath}_j(I_0)| \le 20\hat{\rho}_j\hat{E}_j$ and the segment lies in $(D_j)_{\rho_j}$ where $\|U_j\| \le M_j$, so $|\omega - \omega_j(I_*)| \le 20 M_j\hat{\rho}_j\hat{E}_j \le 7\gamma_1\hat{E}_j$, using $M_j\hat{\rho}_j \le \gamma_1/(3K_j^{\tau+1}) \le \gamma_1/3$. (2) $I_* \in \mathrm{Re}(D_j)_{\rho_j/2}$, so the polydisc of radius $\rho_j/2$ about I_* is in $(D_j)_{\rho_j}$ and [K:`bp:p1:cauchy-open`] gives $|\partial_I f_j| \le 2E_j/\rho_j \le 2E_j/\hat{\rho}_j = 2\hat{E}_j\hat{\rho}_j/L_j \le 2\hat{E}_j M_j\hat{\rho}_j \le \gamma_1\hat{E}_j$ ($L_j \ge M_j^{-1}$). (3) For real θ the polydisc of radius $\bar{s}/4\pi$ in θ lies in $\mathbb{T}_{\bar{s}/2}$, where $|f_j| \le E_j$ (as $\bar{s}/2 \le \bar{s} + s_j$), so $|\partial_\theta f_j| \le 4\pi E_j/\bar{s}$ ([K:`bp:p1:cauchy-open`]); and $E_j = \hat{E}_j\hat{\rho}_j^2/L_j \le \hat{E}_j\hat{\rho}_j M_j\hat{\rho}_j \le \hat{E}_j\gamma_1\hat{\rho}_0$. All three tend to 0, uniformly in (ϑ, t), as $\hat{E}_j \le \hat{E}_0^{2^j}$. Hence $\sup_t|\gamma_j' - X_H(\gamma_j)| \to 0$.

Step S5.27: Passing to the limit. $\delta_j := \sup_t|\gamma_j(t) - \gamma(t)| \to 0$ (Lemma 5.5; unscaled distances are comparable to scaled ones with fixed constants). Only nearby pairs are needed (review P3-08). The action of $\gamma(t)$ lies in $\mathrm{Re}(D_r) \subset \mathrm{Re}(D_{\rho/4})$ ($r = 32\hat{E}\hat{\rho} \le \rho/4$ since $\hat{E} < 2^{-13}$, $\hat{\rho} \le \rho$) and its angle is real; for j so large that $\delta_j < \min\{\rho/4, \bar{s}/4\pi\}$, the real segment $[\gamma_j(t), \gamma(t)]$ lies in the convex polydisc of radius δ_j about $\gamma(t)$, hence in $D_{\rho/2} \times \mathbb{T}_{\bar{s}/2}$. There $\|DX_H\| \le B_0$, where DX_H consists of $U + \partial_J^2 f$, $\partial_\theta\partial_J f$, $\partial_\theta^2 f$ and $B_0 = M + C(E, \rho, \bar{s})$ by [K:`bp:p1:cauchy-open`] (applied to $\sup|f| \le E$ on $D_\rho \times \mathbb{T}_{\bar{s}}$). By the mean value inequality, $|X_H(\gamma_j(t)) - X_H(\gamma(t))| \le B_0\delta_j \to 0$ uniformly in t. A uniform limit of C^1 functions $\gamma_j \to \gamma$ with $\gamma_j' \to X_H(\gamma)$ uniformly is C^1 with $\gamma' = X_H(\gamma)$. Since $\gamma(t) = \varphi_\omega(\vartheta + t\omega)$, this is the claim. (Conjugation identity of Step S5.24: [N:`bpN:p5:symplectic`], [N:`bpN:p5:canonical`].) □

5.4. **Measure estimate (CP10 Step 4).** Let $K_\Omega = \bigcup_{\omega\in\Omega_*} \varphi_\omega(\mathbb{R}^n)$; by definition $K_\Omega = \Phi(D_* \times \mathbb{R}^n)$ ($\check{\imath}$ maps $D_0 = D^{\mathrm{dio}}$ onto D_* and $\omega \mapsto I_0(\omega)$ is a bijection $\Omega_* \to D_0$). Here and below $N_\rho(X)$ is the *closed* real max-norm ρ-neighbourhood $\{x : \mathrm{dist}_\infty(x, X) \le \rho\}$; every radius used is $< r$, so $N_\rho(X) \subset \mathrm{Re}(X_r)$. Put $a := \mathrm{meas}(D \setminus D_0)$, $b := \mathrm{meas}(\mathrm{Re}(D_r) \setminus D)$, $\lambda \le \frac{1}{3}$ as in Lemma 5.2, $\eta := \sup|\check{\imath} - \mathrm{id}| \le 20\hat{E}_0\hat{\rho}_0$.

Lemma 5.9 (Bi-Lipschitz extension and covering)**.** *Let $X \subset \mathbb{R}^N$ be arbitrary, $g : X \to \mathbb{R}^N$ with $L(g) \le \lambda < 1$ in the max norm and $\sup|g| \le \eta$. Then there is a bijection $F = \mathrm{id} + g_e : \mathbb{R}^N \to \mathbb{R}^N$, $F|_X = \mathrm{id}+g$, with $(1-\lambda)|x-y| \le |F(x)-F(y)| \le (1+\lambda)|x-y|$, and for $y \in X$: $\mathrm{dist}_\infty(F^{-1}(y), X) \le |F^{-1}(y) - y| \le \eta/(1-\lambda)$ (a distance inequality, valid also for $\eta = 0$).*

Proof. **Step S5.28: Extension.** g_e: McShane [K:bp:p1:mcshane], componentwise, with the same λ for the max norm.

Step S5.29: Bijectivity. $|F(x) - F(y)| \ge |x - y| - \lambda|x - y|$ gives injectivity and the lower bound; for $y \in \mathbb{R}^N$ the map $x \mapsto y - g_e(x)$ is a λ-contraction ((X4)), with fixed point $x = F^{-1}(y)$.

Step S5.30: Location of preimages. For $y \in X$ and $x = F^{-1}(y)$: $|x - y| = |g_e(x)| \le |g_e(y)| + \lambda|x - y| \le \eta + \lambda|x - y|$ so $|x - y| \le \eta/(1 - \lambda)$. □

Lemma 5.10 (Measure of the complement of D_*)**.** $\mathrm{meas}(D_0 \setminus D_*) \le (4/3)^n(a + b)$ *and hence* $\mathrm{meas}(D \setminus D_*) \le a + (4/3)^n(a + b)$.

Proof. **Step S5.31: Extension and preimage.** Apply Lemma 5.9 with $X = D_0$, $g = \check{\imath} - \mathrm{id}$, $N = n$: $F = \check{\imath}_e$. Let $\rho_1 := \eta/(1 - \lambda) \le 30\hat{E}_0\hat{\rho}_0 < r$. If $y \in D_0 \setminus D_*$, then $x = F^{-1}(y) \in N_{\rho_1}(D_0)$ and $x \notin D_0$ (else $y = F(x) = \check{\imath}(x) \in D_*$). So $D_0 \setminus D_* \subset F\big(N_{\rho_1}(D_0) \setminus D_0\big)$.

Step S5.32: Measure. By [K:bp:p1:lip-image] (F is $(1 + \lambda)$-Lipschitz), $\mathrm{meas}(D_0 \setminus D_*) \le (1 + \lambda)^n \,\mathrm{meas}(N_{\rho_1}(D_0) \setminus D_0)$, $(1 + \lambda) \le \frac{4}{3}$. As $D_0 \subset D$: $N_{\rho_1}(D_0) \setminus D_0 \subset (N_{\rho_1}(D) \setminus D) \cup (D \setminus D_0)$, of measure $\le b + a$ ($\rho_1 < r$). The second claim: $D \setminus D_* \subset (D \setminus D_0) \cup (D_0 \setminus D_*)$. □

Lemma 5.11 (Measure of the complement of K_Ω over D)**.** $\mathrm{meas}\big((D \times \mathbb{T}^n) \setminus K_\Omega\big) \le \big[1 + (4/3)^n + 2^n\big(1 + (4/3)^n\big)\big](a + b)$.

Proof. Identify $\mathbb{T}^n$-measure with the unit cell; K_Ω is $\mathbb{Z}^n$-invariant in the angle.

Step S5.33: Splitting. $(D \times \mathbb{T}^n) \setminus K_\Omega \subset \big((D \setminus D_*) \times \mathbb{T}^n\big) \cup \big(((D_* \cap D) \times \mathbb{T}^n) \setminus K_\Omega\big)$. The first set has measure $\le a + (4/3)^n(a + b)$ by Lemma 5.10.

Step S5.34: Rescaled extension. Work in scaled coordinates $x \mapsto Sx$; S is linear, multiplies the Lebesgue measure of every subset of $\mathbb{R}^n_I \times \mathbb{R}^n_\theta$ by the constant $\hat{\rho}_0^{-n}(2\pi/s_0)^n$ and maps products to products, so all comparisons below between measures of subsets of $\mathbb{R}^n \times \mathbb{R}^n$ are unaffected. By Lemma 5.5, $g = \check{\Phi} - \mathrm{id}$ on $X = S(D_* \times \mathbb{R}^n) \subset \mathbb{R}^{2n}$ has $L_*(g) \le \lambda' := 768n\hat{E}_0 < \frac{1}{10}$ and $\sup|g| \le \eta' := \frac{3}{2}\hat{E}_0$ (Lemma 5.5(i), scaled). Lemma 5.9 ($N = 2n$) gives the bijection $\check{\Phi}_e = \mathrm{id} + g_e$ of $\mathbb{R}^{2n}$ extending $\check{\Phi}$ on X, $(1 + \lambda')$-Lipschitz, with $|\check{\Phi}_e^{-1}(y) - y|_* \le \eta'/(1 - \lambda') \le 2\hat{E}_0$ for $y \in X$.

Step S5.35: Preimage structure. Let $y = (I_y, \vartheta_y)$ with $I_y \in D_* \cap D$ and $\vartheta_y \in [0, 1)^n$ be a point of the second set, i.e. $y \notin K_\Omega$ (mod $\mathbb{Z}^n$ in the angle), and let $x = (I_x, \vartheta_x) := \Phi_e^{-1}(y)$ (unscaled notation; $y \in D_* \times \mathbb{R}^n = S^{-1}X$). Then $|I_x - I_y| \le \rho_2 := 2\hat{E}_0\hat{\rho}_0$ and $|\vartheta_x - \vartheta_y| \le \hat{E}_0 s_0/\pi \le \hat{E}_0/(4\pi) =: \epsilon$ (as $s_0 \le \frac{1}{4}$). If $I_x \in D_*$, then $x \in D_* \times \mathbb{R}^n$, $\Phi_e(x) = \Phi(x) = y$ and $y \in K_\Omega$, a contradiction. Hence $I_x \in N_{\rho_2}(D_*) \setminus D_*$ and $\vartheta_x \in [-\epsilon, 1 + \epsilon)^n$.

Step S5.36: Measure. Thus the second set is contained in $\Phi_e\big((N_{\rho_2}(D_*) \setminus D_*) \times [-\epsilon, 1 + \epsilon)^n\big)$. Φ_e is $(1 + \lambda')$-Lipschitz for $|\cdot|_*$ on $\mathbb{R}^{2n}$, so by [K:bp:p1:lip-image] in scaled coordinates (the scaling constants cancel on both sides) its measure is $\le (1 + \lambda')^{2n}(1 + 2\epsilon)^n \,\mathrm{meas}(N_{\rho_2}(D_*) \setminus D_*) \le 2^n \,\mathrm{meas}(N_{\rho_2}(D_*) \setminus D_*)$, because $(1.1)^{2n}(1 + \hat{E}_0/2\pi)^n \le (1.21 \cdot 1.01)^n < 2^n$.

Step S5.37: The shell. $D_* \subset N_\eta(D_0)$, so $N_{\rho_2}(D_*) \subset N_{\rho_2+\eta}(D_0) \subset N_r(D)$ ($\rho_2 + \eta \le 22\hat{E}_0\hat{\rho}_0 < r$), and $N_{\rho_2}(D_*) \setminus D_* \subset (N_r(D) \setminus D) \cup (D_0 \setminus D_*) \cup (D \setminus D_0)$: if $x \in N_{\rho_2}(D_*) \setminus D_*$ then either $x \notin D$ (first set), or $x \in D \setminus D_0$, or $x \in D_0 \setminus D_*$. Its measure is $\le b + a + (4/3)^n(a + b) = (1 + (4/3)^n)(a + b)$ by Lemma 5.10.

Step S5.38: Sum. Total $\le [a + (4/3)^n(a+b)] + 2^n(1+(4/3)^n)(a+b) \le [1 + (4/3)^n + 2^n(1 + (4/3)^n)](a+b)$. □

5.5. **Proof of Proposition 4.3.**

Proof of Proposition 4.3. **Step S5.39: Construction.** The sequence of Lemma 4.12 exists (hypotheses (S1)–(S3) at $j=0$ from the proposition's hypotheses and the definition of D_0); Lemma 5.2 defines $\check{\imath}$, Lemma 5.5 defines Φ, and φ_ω is defined in §5.3.

Step S5.40: (a), (d). Lemmas 5.5, 5.6.

Step S5.41: (b), (c). Lemmas 5.8, 5.7.

Step S5.42: Injectivity and Lipschitz structure. If $\varphi_\omega(\vartheta) - \varphi_{\omega'}(\vartheta') = (0,k)$, then $\Phi(I_*,\vartheta) = \Phi(I_*', \vartheta'+k)$ by periodicity (Lemma 5.5(iii)), so $(I_*,\vartheta) = (I_*', \vartheta'+k)$ by (ii); $I_* = I_*'$ gives $I_0 = I_0'$ ($\check{\imath}$ injective), so $\omega = \omega_0(I_0) = \omega'$. The factorization and the constants: Lemma 5.2(iii) and Lemma 5.5(ii). For ω_0^{-1}: if $\omega, \omega' \in \omega_0(D)$ with $|\omega - \omega'| < \rho/\bar{M}$, then Lemma 4.6 (with $F = \omega_0$ on D_ρ, $\delta = \rho$, $x_0 = \omega_0^{-1}(\omega)$) gives $|\omega_0^{-1}\omega - \omega_0^{-1}\omega'| \le \bar{M}|\omega - \omega'|$; for far pairs $|\omega_0^{-1}\omega - \omega_0^{-1}\omega'| \le \operatorname{diam} D \le \operatorname{diam} D \cdot \frac{\bar{M}}{\rho}|\omega - \omega'|$.

Step S5.43: Measure. $K_\Omega \subset \mathrm{Re}(D_r) \times \mathbb{T}^n$ (Lemma 5.6), so $(\mathrm{Re}(D_r)\times\mathbb{T}^n)\setminus K_\Omega \subset ((\mathrm{Re}(D_r)\setminus D)\times \mathbb{T}^n) \cup ((D\times\mathbb{T}^n)\setminus K_\Omega)$ (an inclusion; the shell may meet K_Ω, which only helps), and Lemma 5.11 gives (3.1) with $c_n = 1 + [1 + (4/3)^n + 2^n(1+(4/3)^n)] = 1 + (1+2^n)(1+(4/3)^n)$ ($a = \mathrm{meas}(D \setminus D^{\mathrm{dio}})$, $b = \mathrm{meas}(\mathrm{Re}(D_r)\setminus D)$, $\mathrm{meas}\,\mathbb{T}^n = 1$). □

5.6. **What Part 5 consumes.** Part 5 ([P:`bpP:p5:kam-application`]) applies Proposition 4.3 to the Hamiltonian $\breve{H}$ of CP10 Step 6 (p. 560) after Part 2's averaging ([P:`bpP:p2:averaged-remainder`]). *Three radii must not be confused:* ϵ is the oscillator radius of CP10 (the quantity renamed r in the frozen cylinder $\mathcal{C}(V,r)$); $\hat{\rho}$ is the action radius of Definition 4.2; $r = 32\hat{E}\hat{\rho}$ is the enlargement radius of this proposition. The identification (CP10 p. 560, review P3-05): $n_1 = a$, $n_2 = b$, $h = H_0(J_1) + \mu\breve{N}(J)$, $f = \mu\,\epsilon^5\breve{P}$ (*not* $\mu^5\breve{P}$), $\rho = \breve{\rho}$, $s = \breve{s}/5$, $\bar{s} = 4\breve{s}/5$ (in coefficient-radius widths of §4.1), $D = D_{\mathrm{CP}}$ of CP10 (48).

(1) *Before* ϵ, μ (depend only on the fixed family): n, τ (hence $\hat{c}, c_n$), and a constant $C \ge 1$ such that the true bounds satisfy $\sup\|U\| \le C$, $\sup\|\hat{U}\| \le C\mu$, $\sup\|U^{-1}\| \le C\mu^{-1}$, $\sup\|T_1\| \le C$, $\sup\|T_2\| \le C\mu^{-1}$, $\|f\| \le C\mu\epsilon^5$ for $0 < \mu \le 1$, $\epsilon \le 1$ (CP10 p. 560; the N-norm of $\mu\epsilon^5\breve{P}$ is $\le C\mu\epsilon^5$).

(2) *Choice of the numerical bounds (review R2-01).* Since $M, \hat{M}, \bar{M}, \bar{M}_i, E$ need only be *upper* bounds, take them *equal* to these values: $M = C$, $\hat{M} = C\mu$, $\bar{M} = \bar{M}_2 = C\mu^{-1}$, $\bar{M}_1 = C$, $E = C\mu\epsilon^5$ (the normalizations $\hat{M} \le M$, $\bar{M}_i \le \bar{M}$ hold for $\mu \le 1$, $C \ge 1$). Then, exactly, $L = \max\{C\mu^{-1}, C^{-1}, (C\mu)^{-1}\} = C\mu^{-1}$ (because $C \ge C^{-1}$ gives $C\mu^{-1} \ge (C\mu)^{-1}$, and $C\mu^{-1} \ge C^{-1}$ as $\mu \le 1$), $EL = C^2\epsilon^5$ and $x = EM^2L/\gamma_1^2 = C^4\epsilon^5/\gamma_1^2$, so $K = \frac{6}{s}\log_+\big(\gamma_1^2/(C^4\epsilon^5)\big)$ *exactly* (no inequality is inferred from an upper bound on x: $\log_+(1/x)$ is nonincreasing in x, so an upper bound on x would only give a lower bound on K). Whenever $\gamma_1^2/(C^4\epsilon^5) \ge e$ this is $K = \frac{6}{s}\log(\gamma_1^2/(C^4\epsilon^5)) \le C'\log(\epsilon^5/\gamma_1^2)^{-1}$ (CP10 p. 560), with C' depending on s, C only. The smaller true E can only be used if one re-chooses the numerical E *before* computing K; Part 5 should keep the choice above. Then $\hat{\rho} = \min\{\gamma_1/(3CK^{\tau+1}), \gamma_2/(3C\mu K^{\tau+1}), \rho\}$, $\hat{E} = C^2\epsilon^5/\hat{\rho}^2$, and the KAM condition $\hat{c}\hat{E} < 1$ is CP10 (53)–(54), checked by Part 5.

(3) *Outputs used:* the tori φ_ω, $\omega \in \Omega_*$, with (a)–(d); the measure inequality (3.1) with the *absolute* constant c_n; $r = 32\hat{E}\hat{\rho} \le C\hat{\rho}$ so that $r \le C\bar{\gamma}_2$ as in CP10 p. 561; $|v - I^0|$, $|u|$ for the reconstruction (CP10 (51), (35)).

(4) *Frozen frequencies and Part 4.* $H = \tilde{H}/2\pi$ (Lemma 4.5); $\omega = \omega_0(I_0) = \omega_\mu/2\pi$ with ω_μ the frequency map of [P:`bpP:p4:freq`]. Hence $\omega \in \mathcal{D}_{\gamma_1,\gamma_2,\tau}$ iff $\omega_\mu \in \mathcal{D}_{\alpha_1,\alpha_2,\tau}$ with $\alpha_i = 2\pi\gamma_i$ (that is the dictionary of [P:`bpP:p4:freq`]), and $D \setminus D^{\mathrm{dio}}$ of (3.1) is the set Part 4 estimates; (3.1) with the *same* $c_n = 1 + (1+2^n)(1+(4/3)^n)$ and $\operatorname{meas}\mathbb{T}^n = 1$ is exactly the right side consumed by [P:`bpP:p4:budget`], whose radius r_K is $r = 32\hat{E}\hat{\rho}$ here. Part 4 needs $\tau \geq n+1$; this is within the hypothesis $\tau > n$ of the proposition (take $\tau \geq n+1$ in Part 5; nothing else of Part 3 depends on τ beyond $\hat{c}$).

(5) *Conversions, with the maps named* ([P:`bpP:p1:order`], [P:`bpP:p1:cyl-pipeline`]; Lemma 4.5). The torus handed to the pipeline is the *unscaled* period-one torus $K_3 := \pi \circ \varphi_\omega : \mathbb{R}^n \to \mathrm{Ph}(n)$, $\pi(J,\theta) = (\theta, J)$ (frozen order); no factor 2π is applied to it. The 2π enters exactly once, in the pipeline's chart $\Psi = \Phi \circ S_{2\pi}$ (Φ the original reconstruction map of Part 5/Part 2): $K = \Psi \circ K_3 = \Phi \circ S_{2\pi} \circ \pi \circ \varphi_\omega$, with frozen frequency $w = \omega$ (CP's angular frequency is $2\pi w$), and the same w is used in both the pipeline and $\mathcal{D}_{\gamma_1,\gamma_2,\tau}$. The map Q of Lemma 4.5 describes the same torus in CP coordinates, $\tilde{\varphi} = Q \circ \varphi_\omega(\cdot/2\pi) = S_{2\pi} \circ (\cdot)$ in Part 1's language; it must not be fed to the pipeline as K_3 (this would scale the angles twice). The output angle $\tilde{u} = 2\pi u$ is used only when estimates are quoted in CP's coordinates.

Not conclusions of Part 3 (obligations of Part 5): the complex neighbourhood and uniform bounds from the literal premises (H1)–(H6), the injectivity/nondegeneracy of ω_μ on a complex neighbourhood, the volume losses, the reconstruction of the oscillators and the seven clauses of [P:`bpP:p0:cyltorus`], measurability and whole-torus saturation of S, positivity and the strict inequalities.

Source gap: CP10 Prop. 3 claims *"a unique real-analytic embedding"*; translations $\vartheta \mapsto \vartheta + c$ preserve all stated properties, so uniqueness is neither proved nor claimed here. The Lipschitz continuity of $(\vartheta,\omega) \mapsto \varphi_\omega(\vartheta)$ passes through ω_0^{-1}, which is globally Lipschitz only for bounded D (locally, at scale $\rho/\bar{M}$, in general: separated action islands at j^2 with local frequency maps centred at j and Hessian 1 have inverse derivative 1 but no global Lipschitz inverse; review E9).

5.7. **Source gaps and suspected misprints (CP10, Prop. 3, Appendix B).** Dispositions follow the independent review (`reviews/p3-review-1.md`): V = verified misprint, R = proof repair or convention change (not an erratum), T = tentative.

(E1) **V.** p. 554: "$\hat{U}$ denote the $n \times n_1$ submatrix of U, i.e. the matrix with entries $\hat{U}_{ij} = U_{ij}$, for $n_1 + 1 \leq i \leq n$, $1 \leq j \leq n$": size and index range disagree; the proof needs $n_2 \times n$ (Remark 4.1).

(E2) **V.** p. 565 (Lemma B.1): "$M_0 := M$, $\hat{M}_0 := M$, $\bar{M}_0 := \bar{M}$"; $\hat{M}_0$ should be $\hat{M}$.

(E3) **V.** p. 566, B.1(i): "$\hat{\imath}_j : I \in D_{j-1} \mapsto \omega_{j-1} \circ \omega_{j-1}^{-1}(I) \in D_j$" is the identity; it should read $\omega_j^{-1} \circ \omega_{j-1}$ (cf. B.2, p. 568, $\hat{\imath}_+ = \omega_+^{-1} \circ \omega|_D$).

(E4) **V, with scope.** (74) uses $\bar{M}_i/\bar{M} \leq 1$ without specifying compatible bounds (compatible bounds can be chosen: normalization, Remark 4.4); and (35), second line, carries a factor $\hat{M}/M$ for $|v_2 - I_2^0|$ that p. 572 (107) does not give (replacing it by 1 gives what is proved). No counterexample to a sharper estimate is claimed.

(E5) **T (not verified).** p. 551 assigns the 1-norm to the action, the angular sup metric, sup norms to p, q and the entrywise sup norm to matrices; the proof uses operator norms and mixed conversions. The blueprint's constants are a legitimate *convention change* (max norms, N-norm). No source constant is exhibited as false; the allegation that the printed constants "cannot all hold" is withdrawn.

(E6) **V (sharpening).** p. 569 (91) prints $\frac{8}{9}(4s)^{\tau+1}\hat{E}^2$; the computation gives exponent $2(\tau+1)$. The printed inequality is true (weaker).

(E7) **R.** pp. 569–570: the mean value step for ω^{-1} needs the frequency segment inside the domain of the inverse; and Lipschitz bounds on non-convex D, D_j need near/far pairs. Repaired by Lemmas 4.6, 4.7; composition domains stated (Lemma 5.3(i)).

(E8) **R (expository supplement).** Step 3 (pp. 572–573) does use (108) ("Using (108)"); the derivative bound on the open domain and analyticity on a fixed strip are not spelled out; supplied by Steps S5.25, S5.18 and Lemma 5.4. Not a missing estimate.

(E9) **V.** pp. 555, 573: "a unique real-analytic embedding" is not true as stated (translations); global Lipschitz dependence on ω fails for unbounded disconnected D (example above); bounded D suffices.

(E10) **R.** p. 573 footnote 32: Kirszbraun (Euclidean) is replaced by max-norm McShane and the cube-covering bound $\operatorname{meas} F(A) \le \Lambda^d \operatorname{meas} A$ ([K:bp:p1:mcshane], [K:bp:p1:lip-image]); c_n is recomputed, not reproduced; not an erratum.

(E11) **V (dependency).** Prop. 1 is proved in CP10 only through the unpinned [4, Lemma A.5] and a finite induction "as in [4]"; supplied by Part 2. The pointer to Arnold [3, p. 144] in Remark 3 is historical (Appendix B has its own induction) and is not a dependency.

Two further verified suspected defects found by the review (for the orchestrator's ledger, not used here): CP10 p. 565 (70) and the following split print "$k \in \Lambda$" where $k \notin \Lambda$ is meant (with $\Lambda = \{0\}$ the zero mode would be divided by zero); CP10 p. 564 (66) prints the target $W_{v,\sigma}$ where the angle width is s, i.e. $W_{v,s}$.

5.8. **References and imported facts.** Written parts cited: [P:bpP:p1:period-convert], [P:bpP:p1:cyl-pipeline], [P:bpP:p2:avg], [P:bpP:p2:nfb], [P:bpP:p2:split], [P:bpP:p2:averaged-remainder]; track N: [N:bpN:p4:norm], [N:bpN:p5:symplectic], [N:bpN:p5:canonical]. Forward reference to an unwritten part: [P:bpP:p5:kam-application]. Track K: [K:bp:p1:cauchy-open], [K:bp:p1:uniform-limits], [K:bp:p1:mcshane], [K:bp:p1:lip-image], [K:bp:p1:ext-mvt], [K:bp:p1:ext-banach], [K:bp:p1:ext-fourier], [K:bp:p1:ext-rectangle]. The only ingredient proved elsewhere is the averaging step (Lemma 4.8, proved in Part 2 from track N); the rest is proved here from the external facts (X1)–(X6). Formalization notes: the ball lemma needs a finite-dimensional path-continuation package; Lemma 5.4 is a bounded stand-alone task (no several-variable Vitali–Montel theorem is assumed).

6. Measure estimates for P-G

This part supplies the three measure inputs of the proof of [P:bpP:p0:pg]: the loss from the boundary collar of V, the loss from the non-Diophantine frequencies (CP10 Lemma 3.1, p. 562, proved in Appendix C, pp. 574–578), and the loss from the singular shell near the oscillator centres, together with the exact volume of the cylinder cell. Part 3 produces a set of whole tori K_Ω with the measure bound (3.1), whose right side contains $\operatorname{meas}(D \setminus D^{\mathrm{dio}})$ and $\operatorname{meas}(\operatorname{Re}(D_r) \setminus D)$ (Proposition 4.3); the lemmas below bound both quantities for the action domain $D = \bar{D}_t \times A$ of CP10 (48), and convert the bound to the relative loss in the frozen cell $\mathcal{F}(V, r)$ (Proposition 6.15). Part 5 chooses the free parameters.

Route. CP10 estimates the non-Diophantine set by covering the frequency image with p sets $U(I_1^{(i)})$ and slicing each one. For the frequency map of P-G the number p grows as the oscillator radius shrinks, so the constants c_1, c_2 of (58) are *not* independent of ε (Remark 6.10, errata (E1), (E2)). We replace the covering by a direct slicing argument. In the variables $y = \omega_{H_0}(I)$ (bi-Lipschitz coordinates, by the frozen hypothesis HasFastDiffeo) and ξ (oscillator actions), a resonance with $k_1 \neq 0$ is a perturbation of the linear form $k_1 \cdot y$ whose sections along y_j are short, as long as $\mu|k|$ is small; for $\mu|k|$ large the resonance is cut by the nondegenerate slow block β, and the sum over such k carries the factor $\mu^{\tau-n}$. This is the use of the full strength $\tau > n$.

Conventions. $n = a + b$, $n_1 = a$, $n_2 = b$; $|\cdot|$ is the max norm on $\mathbb{R}^m$, $|k|_1$ the ℓ^1 norm of $k \in \mathbb{Z}^m$; meas is Lebesgue outer measure (as in [K:bp:p1:ext-lebesgue]); sets to which Tonelli is applied are Borel. $\mathcal{D}_{\alpha_1,\alpha_2,\tau}$ is the two-scale set of Definition 2.14. The frozen radius of the cylinder is r (Definition 1.10); *Part 3's enlargement radius is written* r_K ($= 32\hat{E}\hat{\rho}$ there). Oscillator actions are $\xi_j = \frac{1}{2}(p_j^2 + q_j^2)$, so the frozen disc condition is $\xi_j < r^2/2$.

6.1. **External facts.**

(X1) *Tonelli.* If $R \subset \mathbb{R}^N$ is Borel, $j \le N$, every section $\{s \in \mathbb{R} : (x', s) \in R\}$ ($x' \in \mathbb{R}^{N-1}$, other coordinates fixed) has outer measure $\le \delta$, and the projection of R forgetting the j-th coordinate lies in a set of measure $\le P$, then $\operatorname{meas} R \le \delta P$. A set of diameter $\le \delta$ in $\mathbb{R}$ has outer measure $\le \delta$.

(X2) Mean value inequality on convex sets ([K:bp:p1:ext-mvt]); Lipschitz images ([K:bp:p1:lip-image]); Borel images of Borel sets under injective continuous maps ((X4)).

(X3) The derivative of a real analytic function on an open set is real analytic (hence C^1); a C^1 function is continuously differentiable and locally Lipschitz.

(X4) A connected set meeting both V and its complement meets ∂V.

6.2. **The boundary collar.** Let V, A, d be as in [P:bpP:p0:pg] ($V \subset \mathbb{R}^a$ open, connected, bounded; $\operatorname{meas}\{I : \exists J \in \partial V,\ |I - J| < t\} \le At$ for $0 < t < d$). Then $0 < \operatorname{meas} V < \infty$ and $\partial V \neq \emptyset$ ($V \neq \emptyset$ is open and not $\mathbb{R}^a$, which is connected). For $t > 0$ put

$$\bar{D}_t := \{I \in V :\ \operatorname{dist}(I, \partial V) \ge t\}, \qquad N_s(X) := \{x \in \mathbb{R}^a : \exists J \in X,\ |x - J| < s\}.$$

Lemma 6.1 (Collar)**.** *(a) $\bar{D}_t$ is compact, $\bar{D}_t \subset V$, and $V \setminus \bar{D}_t = \{I \in V : \operatorname{dist}(I, \partial V) < t\}$.*
(b) If $0 < t < d$ then $\operatorname{meas}(V \setminus \bar{D}_t) \le At$; hence $\operatorname{meas} \bar{D}_t \ge \operatorname{meas} V - At$.
(c) If $0 < s < t$ then $N_s(\bar{D}_t) \subset V$, and $N_s(\bar{D}_t) \setminus \bar{D}_t \subset V \setminus \bar{D}_t$; so for $0 < s < t < d$, $\operatorname{meas}(N_s(\bar{D}_t) \setminus \bar{D}_t) \le At$.

Source: Ours. The frozen hypothesis `hboundary` *replaces CP10 p. 561's unproved collar estimate "*$\operatorname{meas}(\mathrm{Re}(V_r) \setminus V) \le C\bar{\gamma}_2 \operatorname{meas} V$*" (audit P3, D-PG-5, errata (E3)); only the* inner *side is needed because every neighbourhood below stays inside V.*

Proof. **Step S6.1: (a) Compact.** $\bar{D}_t$ is bounded. If $J_m \in \bar{D}_t$, $J_m \to J$, then $J \in \overline{V}$ and $\operatorname{dist}(J, \partial V) \ge t > 0$ by continuity of the distance, so $J \notin \partial V$, i.e. $J \in V$ (V open): closed. The description of $V \setminus \bar{D}_t$ is the definition.

Step S6.2: (b) Measure. If $I \in V \setminus \bar{D}_t$ then $\operatorname{dist}(I, \partial V) < t$, so there is $J \in \partial V$ with $|I - J| < t$: $V \setminus \bar{D}_t$ lies in the set of the hypothesis. Then $\operatorname{meas} \bar{D}_t = \operatorname{meas} V - \operatorname{meas}(V \setminus \bar{D}_t)$.

Step S6.3: (c) Neighbourhoods. Let $J \in \bar{D}_t$, $B = \{x : |x - J| < s\}$, $s < t$. B is convex, hence connected, contains $J \in V$ and misses ∂V; by (X4) it misses V^c, so $B \subset V$. If $x \in N_s(\bar{D}_t) \setminus \bar{D}_t$ then $x \in V$ and $x \notin \bar{D}_t$, which is $x \in V \setminus \bar{D}_t$. □

6.3. **Uniform Lipschitz facts.**

Lemma 6.2 (Uniform Lipschitz bounds on compact sets)**.** *Let $\Theta \subset \mathbb{R}^p$ and $K \subset \mathbb{R}^m$ be compact, $O \subset \mathbb{R}^p \times \mathbb{R}^m$ open with $\Theta \times K \subset O$, and $f : O \to \mathbb{R}^q$ continuous with $x \mapsto f(\theta, x)$ of class C^1 and $D_x f$ jointly continuous on O ($p = 0$ allowed). Then there is Λ with $|f(\theta, x) - f(\theta, x')| \le \Lambda |x - x'|$ for $\theta \in \Theta$, $x, x' \in K$.*

Proof. **Step S6.4: Contradiction.** If not, there are $\theta_m, x_m \neq x'_m$ with $|f(\theta_m, x_m) - f(\theta_m, x'_m)| > m|x_m - x'_m|$. Since f is bounded on $\Theta \times K$, $|x_m - x'_m| \to 0$; pass to a subsequence with $\theta_m \to \theta^*$, $x_m, x'_m \to x^*$.

Step S6.5: Local bound. Choose $\delta > 0$ with $\overline{B_\delta(\theta^*)} \times \overline{B_\delta(x^*)} \subset O$ and $M_* := \sup\|D_x f\|$ over it (finite, continuity). For large m the segment $[x_m, x'_m]$ lies in $B_\delta(x^*)$ and $\theta_m \in B_\delta(\theta^*)$, so (X2) gives $|f(\theta_m, x_m) - f(\theta_m, x'_m)| \le M_*|x_m - x'_m|$, contradicting the choice for $m > M_*$. □

Lemma 6.3 (Lipschitz inverse of the fast frequency)**.** *Assume* HasFastDiffeo(H_0, W) *(Def. 1.13) and $\overline{V} \subset W$ with V bounded. Write $\omega_{H_0} =$ `fastFrequency` H_0, g the frozen inverse. Then ω_{H_0} is continuous on W; $Y_0 := \omega_{H_0}(\overline{V})$ is compact and contained in the open set $\omega_{H_0}(W)$; $g(\omega_{H_0}(I)) = I$ on W; and there is $L_g < \infty$ with $|g(y) - g(y')| \le L_g|y - y'|$ for $y, y' \in Y_0$. Put $R_y := \max_{Y_0}|y|$ and $\Lambda_g := \max\{1, L_g\}$. These depend only on (H_0, V, W).*
Source: Ours; this is the "Lipschitz inverse of ω_0" used in CP10 Lemma 3.1 ("extended to a diffeomorphism"), made explicit from the frozen C^1 inverse.

Proof. **Step S6.6: Continuity and compactness.** H_0 is analytic on W, so ω_{H_0} is analytic on W ((X3)); $\overline{V}$ is compact (bounded, closed), so Y_0 is compact, and it lies in $\omega_{H_0}(W)$, which is open by the frozen clause (ii).

Step S6.7: Lipschitz. g is C^1 on the open set $\omega_{H_0}(W) \supset Y_0$ (frozen clause (iii)): apply Lemma 6.2 with $p = 0$. □

Lemma 6.4 (Uniform inverse of the torsion block)**.** *Let $\beta(\mu, I)$ be symmetric $b \times b$ with $|\det \beta(\mu, I)| \ge B > 0$ and entries bounded by M_β for $\mu \in (0, \mu_0]$, $I \in V$. Then $|\nu| \le L_\beta|\beta(\mu, I)^\top \nu|$ for all $\nu \in \mathbb{R}^b$, with $L_\beta := b!\, M_\beta^{b-1}/B$.*

Proof. Cramer ([K:`bp:p1:ext-cramer`]): $(\beta^\top)^{-1} = \mathrm{adj}(\beta^\top)/\det\beta$; entries of the adjugate are $(b-1) \times (b-1)$ minors, bounded by $(b-1)!M_\beta^{b-1}$ (Leibniz); so the row-sum operator norm is $\le b!M_\beta^{b-1}/B$. □

The boundedness $M_\beta < \infty$ comes from joint analyticity of the family (Part 5, [P:`bpP:p5:uniform-remainder`]), which also gives the regularity (N) below by Lemma 6.2.

6.4. The two-scale Diophantine measure (CP10 Lemma 3.1).

Definition 6.5 (Frequency map of the averaged system)**.** Let $N_\mu(I, \xi) = P_0(\mu, I) + \Omega(\mu, I)\cdot\xi + \frac{1}{2}\xi \cdot \beta(\mu, I)\xi$ ($I \in \mathbb{R}^a$, $\xi \in \mathbb{R}^b$), assume $P_0(\mu, \cdot)$, $\Omega(\mu, \cdot)$, $\beta(\mu, \cdot)$ are C^1 near $\bar{D}$, and put $G_\mu := \nabla_I N_\mu \in \mathbb{R}^a$, $\omega^{\mathrm{s}}_\mu(I, \xi) := \Omega(\mu, I) + \beta(\mu, I)\xi = \nabla_\xi N_\mu \in \mathbb{R}^b$. For $\mu > 0$ the *frequency map* (CP angular convention) is

$$\omega_\mu(I, \xi) := \big(\omega_{H_0}(I) + \mu\, G_\mu(I, \xi),\ \mu\, \omega^{\mathrm{s}}_\mu(I, \xi)\big) = \nabla_{(I,\xi)}\big(H_0 + \mu N_\mu\big).$$

Dictionary with Part 3. Part 3's $h = (H_0 + \mu N_\mu)/2\pi$ and $\omega_0 = \partial h = \omega_\mu/2\pi$ (Lemma 4.5); since $\mathcal{D}_{\gamma_1,\gamma_2,\tau}$ is a cone condition, $\omega_\mu/2\pi \in \mathcal{D}_{\gamma_1,\gamma_2,\tau}$ iff $\omega_\mu \in \mathcal{D}_{2\pi\gamma_1,2\pi\gamma_2,\tau}$ (Lemma 2.15(c) for each line of the definition). So with $\alpha_i = 2\pi\gamma_i$: $D \setminus D^{\mathrm{dio}} = \{J \in D : \omega_\mu(J) \notin \mathcal{D}_{\alpha_1,\alpha_2,\tau}\}$.

Definition 6.6 (Standing data (S4))**.** $\tau > n$. $\bar{D} \subset V$ compact with $\mathrm{meas}\,\bar{D} > 0$. $\lambda_0, \mu_1 > 0$, and constants $\Lambda_N \ge 1$, $L_\beta \ge 1$ such that for $\mu \in (0, \mu_1]$, $I, I' \in \bar{D}$, $|\xi|, |\xi'| \le \lambda_0$:

(N1) $|G_\mu(I, \xi) - G_\mu(I', \xi)| \le \Lambda_N|I - I'|$ and $|\omega^{\mathrm{s}}_\mu(I, \xi) - \omega^{\mathrm{s}}_\mu(I', \xi)| \le \Lambda_N|I - I'|$;
(N2) $|G_\mu(I, \xi) - G_\mu(I, \xi')| \le \Lambda_N|\xi - \xi'|$;
(N3) $|\nu| \le L_\beta|\beta(\mu, I)^\top \nu|$ for $\nu \in \mathbb{R}^b$ (Lemma 6.4).

Constants of Lemma 6.3: R_y, L_g, Λ_g. Put $M := 2b\Lambda_N L_\beta$ and require $\mu \le \mu_1 \le \big(2(1+M)a\Lambda_N\Lambda_g\big)^{-1}$.

Lemma 6.7 (Slab with offset)**.** *For $\ell \in \mathbb{R}^N \setminus \{0\}$, $c \in \mathbb{R}$, $\lambda, a' > 0$: $\mathrm{meas}\{x \in [-\lambda, \lambda]^N : |\ell \cdot x + c| < a'\} \le 2a'(2\lambda)^{N-1}/|\ell|$.*

Proof. Take i with $|\ell_i| = |\ell|$. Each section in x_i lies in an interval of length $2a'/|\ell_i|$; apply (X1) (cf. [K:`bp:p4:slab`]). □

Lemma 6.8 (Lattice sums). *(a)* $\sum_{k\in\mathbb{Z}^n\setminus 0}|k|_1^{-\tau}\le S_1:=2\cdot 3^{n-1}\frac{\tau+1-n}{\tau-n}$ *($\tau>n$);* $\sum_{k_2\in\mathbb{Z}^b\setminus 0}|k_2|_1^{-\tau-1}\le S_2:=2\cdot 3^{b-1}\frac{\tau+2-b}{\tau+1-b}$. *(b) For real $m\ge 1$:* $\sum_{|k|_1>m}|k|_1^{-\tau-1}\le C_T(2/m)^{\tau+1-n}$, $C_T:=2\cdot 3^{n-1}/(\tau+1-n)$.

Proof. **Step S6.8: (a).** [K:bp:p4:lattice-sum] with $(n,\tau-1)$ (valid as $\tau-1>n-1$) and with (b,τ).

Step S6.9: (b). By [K:bp:p4:lattice-count] the sum is $\le 2\cdot 3^{n-1}\sum_{\ell>m}\ell^{-p}$, $p=\tau+2-n>1$. By convexity $\ell^{-p}\le\int_{\ell-1/2}^{\ell+1/2}x^{-p}dx$, and $\ell>m\ge 1$ gives $\ell-\frac12\ge m/2$: the sum is $\le\int_{m/2}^{\infty}x^{-p}dx=(m/2)^{1-p}/(p-1)$. □

Proposition 6.9 (Two-scale Diophantine measure). *Assume (S4), $0<\lambda\le\lambda_0$, $Q\subset(-\lambda,\lambda)^b$ Borel, $\mu\in(0,\mu_1]$, $\alpha_1,\alpha_2>0$, and set*

$$\mathcal{R}:=\{(I,\xi)\in\bar D\times Q:\ \omega_\mu(I,\xi)\notin\mathcal{D}_{\alpha_1,\alpha_2,\tau}\},\qquad\bar\gamma:=\alpha_2/\mu.$$

Then

$$\operatorname{meas}\mathcal{R}\le\Big[c_1\alpha_1\Big(1+\frac{\mu^{\tau-n}}{\lambda}\Big)+c_2\frac{\bar\gamma}{\lambda}\Big]\operatorname{meas}(\bar D)\,(2\lambda)^b,$$

$$c_1:=\frac{\Lambda_g^n}{\operatorname{meas}\bar D}\max\Big\{4(2R_y)^{a-1}S_1,\ 4b^2L_\beta(2R_y)^aC_T(4\Lambda_N\Lambda_g)^{\tau+1-n}\Big\},\qquad c_2:=bL_\beta S_2.$$

The constants depend on $(a,b,\tau,H_0,V,W,\Lambda_N,L_\beta,\bar D)$ only, not on μ,λ,Q,α_i.
Source: CP10 Lemma 3.1, p. 562, proof pp. 574–577, in the form needed for (59). The statement and proof are not *CP10's: see Remark 6.10 and errata (E1), (E2). The constant $c(n,\tau)=\sum|k|^{-\tau}$ of (58) is S_1-type; the factor $\bar\gamma/\lambda=\gamma_2/(\mu\lambda)$ is CP10's $\gamma_2/\hat r$.*

Proof. For $k=(k_1,k_2)\neq 0$ put $F_k:=k\cdot\omega_\mu=k_1\cdot\omega_{H_0}(I)+\mu\big[k_1\cdot G_\mu+k_2\cdot\omega_\mu^{\mathrm s}\big]$ and $\alpha_k:=\alpha_1|k|_1^{-\tau}$ if $k_1\neq 0$, $\alpha_k:=\alpha_2|k|_1^{-\tau}$ if $k_1=0$. Then $\mathcal{R}=\bigcup_k\mathcal{R}_k$, $\mathcal{R}_k=\{(I,\xi)\in\bar D\times Q:|F_k|<\alpha_k\}$ (Borel: F_k is continuous by (N1)–(N2) and Lemma 6.3). Split $\mathcal{R}=\mathcal{R}^{(0)}\cup\mathcal{R}^{(A)}\cup\mathcal{R}^{(B)}$: $k_1=0$; $k_1\neq 0$ in $\mathcal{K}_A$; $k_1\neq 0$ in $\mathcal{K}_B$, where

$$\mathcal{K}_A:=\{k_1\neq 0:\ \mu\Lambda_N\Lambda_g|k|_1\le|k_1|_\infty/2\},\qquad\mathcal{K}_B:=\{k_1\neq 0\}\setminus\mathcal{K}_A.$$

(Here $|k_1|_\infty$ is the max norm of the block $k_1\in\mathbb{Z}^a$, $|k_1|_1$ its ℓ^1 norm, and $|k|_1$ the ℓ^1 norm of k.)

Step S6.10: Case $k_1=0$ (slow resonances). Here $F_k=\mu k_2\cdot(\Omega+\beta\xi)$, $|k|_1=|k_2|_1$. For fixed I, $\{\xi\in[-\lambda,\lambda]^b:|(\beta^\top k_2)\cdot\xi+k_2\cdot\Omega|<\bar\gamma|k_2|_1^{-\tau}\}$ has measure $\le 2\bar\gamma|k_2|_1^{-\tau}(2\lambda)^{b-1}/|\beta^\top k_2|$ by Lemma 6.7 ($\beta^\top k_2\neq 0$ by (N3)). By (N3) and $|k_2|\ge|k_2|_1/b$, $|\beta^\top k_2|\ge|k_2|_1/(bL_\beta)$. Tonelli in I gives $\operatorname{meas}\mathcal{R}_k\le\operatorname{meas}\bar D\,2\bar\gamma bL_\beta(2\lambda)^{b-1}|k_2|_1^{-\tau-1}$, and by Lemma 6.8(a) $\operatorname{meas}\mathcal{R}^{(0)}\le\operatorname{meas}\bar D(2\lambda)^b\,bL_\beta S_2\,\bar\gamma/\lambda$. (No injectivity of ω_μ is used; μ cancels because the threshold is $\mu\bar\gamma$.)

Step S6.11: Change to frequency coordinates. Let $\Phi(I,\xi)=(\omega_{H_0}(I),\xi)$, injective and continuous on $\bar D\times[-\lambda,\lambda]^b$ with inverse $(y,\xi)\mapsto(g(y),\xi)$, which is Λ_g-Lipschitz on $Y_0\times\mathbb{R}^b$ (Lemma 6.3, max norm). Put $Y=\omega_{H_0}(\bar D)\subset[-R_y,R_y]^a$, $\mathcal{R}_k'=\Phi(\mathcal{R}_k)$ (Borel, (X2)). By [K:bp:p1:lip-image] in $\mathbb{R}^n$: $\operatorname{meas}\mathcal{R}_k\le\Lambda_g^n\operatorname{meas}\mathcal{R}_k'$. On $\mathcal{R}_k'$ the function is $\tilde F_k(y,\xi)=k_1\cdot y+\mu\tilde G_k(y,\xi)$, $\tilde G_k:=k_1\cdot G_\mu(g(y),\xi)+k_2\cdot\omega_\mu^{\mathrm s}(g(y),\xi)$, and by (N1) and Lemma 6.3

$$|\tilde G_k(y,\xi)-\tilde G_k(y',\xi)|\le\Lambda_N\Lambda_g|k|_1|y-y'|\quad(y,y'\in Y).$$

Step S6.12: Case $\mathcal{K}_A$ (slicing in y_j). Let $k\in\mathcal{K}_A$ and j with $|k_{1,j}|=|k_1|_\infty\ge 1$. For two points $(y+se_j,\xi)$, (y,ξ) of $\mathcal{R}_k'$: $|\tilde F_k(y+se_j,\xi)-\tilde F_k(y,\xi)|\ge|k_{1,j}||s|-\mu\Lambda_N\Lambda_g|k|_1|s|\ge|k_{1,j}||s|/2$. Both values have modulus $<\alpha_k$, hence $|s|<4\alpha_k/|k_{1,j}|\le 4\alpha_k$: every y_j-section of $\mathcal{R}_k'$ has diameter $\le 4\alpha_k$ (no convexity of $\bar D$ is needed). The projection forgetting y_j lies in $[-R_y,R_y]^{a-1}\times[-\lambda,\lambda]^b$. By (X1), $\operatorname{meas}\mathcal{R}_k'\le 4\alpha_1|k|_1^{-\tau}(2R_y)^{a-1}(2\lambda)^b$, and summing (Lemma 6.8(a)), $\operatorname{meas}\mathcal{R}^{(A)}\le\Lambda_g^n\,4S_1(2R_y)^{a-1}\alpha_1(2\lambda)^b$.

Step S6.13: Structure of $\mathcal{K}_B$. Let $k \in \mathcal{K}_B$. (i) $|k|_1 > K_\mu := (2\mu\Lambda_N\Lambda_g)^{-1}$, because $\mu\Lambda_N\Lambda_g|k|_1 > |k_1|_\infty/2 \geq \frac{1}{2}$; and $K_\mu \geq 1$ since $\mu \leq \mu_1$. (ii) $|k_2|_1 > M|k_1|_1$: otherwise $|k|_1 \leq (1+M)|k_1|_1 \leq (1+M)a|k_1|_\infty$ and $\mu\Lambda_N\Lambda_g|k|_1 \leq |k_1|_\infty/2$ by $\mu \leq \mu_1$, i.e. $k \in \mathcal{K}_A$. Consequently $|k|_1 < (1+M^{-1})|k_2|_1 \leq 2b|k_2|_\infty$ and $\Lambda_N|k_1|_1 \leq \Lambda_N b|k_2|_\infty/M = |k_2|_\infty/(2L_\beta)$.

Step S6.14: Case $\mathcal{K}_B$ (slicing in ξ_j). Fix $k \in \mathcal{K}_B$ and $j \leq b$; let $\mathcal{R}_{k,j}$ be the part of $\mathcal{R}'_k$ with $|(\beta(\mu, g(y))^\top k_2)_j| \geq |k_2|_\infty/L_\beta$; by (N3) $\mathcal{R}'_k = \bigcup_{j\leq b} \mathcal{R}_{k,j}$ (take j maximizing $|(\beta^\top k_2)_j|$). Along $\xi \mapsto \xi + se_j$ (y and the other ξ's fixed, so $I = g(y)$ fixed)

$$\tilde{F}_k(y, \xi + se_j) - \tilde{F}_k(y, \xi) = \mu s\,(\beta^\top k_2)_j + \mu\, k_1 \cdot \big(G_\mu(I, \xi + se_j) - G_\mu(I, \xi)\big),$$

because ω^{s}_μ is affine in ξ. By (N2) and S6.13 the modulus is $\geq \mu|s|\big(|k_2|_\infty/L_\beta - \Lambda_N|k_1|_1\big) \geq \mu|s||k_2|_\infty/(2L_\beta)$. Two points of $\mathcal{R}_{k,j}$ on one line thus satisfy $|s| < 4L_\beta\alpha_k/(\mu|k_2|_\infty) \leq 8bL_\beta\alpha_1|k|_1^{-\tau-1}/\mu$. The projection forgetting ξ_j lies in $[-R_y, R_y]^a \times [-\lambda, \lambda]^{b-1}$. Summing over $j \leq b$, then over $|k|_1 > K_\mu$, with Lemma 6.8(b):

$$\operatorname{meas}\mathcal{R}^{(B)} \leq \Lambda_g^n\, b\,(2R_y)^a(2\lambda)^{b-1}\frac{8bL_\beta\alpha_1}{\mu}C_T\big(2/K_\mu\big)^{\tau+1-n} = \Lambda_g^n(2\lambda)^b\,4b^2L_\beta(2R_y)^aC_T(4\Lambda_N\Lambda_g)^{\tau+1-n}\alpha_1\frac{\mu^{\tau-n}}{\lambda}.$$

(here $2/K_\mu = 4\mu\Lambda_N\Lambda_g$.)

Step S6.15: Sum. Add the three bounds; each of the Λ_g^n-terms is $\leq c_1 \operatorname{meas}\bar{D}\,\alpha_1(2\lambda)^b$ times 1, resp. $\mu^{\tau-n}/\lambda$. □

Remark 6.10 (What is and is not claimed; status of CP10 Lemma 3.1). CP10 Lemma 3.1 (p. 562) states $\operatorname{meas}\mathcal{R} \leq (c_1\gamma_1 + c_2\gamma_2/\hat{r})\operatorname{meas}(\bar{D} \times B_{\hat{r}})$, with c_1 containing an integer p, "for a suitable integer p depending on $\bar{D}$ and ω_1", the number of sets $U(I_1^{(i)})$ of (111) needed to cover $\omega(\bar{D} \times B_{\hat{r}})$. The lemma is true for each $\hat{r}$, but $p = p(\hat{r})$, and in general $p \gtrsim \hat{r}^{-1}$: for $n_1 = n_2 = 1$, $\bar{D} = [0,1]$, $\omega(I_1, I_2) = (I_1, I_1 + I_2)$ (a diffeomorphism, $\beta = 1$), the resonance $k = (1, -1)$ gives $|k\cdot\omega| = |I_2| < \hat{r}$, so the whole domain is resonant as soon as $\gamma_1 \geq 2^\tau\hat{r}$ (CP10's $|k| = |k|_1$) while γ_2 is arbitrary small: the lemma forces $c_1 \geq 2^{-\tau}\hat{r}^{-1}$. CP10 then applies it with $\hat{r} = \check{c}_2\varepsilon^2$ and asserts "c_i independent of ε and μ" (p. 563, first line, continuing (59) from p. 562). That is false for the abstract lemma; Proposition 6.9 proves the needed estimate directly, using the *structure* of the frequency map (the fast Hessian is invertible because ω_{H_0} is bi-Lipschitz, and the slow block is $\mu(\Omega + \beta\xi)$ with β nondegenerate), and so the covering lemma is not used. The counting lemmas of track K are reused as follows: [K:`bp:p4:lattice-count`], [K:`bp:p4:lattice-sum`] (Lemma 6.8); [K:`bp:p4:slab`] (model of Lemma 6.7, which needs a real normal and an offset); [K:`bp:p4:preimage-measure`] is *not* reusable (its hypotheses are the pullback by the inverse frequency map g of a full-dimensional single-scale setting with one threshold; it is replaced by Step S6.11, which is the same Lipschitz-image argument with the product map $(y, \xi) \mapsto (g(y), \xi)$); [K:`bp:p4:cover-mono`] concerns the specific cover `kamCover` of track K and [K:`bp:p4:dyadic-approx`] serves HEAD-Q (no boundary hypothesis): neither is needed since `hboundary` is a hypothesis of P-G.

Remark 6.11 (The hypothesis $\tau > n$ and the factor $\mu^{\tau-n}$). The sums $\sum|k|_1^{-\tau}$ and $\sum|k|_1^{-\tau-1}$ converge iff $\tau > n$ and $\tau > n-1$ resp. The extra term $c_1\alpha_1\mu^{\tau-n}/\lambda$ is harmless when $\mu^{\tau-n} \leq \lambda$; since $\mu < r^6$ and $\lambda = r^2/2$, taking $\tau \geq n+1$ gives $\mu^{\tau-n}/\lambda \leq 2r^4$. Part 5 chooses τ (the frozen theorem only needs some $\tau > a + b$, Lemma 2.16).

6.5. **Collar of the domain $D = \bar{D}_t \times A$.** Fix $0 < \lambda_1 < \lambda_2$ and $A_{\mathrm{sh}} := (\lambda_1, \lambda_2)^b$ (the oscillator shell $\check{c}_1\varepsilon^{5/2} < J_{2,i} < \check{c}_2\varepsilon^2$ of (47); $\lambda_2 = r^2/2$ for the frozen radius r), $\ell := \lambda_2 - \lambda_1$, and

$$D := \bar{D}_t \times A_{\mathrm{sh}} \subset \mathbb{R}^a \times \mathbb{R}^b.$$

Lemma 6.12 (Enlargement of D)**.** *Let $0 < t < d$ and $0 < r_K < \min\{t, \lambda_1\}$. Then $\mathrm{Re}(D_{r_K}) = N_{r_K}(\bar D_t) \times (\lambda_1 - r_K, \lambda_2 + r_K)^b \subset V \times (0,\infty)^b$, and, without any division,*

$$\operatorname{meas}\big(\mathrm{Re}(D_{r_K}) \setminus D\big) \le At(\ell + 2r_K)^b + \operatorname{meas} \bar D_t \big[(\ell + 2r_K)^b - \ell^b\big].$$

If moreover $0 < \operatorname{meas} \bar D_t$ (for instance if $At \le \frac12 \operatorname{meas} V$, since $\operatorname{meas} \bar D_t \ge \operatorname{meas} V - At$ by Lemma 6.1(b); this is how Part 5 ensures it, [P:`bpP:p5:param-facts`]*), then*

$$\operatorname{meas}\big(\mathrm{Re}(D_{r_K}) \setminus D\big) \le \operatorname{meas}(D) \Big[\frac{At}{\operatorname{meas} \bar D_t}\Big(1 + \frac{2r_K}{\ell}\Big)^b + \frac{2br_K}{\ell}\Big(1 + \frac{2r_K}{\ell}\Big)^{b-1}\Big] =: \operatorname{meas}(D)\, \eta_2.$$

Proof. **Step S6.16: Product structure.** For the max norm, $\operatorname{dist}(x, X\times Y) = \max(\operatorname{dist}(x_1, X), \operatorname{dist}(x_2, Y))$, so the real neighbourhood of a product is the product of the real neighbourhoods; $N_{r_K}(\bar D_t) \subset V$ by Lemma 6.1(c) and $\lambda_1 - r_K > 0$.

Step S6.17: Difference. $(X'\times Y')\setminus(X\times Y) \subset ((X'\setminus X)\times Y')\cup(X\times(Y'\setminus Y))$. With $\operatorname{meas}(N_{r_K}(\bar D_t)\setminus \bar D_t) \le At$ (Lemma 6.1(c)), $\operatorname{meas} Y' = (\ell + 2r_K)^b$ and $(\ell + 2r_K)^b - \ell^b \le 2br_K(\ell + 2r_K)^{b-1}$ (mean value), this is the unnormalized bound; under $\operatorname{meas} \bar D_t > 0$ divide by $\operatorname{meas} D = \operatorname{meas} \bar D_t\, \ell^b > 0$. (Without positivity the normalized form is meaningless: for $a = b = 1$, $V = (-1, 1)$, $A = 4$, $d = 2$, $t = 1$, $\bar D_t = \{0\}$, the enlargement has area > 0 while $\operatorname{meas} D = 0$.) □

6.6. **The cylinder cell and the singular shell.** Define the action–angle chart of the oscillators in the frozen coordinates $z = ((\varphi, q), (I, p))$, $\varphi \in \mathbb{R}^a$, $I \in \mathbb{R}^a$, $(p_j, q_j) \in \mathbb{R}^2$:

$$\Pi(I, \xi; \varphi, \psi) := \big(\varphi, q = \sqrt{2\xi}\sin\psi\big),\ \big(I,\ p = \sqrt{2\xi}\cos\psi\big), \qquad (I, \xi) \in \mathbb{R}^a \times (0,\infty)^b,\ \varphi \in \mathbb{R}^a,\ \psi \in \mathbb{R}^b,$$

and, for a period-one angle $\vartheta = (\vartheta_f, \vartheta_s) \in [0, 1)^n$, $\mathsf{E}(I, \xi, \vartheta) := \Pi(I, \xi; 2\pi\vartheta_f, 2\pi\vartheta_s)$ (the dictionary of Lemma 4.5 with Q; Part 3's $(J, \vartheta) = ((I, \xi), \vartheta)$).

Lemma 6.13 (Cell and polar coordinates)**.** *(a) Π is real analytic and injective on $\mathbb{R}^a \times (0,\infty)^b \times \mathbb{R}^a \times [0, 2\pi)^b$ modulo $\varphi \mapsto \varphi + 2\pi k$; $|\det D\Pi| = 1$. Hence E is injective on $\mathbb{R}^a \times (0,\infty)^b \times [0, 1)^n$ and $\operatorname{meas} \mathsf{E}(X) = (2\pi)^n \operatorname{meas} X$ for measurable X there. (b) $\mathcal{F}(V, r) \setminus \mathsf{E}(V \times (0, r^2/2)^b \times [0, 1)^n)$ is null, and*

$$\operatorname{meas} \mathcal{F}(V, r) = (2\pi)^a \operatorname{meas} V\, (\pi r^2)^b = (2\pi)^n \operatorname{meas} V\, (r^2/2)^b.$$

(c) (Singular shell.) For $0 < \lambda_1 < \lambda_2 = r^2/2$, the set of $z \in \mathcal{F}(V, r)$ with $\xi_j \le \lambda_1$ for some j has measure $\le b\frac{\lambda_1}{\lambda_2} \operatorname{meas} \mathcal{F}(V, r)$; with $\lambda_1 = \kappa r^{5/2}$ this is $2b\kappa\sqrt{r} \operatorname{meas} \mathcal{F}$.

Proof. **Step S6.18: Jacobian.** For one oscillator, $(\xi, \psi) \mapsto (q, p) = \sqrt{2\xi}(\sin\psi, \cos\psi)$ has Jacobian determinant of modulus 1 ($dp \wedge dq = d\xi \wedge d\psi$, Remark 2.10(1); directly, $\partial_\xi = (\sin\psi, \cos\psi)/\sqrt{2\xi}$, $\partial_\psi = \sqrt{2\xi}(\cos\psi, -\sin\psi)$). It is a bijection of $(0,\infty) \times [0, 2\pi)$ onto $\mathbb{R}^2 \setminus 0$. Π is the identity on (I, φ), so $|\det D\Pi| = 1$ after the permutation of coordinates (measure preserving), and the area formula ([P:`bpP:p1:liouville`], (X3)) gives equality of measures on injectivity sets. $\vartheta \mapsto 2\pi\vartheta$ has determinant $(2\pi)^n$.

Step S6.19: (b). Let $z \in \mathcal{F}(V, r)$ with $(p_j, q_j) \ne 0$ for all j (the excluded set lies in finitely many subspaces of codimension 2, hence is null). Then $\xi_j \in (0, r^2/2)$, $\psi_j \in [0, 2\pi)$ is unique, $\varphi \in [0, 2\pi)^a$, $I \in V$: $z = \mathsf{E}(I, \xi, \vartheta)$ with $\vartheta \in [0, 1)^n$; the converse inclusion is the same computation. So $\operatorname{meas} \mathcal{F} = (2\pi)^n \operatorname{meas} V (r^2/2)^b$.

Step S6.20: (c). In the chart, the set is $\mathsf{E}(V \times \{\xi : \exists j,\ \xi_j \le \lambda_1\} \times [0, 1)^n)$ with $\{\xi \in (0, \lambda_2)^b : \exists j, \xi_j \le \lambda_1\}$ of measure $\le b\lambda_1\lambda_2^{b-1}$. Divide by (b). □

6.7. The budgets consumed by Parts 3 and 5.

Proposition 6.14 (Budget on the action side (the right side of (3.1))). *Assume (S4); $0 < t < d$; $0 < \lambda_1 \le \lambda_2/2$, $\lambda_2 \le \lambda_0$; $0 < r_K < \min\{t, \lambda_1\}$; $\mu \in (0, \mu_1]$; and $D = \bar{D}_t \times A_{\mathrm{sh}}$ with $\bar{D}_t \subset \bar{D}$-admissible, i.e. (S4) holds with $\bar{D} := \bar{D}_t$. Let $\gamma_1 \ge \gamma_2 > 0$ be the Diophantine constants of Proposition 4.3 (for $h = (H_0 + \mu N_\mu)/2\pi$), D^{dio} as there, and K_Ω any set with (3.1) for $r = r_K$. Then*

$$\operatorname{meas}\big((D \times \mathbb{T}^n) \setminus K_\Omega\big) \le c_n(\eta_1 + \eta_2) \operatorname{meas} D, \qquad \eta_1 := 4^b \Big[c_1\, 2\pi\gamma_1 \Big(1 + \frac{\mu^{\tau-n}}{\lambda_2}\Big) + c_2 \frac{2\pi\gamma_2}{\mu\lambda_2} \Big],$$

η_2 of Lemma 6.12, $c_n = 1 + (1+2^n)(1+(4/3)^n)$ of Proposition 4.3, c_1, c_2 with $\bar{D} = \bar{D}_t$.

Proof. **Step S6.21: Reading (3.1).** $D \subset \mathrm{Re}(D_{r_K})$, so $(D \times \mathbb{T}^n) \setminus K_\Omega \subset (\mathrm{Re}(D_{r_K}) \times \mathbb{T}^n) \setminus K_\Omega$, of measure $\le c_n(\operatorname{meas}(D \setminus D^{\mathrm{dio}}) + \operatorname{meas}(\mathrm{Re}(D_{r_K}) \setminus D))$ ($\operatorname{meas} \mathbb{T}^n = 1$).

Step S6.22: Non-Diophantine part. By the dictionary of Definition 6.5, $D \setminus D^{\mathrm{dio}} = \mathcal{R}$ for $Q = A_{\mathrm{sh}} \subset (-\lambda_2, \lambda_2)^b$, $\lambda = \lambda_2$, $\alpha_i = 2\pi\gamma_i$. Proposition 6.9 gives $\operatorname{meas} \mathcal{R} \le [\dots] \operatorname{meas}(\bar{D}_t)(2\lambda_2)^b$, and $(2\lambda_2)^b = (2\lambda_2/\ell)^b \ell^b \le 4^b \ell^b$ since $\ell \ge \lambda_2/2$; $\operatorname{meas} \bar{D}_t\, \ell^b = \operatorname{meas} D$.

Step S6.23: Collar part. Lemma 6.12. □

Proposition 6.15 (Budget in the frozen cell). *In the situation of Proposition 6.14 with $\lambda_2 = r^2/2$ (r the frozen radius), let $K \subset \mathbb{R}^a \times (0,\infty)^b \times \mathbb{T}^n$ be any measurable set with $\operatorname{meas}((D \times \mathbb{T}^n) \setminus K) \le c_n(\eta_1 + \eta_2) \operatorname{meas} D$ and let $\mathsf{S} := \mathsf{E}(K \cap (\mathbb{R}^a \times (0,\infty)^b \times [0,1)^n))$. Then*

$$\operatorname{meas}\big(\mathcal{F}(V,r) \setminus \mathsf{S}\big) \le \Big[\frac{At}{\operatorname{meas} V} + b\frac{\lambda_1}{\lambda_2} + c_n(\eta_1 + \eta_2) \Big] \operatorname{meas} \mathcal{F}(V,r).$$

Proof. **Step S6.24: Decomposition.** Up to a null set, $\mathcal{F} = \mathsf{E}(V \times (0,\lambda_2)^b \times [0,1)^n)$ (Lemma 6.13(b)), and $\mathcal{F}_{\mathrm{good}} := \mathsf{E}(D \times [0,1)^n) \subset \mathcal{F}$ (since $\bar{D}_t \subset V$, $A_{\mathrm{sh}} \subset (0,\lambda_2)^b$). So $\mathcal{F} \setminus \mathsf{S} \subset (\mathcal{F} \setminus \mathcal{F}_{\mathrm{good}}) \cup \mathsf{E}((D \times \mathbb{T}^n) \setminus K)$ up to a null set.

Step S6.25: Bad action set. By Lemma 6.13, $\operatorname{meas} \mathcal{F}_{\mathrm{good}} = (2\pi)^n \operatorname{meas} \bar{D}_t\, \ell^b$ and $\operatorname{meas} \mathcal{F} = (2\pi)^n \operatorname{meas} V \lambda_2^b$, so $\operatorname{meas} \mathcal{F}_{\mathrm{good}} / \operatorname{meas} \mathcal{F} = (\operatorname{meas} \bar{D}_t / \operatorname{meas} V)(1 - \lambda_1/\lambda_2)^b \ge 1 - \frac{At}{\operatorname{meas} V} - b\frac{\lambda_1}{\lambda_2}$ (Lemma 6.1(b); $(1-x)^b \ge 1 - bx$; $(1-u)(1-v) \ge 1 - u - v$).

Step S6.26: Lost tori. $\operatorname{meas} \mathsf{E}((D \times \mathbb{T}^n) \setminus K) = (2\pi)^n \operatorname{meas}((D \times \mathbb{T}^n) \setminus K) \le c_n(\eta_1 + \eta_2) \operatorname{meas} \mathcal{F}_{\mathrm{good}} \le c_n(\eta_1 + \eta_2) \operatorname{meas} \mathcal{F}$. □

Remark 6.16 (What Part 5 must supply and check). Part 5 chooses, as functions of r and in the order of Theorem 1.28: $\tau \ge n+1$; $t = \sqrt{r}$ (so $At/\operatorname{meas} V \le A\sqrt{r}/\operatorname{meas} V$, $t < d$ for $r < r_*$); $\lambda_2 = r^2/2$, $\lambda_1 = \kappa r^{5/2}$ (shell loss $2b\kappa\sqrt{r}$ by Lemma 6.13(c)); and γ_1, γ_2 so that $\eta_1 \le C\sqrt{r}$: $\gamma_1 \le Cr^{5/2}$ and $\gamma_2/\mu \le Cr^{5/2}$ (CP10 (20): $\gamma_1 = \bar{\gamma}_2 = \gamma_*^2 \varepsilon^{5/2}$), and $r_K < \min\{t, \lambda_1\}$ (strict, as Lemma 6.12 requires) with $r_K/\ell \le C\sqrt{r}$, from $r_K = 32\hat{E}\hat{\rho} \le C\hat{\rho} \le C\min\{\gamma_1, \gamma_2/\mu\}$ (the slow Hessian bound is $\hat{M} = C\mu$, so $\hat{\rho} \le \gamma_2/(3C\mu K^{\tau+1})$: the earlier chain "$\le C\gamma_2$" is false, review P4-02), so $\eta_2 \le C\sqrt{r}$; the bound actually used is Part 5's [P:`bpP:p5:rK`]. Part 5 also ensures $\operatorname{meas} \bar{D}_t > 0$ (e.g. $A\sqrt{r} \le \frac{1}{2} \operatorname{meas} V$) for Lemma 6.12 (review P4-01). It also checks (S4): (N1)–(N2) with Λ_N from joint analyticity of P_0, Ω, β near $[0,\mu_0] \times \overline{V}$ (Lemma 6.2), (N3) from `htorsion` (Lemma 6.4), $\mu \le \mu_1$ from $\mu < r^6$, injectivity and nondegeneracy of ω_μ on the complex neighbourhood required by Proposition 4.3, and Borelness of K_Ω (Lemma 2.21) for the frozen `MeasurableSet`. Proposition 6.15 then gives $\operatorname{meas}(\mathcal{F} \setminus \mathsf{S}) \le C\sqrt{r} \operatorname{meas} \mathcal{F}$. The strict inequalities and positivity are Part 5 ([P:`bpP:p5:strictness`]).

6.8. **Suspected misprints and gaps in CP10 (Lemma 3.1, Appendix C, §3 end).**

(E1) **V (gap).** p. 563, first line (continuing (59) of p. 562): "with c_i independent of ε and μ". The constant c_1 of (58) contains the covering number p, which depends on $\hat{r} = \check{c}_2\varepsilon^2$ (the lemma, pp. 562–563, says "for a suitable integer p depending on $\bar{D}$ and ω_1"; the pointwise-in-$\hat{r}$ estimate (58)/(59), p. 562, allows $p = p(\hat{r})$, and the printed restricted-dependence claim, which would make p independent of $\hat{r}$, is not accepted); Remark 6.10 gives a counterexample to $\hat{r}$-independence for the abstract lemma. Repaired by Proposition 6.9.

(E2) **V.** p. 562, (59) and the line before: Lemma 3.1 is applied to $\omega_{\mathrm{resc}} = (\partial_{J_1}(H_0 + \mu\breve{N}), \partial_{J_2}\breve{N})$, whose $k_1 \neq 0$ resonances $|k_1 \cdot \omega_1 + k_2 \cdot \omega_2|$ differ from those of $\omega_\mu = (\omega_1, \mu\omega_2)$ (where the term is $\mu k_2 \cdot \omega_2$), which define $\mathcal{D}_{\gamma_1,\mu\bar{\gamma}_2,\tau}$. The two resonant sets are different; our proof treats ω_μ itself.

(E3) **T.** p. 561, below (55), footnote 25 and (57): "$\mathrm{meas}(\mathrm{Re}(V_r)\setminus V) \leq C\bar{\gamma}_2 \,\mathrm{meas}\, V$" (as extracted from the pinned PDF; the exponent/subscript may be a typesetting artifact) is asserted without proof and cannot hold for an arbitrary open V; it is replaced by `hboundary` (audit P3, D-PG-5), and only the inner collar is needed.

(E4) **T (not verified).** p. 561 (56): the loss $C\varepsilon^{n_2/2}$ for the shell $\check{c}_1\varepsilon^{5/2} < J_{2,i}$. Our computation (Lemma 6.13(c)) gives $n_2\check{c}_1\varepsilon^{1/2}/\check{c}_2$, which is larger than $\varepsilon^{n_2/2}$ for $n_2 \geq 2$; CP10's displayed (18) and (20) are consistent with the $\sqrt{\varepsilon}$ rate only.

(E5) **R.** p. 575 (113) and p. 576: after the change of variables the Jacobian $1/|k_j|$ and the cover $\mathcal{C}^i_k$ are fine, and the integrals over the slice use the norm $|\cdot|$; the sum $c(n,\tau) = \sum|k|^{-\tau}$ needs $\tau > n$, which is why CP10 assumes $\tau_* > n$ (p. 550, footnote 12); Lemma 6.8 makes this explicit.

6.9. **References and imported facts.** Part 1: [P:`bpP:p1:two-scale`], [P:`bpP:p1:dio-basic`], [P:`bpP:p1:dio-sources`], [P:`bpP:p1:liouville`], [P:`bpP:p1:torus-union`], [P:`bpP:p1:typical`]. Part 3: [P:`bpP:p3:two-scale-kam`] (inequality (3.1), c_n), [P:`bpP:p3:period-dictionary`], §5.6. Track K: [K:`bp:p4:lattice-count`], [K:`bp:p4:lattice-sum`], [K:`bp:p4:slab`] (model), [K:`bp:p1:lip-image`], [K:`bp:p1:ext-cramer`], [K:`bp:p1:ext-lebesgue`], [K:`bp:p1:ext-mvt`]. Forward references to Part 5: [P:`bpP:p5:uniform-remainder`], [P:`bpP:p5:strictness`]. Formalization notes: Lemma 6.2 is a stand-alone compactness argument; Proposition 6.9 is three slicing estimates plus Tonelli on sets of short sections; the polar chart is a bounded stand-alone task.

7. P-G assembly: uniform remainder, parameters, application of Parts 2–4, tori, measure, strictness

This part proves the frozen theorem [P:`bpP:p0:pg`] (Theorem 7.29) from Parts 1–4. Write $n = a+b$. The route is CP10 §4, pp. 562–564, but the *hypotheses are the frozen ones* and everything the source asserts without proof is supplied: the uniform remainder (CP10 (60)), the nondegeneracy of the frequency map on a complex neighbourhood, the explicit parameter choice with the numerical bounds of [P:`bpP:p3:interface`] tracked, the conversion of tori to `IsCylinderTorus`, the cell measure, and the strict inequalities.

What is different from CP10. (1) Hypothesis (A_2'') is the Birkhoff form *already*, so Steps 2–4 of CP10 §3 (equilibrium, diagonalization, Birkhoff) are not needed: the averaged secular function N of [P:`bpP:p0:birkhoff`] is used as it stands, and the remainder after averaging is $\mu Q + \widetilde{P}$, with $Q = P_{\mathrm{av}} - N = O(|(p,q)|^5)$ and $\widetilde{P}$ of Part 2. (2) The KAM step is applied on the shell $\lambda_1 < \xi_j < \lambda_2'$ with the constant $K = K_0$ *bounded* (the two Diophantine scales $\gamma_1 = \bar{\gamma}_2 = \kappa_0 r^{5/2}$ make $K = \frac{6}{s}\log_+(\kappa_0^2/C^4)$ independent of r); this is why no power of $\log(1/r)$ enters the $\sqrt{r}$ loss of the measure. (3) The set of $\bar{\omega}_0$-Diophantine actions is a Cantor-type set $\check{D}$; its complex neighbourhood is not controlled by the collar, so Lemma 7.26 and Proposition 7.27 re-run the budget

of [P:bpP:p4:budget], [P:bpP:p4:cell-budget] for $D = \check{D} \times A_{\mathrm{sh}}$ (replacing $\bar{D}_t$ by $\check{D} \subset \bar{D}_t$ in the *domain*, but not in the Diophantine count).

7.1. **Conventions and the order of the constants.** All coordinates are CP's: angles φ of period 2π, actions I, oscillators (p, q); the frozen fast frequency map is $\bar{\omega}_0 = \omega_{H_0} = \partial H_0$ (CP angular convention). Part 3 works with period-one angles and $H_3 = \tilde{H}(J, 2\pi\theta)/2\pi$ ([P:bpP:p3:period-dictionary]); all quantities of Part 3 with 2π factors are absorbed in the constants C below, and the dictionary $\gamma_i = \alpha_i/2\pi$ of [P:bpP:p4:freq] is used. Norms are max norms; $|\cdot|_*$ is the weighted norm of [N:bpN:p4:norm] with coefficient radius (CP's s). λ_0 denotes the sup over real points of the complex bounds that appear.

Remark 7.1 (Order in which every constant is chosen). In the proof of Theorem 7.29 the constants are chosen in this order; each depends only on the earlier ones.

(1) *Data and hypotheses* (fixed first): $a, b, V, W, A, d, r_0, \mu_0, B, H_0, P, P_0, \Omega, \beta$.
(2) *Analytic constants* (Lemmas 7.2, 7.4, 7.9): δ_c, then s_* and $\varepsilon_{\mathrm{an}} = \min\{r_0/2, \delta_c/4\}$ (the complex oscillator radius is chosen after δ_c), r_1, Λ_P, C_Q, ρ_c, M_β, L_β, Λ_N, c_0, $\bar{M}_0$, R_y, L_g, μ_* (data only; μ and r do not enter).
(3) $\tau := n + 1$; the Part 3 numerical constant $C_1 \geq 1$ of [P:bpP:p3:interface] (Lemma 7.18); the widths $s, \bar{s}$ (functions of s_*).
(4) The scale constants κ_0, then κ_1, then G (Lemma 7.19, Definition 7.12); then the measure constant C_* (Prop. 7.27).
(5) r_* (finite list of smallness conditions, Definition 7.13); $C := \max\{C_* + 1, C_w\}$ with C_w the frequency bound; $c := 2\tau + 2 = 2n + 4$.
(6) Only *then* $r \in (0, r_*)$ and $\mu \in (0, \min\{\mu_0, r^6/(\log \frac{1}{r})^c\})$ are given; $K, \bar{K}, \bar{\gamma}, \bar{\rho}, \rho, \hat{\rho}, \hat{E}, r_K, t, \lambda_1, \lambda_2, \lambda_2'$ are explicit functions of (r, μ) (Definition 7.12); finally the set S (with the deleted tori of §7.10).

7.2. **Uniform complex extension and uniform remainder.** *Warning (review P5-01).* Frozen real analyticity gives only a thin complex tube around the real box; it does *not* give a holomorphic extension to the full complex oscillator polydisc of radius $r_0/2$ (a perturbation with a pole at $p = i\delta$, $0 < \delta < r_0/4$, satisfies every frozen premise). The radius $\varepsilon_{\mathrm{box}} = r_0/2$ is used only for the *real* box; the complex oscillator radius $\varepsilon_{\mathrm{an}}$ is chosen *after* δ_c (Definition 7.3), from the data only.

Lemma 7.2 (Uniform holomorphic extension of the family)**.** *Assume (H1), (H2) of* [P:bpP:p0:pg] *and* HasFastDiffeo(H_0, W). *Put* $\varepsilon_{\mathrm{box}} := r_0/2$ *(a* real *box radius only) and, for* $\delta > 0$, *let* $\mathcal{X}_\delta \subset \mathbb{C}^a \times \mathbb{C}^a \times \mathbb{C}^b \times \mathbb{C}^b$ *be the set of* (I, φ, p, q) *with* $\operatorname{dist}_\infty\big((I, \operatorname{Re}\varphi, \operatorname{Re} p, \operatorname{Re} q),\ \overline{V} \times \mathbb{R}^a \times \{|(p,q)|_\infty \leq \varepsilon_{\mathrm{box}}\}\big) < \delta$ *and* $|\operatorname{Im}(I, \varphi, p, q)|_\infty < \delta$. *Then there are* $\delta_c \in (0, 1)$ *and* $M_P < \infty$ *and a map* $\tilde{P} : [0, \mu_0] \times \mathcal{X}_{\delta_c} \to \mathbb{C}$ *with:* $\tilde{P}(\mu, \cdot)$ *holomorphic,* $\tilde{P} = P$ *at real points,* $\tilde{P}(\mu, \cdot + 2\pi k e_{\mathrm{f}}) = \tilde{P}(\mu, \cdot)$ *(e_{f} the fast angles),* $|\tilde{P}| \leq M_P$; $\tilde{H}_0$ *holomorphic on* $\{I : \operatorname{dist}_\infty(I, \overline{V}) < \delta_c\}$ *with* $|\tilde{H}_0|, \|\partial \tilde{H}_0\|, \|\partial^2 \tilde{H}_0\| \leq M_0$, *and* $\tilde{P}$ *jointly continuous in* μ.

Proof. **Step S7.1: Local extensions.** Let $X_0 = [0, \mu_0] \times \overline{V} \times [0, 2\pi]^a \times \{|(p,q)|_\infty \leq \varepsilon_{\mathrm{box}}\}$, compact. Since $2\varepsilon_{\mathrm{box}}^2 < r_0^2$ for the sup-norm ball, $X_0 \subset [0, \mu_0] \times \overline{\mathcal{C}(V, r_0)}$. By (H1) P is analytic at each point of X_0: a convergent power series in a polydisc, which defines a holomorphic extension there. By compactness a Lebesgue number gives δ_1 such that for every $x \in X_0$ the polydisc $B(x, \delta_1)$ carries a holomorphic extension P_x, with $|P_x| \leq M$ uniformly (shrink δ_1 so that closed polydiscs lie in the domains of finitely many series).

Step S7.2: Gluing. If $x, y \in X_0$ and $B(x, \delta_1) \cap B(y, \delta_1) \neq \emptyset$ then $|x - y| < 2\delta_1$ and the real midpoint lies in both; the intersection is convex hence connected, $P_x = P_y$ on its real points (both equal P), so $P_x = P_y$ on the intersection (identity theorem). Thus the P_x define one holomorphic $\tilde{P}$ on $\bigcup_x B(x, \delta_1)$ (for each real μ; jointly holomorphic in $(\mu, \cdot)$ near real μ).

Step S7.3: Periodicity. By (H2) and continuity of P on $\overline{\mathcal{C}(V,r_0)}$ (density of $\mathcal{C}(V,r_0)$ in its closure; P analytic hence continuous) periodicity holds on the closure. For φ outside $[0,2\pi]^a$ define $\tilde{P}$ by translation by $2\pi k_{\mathrm{f}}$ from the fundamental box; two such definitions agree on overlaps by the argument of Step S7.2 (agreement at real points, convex intersection). This gives $\mathcal{X}_{\delta_c}$ for $\delta_c = \delta_1$.

Step S7.4: H_0**.** H_0 is analytic on $W \supset \overline{V}$ (compact): Steps S7.1–S7.2 (without periodicity) give $\tilde{H}_0$ on a complex δ_c-neighbourhood of $\overline{V}$ (decrease δ_c); derivatives bounded by shrinking and Cauchy ([K:bp:p1:cauchy-open]). □

Definition 7.3 (Strip, averaged function, constants)**.** $s_* := \min\{\delta_c, 1\}/2$ (a coefficient radius $\le 1/2$), $\varepsilon_{\mathrm{an}} := \min\{\varepsilon_{\mathrm{box}}, \delta_c/4\}$ (the *complex oscillator radius*), $\hat{D}_c := \{I \in \mathbb{C}^a : \mathrm{dist}_\infty(I, \overline{V}) < \delta_c/2\}$. For (μ, I, p, q) with $I \in \hat{D}_c$, $|(p,q)|_\infty < \varepsilon_{\mathrm{an}}$ complex (such points, with $|\mathrm{Im}\,\varphi| < s_*/6$, lie in $\mathcal{X}_{\delta_c}$: $|\mathrm{Re}(p,q)| < \varepsilon_{\mathrm{an}} \le \varepsilon_{\mathrm{box}}$, $|\mathrm{Im}(p,q)| < \delta_c/4$), put

$$\tilde{P}_{\mathrm{av}}(\mu, I; p, q) = (2\pi)^{-a} \int_{[0,2\pi]^a} \tilde{P}(\mu, I, \varphi, p, q)\, d\varphi$$

(real φ; $= P_{\mathrm{av}}$ of [P:bpP:p0:secavg] at real points of the cylinder). The N-norm of $\tilde{P}(\mu,\cdot)$ (Fourier in φ, coefficient radius s_*, sup over (I,p,q) in the complex domain $\hat{D}_c \times B_{\varepsilon_{\mathrm{an}}} \times B_{\varepsilon_{\mathrm{an}}}$) satisfies

$$\Lambda_P := \sup_{\mu \in [0,\mu_0]} |\tilde{P}(\mu,\cdot)|_{,s_*} \le M_P \coth(\delta_c/4)^a < \infty \quad (\ge 1 \text{ after enlarging}).$$

Justification of finiteness. $|\hat{f}_k| \le \sup|f|\, e^{-\sigma_1|k|_1}$ on a strip of width σ_1 (Cauchy in the angle, $\hat{f}_k = (2\pi)^{-a}\int f(\varphi)e^{-ik\cdot\varphi}d\varphi$ shifted to $\mathrm{Im}\,\varphi = \mp\sigma_1 \operatorname{sgn} k$), so $\sum_k |\hat{f}_k| e^{s_*|k|_1} \le \sup|f| \sum_k e^{-(\delta_c - s_*)|k|_1} \le M_P \coth(\delta_c/4)^a$ since $\delta_c - s_* \ge \delta_c/2$. This is the elementary *sup-to-weighted conversion*, used again in Lemma 7.18 (cf. [N:bpN:p4:sup-weighted], [K:bp:p1:decay]). □

Lemma 7.4 (Birkhoff coefficients as derivatives; analytic extension to $\overline{V}$)**.** *Assume (H1), (H2), (H4), (H5), (H6) of* [P:bpP:p0:pg] *((H5) is used only for (c)). For $(\mu, I) \in [0,\mu_0] \times \hat{D}_c$ define, with the holomorphic extension $\tilde{P}_{\mathrm{av}}$ (and $R_j = (p_j^2 + q_j^2)/2$)*

$$\tilde{P}_0 = \tilde{P}_{\mathrm{av}}\big|_{x=0}, \quad \tilde{\Omega}_j = \partial^2_{p_j} \tilde{P}_{\mathrm{av}}\big|_{x=0}, \quad \tilde{\beta}_{ij} = \partial^2_{p_i}\partial^2_{p_j} \tilde{P}_{\mathrm{av}}\big|_{x=0}\ (i \ne j), \quad \tilde{\beta}_{jj} = \tfrac{1}{3}\partial^4_{p_j} \tilde{P}_{\mathrm{av}}\big|_{x=0},$$

$x = (p,q)$, and $\tilde{N} = \tilde{P}_0 + \tilde{\Omega}\cdot R + \frac{1}{2} R \cdot \tilde{\beta} R$, $Q := \tilde{P}_{\mathrm{av}} - \tilde{N}$. Then: (a) $\tilde{P}_0, \tilde{\Omega}, \tilde{\beta}$ are holomorphic in $I \in \hat{D}_c$, continuous in $\mu \in [0,\mu_0]$, $\tilde{\beta}$ is symmetric, and they equal *P_0, Ω, β on $[0,\mu_0] \times V$; (b) for each $\mu \in [0,\mu_0]$ and $I \in \hat{D}_c$, the jet of $Q(\mu, I; \cdot)$ at $x = 0$ vanishes to order 4 (all derivatives of order ≤ 4); (c) $|\det \tilde{\beta}(\mu, I)| \ge B$ for all $(\mu, I) \in [0,\mu_0] \times \overline{V}$.*

Proof. **Step S7.5: Coefficients on** V**.** Fix $\mu \in [0,\mu_0]$, $I \in V$. By [P:bpP:p0:birkhoff](ii), $P_{\mathrm{av}} - N = o(|x|^4)$, and $P_{\mathrm{av}} - N$ is real analytic near $x = 0$ (analytic by Lemma 7.2 and differentiation under the integral; N polynomial). Hence the Taylor polynomial of $P_{\mathrm{av}}(\mu, I; \cdot)$ of degree ≤ 4 equals that of N, which is N. Compute from $R_j = (p_j^2 + q_j^2)/2$: $\partial^2_{p_j} N(0) = \Omega_j$; $R_i R_j = \frac{1}{4}(p_i^2 + q_i^2)(p_j^2 + q_j^2)$ gives $\partial^2_{p_i}\partial^2_{p_j} N(0) = \frac{1}{2}(\beta_{ij} + \beta_{ji}) = \beta_{ij}$ for $i \ne j$ *using* (H6); $\frac{1}{2}\beta_{jj} R_j^2 = \frac{1}{8}\beta_{jj}(p_j^2 + q_j^2)^2$ gives $\partial^4_{p_j} N(0) = 3\beta_{jj}$; $P_0 = P_{\mathrm{av}}|_{x=0}$. So (a) on V.

Step S7.6: Analyticity. The listed derivatives of $\tilde{P}_{\mathrm{av}}$ at $x = 0$ are holomorphic in I (integral of holomorphic functions, uniform bounds by Lemma 7.2), continuous in μ; symmetry of $\tilde{\beta}$ is symmetry of mixed derivatives.

Step S7.7: (b) on V**, then everywhere.** On V, $\tilde{N} = N$ and Step S7.5 give the vanishing jet. Each derivative $\partial_x^\alpha Q(\mu, \cdot; 0)$, $|\alpha| \le 4$, is holomorphic on $\hat{D}_c$, which is connected ($\overline{V}$ is connected as the closure of the connected V (hVc), and $\hat{D}_c$ is a union of balls centered on it) and vanishes on the real open set V; by the identity theorem it vanishes on $\hat{D}_c$.

Step S7.8: (c). On V by (H5) and (a); $\overline{V}$ by continuity of $\det\tilde\beta$ in I and V dense in $\overline{V}$. □

Proposition 7.5 (Uniform remainder: CP10 (60) from the frozen hypotheses)**.** *Under the hypotheses of Lemma 7.4 (that is, (H1), (H2), (H4)–(H6)) there are $r_1 \in (0, \varepsilon_{\mathrm{an}}]$ and $C_Q < \infty$ (depending on the data only) such that for all $\mu \in [0, \mu_0]$, $I \in \hat D_c'$ (:= $\{\mathrm{dist}_\infty(I, \overline{V}) < \delta_c/4\}$) and complex $x = (p, q)$ with $|x|_\infty \le r_1$:*

$$|Q(\mu, I; x)| \le C_Q |x|_\infty^5.$$

In particular $|P_{\mathrm{av}} - \tilde N| \le C_Q |(p,q)|_\infty^5$ on $[0,\mu_0] \times \overline{V} \times \{|x|_\infty \le r_1\}$ (real), where $\tilde N, \tilde P_0, \tilde\Omega, \tilde\beta$ are the constructed extensions *of Lemma 7.4, analytic on $\hat D_c$, with $|\det\tilde\beta| \ge B$ on $\overline{V}$. They equal the original P_0, Ω, β, N on V* only *(the frozen total functions are unconstrained on ∂V and outside, so no statement about their boundary values is made); on V the bound reads $|P_{\mathrm{av}} - N| \le C_Q|(p,q)|_\infty^5$.* Hypothesis ($\mathrm{A}_2''$) with a δ depending on (μ, I, ϵ) is thus upgraded to a uniform $O(|x|^5)$ bound.
Source: CP10 (60), p. 563, asserted "for a suitable $C > 0$" from (A_2''); proof not given.
Source gap: CP10 (60) is asserted without the compactness and analyticity argument; supplied here.

Proof. **Step S7.9: Bound.** Take $r_1 := \varepsilon_{\mathrm{an}}/2$. Q is holomorphic in x on $\{|x|_\infty < \varepsilon_{\mathrm{an}}\}$ and $|Q| \le M_P + |\tilde N| \le M_1$ on $\hat D_c' \times \{|x|_\infty \le r_1\}$, $r_1 = \varepsilon_{\mathrm{an}}/2$, with M_1 from the bounds of Lemma 7.2 and Cauchy estimates for $\tilde\Omega, \tilde\beta$ (derivatives at 0 of a function bounded by M_P on a polydisc of radius $\varepsilon_{\mathrm{an}}$).

Step S7.10: Schwarz. Fix (μ, I) and $\hat x$ with $|\hat x|_\infty = 1$; $g(t) = Q(\mu, I; t\hat x)$ is holomorphic on $|t| < r_1$, $|g| \le M_1$, and by Lemma 7.4(b) g vanishes to order ≥ 5 at 0 (derivatives of g at 0 of order ≤ 4 are combinations of derivatives of Q of order ≤ 4). So $g(t)/t^5$ is holomorphic and bounded by M_1/r_1^5 on the boundary circles, hence on the disc (maximum principle): $|g(t)| \le M_1 |t|^5/r_1^5$. Take $C_Q = M_1 r_1^{-5}$. □

7.3. **The fast frequency map on a complex neighbourhood; the nondegenerate frequency map of the averaged system (domain reduction, CP10 (58)).** The frozen hypothesis HasFastDiffeo(H_0, W) replaces CP10's assumption that $\bar\omega_0 = \partial H_0$ is a diffeomorphism of V_{ρ_0} with bounded inverse Jacobian (58). Write $Z_c := \hat D_c$-type neighbourhoods $X_\rho := \{z \in \mathbb{C}^N : \mathrm{dist}_\infty(z, X) < \rho\}$ for $X \subset \mathbb{R}^N$.

Lemma 7.6 (Lipschitz bound on a complex neighbourhood)**.** *Let $X \subset \mathbb{R}^N$, $\rho > 0$, $\Delta : X_{2\rho} \to \mathbb{C}^m$ holomorphic with* $\sup|\Delta| \le A$. *Then $|\Delta(z) - \Delta(z')| \le \frac{2NA}{\rho}|z - z'|$ for all $z, z' \in X_\rho$.*

Proof. Close pairs $|z - z'| < \rho$: every point w of the segment $[z, z']$ is within $|z - z'|/2 < \rho/2$ of the nearer endpoint, which lies in X_ρ, so $w \in X_{3\rho/2}$ and the polydisc $B(w, \rho/2) \subset X_{2\rho}$; Cauchy ([K:`bp:p1:cauchy-open`]) gives $\|D\Delta(w)\| \le 2NA/\rho$; mean value inequality ([K:`bp:p1:ext-mvt`]). Distant pairs: $|\Delta(z) - \Delta(z')| \le 2A \le (2A/\rho)|z - z'|$.
Source: Ours; the complex-point version of [P:`bpP:p3:lip-sup`].

□

Lemma 7.7 (Complex bi-Lipschitz bound for $\bar\omega_0$)**.** *Under* HasFastDiffeo(H_0, W) *there are $\rho_c \in (0, \delta_c/16)$ and $c_0 > 0$ with $|\tilde\omega_0(z) - \tilde\omega_0(z')| \ge c_0|z - z'|$ for all $z, z' \in (\overline{V})_{\rho_c}$, where $\tilde\omega_0 = \partial\tilde H_0$; hence $\|(\partial\tilde\omega_0(z))^{-1}\| \le c_0^{-1}$ there. Moreover $\bar M_0 := \sup_{(\overline{V})_{2\rho_c}} \|\partial\tilde\omega_0\| < \infty$.*

Proof. **Step S7.11: Real facts.** By [P:`bpP:p0:fastdiffeo`], $g(\bar\omega_0(I)) = I$ on W, so $\bar\omega_0$ is injective on $W \supset \overline{V}$; differentiating (chain rule, $g \in C^1$, $\bar\omega_0$ analytic) $Dg\, D\bar\omega_0 = 1$, so $\partial^2 H_0 = D\bar\omega_0$ is invertible on W.

Step S7.12: Compactness. Suppose not: $z_m \ne z_m' \in (\overline{V})_{1/m}$ with $|\tilde\omega_0(z_m) - \tilde\omega_0(z_m')| < |z_m - z_m'|/m$. Pass to subsequences with $z_m \to x$, $z_m' \to x' \in \overline{V}$ (compact). If $x \ne x'$, continuity

gives $\bar\omega_0(x)=\bar\omega_0(x')$, contradicting injectivity. If $x=x'$: let $m_0=\|(\partial\tilde\omega_0(x))^{-1}\|$; choose δ with $\|\partial\tilde\omega_0(w)-\partial\tilde\omega_0(x)\|\le 1/(2m_0)$ on $B(x,\delta)$ (continuity); for $z,z'\in B(x,\delta)$, $|\tilde\omega_0(z)-\tilde\omega_0(z')|\ge |\partial\tilde\omega_0(x)(z-z')|-\frac{1}{2m_0}|z-z'|\ge\frac{1}{2m_0}|z-z'|$, contradicting the choice for $m>2m_0$. The operator bound: differentiate along $z'\to z$. $\bar M_0<\infty$ by Lemma 7.2. □

Lemma 7.8 (One-scale Diophantine measure on V (CP10 (38))). *Let L_g,R_y,Λ_g be as in* [P:bpP:p4:lip-inverse] *and S_1 as in* [P:bpP:p4:lattice-tail] *(with $n=a+b$, $\tau>n$). For $\bar\gamma>0$ put $\mathcal{D}^1_{\bar\gamma}:=\{y\in\mathbb{R}^a:|k\cdot y|\ge\bar\gamma|k|_1^{-\tau}\ \forall k\in\mathbb{Z}^a\setminus 0\}$ (closed). Then* $\operatorname{meas}\big(V\setminus\bar\omega_0^{-1}(\mathcal{D}^1_{\bar\gamma})\big)\le c_{\rm dio}\bar\gamma$ *with $c_{\rm dio}:=2\Lambda_g^a S_1(2R_y)^{a-1}$.*

Proof. $V\setminus\bar\omega_0^{-1}(\mathcal{D}^1_{\bar\gamma})\subset\bigcup_k g(Y_k)$, $Y_k=\{y\in Y_0:|k\cdot y|<\bar\gamma|k|_1^{-\tau}\}$ (a point $I\in V$ with $\bar\omega_0(I)\notin\mathcal{D}^1_{\bar\gamma}$ has $\bar\omega_0(I)\in Y_k$ for some k and $I=g(\bar\omega_0(I))$, [P:bpP:p4:lip-inverse]; containment suffices). By [P:bpP:p4:affine-slab] (with $c=0$, $N=a$, $\lambda=R_y$, $\ell=k$, $|\ell|\ge1$), $\operatorname{meas}Y_k\le 2\bar\gamma|k|_1^{-\tau}(2R_y)^{a-1}$; g is Λ_g-Lipschitz on Y_0, so [K:bp:p1:lip-image] (after McShane extension [K:bp:p1:mcshane]) gives $\operatorname{meas}g(Y_k)\le\Lambda_g^a\operatorname{meas}Y_k$. Sum over k using $\sum_{k\in\mathbb{Z}^a\setminus0}|k|_1^{-\tau}\le S_1$ (the lattice $\mathbb{Z}^a\subset\mathbb{Z}^n$).
Source: CP10 (38), p. 556, "standard measure estimate"; proof ours.

□

Lemma 7.9 (Constants of the averaged frequency map). *Let $\tilde N$ be as in Lemma 7.4. There are constants $A_N,\Lambda_N,M_\beta,L_\beta\ge1$ (data only) such that on $\mathcal{E}:=(\overline{V})_{\rho_c}\times\{|J_2|_\infty<2\}$, $\tilde N$, $\partial_J\tilde N$, $\partial_J^2\tilde N$ are bounded by A_N; with $G:=\partial_{J_1}\tilde N$ and $\Theta(J_1,\xi):=\tilde\Omega(J_1)+\tilde\beta(J_1)\xi$, for $J_1,J_1'\in(\overline{V})_{\rho_c}$, $|\xi|,|\xi'|\le1$:*

$$|G(J_1,\xi)-G(J_1',\xi)|,\ |\Theta(J_1,\xi)-\Theta(J_1',\xi)|\le\Lambda_N|J_1-J_1'|,\quad |G(J_1,\xi)-G(J_1,\xi')|\le\Lambda_N|\xi-\xi'|,$$

and $|\nu|\le L_\beta|\tilde\beta(J_1)\nu|$ for $J_1\in(\overline{V})_{\rho_c}$, $\nu\in\mathbb{C}^b$ (after decreasing ρ_c). The real-variable constants of [P:bpP:p4:data] *((N1)–(N3) with $\bar D=\bar D_t$, $\lambda_0=1$) hold with the same Λ_N,L_β and for all $\mu\in(0,\mu_0]$. (On $\bar D_t\subset V$ the extensions $\tilde P_0,\tilde\Omega,\tilde\beta$ coincide with the frozen P_0,Ω,β, Lemma 7.4(a), so Part 4 is applied to the original functions only at points of V.)*

Proof. **Step S7.13: Bounds.** $\tilde N$ is a polynomial in J_2 with coefficients $\tilde P_0,\tilde\Omega,\tilde\beta$ holomorphic and bounded on $\hat D_c$ (Lemma 7.4, Cauchy from M_P); derivatives in J_1 up to order 3 are bounded on $(\overline{V})_{2\rho_c}$ for $\rho_c\le\delta_c/16$.

Step S7.14: Lipschitz. In J_1: Lemma 7.6 applied to $\partial_{J_1}\tilde N(\cdot,\xi)$ and $\Theta(\cdot,\xi)$ ($|\xi|\le1$ makes the sup bound uniform). In J_2: polynomial of degree ≤2 with bounded coefficients on the convex set $|\xi|\le1$.

Step S7.15: Inverse of $\tilde\beta$. $M_\beta:=\sup|\tilde\beta_{ij}|$ on $(\overline{V})_{\rho_c}$; $|\det\tilde\beta|\ge B$ on $\overline{V}$ (Lemma 7.4(c)), and det is Lipschitz, so $|\det\tilde\beta|\ge B/2$ on $(\overline{V})_{\rho_c}$ after decreasing ρ_c. Cramer as in [P:bpP:p4:beta-inverse] (the adjugate bound holds for complex matrices): $L_\beta:=2b!M_\beta^{b-1}/B$ (and $\tilde\beta^\top=\tilde\beta$). □

Lemma 7.10 (Nondegeneracy and injectivity of the frequency map on the complex neighbourhood). *For $\mu\in(0,\mu_*]$, where $\mu_*\le1$ is defined by*

$$\mu_*A_N\le\tfrac{c_0}{2},\qquad 4\mu_*A_N^2L_\beta\le c_0,\qquad 2\mu_*\Lambda_N^2L_\beta<c_0,$$

let $\tilde h_\mu(J)=\tilde H_0(J_1)+\mu\tilde N(J)$ on $\mathcal{E}_1:=(\overline{V})_{\rho_c}\times\{|J_2|<1\}$ and $F_\mu=\partial\tilde h_\mu=\big(\tilde\omega_0(J_1)+\mu G(J),\ \mu\Theta(J_1,J_2)\big)$, $U=\partial^2\tilde h_\mu$. Then

(a) F_μ is injective on $\mathcal{E}_1$;

(b) U is invertible on $\mathcal{E}_1$, and with $U=\begin{pmatrix}U_{11}&U_{12}\\U_{21}&U_{22}\end{pmatrix}$ ($a+b$ blocks), $\hat U=(U_{21},U_{22})$, $U^{-1}=\binom{T_1}{T_2}$:

$$\|U\|\le C_U,\quad \|\hat U\|\le C_U\mu,\quad \|U^{-1}\|\le C_U\mu^{-1},\quad \|T_1\|\le C_U,\quad \|T_2\|\le C_U\mu^{-1},$$

with $C_U:=\max\{\bar M_0+A_N+1,\ 4(A_N+1)^2L_\beta c_0^{-1}+4,\ 8L_\beta+4\}$ (depends on data only).

Proof. **Step S7.16: (a).** Let $F_\mu(J) = F_\mu(J')$. First components: $c_0|J_1 - J_1'| \le |\tilde\omega_0(J_1) - \tilde\omega_0(J_1')| = \mu|G(J) - G(J')| \le \mu\Lambda_N(|J_1 - J_1'| + |J_2 - J_2'|)$ (Lemma 7.7, Lemma 7.9, splitting $G(J_1, J_2) - G(J_1', J_2')$ through $G(J_1', J_2)$). Since $\mu\Lambda_N \le c_0/2$ ($\Lambda_N \le A_N$ w.l.o.g.), $|J_1 - J_1'| \le (2\mu\Lambda_N/c_0)|J_2 - J_2'|$. Second components: $\tilde\beta(J_1)(J_2 - J_2') = -[\Theta(J_1, J_2') - \Theta(J_1', J_2')]$, so $|J_2 - J_2'| \le L_\beta\Lambda_N|J_1 - J_1'| \le (2\mu\Lambda_N^2 L_\beta/c_0)|J_2 - J_2'|$, hence $J_2 = J_2'$, then $J_1 = J_1'$.

Step S7.17: (b) Blocks. $U_{11} = \partial\tilde\omega_0 + \mu\tilde N_{11} =: A$, $U_{12} = \mu\tilde N_{12}$, $U_{21} = \mu\tilde N_{21} = U_{12}^\top$, $U_{22} = \mu\tilde\beta$. $\|A^{-1}\| \le 2/c_0$ (Lemma 7.7; Neumann series since $\mu A_N \le c_0/2$). Schur complement $S = \mu(\tilde\beta - \mu\tilde N_{21}A^{-1}\tilde N_{12})$; as $\mu A_N^2 \cdot \frac{2}{c_0}L_\beta \le \frac12$, $\|\tilde\beta^{-1}\| \le L_\beta$ gives $\|S^{-1}\| \le 2L_\beta/\mu$.

Step S7.18: Inverse. Block inversion: $T_{11} = A^{-1} + \mu^2 A^{-1}\tilde N_{12}S^{-1}\tilde N_{21}A^{-1}$, $T_{12} = -\mu A^{-1}\tilde N_{12}S^{-1}$, $T_{21} = -\mu S^{-1}\tilde N_{21}A^{-1}$, $T_{22} = S^{-1}$. Their norms are $\le 2/c_0 + 8\mu A_N^2 L_\beta c_0^{-2}$, $4A_N L_\beta/c_0$, $4A_N L_\beta/c_0$, $2L_\beta/\mu$. Row sums of $T_1 = (T_{11}, T_{12})$ are $O(1)$, of T_2 are $O(\mu^{-1})$, and the entries of $\hat U$ are $\mu\times$ bounded. The constant C_U dominates all of these. □

Remark 7.11. (a) gives injectivity on the *complex* neighbourhood that [P:`bpP:p3:interface`] lists as an obligation of Part 5 ("injectivity/nondegeneracy of ω_μ on a complex neighbourhood"); it uses only HasFastDiffeo (through Lemma 7.7), joint analyticity and (H5). The same computation, with $H_3 = \tilde H(\cdot, 2\pi\cdot)/2\pi$, gives the bounds for Part 3 with every number divided/multiplied by 2π as in [P:`bpP:p3:period-dictionary`]; those factors are absorbed in C_1 of Lemma 7.18.

7.4. The parameters.

Definition 7.12 (Parameters as functions of (r, μ))**.** Let $\tau := n + 1$ and $c := 2\tau + 2$. Fix the data constants of Remark 7.1 (2), $\sigma_{\mathrm{av}} := s_*/6$, $\sigma_{\mathrm{tot}} := s_*/12$, $s := \sigma_{\mathrm{tot}}/5$, $\bar s := 4\sigma_{\mathrm{tot}}/5$ (so $4s = \bar s < 1$ and $\bar s + s = \sigma_{\mathrm{tot}}$), and C_1, κ_0, κ_1, G as in Lemmas 7.18, 7.19. For $r \in (0,1)$ and $\mu \in (0, \mu_0)$ put ($\mu_2 := \Lambda_P\mu$)

$$\bar K := \tfrac{30}{s_*}\log\tfrac1r, \quad t := \sqrt r, \quad \lambda_2 := \tfrac{r^2}{2}, \quad \lambda_1 := \kappa_1 r^{5/2}, \quad \lambda_2' := \lambda_2 - 2\lambda_1, \quad \ell := \lambda_2' - \lambda_1 = \lambda_2 - 3\lambda_1, \quad \rho := \tfrac{\lambda_1}{2},$$

$$\bar\gamma := G\max\Big\{\sqrt{\mu_2}\,\bar K^{\tau+\frac12} r^{-5/2},\ r^{5/2}\bar K^{\tau+1}\Big\}, \quad \alpha_{\mathrm{av}} := \frac{\bar\gamma}{2\bar K^\tau}, \quad \bar\rho := \frac{\bar\gamma}{2\bar M_0 \bar K^{\tau+1}},$$

$$\gamma_1 := \kappa_0 r^{5/2}, \quad \gamma_2 := \mu\kappa_0 r^{5/2}, \quad K := K_0 := \tfrac{6}{s}\log_+(\kappa_0^2/C_1^4),$$

$\hat\rho, \hat E, r_K$ as in [P:`bpP:p3:constants`] with the numerical bounds of Lemma 7.18. The sets: $\bar D_t$ of [P:`bpP:p4:s-collar`], $\check D := \bar D_t \cap \bar\omega_0^{-1}(\mathcal D^1_{\bar\gamma})$ (compact: $\mathcal D^1_{\bar\gamma}$ is closed, $\bar\omega_0$ continuous on $W \supset \bar D_t$), $A_{\mathrm{sh}} := (\lambda_1, \lambda_2')^b$, $D := \check D \times A_{\mathrm{sh}}$. The 2π-dictionary: $\alpha_i := 2\pi\gamma_i$.

Definition 7.13 (Smallness thresholds)**.** $r_* \in (0, \min\{r_0, e^{-1/5}, 1\})$ is so small that for $0 < r < r_*$ (all conditions involve only the constants of Remark 7.1, items 2–4, and C_* of Proposition 7.27):

- (r1) $\log\frac{1}{r_*} \ge \Lambda_P$ and $r_*^2(\log\frac{1}{r_*})^{\tau+1} \le 1$, so that $\mu < r^6/(\log\frac1r)^c$ implies $\mu_2 \le r^6(\log\frac1r)^{-(2\tau+1)}$; and $r_*^6 \le \mu_1 := \min\{\mu_*,\ (2(1+M)a\Lambda_N\Lambda_g)^{-1}\}$, $M = 2b\Lambda_N L_\beta$ ([P:`bpP:p4:data`]);
- (r2) $\sqrt{r_*} < d$, $A\sqrt{r_*} \le \frac12 \operatorname{meas} V$ (so $\operatorname{meas}\bar D_t \ge \frac12\operatorname{meas} V$: positive inner core), $r_* \le r_1/2$, $r_* \le \varepsilon_{\mathrm{an}}/4$;
- (r3) with $C_\gamma := G(30/s_*)^{\tau+1}$: $C_\gamma\sqrt{r_*} \le 2\bar M_0\rho_c$ and $c_{\mathrm{dio}}C_\gamma\sqrt{r_*} \le \frac14\operatorname{meas} V$;
- (r4) $6\kappa_1\sqrt{r_*} \le \frac12$;
- (r5) the displacement margin $\sqrt2\,\Delta_{\max} \le \kappa_1 r^{3/2}$ of Lemma 7.22, and $(C_* + 1)\sqrt{r_*} \le \frac12$.

Lemma 7.14 (Consequences of the parameter choice)**.** *For $0 < r < r_*$, $0 < \mu < \min\{\mu_0, r^6/(\log\frac1r)^c\}$: (a) $\mu_2 \le r^6(\log\frac1r)^{-(2\tau+1)}$; $\mu < \mu_1$; $\bar K s_* = 30\log\frac1r \ge 6$; $\bar\gamma \le C_\gamma\sqrt r$. (b) $\bar\rho \le \rho_c$, $\bar\rho \ge \lambda_1$ if $G \ge 2\bar M_0\kappa_1$ (so $\rho = \lambda_1/2 \le \bar\rho/2$), and $\ell \ge \lambda_2/2 \ge r^2/4$. (c) $\operatorname{meas}\bar D_t \ge \frac12\operatorname{meas} V$, and $\operatorname{meas}\check D \ge \frac14\operatorname{meas} V$, $\operatorname{meas}\bar D_t \le 4\operatorname{meas}\check D$.*

Proof. (a) $\mu < r^6(\log\frac{1}{r})^{-c}$ and $\Lambda_P \le \log\frac{1}{r}$ give the first; $\mu_2\bar K^{2\tau+1} \le r^6(30/s_*)^{2\tau+1}$, so $\sqrt{\mu_2}\bar K^{\tau+1/2}r^{-5/2} \le (30/s_*)^{\tau+1/2}\sqrt{r}$, and $r^{5/2}\bar K^{\tau+1} \le (30/s_*)^{\tau+1}r^{1/2}(r^2\log^{\tau+1}\frac{1}{r})$, both $\le (30/s_*)^{\tau+1}\sqrt{r}$ by (r1). (b) $\bar\rho \le \bar\gamma/(2\bar M_0) \le C_\gamma\sqrt{r}/(2\bar M_0) \le \rho_c$ by (r3); $\bar\rho \ge Gr^{5/2}/(2\bar M_0) \ge \lambda_1$ from the second branch of $\bar\gamma$; $\ell = \lambda_2(1-6\kappa_1\sqrt{r}) \ge \lambda_2/2$ by (r4). (c) [P:`bpP:p4:collar`](b): $\operatorname{meas}\bar D_t \ge \operatorname{meas}V - At \ge \frac12\operatorname{meas}V$ (r2); $\operatorname{meas}(\bar D_t\setminus\check D) \le \operatorname{meas}(V\setminus\bar\omega_0^{-1}\mathcal D^1_{\bar\gamma}) \le c_{\rm dio}\bar\gamma \le \frac14\operatorname{meas}V$ (Lemma 7.8, (r3)); here $\frac12-\frac14=\frac14$. □

7.5. **Step 1 of CP10 §4: averaging (Part 2).**

Proposition 7.15 (Averaged system in CP variables)**.** *For r,μ as above there is a map $\bar\Psi$ on $\mathcal W := \{(I,\varphi,p,q)\in\mathbb C^{2n} : I\in\check D_{\bar\rho/2},\ |\operatorname{Im}\varphi|<\sigma_{\rm av},\ |(p,q)|_\infty<\varepsilon_{\rm an}/2\}$ (angles of period 2π), holomorphic, injective, real at real points, with $\bar\Psi-\mathrm{id}$ 2π-periodic in φ, mapping $\mathcal W$ into the domain of $\tilde P$ of Lemma 7.2, with $D\bar\Psi^\top M_0 D\bar\Psi = M_0$ (it preserves the frozen Ω; this is the symplectic conclusion of* [P:`bpP:p2:osc-averaging`] *in the frozen order; the condition is unchanged if the form is multiplied by -1), and*

$$H_\mu\circ\bar\Psi = \tilde H_0(I) + \mu\tilde N(I,R) + \mu Q(\mu,I;p,q) + \widetilde P_{\rm rem},\qquad |\widetilde P_{\rm rem}|_{,\sigma_{\rm av}} \le 2\Lambda_P\mu r^5 \quad (\text{on } \mathcal W),$$

and, writing $(\bar I,\bar\varphi,\bar p,\bar q)\mapsto(I,\varphi,p,q)$: $|I-\bar I|\le\Delta_I := \dfrac{12\mu_2\bar K^\tau}{s_*\bar\gamma}$, $|p-\bar p|,|q-\bar q|\le\Delta_{\max} := \dfrac{18\mu_2\bar K^\tau}{\varepsilon_{\rm an}\bar\gamma}$, where, by the choice of $\bar\gamma$, $\Delta_{\max}\le\dfrac{18}{G\varepsilon_{\rm an}}r^{11/2}$, $\Delta_I\le\dfrac{12}{Gs_*}r^{11/2}$.

Proof. **Step S7.19: Dictionary.** Apply [P:`bpP:p2:averaged-remainder`] with $n=a$, $m=b$, $\bar D=\check D$, $\bar\rho=\bar\rho_{\rm CP}$, $\epsilon=r$, $\mu\to\mu_2$, $P\to\tilde P/\Lambda_P$ (so $|\cdot|,\le1$ by Definition 7.3, and $H_\mu = H_0+\mu_2(\tilde P/\Lambda_P)$), in CP variables through the canonical chart $(\theta,I,p,q)=(\varphi/2\pi,2\pi I_{\rm CP},p,q)$ of [P:`bpP:p2:use5`] (3): $s_0=s_*/2\pi$, $\bar\alpha=2\pi\alpha=\alpha_{\rm av}$, $d=\min\{\bar\rho s_*,\varepsilon_{\rm an}^2\}=\bar\rho s_*$ ($\bar\rho\le\rho_c<\varepsilon_{\rm an}^2/s_*$ after decreasing ρ_c), $\kappa=\bar K s_*=30\log\frac1r$ and $e^{-\kappa/6}=r^5$.

Step S7.20: (R1). Let $I'\in\check D_{\bar\rho}$, $I\in\check D$ with $|I'-I|<\bar\rho$ and $0<|k|_1\le\bar K$. As $\bar\omega_0(I)\in\mathcal D^1_{\bar\gamma}$, $|k\cdot\bar\omega_0(I)|\ge\bar\gamma\bar K^{-\tau}$; the mean value inequality on the segment $[I,I']\subset(\overline V)_{\rho_c}$ gives $|k\cdot(\tilde\omega_0(I')-\bar\omega_0(I))|\le|k|_1\bar M_0\bar\rho\le\bar K\bar M_0\bar\rho=\bar\gamma\bar K^{-\tau}/2$. So $|k\cdot\tilde\omega_0(I')|\ge\bar\gamma/(2\bar K^\tau)=\alpha_{\rm av}$ (CP10 (39)).

Step S7.21: (R2) and the remainder. $\alpha_{\rm av}d=\bar\gamma^2s_*/(4\bar M_0\bar K^{2\tau+1})\ge G^2s_*\mu_2r^{-5}/(4\bar M_0)$ (first branch of $\bar\gamma$). (R2): $2^{12}(1+b)\bar K s_*\mu_2\le\alpha_{\rm av}d$ holds if $G^2\ge2^{14}(1+b)\bar M_0$ (as $\bar Kr^5\le1$); and $400(1+b)\mu_2/(\alpha_{\rm av}d)\le r^5$ holds if $G^2\ge1600(1+b)\bar M_0/s_*$. Both are conditions on G (Lemma 7.19). Then $|\widetilde P_{\rm rem}|,\le\mu_2r^5+\mu_2r^5=2\Lambda_P\mu r^5$ ([P:`bpP:p2:averaged-remainder`]).

Step S7.22: Identification. In [P:`bpP:p2:averaged-remainder`], $\mu_2\langle P_2\rangle=\mu\tilde P_{\rm av}=\mu\tilde N+\mu Q$ (Lemma 7.4), which gives the displayed identity. **Displacements:** [P:`bpP:p2:use5`] (3) converts [P:`bpP:p2:averaged-remainder`]'s bounds to $\alpha_{\rm av}s_*|\Delta I|\le6\mu_2$, $\alpha_{\rm av}\varepsilon_{\rm an}|\Delta p|,|\Delta q|\le9\mu_2$; insert $\alpha_{\rm av}=\bar\gamma/2\bar K^\tau$ and $\bar\gamma\ge G\sqrt{\mu_2}\bar K^{\tau+1/2}r^{-5/2}$: $\mu_2\bar K^\tau/\bar\gamma\le\sqrt{\mu_2}\,r^{5/2}\bar K^{-1/2}/G\le r^{11/2}/G$. □

7.6. **The KAM Hamiltonian in oscillator action–angle variables.**

Definition 7.16 (Polar chart, composite chart Φ)**.** $\operatorname{Pol}(\psi,J)=\big((\varphi,q),(I,p)\big)$ with $\varphi=\psi_{\rm f}$, $I=J_1$, $(q_j,p_j)=\sqrt{2J_{2,j}}(\sin\psi_{{\rm s},j},\cos\psi_{{\rm s},j})$ ($J_2\in(0,\infty)^b$): the map Π of [P:`bpP:p4:s-disc`], in the frozen order (position, momentum); it extends holomorphically to $\{\operatorname{Re}J_2>0\}\times\mathbb C^n$ (principal square root). Let $O_0:=\{(\psi,J)\in\operatorname{Ph}(n) : J_1\in\operatorname{Re}(\check D_{\bar\rho/2}),\ J_2\in(\lambda_1/2,2\lambda_2)^b\}$ (open, invariant under $\psi\mapsto\psi+2\pi k$) and $\Phi:=\bar\Psi\circ\operatorname{Pol}$ (real restriction on O_0, any extension elsewhere), $\Phi:O_0\to\operatorname{Ph}(n)$.

Lemma 7.17 (Φ is a 1-chart with the lattice properties of [P:`bpP:p1:cyl-pipeline`])**.** *Φ is a symplectic chart on O_0 (Definition 2.3, $c=1$), $O_0+(2\pi k,0)=O_0$, $\Phi(z+(2\pi k,0))=\Phi(z)+\operatorname{shift}(k_{\rm f})$*

for $k \in \mathbb{Z}^n$, $z \in O_0$, and $\Phi(z') = \Phi(z) + \mathrm{shift}(k_\mathrm{f}) \Rightarrow z' = z + (2\pi m, 0)$ for some $m \in \mathbb{Z}^n$. Moreover $H_\mu \circ \Phi = h_\mu + f_\mu$ on O_0, where, with $N(J) = \tilde{N}(J_1, J_2)$,

$$h_\mu(J) = \tilde{H}_0(J_1) + \mu\tilde{N}(J), \qquad f_\mu = \big(\mu Q + \widetilde{P}_\mathrm{rem}\big) \circ \mathrm{Pol}.$$

Proof. **Step S7.23: Image.** For $z \in O_0$: $|\bar{p}|, |\bar{q}| \le \sqrt{2J_2} < 2\sqrt{\lambda_2} \le 2r < \varepsilon_\mathrm{an}/2$ and $I \in \check{D}_{\bar{\rho}/2}$: $\mathrm{Pol}(O_0) \subset \mathcal{W} \cap \mathbb{R}^{2n}$, so $\bar{\Psi}$ is defined and holomorphic there. $\bar{\Psi}$ real analytic and symplectic: [P:`bpP:p1:hol-chart`] (with Proposition 7.15); Pol is a 1-chart on $\{J_2 > 0\}$: [P:`bpP:p1:typical`](1) for each oscillator and the identity on (φ, I); composition: [P:`bpP:p1:typical`](3).

Step S7.24: Lattice. $\mathrm{Pol}(\psi + 2\pi k, J) = \mathrm{Pol}(\psi, J) + \mathrm{shift}(k_\mathrm{f})$ (ψ_s enters through 2π-periodic functions; shift moves only the fast angles) and $\bar{\Psi}(w + \mathrm{shift}(k_\mathrm{f})) = \bar{\Psi}(w) + \mathrm{shift}(k_\mathrm{f})$ (Proposition 7.15: $\bar{\Psi} - \mathrm{id}$ periodic). Injectivity modulo the lattice: if $\Phi(z') = \Phi(z) + \mathrm{shift}(k_\mathrm{f}) = \Phi(z + (2\pi(k_\mathrm{f}, 0), 0))$, injectivity of $\bar{\Psi}$ gives $\mathrm{Pol}(z') = \mathrm{Pol}(z + (2\pi(k_\mathrm{f}, 0), 0))$, and Pol is injective modulo $\psi \mapsto \psi + 2\pi k$ on $\{J_2 > 0\}$ ([P:`bpP:p4:disc-measure`](a)).

Step S7.25: Hamiltonian. $H_\mu \circ \bar{\Psi}$ is given by Proposition 7.15; compose with Pol, $R_j \circ \mathrm{Pol} = J_{2,j}$ ([P:`bpP:p0:oscaction`]). □

Lemma 7.18 (Hypotheses of the two-scale KAM theorem)**.** *Let $H_3(J, \theta) := \tilde{H}(J, 2\pi\theta)/2\pi$ where $\tilde{H} = h_\mu + f_\mu$ in (J, ψ), i.e. $h_3 = h_\mu/2\pi$, $f_3 = f_\mu(\cdot, 2\pi\theta)/2\pi$. There is $C_1 \ge 1$, depending on the data only (Remark 7.1, items 2–3), such that for $0 < r < r_*$, $0 < \mu < \min\{\mu_0, r^6/(\log \frac{1}{r})^c\}$, the hypotheses of* [P:`bpP:p3:two-scale-kam`] *hold with: $n_1 = a$, $n_2 = b$, $\tau = n + 1$, $D = \check{D} \times A_\mathrm{sh}$, $\rho = \lambda_1/2$, $A = D_\rho$, $s, \bar{s}$ of Definition 7.12, and*

$$M = C_1, \quad \hat{M} = C_1\mu, \quad \bar{M} = \bar{M}_2 = C_1\mu^{-1}, \quad \bar{M}_1 = C_1, \quad E = C_1\mu r^5 \qquad \text{(the values of [P:bpP:p3:interface] (2}$$

Explicitly $C_1 := \max\{1,\ 2\pi C_U,\ C_R \coth(\sigma_\mathrm{av}/4)^n/2\pi\}$ with $C_R := 2^5 C_Q + 2\Lambda_P$ and C_U of Lemma 7.10.

Proof. **Step S7.26: Domain.** $D_\rho = \check{D}_\rho \times (A_\mathrm{sh})_\rho$ (max norm) $\subset \mathcal{E}_1$, since $\rho \le \bar{\rho}/2 \le \rho_c$ and $|J_2| < \lambda_2' + \rho < 1$; also $\mathrm{Re}\, J_2 > \lambda_1/2$. The polar image of $D_\rho \times \{|\mathrm{Im}\,\psi| < \sigma_\mathrm{av}\}$: $|(\bar{p}, \bar{q})|_\infty \le \sqrt{2|J_2|}\, e^{\sigma_\mathrm{av}} < 2r \le \varepsilon_\mathrm{an}/2$ (as $|J_2| < \lambda_2$, $\sigma_\mathrm{av} \le \frac{1}{12}$), $|\mathrm{Im}\,\varphi| < \sigma_\mathrm{av}$, $J_1 \in \check{D}_{\bar{\rho}/2}$: inside $\mathcal{W}$. So f_μ is holomorphic on $D_\rho \times \{|\mathrm{Im}\,\psi| < \sigma_\mathrm{av}\}$, 2π-periodic, real on real points; h_μ is holomorphic on D_ρ.

Step S7.27: h. Lemma 7.10: $\partial h_3 = F_\mu/2\pi$ injective, $U_3 = U/2\pi$ nonsingular on A, $\|U_3\| \le C_U$, $\|\hat{U}_3\| \le C_U\mu$, $\|U_3^{-1}\| \le 2\pi C_U\mu^{-1}$, $\|T_1\| \le 2\pi C_U$, $\|T_2\| \le 2\pi C_U/\mu$ (the factors 2π from $U_3^{-1} = 2\pi U^{-1}$). $\mu < \mu_*$ by Lemma 7.14(a).

Step S7.28: f. On the polar image $|x|_\infty < 2r \le r_1$: $|\mu Q| \le \mu C_Q (2r)^5$ (Proposition 7.5; $I \in \hat{D}_c'$), $|\widetilde{P}_\mathrm{rem}| \le |\widetilde{P}_\mathrm{rem}|_* \le 2\Lambda_P \mu r^5$ ([N:`bpN:p4:sup-weighted`]); so $\sup|f_\mu| \le C_R\mu r^5$ on the strip σ_av. Sup-to-weighted conversion (proof of Definition 7.3, all n angles, loss $\sigma_\mathrm{av} - \sigma_\mathrm{tot} = \sigma_\mathrm{av}/2$): $\|f_3\|_{\rho,\sigma_\mathrm{tot}} \le C_R \coth(\sigma_\mathrm{av}/4)^n \mu r^5/2\pi \le C_1\mu r^5 = E$.

Step S7.29: Normalizations. $\hat{M} = C_1\mu \le M = C_1$, $\bar{M}_i \le \bar{M}$ since $\mu \le 1$. All numerical constants are upper bounds as required ([P:`bpP:p3:rem-norm`]). □

7.7. The scale constants, the KAM condition, and r_K (replacing [P:`bpP:p4:use5`]).

Lemma 7.19 (Scale constants and the KAM condition)**.** *Let $\hat{c} = \hat{c}(n, \tau) = \max\{2^{13}n, 24^{\tau+1}/3\}$ (*[P:`bpP:p3:constants`]*) and C_1, s as above. There are constants, chosen* in this order*:*

(1) $\kappa_0 \ge C_1^2$ with $\kappa_0^2 > 9\hat{c}C_1^4 K_0^{2\tau+2}$, $K_0 = \frac{6}{s}\log_+(\kappa_0^2/C_1^4)$ (exists: $\kappa_0^2/(\log\kappa_0)^{2\tau+2} \to \infty$);
(2) κ_1 with $\kappa_1^2 > 4\hat{c}C_1^2$;
(3) $G \ge \max\{2\bar{M}_0\kappa_1,\ (2^{14}(1+b)\bar{M}_0)^{1/2},\ (1600(1+b)\bar{M}_0/s_*)^{1/2}\}$.

With them, for all $r \in (0,1)$ small enough that $\rho = \lambda_1/2$ makes sense and all $\mu \in (0,1]$ (Part 3's formulas, with $E = C_1\mu r^5$ etc. of Lemma 7.18): $L = C_1/\mu$, $x = C_1^4 r^5/\gamma_1^2 = C_1^4/\kappa_0^2$, $K = K_0$ independent of r, μ,

$$\hat\rho = \min\Big\{\frac{\kappa_0 r^{5/2}}{3C_1 K_0^{\tau+1}},\ \frac{\kappa_1 r^{5/2}}{2}\Big\}, \qquad \hat E = \frac{EL}{\hat\rho^2} = \frac{C_1^2 r^5}{\hat\rho^2} = \max\Big\{\frac{9C_1^4 K_0^{2\tau+2}}{\kappa_0^2}, \frac{4C_1^2}{\kappa_1^2}\Big\} < \frac{1}{\hat c},$$

*that is, the **KAM condition** $\hat c\hat E < 1$ holds, $\gamma_1 \ge \gamma_2$, and $\tau > n$.*

Proof. **Step S7.30: Evaluation.** $L = \max\{\bar M, M^{-1}, \hat M^{-1}\} = \max\{C_1/\mu, 1/C_1, 1/(C_1\mu)\} = C_1/\mu$ ($C_1 \ge 1$, $\mu \le 1$). $EM^2L = C_1^4 r^5$, so $x = C_1^4 r^5/\gamma_1^2 = C_1^4/\kappa_0^2 \le 1$ and $K = \frac{6}{s}\log_+(x^{-1}) = K_0$ exactly ([P:bpP:p3:interface] (2)). Since $\gamma_2/(3\hat M K^{\tau+1}) = \mu\kappa_0 r^{5/2}/(3C_1\mu K_0^{\tau+1}) = \gamma_1/(3MK^{\tau+1})$, $\hat\rho$ is as displayed; $\hat E = EL/\hat\rho^2 = C_1^2 r^5/\hat\rho^2$; $\hat\rho^{-2} r^5 = \max\{(3C_1 K_0^{\tau+1}/\kappa_0)^2, 4/\kappa_1^2\}$.

Step S7.31: Smallness. Items 1, 2 give $\hat c\hat E < 1$. $\gamma_2 = \mu\gamma_1 \le \gamma_1$. □

Lemma 7.20 (The radius r_K (what Part 4 should state instead of $r_K \le C\hat\rho \le C\gamma_2$)). *In the situation of Lemma 7.19, $r_K := 32\hat E\hat\rho$ (the enlargement radius of [P:bpP:p3:two-scale-kam]) satisfies, for all admissible (r,μ),*

$$r_K < \tfrac{1}{256n}\hat\rho \le \tfrac{1}{512n}\lambda_1, \qquad r_K < \min\{t, \lambda_1\}, \qquad r_K < \tfrac{\bar\rho}{2}, \qquad \frac{2r_K}{\ell} \le \frac{\kappa_1}{64n}\sqrt{r}, \qquad \Big(1 + \frac{2r_K}{\ell}\Big)^b < 2.$$

Proof. **Step S7.32: Direct.** $\hat c \ge 2^{13} n$ and $\hat c\hat E < 1$ give $\hat E < 2^{-13}/n$, so $r_K = 32\hat E\hat\rho < \hat\rho/(256n)$, and $\hat\rho \le \rho = \lambda_1/2$. Further $\lambda_1 = \kappa_1 r^{5/2} < \sqrt{r} = t$ for $r < 1$ ($\kappa_1 r^2 < 1$ by (r4)), and $\lambda_1 \le \bar\rho$ (Lemma 7.14(b)). Finally $\ell \ge r^2/4$ gives $2r_K/\ell \le 2\kappa_1 r^{5/2}\cdot 4/(512n\, r^2) = \kappa_1\sqrt{r}/(64n)$, and $(1+x)^b \le e^{bx}$ with $x = 2r_K/\ell$, $b \le n$, $\kappa_1\sqrt{r} \le \frac{1}{12}$ (r4): $bx \le 1/768$, so $(1+x)^b < 2$. □

Remark 7.21 (Interface with Part 4 (reviews P4-01, P4-02)). Part 4's [P:bpP:p4:use5] and [P:bpP:p4:domain-collar] have been corrected in the meantime (chain $r_K \le C\hat\rho \le C\min\{\gamma_1, \gamma_2/\mu\}$; unnormalised collar bound with a positive-volume premise). Part 5 does not use the $\bar D_t$-version of [P:bpP:p4:domain-collar] as a black box: for the Cantor-type domain $\check D \times A_{\mathrm{sh}}$ it re-derives the corresponding bound in Lemma 7.26, Step S7.47 (the same Steps S6.16–S6.17, with the extra piece $\bar D_t \setminus \check D$), and gets r_K from Lemma 7.20. Positive volume is ensured by $A\sqrt{r} \le \frac{1}{2}\operatorname{meas} V$ (Lemma 7.14(c)).

7.8. Location, the tori, and the Kolmogorov set.

Lemma 7.22 (Location of the tori in the cylinder). *Let $\varphi_\omega = (v, \vartheta + u)$ be a torus of [P:bpP:p3:two-scale-kam] (so $v \in \mathrm{Re}(D_{r_K})$), and let $z = \Phi(2\pi(\vartheta+u), v) \in \mathrm{Ph}(n)$. Then $v \in O_0$-range ($v_1 \in \mathrm{Re}(\check D_{\bar\rho/2})$, $v_2 \in (\lambda_1/2, 2\lambda_2)^b$), $I(z) \in V$ and $p_j^2 + q_j^2 < r^2$ for all j; thus $z \in \mathcal{C}(V,r)$. Here $\Delta_{\max}$ is that of Proposition 7.15, and (r5) is the condition $\sqrt{2}\Delta_{\max} \le \kappa_1 r^{3/2}$ (satisfied for small r since $\Delta_{\max} \le \frac{18}{G\varepsilon_{\mathrm{an}}} r^{11/2}$).*

Proof. **Step S7.33: Range.** $v \in \mathrm{Re}(D_{r_K})$ means $\operatorname{dist}(v_1, \check D) < r_K < \bar\rho/2$ (Lemma 7.20) and $v_2 \in (\lambda_1 - r_K, \lambda_2' + r_K)^b \subset (\lambda_1/2, 2\lambda_2)^b$.

Step S7.34: Actions. $I = \bar I + (I - \bar I)$ with $\bar I = v_1$, $\operatorname{dist}(\bar I, \bar D_t) < r_K$ and $|I - \bar I| \le \Delta_I$ (Proposition 7.15); $r_K + \Delta_I < t$, hence $I \in N_{r_K+\Delta_I}(\bar D_t) \subset V$ by [P:bpP:p4:collar](c).

Step S7.35: Discs. For the polar point $(\bar p, \bar q)$: $\bar p_j^2 + \bar q_j^2 = 2v_{2,j} < 2(\lambda_2' + r_K) = r^2 - 4\lambda_1 + 2r_K \le r^2 - 3\lambda_1$ (Lemma 7.20), so $\sqrt{\bar p_j^2 + \bar q_j^2} < r - \frac{3\lambda_1}{2r} = r - \frac{3}{2}\kappa_1 r^{3/2}$. With $|p - \bar p|, |q - \bar q| \le \Delta_{\max}$: $\sqrt{p_j^2 + q_j^2} \le \sqrt{\bar p_j^2 + \bar q_j^2} + \sqrt{2}\Delta_{\max} < r - \frac{3}{2}\kappa_1 r^{3/2} + \kappa_1 r^{3/2} < r$. □

Definition 7.23 (Torus family and Kolmogorov set)**.** Let $\Omega_* := \omega_0(D) \cap \mathcal{D}_{\gamma_1,\gamma_2,\tau}$ ($\omega_0 = \partial h_3 = F_\mu/2\pi$), $D_0 := D^{\mathrm{dio}} = \omega_0^{-1}(\Omega_*) \cap D$, and for each Borel $B \subset D_0$ with $\Omega_B := \omega_0(B)$ and φ_ω as in [P:bpP:p3:two-scale-kam],

$$K_\omega(\theta) := \Phi\big(S_{2\pi}\,\pi\,\varphi_\omega(\theta)\big) \quad (\omega \in \Omega_*), \qquad S_B := \bigcup_{\omega\in\Omega_B} K_\omega(\mathbb{R}^n), \qquad \mathsf{K}_B := \bigcup_{\omega\in\Omega_B} \varphi_\omega(\mathbb{R}^n) \subset \mathbb{R}^n\times\mathbb{R}^n \text{ (coordinates } (J,$$

$(\pi(J,\vartheta) = (\vartheta, J)$, $S_{2\pi}$ of [P:bpP:p1:period-convert].) D_0 is relatively closed in D ($\mathcal{D}_{\gamma_1,\gamma_2,\tau}$ closed, F_μ continuous), $D = \check{D} \times A_{\mathrm{sh}}$ is σ-compact, so D_0 is σ-compact.

Proposition 7.24 (Each torus is a frozen cylinder torus with frozen frequencies)**.** *For $0 < r < r_*$ and admissible μ, $\omega \in \Omega_*$, $w := \omega$:* $\mathrm{IsCylTorus}(H_\mu, \mathcal{C}(V,r), w, K_\omega)$*; w is $(\min\{\gamma_1,\gamma_2\},\tau)$-Diophantine with $\tau = n+1 > a+b$; $|w^{\mathrm{f}}|_\infty \le C_w$, $|w^{\mathrm{s}}|_\infty \le C_w\mu$ with $C_w := \frac{M_\omega + A_N}{2\pi} + 1$, $M_\omega := \sup_{\overline{V}}|\bar\omega_0|$; $K_\omega(\mathbb{R}^n) \subset \mathcal{C}(V,r)$.*

Proof. **Step S7.36: Chart data.** Lemma 7.17 supplies O_0, Φ with the hypotheses of [P:bpP:p1:cyl-pipeline] ($d = n$); $G_3 = (2\pi)^{-1}(H_\mu \circ \Phi)(2\pi z_1, z_2)$, which by Lemma 7.17 equals H_3 in the frozen order π on $O = S_{2\pi}^{-1}(O_0)$.

Step S7.37: $\mathrm{LT}(G_3, O, w, K_3)$**,** $K_3 = \pi \circ \varphi_\omega$**.** (L1): φ_ω is real analytic ([P:bpP:p3:two-scale-kam](a)); $K_3(\mathbb{R}^n) \subset O$ by Lemma 7.22, Step S7.33; G_3 is differentiable at $K_3(\theta)$ (chart-field Lemma 2.4: H_μ is analytic near $\Phi(\dots) \in \mathcal{C}(V,r)$ since $r < r_0$, (H1), H_0 analytic on W). (L2): v, u are 1-periodic. (L3): injectivity modulo translations (Proposition 4.3, with equal ω). (L4), (L5): Proposition 4.3(c); the displayed vanishing sum is $\Omega(DK_3\,\cdot, DK_3\,\cdot)$ up to sign. (L6): Proposition 4.3(b), with π ([P:bpP:p1:order]), where $H_3 = G_3$ on the open set $\mathrm{Re}(D_\rho) \times \mathbb{R}^n \ni K_3(\theta)$ ($r_K < \rho$) so the fields agree.

Step S7.38: Pipeline. [P:bpP:p1:cyl-pipeline] with $U = \mathcal{C}(V,r) \supset \Psi(K_3(\mathbb{R}^n)) = K_\omega(\mathbb{R}^n)$ (Lemma 7.22) gives $\mathrm{IsCylTorus}(H_\mu, \mathcal{C}(V,r), w, K_\omega)$ with the same $w = \omega$.

Step S7.39: Frequencies. $\omega = \omega_0(I_0)$, $I_0 = (J_1, J_2) \in D$ real, $|J_2| \le \lambda_2 \le 1$: $w^{\mathrm{f}} = (\bar\omega_0(J_1) + \mu G)/2\pi$, $w^{\mathrm{s}} = \mu\Theta/2\pi$, bounded using $|G|, |\Theta| \le A_N$. Diophantine: [P:bpP:p1:dio-sources](a) ($\gamma_1, \gamma_2 > 0$, $\tau > n$). □

Lemma 7.25 (Borelness, whole tori, cell measure)**.** *Let $B \subset D_0$ be σ-compact and $\mathsf{A}_B := \mathsf{K}_B \cap (\mathbb{R}^n \times [0,1)^n)$ (coordinates (J,ϑ); the* measured representative*, an* output*-angle cell, not the image of the parameter cell). (a) A_B is σ-compact; S_B is Borel, $S_B \subset \mathcal{C}(V,r)$, and every point of S_B lies on some $K_\omega(\mathbb{R}^n) \subset S_B$. (b)* $\mathrm{meas}(S_B \cap \mathcal{F}(V,r)) = (2\pi)^n \,\mathrm{meas}\,\mathsf{A}_B$.

Proof. **Step S7.40: σ-compactness.** Let $\mathsf{K}_B^{\mathrm{c}} := \Phi_3(\check\imath(B)\times[0,1)^n)$ (notation of Part 3), the image of the *parameter* cell. $\check\imath$ and Φ_3 are continuous ([P:bpP:p3:limit-actions], [P:bpP:p3:limit-phi]) and $B \times [0,1)^n$ is σ-compact, so $\mathsf{K}_B^{\mathrm{c}}$ is σ-compact. Since $\varphi_\omega(\vartheta + k) = \varphi_\omega(\vartheta) + (0,k)$, $\mathsf{K}_B = \bigcup_{k\in\mathbb{Z}^n}(\mathsf{K}_B^{\mathrm{c}} + (0,k))$ is σ-compact and invariant under $(0,k)$. A_B is the intersection of a σ-compact set with an F_σ set, hence σ-compact. (The output angles of $\mathsf{K}_B^{\mathrm{c}}$ are $\vartheta + u$ and need not lie in $[0,1)^n$, so $\mathsf{K}_B^{\mathrm{c}} \ne \mathsf{A}_B$ in general; only A_B is measured below.)

Step S7.41: Decomposition. Put $T := \Phi(S_{2\pi}\pi\,\mathsf{A}_B)$; $S_{2\pi}\pi\mathsf{A}_B \subset Q'' := \{\psi \in [0,2\pi)^n\} \cap O_0$ (Lemma 7.22), σ-compact, and Φ is continuous on O_0, so T is σ-compact. If $(J,\vartheta) = \varphi_\omega(\theta) \in \mathsf{K}_B$ write $\vartheta = \vartheta' + m$, $\vartheta' \in [0,1)^n$, $m \in \mathbb{Z}^n$: $(J,\vartheta') \in \mathsf{A}_B$ and, by Lemma 7.17, $K_\omega(\theta) = \Phi(2\pi\vartheta' + 2\pi m, J) = \Phi(2\pi\vartheta', J) + \mathrm{shift}(m_{\mathrm{f}}) \in T + \mathrm{shift}(m_{\mathrm{f}})$. Conversely every point of T is some $K_\omega(\theta)$, and S_B is invariant under $\mathrm{shift}(k)$ ($K_\omega(\theta+k) = K_\omega(\theta) + \mathrm{shift}(k_{\mathrm{f}})$). So $S_B = \bigcup_{k\in\mathbb{Z}^a}(T + \mathrm{shift}(k))$, Borel; (a) follows with Lemma 7.22.

Step S7.42: Disjointness and injectivity. If $z, z' \in Q''$ and $\Phi(z) + \mathrm{shift}(k) = \Phi(z') + \mathrm{shift}(k')$, then $\Phi(z') = \Phi(z) + \mathrm{shift}(k - k')$ and (Q2, Lemma 7.17) $z' = z + (2\pi m, 0)$ with $m \in \mathbb{Z}^n$; both

lie in $[0,2\pi)^n$, so $m=0$, $z'=z$ and $k=k'$. Thus Φ is injective on $S_{2\pi}\pi\mathsf{A}_B$ and the translates $T+\mathrm{shift}(k)$, $k\in\mathbb{Z}^a$, are pairwise disjoint.

Step S7.43: Counting. $S_B\subset\mathcal{C}(V,r)$, so $S_B\cap\mathcal{F}=S_B\cap\{\varphi\in[0,2\pi)^a\}$ and by translation invariance of meas, $\mathrm{meas}(S_B\cap\{\varphi\in[0,2\pi)^a\})=\sum_k\mathrm{meas}\big((T+\mathrm{shift}(k))\cap\{\varphi\in[0,2\pi)^a\}\big)=\sum_k\mathrm{meas}\big(T\cap\{\varphi\in[-2\pi k,2\pi(1-k))\}\big)=\mathrm{meas}\,T$ (the slabs partition $\mathbb{R}^a$; the averaging map moves the fast angles, but no physical cell is required to be preserved).

Step S7.44: Liouville. Φ is a 1-chart, injective on the measurable set $S_{2\pi}\pi\mathsf{A}_B$ (Step S7.42): [P:`bpP:p1:liouville`] gives $\mathrm{meas}\,T=\mathrm{meas}(S_{2\pi}\pi\mathsf{A}_B)=(2\pi)^n\,\mathrm{meas}\,\mathsf{A}_B$ ($S_{2\pi}$ multiplies volume by $(2\pi)^n$, [P:`bpP:p1:period-convert`](a); π permutes coordinates). □

7.9. **The measure budget (replacing [P:`bpP:p4:budget`] and [P:`bpP:p4:cell-budget`] for $D=\check{D}\times A_{\mathrm{sh}}$).**

Lemma 7.26 (Budget on the action side)**.** *There is $C_{\mathrm{b}}<\infty$ (data and the constants κ_0,κ_1,G only) such that for admissible (r,μ), with $\mathsf{K}=\mathsf{K}_{D_0}$ and $c_n=1+(1+2^n)(1+(4/3)^n)$,*

$$\mathrm{meas}\big((D\times\mathbb{T}^n)\setminus\mathsf{K}\big)\le c_nC_{\mathrm{b}}\sqrt{r}\ \mathrm{meas}\,D,\qquad \mathrm{meas}\,D=\mathrm{meas}\,\check{D}\,\ell^b>0.$$

Proof. **Step S7.45: Reading (3.1).** $\mathsf{K}=K_\Omega$ of [P:`bpP:p3:two-scale-kam`] for (Ω_*,r_K); $D\subset\mathrm{Re}(D_{r_K})$, so $(D\times\mathbb{T}^n)\setminus\mathsf{K}\subset(\mathrm{Re}(D_{r_K})\times\mathbb{T}^n)\setminus\mathsf{K}$, of measure $\le c_n\big(\mathrm{meas}(D\setminus D_0)+\mathrm{meas}(\mathrm{Re}(D_{r_K})\setminus D)\big)$.

Step S7.46: Diophantine part. $D\setminus D_0\subset\mathcal{R}:=\{(I,\xi)\in\bar{D}_t\times A_{\mathrm{sh}}:F_\mu(I,\xi)\notin\mathcal{D}_{\alpha_1,\alpha_2,\tau}\}$, $\alpha_i=2\pi\gamma_i$ (dictionary [P:`bpP:p4:freq`]). The standing data (S4) of [P:`bpP:p4:data`] hold with $\bar{D}=\bar{D}_t$ (compact, meas >0 by Lemma 7.14(c)), $\lambda_0=1$, μ_1 of (r1), and the constants of Lemma 7.9. [P:`bpP:p4:two-scale-diophantine-measure`] with $\lambda=\lambda_2'\in[\lambda_2/2,1]$, $Q=A_{\mathrm{sh}}$, $\bar{\gamma}_{\mathrm{P4}}=\alpha_2/\mu=2\pi\kappa_0r^{5/2}$: $\mathrm{meas}\,\mathcal{R}\le[c_1\alpha_1(1+\mu^{\tau-n}/\lambda)+c_2\bar{\gamma}_{\mathrm{P4}}/\lambda]\,\mathrm{meas}\,\bar{D}_t(2\lambda)^b$. Here c_1,c_2 (which depend on $\bar{D}$ only through $1/\mathrm{meas}\,\bar{D}$) are $\le\bar{c}_1,\bar{c}_2$, data-dependent, as $\mathrm{meas}\,\bar{D}_t\ge\frac12\,\mathrm{meas}\,V$; $\tau-n=1$, $\mu\le r^6$, $\lambda\ge r^2/4$ give $\mu/\lambda\le4$ and $\bar{\gamma}_{\mathrm{P4}}/\lambda\le8\pi\kappa_0r^{1/2}$. Also $(2\lambda)^b\le(4\ell)^b$ ($\lambda_1\le\lambda_2'/2$), $\mathrm{meas}\,\bar{D}_t\le4\,\mathrm{meas}\,\check{D}$. So $\mathrm{meas}(D\setminus D_0)\le4^{b+1}2\pi\kappa_0(5\bar{c}_1+4\bar{c}_2)\sqrt{r}\ \mathrm{meas}\,D$.

Step S7.47: Collar part. $\mathrm{Re}(D_{r_K})=N_{r_K}(\check{D})\times(\lambda_1-r_K,\lambda_2'+r_K)^b$ (max norm, [P:`bpP:p4:domain-collar`], Step S6.16). $N_{r_K}(\check{D})\setminus\check{D}\subset(N_{r_K}(\bar{D}_t)\setminus\bar{D}_t)\cup(\bar{D}_t\setminus\check{D})$, of measure $\le At+c_{\mathrm{dio}}\bar{\gamma}\le(A+c_{\mathrm{dio}}C_\gamma)\sqrt{r}$ ([P:`bpP:p4:collar`](c) with $r_K<t<d$; Lemma 7.8). As in [P:`bpP:p4:domain-collar`], Step S6.17, and with $\mathrm{meas}\,\check{D}\ge\frac14\,\mathrm{meas}\,V$ and $(1+2r_K/\ell)^b<2$ (Lemma 7.20): $\mathrm{meas}(\mathrm{Re}(D_{r_K})\setminus D)\le\mathrm{meas}\,D\,[8(A+c_{\mathrm{dio}}C_\gamma)/\mathrm{meas}\,V+b\kappa_1/16]\sqrt{r}$. (The denominator $\mathrm{meas}\,\check{D}>0$ is part of the statement: review P4-01.) □

Proposition 7.27 (Lower bound for $\mathrm{meas}(S\cap\mathcal{F})$)**.** *Let* $C_*:=\dfrac{A+c_{\mathrm{dio}}C_\gamma}{\mathrm{meas}\,V}+6b\kappa_1+c_nC_{\mathrm{b}}$. *For $0<r<r_*$ and admissible μ the set $S:=S_{D_0}$ satisfies $\mathrm{meas}(\mathcal{F}(V,r)\setminus S)\le C_*\sqrt{r}\ \mathrm{meas}\,\mathcal{F}(V,r)$, i.e. $\mathrm{meas}(S\cap\mathcal{F})\ge(1-C_*\sqrt{r})\,\mathrm{meas}\,\mathcal{F}$.*
Source: CP10 §3, (55)–(59) and §4, p. 564: $\mathrm{meas}(P\setminus K_*)\le C(\bar{\gamma}+\gamma_1+\bar{\gamma}_2/\epsilon^2+\epsilon^{n_2/2})\,\mathrm{meas}\,P$, here with the shell loss $6b\kappa_1\sqrt{r}$ (Part 4 erratum E4) and the collar term from `hboundary`.

Proof. **Step S7.48: Set up.** By Lemma 7.25(b), $\mathrm{meas}(S\cap\mathcal{F})=(2\pi)^n\,\mathrm{meas}\,\mathsf{L}$, $\mathsf{L}:=\mathsf{K}\cap(\mathbb{R}^n\times[0,1)^n)$; by Lemma 7.22 (the (v_1,v_2) part) $\mathsf{L}\subset V\times(0,\lambda_2)^b\times[0,1)^n=:\mathsf{F}'$, and $\mathrm{meas}\,\mathcal{F}=(2\pi)^n\,\mathrm{meas}\,\mathsf{F}'=(2\pi)^n\,\mathrm{meas}\,V\lambda_2^b$ ([P:`bpP:p4:disc-measure`](b)). So $\mathrm{meas}(\mathcal{F}\setminus S)=(2\pi)^n\,\mathrm{meas}(\mathsf{F}'\setminus\mathsf{L})$ (no passage through E is needed: the two measures are compared in the (J,ϑ) coordinates).

Step S7.49: Bad actions. $D\times[0,1)^n\subset\mathsf{F}'$ and $\mathrm{meas}\,V\lambda_2^b-\mathrm{meas}\,\check{D}\ell^b\le\lambda_2^b(\mathrm{meas}\,V-\mathrm{meas}\,\check{D})+\mathrm{meas}\,V\,b\lambda_2^{b-1}\cdot3\lambda_1$; $\mathrm{meas}\,V-\mathrm{meas}\,\check{D}\le At+c_{\mathrm{dio}}\bar{\gamma}$ ([P:`bpP:p4:collar`](b), Lemma 7.8); $3\lambda_1/\lambda_2=6\kappa_1\sqrt{r}$. Divide by $\mathrm{meas}\,V\lambda_2^b$: $\le[(A+c_{\mathrm{dio}}C_\gamma)/\mathrm{meas}\,V+6b\kappa_1]\sqrt{r}$.

Step S7.50: Lost tori. $\operatorname{meas}((D\times[0,1)^n)\setminus \mathsf{L}) = \operatorname{meas}((D\times\mathbb{T}^n)\setminus \mathsf{K}) \le c_n C_{\mathrm{b}}\sqrt{r}\operatorname{meas} D \le c_n C_{\mathrm{b}}\sqrt{r}\operatorname{meas}\mathsf{F}'$ (K is $\mathbb{Z}^n$-invariant in ϑ, Lemma 7.26). Add. □

7.10. Strictness and positivity (ours).

Proposition 7.28 (Deleting a small positive-measure family of tori)**.** *For $0<r<r_*$ and admissible μ there is a σ-compact $B\subset D_0$ such that $S' := S_{D_0\setminus B}$ satisfies*

$$0<\operatorname{meas}(S'\cap\mathcal{F}),\qquad (1-(C_*+\tfrac12)\sqrt{r})\operatorname{meas}\mathcal{F}\le\operatorname{meas}(S'\cap\mathcal{F})<\operatorname{meas}\mathcal{F}.$$

Source: Ours; not in CP10 (12) or CP11 Thm. 11.1, where $\operatorname{meas}P>\operatorname{meas}K$ *is asserted without proof.*
Source gap: The strict inequality is obtained here from the slack $\frac12\sqrt{r}\operatorname{meas}\mathcal{F}>0$ in the lower bound and the bi-Lipschitz parametrisation of Part 3.

Proof. **Step S7.51: Additivity.** For σ-compact $B\subset D_0$ put $m(B) := \operatorname{meas}(S_B\cap\mathcal{F}) = (2\pi)^n\operatorname{meas}\mathsf{A}_B$ (Lemma 7.25). The tori $\varphi_\omega(\mathbb{R}^n)$, $\omega\in\Omega_*$, are pairwise disjoint (Proposition 4.3, injectivity modulo translations), so m is countably additive on σ-compact subsets of D_0.

Step S7.52: Slices are null. Let $x_1(I)=I_1$ (first coordinate) and $H_s = D_0\cap\{x_1=s\}$ (a null set). $\mathsf{K}^{\mathrm{c}}_{H_s} = \Phi_3(\check\imath(H_s)\times[0,1)^n)$ is the image of $\check\imath(H_s)\times[0,1)^n$ under a Lipschitz map ($\check\imath$ is Lipschitz, Φ_3 Lipschitz in the rescaled norm: [P:`bpP:p3:limit-actions`], [P:`bpP:p3:limit-phi`]); [P:`bpP:p1:torus-union`] gives $\operatorname{meas}\mathsf{K}^{\mathrm{c}}_{H_s}\le\Lambda^{2n}\operatorname{meas}\check\imath(H_s)=0$ (Lipschitz image of a null set); hence $\mathsf{K}_{H_s}=\bigcup_k(\mathsf{K}^{\mathrm{c}}_{H_s}+(0,k))$ and $\mathsf{A}_{H_s}\subset\mathsf{K}_{H_s}$ are null, i.e. $m(H_s)=0$.

Step S7.53: Intermediate value. $f(s) := m(D_0\cap\{x_1\le s\})$ is nondecreasing, $f(-\infty)=0$, $f(+\infty)=m(D_0)=\operatorname{meas}(S\cap\mathcal{F})\ge(1-C_*\sqrt{r})\operatorname{meas}\mathcal{F}>0$ (Proposition 7.27, (r5)). It is right-continuous (decreasing intersections, finite measure, additivity) and left-continuous by Step S7.52. Put $\theta_0 := \min\{\frac12 m(D_0),\frac12\sqrt{r}\operatorname{meas}\mathcal{F}\}>0$ and choose s with $f(s)=\theta_0$; $B := D_0\cap\{x_1\le s\}$ is closed in D_0, hence σ-compact, and $D_0\setminus B = D_0\cap\{x_1>s\}$ is σ-compact.

Step S7.54: Conclusion. $\operatorname{meas}(S'\cap\mathcal{F}) = m(D_0)-\theta_0$. Upper: $\le\operatorname{meas}(S\cap\mathcal{F})-\theta_0\le\operatorname{meas}\mathcal{F}-\theta_0<\operatorname{meas}\mathcal{F}$ ($S\cap\mathcal{F}\subset\mathcal{F}$). Lower: $\ge(1-C_*\sqrt{r})\operatorname{meas}\mathcal{F}-\frac12\sqrt{r}\operatorname{meas}\mathcal{F}$; positive since $(C_*+\frac12)\sqrt{r}\le\frac12$ (r5) and $\operatorname{meas}\mathcal{F}>0$. □

7.11. Proof of the frozen theorem.

Theorem 7.29 (P-G; [P:`bpP:p0:pg`])**.** *Frozen Lean declaration:* `KAM.Planetary.properlyDegenerateKAM`

Under the hypotheses and data of [P:`bpP:p0:pg`]*, with $c := 2n+4$, $C := \max\{C_*+1, C_w\}$, and r_* of Definition 7.13, the conclusion holds with exactly the printed quantifier order.*

Proof. **Step S7.55: Constants first.** Following Remark 7.1: the data constants of Lemmas 7.2–7.10 (Proposition 7.5); C_U, C_1; κ_0,κ_1,G (Lemma 7.19); $C_\gamma, C_{\mathrm{b}}, C_*$ (Lemma 7.26, Proposition 7.27); C_w; r_* (Definition 7.13, which is a finite list of conditions on these constants; $r_*<r_0$, $r_*<1$ included); $c=2n+4$ and C. None depends on r or μ, and $0<r_*<r_0$, $r_*<1$, $C>0$, $c>0$.

Step S7.56: Given (r,μ). Let $0<r<r_*$, $0<\mu<\mu_0$, $\mu<r^6/(\log\frac1r)^c$ (positive base: $r<1$). All objects of Definition 7.12 are defined, Lemma 7.14 holds; Propositions 7.15, Lemmas 7.18, 7.19 verify the hypotheses of [P:`bpP:p3:two-scale-kam`] (including the complex injectivity, Lemma 7.10).

Step S7.57: The set. $S := S'$ of Proposition 7.28. By Lemma 7.25(a) it is Borel (so Lebesgue measurable), $S\subset\mathcal{C}(V,r)$, and every $z\in S$ is $K_\omega(\theta)$ for some $\omega\in\Omega_*$, $\theta\in\mathbb{R}^n$, with $K_\omega(\mathbb{R}^n)\subset S$; Proposition 7.24 supplies $\mathrm{IsCylTorus}(H_\mu,\mathcal{C}(V,r),w,K_\omega)$, $\gamma>0$, $\tau>a+b$, $|w^{\mathrm{f}}|_\infty\le C_w\le C$, $|w^{\mathrm{s}}|_\infty\le C_w\mu\le C\mu$. Hence $\mathrm{IsCylKol}(H_\mu,V,r,\mu,C,S)$ ([P:`bpP:p0:cylkol`]).

Step S7.58: Measures. Proposition 7.28: $0<\operatorname{meas}(S\cap\mathcal{F})<\operatorname{meas}\mathcal{F}$ and $\operatorname{meas}(S\cap\mathcal{F})\ge(1-(C_*+\frac12)\sqrt{r})\operatorname{meas}\mathcal{F}>(1-C\sqrt{r})\operatorname{meas}\mathcal{F}$ since $C\ge C_*+1$ and $\operatorname{meas}\mathcal{F}>0$. If $1-C\sqrt{r}\le0$,

ofReal$(1 - C\sqrt{r}) = 0 <$ meas$(S \cap \mathcal{F})$; otherwise ofReal$(1 - C\sqrt{r})$ meas $\mathcal{F} = (1 - C\sqrt{r})$ meas $\mathcal{F} <$ meas$(S \cap \mathcal{F})$ ([P:`bpP:p0:totality`]). meas $\mathcal{F} < \infty$ because V is bounded. □

Remark 7.30 (Quantifier audit). Frozen order: $\exists\, r_*, C, c\ \forall r, \mu\ \exists S$. Constants chosen before r: r_*, C, c and every constant of Remark 7.1, items 1–5 (all functions of the data). Chosen after (r, μ): $\bar{K}, t, \lambda_i, \bar{\gamma}, \bar{\rho}, \rho, \gamma_i, \hat{\rho}, \hat{E}, r_K$, the set D, the family φ_ω, the deleted parameters B, hence S. $K = K_0$ does not depend on (r, μ) at all. The Diophantine exponent $\tau = n + 1$ is data-independent. The frequency bound C and the measure constant coincide, as in the frozen statement.

7.12. **Errata, deviations, external facts, forward references.**
Suspected defects in the sources. (1) CP10 p. 563, (60): "$\sup|\tilde{P}_{\mathrm{av}}| \le C\epsilon^5$" is asserted from (A_2'') (a little-o statement with δ depending on (μ, I, ϵ)); supplied as Proposition 7.5 via joint analyticity, periodicity, the jet identification and Schwarz. (2) CP10 p. 563, Step 2: "we proceed as in Steps 5 and 6" and the measure estimate (56)–(57) uses meas$(\mathrm{Re}(V_r) \setminus V) \le C\bar{\gamma}_2$ meas V, which cannot hold for arbitrary V; replaced by `hboundary`. (3) The neighbourhood of the Cantor set $\check{D}$ of $\bar{\omega}_0$-Diophantine actions is not controlled by the collar of V; CP10's (57) writes $\mathrm{Re}(D_r) \setminus D \subset (\mathrm{Re}(V_r) \setminus V) \cup (V \setminus D)$ without proof; Lemma 7.26, Step S7.47 proves it. (4) CP10 (20) $\bar{\gamma}_2 = \gamma_*^2\epsilon^{5/2}$ vs (17) lower bound with $(\log(\epsilon^5/\gamma_1^2)^{-1})^{\tau_*+1}$: consistent only because $\gamma_1 = \gamma_*^2\epsilon^{5/2}$ makes this logarithm a constant (our K_0); not an erratum, but the dependence must be tracked, as done in Lemma 7.19. (5) CP10 p. 564 "replacing ϵ by $\sqrt{2\check{c}_2}\epsilon$": in our frozen normalisation the disc is $\xi_j < r^2/2$ and the shell loss constant is $6b\kappa_1$.
Deviations. Steps 2–4 of CP10 §3 are skipped (the hypothesis is already Birkhoff form); the polynomial $\tilde{N}$ is defined from $\tilde{P}_{\mathrm{av}}$ through its fourth jet; $\check{D}$ replaces $\bar{D} \times A$ with an explicit Cantor-set measure estimate; the top of the shell is cut at $\lambda_2' = \lambda_2 - 2\lambda_1$ so that the tori stay in the open cylinder; $\mu_2 = \Lambda_P\mu$ absorbs the sup-norm of P (CP10 normalises $|P|, = 1$).
External facts used (beyond Parts 1–4). Identity theorem and Cauchy estimates for holomorphic functions of several variables; compactness of closed bounded sets; Cramer's rule; block inversion (Schur complement) and Neumann series; Schwarz lemma in one variable; continuity of measure along monotone sequences; Lebesgue number lemma. Tracks K, N as cited in Parts 1–4.
Forward and cross-track references. Cited from Part 1: `cyl-pipeline`, `hol-chart`, `chart-field`, `typical`, `liouville`, `torus-union`, `dio-sources`, `period-convert`; Part 2: `averaged-remainder`, `use5`; Part 3: `two-scale-kam`, `interface`, `period-dictionary`, `constants`, `rem-norm`, `limit-actions`, `limit-phi`; Part 4: `collar`, `lip-inverse`, `affine-slab`, `lattice-tail`, `two-scale-diophantine-measure`, `data`, `freq`, `domain-collar`, `disc-measure`. Tracks N: `bpN:p4:norm`, `bpN:p4:sup-weighted`. Parts 12–13 are not used by P-G. Part 4 must repair `use5` and `domain-collar` as stated in Remark 7.21.
Label index. `bpP:p5:ext`, `consts`, `birkhoff-ext`, `uniform-remainder`, `lip-complex`, `bilip`, `dio-one`, `freq-data`, `freq-inj`, `params`, `rstar`, `param-facts`, `avg`, `phi`, `phi-chart`, `kam-hyp`, `kam-cond`, `rK`, `location`, `kset`, `torus-conv`, `cell-transfer`, `budget`, `lower`, `strictness`, `main`.

8. Newtonian models and physical reconstruction

This part connects the Hamiltonians H_{mov}, H_{fix} of Part 0 (Definitions 1.22, 1.23) to the physical $(n + 1)$-body problem, and supplies the three physical clauses of the frozen Kolmogorov sets (Definitions 1.25, 1.26) once the torus predicate is known. The results are:

- `reduction` (Prop. 8.8) and `split` (Lemma 8.3): H_{mov} is the barycentric N-body Hamiltonian divided by μ, and $H_{\mathrm{mov}} - H_{\mathrm{fix}}|_{\varepsilon=\mu} = \mu|\sum_i p^{(i)}|_2^2/(2m_0)$;
- `hamilton-eqs` (Prop. 8.6): explicit Hamilton equations of both models on the collision-free set;
- `reconstruction`, `reconstruction-fixed` (Props. 8.11, 8.13): an H_{mov}-orbit gives, through the frozen barycentric maps, a solution of Newton's equations with the actual masses; an

H_{fix}-orbit gives a fixed-centre solution. *Both are stated pointwise in time, for an arbitrary set $J \subset \mathbb{R}$ of times*, so that the later additions (D22/D25: stopped solutions on $[-T,T]$, `KAM.Planetary.IsNewtonSolutionOn`, `KAM.Planetary.IsMovingSolutionOn`) reuse them verbatim; the all-time frozen predicates are the case $J = \mathbb{R}$;

- `tube-geometry` (Lemma 8.16): explicit constants d, D (chosen from the axes alone, before any coupling) and a tube radius ζ_* such that every point of U_{ζ_*} is collision free with all Sun–planet and planet–planet distances in $[d, D]$, *exactly independently of* μ;
- `orbit-transfer`, `movkol-assembly`, `fixkol-assembly`: the physical clauses of MovKol and FixKol follow from $S \subset U_{\zeta_*}$ and the torus predicate alone;
- `measure-positive` (Lemma 8.19): $0 < \operatorname{meas} S < \infty$.

Nothing here is proved by a hypothesis shaped like its conclusion: the inputs are the Hamilton equations along the orbit, the collision-free clause, and (for the tube) the position of the orbit. *Units.* The frozen definitions keep the gravitational constant G and the Sun's mass m_0 as parameters; Fej04r uses $G = 1$. The statements below are for general $G, m_0 > 0$ (the frozen `IsMovingSolutionOn` fixes $G = m_0 = 1$, whereas `IsNewtonSolutionOn` uses $G = 1$ in its force formula with an arbitrary mass array; see Remark 8.12).

8.1. **Conventions and external facts.** Throughout $n \geq 1$, $G, m_0 > 0$, $m_i > 0$ $(i < n)$, $\mu, \varepsilon \geq 0$ unless said otherwise; $\ell = |\cdot|_2$ is the planar length (Definition 1.4). For $q, p \in \mathbb{R}^{2n}$ write $q_i = q^{(i)}$, $p_i = p^{(i)} \in \mathbb{R}^2$ and $P = \sum_i p_i$. The set of *collision-free phase points* is

$$W = \{z = (q,p) \in \mathrm{Ph}(2n) : \mathrm{NoColl}(q)\} \qquad \text{(Definition 1.18)}.$$

For q with $\mathrm{NoColl}(q)$ put

$$g_i(q) = \frac{q_i}{\ell(q_i)^3}, \qquad f_i(q) = \sum_{j \neq i} m_j \frac{q_j - q_i}{\ell(q_j - q_i)^3} \qquad (i < n),$$

the central and the planet–planet force kernels. Planet i of the $(n+1)$-body system sits at index $\mathrm{succ}(i) \in \mathtt{Fin}\,(n+1)$ and the Sun at index 0 (the order of `Fin.cons`); I write Q_0, Q_i for the entries of a configuration $Q : \mathtt{Fin}\,(n+1) \to \mathbb{R}^2$.

External facts (Mathlib level, used by label).

(E1) Chain, product, quotient and finite-sum rules for `HasDerivAt` and `HasFDerivAt`; a map into a finite product $\mathbb{R}^N$ has a derivative iff every component does (`hasDerivAt_pi`); `fderiv` of a differentiable function is the derivative; a polynomial is C^∞.

(E2) $\sqrt{\cdot}$ is C^∞ on $(0,\infty)$ with derivative $x \mapsto 1/(2\sqrt{x})$; $x \mapsto 1/x$ is C^∞ on $\mathbb{R} \setminus \{0\}$.

(E3) Preimages of open sets under continuous maps are open; finite intersections of open sets are open; $\mathbb{R}$ is totally ordered and finite minima/maxima exist.

(E4) Finite-sum algebra: `Fin.sum_univ_succ`, `Finset.sum_comm`, splitting $\sum_j = \sum_{j<i} + \sum_{j>i}$, and $\sum_{j \neq i} = \sum_j -(\text{term } j = i)$.

(E5) The reverse triangle inequality $|\,|u| - |w|\,| \leq |u - w|$ in a normed space.

(E6) Bounded sets of $\mathbb{R}^N$ have finite Lebesgue measure; the product measure on $\mathrm{Ph}(d) = \mathbb{R}^d \times \mathbb{R}^d$ (`volume`) is Lebesgue measure on $\mathbb{R}^{2d}$ and a box $\prod[-r_k, r_k]$ has measure $\prod 2r_k$; measure is monotone.

8.2. **Planar length.**

Lemma 8.1 (Planar length)**.** *For $v, w \in \mathbb{R}^2$, $c \in \mathbb{R}$:*

(a) $\ell(v) \geq 0$, $\ell(v)^2 = \mathrm{sq}(v) = v \cdot v$, $\ell(v) = 0 \iff v = 0$, $\ell(-v) = \ell(v)$, $\ell(cv) = |c|\ell(v)$, $\ell(v+w) \leq \ell(v) + \ell(w)$.

(b) $|v|_\infty \leq \ell(v) \leq \sqrt{2}\,|v|_\infty$.

(c) On $\mathbb{R}^2 \setminus \{0\}$ the functions ℓ, $1/\ell$ are C^∞ with $D\ell(v)h = \dfrac{v \cdot h}{\ell(v)}$ and $D(1/\ell)(v)h = -\dfrac{v \cdot h}{\ell(v)^3}$.

Proof. **Step S8.1: (a).** Algebra of $\sqrt{v_1^2+v_2^2}$; the triangle inequality is Cauchy–Schwarz $v\cdot w\le \ell(v)\ell(w)$ in $\ell(v+w)^2=\ell(v)^2+2v\cdot w+\ell(w)^2$.

Step S8.2: (b). $v_k^2\le v_1^2+v_2^2\le 2\max_k v_k^2$; take square roots.

Step S8.3: (c). sq is a polynomial with $D\,\mathrm{sq}(v)h=2v\cdot h$, positive on $v\neq 0$; apply (E2) and (E1): $D\ell(v)h=\frac{2v\cdot h}{2\ell(v)}$, and $D(1/\ell)=-D\ell/\ell^2$. □

8.3. **The two reduced models.**

Lemma 8.2 (Collision-free domain; regularity)**.** *W is open, and $H_{\mathrm{mov}}=$ `movingHamiltonian`$G\,m_0\,m\,\mu$ and $H_{\mathrm{fix}}=$ `fixedHamiltonian`$G\,m_0\,m\,\varepsilon$ are C^∞ on W (hence differentiable at every point of W, with continuous Fréchet derivative). The Lean value of H off W is irrelevant to everything below, because every use is guarded by* NoColl *(Remark 1.27).*

Proof. **Step S8.4: Openness.** NoColl(q) is the finite conjunction of $\mathrm{sq}(q_i)>0$ and $\mathrm{sq}(q_i-q_j)>0$ $(i\neq j)$ (Lemma 8.1(a)); each sq $\circ$(linear map of q) is continuous, so W is open by (E3).

Step S8.5: Smoothness. On W each term of H_{mov}, H_{fix} is a polynomial in p or q-dependent $1/\ell(q_i)$, $1/\ell(q_i-q_j)$, which are C^∞ by Lemma 8.1(c) and (E1); finite sums and products of C^∞ maps are C^∞. □

Lemma 8.3 (Splitting; the identity of Part 0)**.** *Let $H_{\mathrm{Kep}}(q,p)=\sum_i\big[\frac{|p_i|_2^2}{2m_i}-\frac{Gm_0m_i}{\ell(q_i)}\big]$ (independent of μ,ε), and for $z\in W$*

$$f_{\mathrm{mov}}(z)=\frac{|P|_2^2}{2m_0}-G\sum_{i<j}\frac{m_im_j}{\ell(q_i-q_j)},\qquad f_{\mathrm{fix}}(z)=-G\sum_{i<j}\frac{m_im_j}{\ell(q_i-q_j)}.$$

Then for all $z\in\mathrm{Ph}(2n)$ (with the Lean total functions) $H_{\mathrm{mov}}=H_{\mathrm{Kep}}+\mu f_{\mathrm{mov}}$, $H_{\mathrm{fix}}=H_{\mathrm{Kep}}+\varepsilon f_{\mathrm{fix}}$, and

$$H_{\mathrm{mov}}(z)-H_{\mathrm{fix}}(z)\big|_{\varepsilon=\mu}=\frac{\mu\,|P|_2^2}{2m_0}\ \Big(=\mu\,\frac{|\sum_i p^{(i)}|_2^2}{2m_0}\Big).$$

Proof. **Step S8.6: Algebra.** $\big(\frac{1}{m_i}+\frac{\mu}{m_0}\big)\frac{|p_i|_2^2}{2}=\frac{|p_i|_2^2}{2m_i}+\frac{\mu|p_i|_2^2}{2m_0}$, and $|P|_2^2=\sum_i|p_i|_2^2+2\sum_{i<j}p_i\cdot p_j$ (Lemma 8.1(a), bilinearity of $\cdot$). Hence the kinetic part of H_{mov} is $\sum_i\frac{|p_i|_2^2}{2m_i}+\frac{\mu}{m_0}\big(\frac12|P|_2^2\big)$; the potential parts agree term by term. At $\varepsilon=\mu$ the pair-potential terms of H_{mov} and H_{fix} are the same, which gives the difference. □

Remark 8.4 (Kepler part; mass dictionary). Lemma 8.3 is why both models share the unperturbed part H_{Kep} and have perturbations μf_{mov}, $\varepsilon f_{\mathrm{fix}}$ with $f_{\mathrm{mov}}, f_{\mathrm{fix}}$ independent of the coupling. Each summand of H_{Kep} is $m_iE_k(q_i,p_i/m_i)$ with $k=Gm_0$ in the notation of [I:`bpI:p7:scaling`], whose "mass dictionary" gives the Kepler orbit, action $m_i\sqrt{ka}$ and the mean values used in Part 7. (In the moving-Sun reduction the *reduced* mass $m_0m_i/(m_0+\mu m_i)$ appears if one chooses to keep the term $\mu|p_i|_2^2/(2m_0)$ in the Kepler part, as Fej04r (28) does with μ_j, M_j; the frozen H_{mov} and Part 7 do not: that term belongs to μf_{mov}.)

Lemma 8.5 (Action–reaction)**.** *For q with* NoColl(q)*: $\sum_i m_if_i(q)=0$.*

Proof. $\sum_i m_if_i=\sum_i\sum_{j\neq i}m_im_j\frac{q_j-q_i}{\ell(q_j-q_i)^3}=:\sum_{i\neq j}a_{ij}$ with $a_{ji}=-a_{ij}$ (Lemma 8.1(a)). Swapping the names of i,j ((E4)) gives $\sum a_{ij}=\sum a_{ji}=-\sum a_{ij}$. □

Proposition 8.6 (Hamilton equations of H_{mov} and H_{fix})**.** *Let $z=(q,p)\in W$. Using the Lean convention $X_H=(\partial_pH,-\partial_qH)$ of Definition 1.2 and the block form $v=(v_0,\dots,v_{n-1})$, $v_i\in\mathbb{R}^2$, of vectors of $\mathbb{R}^{2n}$ (index $2i+c$):*

$$X_{H_{\mathrm{mov}}}(z)=\Big(\big(\tfrac{p_i}{m_i}+\tfrac{\mu}{m_0}P\big)_i,\ \big(-Gm_0m_i\,g_i(q)+\mu Gm_i\,f_i(q)\big)_i\Big),$$

$$X_{H_{\mathrm{fix}}}(z) = \Big(\big(\tfrac{p_i}{m_i}\big)_i,\ \big(-Gm_0m_i\, g_i(q) + \varepsilon G m_i\, f_i(q)\big)_i \Big).$$

In particular a curve $z = (q,p) : J \to W$ has derivative $X_H(z(t))$ at t (with H differentiable there, automatic by Lemma 8.2) iff $q_i' = \frac{p_i}{m_i} + \frac{\mu}{m_0} P$ (resp. $\frac{p_i}{m_i}$) and $p_i' = -Gm_0m_i g_i(q) + \mu G m_i f_i(q)$ (resp. with ε), componentwise.

Proof. **Step S8.7: Directional derivative.** By Lemma 8.2 H_{mov} is differentiable at z. Split $H_{\mathrm{mov}} = T(p) + V(q)$, $T = \sum_i c_i |p_i|_2^2/2 + \frac{\mu}{m_0}\sum_{i<j} p_i \cdot p_j$ with $c_i = \frac{1}{m_i} + \frac{\mu}{m_0}$, a quadratic polynomial, and $V(q) = -Gm_0 \sum_i m_i/\ell(q_i) - \mu G \sum_{i<j} m_i m_j/\ell(q_i - q_j)$. For $(\delta q, \delta p) \in \mathrm{Ph}(2n)$, by (E1) and Lemma 8.1(c),

$$DT(p)\delta p = \sum_i c_i p_i \cdot \delta p_i + \tfrac{\mu}{m_0}\sum_{i<j}(p_i \cdot \delta p_j + \delta p_i \cdot p_j) = \sum_i \delta p_i \cdot \big(\tfrac{p_i}{m_i} + \tfrac{\mu}{m_0} P\big),$$

using $c_i p_i + \frac{\mu}{m_0}\sum_{j\neq i} p_j = \frac{p_i}{m_i} + \frac{\mu}{m_0} P$, and

$$DV(q)\delta q = Gm_0 \sum_i m_i \frac{q_i \cdot \delta q_i}{\ell(q_i)^3} + \mu G \sum_{i<j} m_i m_j \frac{(q_i - q_j)\cdot(\delta q_i - \delta q_j)}{\ell(q_i - q_j)^3}.$$

Step S8.8: Collect coefficients. The coefficient of δq_i in the double sum is $\mu G m_i \sum_{j\neq i} m_j \frac{q_i - q_j}{\ell(q_i-q_j)^3} = -\mu G m_i f_i(q)$ (a pair $\{i,j\}$ with $j<i$ contributes with δq_i the same way, because $\ell(q_i - q_j) = \ell(q_j - q_i)$). So $\partial_{q_i} H_{\mathrm{mov}} = Gm_0m_i g_i - \mu G m_i f_i$, $\partial_{p_i} H_{\mathrm{mov}} = \frac{p_i}{m_i} + \frac{\mu}{m_0} P$. The first component of X_H at (i,c) is $DH(z)(0, e_{2i+c})$ and the second is $-DH(z)(e_{2i+c}, 0)$ (Definition 1.2); this is the claim.

Step S8.9: H_{fix}. The same computation: the kinetic part is $\sum_i |p_i|_2^2/(2m_i)$ with $\partial_{p_i} = p_i/m_i$, and the potential is V with μ replaced by ε.

Step S8.10: The curve statement. A curve into $\mathrm{Ph}(2n) = \mathbb{R}^{4n}$ has derivative (q', p') at t iff each coordinate does ((E1)); compare with the displayed vector field. □

8.4. Derivation of H_{mov} from the barycentric Hamiltonian.

Definition 8.7 (The $(n+1)$-body Hamiltonian and Newton equations)**.** Let $\mu > 0$, $M = \mathrm{mass}_\mu = (m_0, \mu m_1, \dots, \mu m_n)$ the actual masses (Definition 1.19), $X = (X_0, \dots, X_n)$, $Y = (Y_0, \dots, Y_n)$ with $X_k, Y_k \in \mathbb{R}^2$ (positions and actual momenta), and

$$F_\mu(X,Y) = \sum_{k=0}^{n} \frac{|Y_k|_2^2}{2M_k} - G \sum_{0\le k<l\le n} \frac{M_k M_l}{\ell(X_k - X_l)}.$$

Newton's equations are $X_k' = Y_k/M_k$, $Y_k' = \sum_{l\neq k} \frac{GM_kM_l}{\ell(X_l - X_k)^3}(X_l - X_k)$; with $V_k = X_k'$ they are $V_k' = \sum_{l\neq k} \frac{GM_l}{\ell(X_l - X_k)^3}(X_l - X_k)$, the equations of `IsNewtonSolution` (Definition 1.20). They are Hamilton's equations of F_μ for $\Omega = \sum_k dY_k \wedge dX_k$ on $\{\ell(X_k - X_l) > 0\}$ (Lemma 8.1(c) as in Prop. 8.6).

Source: Fej04r §6.1, pp. 45–46 (Newton equations (1), Hamiltonian with $\|y_0\|^2/2m_0$, $\epsilon\sum\|y_j\|^2/2m_j$, potential m_0m_j and $\epsilon^2 m_j m_k$ terms), G restored; Fej04r writes momenta $y_0, \epsilon y_j$ and the symplectic form $\sum dx_0 \wedge dy_0 + \epsilon \sum dx_j \wedge dy_j$, so $Y_j = \epsilon y_j = \mu p_j$ below.

Proposition 8.8 (Heliocentric symplectic coordinates and reduction)**.** *Let $\mu > 0$. Define the linear map $\Theta(X,Y) = (Q_0, q, P_0, P)$ with $Q_0 = X_0$, $q_i = X_i - X_0$, $P_0 = \sum_{k=0}^n Y_k$, $P_i = Y_i$ ($1 \le i \le n$; planet i has index i here).*

(a) Θ is a linear isomorphism with $\sum_k X_k \cdot Y_k = Q_0 \cdot P_0 + \sum_i q_i \cdot P_i$ (the canonical pairing is preserved), hence it is symplectic: $\sum_k dY_k \wedge dX_k = dP_0 \wedge dQ_0 + \sum_i dP_i \wedge dq_i$.

(b) On $\{\mathrm{NoColl}(q)\}$ *one has*

$$F_\mu \circ \Theta^{-1}(Q_0, q, P_0, P) = \frac{|P_0 - \sum_i P_i|_2^2}{2m_0} + \sum_i \frac{|P_i|_2^2}{2\mu m_i} - G\mu \sum_i \frac{m_0 m_i}{\ell(q_i)} - G\mu^2 \sum_{i<j} \frac{m_i m_j}{\ell(q_i - q_j)},$$

independent of Q_0*; so* P_0 *(total momentum) is conserved. On the invariant set* $P_0 = 0$*, writing* $P_i = \mu p_i$*:*

$$F_\mu \circ \Theta^{-1}(Q_0, q, 0, \mu p) = \mu\, H_{\mathrm{mov}}(q, p).$$

*(c) (*Solutions correspond.*) If* (X, Y) *solves Newton's equations on a set* $J \subset \mathbb{R}$ *of times, with total momentum* $P_0(t) = 0$ *and* $\mathrm{NoColl}(q(t))$ *for* $t \in J$*, then* $(q, p) = (X_i - X_0,\, Y_i/\mu)$ *satisfies* $(q, p)' = X_{H_{\mathrm{mov}}}(q, p)$ *on* J *(the converse direction of Proposition 8.11, where* X_0 *is then recovered as* $x_0(q)$ *if the barycentre is* 0*).*

Source: Fej04r (28), pp. 46–47 ("coordonnées héliocentriques symplectiques", reduction to $Y_0 = 0$*,* $F = \sum(\|Y_j\|^2/2\mu_j - \mu_j M_j/\|X_j\|) + \epsilon \sum(-m_j m_k/\|X_j - X_k\| + Y_j \cdot Y_k/m_0))$*, with the index correction of Fej07e p. 3 (*$m_i m_j \to m_j m_k$*); the revised printing already has* $m_j m_k$ *(read on p. 46);* G *restored. Ours: the statement as a lemma about* H_{mov} *and with* μ*-scalings written out.*

Proof. **Step S8.11: (a).** The inverse is $X_0 = Q_0$, $X_i = q_i + Q_0$, $Y_i = P_i$, $Y_0 = P_0 - \sum_i P_i$. Then $X_0 \cdot Y_0 + \sum_i X_i \cdot Y_i = Q_0 \cdot (P_0 - \sum P_i) + \sum_i (q_i + Q_0) \cdot P_i = Q_0 \cdot P_0 + \sum_i q_i \cdot P_i$. Moreover the Liouville forms agree identically: $\sum_k Y_k\, dX_k = (P_0 - \sum_i P_i)\, dQ_0 + \sum_i P_i\, d(q_i + Q_0) = P_0\, dQ_0 + \sum_i P_i\, dq_i$; taking d gives the symplectic statement.

Step S8.12: (b). Substitute $Y_0 = P_0 - \sum P_i$, $X_i - X_0 = q_i$, $X_i - X_j = q_i - q_j$, $M_0 = m_0$, $M_i = \mu m_i$: the kinetic energy is $\frac{|P_0 - \sum P_i|_2^2}{2m_0} + \sum_i \frac{|P_i|_2^2}{2\mu m_i}$ and the potential is $-G\mu \sum_i \frac{m_0 m_i}{\ell(q_i)} - G\mu^2 \sum_{i<j} \frac{m_i m_j}{\ell(q_i - q_j)}$. At $P_0 = 0$, $P_i = \mu p_i$ the total is

$$\frac{\mu^2 |\sum_i p_i|_2^2}{2m_0} + \mu \sum_i \frac{|p_i|_2^2}{2m_i} - G\mu \sum_i \frac{m_0 m_i}{\ell(q_i)} - G\mu^2 \sum_{i<j} \frac{m_i m_j}{\ell(q_i - q_j)} = \mu\big(H_{\mathrm{Kep}} + \mu f_{\mathrm{mov}}\big) = \mu H_{\mathrm{mov}}$$

by Lemma 8.3. $F_\mu \circ \Theta^{-1}$ does not involve Q_0, so $P_0' = -\partial_{Q_0}(\cdot) = 0$.

Step S8.13: (c). With $P_0 = \sum_k Y_k = 0$ we have $Y_0 = -\sum_i Y_i = -\mu P$, $P = \sum_i p_i$, hence $q_i' = X_i' - X_0' = \frac{Y_i}{\mu m_i} - \frac{Y_0}{m_0} = \frac{p_i}{m_i} + \frac{\mu}{m_0} P$ and, from $Y_i' = \sum_{l \neq i} \frac{G M_i M_l}{\ell(X_l - X_i)^3}(X_l - X_i)$ with $X_0 - X_i = -q_i$, $X_j - X_i = q_j - q_i$, $p_i' = \mu^{-1} Y_i' = G m_i \big[-m_0 \frac{q_i}{\ell(q_i)^3} + \mu \sum_{j \neq i} m_j \frac{q_j - q_i}{\ell(q_j - q_i)^3}\big]$. This is Proposition 8.6. □

Remark 8.9 (Role of this proposition). Part (b) is the derivation of Definition 1.22: H_{mov} is F_μ on the reduced space divided by μ (Fej04r's F with $\epsilon = \mu$). Nothing below uses (a)–(b): the forward reconstruction (Prop. 8.11) is verified directly from the Hamilton equations, so the dependence of the physical clauses on the Hamiltonian reduction is only through Prop. 8.6. Part (c) is needed only by consumers who start from a Newton solution (the uniqueness predicate `MovingUniqueOn` of D25 is the reverse direction).

8.5. **Reconstruction of the physical motion.**

Lemma 8.10 (Algebra of the barycentric maps)**.** *Let* $\mu \geq 0$*,* $q, p \in \mathbb{R}^{2n}$*,* $M_* = m_0 + \mu \sum_i m_i > 0$*,* $M = \mathrm{mass}_\mu$ *(the frozen* `actualMasses`*),* $Q =$ `barycentricPositions` $m_0\, m\, \mu\, q$*,* $V =$ `barycentricVelocities` $m_0\, m\, \mu\, p$*. Then, with* $P = \sum_i p_i$*:*

(a) $Q_0 = x_0 = -\frac{\mu}{M_*} \sum_i m_i q_i$*,* $Q_i - Q_0 = q_i$ *and* $Q_j - Q_i = q_j - q_i$ *for planets* i, j*; hence* $\ell(Q_i - Q_0) = \ell(q_i)$ *and* $\ell(Q_j - Q_i) = \ell(q_j - q_i)$ for every μ.

(b) $V_0 = -\frac{\mu}{m_0} P$*,* $V_i = p_i/m_i$*, and* $V_i - V_0 = \frac{p_i}{m_i} + \frac{\mu}{m_0} P$ *(the first component of* $X_{H_{\mathrm{mov}}}$*, Prop. 8.6).*

(c) (barycentre and momentum) $\sum_{k=0}^n M_k Q_k = 0$ *and* $\sum_{k=0}^n M_k V_k = 0$.

(d) $\ell(x_0) \leq \frac{\mu}{m_0} \sum_i m_i \ell(q_i)$.

Proof. **Step S8.14: (a),(b).** Unfold the definitions (`Fin.cons`, `Fin.cons_zero`, `Fin.cons_succ`). $Q_j - Q_i = (x_0 + q_j) - (x_0 + q_i)$, and ℓ is even (Lemma 8.1(a)).

Step S8.15: (c). $\sum_k M_k Q_k = m_0 x_0 + \sum_i \mu m_i (x_0 + q_i) = M_* x_0 + \mu \sum m_i q_i = 0$; $\sum_k M_k V_k = m_0 V_0 + \mu \sum_i p_i = -\mu P + \mu P = 0$.

Step S8.16: (d). $M_* \geq m_0$ and Lemma 8.1(a). □

Proposition 8.11 (Reconstruction: moving Sun ⇒ Newton, on any set of times)**.** *Let $G, m_0 > 0$, $m_i > 0$, $\mu \geq 0$, $J \subset \mathbb{R}$ arbitrary, and $z = (q, p) : \mathbb{R} \to \mathrm{Ph}(2n)$ such that for all $t \in J$:*

(R1) $\mathrm{NoColl}(q(t))$*;*

(R2) z has derivative $X_{H_{\mathrm{mov}}}(z(t))$ at t, $H_{\mathrm{mov}} =$ `movingHamiltonian` $G\, m_0\, m\, \mu$.

(Differentiability of H_{mov} at $z(t)$ is automatic, Lemma 8.2; for $G = m_0 = 1$ conditions (R1)–(R2) say exactly `IsMovingSolutionOn` *$m\, \mu\, J\, z$ of D25.) Put $M = \mathrm{mass}_\mu$, $Q(t) =$* `barycentricPositions` *$m_0\, m\, \mu\, q(t)$, $V(t) =$* `barycentricVelocities` *$m_0\, m\, \mu\, p(t)$. Then for every $t \in J$ and every pair $k \neq l$ of indices of* `Fin` *$(n+1)$:*

(a) $\ell(Q_l(t) - Q_k(t)) > 0$;

(b) Q_k has derivative $V_k(t)$ at t;

(c) V_k has derivative $\displaystyle\sum_{l \neq k} \frac{G M_l}{\ell(Q_l(t) - Q_k(t))^3} \big(Q_l(t) - Q_k(t)\big)$ at t.

Consequently: `IsNewtonSolutionOn` *$(G \cdot M)\, J\, Q\, V$ (the frozen interval predicate, whose mass argument absorbs G; for $G = 1$ it is* `IsNewtonSolutionOn` *(*`actualMasses` *$1\, m\, \mu)\, J\, Q\, V$ as in D25), and for $J = \mathbb{R}$ the frozen all-time predicate* `IsNewtonSolution` *G (*`actualMasses` *$m_0\, m\, \mu)\, Q\, V$ (Definition 1.20). The conclusions are literally the three clauses of those predicates at each $t \in J$, with no continuity or uniformity in t used.*

Frozen Lean declaration: `KAM.Planetary.IsNewtonSolution`, `KAM.Planetary.IsNewtonSolutionOn`, `KAM.Planetary.IsMovingSolutionOn`

Source: Fej04r §6.1, p. 46 (the Newton curves are projections of the integral curves of the Hamiltonian; "Si l'on exclut les collisions"); G restored; ours as stated (pointwise for arbitrary J).

Proof. **Step S8.17: Coordinates of the equations.** Fix $t \in J$. By (R1)–(R2) and Prop. 8.6 (curve statement), at t: $q_i' = \frac{p_i}{m_i} + \frac{\mu}{m_0} P =: a_i$ and $p_i' = -G m_0 m_i g_i(q) + \mu G m_i f_i(q) =: b_i$, for all i, where $q = q(t)$, $p = p(t)$. Since `bodyVector` is a coordinate projection, `bodyVector`$(q(\cdot), i)$ has derivative a_i at t ((E1)), and so does every linear combination.

Step S8.18: (a). By Lemma 8.10(a), $\ell(Q_i - Q_0) = \ell(q_i) > 0$, $\ell(Q_j - Q_i) = \ell(q_j - q_i) > 0$ by (R1) (the Lean `NoCollision` gives positivity of $\ell(q_i - q_j)$ for $i \neq j$, and ℓ is even); and $\ell(Q_0 - Q_i) = \ell(Q_i - Q_0)$.

Step S8.19: (b), Sun. $Q_0 = x_0 = -\frac{\mu}{M_*} \sum_i m_i q_i$ is a constant multiple of a sum of the q_i, with M_* independent of t; so $Q_0' = -\frac{\mu}{M_*} \sum_i m_i a_i = -\frac{\mu}{M_*} \big(P + \frac{\mu}{m_0} P \sum_i m_i\big) = -\frac{\mu}{M_*} \cdot \frac{M_*}{m_0} P = -\frac{\mu}{m_0} P = V_0$, because $\sum_i m_i a_i = P + \frac{\mu}{m_0} P \sum_i m_i = P\, M_*/m_0$.

Step S8.20: (b), planets. $Q_i = Q_0 + q_i$ has derivative $V_0 + a_i = V_0 + (V_i - V_0) = V_i$ by Lemma 8.10(b).

Step S8.21: (c), planets. $V_i = p_i/m_i$ has derivative $b_i/m_i = -G m_0 g_i(q) + \mu G f_i(q) = \frac{G m_0 (Q_0 - Q_i)}{\ell(Q_0 - Q_i)^3} + \sum_{j \neq i} \frac{G \mu m_j (Q_j - Q_i)}{\ell(Q_j - Q_i)^3}$, using $Q_0 - Q_i = -q_i$ and (a); the right-hand side is the sum over $l \neq i$ of $G M_l (Q_l - Q_i)/\ell(Q_l - Q_i)^3$ (with $M_0 = m_0$, $M_j = \mu m_j$; the sum over $l \in$ `Fin`$(n+1)$ is split as $l = 0$ and $l = \mathrm{succ}(j)$, `Fin.sum_univ_succ`, (E4)).

Step S8.22: (c), Sun. $V_0 = -\frac{\mu}{m_0} \sum_i p_i$ has derivative $-\frac{\mu}{m_0} \sum_i b_i = \mu G \sum_i m_i g_i(q) - \frac{\mu^2 G}{m_0} \sum_i m_i f_i(q) = \mu G \sum_i m_i g_i(q)$ by Lemma 8.5. By Lemma 8.10(a) ($Q_i - Q_0 = q_i$) this is $\sum_{l \geq 1} G M_l (Q_l - Q_0)/\ell(Q_l - Q_0)^3$ with $M_l = \mu m_l$.

Step S8.23: Packaging. The three bullet clauses are conjunctions over $t \in J$ of (a), (b), (c). For $J = \mathbb{R}$ and the frozen `IsNewtonSolution` (Definition 1.20) they are literally the three conjuncts; `IsNewtonSolutionOn` differs by the restriction to $t \in J$ and by having G absorbed in the mass, replace M by $G \cdot M$. The case $\mu = 0$ is included: then $M_l = 0$ for planets, $Q_0 = 0$ and the equations are the test-particle equations. □

Remark 8.12 (Units and the D25 predicates). The frozen `IsMovingSolutionOn` fixes the Hamiltonian parameters $G = m_0 = 1$ (the D25 normalisation, `OrbitalUnit`). The frozen `IsNewtonSolutionOn` instead uses $G = 1$ in its force formula and accepts an *arbitrary mass array*, with no restriction making its Sun entry 1: it absorbs G into the masses, so the general-G application uses the array $(k \mapsto G\,M_k)$, and only the normalised D25 application has the array $\texttt{actualMasses}\,1\,m\,\mu$. Proposition 8.11 then states: if z satisfies $\texttt{IsMovingSolutionOn}\ m\,\mu\,J$ (Hamiltonian solution on J in $B =$ univ, plus `NoCollision`), then $\texttt{IsNewtonSolutionOn}\,(\texttt{actualMasses}\,1\,m\,\mu)\,J\,Q\,V$ for $Q = \texttt{barycentricPositions}\,1\,m\,\mu\,q$, $V = \texttt{barycentricVelocities}\,1\,m\,\mu\,p$. The interval need not be open or connected; each time is treated separately. Track N's Prop. [N:`bpN:p3:newton-reduction`] covers only the all-time predicates (a `IsSolIn` reformulation) and is not needed here; the equivalence it gives for `IsNewtonSolutionOn` is the same two-line argument with J in place of $\mathbb{R}$.

Proposition 8.13 (Reconstruction: fixed Sun)**.** *Let $G, m_0 > 0$, $m_i > 0$, $\varepsilon \in \mathbb{R}$, $J \subset \mathbb{R}$ arbitrary, and $z = (q,p) : \mathbb{R} \to \mathrm{Ph}(2n)$ with, for $t \in J$:* $\mathrm{NoColl}(q(t))$ *and z has derivative $X_{H_{\mathrm{fix}}}(z(t))$ at t ($H_{\mathrm{fix}} = \texttt{fixedHamiltonian}\,G\,m_0\,m\,\varepsilon$). Put $Q_i(t) = q_i(t)$, $V_i(t) = \frac{1}{m_i} p_i(t)$. Then for all $t \in J$: for every i, $\ell(Q_i(t)) > 0$; for all i, j with $i \neq j$, $\ell(Q_j(t) - Q_i(t)) > 0$; for every i, Q_i has derivative $V_i(t)$ at t, and V_i has derivative*

$$-\frac{G m_0}{\ell(Q_i)^3} Q_i + \varepsilon \sum_{j \neq i} \frac{G m_j}{\ell(Q_j - Q_i)^3} (Q_j - Q_i)$$

at t. For $J = \mathbb{R}$ this is $\texttt{IsFixedSolution}\,G\,m_0\,m\,\varepsilon\,Q\,V$ (Definition 1.21).
Frozen Lean declaration: `KAM.Planetary.IsFixedSolution`
Source: Ours (differentiating H_{fix}); after Fej04r (28).

Proof. **Step S8.24: Equations.** Prop. 8.6 gives $q_i' = p_i/m_i$ and $p_i' = -G m_0 m_i g_i + \varepsilon G m_i f_i$. Positivity: `NoCollision` and evenness of ℓ. $V_i = p_i/m_i$ has derivative $p_i'/m_i = -G m_0\, q_i/\ell(q_i)^3 + \varepsilon G f_i(q)$, which is the displayed expression since $f_i = \sum_{j \neq i} m_j (q_j - q_i)/\ell(q_j - q_i)^3$. □

Corollary 8.14 (Orbit transfer for tori)**.** *Let $d = 2n$, $K : \mathbb{R}^d \to \mathrm{Ph}(2n)$ and $w \in \mathbb{R}^d$.*

(a) If $\mathrm{IsPhysTorus}(H_{\mathrm{mov}}, w, K)$ *and* $\mathrm{NoColl}(K(\phi)_1)$ *for all ϕ, then for every ϕ*

$$\texttt{IsNewtonSolution}\,G\,(\texttt{actualMasses}\,m_0\,m\,\mu)\,(t \mapsto Q(K(\phi + tw)_1))\,(t \mapsto V(K(\phi + tw)_2)),$$

clause (b) of MovKol *(Definition 1.25); here $Q = \texttt{barycentricPositions}\,m_0\,m\,\mu$, $V = \texttt{barycentricVelocities}\,m_0\,m\,\mu$.*

(b) If $\mathrm{IsPhysTorus}(H_{\mathrm{fix}}, w, K)$ *and* $\mathrm{NoColl}(K(\phi)_1)$ *for all ϕ, then $\texttt{IsFixedSolution}\,G\,m_0\,m\,\varepsilon\,(t \mapsto \texttt{body}(K(\phi + tw)_1))\,(t \mapsto \frac{1}{m_i}\texttt{body}(K(\phi + tw)_2))$ for all ϕ.*

Proof. **Step S8.25: The curve.** Fix ϕ and put $z(t) = K(\phi + tw)$. Clause (viii) of Definition 1.24 (applied at $\theta = \phi$ and the time t) says precisely that $s \mapsto K(\phi + sw)$ has derivative $X_H(K(\phi + tw))$ at $s = t$, i.e. (R2) of Prop. 8.11 for every $t \in \mathbb{R}$; (R1) is the hypothesis at the angle $\phi + tw$.

Step S8.26: Apply. Proposition 8.11 (resp. 8.13) with $J = \mathbb{R}$. □

8.6. **Tube geometry: collision-free, uniform pair bounds.**

Definition 8.15 (Axis constants; tube radius; d, D)**.** Let $a \in \mathbb{R}^n$ with $a_i > 0$ pairwise distinct (in P-PLANAR $n = 2$, $0 < a_0 < a_1$; in P-FIXED $0 < a_0 < \cdots < a_{n-1}$), and $\eta > 0$. Put

$$a_{\max} = \max_i a_i, \qquad \gamma_a = \min\Big\{\min_i a_i,\ \min_{i\neq j}|a_i - a_j|\Big\} > 0$$

(for $n = 1$ the inner minimum over pairs is omitted), and

$$\zeta_* = \min\{\eta, \gamma_a/6\}, \qquad d = \gamma_a/2, \qquad D = 2a_{\max} + \gamma_a.$$

Then $0 < d < D$; d, D depend only on a (not on $\eta, G, m_0, m, \mu, \varepsilon$), and ζ_* only on (a, η).

Lemma 8.16 (Tube geometry)**.** *Let $G, m_0 > 0$, $m_i > 0$, a, η, ζ_*, d, D as in Definition 8.15 and $0 < \zeta \le \zeta_*$. For every $z = (q, p) \in U_\zeta$ (Definition 1.17, tube radius ζ):*

(a) $U_\zeta \subset U_{\zeta_*} \subset U_\eta$.

*(b) (*Euclidean positions*) there is $\phi \in \mathbb{R}^n$ with $\ell\big(q_i - a_i e(\phi_i)\big) < \sqrt{2}\,\zeta$ for all i, $e(\phi) = (\cos 2\pi\phi, \sin 2\pi\phi)$.*

*(c) (*distances*) for all i and all $i \neq j$:*

$$a_i - \sqrt{2}\zeta < \ell(q_i) < a_i + \sqrt{2}\zeta, \qquad |a_i - a_j| - 2\sqrt{2}\zeta < \ell(q_j - q_i) < a_i + a_j + 2\sqrt{2}\zeta.$$

(d) $d \le \ell(q_i) \le D$ and $d \le \ell(q_j - q_i) \le D$; in particular NoColl$(q)$.

(e) For every $\mu \ge 0$*:* PairBounds$\big(d, D;\ \texttt{barycentricPositions}\, m_0\, m\, \mu\, q\big)$ *(all pairs including the Sun), with the same d, D* for all μ*; and* PairBounds$\big(d, D;\ (0, q_0, \ldots, q_{n-1})\big)$ *(the Sun at the origin, fixed-Sun clause of* FixKol*).*

Proof. **Step S8.27: (a).** The tube is defined by a strict bound on $\|z - \mathrm{circ}(\phi)\|$, so it is monotone in the radius.

Step S8.28: (b) Max norm to Euclidean norm. The phase norm is the maximum of the two coordinate max norms (Part 0, conventions), so $\|z - \mathrm{circ}(\phi)\| < \zeta$ gives $|q_{(i,c)} - a_i e(\phi_i)_c| < \zeta$ for $c = 1, 2$, i.e. $|q_i - a_i e(\phi_i)|_\infty < \zeta$; Lemma 8.1(b) gives the Euclidean bound with the factor $\sqrt{2}$. *This factor is not optional*: the planar Euclidean distance of a point of the max-norm tube from the circle may be $\sqrt{2}\zeta$.

Step S8.29: (c). Let $u_i = a_i e(\phi_i)$, so $\ell(u_i) = a_i$ ($\cos^2 + \sin^2 = 1$). Reverse triangle inequality (E5): $|\ell(q_i) - a_i| \le \ell(q_i - u_i) < \sqrt{2}\zeta$. For $i \neq j$: $\ell(q_j - q_i) \ge \ell(u_j - u_i) - \ell(q_j - u_j) - \ell(q_i - u_i) > \ell(u_j - u_i) - 2\sqrt{2}\zeta$ and $\ell(q_j - q_i) < \ell(u_j - u_i) + 2\sqrt{2}\zeta$; finally $|a_j - a_i| = |\ell(u_j) - \ell(u_i)| \le \ell(u_j - u_i) \le \ell(u_j) + \ell(u_i) = a_i + a_j$.

Step S8.30: (d) Numerics. Since $\zeta \le \gamma_a/6$: $\sqrt{2}\zeta \le 0.2358\,\gamma_a < \gamma_a/4$ and $2\sqrt{2}\zeta \le 0.4715\,\gamma_a < \gamma_a/2$ (as $\sqrt{2} < 1.4143$). As $a_i \ge \gamma_a$ and $|a_i - a_j| \ge \gamma_a$: $\ell(q_i) > a_i - \gamma_a/4 \ge \frac{3}{4}\gamma_a \ge d$ and $\ell(q_j - q_i) > \gamma_a - \gamma_a/2 = d$. Upper: $\ell(q_i) < a_{\max} + \gamma_a/4 < D$ and $\ell(q_j - q_i) < 2a_{\max} + \gamma_a/2 < D$. Positivity of d gives NoColl(q) (Definition 1.18; $\ell(q_i - q_j) = \ell(q_j - q_i)$).

Step S8.31: (e) Exact μ-independence. By Lemma 8.10(a), for $Q = \texttt{barycentricPositions}\, m_0\, m\, \mu\, q$: $\ell(Q_i - Q_0) = \ell(q_i)$ and $\ell(Q_j - Q_i) = \ell(q_j - q_i)$ for all planets $i \neq j$ and *all* μ; and PairBounds asks $d \le \ell(Q_l - Q_k) \le D$ for all ordered pairs $k \neq l$ of $\texttt{Fin}(n+1)$, which splits into the three cases (Sun–planet in both orders, planet–planet) covered by (d) and evenness. The fixed-Sun configuration has $Q_0 = 0$, $Q_i = q_i$, the same argument. No estimate $|x_0| = O(\mu)$ is used; the identity $x_i - x_j = q_i - q_j$ makes the bounds independent of the position of the barycentre. □

Remark 8.17 (On collision exclusion from track I). Track I's lemma [I:`bpI:p8:collision-exclusion`] (distances to points that move by $\le c|\mu|$ are bounded below by half the unperturbed distance for $|\mu| \le \delta/2c$) and corollary [I:`bpI:p8:min-distance`] (compact collision-free sets of configurations have a positive minimal separation) are *not* needed for Parts 12–13: after reducing to heliocentric coordinates q the only displaced mass is the Sun, and $Q_i - Q_0 = q_i$ holds exactly. They would be needed for distances to the *barycentre* (not a frozen clause), for which Lemma 8.10(d) gives $\ell(x_0) \le \mu \sum_i m_i \ell(q_i)/m_0 = O(\mu)$. The explicit constants of Definition 8.15 replace the compactness argument of [I:`bpI:p8:min-distance`] for the same purpose on the tube.

Corollary 8.18 (Newton solutions staying in the tube, on any set of times)**.** *Let $z : \mathbb{R} \to \mathrm{Ph}(2n)$, $J \subset \mathbb{R}$, $\mu \ge 0$, $0 < \zeta \le \zeta_*$ with $z(t) \in U_\zeta$ and z having derivative $X_{H_{\mathrm{mov}}}(z(t))$ at t, for all $t \in J$. Then (R1)–(R2) of Prop. 8.11 hold, so* `IsNewtonSolutionOn` $(G \cdot \mathrm{mass}_\mu)\, J\, Q\, V$ *holds, and moreover* $\mathrm{PairBounds}(d, D; Q(t))$ *for every $t \in J$ with the constants of Def. 8.15. The analogous statement holds for H_{fix} with Prop. 8.13.*

Proof. (R1) is Lemma 8.16(d) at each $t \in J$, (R2) is a hypothesis; apply Prop. 8.11 and Lemma 8.16(e). □

8.7. **Positive finite measure.**

Lemma 8.19 ($0 < \mathrm{meas}\, S < \infty$)**.** *Let $G, m_0 > 0$, $m_i > 0$, $a_i > 0$, $\eta > 0$.*

*(a) (*finiteness*) U_η is bounded: every $z \in U_\eta$ has $|z_1|_\infty < a_{\max} + \eta$ and $|z_2|_\infty < P_{\max} + \eta$, $P_{\max} = \max_i m_i \sqrt{G m_0 / a_i}$. Hence every $S \subset U_\eta$ has $\mathrm{meas}\, S \le \big(2(a_{\max}+\eta)\big)^{2n} \big(2(P_{\max}+\eta)\big)^{2n} < \infty$.*

*(b) (*positivity from the chart*) Let Ψ be a c-chart on an open $O \subset \mathrm{Ph}(2n)$ with the lattice compatibility (Q1)–(Q2) of Theorem 2.9 for $\sigma_k = 0$ ($d = 2n$), and let $A \subset O$ be Lebesgue measurable with $A + \iota_k = A$ for all $k \in \mathbb{Z}^{2n}$, and $\mathrm{meas}\, A > 0$. Then $S = \Psi(A)$ is measurable and $0 < \mathrm{meas}\, S = c^{2n}\, \mathrm{meas}(A \cap Q)$ (Q the unit cell, Lemma 2.19); if A is Borel then S is Borel (the frozen* `MeasurableSet`*). If moreover $S \subset U_\eta$ then $0 < \mathrm{meas}\, S < \infty$.*

Proof. **Step S8.32: (a).** For $z \in U_\eta$ pick ϕ with $\|z - \mathrm{circ}(\phi)\| < \eta$. The coordinates of $\mathrm{circ}(\phi)$ are bounded by a_i (position, $|a_i \cos 2\pi\phi_i| \le a_i$) and $m_i\sqrt{G m_0/a_i}$ (momentum); the triangle inequality gives the bounds. So U_η lies in a box and (E6) gives the measure bound.

Step S8.33: (b). Lemma 2.19 (with its Borel version) gives measurability, the formula and $0 < \mathrm{meas}\, \Psi(A) \iff 0 < \mathrm{meas}\, A$; the last sentence combines with (a). □

Remark 8.20 (Use). In P-PLANAR/P-FIXED, S is the image under the symplectic chart Φ of the chart-level union of whole tori of positive measure (Part 11, [P:`bpP:p1:measure-use`]; the Borel version of the union is Lemma 2.21(B1)); the containment $S \subset U_\eta$ is the tube containment of Part 12 and the finiteness is then free.

8.8. **Assembly of the physical clauses.**

Proposition 8.21 (Physical clauses of MovKol)**.** *Let $G, m_0 > 0$, $m_i > 0$, $a_i > 0$ pairwise distinct, $\eta > 0$, and ζ_*, d, D as in Definition 8.15. Let $\mu \ge 0$ and $S \subset \mathrm{Ph}(2n)$ satisfy:*

(S1) S is measurable (the frozen `MeasurableSet` *S, i.e. Borel measurable in the standard ambient measurable space; the Borel version of Lemma 8.19, not mere Lebesgue-completion measurability, is the one used) and $S \subset U_{\zeta_*}$;*

(S2) every $z \in S$ has $w \in \mathbb{R}^{2n}$, K, θ with $K(\theta) = z$, $K(\mathbb{R}^{2n}) \subset S$ and $\mathrm{IsPhysTorus}(H_{\mathrm{mov}}, w, K)$.

Then $\mathrm{MovKol}(G, m_0, m, a, \mu, \eta, d, D, S)$ (Definition 1.25). The constants d, D depend on a only, so in P-PLANAR they may be fixed before *μ_0 and μ; the dependence on the coupling is confined to (S1)–(S2).*

Proof. **Step S8.34: Tube and set.** $S \subset U_{\zeta_*} \subset U_\eta$ (Lemma 8.16(a)); S is measurable by (S1).

Step S8.35: Collision. For $z \in S$ and $\phi \in \mathbb{R}^{2n}$: $K(\phi) \in K(\mathbb{R}^{2n}) \subset S \subset U_{\zeta_*}$, so $\mathrm{NoColl}(K(\phi)_1)$ by Lemma 8.16(d).

Step S8.36: Newton. Corollary 8.14(a).

Step S8.37: Pair bounds. Lemma 8.16(e) at $K(\phi)$; as ϕ is arbitrary this covers every point $K(\phi + tw)$ of every orbit. □

Proposition 8.22 (Physical clauses of FixKol)**.** *Under the same hypotheses with H_{fix} and $\varepsilon \in \mathbb{R}$ in place of H_{mov} and μ in (S2),* $\mathrm{FixKol}(G, m_0, m, a, \varepsilon, \eta, d, D, S)$ *holds. The proof is that of Prop. 8.21 with Cor. 8.14(b) and the fixed-Sun case of Lemma 8.16(e).*

8.9. **Totality, sources, deviations, forward references.**
Totality. Every division by $\ell(\cdot)^3$ or $\ell(\cdot)$ is guarded by `NoCollision` or by $\ell > 0$ from the Newton predicates (Remark 1.27); the derivative of H is used only on W where H is C^∞ (Lemma 8.2); $M_* = m_0 + \mu \sum m_i > 0$ and $m_i > 0$ guard the barycentric divisions; $1/\ell(q_j - q_i)$ appears in the tube bounds only after $d > 0$.
Sources read. Fej04r pp. 46–47 (Newton equations, Hamiltonian, heliocentric symplectic coordinates, reduction, formula (28) with μ_j, M_j), Fej07e p. 3 (`Formule (28). Remplacer` $m_i m_j$ `par` $m_j m_k$); the revised printing's (28) already has $m_j m_k$.
Deviations. (1) Fej04r works with $G = 1$; G is restored throughout, with the frozen H_{mov} (in which the Kepler part uses planet mass m_i, *not* the reduced mass; the term $\mu|p_i|_2^2/2m_0$ is part of the perturbation, Remark 8.4). (2) The tube sketch of `PLAN.md` ($|q^{(i)}| \in [a_i - \eta, a_i + \eta]$, "barycentric Sun displacement $\le \mu C$") is replaced by the exact statement with the factor $\sqrt{2}$ of the max-to-Euclidean conversion (route study, `p11-route-study.md`, "Uniform tube and pair bounds"), and the μ-dependence of the barycentre is removed altogether by Lemma 8.10(a). (3) The statements are for arbitrary $J \subset \mathbb{R}$, not only $\mathbb{R}$.
Suspected source errata. None new. Fej04r (28) was printed with $m_i m_j$ in the original and corrected in Fej07e p. 3 (and in the revised printing); in the original-formula direction the reduction (Prop. 8.8(b)) shows the pair potential is $m_j m_k/\ell(X_j - X_k)$. One bookkeeping remark: Fej04r p. 46 writes $\epsilon\omega$ for the reduced symplectic form and F for the Hamiltonian "relative to ω"; this is the scaling $P_i = \mu p_i$, $F_\mu = \mu H_{\mathrm{mov}}$ of Prop. 8.8(b).
Doubts about frozen statements. None. The frozen `IsNewtonSolution` is formulated with the actual masses and G, and `MovingKolmogorovSet` (c) requires `PairBounds` for the barycentric positions of every orbit point; both are proved above. Note the frozen `IsMovingSolutionOn` fixes $G = m_0 = 1$ (D25) while P-PLANAR keeps G, m_0 general; Prop. 8.11 covers both.
Forward references. Part 12 ([P:`bpP:p12:physical`], [P:`bpP:p12:quantifiers`]), Part 13 ([P:`bpP:p13:assembly`]) cite Props. 8.21, 8.22 and Lemma 8.19; the D22/D25 additions (Part 14) cite Props. 8.11, 8.13 and Prop. 8.8(c). Track I cites Lemma 8.5 and Prop. 8.6 if it needs the planar n-body gradient.
Label index. `bpP:p6:plane`, `domain`, `split`, `kepler-part`, `action-reaction`, `hamilton-eqs`, `nbody`, `reduction`, `bary-algebra`, `reconstruction`, `units`, `reconstruction-fixed`, `orbit-transfer`, `tube-constants`, `tube-geometry`, `vs-I`, `newton-in-tube`, `measure-positive`, `movkol-assembly`, `fixkol-assembly`. The plan's names `bpP:p6:tube-geometry`, `measure-positive` are kept; `bpP:p6:reduction` and `bpP:p6:reconstruction` are the plan's.

9. Kepler problem and Poincaré coordinates

The unperturbed part of both physical models is a sum of n decoupled planar Kepler problems (Lemma 8.3). This part gives the coordinates in which that sum is integrable *and* regular at the circular orbits: the Poincaré chart $(\lambda, \Lambda, \xi, \eta)$. The results are:

- `kepler` (Prop. 9.3): the planar Kepler problem of a planet of mass m in the field of a centre of strength $\kappa = Gm_0$, through the mass/strength dictionary of [I:`bpI:p7:scaling`]: the Delaunay map with mass is symplectic (factor one) in the variables (ℓ, g, Λ, C) with $\Lambda = m\sqrt{\kappa a}$, $H_{\mathrm{Kep}} = -\kappa^2 m^3/(2\Lambda^2)$, and the mean values $\langle 1/|q|\rangle$, $\langle |p|^2\rangle$, $\langle |q|\rangle$;
- `regular-kepler` (Prop. 9.4): Kepler's equation in the eccentric longitude $F - v_0 \sin F + v_1 \cos F = \lambda$, with the eccentricity *vector* v as parameter, solved analytically (real, and holomorphically near $v = 0$), with no angle of perihelion;
- `poincare` (Thm. 9.7): the Poincaré chart, real analytic through $\xi = \eta = 0$ (circular orbits), a 1-chart in the sense of Definition 2.3, bijective modulo 2π in λ onto the prograde bound region, with the inverse formulas; Prop. 9.8 is its holomorphic extension near the circular torus;
- `product-chart` (Thm. 9.12): n independent charts; the *same* chart serves H_{mov} and H_{fix}, with $H^{\Psi} = h(\Lambda) + \mu f_{\mathrm{mov}} \circ \Psi$, $h(\Lambda) + \varepsilon f_{\mathrm{fix}} \circ \Psi$; the tube lemma, and the lattice hypotheses (Q1), (Q2) of Theorem 2.9 (Lemmas 9.14, 9.15);
- `eccentric-longitude` (Prop. 9.16): the D25 secular definitions (`ellipsePosition`, `ellipseMomentum`, `keplerMeanLongitude`, `longitudeJacobian`, `secularEccVector`, ...) are exactly the chart in the eccentric longitude F; and the change of variable $\lambda \leftrightarrow F$ in angular averages (Cor. 9.17).

Ownership. The Kepler fundamentals are owned by track I Part 7 and are cited, not repeated: Kepler's equation on a complex tube around the real eccentricity interval, including $e = 0$ ([I:`bpI:p7:kepler-eq`]), the Delaunay orbit maps ([I:`bpI:p7:orbit-maps`], [I:`bpI:p7:orbit-maps-e`]), their identities ([I:`bpI:p7:orbit-identities`]), the circular endpoint ([I:`bpI:p7:circular`]), the mean values ([I:`bpI:p7:means`]), the dictionary ([I:`bpI:p7:scaling`]), and the Delaunay chart ([I:`bpI:p7:symplectic`], [I:`bpI:p7:chart`]). Part 7 owns the Poincaré chart and the product chart.

9.1. **Conventions, external facts, notation.** Throughout $m > 0$ is the planet mass and $\kappa = Gm_0 > 0$ the strength of the centre; $|\cdot|_2$ is the planar length (Definition 1.4); angles λ, ℓ, g, F are in *radians* (period 2π), as in the D25 secular definitions; the passage to the frozen period-one torus parameter is Lemma 2.11 and is made in Lemma 9.15. The frozen sign convention is that of Definition 1.2: a phase point is $z = (z_1, z_2)$ with positions first, and $\Omega = \sum dz_{2,i} \wedge dz_{1,i}$. In the Poincaré coordinates of one planet $(\lambda, \xi; \Lambda, \eta)$ the positions are (λ, ξ) and the momenta (Λ, η) (this is the order of `secularGamma`: $\Gamma = (\xi^2 + \eta^2)/2 = R(\eta, \xi)$, Definition 1.7), so

$$\Omega_{\mathrm{chart}} = d\Lambda \wedge d\lambda + d\eta \wedge d\xi.$$

Write $J_0 v = (-v_1, v_0)$ (`quarterTurn`), $\mathsf{R}(g)$ for the rotation by g, $e_g = \mathsf{R}(g)(1,0) = (\cos g, \sin g)$, and $\mathrm{cr}(q,p) = q_0 p_1 - q_1 p_0$ (`planeCross`). The Delaunay map of track I (strength one, unit mass) is written $\mathsf{Q}^{\mathrm{I}}(\ell, g, L, G)$ ([I:`bpI:p7:orbit-maps`]); it satisfies $(\mathsf{Q}^{\mathrm{I}})^*\big(\sum dq_i \wedge dp_i\big) = d\ell \wedge dL + dg \wedge dG$ ([I:`bpI:p7:symplectic`]).

External facts.

(E1) *Real-analytic inverse function theorem* (as (X2), with analytic regularity): a real-analytic map with invertible differential at z_0 is a real-analytic diffeomorphism from a neighbourhood of z_0 onto an open set; in particular an implicit equation $\Phi(F, x) = 0$ with $\partial_F \Phi \neq 0$ has a locally unique real-analytic solution $F(x)$. A bijective real-analytic map between open sets with everywhere invertible differential is a real-analytic diffeomorphism.

(E2) *Holomorphic calculus.* Sums, products, compositions and quotients (nonvanishing denominator) of holomorphic functions of several variables are holomorphic; $\sqrt{\cdot}$ is holomorphic on $\{\mathrm{Re} > 0\}$; a locally uniform limit of holomorphic functions is holomorphic ([I:`bpI:p7:x-hol`], [I:`bpI:p7:x-weier`]).

(E3) Banach's fixed point theorem, the mean value inequality on a convex set ([K:bp:p1:ext-banach], [K:bp:p1:ext-mvt]).
(E4) *Exterior calculus on open sets of* $\mathbb{R}^N$ ([I:bpI:p7:x-forms]): d commutes with pull-back; the identity $\Omega(Au, Av) = c\,\Omega(u, v)$ for the differential $A = D\Psi$ holds iff the pulled-back 2-forms satisfy $\Psi^*\Omega = c\,\Omega$ (Lemma 2.1(c)).
(E5) A continuous identity between continuous functions which holds on a dense subset of an open set holds on the whole set.
(E6) *Change of variables in* $\mathbb{R}$ ([I:bpI:p7:x-change]).
(E7) Compact sets and open sets: if $K \subset V \subset \mathbb{R}^N$ with K compact and V open, there is $\delta > 0$ such that every point within max-norm distance $< \delta$ of some point of K lies in V (the neighbourhood form; it holds for empty K and for $V = \mathbb{R}^N$, so no totalized distance to the complement is used).

9.2. The planar Kepler problem with mass m and strength κ.

Definition 9.1 (Kepler data)**.** For $(q, p) \in \mathbb{R}^2 \times \mathbb{R}^2$, $q \neq 0$, put

$$E_{m,\kappa}(q,p) = \frac{|p|_2^2}{2m} - \frac{\kappa m}{|q|_2}, \qquad C(q,p) = \mathrm{cr}(q,p), \qquad a = -\frac{\kappa m}{2E_{m,\kappa}}, \qquad \Lambda = m\sqrt{\kappa a} \ \ (E_{m,\kappa} < 0).$$

The *prograde bound region* is $\mathcal{K}_{m,\kappa} = \{(q,p) : q \neq 0,\ E_{m,\kappa} < 0,\ C > 0\}$ (open: strict inequalities between continuous functions on $\{q \neq 0\}$). On it the eccentricity is $e = \sqrt{1 - C^2/\Lambda^2}$ (Lemma 9.2: $0 < C \leq \Lambda$). The *Delaunay map with mass* is defined on $\mathcal{D}^m = \{(\ell, g, \Lambda, C) \in \mathbb{R}^4 : \Lambda > C > 0\}$ by

$$\mathsf{Q}_{m,\kappa}(\ell, g, \Lambda, C) = \mathsf{S}_{m,\kappa} \circ \mathsf{Q}^{\mathrm{I}}\Big(\ell, g, \frac{\Lambda}{m}, \frac{C}{m}\Big), \qquad \mathsf{S}_{m,\kappa}(Q,P) = \Big(\frac{Q}{\kappa},\ m\kappa P\Big).$$

For $\kappa = m = 1$ this is Q^{I}. For $E_{m,\kappa} < 0$ and $m = 1$ the quantity $\Lambda = \sqrt{\kappa a}$ and the orbit is the one of E_κ in [I:bpI:p7:scaling].
Frozen Lean declaration: `KAM.Planetary.keplerEnergy, keplerAxis, keplerAction, planeCross (the case` $\kappa = 1$`, ` n `planets componentwise)`
Source: Fej04r (31)–(32), p. 47–48 ($\Lambda_j = \mu_j\sqrt{M_j a_j}$, $F_{\mathrm{Kep}} = -\mu_j^3 M_j^2/(2\Lambda_j^2)$; here the mass m_j and strength κ replace the reduced mass and total mass, see the deviations); [I:bpI:p7:scaling] for $\mathsf{S}_{m,\kappa}$ (this is its mass dictionary).

Lemma 9.2 (Range of the elements)**.** *For $(q,p) \in \mathcal{K}_{m,\kappa}$: $0 < C \leq \Lambda$, with equality iff the orbit is circular ($|q|_2$ constant). Equivalently $\Gamma := \Lambda - C \in [0, \Lambda)$.*

Proof. **Step S9.1: Scaling to strength one.** With $(Q,P) = \mathsf{S}_{m,\kappa}^{-1}(q,p) = (\kappa q, p/(m\kappa))$: $E_1(Q,P) = \frac{1}{m\kappa^2} E_{m,\kappa}(q,p)$ ([I:bpI:p7:scaling]: $E_\kappa \circ \mathsf{S}_\kappa = \kappa^2 E_1$ and $E_{m,\kappa} = mE_\kappa(q, p/m)$), $\mathrm{cr}(Q,P) = C/m$, and $L := \Lambda/m = \sqrt{-1/(2E_1(Q,P))}$.

Step S9.2: Lagrange identity. With $r = |Q|_2$, $|Q|_2^2|P|_2^2 = (Q\cdot P)^2 + \mathrm{cr}(Q,P)^2$ and $\frac{1}{2}|P|_2^2 = E_1 + 1/r = -\frac{1}{2L^2} + \frac{1}{r}$. Hence $\mathrm{cr}(Q,P)^2 = 2r - r^2/L^2 - (Q\cdot P)^2 \leq 2r - r^2/L^2 = L^2 - (r-L^2)^2/L^2 \leq L^2$, i.e. $C \leq \Lambda$; equality forces $Q \cdot P = 0$ and $r = L^2$, i.e. constant radius, and conversely. □

Proposition 9.3 (Planar Kepler problem, mass m, strength κ)**.** *Let $\mathcal{D}^m$ and $\mathsf{Q}_{m,\kappa}$ be as in Definition 9.1.*

(a) (Delaunay chart) *$\mathsf{Q}_{m,\kappa}$ is real analytic, 2π-periodic in ℓ and in g, induces a real-analytic diffeomorphism $(\mathbb{R}/2\pi\mathbb{Z})^2 \times \{\Lambda > C > 0\} \to \mathcal{K}^*_{m,\kappa} := \{(q,p) \in \mathcal{K}_{m,\kappa} : C < \Lambda\}$ (the non-circular part), and $\mathsf{Q}^*_{m,\kappa}\Big(\sum dp_i \wedge dq_i\Big) = d\Lambda \wedge d\ell + dC \wedge dg$.*

(b) (Identities) *For* $(q,p) = \mathsf{Q}_{m,\kappa}(\ell, g, \Lambda, C)$, $a = \Lambda^2/(m^2\kappa)$, $e = \sqrt{1 - C^2/\Lambda^2}$, $u = u_{\mathbb{R}}(\ell, e)$ ([I:`bpI:p7:kepler-eq`](c)): $q = a\big((\cos u - e)e_g + \sqrt{1-e^2}\,\sin u\, J_0 e_g\big)$, $|q|_2 = a(1 - e\cos u)$, $\mathrm{cr}(q,p) = C$, $E_{m,\kappa}(q,p) = -\dfrac{\kappa^2 m^3}{2\Lambda^2}$, *and the elements of Definition 9.1 are* Λ, C, a, e.

(c) (Mean values; includes the circular case $e = 0$) *For* $\Lambda > 0$, $0 < C \le \Lambda$ *and* $g \in \mathbb{R}$, *with* $(q,p) = \mathsf{Q}_{m,\kappa}(\ell, g, \Lambda, C)$ (extended to $C = \Lambda$ by the same formulas, [I:`bpI:p7:circular`]):

$$\frac{1}{2\pi}\int_0^{2\pi} \frac{d\ell}{|q|_2} = \frac{1}{a}, \qquad \frac{1}{2\pi}\int_0^{2\pi} |p|_2^2 d\ell = \frac{m^2\kappa}{a}, \qquad \frac{1}{2\pi}\int_0^{2\pi} |q|_2\, d\ell = a\Big(1 + \frac{e^2}{2}\Big).$$

(d) (Hamiltonian) $H_{\mathrm{Kep},1} := E_{m,\kappa}$ *satisfies* $H_{\mathrm{Kep},1} \circ \mathsf{Q}_{m,\kappa} = h_m(\Lambda) := -\kappa^2 m^3/(2\Lambda^2)$, *and* $h_m'(\Lambda) = \kappa^2 m^3/\Lambda^3 = \sqrt{\kappa}\, a^{-3/2}$ (Kepler's third law)*; the Hamilton equations of* $h_m(\Lambda)$ *in the variables* $(\ell, g; \Lambda, C)$ *are* $\dot\ell = h_m'(\Lambda)$, $\dot g = \dot\Lambda = \dot C = 0$.

Source: Fej04r Thm. 62 and (32)–(33), p. 48 (energy $-\mu^3 M^2/(2\Lambda^2)$*, mean motion* $\nu = M/a^{3/2}$*-type formula); Chi05 §2 for the Delaunay chart (via track I); the dictionary is* [I:`bpI:p7:scaling`]*; the mean values are standard* (track I states them for $\kappa = m = 1$).

Proof. Throughout, $L = \Lambda/m$, $G = C/m$, $(Q,P) = \mathsf{Q}^{\mathrm{I}}(\ell, g, L, G)$, $(q,p) = \mathsf{S}_{m,\kappa}(Q,P)$; note $\Lambda > C > 0$ iff $L > G > 0$, i.e. $(L,G) \in \mathcal{C}_0$ of [I:`bpI:p7:orbit-maps`].

Step S9.3: (a) Chart. $\mathsf{S}_{m,\kappa}$ is linear and invertible, and $(\Lambda, C) \mapsto (L,G) = (\Lambda, C)/m$ is a linear isomorphism of $\{\Lambda > C > 0\}$ onto $\mathcal{C}_0$. By [I:`bpI:p7:chart`], Q^{I} induces a real-analytic diffeomorphism of $(\mathbb{R}/2\pi\mathbb{Z})^2 \times \mathcal{C}_0$ onto $\mathcal{C} = \{E_1 < 0,\ \mathrm{cr} > 0,\ \mathrm{cr} < L\}$ ([I:`bpI:p7:elements`]). Under $(q,p) = \mathsf{S}_{m,\kappa}(Q,P)$ one has $E_{m,\kappa}(q,p) = m\kappa^2 E_1(Q,P)$, $\mathrm{cr}(q,p) = m\,\mathrm{cr}(Q,P)$ and (Step S9.1) $\Lambda = mL(Q,P)$; so $\mathsf{S}_{m,\kappa}$ maps $\mathcal{C}$ bijectively onto $\{E_{m,\kappa} < 0,\ C > 0,\ C < \Lambda\} = \mathcal{K}^*_{m,\kappa}$.

Step S9.4: (a) The form. $\sum dp_i \wedge dq_i = \sum (m\kappa\, dP_i) \wedge (dQ_i/\kappa) = -m \sum dQ_i \wedge dP_i$. By [I:`bpI:p7:symplectic`], pulling back by Q^{I} gives $-m(d\ell \wedge dL + dg \wedge dG) = m\, dL \wedge d\ell + m\, dG \wedge dg = d\Lambda \wedge d\ell + dC \wedge dg$.

Step S9.5: (b). [I:`bpI:p7:orbit-identities`] for (Q,P): $|Q|_2 = L^2(1 - e\cos u)$, $E_1(Q,P) = -1/(2L^2)$, $\mathrm{cr}(Q,P) = G$, with $e = \sqrt{1 - G^2/L^2} = \sqrt{1 - C^2/\Lambda^2}$. With $q = Q/\kappa$: $|q|_2 = a(1 - e\cos u)$ for $a = L^2/\kappa = \Lambda^2/(m^2\kappa)$ (this is the a of Definition 9.1: $a = -\kappa m/(2E_{m,\kappa})$ with $E_{m,\kappa} = m\kappa^2 E_1 = -m\kappa^2/(2L^2)$ gives $a = L^2/\kappa$). The formula for q is Q/κ with $Q = \mathsf{R}(g)(X,Y)$, $X = L^2(\cos u - e)$, $Y = L^2\sqrt{1-e^2}\,\sin u$ and $\mathsf{R}(g)(x,y) = xe_g + yJ_0e_g$. $\mathrm{cr}(q,p) = m\,\mathrm{cr}(Q,P) = mG = C$; $E_{m,\kappa}(q,p) = -m\kappa^2/(2L^2) = -\kappa^2 m^3/(2\Lambda^2)$.

Step S9.6: (c). For $C < \Lambda$ and $C = \Lambda$ (circular), [I:`bpI:p7:means`] and [I:`bpI:p7:circular`] give $\langle 1/|Q|_2\rangle = 1/L^2$, $\langle |P|_2^2\rangle = 1/L^2$, $\langle |Q|_2\rangle = L^2(1 + e^2/2)$. Then $1/|q|_2 = \kappa/|Q|_2$ has mean $\kappa/L^2 = 1/a$; $|p|_2^2 = m^2\kappa^2|P|_2^2$ has mean $m^2\kappa^2/L^2 = m^2\kappa/a$; $|q|_2 = |Q|_2/\kappa$ has mean $a(1 + e^2/2)$.

Step S9.7: (d). The value of $E_{m,\kappa} \circ \mathsf{Q}_{m,\kappa}$ is (b). $h_m' = \kappa^2 m^3/\Lambda^3$, and $\Lambda^3 = m^3\kappa^{3/2}a^{3/2}$ gives $\sqrt{\kappa}\, a^{-3/2}$. By (a) the chart $\mathsf{Q}_{m,\kappa}$ is symplectic with factor one in the variables $(\ell, g; \Lambda, C)$ (positions (ℓ, g), momenta (Λ, C), $\Omega = d\Lambda \wedge d\ell + dC \wedge dg$), so the Hamilton equations of $h_m(\Lambda)$ in these variables are $\dot\ell = \partial_\Lambda h_m$, $\dot g = \partial_C h_m = 0$, $\dot\Lambda = -\partial_\ell h_m = 0$, $\dot C = -\partial_g h_m = 0$, and Lemma 2.4 transports them to the Kepler flow. □

9.3. **Kepler's equation in the eccentric longitude.** The Delaunay map is singular at $e = 0$ (g is undefined). The D25 formulas and the Poincaré chart use instead the *eccentric longitude* $F = u + g$ and the *eccentricity vector* $v = e\, e_g$. Substituting $\ell = \lambda - g$, $u = F - g$ in $u - e\sin u = \ell$ gives

$$F - e\sin(F - g) = \lambda, \qquad e\sin(F - g) = v_0 \sin F - v_1 \cos F \quad (v = e(\cos g, \sin g)),$$

an equation *regular in* $v \in \mathbb{R}^2$: $F - v_0 \sin F + v_1 \cos F = \lambda$ (`keplerMeanLongitude`).

Proposition 9.4 (Regular Kepler equation)**.** *For $v \in \mathbb{R}^2$, $F \in \mathbb{R}$ put $\Phi(F, v) = F - v_0 \sin F + v_1 \cos F$ (so Φ is `keplerMeanLongitude` with $v =$ `secularEccVector`) and $J(F, v) = \partial_F \Phi = 1 - v_0 \cos F - v_1 \sin F = 1 - v \cdot (\cos F, \sin F)$ (`longitudeJacobian`).*

(a) (real) *For $|v|_2 < 1$: $J \geq 1 - |v|_2 > 0$; $F \mapsto \Phi(F, v)$ is a real-analytic increasing bijection of $\mathbb{R}$ with $\Phi(F + 2\pi, v) = \Phi(F, v) + 2\pi$. So for each $(\lambda, v) \in \mathbb{R} \times \{|v|_2 < 1\}$ there is exactly one real $F = F(\lambda, v)$ with $\Phi(F, v) = \lambda$; F is real analytic on $\mathbb{R} \times \{|v|_2 < 1\}$, $F(\lambda + 2\pi, v) = F(\lambda, v) + 2\pi$, $F(\lambda, 0) = \lambda$, and*

$$\partial_\lambda F = \frac{1}{J}, \qquad \partial_{v_0} F = \frac{\sin F}{J}, \qquad \partial_{v_1} F = -\frac{\cos F}{J} \qquad (J = J(F(\lambda, v), v)).$$

(b) (complex) *Let $\mathcal{N}_0 = \{(\lambda, v) \in \mathbb{C} \times \mathbb{C}^2 : \ |\operatorname{Im} \lambda| < \frac{1}{4}, \ |v_0| + |v_1| < \frac{1}{4}\}$ and $\mathcal{S} = \{F \in \mathbb{C} : |\operatorname{Im} F| \leq 1\}$. There is a unique holomorphic $F : \mathcal{N}_0 \to \mathcal{S}$ with $\Phi(F(\lambda, v), v) = \lambda$ (with Φ, J extended to complex arguments by the same formulas); $|J| \geq 1 - \frac{1}{4}\cosh 1 > \frac{3}{5}$ on $\mathcal{N}_0$, and the formulas of (a) hold. At real points F is real and equals the F of (a).*

(c) (relation to track I) *For $v = e\, e_g$ with $e \in (0, 1)$, $g \in \mathbb{R}$: $F(\lambda, v) = g + u_{\mathbb{R}}(\lambda - g, e)$, where $u_{\mathbb{R}}$ is the solution of Kepler's equation of* [I:`bpI:p7:kepler-eq`]*(c). (The composition of* [I:`bpI:p7:kepler-eq`] *with $v \mapsto (g, e)$ is not defined at $v = 0$, which is why (b) is proved directly: it is the holomorphic continuation of this branch through $v = 0$.)*

Source: [I:`bpI:p7:kepler-eq`] *(Kepler's equation near $e = 0$); the regular form in v is the D25 definition* `keplerMeanLongitude` *(derived from LP17 (10)–(11)); the contraction proof is ours.*

Proof. **Step S9.8: (a) Monotonicity.** $|v \cdot (\cos F, \sin F)| \leq |v|_2$ (Cauchy–Schwarz), so $J \geq 1 - |v|_2 > 0$. Hence $\Phi(\cdot, v)$ is strictly increasing with $\Phi(F, v) - F$ bounded ($\leq |v|_2$), so $\Phi(F, v) \to \pm\infty$ as $F \to \pm\infty$: a bijection of $\mathbb{R}$. $\Phi(F + 2\pi, v) = \Phi(F, v) + 2\pi$ by periodicity of $\sin, \cos$; $\Phi(F, 0) = F$.

Step S9.9: (a) Analyticity. Φ is real analytic on $\mathbb{R} \times \mathbb{R}^2$ with $\partial_F \Phi = J \neq 0$ on $\{|v|_2 < 1\}$; (E1) gives a local real-analytic solution near any (λ_0, v_0), which coincides with the unique global solution; so F is real analytic. Differentiating $\Phi(F(\lambda, v), v) = \lambda$: $J\, \partial_\lambda F = 1$; $J\, \partial_{v_0} F - \sin F = 0$; $J\, \partial_{v_1} F + \cos F = 0$.

Step S9.10: (b) Contraction. Let $\beta_{\lambda,v}(F) = \lambda + v_0 \sin F - v_1 \cos F$, so $\Phi(F, v) = \lambda \iff \beta_{\lambda,v}(F) = F$. For $F = x + iy \in \mathcal{S}$: $|\sin F|^2 = \sin^2 x + \sinh^2 y$, $|\cos F|^2 = \cos^2 x + \sinh^2 y$, both $\leq \cosh^2 y \leq \cosh^2 1$ ([I:`stepI:p7:ke-trig`]). For $(\lambda, v) \in \mathcal{N}_0$ then $|\operatorname{Im} \beta| \leq |\operatorname{Im} \lambda| + (|v_0| + |v_1|) \cosh 1 < \frac{1}{4} + \frac{1}{4} \cdot 1.55 < 1$, so $\beta_{\lambda,v}(\mathcal{S}) \subset \mathcal{S}$, and $|\beta'| = |v_0 \cos F + v_1 \sin F| \leq \frac{1}{4}\cosh 1 < 0.39$, so $\beta_{\lambda,v}$ is a 0.39-contraction of the closed convex set $\mathcal{S}$ ((E3)). It has a unique fixed point $F(\lambda, v) \in \mathcal{S}$.

Step S9.11: (b) Holomorphy, real points. $F_0 = \lambda$ (which lies in $\mathcal{S}$) and $F_{j+1} = \beta_{\lambda,v}(F_j)$ are holomorphic on $\mathcal{N}_0$ with values in $\mathcal{S}$ and converge uniformly (geometric rate 0.39, Banach), so F is holomorphic ((E2)). For real $(\lambda, v) \in \mathcal{N}_0$ the real map $\beta_{\lambda,v}$ preserves $\mathbb{R} \subset \mathcal{S}$ and is a contraction there, so its fixed point is real, equals the real solution of (a) (uniqueness there), and F is real. $|J| \geq 1 - (|v_0| + |v_1|) \cosh 1 > 1 - 0.39 > \frac{3}{5}$. The derivative formulas are the same implicit differentiation as in (a), valid for holomorphic maps.

Step S9.12: (c) Relation. For $v = e\, e_g$: $v_0 \sin F - v_1 \cos F = e(\cos g \sin F - \sin g \cos F) = e \sin(F - g)$, so $F' = g + u_{\mathbb{R}}(\lambda - g, e)$ solves $\Phi(F', v) = F' - e \sin(F' - g) = g + u_{\mathbb{R}} - e \sin u_{\mathbb{R}} = g + (\lambda - g) = \lambda$; uniqueness in (a) gives $F' = F$. □

9.4. **The Poincaré chart.**

Definition 9.5 (Poincaré chart of one planet)**.** Let $m > 0$ and $\kappa > 0$. The *Poincaré domain* is $\mathcal{O} = \{(\lambda, \xi, \Lambda, \eta) \in \mathbb{R}^4 : \ \Lambda > 0, \ \Gamma < \Lambda\}$ with $\Gamma = \Gamma(\xi, \eta) = (\xi^2 + \eta^2)/2$ (open, invariant under

$\lambda \mapsto \lambda + 2\pi$). On $\mathcal{O}$ put

$$k = 1 - \frac{\Gamma}{\Lambda} \in (0,1], \qquad v = \sqrt{\frac{1-\Gamma/(2\Lambda)}{\Lambda}}\,(\xi,\eta), \qquad a = \frac{\Lambda^2}{m^2\kappa}, \qquad F = F(\lambda, v) \ \text{(Prop. 9.4(a))},$$
$$c = (\cos F, \sin F), \qquad J = 1 - v\cdot c,$$
$$q = a\Big(k\,c - v + \frac{v\cdot c}{1+k}\,v\Big), \qquad p = \frac{m^2\kappa}{\Lambda J}\Big(J_0 c - \frac{v\cdot c}{1+k}\,J_0 v\Big),$$

and $\Psi_{m,\kappa}(\lambda,\xi,\Lambda,\eta) = (q,p) \in \mathbb{R}^2 \times \mathbb{R}^2$. For $\kappa = 1$ these are `secularEccVector`, `secularRoot` ($= k = \sqrt{1-e^2}$), `secularAxis`, `longitudeJacobian`, `ellipsePosition`, `ellipseMomentum`, and $F =$ the solution of `keplerMeanLongitude` $= \lambda$ (Proposition 9.16).

Frozen Lean declaration: `KAM.Planetary.secularEccVector, secularRoot, secularGamma, secularAxis, ellipsePosition, ellipseMomentum, keplerMeanLongitude, longitudeJacobian (for` $\kappa = 1$`)`

Source: Fej04r (31), Thm. 62, pp. 47–48, with the normalisation of Fej07e p. 3: $r_j = \xi_j + i\eta_j = \sqrt{2\Lambda_j}\sqrt{1-\sqrt{1-e_j^2}}\,e^{i(g_j+\theta_j)}$ (planar: $\theta_j = 0$); explicit regular form: LP17 (10)–(11) via the D25 definitions.

Lemma 9.6 (Elementary facts on $\mathcal{O}$)**.** *On $\mathcal{O}$: $\Gamma \geq 0$, $0 < k \leq 1$, $|v|_2^2 = 1 - k^2$ (so $|v|_2 < 1$ and F is defined, real analytic, Proposition 9.4(a)); $J \geq 1 - |v|_2 > 0$; the radicand of v is positive; $v\cdot c = e\cos(F-g)$ and $|v|_2 = e$ with $e = \sqrt{1-k^2}$. For $(\xi,\eta) \neq 0$ write $(\xi,\eta) = \rho\, e_g$ with $\rho = \sqrt{2\Gamma}$: then $v = e\, e_g$ and $\Gamma = \Lambda - C$ with $C = \Lambda k$.*

Proof. $\Gamma < \Lambda$ and $\Lambda > 0$ give $k = 1-\Gamma/\Lambda > 0$ and $1-\Gamma/(2\Lambda) > \frac{1}{2}$. $|v|_2^2 = \frac{1-\Gamma/(2\Lambda)}{\Lambda} 2\Gamma = \frac{2\Gamma}{\Lambda}\big(1-\frac{\Gamma}{2\Lambda}\big) = 1 - (1-\frac{\Gamma}{\Lambda})^2 = 1 - k^2$. The rest is Proposition 9.4(a), and for $(\xi,\eta) \neq 0$: $|v|_2/\rho = \sqrt{(1-\Gamma/2\Lambda)/\Lambda}$ with $|v|_2 = \sqrt{1-k^2}$ the eccentricity, and the direction of v is that of (ξ,η). □

Theorem 9.7 (Poincaré chart, analytic through $e = 0$)**.** *Let $\Psi = \Psi_{m,\kappa} : \mathcal{O} \to \mathbb{R}^2 \times \mathbb{R}^2$ be as in Definition 9.5.*

- *(a)* (analytic) *Ψ is real analytic on $\mathcal{O}$, including $\xi = \eta = 0$; $\Psi(\lambda + 2\pi, \xi, \Lambda, \eta) = \Psi(\lambda,\xi,\Lambda,\eta)$.*
- *(b)* (Delaunay on $\Gamma > 0$) *Let $(\xi,\eta) = \rho\, e_g$, $\rho > 0$. Then $\Psi(\lambda,\xi,\Lambda,\eta) = \mathsf{Q}_{m,\kappa}(\lambda - g,\ g,\ \Lambda,\ \Lambda - \Gamma)$, with the eccentric longitude $F = g + u$, $u = u_{\mathbb{R}}(\lambda - g, e)$.*
- *(c)* (symplectic) *$\Omega(D\Psi\, u, D\Psi\, w) = \Omega(u,w)$ on all of $\mathcal{O}$, i.e. $\Psi^*\Omega_{\mathrm{cart}} = d\Lambda \wedge d\lambda + d\eta \wedge d\xi$ with $\Omega_{\mathrm{cart}} = \sum dp_i \wedge dq_i$. Thus Ψ is a 1-chart on $\mathcal{O}$ in the sense of Definition 2.3, and $D\Psi$ is invertible.*
- *(d)* (elements) *$\Psi(\mathcal{O}) \subset \mathcal{K}_{m,\kappa}$ and for $(q,p) = \Psi(\lambda,\xi,\Lambda,\eta)$: $E_{m,\kappa}(q,p) = -\kappa^2 m^3/(2\Lambda^2)$, $a(q,p) = \Lambda^2/(m^2\kappa)$, the action of Definition 9.1 is Λ, $\mathrm{cr}(q,p) = \Lambda - \Gamma = \Lambda k$ and, the Laplace–Runge–Lenz vector satisfies $\kappa^{-1}\big[\big(\frac{|p|_2^2}{m^2} - \frac{\kappa}{|q|_2}\big)q - \frac{q\cdot p}{m^2}p\big] = v$ (for $\kappa = 1$ this is `keplerEccVector` $= v$).*
- *(e)* (bijectivity modulo 2π) *$\Psi : \mathcal{O} \to \mathcal{K}_{m,\kappa}$ is surjective, and $\Psi(\lambda',\xi',\Lambda',\eta') = \Psi(\lambda,\xi,\Lambda,\eta)$ iff $(\xi',\Lambda',\eta') = (\xi,\Lambda,\eta)$ and $\lambda' - \lambda \in 2\pi\mathbb{Z}$. The induced map $\mathcal{O}/(2\pi\mathbb{Z}\times\{0\}) \to \mathcal{K}_{m,\kappa}$ (quotient by the λ-translations) is a real-analytic diffeomorphism. Its inverse is $\Lambda = m\sqrt{\kappa a}$, $\Gamma = \Lambda - \mathrm{cr}(q,p)$, $v =$ the rescaled LRL vector of (d), $(\xi,\eta) = v/\sqrt{(1-\Gamma/(2\Lambda))/\Lambda}$, $c = \big(k\,\mathrm{Id} + \frac{1}{1+k}vv^\top\big)^{-1}(q/a + v)$, $F = \arg c$, $\lambda = \Phi(F,v)$.*
- *(f)* (circular torus) *At $\xi = \eta = 0$: $v = 0$, $k = 1$, $F = \lambda$, $q = a(\cos\lambda, \sin\lambda)$, $p = m\sqrt{\kappa/a}\,(-\sin\lambda, \cos\lambda)$, $a = \Lambda^2/(m^2\kappa)$: for $\Lambda = m\sqrt{\kappa a}$ this is `circularPoint` (with $\phi = \lambda/2\pi$, one planet).*

Proof. **Step S9.13: (a).** k, v (positive radicand, Lemma 9.6) and J are real analytic, F is real analytic (Proposition 9.4(a) composed with the analytic $v(\xi,\eta,\Lambda)$, $|v|_2 < 1$), $1 + k \geq 1$, $\Lambda J > 0$;

q, p are rational expressions in these and in $\cos F, \sin F$. Periodicity: $F(\lambda + 2\pi, v) = F + 2\pi$ and c is 2π-periodic in F.

Step S9.14: (b) Delaunay on $\Gamma > 0$. Put $\ell = \lambda - g$, $C = \Lambda - \Gamma = \Lambda k$, $L = \Lambda/m$, $G = C/m = Lk$. Then $\Lambda > C > 0$, $e = \sqrt{1 - C^2/\Lambda^2} = \sqrt{1 - k^2}$ and $v = e\, e_g$ (Lemma 9.6), so by Proposition 9.4(c) $F = g + u$, $u = u_{\mathbb{R}}(\ell, e)$. Write $\hat{e} = e_g$, $\hat{e}^{\perp} = J_0 e_g$ and $c = \cos u\, \hat{e} + \sin u\, \hat{e}^{\perp}$ ($F = u + g$). Then $v \cdot c = e \cos u$, $J = 1 - e\cos u$ and, using $e^2/(1+k) = (1-k^2)/(1+k) = 1 - k$,

$$kc - v + \frac{v \cdot c}{1+k} v = (k\cos u - e + (1-k)\cos u)\hat{e} + k \sin u\, \hat{e}^{\perp} = (\cos u - e)\hat{e} + k \sin u\, \hat{e}^{\perp},$$

so $q = a((\cos u - e)\hat{e} + k \sin u\, \hat{e}^{\perp})$, which is the position of Proposition 9.3(b). Likewise $J_0 c = \cos u\, \hat{e}^{\perp} - \sin u\, \hat{e}$, $J_0 v = e\hat{e}^{\perp}$ and

$$J_0 c - \frac{v \cdot c}{1+k} J_0 v = -\sin u\, \hat{e} + \left(1 - \tfrac{e^2}{1+k}\right) \cos u\, \hat{e}^{\perp} = -\sin u\, \hat{e} + k \cos u\, \hat{e}^{\perp} = \mathsf{R}(g) \begin{pmatrix} -\sin u \\ k \cos u \end{pmatrix}.$$

By [I:bpI:p7:orbit-maps], $P = \mathsf{R}(g)\left(\begin{smallmatrix} -\sin u \\ k\cos u \end{smallmatrix}\right)/(LJ)$ and $p = m\kappa P = \frac{m\kappa}{LJ}(\cdots) = \frac{m^2\kappa}{\Lambda J}(\cdots)$, which is the p of the definition. Hence $\Psi = \mathsf{S}_{m,\kappa} \circ \mathsf{Q}^{\mathrm{I}}(\ell, g, L, G) = \mathsf{Q}_{m,\kappa}(\ell, g, \Lambda, C)$.

Step S9.15: (c) Symplecticity on $\{\Gamma > 0\}$. On the open set $\mathcal{A} = \{(\xi, \eta) \neq 0\} \subset \mathcal{O}$ choose locally a real-analytic branch $g(\xi, \eta) = \arg(\xi + i\eta)$ and put $\rho = \sqrt{2\Gamma}$. By (b), $\Psi = \mathsf{Q}_{m,\kappa} \circ T$ with $T(\lambda, \xi, \Lambda, \eta) = (\lambda - g, g, \Lambda, \Lambda - \Gamma)$ (real analytic, values in $\mathcal{D}^m$). By Proposition 9.3(a), $\Psi^* \Omega_{\mathrm{cart}} = T^*(d\Lambda \wedge d\ell + dC \wedge dg) = d\Lambda \wedge (d\lambda - dg) + (d\Lambda - d\Gamma) \wedge dg = d\Lambda \wedge d\lambda - d\Gamma \wedge dg$. Since $\xi = \rho \cos g$, $\eta = \rho \sin g$: $d\xi \wedge d\eta = \rho\, d\rho \wedge dg = d\Gamma \wedge dg$, i.e. $d\eta \wedge d\xi = -d\Gamma \wedge dg$. So $\Psi^* \Omega_{\mathrm{cart}} = d\Lambda \wedge d\lambda + d\eta \wedge d\xi$ on $\mathcal{A}$.

Step S9.16: (c) All of $\mathcal{O}$. The matrix identity $D\Psi^{\top} M_0 D\Psi = M_0$ (equivalent to the form identity, (E4)) is an identity between continuous functions on $\mathcal{O}$ (Ψ is C^1 there by (a)), valid on the dense subset $\mathcal{A}$ (the complement $\{\xi = \eta = 0\}$ has empty interior); (E5). This is the chart property of Definition 2.3 with $c = 1$ (the factor is 1 because the source form is the frozen Ω in the order $(\lambda, \xi; \Lambda, \eta)$). Invertibility: Lemma 2.1(b).

Step S9.17: (d) Elements. On $\mathcal{A}$ by (b) and Proposition 9.3(b): $E_{m,\kappa} = -\kappa^2 m^3/(2\Lambda^2) < 0$, $\mathrm{cr} = \Lambda k > 0$, $q \neq 0$ (as $|q|_2 = a(1 - e\cos u) > 0$), hence $\Psi(\mathcal{A}) \subset \mathcal{K}_{m,\kappa}$, and $a(q,p) = -\kappa m/(2E) = \Lambda^2/(m^2\kappa)$, action $m\sqrt{\kappa a} = \Lambda$. At $\xi = \eta = 0$ the same identities are [I:bpI:p7:circular] (with $e = 0$, $k = 1$, $u = F = \lambda$) through the scaling $\mathsf{S}_{m,\kappa}$ of Step S9.5; they can also be checked directly from (f). The LRL identity: for $(Q, P) = \mathsf{S}_{m,\kappa}^{-1}(q, p)$ one computes $\kappa^{-1}\left[(\frac{|p|_2^2}{m^2} - \frac{\kappa}{|q|_2})q - \frac{q \cdot p}{m^2} p\right] = (|P|_2^2 - \frac{1}{|Q|_2})Q - (Q \cdot P)P =: A$. On $\mathcal{A}$, in the orbit frame $(\hat{e}, \hat{e}^{\perp})$ of Step S9.14 with $c_u = \cos u$, $s_u = \sin u$, $J = 1 - ec_u$, $L = \Lambda/m$ ([I:bpI:p7:orbit-identities]): $|P|_2^2 = \frac{1+ec_u}{L^2 J}$, $\frac{1}{|Q|_2} = \frac{1}{L^2 J}$, $Q = L^2(c_u - e,\, k s_u)$, $Q \cdot P = L e s_u$, $P = \frac{1}{LJ}(-s_u, kc_u)$, hence $A = \frac{ec_u}{J}(c_u - e,\, k s_u) - \frac{e s_u}{J}(-s_u,\, kc_u) = \left(\frac{e}{J}(c_u^2 - ec_u + s_u^2),\, 0\right) = (e, 0)$ in the frame, i.e. $A = e\,\hat{e} = v$. At $\xi = \eta = 0$ both sides vanish ($|Q|_2 = L^2$, $|P|_2^2 = 1/L^2$, $Q \cdot P = 0$ by (f)).

Step S9.18: (e) Bijectivity. *Injectivity modulo 2π in λ.* Let $\Psi(w') = \Psi(w)$, $w, w' \in \mathcal{O}$. By (d) $\Lambda' = \Lambda$ ($\Lambda = m\sqrt{\kappa a}$ is a function of (q,p)) and $\Gamma' = \Lambda - \mathrm{cr} = \Gamma$. If $\Gamma = 0$: $\xi = \eta = \xi' = \eta' = 0$ and by (f) $\Psi = (a(\cos\lambda, \sin\lambda), \cdots)$ so $\lambda' - \lambda \in 2\pi\mathbb{Z}$. If $\Gamma > 0$: $w, w' \in \mathcal{A}$, (ℓ, g), (ℓ', g') as in (b), and by [I:bpI:p7:chart] (injectivity modulo 2π of Q^{I}, transported by Proposition 9.3(a)) $g' - g \in 2\pi\mathbb{Z}$, $\ell' - \ell \in 2\pi\mathbb{Z}$. Then $(\xi', \eta') = \rho\, e_{g'} = \rho\, e_g = (\xi, \eta)$ and $\lambda' = \ell' + g' \in \lambda + 2\pi\mathbb{Z}$. The converse is (a). *Surjectivity.* Let $(q,p) \in \mathcal{K}_{m,\kappa}$, Λ, C its elements, $\Gamma = \Lambda - C \in [0, \Lambda)$ (Lemma 9.2). If $C < \Lambda$: $(q,p) \in \mathcal{K}_{m,\kappa}^*$ is $\mathsf{Q}_{m,\kappa}(\ell, g, \Lambda, C)$ by Proposition 9.3(a), and with $(\xi, \eta) = \sqrt{2\Gamma}\, e_g$, $\lambda = \ell + g$, (b) gives $\Psi(\lambda, \xi, \Lambda, \eta) = (q, p)$. If $C = \Lambda$ (so $\Gamma = 0$): in the scaling of Step S9.1, (Q, P) has $\mathrm{cr} = L$ and the equality case of Step S9.2 gives $|Q|_2 = L^2$, $Q \cdot P = 0$, $|P|_2 = \mathrm{cr}/|Q|_2 = 1/L$; so $Q = L^2(\cos\lambda, \sin\lambda)$ for some λ and $P = \frac{1}{L}(-\sin\lambda, \cos\lambda)$ (orthogonal to Q, of length $1/L$, positive cross product

$\mathrm{cr}(Q,P) = L > 0$). That is $(q,p) = \Psi(\lambda, 0, \Lambda, 0)$ by (f). *Diffeomorphism.* $D\Psi$ is invertible (c); Ψ is a bijective, real-analytic local diffeomorphism from the quotient by the lattice $2\pi\mathbb{Z}$ in λ onto the open set $\mathcal{K}_{m,\kappa}$ ((E1)). Inverse formulas: Λ, Γ as in (d); v by the LRL identity (d); $(\xi,\eta) = v/s$ with $s = \sqrt{(1-\Gamma/(2\Lambda))/\Lambda} > 0$ (defined also at $v=0$); $\frac{q}{a} = kc - v + \frac{v\cdot c}{1+k}v = (k\,\mathrm{Id} + \frac{1}{1+k}vv^\top)c - v$ and $k\,\mathrm{Id} + \frac{1}{1+k}vv^\top$ has eigenvalues k and $k + |v|_2^2/(1+k) = 1$, hence is invertible; c is then the unit vector $\cos F, \sin F$, and $\lambda = \Phi(F, v)$ modulo 2π.

Step S9.19: (f) Circular points. $\Gamma = 0$ gives $v = 0$, $k = 1$, $F = \lambda$ (Proposition 9.4(a)), $J = 1$, $q = a(c - 0 + 0) = a(\cos\lambda, \sin\lambda)$, $p = \frac{m^2\kappa}{\Lambda} J_0 c = m\sqrt{\kappa/a}\, J_0 c$ since $\Lambda = m\sqrt{\kappa a}$. Compare Definition 1.17. □

Proposition 9.8 (Holomorphic extension near the circular orbits)**.** *Let* $\Lambda^\circ > 0$ *and*

$$\mathcal{N}(\Lambda^\circ) = \Big\{(\lambda,\xi,\Lambda,\eta) \in \mathbb{C}^4 :\ |\mathrm{Im}\,\lambda| < \tfrac{1}{4},\ |\Lambda - \Lambda^\circ| < \tfrac{1}{2}\Lambda^\circ,\ |\xi|, |\eta| < \tfrac{1}{16}\sqrt{\Lambda^\circ}\Big\}.$$

The formulas of Definition 9.5 (with $\Gamma = (\xi^2+\eta^2)/2$, $F = F(\lambda, v)$ *of Proposition 9.4(b)) define a holomorphic map* $\Psi : \mathcal{N}(\Lambda^\circ) \to \mathbb{C}^4$, 2π*-periodic in* λ*, real at real points, with* $\mathcal{N}(\Lambda^\circ) \cap \mathbb{R}^4 \subset \mathcal{O}$ *and* $D\Psi^\top M_0 D\Psi = M_0$ *at real points. Hence it satisfies the hypotheses of Lemma 2.7 with* $c = 1$.

Proof. **Step S9.20: Sizes.** On $\mathcal{N}(\Lambda^\circ)$: $|\Lambda| > \frac{1}{2}\Lambda^\circ$ and $|\arg\Lambda| \le \arcsin\frac{1}{2} = \frac{\pi}{6}$; $|\Gamma| < \frac{1}{256}\Lambda^\circ$, so $|\Gamma/\Lambda| < \frac{1}{128}$, hence $|\arg(1-\Gamma/2\Lambda)| < 0.01$, $|1-\Gamma/2\Lambda| < 1.01$, $\mathrm{Re}\,k > 0.99$ ($k = 1-\Gamma/\Lambda$), $|1+k| > 1.9$. The radicand $(1-\Gamma/2\Lambda)/\Lambda$ has argument of modulus $< \frac{\pi}{6} + 0.01 < \frac{\pi}{2}$, so its principal square root s is holomorphic ((E2)), and $|s| \le \sqrt{1.01\cdot 2/\Lambda^\circ} < 1.5/\sqrt{\Lambda^\circ}$. Therefore $|v_0| + |v_1| \le |s|(|\xi| + |\eta|) < 1.5 \cdot \frac{2}{16} < \frac{1}{4}$, $(\lambda, v) \in \mathcal{N}_0$, and F is holomorphic with $|J| > \frac{3}{5}$ (Proposition 9.4(b)).

Step S9.21: Holomorphy, reality, periodicity. q, p are rational in $(\Lambda, k, v, \cos F, \sin F, J^{-1}, (1+k)^{-1})$, all holomorphic with nonvanishing denominators; periodicity as in Step S9.13; at real points everything is real and $\Gamma < \Lambda$, so the point lies in $\mathcal{O}$ and the real maps agree. The matrix identity at real points is Theorem 9.7(c) (the complex derivative restricts to the real one, [K:`bp:p1:real-restriction`]). □

Remark 9.9. The neighbourhood is small in v by design (a contraction in $|v_0| + |v_1| < \frac{1}{4}$); this is all that the circular-orbit analysis (Parts 2, 5, 8) needs: $|\xi|, |\eta|$ of size $\ll \sqrt{\Lambda}$ is eccentricity $e \lesssim \frac{1}{4}$ (cf. $|v|_2 = e$, Lemma 9.6). For larger eccentricities (not needed in this track) the complex tube of [I:`bpI:p7:kepler-eq`] covers $e \le e_1 < 1$ in the Delaunay form away from $e = 0$.

Remark 9.10 (Normalisation, sign conventions, what the source does not prove). (1) *Normalisation.* Fej04r (31) in the original printing, as reported by Fej07e (the pinned January-2007 revised Fej04r, p. 47, already prints $\sqrt{2\Lambda_j}$ and $\sqrt{2G_j}$), had $r_j = \xi_j + i\eta_j = \sqrt{\Lambda_j}\sqrt{1-\sqrt{1-\varepsilon_j^2}}\, e^{i(g_j+\theta_j)}$ with only $\sqrt{\Lambda_j}$, for which the chart is merely conformally symplectic (factor 2 in $d\xi \wedge d\eta$). Fej07e p. 3 ("sans le facteur $\sqrt{2}$, elles ne sont que conformément symplectiques") corrects it to $\sqrt{2\Lambda_j}$, i.e. $\frac{1}{2}(\xi^2+\eta^2) = \Lambda - C = \Gamma$ (Lemma 9.6), which is exactly `secularGamma`: $\Gamma = R(\eta,\xi)$ with Definition 1.7. Omitting $\sqrt{2}$ would change only the coefficient of the slow block of the form (the fast actions being unchanged), not the whole form by one global conformal factor. After canonical diagonalisation $H_2 = \sum \sigma_j(\xi_j^2+\eta_j^2) = \sum 2\sigma_j\Gamma_j$, so the coefficient of Γ_j, i.e. $\partial H_2/\partial\Gamma_j$, is the oscillator frequency $\Omega_j = 2\sigma_j$ (the second Γ-derivatives vanish); this fixes the frozen $\Omega = 2\sigma$. (2) *Sign.* Fej04r Thm. 62 writes $\omega = \sum d\lambda_j \wedge d\Lambda_j + d\xi_j \wedge d\eta_j$ (position∧momentum), and Fej07e p. 3 warns that signs vary ("conventions de signe variables quant à la forme symplectique") and gives the a-posteriori test (the perihelion of the larger planet precesses forward). The frozen order $(\lambda, \xi; \Lambda, \eta)$ makes ξ the position and η the momentum, $\Omega = \sum dz_2 \wedge dz_1$; with that order the identity of Theorem 9.7(c) forces $(\xi,\eta) = \rho(\cos g, \sin g)$ (the argument of $\xi + i\eta$ is $+g$, as in Fej07e). If one writes the same

chart with the roles of ξ,η exchanged (η position), the formula becomes $(\xi,\eta)=\rho(\cos g,-\sin g)$; this is only a relabelling and is not used. (3) *Source gap.* Fej04r Thm. 62 gives no proof ("Voir [Poin05, Chap. III], ou [Che88, Appendice]", neither pinned); in particular the analyticity *through the circular orbits* is the whole content of the theorem. The proof above is ours and uses only explicit formulas: the regular Kepler equation (Proposition 9.4), the Delaunay chart of track I on $\Gamma>0$, and a density argument for the symplectic identity at $\Gamma=0$.

Source gap: Fej04r Thm. 62 (p. 48) cites Poincaré and Chenciner, not pinned; analyticity at $e=0$ and bijectivity at the circular orbits are proved here from explicit formulas (Theorem 9.7(a),(c),(e)).

9.5. **The product chart for the two physical models.** Fix $n\ge 1$, $G,m_0>0$ ($\kappa=Gm_0$) and $m=(m_i)\in(0,\infty)^n$. Coordinates $w=(w_1,w_2)\in\mathrm{Ph}(2n)$ with $w_1=(\lambda,\xi)$ and $w_2=(\Lambda,\eta)$, $\lambda,\xi,\Lambda,\eta\in\mathbb{R}^n$ (fast block first: the order of `assemble`, Definition 1.6, with $a=b=n$). Physical points are $(q,p)\in\mathrm{Ph}(2n)$ with $q^{(i)}=\mathrm{body}(q,i)$.

Definition 9.11 (Product chart)**.** $\mathcal{O}_n=\{w:\ (\lambda_i,\xi_i,\Lambda_i,\eta_i)\in\mathcal{O}\ \forall i\}=\{\Lambda_i>0,\ \frac12(\xi_i^2+\eta_i^2)<\Lambda_i\ \forall i\}$ and $\Psi^{(n)}(w)=(q,p)$ with $(q^{(i)},p^{(i)})=\Psi_{m_i,\kappa}(\lambda_i,\xi_i,\Lambda_i,\eta_i)$. $h(\Lambda)=-\sum_i\kappa^2m_i^3/(2\Lambda_i^2)$ and $\mathcal{K}^{(n)}=\{(q,p):\ (q^{(i)},p^{(i)})\in\mathcal{K}_{m_i,\kappa}\ \forall i\}$; for $\kappa=1$, $\mathcal{K}^{(n)}=\{\mathrm{ProgradeElliptic}\ m\ z\}$ (`ProgradeElliptic`: $0<m_i$, $|q^{(i)}|_2>0$, `keplerEnergy` <0, `planeCross` >0).

Theorem 9.12 (Product Poincaré chart)**.** *(a) $\Psi^{(n)}$ is a real-analytic 1-chart on the open set $\mathcal{O}_n$ (Definition 2.3, $c=1$), $\mathcal{O}_n+((2\pi k,0),0)=\mathcal{O}_n$, $\Psi^{(n)}(w+((2\pi k,0),0))=\Psi^{(n)}(w)$ for $k\in\mathbb{Z}^n$, and $\Psi^{(n)}(w')=\Psi^{(n)}(w)$ iff $w'=w+((2\pi k,0),0)$ for some $k\in\mathbb{Z}^n$. It maps $\mathcal{O}_n$ onto $\mathcal{K}^{(n)}$, and induces a real-analytic diffeomorphism of $\mathcal{O}_n/\{(2\pi k,0,0,0)\}$ onto $\mathcal{K}^{(n)}$.*

(b) (Decoupling) *$H_{\mathrm{Kep}}\circ\Psi^{(n)}=h(\Lambda)$ on $\mathcal{O}_n$, with H_{Kep} of Lemma 8.3; h does not depend on λ,ξ,η, nor on any coupling parameter. $\partial h/\partial\Lambda_i=\nu_i(\Lambda_i):=\kappa^2m_i^3/\Lambda_i^3$ and $\partial^2h/\partial\Lambda_i^2=-3\kappa^2m_i^3/\Lambda_i^4\neq0$; h is real analytic on $\{\Lambda_i>0\}$.*

(c) (The two models share the chart) *Let $\Psi^{-1}(W)=\{w\in\mathcal{O}_n:\Psi^{(n)}(w)\in W\}$ (open, W of Lemma 8.2) and*

$$\hat f_{\mathrm{mov}}=f_{\mathrm{mov}}\circ\Psi^{(n)},\qquad \hat f_{\mathrm{fix}}=f_{\mathrm{fix}}\circ\Psi^{(n)}\qquad (f\text{ of Lemma 8.3}).$$

These are real analytic on $\Psi^{-1}(W)$ and do not depend on μ or ε. For all μ,ε and $w\in\mathcal{O}_n$: $H^\Psi_{\mathrm{mov}}(w)=h(\Lambda)+\mu\hat f_{\mathrm{mov}}(w)$ and $H^\Psi_{\mathrm{fix}}(w)=h(\Lambda)+\varepsilon\hat f_{\mathrm{fix}}(w)$ (the Lean total functions; $H^\Psi=H\circ\Psi^{(n)}$ because $c=1$). On $\Psi^{-1}(W)$ both are differentiable and Lemma 2.4 gives $D\Psi^{(n)}X_{H^\Psi}=X_H\circ\Psi^{(n)}$; orbits transfer by Corollary 2.5.

(d) (Hamilton equations in the chart) *On $\Psi^{-1}(W)$ the equations $\dot w=X_{H^\Psi}(w)$ read $\dot\lambda_i=\nu_i(\Lambda_i)+\partial_{\Lambda_i}(\mu\hat f)$, $\dot\Lambda_i=-\mu\partial_{\lambda_i}\hat f$, $\dot\xi_i=\mu\partial_{\eta_i}\hat f$, $\dot\eta_i=-\mu\partial_{\xi_i}\hat f$ (with $\mu\hat f$ replaced by $\varepsilon\hat f_{\mathrm{fix}}$ for H_{fix}).*

Proof. **Step S9.22: Product structure.** The i-th pair $(q^{(i)},p^{(i)})$ depends only on $(\lambda_i,\xi_i,\Lambda_i,\eta_i)$, and $\Omega(v,w)=\sum_i\Omega_i$ with $\Omega_i(v,w)=v_{2,i}\cdot w_{1,i}-v_{1,i}\cdot w_{2,i}$ over the coordinates $(\lambda_i,\xi_i;\Lambda_i,\eta_i)$ (positions λ_i,ξ_i) resp. $(q^{(i)};p^{(i)})$; the index bookkeeping ($k+2i$, `finProdFinEquiv`) is a permutation applied to positions and to momenta alike, which does not change $\Omega=\sum dz_2\wedge dz_1$. Hence $D\Psi^{(n)}$ is block diagonal with blocks $D\Psi_{m_i,\kappa}$ each satisfying the identity of Theorem 9.7(c); the sum gives $\Omega(D\Psi u,D\Psi w)=\Omega(u,w)$. Analyticity, lattice invariance, bijectivity, the diffeomorphism and $\mathcal{O}_n$ open are blockwise Theorem 9.7(a),(c),(e) (the lattice $2\pi\mathbb{Z}^n$ in λ acts coordinatewise).

Step S9.23: Decoupling. $H_{\mathrm{Kep}}(q,p)=\sum_i\big[|p_i|_2^2/(2m_i)-\kappa m_i/|q_i|_2\big]=\sum_i E_{m_i,\kappa}(q_i,p_i)$ (Remark 8.4), and $E_{m_i,\kappa}\circ\Psi_{m_i,\kappa}=-\kappa^2m_i^3/(2\Lambda_i^2)$ by Theorem 9.7(d). Derivatives are elementary.

Step S9.24: Shared chart. By Lemma 8.3, $H_{\mathrm{mov}}=H_{\mathrm{Kep}}+\mu f_{\mathrm{mov}}$, $H_{\mathrm{fix}}=H_{\mathrm{Kep}}+\varepsilon f_{\mathrm{fix}}$ on all of $\mathrm{Ph}(2n)$ (total functions), with H_{Kep}, f_{mov}, f_{fix} independent of μ,ε; composing with $\Psi^{(n)}$ gives the formulas. $f_{\mathrm{mov}},f_{\mathrm{fix}}$ are real analytic on W (finite sums of polynomials and $1/|\cdot|_2$ with positive radicand, Lemma 8.2, whose proof gives analyticity as well as C^∞); $\Psi^{(n)}$ is analytic; the composition

is analytic by (X1). The domain $\Psi^{-1}(W)$ is open since W is open and $\Psi^{(n)}$ continuous. The field statements are Lemma 2.4, Corollary 2.5.

Step S9.25: (d). Definition 1.2: $X_G = (\partial_{z_2} G, -\partial_{z_1} G)$ with $z_1 = (\lambda, \xi)$, $z_2 = (\Lambda, \eta)$. □

Remark 9.13 (Why the same chart serves both models). The Kepler part of H_{mov} is H_{Kep} with the *planet* mass m_i and centre mass m_0, not Fej04r's reduced mass $\mu_i = m_0 m_i/(m_0 + \mu m_i)$ and $M_i = m_0 + \mu m_i$ (Remark 8.4): the term $\mu|p_i|_2^2/(2m_0)$ is part of μf_{mov}. The chart is therefore independent of μ and equal to the chart of the fixed-Sun problem; this is what makes $H_{\mathrm{mov}} = h + \mu f$ and $H_{\mathrm{fix}} = h + \varepsilon f'$ with coupling-independent f, f'. The difference $f_{\mathrm{mov}} - f_{\mathrm{fix}} = |P|_2^2/(2m_0)$, $P = \sum_i p_i$, is analytic on all of $\mathrm{Ph}(2n)$ (a polynomial) and its λ-average is computed in Part 8 from Proposition 9.3(c) and Fej04r Lemma 64.

Lemma 9.14 (The tube in chart coordinates). *Let $a_i > 0$ and $\Lambda_i^* = m_i\sqrt{\kappa a_i}$. For every $\delta > 0$ there is $\eta_\delta > 0$ such that for all $\eta \in (0, \eta_\delta]$: the circular tube U_η (Definition 1.17, with G, m_0, a as in the definition) is contained in $\mathcal{K}^{(n)}$, and every $w \in \mathcal{O}_n$ with $\Psi^{(n)}(w) \in U_\eta$ satisfies $|\Lambda_i - \Lambda_i^*| < \delta$ and $\frac{1}{2}(\xi_i^2 + \eta_i^2) < \delta$ for all i. Moreover* $\mathrm{circ}(\phi) = \Psi^{(n)}\big((2\pi\phi, 0), (\Lambda^*, 0)\big)$.

Proof. **Step S9.26: Circular points.** Theorem 9.7(f) blockwise: with $\Lambda_i = \Lambda_i^*$, $a = a_i$, $\xi = \eta = 0$ and $\lambda_i = 2\pi\phi_i$ the image is $\mathrm{circ}(\phi)$.

Step S9.27: Continuity of the elements. On the open set $\mathcal{K}^{(n)}$ the functions $\Lambda_i(z) = m_i\sqrt{-\kappa^2 m_i/(2E_{m_i,\kappa})}$ and $\Gamma_i(z) = \Lambda_i(z) - \mathrm{cr}(q^{(i)}, p^{(i)})$ are continuous, and by Theorem 9.7(d) they equal Λ_i, Γ_i of any preimage. The set $V_\delta = \{z \in \mathcal{K}^{(n)} : |\Lambda_i(z) - \Lambda_i^*| < \delta,\ \Gamma_i(z) < \delta\ \forall i\}$ is open and contains the compact torus $T = \{\mathrm{circ}(\phi)\}$ (Step S9.26: there $\Lambda_i = \Lambda_i^*$, $\Gamma_i = 0$; T is the continuous image of $[0,1]^n$). By (E7) there is $\eta_\delta > 0$ with $\{z : \mathrm{dist}_\infty(z, T) < \eta_\delta\} \subset V_\delta$; this set is U_{η_δ} (Definition 1.17). For $\eta \le \eta_\delta$, $U_\eta \subset V_\delta \subset \mathcal{K}^{(n)}$, and a preimage w of $z \in U_\eta$ has $\Lambda_i(w) = \Lambda_i(z)$, $\frac{1}{2}(\xi_i^2 + \eta_i^2) = \Gamma_i(z) < \delta$. □

Lemma 9.15 (Lattice compatibility for Part 1). *Let $\Theta_n : \{(\varphi, \psi; I, R) :\ I_i > 0,\ 0 < R_i < I_i\} \to \mathcal{O}_n$ be the oscillator polar chart*

$$\Theta_n((\varphi, \psi), (I, R)) = \big((\varphi,\ \sqrt{2R}\sin\psi),\ (I,\ \sqrt{2R}\cos\psi)\big)$$

(componentwise square roots and sines; the formula of Remark 2.10(1): $\xi = \sqrt{2R}\sin\psi$ is the position, $\eta = \sqrt{2R}\cos\psi$ the momentum; $\Gamma_i = R_i$, $\Lambda_i = I_i$), and $\Phi_n = \Psi^{(n)} \circ \Theta_n$ on $O_0 = \{I_i > 0, 0 < R_i < I_i\}$. Then:

- *(a) Θ_n is a 1-chart on O_0; Φ_n is a 1-chart on O_0; $O_0 + \iota'_k = O_0$ for the full lattice shift $\iota'_k = ((2\pi k, 0), 0)$ of all $2n$ positions, $k \in \mathbb{Z}^{2n}$ (ι'_k adds $2\pi k$ to (φ, ψ)).*
- *(b) $\Phi_n(z + \iota'_k) = \Phi_n(z)$ for $z \in O_0$, $k \in \mathbb{Z}^{2n}$; and if $z, z' \in O_0$, $\Phi_n(z') = \Phi_n(z)$ then $z' = z + \iota'_k$ for some $k \in \mathbb{Z}^{2n}$. This is (Q1) (with $\sigma_k = 0$) and (Q2) of Theorem 2.9 in the radian convention; the period-one version is (c).*
- *(c) In the period-one torus parameter of the frozen predicates: $\Phi_n^{(1)} := \Phi_n \circ S_{2\pi}$ (with $S_{2\pi}$ of Lemma 2.11) is a 2π-chart on $S_{2\pi}^{-1}(O_0)$ satisfying (Q1) with $\sigma_k = 0$ for the lattice $\iota_k = (k, 0)$, $k \in \mathbb{Z}^{2n}$, and (Q2); in particular Corollary 2.13 applies to Φ_n and Theorem 2.9(b) to $\Phi_n^{(1)}$.*
- *(d) Under Θ_n the pulled-back Hamiltonians are $H^{\Phi_n} = H \circ \Psi^{(n)} \circ \Theta_n = h(I) + \mu\hat{f} \circ \Theta_n$ ($I = \Lambda$), i.e. in the variables $(\varphi, \psi; I, R)$ the unperturbed part is $h(I)$, independent of ψ, R.*

Proof. **Step S9.28: (a) Polar chart.** Θ_n is blockwise $(\psi, R) \mapsto (\xi, \eta) = \sqrt{2R}(\sin\psi, \cos\psi)$, with $d\eta \wedge d\xi = dR \wedge d\psi$ (computed: $d\eta \wedge d\xi = \rho\, d\rho \wedge d\psi = dR \wedge d\psi$ for $\rho = \sqrt{2R}$), real analytic for $R > 0$, mapping into $\mathcal{O}_n$ since $\Gamma_i = R_i < I_i = \Lambda_i$, and the identity on the φ, I blocks; so Θ_n is a 1-chart. The composition of 1-charts is a 1-chart (Remark 2.10(3)); $\Theta_n(O_0) \subset \mathcal{O}_n$ (indeed $= \{\Gamma > 0\} \cap \mathcal{O}_n$).

Step S9.29: (b) Invariance. Θ_n is 2π-periodic in ψ and equals Θ_n in φ shifted; precisely $\Theta_n(z+\iota'_k) = \Theta_n(z) + ((2\pi k_\varphi, 0), 0)$ $(k = (k_\varphi, k_\psi))$, and $\Psi^{(n)}$ is invariant under $((2\pi k_\varphi, 0), 0)$ (Theorem 9.12(a)). If $\Phi_n(z') = \Phi_n(z)$ then $\Theta_n(z') = \Theta_n(z) + ((2\pi k_\varphi, 0), 0)$ by the same theorem, so $\varphi' = \varphi + 2\pi k_\varphi$, $I' = I$, and $\sqrt{2R'}(\sin\psi', \cos\psi') = \sqrt{2R}(\sin\psi, \cos\psi)$ with $R, R' > 0$: then $R' = R$ and $\psi' - \psi \in 2\pi\mathbb{Z}^n$.

Step S9.30: (c) Period one. Lemma 2.11(a): $S_{2\pi}$ is a 2π-chart with $S_{2\pi}(z+\iota_k) = S_{2\pi}(z) + (2\pi k, 0)$; compose with (a), (b) (Remark 2.10(3): $\Phi_n \circ S_{2\pi}$ is a 2π-chart), and note $S_{2\pi}^{-1}(O_0) + \iota_k = S_{2\pi}^{-1}(O_0 + \iota'_k) = S_{2\pi}^{-1}(O_0)$; the injectivity statement (Q2) for $\Phi_n \circ S_{2\pi}$ is that of (b) read through $S_{2\pi}$.

Step S9.31: (d). Theorem 9.12(b),(c) and $c = 1$. □

9.6. **The D25 secular definitions in terms of the chart.** The frozen D25 definitions (`EllipsePosition.lean`, `EllipseMomentum.lean`, `KeplerMeanLongitude.lean`, `LongitudeJacobian.lean`, `SecularEccVector.lean`, `SecularRoot.lean`, `SecularAxis.lean`, `SecularGamma.lean`, `EllipsePhase.lean`, `KeplerAction.lean`, `KeplerAxis.lean`, `KeplerEnergy.lean`, `KeplerEccVector.lean`) are stated for $n = 2$ and $G = m_0 = 1$ $(\kappa = 1)$ in the phase of one chart point $w \in \mathtt{Phase}(2+2)$ with $w_1 = (\lambda_1, \lambda_2, \xi_1, \xi_2)$, $w_2 = (\Lambda_1, \Lambda_2, \eta_1, \eta_2)$ (so $\mathtt{fast}\, w_1 = \lambda$, $\mathtt{slow}\, w_1 = \xi$, $\mathtt{fast}\, w_2 = \Lambda$, $\mathtt{slow}\, w_2 = \eta$) and a vector of eccentric longitudes $F \in \mathbb{R}^2$. (The same statement holds for every n componentwise.)

Proposition 9.16 (D25 definitions = the chart in the eccentric longitude)**.** *Let $n = 2$, $\kappa = 1$, $m \in (0,\infty)^2$, $w \in \mathcal{O}_2$, $i < 2$, and write $\Lambda_i = \mathtt{fast}\, w_2\, i$, $\Gamma_i = \mathtt{secularGamma}\, w\, i$. Then (all formulas for the Lean functions, which are total, are guarded by $\Lambda_i > 0$, $\Gamma_i < \Lambda_i$):*

- *(a)* (vector, root, axis) $\Gamma_i = \frac{1}{2}(\xi_i^2 + \eta_i^2) \geq 0$; $\mathtt{secularRoot} = k_i = 1 - \Gamma_i/\Lambda_i \in (0,1]$; $\mathtt{secularEccVector} = v_i$ *of Definition 9.5 with* $|v_i|_2^2 = 1 - k_i^2$ *(eccentricity* $e_i = |v_i|_2$*);* $\mathtt{secularAxis} = (\Lambda_i/m_i)^2 = a_i$.
- *(b)* (Kepler's equation, Jacobian) $F \mapsto \mathtt{keplerMeanLongitude}\, w\, i\, F = \Phi(F, v_i)$ *is a real-analytic increasing bijection of* $\mathbb{R}$ *commuting with* $F \mapsto F + 2\pi$*, with derivative* $\mathtt{longitudeJacobian} = J(F, v_i) \geq 1 - e_i > 0$*. Let* $F_i(w)$ *be the unique solution of* $\Phi(F, v_i) = \lambda_i$ *(Proposition 9.4(a);* $\lambda_i = \mathtt{fast}\, w_1\, i$*);* F_i *is real analytic on* $\mathcal{O}_2$.
- *(c)* (positions and momenta) *For all* $F \in \mathbb{R}$ *and* $i < 2$*, with* $\tilde{q}_i(F) = \mathtt{ellipsePosition}\, m\, w\, i\, F$*,* $\tilde{p}_i(F) = \mathtt{ellipseMomentum}\, m\, w\, i\, F$*:* $(\tilde{q}_i(F_i(w)), \tilde{p}_i(F_i(w))) = \Psi_{m_i,1}(\lambda_i, \xi_i, \Lambda_i, \eta_i)$*; i.e.* $\mathtt{ellipsePhase}\, m\, w\, (F_1(w), F_2(w)) = \Psi^{(2)}(w)$*. More generally* $\tilde{q}_i(F)$*,* $\tilde{p}_i(F)$ *for* arbitrary F *are the point of the same osculating Kepler ellipse (same* E*,* cr*,* v*) at the eccentric longitude* F*, i.e.* $\Psi(\Phi(F, v_i), \xi_i, \Lambda_i, \eta_i)$.
- *(d)* (Kepler functions of the physical point) *For* $z = \Psi^{(2)}(w)$*:* $\mathtt{PrograderElliptic}\, m\, z$*;* $\mathtt{keplerEnergy}\, m\, z\, i = -m_i^3/(2\Lambda_i^2)$*;* $\mathtt{keplerAxis}\, m\, z\, i = a_i = \mathtt{secularAxis}\, m\, w\, i$*;* $\mathtt{keplerAction}\, m\, z\, i = \Lambda_i$*;* $\mathtt{planeCross}(q^{(i)}, p^{(i)}) = \Lambda_i - \Gamma_i$*;* $\mathtt{keplerEccVector}\, m\, z\, i = v_i = \mathtt{secularEccVector}\, w\, i$*. Conversely every* z *with* $\mathtt{PrograderElliptic}\, m\, z$ *is* $\Psi^{(2)}(w)$ *for a* $w \in \mathcal{O}_2$*, unique modulo* $\lambda \mapsto \lambda + 2\pi k$.
- *(e)* (elements) *For* $\Gamma_i > 0$*:* $(\xi_i, \eta_i) = \rho_i e_{g_i}$*,* g_i *the perihelion argument,* $F_i = g_i + u_i$*,* u_i *the eccentric anomaly and* $\lambda_i = \ell_i + g_i$ *(mean longitude = mean anomaly + perihelion argument).*

Proof. **Step S9.32: (a),(b).** The Lean definitions are exactly $\Gamma = R(\eta, \xi)$, $k = 1 - \Gamma/\Lambda$, $v = \sqrt{(1 - \Gamma/(2\Lambda))/\Lambda}\,(\xi, \eta)$, $\Phi(F, v) = F - v_0 \sin F + v_1 \cos F$, $J = 1 - v \cdot (\cos F, \sin F)$, $a = (\Lambda/m)^2$ (`planeDot`, `orbitalUnit`); Lemma 9.6 and Proposition 9.4(a).

Step S9.33: (c). $\mathtt{ellipsePosition} = a(kc - v + \frac{v \cdot c}{1+k} v)$ and $\mathtt{ellipseMomentum} = \frac{m^2}{\Lambda J}(J_0 c - \frac{v \cdot c}{1+k} J_0 v)$, $c = \mathtt{orbitalUnit}\, F$ (note $\mathtt{fast}\, w_2\, i \cdot \mathtt{longitudeJacobian} = \Lambda J$): these are the displayed q, p of

Definition 9.5 with $\kappa = 1$ and F arbitrary. At $F = F_i(w)$ they are Ψ. For arbitrary F put $\lambda' = \Phi(F, v_i)$: then $F = F(\lambda', v_i)$ by uniqueness, so the same formulas give $\Psi(\lambda', \xi_i, \Lambda_i, \eta_i)$.

Step S9.34: (d). Theorem 9.7(d): $\Psi(w) \in \mathcal{K}_{m_i,1}$ blockwise, i.e. $m_i > 0$, $|q^{(i)}|_2 > 0$ (`planeLength`> 0), $E < 0$ (`keplerEnergy` is $E_{m_i,1}$), cr > 0: this is `Prograde Elliptic`. The values: $E = -m_i^3/(2\Lambda_i^2)$; `keplerAxis` $= -m_i/(2E) = \Lambda_i^2/m_i^2$; `keplerAction` $= m_i\sqrt{a} = \Lambda_i$; the LRL identity of Theorem 9.7(d) is `keplerEccVector` $= v_i$. Converse and uniqueness: Theorem 9.12(a).

Step S9.35: (e). Step S9.14 and Proposition 9.4(c). □

Corollary 9.17 (Angular averages via the eccentric longitude)**.** *Let $w \in \mathcal{O}_2$, $i \in \{1, 2\}$ and φ : `Phase`$(2 \cdot 2) \to \mathbb{R}$ (equivalently $\mathbb{R}^4 \times \mathbb{R}^4 \to \mathbb{R}$, the full two-planet phase space; $\Psi^{(2)}(w)$ and `ellipsePhase`$m\,w\,F$ are 8-dimensional points) be continuous. Then for each fixed (ξ, Λ, η), as functions of λ_i with the other λ_j fixed,*

$$\frac{1}{2\pi}\int_0^{2\pi} \varphi\big(\Psi^{(2)}(\lambda, \xi, \Lambda, \eta)\big)\, d\lambda_i = \frac{1}{2\pi}\int_0^{2\pi} \varphi\big(\texttt{ellipsePhase}\, m\, w\, F\big)\Big|_{F_i = s} J(s, v_i)\, ds$$

for the unit speed $F_i = s$ and the other $F_j = F_j(w)$ fixed (the integrand is the point of the i-th osculating ellipse at eccentric longitude s), with $J =$ `longitudeJacobian`. In particular, with a_i, e_i and (Proposition 9.3(c)): $\langle 1/|q^{(i)}|_2\rangle_{\lambda_i} = 1/a_i$, $\langle |p^{(i)}|_2^2\rangle_{\lambda_i} = m_i^2/a_i$, $\langle |q^{(i)}|_2\rangle_{\lambda_i} = a_i(1 + e_i^2/2)$, with $a_i = (\Lambda_i/m_i)^2$ and $e_i^2 = |v_i|_2^2 = 1 - k_i^2$.

Proof. **Step S9.36: Change of variable.** For fixed $v = v_i$, $s \mapsto \Phi(s, v)$ is C^1, increasing, with $\Phi(s + 2\pi) = \Phi(s) + 2\pi$ and $\Phi' = J > 0$ (Proposition 9.4(a)); its inverse is $\lambda \mapsto F(\lambda, v)$. Over one period, $\int_{\lambda_*}^{\lambda_*+2\pi} \Xi(F(\lambda, v))\, d\lambda = \int_{F_*}^{F_*+2\pi} \Xi(s) J(s, v)\, ds$ for continuous 2π-periodic Ξ ((E6)), with $\Xi(s) = \varphi(\Psi$ at eccentric longitude $s)$ by Proposition 9.16(c); periodic integrals do not depend on the starting point.

Step S9.37: Means. The mean values: the average over λ_i over a period equals that over $\ell_i = \lambda_i - g_i$ for $\Gamma_i > 0$ (shift by a constant of a periodic function), and Proposition 9.3(c) applies ($\kappa = 1$, $m = m_i$, $\Lambda = \Lambda_i$, $C = \Lambda_i k_i$, $a = a_i$); for $\Gamma_i = 0$ the circular case of the same item ([I:`bpI:p7:circular`]). □

9.7. Sources, deviations, errata, forward references.

Sources read. Fej04r pp. 46–48 (Hamiltonian (28), the heliocentric coordinates, (31) and Thm. 62, (32)–(33)), Fej07e p. 3 (`Formule (31)`: factor $\sqrt{2}$; `Formules (35-37)`: sign conventions and the sign test). Chi05 §2 via track I's Part 7 (not re-read here). The D25 frozen Lean files listed in §9.6.
Totality. The chart uses $1/\Lambda$, $\sqrt{(1 - \Gamma/2\Lambda)/\Lambda}$, $1/(1 + k)$, $1/J$: guarded on $\mathcal{O}$ by $\Lambda > 0$, $\Gamma < \Lambda$ (so radicand$> \frac{1}{2\Lambda} > 0$, $k \in (0, 1]$, $J \geq 1 - e > 0$; Lemma 9.6). The frozen Lean functions `secularEccVector`, `ellipsePosition`, `ellipseMomentum` use total `Real.sqrt` and division; on $\mathcal{O}_2$ nothing is truncated. The inverse of the matrix $k\mathrm{Id} + vv^\top/(1 + k)$ is used only in the formula of Theorem 9.7(e), where its eigenvalues are $k, 1 > 0$.
Deviations from the source. (1) Fej04r (31) and (32) use the reduced mass μ_j and the corresponding total mass M_j in the Kepler part; the frozen H_{mov} (and Part 6) put $\mu|p_i|_2^2/(2m_0)$ into the perturbation, so the unperturbed Kepler problem has mass m_i and strength Gm_0 and Fej04r's Λ_j is replaced by $m_i\sqrt{Gm_0 a_i}$ (Definition 9.1, Remark 9.13). (2) The planar chart has no nodal angle: $\theta_j = 0$ and the spatial coordinates (p_j, q_j) of (31) are absent. (3) The proof of Thm. 62 is ours (not Poincaré/Chenciner). (4) The D25 secular definitions are stated for $G = m_0 = 1$; the chart is proved for general $\kappa = Gm_0$ (Theorem 9.7), and Proposition 9.16 is the case $\kappa = 1$.

Suspected errata. (1) Fej04r (31), original printing as reported by Fej07e p. 3 (not the pinned revised p. 47, which already has the factor $\sqrt{2}$): "$r_j = \xi_j + i\eta_j = \sqrt{\Lambda_j}\sqrt{1-\sqrt{1-\varepsilon_j^2}}\ldots$" without the factor $\sqrt{2}$ makes the chart only conformally symplectic: recorded in Fej07e p. 3, reproduced here as Remark 9.10(1). (2) The orchestrator's brief wrote the regularising change as $(\xi,\eta) = \sqrt{2(\Lambda - G)}(\cos, -\sin)(g)$; with the frozen order $(\lambda,\xi;\Lambda,\eta)$ (position ξ) the correct sign is $(\cos g, +\sin g)$ (derived in Step S9.15 and checked numerically in a scratch script); the minus sign belongs to the exchanged-roles convention (Remark 9.10(2)). Not an error in a pinned source.
Doubts about frozen statements. None. The frozen D25 definitions agree with the chart (checked above: formulas, and by a numerical scratch check at one sample point that $\Psi^*\Omega = \Omega$, $E = -m^3/2\Lambda^2$, $\mathrm{cr} = \Lambda - \Gamma$, $\mathrm{LRL} = v$ at a sample point). `secularAxis`$= (\Lambda/m)^2$ requires $Gm_0 = 1$ (D25 normalisation), consistent with Remark 8.12.
External facts used. (E1)–(E7) of §9.1, plus [K:`bp:p1:real-restriction`] and the results of Parts 1 and 6 and track I quoted by label.
Forward references and consumers. Part 8: Corollary 9.17, Proposition 9.3(c), Theorem 9.12(b),(c), Remark 9.13; Parts 2, 5: Proposition 9.8; Parts 12, 13: Lemma 9.15, Lemma 9.14, Theorem 9.12; Part 14: Proposition 9.16, Corollary 9.17. Needed from other parts and not yet written (forward): none beyond the labels of Parts 1, 6 and of track I Part 7 already in the repository.
Label index. `bpP:p7:kepler-data`, `range`, `kepler`, `regular-kepler`, `poincare-def`, `o-facts`, `poincare`, `holomorphic`, `normalisation`, `product-def`, `product-chart`, `same-chart`, `tube`, `lattice`, `d25` (= `eccentric-longitude`), `average`.

10. Averaged perturbation and the secular quadratic form

In the Poincaré chart of Part 7 both physical models are $h(\Lambda) + \mu f$ with a coupling-independent perturbation f (Theorem 9.12(c)). Averaging f over the mean longitudes $\lambda \in \mathbb{T}^n$ gives the *secular function* $S(\xi,\Lambda,\eta)$, whose Taylor expansion in the slow variables (ξ,η) at the circular torus feeds the Birkhoff form (A_2'') of Part 0. This part proves what that expansion needs, and the exact relation between the two models.

- `perturbation-analytic` (Prop. 10.2): on an explicit complex polydisc around the circular torus the pair interaction in the chart is holomorphic for every real λ, and S is holomorphic (real analytic) there;
- `evenness` (Prop. 10.3 and Cor. 10.4): three exact symmetries of the chart; S is even in η and in ξ and invariant under simultaneous rotation, so every odd Taylor coefficient vanishes and the quadratic and quartic parts have a prescribed invariant form;
- `lemma64` (Prop. 10.5): $\langle p^{(j)} \cdot p^{(k)} \rangle_\lambda = 0$ for $j \neq k$ *exactly*, and $\langle |p^{(i)}|^2 \rangle = m_i^2 \kappa / a_i$;
- `laplace` (Def. 10.6, Lemma 10.7, Lemma 10.9): Laplace coefficients, locally holomorphic on $|\alpha| < 1$ from the series, and two series identities;
- `quadratic-form` (Thm. 10.11): C_0 and Q_h with the P5 corrections *and* a further overall factor $\frac{1}{2}$ forced by the erratum's normalisation (suspected erratum E-P-Fej04r-3);
- `identification` (Thm. 10.13): $\langle f_{\mathrm{mov}} \rangle - \langle f_{\mathrm{fix}} \rangle = \delta(\Lambda)$ is a function of Λ alone, *exactly*, so all (ξ,η)-Taylor coefficients of order ≥ 1 agree (Cor. 10.14); this is what lets $n = 2$ P-FIXED follow route (c);
- `quartic-jet` (Prop. 10.16): the interface for the planar torsion (Part 11b, CP11 §8.1).

10.1. **Conventions, external facts, the secular polydisc.** Fix $n \geq 1$, $G, m_0 > 0$, $\kappa = Gm_0$, $m \in (0,\infty)^n$, and the chart $\Psi = \Psi^{(n)}$ with $\hat{f}_{\mathrm{fix}} = f_{\mathrm{fix}} \circ \Psi$, $\hat{f}_{\mathrm{mov}} = f_{\mathrm{mov}} \circ \Psi$ (Theorem 9.12). Coordinates $w = (\lambda,\xi,\Lambda,\eta)$; $u_i = (\xi_i,\eta_i) \in \mathbb{R}^2$; $\Gamma_i = |u_i|^2/2$; $a_i = \Lambda_i^2/(m_i^2\kappa)$. Angles have period

2π as in Part 7. For continuous φ on $\{(\lambda, w')\}$, $w' = (\xi, \Lambda, \eta)$,

$$\langle\varphi\rangle(w') = (2\pi)^{-n}\int_{[0,2\pi]^n}\varphi(\lambda, w')\,d\lambda, \qquad S_{\mathrm{fix}} = \langle\hat{f}_{\mathrm{fix}}\rangle, \quad S_{\mathrm{mov}} = \langle\hat{f}_{\mathrm{mov}}\rangle.$$

Dictionary to the frozen average. Since $\mathrm{assemble}(I, \varphi, p, q) = ((\varphi, q), (I, p))$ and $w = ((\lambda, \xi), (\Lambda, \eta))$ (Part 7), $\texttt{secularAverage}\,P\,\mu\,I\,p\,q = \langle P \circ \cdot\rangle$ at $(\xi, \Lambda, \eta) = (q, I, p)$: the frozen momentum p is η, the frozen position q is ξ, $I = \Lambda$, and $R_j = \Gamma_j$.

For the axes $0 < a_1^* < \cdots < a_n^*$ (the *increasing* order of the Lean statements; Fej04r uses the decreasing one, see Remark 10.12) put $\Lambda_i^* = m_i\sqrt{\kappa a_i^*}$ and, for $\rho > 0$,

$$\mathcal{N}(\rho) = \{(\xi, \Lambda, \eta) \in \mathbb{C}^{3n} : \ |\xi_i|, |\eta_i| < \rho, \ |\Lambda_i - \Lambda_i^*| < \rho \ \ \forall i\}, \qquad \mathcal{N}_{\mathbb{R}}(\rho) = \mathcal{N}(\rho) \cap \mathbb{R}^{3n}.$$

External facts.

(E1) *Holomorphic parameter integrals.* If $\varphi(\lambda, w')$ is continuous on $[0, 2\pi]^n \times U$ and holomorphic in $w' \in U \subset \mathbb{C}^N$ open for each λ, then $w' \mapsto \int \varphi\, d\lambda$ is holomorphic on U (Morera/Fubini, differentiation under the integral); likewise real-analytic parameter dependence follows by restriction ([I:`bpI:p7:x-param`], (E2)).

(E2) *Differentiation under the integral:* if φ is C^k in w' with derivatives jointly continuous on $[0, 2\pi]^n \times U$, then $\partial^\alpha\langle\varphi\rangle = \langle\partial^\alpha\varphi\rangle$, $|\alpha| \le k$. Fubini on products of intervals; translation invariance of $\int_0^{2\pi}$ of a 2π-periodic continuous function.

(E3) Taylor expansion of a real-analytic function: the Taylor polynomial of order N at a point has remainder $O(|x|^{N+1})$ locally uniformly; Taylor coefficients are the iterated derivatives; a composition of an analytic function with a real-analytic map is real analytic.

(E4) *First fundamental theorem for $O(2)$* (Weyl): a polynomial in vectors $u_1, \dots, u_r \in \mathbb{R}^2$ invariant under the diagonal action of $O(2)$ is a polynomial in the dot products $u_i \cdot u_j$.

(E5) Pochhammer symbols $(s)_p = s(s+1)\cdots(s+p-1)$, $(s)_{p+1} = (s)_p(s+p)$; binomial series $(1-z)^{-s} = \sum_p \frac{(s)_p}{p!}z^p$ for $|z| < 1$, and $\frac{(s)_p}{p!} = O(p^{s-1})$.

Definition 10.1 (Secular polydisc constants)**.** $\rho_0 := \min_i \min\{\Lambda_i^*/2, \ \sqrt{\Lambda_i^*}/16\}$, so that $\mathcal{N}(\rho)$, $\rho < \rho_0$, is inside the domain of Proposition 9.8 for each planet. $\rho_* = \rho_*(m, \kappa, a^*) \in (0, \rho_0)$ is the constant of Proposition 10.2.

10.2. **Analyticity of the interaction and of its average.**

Proposition 10.2 (Analytic perturbation near the circular torus)**.** *There is $\rho_* \in (0, \rho_0)$ depending only on (m, κ, a^*) such that, with $s_{ij}(\lambda, w') = \sum_{c=0,1}(q_c^{(i)} - q_c^{(j)})^2$ (complex bilinear square, $q = \Psi$ of Prop. 9.8), for all real λ and $w' \in \mathcal{N}(\rho_*)$ and all $i < j$:*

(a) $\operatorname{Re} s_{ij} \ge \frac{1}{2}(a_j^* - a_i^*)^2$. *Hence for real points $|q^{(i)} - q^{(j)}| \ge (a_j^* - a_i^*)/\sqrt{2}$ for* all $\lambda \in \mathbb{R}^n$*: $\Psi(\lambda, w') \in W$ (Lemma 8.2) and $\mathcal{N}_{\mathbb{R}}(\rho_*) \times \mathbb{R}^n \subset \Psi^{-1}(W)$.*

(b) $\hat{f}_{\mathrm{fix}}(\lambda, w') = -G\sum_{i<j} m_i m_j\, s_{ij}^{-1/2}$ (principal root) is holomorphic in $w' \in \mathcal{N}(\rho_*)$ for each real λ, jointly continuous, 2π-periodic in λ, real at real points; the same holds for $\hat{f}_{\mathrm{mov}} = \hat{f}_{\mathrm{fix}} + \sum_{c=0,1}\big(\sum_i p_c^{(i)}\big)^2/(2m_0)$, where on complex w' the square is the bilinear (holomorphic) square, never a Hermitian norm and with no complex conjugation; it restricts to the real squared Euclidean norm $|\sum_i p^{(i)}|^2/(2m_0)$ at real points.

(c) $S_{\mathrm{fix}}, S_{\mathrm{mov}}$ are holomorphic on $\mathcal{N}(\rho_*)$, real at real points, hence real analytic on $\mathcal{N}_{\mathbb{R}}(\rho_*)$, and $\partial^\alpha_{(\xi,\eta,\Lambda)}S = \langle\partial^\alpha \hat{f}\rangle$ for all α.

Proof. **Step S10.1: Value at the circular torus.** At $w'^* = (0, \Lambda^*, 0)$, Theorem 9.7(f) gives $q^{(i)} = a_i^*(\cos\lambda_i, \sin\lambda_i)$, so $s_{ij} = a_i^{*2} + a_j^{*2} - 2a_i^*a_j^*\cos(\lambda_i - \lambda_j) \ge (a_j^* - a_i^*)^2$.

Step S10.2: Uniformity. s_{ij} is holomorphic in w' on $\mathcal{N}(\rho_0)$ (Prop. 9.8 and $\bar\Psi$ real at real points, polynomial in q) and continuous and 2π-periodic in $\lambda \in \mathbb{R}^n$; hence uniformly continuous on $[0,2\pi]^n \times \overline{\mathcal{N}(\rho_0/2)}$ (compact). By Step S10.1 there is $\rho_* < \rho_0/2$ with $\operatorname{Re} s_{ij} \geq \frac{1}{2}(a_j^* - a_i^*)^2$ on $[0,2\pi]^n \times \mathcal{N}(\rho_*)$; it depends only on the data (m, κ, a^*) (through Λ^* and Ψ). This is (a); at real points $|q^{(i)} - q^{(j)}|^2 = s_{ij}$, and NoColl also needs $|q^{(i)}| > 0$, which holds on $\mathcal{O}_n$ (Lemma 9.6, Theorem 9.7(d)).

Step S10.3: Holomorphy of $\hat f$. On $\{\operatorname{Re} s > 0\}$ the principal $s^{-1/2}$ is holomorphic ((E2)); composing with the holomorphic s_{ij} gives (b); at real points $s_{ij} > 0$ and $s_{ij}^{-1/2} = 1/|q^{(i)} - q^{(j)}|$ so the formula is $f_{\text{fix}} \circ \Psi$ (Lemma 8.3). The extra term of $\hat f_{\text{mov}}$ is a polynomial in the components of $p^{(i)} = \Psi_{m_i,\kappa}$-momenta, holomorphic by Prop. 9.8.

Step S10.4: The average. By (E1) S is holomorphic on $\mathcal{N}(\rho_*)$; real at real points; the derivative formula is (E2) (the w'-derivatives of a holomorphic function are jointly continuous in (λ, w') by Cauchy's formula on a slightly smaller polydisc, uniformly in the compact λ-torus). □

10.3. **Symmetries and evenness.** Write R_θ for the rotation of $\mathbb{R}^2$ by θ, $\sigma = \operatorname{diag}(1,-1)$, $\mathbb{1} = (1,\dots,1)$, and for $\xi, \eta \in \mathbb{R}^n$ put $R_\theta(\xi,\eta) = (\xi\cos\theta - \eta\sin\theta,\ \xi\sin\theta + \eta\cos\theta)$ (planetwise; this rotates each u_i by θ).

Proposition 10.3 (Symmetries of the chart)**.** *For $w \in \mathcal{O}_n$ write $\Psi(w) = (q,p)$ (blocks $q^{(i)}, p^{(i)}$). Then:*

(S1) $\Psi(\lambda + \pi\mathbb{1}, -\xi, \Lambda, -\eta) = (-q, -p)$;
(S2) $\Psi(\lambda + \theta\mathbb{1}, R_\theta(\xi,\eta), \Lambda) = (R_\theta q^{(i)}, R_\theta p^{(i)})_i$ *for every* $\theta \in \mathbb{R}$;
(S3) $\Psi(-\lambda, \xi, \Lambda, -\eta) = (\sigma q^{(i)}, -\sigma p^{(i)})_i$.

(The domain $\mathcal{O}_n$ is invariant under all three maps.) The Kepler functions $f_{\text{fix}}, f_{\text{mov}}$ of Lemma 8.3 satisfy $f(-q,-p) = f(q,p) = f(Rq, \pm Rp)$ for $R \in O(2)$ acting blockwise.

Proof. It suffices to treat one planet. Put $\Phi(F, v) = F - v_0 \sin F + v_1 \cos F$, $c = (\cos F, \sin F)$.

Step S10.5: (S1). $\Phi(F + \pi, -v) = \Phi(F, v) + \pi$, so $F(\lambda + \pi, -v) = F(\lambda, v) + \pi$ (uniqueness, Prop. 9.4(a)), $c \mapsto -c$, J and k are unchanged, $v \mapsto -v$. In $q = a(kc - v + \frac{v\cdot c}{1+k} v)$ and $p = \frac{m^2\kappa}{\Lambda J}(J_0 c - \frac{v\cdot c}{1+k} J_0 v)$ every term is odd under $(c, v) \mapsto (-c, -v)$ ($v \cdot c$ is even, the remaining factor is linear in c or v, resp. quadratic factor $v \cdot c\, v$ odd).

Step S10.6: (S2). $v \mapsto R_\theta v$ for $(\xi,\eta) \mapsto R_\theta(\xi,\eta)$ (v is a positive multiple of (ξ,η) with Γ, Λ unchanged); expanding, $\Phi(F + \theta, R_\theta v) = \Phi(F, v) + \theta$ (the $\cos\theta, \sin\theta$ terms combine to $-v_0 \sin F + v_1 \cos F$), so $F(\lambda + \theta, R_\theta v) = F + \theta$, $c \mapsto R_\theta c$, J unchanged, and $R_\theta J_0 = J_0 R_\theta$, $(R_\theta v)\cdot(R_\theta c) = v \cdot c$: $q, p \mapsto R_\theta q, R_\theta p$.

Step S10.7: (S3). With $\bar v = \sigma v$ ($\eta \mapsto -\eta$): $\Phi(-F, \bar v) = -\Phi(F, v)$, so $F(-\lambda, \bar v) = -F(\lambda, v)$, $c \mapsto \sigma c$, J unchanged, $\bar v \cdot \sigma c = v \cdot c$, $q \mapsto \sigma q$; $J_0 \sigma = -\sigma J_0$ gives $p \mapsto -\sigma p$.

Step S10.8: The Kepler functions. f_{fix} depends on $|q_i - q_j|$ and $f_{\text{mov}} - f_{\text{fix}} = |\sum p_i|^2/(2m_0)$; both are invariant under $(q,p) \mapsto (\pm Rq, \pm' Rp)$ for any isometry R. □

Corollary 10.4 (Evenness and invariance of the secular function)**.** *Let $S \in \{S_{\text{fix}}, S_{\text{mov}}\}$ on $\mathcal{N}_{\mathbb{R}}(\rho_*)$. Then $S(-\xi,\Lambda,-\eta) = S(\xi,\Lambda,\eta) = S(\xi,\Lambda,-\eta) = S(-\xi,\Lambda,\eta)$ and $S(R_\theta(\xi,\eta),\Lambda) = S(\xi,\Lambda,\eta)$. Consequently the Taylor expansion of S at $(\xi,\eta) = 0$ has only terms of even total degree, $(\xi,\eta) = 0$ is a critical point (the origin is the circular torus, Fej04r p. 50), and the degree-$2d$ homogeneous part is an $O(2)$-invariant polynomial in $(u_1,\dots,u_n)$ with coefficients real-analytic in Λ; the degree-two part is*

$$\xi^\top A(\Lambda)\xi + \eta^\top A(\Lambda)\eta, \qquad A = A^\top. \tag{1}$$

Proof. **Step S10.9: Invariances.** Each of (S1)–(S3) shifts or reflects the λ variables and maps $\hat{f}$ to itself (Prop. 10.3); the λ-average of a 2π-periodic continuous function is invariant under translation $\lambda \mapsto \lambda + c\mathbb{1}$ and under $\lambda \mapsto -\lambda$ ((E2)). This gives the invariances of S; the fourth equality combines the first and second.

Step S10.10: Taylor structure. S is real analytic in (ξ, η) (Prop. 10.2(c)); evenness under $(\xi, \eta) \mapsto -(\xi, \eta)$ kills odd-degree parts. The degree-$2d$ part of an $O(2)$-invariant analytic function is invariant, because $O(2)$ acts linearly and preserves homogeneity.

Step S10.11: Quadratic part. Write the degree-two part as $\xi^\top A\xi + 2\xi^\top B\eta + \eta^\top C\eta$. Evenness in η alone gives $B = 0$ (it kills terms odd in η); the rotation $\theta = \pi/2$, $(\xi, \eta) \mapsto (-\eta, \xi)$, gives $\eta^\top A\eta + \xi^\top C\xi = \xi^\top A\xi + \eta^\top C\eta$, so $C = A$. □

10.4. **Exact orthogonality of momenta (Fej04r Lemma 64).**

Proposition 10.5 (Lemma 64, exactly)**.** *For every $w \in \mathcal{O}_n$ (no neighbourhood, no collision hypothesis), $\nu_i = \kappa^2 m_i^3/\Lambda_i^3$:*

(a) $m_i\nu_i\, \partial_{\lambda_i} q^{(i)}(w) = p^{(i)}(w)$ and $\partial_{\lambda_k} q^{(i)} = 0$ for $k \neq i$;
(b) $\frac{1}{2\pi}\int_0^{2\pi} p^{(i)}\, d\lambda_i = 0 \in \mathbb{R}^2$;
(c) for $j \neq k$: $(2\pi)^{-n}\int_{[0,2\pi]^n} p^{(j)} \cdot p^{(k)}\, d\lambda = 0$;
(d) $(2\pi)^{-n}\int |p^{(i)}|^2 d\lambda = \frac{1}{2\pi}\int_0^{2\pi} |p^{(i)}|^2 d\lambda_i = m_i^2\kappa/a_i = m_i^4\kappa^2/\Lambda_i^2$;
(e) hence $(2\pi)^{-n}\int \big|\sum_i p^{(i)}\big|^2 d\lambda = \sum_i m_i^4\kappa^2/\Lambda_i^2$.

Source: Fej04r Lemme 64, pp. 49–50 (cross terms; proof: $Y_j = \mu_j\, dX_j/dt \propto \partial X_j/\partial\lambda_j$ and Fubini). The virial value (d) is ours (the source drops the diagonal term, which is not needed for $\langle F_{\mathrm{per}}\rangle = \langle F_{\mathrm{princ}}\rangle$; here it is).

Proof. **Step S10.12: (a).** The decoupled Kepler Hamiltonian $H_{\mathrm{Kep}}(q,p) = \sum_i \big[|p^{(i)}|^2/(2m_i) - \kappa m_i/|q^{(i)}|\big]$ (Remark 8.4; no pair term) is differentiable on the set where each $q^{(i)} \neq 0$, which contains $\Psi(w)$ for $w \in \mathcal{O}_n$; no pair-distance guard is needed and no collision premise is added. Its position derivative is $p^{(i)}/m_i$ in the i-th block, so the position part of $X_{H_{\mathrm{Kep}}}$ is $(p^{(i)}/m_i)_i$ by direct differentiation (this does not use Prop. 8.6, whose hypotheses include the pair-collision guard). With $H_{\mathrm{Kep}}^{\Psi} = h(\Lambda)$ (Thm. 9.12(b)) and Lemma 2.4 (applied to the chart, not to the interaction), $D\Psi\, X_h = X_{H_{\mathrm{Kep}}} \circ \Psi$ with X_h having entries ν_i in the λ_i slots and 0 elsewhere (Thm. 9.12(b), the form of X_h computed directly from h), the position part of $D\Psi X_h$ is $(\sum_k \nu_k \partial_{\lambda_k} q^{(i)})_i = \nu_i \partial_{\lambda_i} q^{(i)}$, since $q^{(i)}$ depends only on $(\lambda_i, \xi_i, \Lambda_i, \eta_i)$.

Step S10.13: (b). $q^{(i)}$ is real analytic and 2π-periodic in λ_i, so $\int_0^{2\pi} \partial_{\lambda_i} q^{(i)} d\lambda_i = 0$ (fundamental theorem of calculus); use (a) and $m_i\nu_i > 0$.

Step S10.14: (c). $p^{(j)} \cdot p^{(k)}$ is continuous, $p^{(j)}$ depends on λ_j only, $p^{(k)}$ on λ_k only; by Fubini the integral is $\big(\frac{1}{2\pi}\int p^{(j)} d\lambda_j\big) \cdot \big(\frac{1}{2\pi}\int p^{(k)} d\lambda_k\big) = 0$.

Step S10.15: (d). The first equality is Fubini. For $\Gamma_i > 0$ write $p^{(i)} = \mathsf{Q}_{m_i,\kappa}(\ell, g, \Lambda, C)$ with $\ell = \lambda - g$ (Step S9.14); the period integral over λ_i equals that over ℓ (shift of a periodic function), and Prop. 9.3(c) gives $m^2\kappa/a$. For $\Gamma_i = 0$ the circular case of the same item. And $a_i = \Lambda_i^2/(m_i^2\kappa)$.

Step S10.16: (e). Expand $|\sum p^{(i)}|^2 = \sum |p^{(i)}|^2 + 2\sum_{j<k} p^{(j)} \cdot p^{(k)}$ and use (c),(d). □

10.5. **Laplace coefficients.**

Definition 10.6 (Laplace coefficients)**.** For $s > 0$, $k \in \mathbb{Z}$ and $\alpha \in [0, 1)$, $b_s^{(k)}(\alpha) = \frac{1}{2\pi}\int_0^{2\pi} \cos(kt)\,(1 + \alpha^2 - 2\alpha\cos t)^{-s} dt$, so $(1 + \alpha^2 - 2\alpha\cos t)^{-s} = \sum_{k\in\mathbb{Z}} b_s^{(k)}(\alpha) e^{ikt}$ (Poincaré normalisation, as in Fej04r Def. 66).

Source: Fej04r Def. 66, p. 50; CP11 App. B footnote 72 ($b_{h,k}$, $\beta_k^{(r)} = b_{r/2,k}$).

Lemma 10.7 (Series; local holomorphy)**.** *For $k \geq 0$ and $\alpha \in [0,1)$,*

$$b_s^{(k)}(\alpha) = \sum_{q \geq 0} \frac{(s)_{q+k}\,(s)_q}{(q+k)!\,q!}\ \alpha^{2q+k}, \qquad b_s^{(-k)} = b_s^{(k)}.$$

The series has radius of convergence ≥ 1; it defines a holomorphic function on $\{|\alpha| < 1\}$, real and nonnegative on $[0,1)$, strictly positive for $0 < \alpha < 1$, with $b_s^{(0)}(0) = 1$ (and $b_s^{(k)}(0) = 0$ for $k > 0$), and the Laplace coefficients are understood as this function wherever a complex argument is used. For $0 < \alpha < 1$ they are real analytic in α, and $\alpha \mapsto b_s^{(k)}(a_i/a_j)$ is real analytic on $\{0 < a_i < a_j\}$.

Proof. **Step S10.17: Expansion.** $1 + \alpha^2 - 2\alpha\cos t = (1 - \alpha e^{it})(1 - \alpha e^{-it})$, so by (E5) $(\cdot)^{-s} = \sum_{p,q\geq 0} \frac{(s)_p (s)_q}{p!q!}\alpha^{p+q} e^{i(p-q)t}$, absolutely and uniformly in t for $\alpha < 1$ (coefficients are $O((p+1)^{s-1}(q+1)^{s-1})$). Integrating against $e^{-ikt}/(2\pi)$ termwise keeps $p - q = k$, i.e. $p = q + k$. Evenness in k is evenness of cos.

Step S10.18: Convergence. The coefficient of α^{2q+k} is $O\big((q+k+1)^{s-1}(q+1)^{s-1}\big)$ as $q \to \infty$ (hence $O(q^{2s-2})$ for fixed k), polynomial growth, so the radius is ≥ 1; all coefficients are ≥ 0 for $s > 0$ and the coefficient of α^{2q+k} is > 0, so $b_s^{(k)} > 0$ on $(0,1)$ (at $\alpha = 0$ only $q = k = 0$ survives). □

Remark 10.8 (Fej04r Lemma 76 is not used). Lemma 76 (p. 56) asserts meromorphy on the Riemann sphere with poles $0, \pm 1, \infty$. This is false as printed: $b_{1/2}^{(0)}$ has a logarithmic singularity at $\alpha = 1$ (suspected erratum E-P-Fej04r-2, recorded in Part 0). Only the local holomorphy of Lemma 10.7 on the unit disc is used here and in Part 9–10.

Lemma 10.9 (Two identities)**.** *For $\alpha \in [0,1)$ with $\beta = b_{1/2}^{(0)}$ and $b = b_{1/2}^{(1)}$:*

(I1) $\alpha\beta'' + 2\beta' = b_{3/2}^{(1)}$;

(I2) $\alpha^2 b'' + 2\alpha b' - 2b = \alpha\, b_{3/2}^{(2)}$.

Proof. Compare coefficients of the series of Lemma 10.7, using $(\frac{3}{2})_p = (\frac{1}{2})_{p+1}/\frac{1}{2} = (2p+1)(\frac{1}{2})_p$ and $(\frac{3}{2})_{p-1} = 2(\frac{1}{2})_p$.

Step S10.19: (I1). $\beta = \sum_q \big(\frac{(1/2)_q}{q!}\big)^2 \alpha^{2q}$ and $(\alpha\partial^2 + 2\partial)\alpha^{2q} = 2q(2q+1)\alpha^{2q-1}$. The coefficient of α^{2q-1} ($q \geq 1$) on the left is $2q(2q+1)\big(\frac{(1/2)_q}{q!}\big)^2$; on the right ($p = q-1$) it is $\frac{(3/2)_q (3/2)_{q-1}}{q!(q-1)!} = \frac{(2q+1)(1/2)_q \cdot 2(1/2)_q}{q!(q-1)!}$, equal.

Step S10.20: (I2). $b = \sum_q c_q \alpha^{2q+1}$, $c_q = \frac{(1/2)_{q+1}(1/2)_q}{(q+1)!q!}$, and $(\alpha^2\partial^2 + 2\alpha\partial - 2)\alpha^m = (m+2)(m-1)\alpha^m$, which for $m = 2q+1$ is $2q(2q+3)$. For $q = 0$ both sides have zero coefficient of α (the multiplier is 0 and $\alpha b_{3/2}^{(2)} = O(\alpha^3)$); for $q \geq 1$, where $(3/2)_{q-1}$ is used, the coefficient of α^{2q+1} in $\alpha b_{3/2}^{(2)}$ is $\frac{(3/2)_{q+1}(3/2)_{q-1}}{(q+1)!(q-1)!} = \frac{(2q+3)(1/2)_{q+1}\cdot 2(1/2)_q}{(q+1)!(q-1)!} = 2q(2q+3)c_q$. □

10.6. The quadratic part: C_0 and Q_h.

Definition 10.10 (Secular coefficients)**.** Let $a_1 < \cdots < a_n$, $\alpha_{ij} = a_i/a_j < 1$ for $i < j$ (planet i inner, j outer), and

$$\mathfrak{a}_{ij} = \frac{G m_i m_j\, a_i}{4a_j^2}\, b_{3/2}^{(1)}(\alpha_{ij}) > 0, \qquad \mathfrak{b}_{ij} = \frac{G m_i m_j\, a_i}{4a_j^2}\, b_{3/2}^{(2)}(\alpha_{ij}) > 0 \qquad (i < j),$$

$\mathfrak{a}_{ji}=\mathfrak{a}_{ij}$, $\mathfrak{b}_{ji}=\mathfrak{b}_{ij}$, and

$$C_0(a)=-\sum_{i<j}\frac{Gm_im_j}{a_j}\,b^{(0)}_{1/2}(\alpha_{ij}),\quad \hat q_{jj}=-\sum_{k\neq j}\mathfrak{a}_{jk},\quad \hat q_{jk}=\mathfrak{b}_{jk}\,(j\neq k),\quad Q_h=\Delta^{-1}\hat q\,\Delta^{-1},\ \Delta=\operatorname{diag}(\sqrt{\Lambda_i}).$$

All are real analytic functions of Λ on $\{a_1<\cdots<a_n\}$ ($a=a(\Lambda)$, Lemma 10.7).

Theorem 10.11 (Quadratic part of the secular function)**.** *In the situation of Prop. 10.2, with a^* increasing, for $(\xi,\Lambda,\eta)\in\mathcal{N}_{\mathbb{R}}(\rho_*)$ (and also $a(\Lambda)$ increasing there, shrinking ρ_* if needed):*

$$S_{\mathrm{fix}}(\xi,\Lambda,\eta)=C_0(a)+\xi^{\top}Q_h\xi+\eta^{\top}Q_h\eta+O\big(|(\xi,\eta)|^4\big),$$

the error being real analytic and $O(|(\xi,\eta)|^4)$ locally uniformly in Λ. Equivalently the matrix A of (1) is Q_h. If $\varrho\in SO(n)$ diagonalises $Q_h=\varrho\operatorname{diag}(\sigma)\varrho^{\top}$ then in the symplectic variables $(\tilde\xi,\tilde\eta)=\varrho^{\top}(\xi,\eta)$ the quadratic part is $\sum_j\sigma_j(\tilde\xi_j^2+\tilde\eta_j^2)=\sum_j 2\sigma_j\tilde\Gamma_j$, i.e. $\Omega_j=2\sigma_j$ in the frozen sense.
Source: Fej04r Formule 67 and (35)–(37), pp. 50–51; Prop. 73 proof, p. 55 (the matrix $\hat q=\Delta^{\top}Q_h\Delta$); CP11 (B.1)–(B.2), p. 62; corrections P5 (`statement.md`) and the factor $\frac12$, see Remark 10.12. The derivation is ours (the source cites Laplace–Lagrange and [LR95], not pinned).
Source gap: Fej04r p. 50: "Le calcul complet est un peu long", detailed only in [LR95], not pinned. The proof below is a complete replacement.

Proof. **Step S10.21: Pair decomposition.** $\hat f_{\mathrm{fix}}=\sum_{i<j}V_{ij}$, $V_{ij}=-Gm_im_j/|q^{(i)}-q^{(j)}|$ depends on (λ_i,λ_j) and on the data $(u_i,\Lambda_i,u_j,\Lambda_j)$ only. So $S_{\mathrm{fix}}=\sum_{i<j}S_{ij}$, $S_{ij}=\langle V_{ij}\rangle$ with the average over (λ_i,λ_j) (Fubini). Corollary 10.4 applies to each S_{ij} (as the $n=2$ function of (u_i,u_j)): its degree-two part is $A_{ij}|u_i|^2+A'_{ij}|u_j|^2+B_{ij}\,u_i\cdot u_j$ (invariants of $O(2)$, (E4); the quadratic invariants of two vectors are spanned by the three dot products).

Step S10.22: Scaling. Let $\Phi(r_1,r_2)=\frac{1}{2\pi}\int_0^{2\pi}(r_1^2+r_2^2-2r_1r_2\cos t)^{-1/2}dt=\frac{1}{r_2}b^{(0)}_{1/2}(r_1/r_2)$ for $r_1<r_2$ and $\Phi(r_1,r_2)=\Phi(r_2,r_1)$: real analytic on $\{r_1\neq r_2\}$ and homogeneous of degree -1.

Step S10.23: C_0**.** At $u=0$ both planets are circular (Thm. 9.7(f)): $S_{ij}(0)=-Gm_im_j\Phi(a_i,a_j)=-Gm_im_ja_j^{-1}b^{(0)}_{1/2}(\alpha_{ij})$.

Step S10.24: Diagonal coefficients. Let $u_j=0$ (j circular, any $i\neq j$, $u_i=(\xi_i,\eta_i)$). Averaging over λ_j first, which by rotation invariance of the circle gives $\Phi(|q^{(i)}|,a_j)$, independent of the direction of $q^{(i)}$. By Prop. 9.3(b), $|q^{(i)}|=a_i(1-e\cos u)$, $d\lambda_i=(1-e\cos u)\,du$ over a period (u the eccentric anomaly, or $F-g$; [I:`bpI:p7:orbit-identities`]); with $e^2=|v_i|^2=2\Gamma_i/\Lambda_i-\Gamma_i^2/\Lambda_i^2=:E$ (Lemma 9.6),

$$S_{ij}\big|_{u_j=0}=-Gm_im_j\,\frac{1}{2\pi}\int_0^{2\pi}h(e\cos u)\,du,\qquad h(\varepsilon)=\Phi\big(a_i(1-\varepsilon),a_j\big)(1-\varepsilon).$$

Only even powers of $e\cos u$ survive and $\langle\cos^2\rangle=\frac12$, so the right side is $-Gm_im_j\big[h(0)+\frac14h''(0)E+O(E^2)\big]$ with $h''(0)=a_i^2\Phi_{11}+2a_i\Phi_1=(r_1^2\partial_{r_1}^2+2r_1\partial_{r_1})\Phi\,|_{(a_i,a_j)}$. For $i<j$ (inner eccentric): with $\beta=b^{(0)}_{1/2}$, $h''(0)=a_j^{-1}(\alpha^2\beta''+2\alpha\beta')=\alpha\,b^{(1)}_{3/2}(\alpha)/a_j$ by (I1). For $i>j$ (outer eccentric, inner circular of radius $c=a_j$): $g(r)=r^{-1}\beta(c/r)$, $r^2g''+2rg'=2c\beta'/r^2+c^2\beta''/r^3$, at $r=a_i$ equal to $\alpha\,b^{(1)}_{3/2}(\alpha)/a_i$ with $\alpha=c/a_i$ again by (I1). Since $E=|u|^2/\Lambda_i+O(|u|^4)$, the coefficient of $|u_i|^2$ is $-\mathfrak{a}_{ij}/\Lambda_i$ in both cases ($\mathfrak{a}_{ij}=Gm_im_j\frac{\alpha}{4a_{\max}}b^{(1)}_{3/2}$ with $a_{\max}$ the larger axis, α the ratio).

Step S10.25: Cross coefficient. Let $i<j$. By (E2), B_{ij} is determined by $\partial_{\xi_i}\partial_{\xi_j}\langle V_{ij}\rangle=\langle\partial_{\xi_i}\partial_{\xi_j}V_{ij}\rangle=B_{ij}$ at 0. Use $v=(1-\Gamma/2\Lambda)^{1/2}\Lambda^{-1/2}u$, so to first order $v_i=u_i/\sqrt{\Lambda_i}$, and write $v_i=(s,0)$, $v_j=(t,0)$. To first order (Prop. 9.4(a): $\partial_vF=(\sin F,-\cos F)$ at $v=0$, $F=\lambda$,

$J=1$) the position of planet i is $q=a[(1-v\cdot c)c-2(v\cdot J_0c)J_0c]$, $c=(\cos\lambda,\sin\lambda)$: in polar coordinates $r_i=a_i(1-s\cos\lambda_i)$, $\theta_i=\lambda_i+2s\sin\lambda_i$, and likewise for j with t. Write $V_{ij}=U(r_i,r_j,\theta_i-\theta_j)$ with $U=-Gm_im_j(r_i^2+r_j^2-2r_ir_j\cos\vartheta)^{-1/2}=-Gm_im_j\big[B_0+2\sum_{k\ge1}B_k\cos k\vartheta\big]$, $B_k(r_i,r_j)=r_j^{-1}b^{(k)}_{1/2}(r_i/r_j)$ (Def. 10.6), homogeneous of degree -1. The mixed derivative at $s=t=0$ is $U_{r_ir_j}r_i'r_j'-U_{r_i\vartheta}r_i'\theta_j'+U_{\vartheta r_j}\theta_i'r_j'-U_{\vartheta\vartheta}\theta_i'\theta_j'$ with $r_i'=-a_i\cos\lambda_i$, $\theta_i'=2\sin\lambda_i$, etc. Averaging over (λ_i,λ_j) with $\vartheta=\lambda_i-\lambda_j$ and $\langle\cos\lambda_i\cos\lambda_j\cos k\vartheta\rangle=\langle\sin\lambda_i\sin\lambda_j\cos k\vartheta\rangle=\frac14\delta_{k1}$, $\langle\sin\lambda_i\cos\lambda_j\sin k\vartheta\rangle=-\langle\cos\lambda_i\sin\lambda_j\sin k\vartheta\rangle=\frac14\delta_{k1}$, only B_1 survives:

$$\partial_s\partial_t\langle V_{ij}\rangle=Gm_im_j\Big[-\tfrac12a_ia_j\partial_i\partial_jB_1-a_i\partial_iB_1-a_j\partial_jB_1-2B_1\Big]=Gm_im_j\Big[-\tfrac12a_ia_j\partial_i\partial_jB_1-B_1\Big]$$

by Euler's relation $r_i\partial_iB_1+r_j\partial_jB_1=-B_1$. With $b=b^{(1)}_{1/2}$, $B_1=r_j^{-1}b(r_i/r_j)$, $\partial_i\partial_jB_1=-r_j^{-3}(2b'+\alpha b'')$; the bracket is $a_j^{-1}\big[\alpha b'+\frac12\alpha^2b''-b\big]=\frac{\alpha}{2a_j}b^{(2)}_{3/2}(\alpha)$ by (I2). Thus $\partial_s\partial_t\langle V_{ij}\rangle=2\mathfrak{b}_{ij}$, and $B_{ij}=\partial_{\xi_i}\partial_{\xi_j}S=2\mathfrak{b}_{ij}/\sqrt{\Lambda_i\Lambda_j}$.

Step S10.26: Assembly. $S_{\mathrm{fix}}-C_0$ has degree-two part $\sum_{i<j}\big[-\mathfrak{a}_{ij}(|u_i|^2/\Lambda_i+|u_j|^2/\Lambda_j)+2\mathfrak{b}_{ij}\,u_i\cdot u_j/\sqrt{\Lambda_i\Lambda_j}\big]=\sum_{j,k}(Q_h)_{jk}(\xi_j\xi_k+\eta_j\eta_k)$ with $Q_h=\Delta^{-1}\hat q\Delta^{-1}$. The remainder is analytic with vanishing jet up to order three (Cor. 10.4), whence $O(|(\xi,\eta)|^4)$ by (E3). For the shrinking of ρ_*: $a(\Lambda)$ is continuous.

Step S10.27: Diagonalisation. $(\xi,\eta)\mapsto(\varrho^\top\xi,\varrho^\top\eta)$ is symplectic for $d\eta\wedge d\xi$ and leaves Λ and the form Γ-sum invariant in the sense $\sum_j(\tilde\xi_j^2+\tilde\eta_j^2)=\sum_j(\xi_j^2+\eta_j^2)$; $\tilde\Gamma_j=(\tilde\xi_j^2+\tilde\eta_j^2)/2$. □

Remark 10.12 (Factor $\frac12$ against the source, dictionary, numerical evidence). (1) *Index order.* Fej04r (34) orders $a_1>\cdots>a_n$ and writes $C_1(a_j,a_k)=-\frac{a_k}{2a_j^2}b^{(1)}_{3/2}(\frac{a_k}{a_j})$, $C_2=\frac{a_k}{2a_j^2}b^{(2)}_{3/2}$, $j<k$ (outer j, inner k). The dictionary to Definition 10.10 is $j\leftrightarrow n+1-j$; each of C_1,C_2 is $-2\mathfrak{a}/(Gm_im_j)$, $+2\mathfrak{b}/(Gm_im_j)$ for the pair. The corrected entries of `statement.md` P5 (decreasing order, $\hat q=\Delta^\top Q_h\Delta$) are $2\hat q$ in the present notation. (2) *Factor $\frac12$ (suspected erratum E-P-Fej04r-3).* (35)–(37) as printed give $\langle F_{\mathrm{per}}\rangle=C_0+Q_h^{\mathrm F}\cdot(\xi^2+\eta^2)$ with $Q_h^{\mathrm F}=2Q_h$. This is the right constant for the normalisation $|r|^2=\Gamma$, i.e. the historical original (31) with $r=\sqrt{\Lambda}\sqrt{1-\sqrt{1-\varepsilon^2}}\,e^{i\cdot}$, whose omitted $\sqrt2$ is reported by Fej07e p. 3. The *pinned revised* (31), p. 47, already carries $\sqrt{2\Lambda_j}$ ($|u|^2=2\Gamma$, the frozen chart), but its (35)–(37) retain the larger coefficients: that is a *new, separate* matrix defect inside the pinned printing (distinct from the original-coordinate normalisation erratum). With the frozen normalisation every coefficient of (35)–(37) is halved, which Theorem 10.11 proves; the present Q_h is half the pinned printed quadratic matrix. In frozen terms $\langle f\rangle=C_0+\sum_j\Omega_j\tilde\Gamma_j$ with $\Omega_j=2\sigma_j=$ the eigenvalues of $2Q_h$, i.e. of the matrix printed in Fej04r (36)–(37) in the ξ-basis; `statement.md` P7 "$\Omega_j=2\sigma_j$" is correct only if σ_j are the eigenvalues of the *present* Q_h (half of the printed). Positivity/definiteness (Part 9) is unaffected. (3) CP11 (B.2), p. 62 writes $C_2=\frac{\alpha}{b}\beta^{(3)}_2$ (no $\frac12$) while Fej04r (37) has $\frac12$ in C_2; with (B.1) read literally the off-diagonal term is twice too large (suspected erratum E-P-CP11-1; the diagonal part of (B.1) agrees with Theorem 10.11). (4) *Numerical check* (scratch script in `.drafts/P/bp-scratch-bpP-p8/`, three planets $m=(0.7,1.9,0.4)$, $a=(4.1,2.3,1.0)$, $G=1.3$, $m_0=0.9$, torus averages by trapezoid rule with 48^2 nodes of the chart of Def. 9.5): $(S_{\mathrm{fix}}-C_0)/(\xi^\top Q_h\xi+\eta^\top Q_h\eta)=1.0000128,\ 1.0000021$ at amplitudes $10^{-3},5\cdot10^{-4}$ (error $O(|u|^2)$); C_0 agrees to $4\cdot10^{-14}$; $\langle p^{(1)}\cdot p^{(2)}\rangle=10^{-18}$; (S1)–(S3) hold to $5\cdot10^{-16}$. This is evidence, not a proof; the proof is Theorem 10.11.

10.7. **The exact identification of the two models.**

Theorem 10.13 (Identification of $\langle H_{\mathrm{mov}}\rangle$ and $\langle H_{\mathrm{fix}}\rangle$)**.** *Let $w'=(\xi,\Lambda,\eta)$ with $(\lambda,\xi,\Lambda,\eta)\in\mathcal{O}_n$, and put*

$$\delta(\Lambda)=\frac{1}{2m_0}\sum_{i=1}^{n}\frac{m_i^4\kappa^2}{\Lambda_i^2}=\frac{G}{2}\sum_i\frac{m_i^2}{a_i}\ \Big(=\tfrac{\kappa G}{2}\sum_i\tfrac{m_i^4}{\Lambda_i^2}\Big).$$

Then $\lambda\mapsto\hat f_{\mathrm{mov}}(\lambda,w')-\hat f_{\mathrm{fix}}(\lambda,w')=|\sum_i p^{(i)}|^2/(2m_0)$ is continuous, and

$$(2\pi)^{-n}\int_{[0,2\pi]^n}\big(\hat f_{\mathrm{mov}}-\hat f_{\mathrm{fix}}\big)\,d\lambda=\delta(\Lambda).$$

Consequently $\hat f_{\mathrm{fix}}(\cdot,w')$ is integrable iff $\hat f_{\mathrm{mov}}(\cdot,w')$ is, and then $S_{\mathrm{mov}}=S_{\mathrm{fix}}+\delta(\Lambda)$; in particular on $\mathcal{N}_{\mathbb{R}}(\rho_*)$ (where both are analytic, Prop. 10.2). With the same average for the full Hamiltonians: $\langle H^{\Psi}_{\mathrm{mov}}\rangle=h(\Lambda)+\mu S_{\mathrm{mov}}=h+\mu S_{\mathrm{fix}}+\mu\delta$ and $\langle H^{\Psi}_{\mathrm{fix}}\rangle=h+\varepsilon S_{\mathrm{fix}}$.
Source: Ours (`statement.md` P7 item 4 gives the plan; Fej04r Lemma 64 gives the cross terms); Lemma 8.3 for the difference.

Proof. **Step S10.28: Difference.** $f_{\mathrm{mov}}-f_{\mathrm{fix}}=|P|^2/(2m_0)$ identically (Lemma 8.3), a polynomial in the continuous functions $p^{(i)}(\lambda_i)$.

Step S10.29: Average. Divide Prop. 10.5(e) by $2m_0$ and use $\kappa/m_0=G$ for the second form.

Step S10.30: Integrability. A bounded continuous function is integrable on the compact torus, so integrability of one of $\hat f_{\mathrm{fix}},\hat f_{\mathrm{mov}}$ is that of the other, and the averages subtract. (Lean: where neither is integrable both Bochner averages are 0 and the difference is 0 rather than δ; this never occurs on $\mathcal{N}_{\mathbb{R}}(\rho_*)$, where $\hat f$ is continuous on the torus, Prop. 10.2(a).) □

Corollary 10.14 (Same jets, same quadratic form, same Hamilton equations)**.** *On $\mathcal{N}_{\mathbb{R}}(\rho_*)$, for every multi-index α in (ξ,η) with $\alpha\neq0$ and every Λ-derivative, $\partial^\alpha_{(\xi,\eta)}S_{\mathrm{mov}}=\partial^\alpha_{(\xi,\eta)}S_{\mathrm{fix}}$ (all orders, not only quadratic). Therefore:*

- *(a) $S_{\mathrm{mov}}=C_0+\delta+\xi^\top Q_h\xi+\eta^\top Q_h\eta+O(|(\xi,\eta)|^4)$: same Q_h, same σ_j, same $\Omega_j=2\sigma_j$; only the constant term $P_0=C_0+\delta$ (resp. C_0) differs, by an action-only function; and the quartic jets (Prop. 10.16) coincide.*
- *(b) For $\mu=\varepsilon$ the secular equations $\dot\xi_i=\mu\partial_{\eta_i}S$, $\dot\eta_i=-\mu\partial_{\xi_i}S$ (Thm. 9.12(d), averaged) of the two models are* identical*; Λ is constant in both, and the frequencies $\dot\lambda_i$ differ by $\mu\partial_{\Lambda_i}\delta$.*
- *(c) (*Transfer to the frozen Birkhoff form.*) Take $P_{\mathrm{mov}}=\hat f_{\mathrm{mov}}$, $P_{\mathrm{fix}}=\hat f_{\mathrm{fix}}$ (no μ-dependence), assembled as in the dictionary of §10.1. If* HasBirkhoff$(P_{\mathrm{mov}},V,r_0,\mu_0,P_0,\Omega,\beta)$ *(Def. 1.14) holds, then* HasBirkhoff$(P_{\mathrm{fix}},V,r_0,\mu_0,P_0-\delta,\Omega,\beta)$ *holds, and conversely, provided the points $(I,p,q)=(\Lambda,\eta,\xi)$ with $I\in V$, $p_j^2+q_j^2<r_0^2$ lie in $\mathcal{N}_{\mathbb{R}}(\rho_*)$; the hypotheses (H5), (H6) on β are literally the same. If a slow-variable symplectic change $\chi(\Lambda;\cdot)$ (rotation ϱ, Birkhoff map of Part 11b), depending on Λ only as a parameter, is applied before averaging, $S_{\mathrm{mov}}\circ\chi-S_{\mathrm{fix}}\circ\chi=\delta(\Lambda)$ still.*

Proof. δ does not depend on (ξ,η) by Theorem 10.13, so all derivatives of order ≥1 in (ξ,η) of the difference vanish. (a),(b) follow from Theorem 10.11 and Theorem 9.12(d). (c): $P_{\mathrm{av}}^{\mathrm{mov}}-P_{\mathrm{av}}^{\mathrm{fix}}=\delta(I)$ (frozen average of Def. 1.12, $I=\Lambda$) on the integrability set, so the integrability clause (i) transfers and the clause (ii) expansion with $P_0\to P_0-\delta$; Ω,β unchanged, χ acts on (ξ,η) only. □

Remark 10.15 (Use in Part 11). For $n=2$ the hypotheses of P-G on $(\Omega,\beta,\det\beta)$ for $H_{\mathrm{fix}}=h+\varepsilon\hat f_{\mathrm{fix}}$ coincide with those for $H_{\mathrm{mov}}=h+\mu\hat f_{\mathrm{mov}}$; hence $n=2$ P-FIXED follows P-PLANAR route (c) with f_{mov} replaced by f_{fix} (the Kepler part h and the fast frequency map are the same, Lemma 8.3). The additional constant is the only change (P_0, not used by the KAM argument beyond the energy normalisation).

10.8. **Quartic data for the planar torsion (interface for Part 11b).** Part 11b evaluates the second-order Birkhoff invariants τ of CP11 §8.1, pp. 32–34, (8.1)–(8.4) (the frozen β of Def. 1.14). The input is the jet of S to order four in $u=(u_1,u_2)$ ($n=2$, planar: no $(\bar p,\bar q)$) as an analytic function of the axes. CP11 (6.9)–(6.10) and (B.5)–(B.6), pp. 21, 62–63, write $\langle f\rangle = C_0+Q_h\cdot\frac{\eta^2+\xi^2}{2}+\frac12 F_h\cdot\eta^2\xi^2$-type quartic with coefficients r_1,r_2,r_3,r_4 (functions of (a_i,a_j) via $\beta_k^{(9)}=b_{9/2}^{(k)}$, $\beta_k^{(r)}:=b_{r/2,k}$); their ordering is $(\eta,\xi)=(p,q)$ and $a_1<a_2$, the Lean order. The statements below fix the interface; the quartic computation is left to Part 11b.

Proposition 10.16 (Quartic jet)**.** *Let $n\ge 2$ and $\Lambda\in\mathcal{N}_{\mathbb{R}}(\rho_*)$-range. The degree-four part of S_{fix} (and of S_{mov}, Cor. 10.14) in (ξ,η) is*

$$\mathcal{Q}_4=\sum_{i<j}\Big[c_1^{ij}|u_i|^4+c_2^{ij}|u_j|^4+c_3^{ij}|u_i|^2|u_j|^2+c_4^{ij}(u_i\cdot u_j)^2+c_5^{ij}|u_i|^2(u_i\cdot u_j)+c_6^{ij}|u_j|^2(u_i\cdot u_j)\Big],$$

with $c_\nu^{ij}=c_\nu^{ij}(\Lambda_i,\Lambda_j)$ real analytic. Moreover:

*(a) (*scaling*) In the eccentricity vectors v of Def. 9.5 the pair average is $S_{ij}=-Gm_im_ja_j^{-1}\mathcal{W}(\alpha_{ij};v_i,v_j)$ with $\mathcal{W}$ real analytic on the open* separated *domain $\mathcal{D}=\{(\alpha,v_i,v_j):0<\alpha<1,\ |v_i|,|v_j|<\frac14,\ \alpha(1+|v_i|)<1-|v_j|\}$ (the inner apocentre radius lies below the outer pericentre radius, so the two ellipses are disjoint; a product $(0,1)\times\{|v|<\frac14\}$ would contain tangent ellipses, e.g. $\alpha=5/6$, $v_i=(1/5,0)$, $v_j=0$, where the average is singular) and independent of m,G,κ. A sufficient sub-domain is $|v_i|,|v_j|<(1-\alpha)/(1+\alpha)$ intersected with $|v|<\frac14$. The Taylor coefficients of $\mathcal{W}$ at $v=0$ are real analytic on the whole interval $\alpha\in(0,1)$; the radius of the neighbourhood of $v=0$ on which the series converges shrinks as $\alpha\to1$. The passage to u is exact: $v_i=(1-|u_i|^2/(4\Lambda_i))^{1/2}u_i/\sqrt{\Lambda_i}$, so if $\mathcal{W}_2=A\,|v_i|^2+A'|v_j|^2+B\,v_i\cdot v_j$ is the quadratic part and $\mathcal{W}_4$ the quartic part in v, then $\mathcal{Q}_4^{(ij)}=\mathcal{W}_4(u/\sqrt{\Lambda})-\big(\frac{A}{4}\frac{|u_i|^4}{\Lambda_i^2}+\frac{A'}{4}\frac{|u_j|^4}{\Lambda_j^2}+\frac{B}{8}\frac{u_i\cdot u_j}{\sqrt{\Lambda_i\Lambda_j}}\big(\frac{|u_i|^2}{\Lambda_i}+\frac{|u_j|^2}{\Lambda_j}\big)\big)$ times the appropriate prefactor $-Gm_im_ja_j^{-1}$.*

*(b) (*diagonal terms, exact*) c_1^{ij} and c_2^{ij} are*

$$c=-Gm_im_j\Big(\frac{h^{(4)}(0)}{64}-\frac{h''(0)}{16}\Big)\frac{1}{\Lambda^2},\qquad h(\varepsilon)=\Phi\big(a(1-\varepsilon),a'\big)(1-\varepsilon),$$

with $(a,\Lambda)=(a_i,\Lambda_i)$, $a'=a_j$ for c_1 and (a_j,Λ_j), $a'=a_i$ for c_2.

*(c) (*relation to CP11; forward obligations*) Define $r_1^{\mathrm{ex}}(a,a')=-\frac{1}{32}h^{(4)}(0)+\frac18h''(0)$ (with h as in (b), $G=1$). Then (b) reads $c_1^{ij}=\frac{m_im_j}{2\Lambda_i^2}r_1^{\mathrm{ex}}(a_i,a_j)$ and $c_2^{ij}=\frac{m_im_j}{2\Lambda_j^2}r_1^{\mathrm{ex}}(a_j,a_i)$ (units $G=m_0=1$). The printed r_1 of CP11 (B.6) is* not *asserted equal to r_1^{ex} (Remark 10.17); the repaired formula, with $37\,\alpha^3\beta_4^{(9)}(\alpha)$ in place of the printed $37\,\alpha^4\beta_4^{(9)}(\alpha)$ in its last term, equals r_1^{ex} (certificate in Remark 10.17). The existence and analyticity of the six invariant coefficients $c_1,\dots,c_6$ is the proved interface; the explicit values of the mixed coefficients $c_3,\dots,c_6$ (CP11's r_2,r_3,r_4, unchecked) are* forward obligations of Part 11b. *Their direct computation by Fourier expansion in the orbit variables as in Theorem 10.11 must keep the* third-order *single-planet variations of the orbit map for the bidegrees $(3,1)$ and $(1,3)$ (the second-order expansion $q=a(kc-v+\frac{v\cdot c}{1+k}v)$ does not suffice), or prove an identity eliminating them; the monomial relations are those of the $O(2)$-structure above (CP11 uses a non-symmetric tensor F_h).*

Source: CP11 §8.1, pp. 32–34 ((8.1) $\tau_{11},\tau_{12},\tau_{22}$ for the planar block, asymptotics (8.2)–(8.4)); App. B (B.5)–(B.6), pp. 62–64 (quartic tensor F_h and $r_1,\dots,r_4$); (6.9)–(6.10), p. 21. Symmetry structure: Cor. 10.4, Weyl.

Source gap: CP11's proof of (B.1)–(B.17) is in App. B only for the structure; the coefficients are printed without derivation of the $\beta^{(9)}$ polynomials.

Proof. **Step S10.31: Structure.** By Cor. 10.4 and Prop. 10.2, the degree-four part is an $O(2)$-invariant polynomial, analytic in Λ (Taylor coefficients of a jointly analytic function). By pair decomposition (Step S10.21) it is a sum over pairs of $O(2)$-invariants of (u_i, u_j), i.e. polynomials in $|u_i|^2, |u_j|^2, u_i \cdot u_j$ ((E4)) of degree two: the six monomials listed.

Step S10.32: (a). $q^{(i)} = a_i \tilde{q}(\lambda_i, v_i)$ with $\tilde{q}$ independent of a, m, κ (Def. 9.5), so $V_{ij} = -Gm_i m_j a_j^{-1} \big|\alpha\tilde{q}(\lambda_i, v_i) - \tilde{q}(\lambda_j, v_j)\big|^{-1}$ and the average over (λ_i, λ_j) is a function of (α, v_i, v_j), real analytic on $\mathcal{D}$: there the two orbit ellipses are disjoint, so $|\alpha\tilde{q}(\lambda_i, v_i) - \tilde{q}(\lambda_j, v_j)|$ is bounded below by a positive continuous function and the argument of Prop. 10.2 applies (the Taylor coefficients at $v = 0$ are those of the circular case, analytic for every $\alpha \in (0,1)$ by Step S10.1). The conversion is the expansion of $(1-t)^{1/2}$ with $t = |u|^2/(4\Lambda)$.

Step S10.33: (b). For $u_j = 0$ the computation of Step S10.24 gives $S_{ij} = -Gm_i m_j \sum_l E^l \frac{h^{(2l)}(0)}{(2l)!} \frac{\binom{2l}{l}}{4^l}$, i.e. the coefficient of E^2 is $h^{(4)}(0)/64$, and $E = |u|^2/\Lambda - |u|^4/(4\Lambda^2)$ gives $|u|^4$ coefficient $-Gm_i m_j\big(h^{(4)}/64 - h''/16\big)/\Lambda^2$.

Step S10.34: (c). The identification of c_1, c_2 with r_1^{ex} is (b). The mixed coefficients are not computed here (forward obligation of Part 11b, see the statement); the comparison of the printed r_1 with r_1^{ex} is in Remark 10.17. □

Remark 10.17 (Check of CP11 (B.6) r_1; suspected erratum E-P-CP11-2). c_1, c_2 of Prop. 10.16(b) were evaluated numerically (mpmath, Φ by quadrature, $m = (0.7, 1.9)$, $G = \kappa = 1$) and independently by the series $\sum_k \binom{2k}{k}^2 4^{-2k} \alpha^{2k} \langle r^{2k} \rangle$ and by brute-force torus averages; the three agree. For $a = (1,3)$: $c_1 = 0.0065937$, $c_2 = -0.0028633$, while $\frac{m_1 m_2}{2\Lambda_i^2} r_1$ with r_1 of (B.6) as printed gives 0.007418 and -0.002975; for $a = (1,10)$ the agreement is $1 + 4 \cdot 10^{-5}$ (the leading terms $\frac{3}{16}\alpha^2/a_2$, (8.2) of CP11, are right), for $a = (1,2)$ the printed value has the wrong sign (0.0574 against -0.0310). So the printed r_1 is correct only asymptotically (as $\alpha \to 0$, which is all (8.2)–(8.4) use) and *must not be used for the torsion at finite axis ratio*. The defect is one exponent: in the last term of r_1 in (B.6), p. 63, the printed $37\,\alpha^4 \beta_4^{(9)}(\alpha)$ must be $37\,\alpha^3 \beta_4^{(9)}(\alpha)$ (the printed formula has an odd α^9 contribution that the inner radial average cannot have). *Certificate* (independent review, exact): with $b = 1$, $x = \alpha$, $z = \cos t$, $D = 1 + x^2 - 2xz$, $H(\epsilon, t) = (1-\epsilon)/\sqrt{x^2(1-\epsilon)^2 + 1 - 2x(1-\epsilon)\cos t}$ and $K_{\mathrm{ex}} = -\frac{1}{32}\partial_\epsilon^4 H(0,t) + \frac{1}{8}\partial_\epsilon^2 H(0,t)$, let K_{corr} be the cosine kernel of the repaired (B.6) (each $\beta_k^{(9)}$ replaced by $\cos(kt) D^{-9/2}$). Then $K_{\mathrm{corr}} - K_{\mathrm{ex}} = \frac{d}{dt}\big[\sin t\, R(x, \cos t) D^{-7/2}\big]$ with $R(x,z) = \frac{3x}{32}(3x^4 - 20x^3 z - 12x^2 z^2 + 38x^2 - 4xz - 5)$, a polynomial identity; $D > 0$ for $x > 0$, $x \neq 1$, so the periodic integral of the derivative vanishes and $r_1^{\mathrm{corr}} = r_1^{\mathrm{ex}}$. Equivalently $r_1^{\mathrm{printed}} - r_1^{\mathrm{ex}} = 37\,\alpha^4(1-\alpha)\beta_4^{(9)}(\alpha)/(256\,b)$. (The identity is an algebraic computation we have not re-derived by hand; the numerical agreement of the repaired formula with r_1^{ex} is the independent evidence, and the statement is marked for Lean as a rational-function identity plus periodic integration.) r_2, r_3, r_4 are *not* verified. Part 11b should recompute $c_1, \dots, c_6$ from the method above (exactly, and checkable by quadrature), not from (B.6). The quartic quantities CP11 uses for τ in (8.1) are all of $r_1, \dots, r_4$ (equivalently $c_1, \dots, c_6$), C_1, C_2 (equivalently $\mathfrak{a}, \mathfrak{b}$) and the eigenvector matrix U_h of Q_h (7.19), (7.27).

10.9. **Sources, deviations, errata, forward references.**
Sources read. Fej04r pp. 46–55 (Thm. 62, §6.2, Lemma 64, Def. 66, Formule 67, Prop. 73 proof incl. $\hat{q}$ on p. 55, Lemma 76 p. 56), Fej07e p. 3, CP11 pp. 21–22, 31–34, 62–65 (rendered pages 62–63 read as images), `docs/tracks/P/statement.md` P5–P7.

Deviations. (1) Lean (increasing) axis order throughout; Fej04r's decreasing order by $j \mapsto n+1-j$. (2) Q_h halved against the printed formula (normalisation of the frozen chart). (3) Identification is proved to all orders and *exactly*, not only at quadratic order, and the diagonal virial term is included. (4) δ is expressed in the action Λ and in the axis a.
Suspected errata (for the ledger). E-P-Fej04r-3: the pinned revised (31), p. 47, already has $\sqrt{2\Lambda}$ (the omitted $\sqrt{2}$ of the original is reported in Fej07e p. 3) but (35)–(37), pp. 50–51, retain the coefficients for $|r|^2 = \Gamma$; every printed matrix coefficient must be halved (Remark 10.12(2)). E-P8-Fej50 (candidate, new): in the proof of Lemma 64, p. 50, the last coefficient of $Y_j = \mu_j\, dX_j/dt$ reads $\mu_j^3 M_j^2/\Lambda_j^3\, \partial_{\lambda_j} X_j$ but by (33) it should be $\mu_j^4 M_j^2/\Lambda_j^3$; the missing mass factor does not affect the zero-average argument, and Prop. 10.5 states the correct $p_i = m_i \nu_i \partial_{\lambda_i} q_i$. E-P-Fej04r-1 (P5) confirmed numerically up to that factor. E-P-Fej04r-2 (Lemma 76) unchanged. E-P-CP11-1: (B.2) C_2 has no $\frac{1}{2}$ (p. 62). E-P-CP11-2: in r_1 of (B.6), p. 63, the last term $37\,\alpha^4\beta_4^{(9)}$ must read $37\,\alpha^3\beta_4^{(9)}$ (integration-by-parts certificate in Remark 10.17); r_2, r_3, r_4 not checked.
Doubts about frozen statements. None. `statement.md` P7 "$\Omega_j = 2\sigma_j$" needs the qualifier of Remark 10.12(2).
External facts used. (E1)–(E5), (E2); Parts 1, 6, 7 and [I:`bpI:p7:orbit-identities`].
Forward references and consumers. Part 9 (definiteness of Q_h, Lemmas 74–75: uses Def. 10.10 and positivity Lemma 10.7); Part 10 (analytic continuation: Lemma 10.7, no global meromorphy); Parts 11, 11a, 11b (Thm. 10.11, Cor. 10.14, Prop. 10.16, Remark 10.15); Part 12 (Prop. 10.2).
Needed from other parts: none.
Label index. `rho`, `perturbation-analytic`, `symmetries`, `evenness`, `lemma64`, `laplace-def`, `laplace`, `identities`, `coeffs`, `quadratic-form`, `identification`, `same-jets`, `quartic`.

11. The spectrum of the secular matrix

Part 8 reduced the secular function to $S = C_0 + \xi^\top Q_h \xi + \eta^\top Q_h \eta + O(|(\xi,\eta)|^4)$ with the real analytic symmetric $n \times n$ matrix Q_h (Definition 10.10, Theorem 10.11). This part studies the spectrum of Q_h and turns it into the frozen first-order Birkhoff data $\Omega_j = 2\sigma_j$ of Definition 1.14.

- `lemmas-70-72` (Lemma 11.1, Lemma 11.2, Corollary 11.3): eigenvalue perturbation of $\mathrm{Diag}(d) + \varepsilon B$. Fej04r's Lemma 71 is re-proved in a sharper, fully explicit form (eigenvectors included, no sorting, no dependence on $\|D\|$);
- `definite` (Lemma 11.5, Lemma 11.4, Proposition 11.6): Q_h is negative definite for $n \geq 2$ with an explicit bound; Lemma 74 is replaced by a one-line integration by parts that gives the one inequality needed, $b_s^{(1)} > b_s^{(2)}$;
- `diagonalize` (Proposition 11.7, Proposition 11.8, Theorem 11.9): on a patch where the spectrum is simple, the eigenvalues σ_j and the rotation $\varrho \in SO(n)$ are real analytic; the rotation of (ξ,η) has an explicit symplectic lift to the full chart, under which the secular function is simply composed with the rotation; relation to $\Omega_j = 2\sigma_j$ and the frozen R_j;
- `interface` (Definition 11.11, Proposition 11.12, Lemma 11.13, Corollary 11.14): a single analytic function Φ_4 of the axes whose non-vanishing means *simple spectrum and non-resonance up to order four*; its zero set, the consequence of one good point, and the explicit case $n = 2$ (the spectrum is always simple there, and non-resonance is the exclusion of four ratios).

Global continuation of the eigenvalues over the whole domain of axes is *not* used (Part 10; Fej04r Lemma 76 is false as printed, Remark 10.8). Everything here is local, or an identity between analytic functions.

11.1. **Conventions and external facts.** $|x|$ is the Euclidean norm of $x \in \mathbb{R}^n$ or $\mathbb{C}^n$, $\|M\|$ the induced operator norm, $\|M\|_F$ the Frobenius norm ($\|M\| \leq \|M\|_F$), e_j the unit vectors, Sym_n the real symmetric matrices. For a matrix, $|M_{jk}| \leq \|M\|$ and every column has Euclidean norm

$\le \|M\|$. *Comparison with the ambient max norms* (labelled `norm-compare`): on $\mathbb{R}^n$ or $\mathbb{C}^n$ (= `Fin n` $\to \mathbb{R}/\mathbb{C}$, never `EuclideanSpace`) $|x|_\infty \le |x| \le \sqrt{n}\,|x|_\infty$, and the induced operator norms satisfy $\|M\|_{\infty\to\infty} \le \sqrt{n}\|M\|$ and $\|M\| \le \sqrt{n}\|M\|_{\infty\to\infty}$ (from the two vector inequalities). All perturbation constants of this part are stated in Euclidean norms and must be converted by this comparison before a max-norm consumer; the matrix-valued spectral theorem (unitary/orthogonal diagonaliser) is used to extract the coordinate functions of ϱ, not Mathlib's `eigenvectorBasis` on `EuclideanSpace`. The data are as in Part 8: $n \ge 1$, G, m_0, m, κ, increasing axes $a_1 < \cdots < a_n$, $\Lambda_i = m_i\sqrt{\kappa a_i}$; $\mathfrak{a}_{ij}, \mathfrak{b}_{ij}, \hat{q}, Q_h = \Delta^{-1}\hat{q}\Delta^{-1}$ are those of Definition 10.10. Note that $\hat{q}$ is a function of a (and G, m) and Δ a function of $\Lambda(a)$, so $Q_h = Q_h(a)$ is a real analytic function of a on $\{a_1 < \cdots < a_n\}$ (Definition 10.10). The eigenvalues of Q_h are labelled in *increasing* order $\sigma_1 < \cdots < \sigma_n$ wherever they are simple; this labelling is unrelated to the order of the planets (Fej04r's decreasing planet order corresponds to $j \mapsto n+1-j$ in the matrix entries only, Remark 10.12(1)).

External facts.

(E1) *Real spectral theorem.* A real symmetric matrix M has an orthonormal basis of eigenvectors: $M = \varrho\,\mathrm{diag}(\lambda)\varrho^\top$ with $\varrho \in O(n)$, and $\det(sI - M) = \prod_j (s - \lambda_j)$. Mathlib: `Matrix.IsHermitian.spectral_theorem`, `Matrix.IsHermitian.eigenvalues`, `eigenvectorBasis`, `eigenvectorUnitary`, `Matrix.IsHermitian.charpoly_eq`, `det_eq_prod_eigenvalues` (for $\mathbb{R}$, `Matrix.isHermitian_iff_isSymm`). Mathlib's `eigenvalues` is indexed by an arbitrary finite type; `eigenvalues`$_0$ is antitone. Neither is analytic in parameters; the blueprint therefore never uses them as functions of a except through Proposition 11.7.

(E2) *Analytic implicit function theorem.* If $p(w, s)$ is real analytic near $(w^0, s^0) \in \mathbb{R}^m \times \mathbb{R}$, $p(w^0, s^0) = 0$, $\partial_s p(w^0, s^0) \ne 0$, then there is a real analytic $s(w)$ near w^0 with $s(w^0) = s^0$ and $p(w, s(w)) = 0$, and it is the only continuous solution near (w^0, s^0). (Mathlib: the parameter version holds at differentiability level ω through `ContDiffAt.contDiffAt_implicitFunction` together with `AnalyticAt.contDiffAt` and `ContDiffAt.analyticAt`, specialised to the scalar condition $\partial_s p \ne 0$, with its value and local-uniqueness lemmas; alternatively it follows from the general analytic inverse function theorem of `Analysis/Analytic/Inverse.lean`. The one-variable inverse is `AnalyticAt.analyticAt_localInverse`.)

(E3) *Fundamental theorem of symmetric polynomials.* Every symmetric polynomial in $\sigma_1, \ldots, \sigma_n$ with real coefficients is a polynomial in the elementary symmetric polynomials; the latter are, up to sign, the coefficients of $\det(sI - M) = \prod(s - \sigma_j)$. Mathlib: `MvPolynomial.esymmAlgEquiv`; `Matrix.charpoly`.

(E4) *Identity theorem.* A real analytic function on a connected open set $\mathcal{A} \subset \mathbb{R}^m$ that vanishes on a nonempty open subset vanishes identically; if it is not identically zero its zero set is closed, has empty interior and Lebesgue measure zero.

(E5) Integration by parts and positivity of the integral of a continuous nonnegative function that is positive somewhere.

11.2. **Eigenvalue perturbation: Fej04r Lemmas 70, 71, Corollary 72.**

Lemma 11.1 (Fej04r Lemma 70)**.** *Let $D = \mathrm{diag}(\delta_1, \ldots, \delta_n)$ with $\delta_j \in \mathbb{R}$, $A \in M_n(\mathbb{C})$, and $\sigma \in \mathbb{C}$ an eigenvalue of $D + A$. Then $\min_j |\sigma - \delta_j| \le \|A\|$.*

Source: Fej04r Lemme 70, p. 53 (the norm "as in (23)" is the Euclidean operator norm, defined on p. 33). Proof ours, by eigenvectors.

Proof. **Step S11.1: Eigenvector.** Let $(D+A)x = \sigma x$, $|x| = 1$. If $\sigma = \delta_j$ for some j there is nothing to prove. Otherwise $(D - \sigma)x = -Ax$, so $x = -(D-\sigma)^{-1}Ax$ and $1 \le \max_j |\sigma - \delta_j|^{-1}\|A\|$. □

Lemma 11.2 (Fej04r Lemma 71, sharpened: second-order perturbation with eigenvectors)**.** *Let $n \ge 2$, $d \in \mathbb{R}^n$ with $g := \min_{j\ne k}|d_j - d_k| > 0$ (no ordering assumed), $D = \mathrm{diag}(d)$, $B \in \mathrm{Sym}_n$ with*

$B_{jj} = 0$ for all j, $\varepsilon \in \mathbb{R}$, and $\tau := |\varepsilon|\|B\|/g$. Assume $8\sqrt{n}\,\tau \le 1$. Then there are $\varrho \in SO(n)$ with columns $v^{(1)}, \dots, v^{(n)}$ and $\lambda \in \mathbb{R}^n$ with $(D + \varepsilon B)\varrho = \varrho\,\mathrm{diag}(\lambda)$ and, for all $j \ne k$,

(a) $|\lambda_j - d_j| \le 2\tau^2 g = 2\varepsilon^2\|B\|^2/g$;
(b) $|v^{(j)} - e_j| \le 3\tau$; hence $\|\varrho - I\|_F \le 3\sqrt{n}\,\tau \le \frac{3}{8}$;
(c) $|\lambda_j - \lambda_k| \ge \frac{15}{16} g$: the spectrum of $D + \varepsilon B$ is simple, and these are all its eigenvalues.

Source: Fej04r Lemme 71, p. 53 (the printed statement has $\frac{3}{\sigma}(1 + \|D\|/\sigma)$, one power of the bracket too many; the $\frac{2}{\sigma}$ at the end of the proof is not inconsistent with it, an additional factor $1 + O(\varepsilon)$ being absorbed by 3 for ε small, see §11.6); the present lemma is an explicit uniform sharpening with an explicit smallness threshold. The statement here and its proof are ours; the idea (eliminate the first-order term) is the source's.

Proof. Put $N = D + \varepsilon B$ (symmetric) and note $\tau \le \frac{1}{8\sqrt{2}} < \frac{1}{8}$.

Step S11.2: Localisation. Let λ be an eigenvalue of N with unit eigenvector v. By Lemma 11.1 (with $A = \varepsilon B$) there is j with $|\lambda - d_j| \le |\varepsilon|\|B\| = \tau g$. For $k \ne j$, $|\lambda - d_k| \ge g - \tau g \ge \frac{7}{8} g$. Write $j = j(\lambda)$.

Step S11.3: Eigenvector equations. Let $b \in \mathbb{R}^{n-1}$ be the j-th column of B without its (zero) j-th entry, B' the submatrix of B with row and column j removed, $\Delta' = \mathrm{diag}(\lambda - d_k)_{k \ne j}$ (invertible by Step S11.2) and $u = (v_k)_{k \ne j}$. Row $k \ne j$ of $Nv = \lambda v$ reads $(\lambda - d_k)v_k = \varepsilon(B_{kj}v_j + \sum_{l \ne j} B_{kl}v_l)$, i.e. $\Delta' u = \varepsilon(v_j b + B'u)$; row j reads $(\lambda - d_j)v_j = \varepsilon\, b \cdot u$ (as $B_{jj} = 0$ and B is symmetric). Since $\|\Delta'^{-1}\| \le \frac{8}{7g}$ and $|b|, \|B'\| \le \|B\|$: $|u| \le \frac{8\tau}{7}(|v_j| + |u|)$, i.e. $|u| \le \frac{8\tau/7}{1 - 8\tau/7}|v_j| \le \frac{4\tau}{3}|v_j|$. In particular $v_j \ne 0$ (else $u = 0$, $v = 0$). Put $w = u/v_j$: $|w| \le \frac{4}{3}\tau \le 2\tau$.

Step S11.4: Estimates (a), (b). From row j: $|\lambda - d_j| = |\varepsilon\, b \cdot w| \le |\varepsilon|\|B\||w| \le \tau g \cdot 2\tau$. Normalise $v_j > 0$: $v = (1 + |w|^2)^{-1/2}(e_j + w)$, so $|v - e_j| \le (1 - v_j) + v_j|w| \le \frac{1}{2}|w|^2 + |w| \le 2\tau + 2\tau^2 \le 3\tau$.

Step S11.5: The map $\lambda \mapsto j(\lambda)$ is a bijection. Take an orthonormal eigenbasis of N ((E1)) and for each member the index j of Step S11.2. If two members v, v' had the same index, then $v \cdot v' = v_j v'_j(1 + w \cdot w')$ with $|w \cdot w'| \le 4\tau^2 < 1$ and $v_j v'_j \ne 0$, so $v \cdot v' \ne 0$, contradicting orthogonality. So the assignment is injective, hence bijective; order the basis so that member j has index j, and fix the signs so that $v_j^{(j)} > 0$.

Step S11.6: (c). $|\lambda_j - \lambda_k| \ge g - 2 \cdot 2\tau^2 g \ge g(1 - \frac{4}{64})$.

Step S11.7: $\varrho \in SO(n)$. ϱ is orthogonal; $\|\varrho - I\| \le \|\varrho - I\|_F \le \sqrt{n}\,3\tau < 2$. If $\det \varrho = -1$, a real orthogonal matrix has an odd number of eigenvalues -1 (non-real ones come in conjugate pairs of product 1), so $\varrho x = -x$ for some unit x and $|(\varrho - I)x| = 2$, a contradiction. Hence $\det \varrho = 1$. □

Corollary 11.3 (Fej04r Corollary 72, quantitative)**.** *Let $n \ge 2$, $\hat{D} \in \mathrm{Sym}_{n-1}$ with eigenvalues $\delta_1, \dots, \delta_{n-1}$, $\delta_n := 0$, and $\sigma := \min_{j \ne k}|\delta_j - \delta_k| > 0$ (so the δ_j are distinct and nonzero for $j < n$). Let $D = \mathrm{diag}(\hat{D}, 0)$, $K > 0$, and $0 < \varepsilon$ with $32\sqrt{n}\,\varepsilon K \le \sigma$. Let $A \in \mathrm{Sym}_n$ (it may depend on ε) with $\|A\| \le K$. Then $D + \varepsilon A$ has n simple eigenvalues $\lambda_1, \dots, \lambda_n$, pairwise at distance $\ge \sigma/4$, with*

$$|\lambda_n - \varepsilon A_{nn}| \le \frac{16\,\varepsilon^2 K^2}{\sigma}.$$

In particular, if $A_{nn} = c_1 + c_2\varepsilon^\beta$ $(0 \le \beta < 2)$, then $\lambda_n = \varepsilon(c_1 + c_2\varepsilon^\beta) + O(\varepsilon^2)$, with the constant in O explicit; and if $|c_1 + c_2\varepsilon^\beta| \ge c > 0$ and $16\varepsilon K^2 < c\,\sigma$ this eigenvalue is nonzero. If $\hat{D}$ is diagonal, $D + \varepsilon A$ is conjugated to a diagonal matrix by some $\varrho \in SO(n)$ with $\|\varrho - I\|_F \le 12\sqrt{n}\,\varepsilon K/\sigma$ (in general $\|\varrho - \varrho_0\|_F$ is bounded by the same quantity, with $\varrho_0 = \mathrm{diag}(\hat{\varrho}, 1)$, $\hat{\varrho} \in SO(n-1)$ diagonalising $\hat{D}$).

Source: Fej04r Corollaire 72, p. 54. The printed statement leaves implicit that A_ε is bounded as $\varepsilon \to 0$ (we assume $\|A\| \le K$) and that β plays no role in the proof; the explicit remainder is ours.

Proof. **Step S11.8: Diagonalise $\hat{D}$.** Take $\hat{\varrho} \in SO(n-1)$ ((E1), flipping a sign if needed) with $\hat{\varrho}^\top \hat{D} \hat{\varrho} = \mathrm{diag}(\delta_1, \dots, \delta_{n-1})$ and $\varrho_0 = \mathrm{diag}(\hat{\varrho}, 1)$. Then $\varrho_0^\top (D + \varepsilon A)\varrho_0 = \mathrm{diag}(\delta) + \varepsilon A'$, $A' = \varrho_0^\top A \varrho_0$, $\|A'\| = \|A\| \le K$, $A'_{nn} = A_{nn}$.

Step S11.9: Split off the diagonal. $A' = \mathrm{Diag}(A') + B$ with $B \in \mathrm{Sym}_n$ of zero diagonal, $\|B\| \le 2K$ (diagonal entries are $\le \|A'\|$). Put $d_j = \delta_j + \varepsilon A'_{jj}$, so $\mathrm{diag}(\delta) + \varepsilon A' = \mathrm{diag}(d) + \varepsilon B$. Its gap is $g \ge \sigma - 2\varepsilon K \ge \sigma/2$.

Step S11.10: Apply Lemma 11.2. $\tau = \varepsilon\|B\|/g \le 4\varepsilon K/\sigma$ and $8\sqrt{n}\tau \le 32\sqrt{n}\varepsilon K/\sigma \le 1$. Part (a) with $j = n$, $d_n = \varepsilon A_{nn}$ gives $|\lambda_n - \varepsilon A_{nn}| \le 2\varepsilon^2\|B\|^2/g \le 2\varepsilon^2 \cdot 4K^2 \cdot \frac{2}{\sigma} = 16\varepsilon^2 K^2/\sigma$. Part (c) gives the separation $\ge \frac{15}{16} g \ge \frac{15}{32}\sigma \ge \sigma/4$. Part (b): $\|\varrho_1 - I\|_F \le 3\sqrt{n}\tau \le 12\sqrt{n}\varepsilon K/\sigma$ for the rotation ϱ_1 of the lemma; $\varrho = \varrho_0\varrho_1$ and $\|\varrho - \varrho_0\|_F = \|\varrho_1 - I\|_F$. □

11.3. **Negative definiteness of Q_h: Fej04r Proposition 73, Lemmas 74, 75.**

Lemma 11.4 (Hadamard; Fej04r Lemma 75)**.** *Let $Q \in \mathrm{Sym}_n$ with $q_{jk} \ge 0$ for $j \ne k$, and $\delta q_j := -q_{jj} - \sum_{l \ne j} q_{jl}$. Then for all $x \in \mathbb{R}^n$*

$$x^\top Q x = -\sum_j \delta q_j \, x_j^2 - \sum_{j<k} q_{jk}(x_j - x_k)^2.$$

Hence if $\delta q_j \ge 0$ for all j then $Q \le 0$, and if $c := \min_j \delta q_j > 0$ then $x^\top Q x \le -c|x|^2$.
Source: Fej04r Lemme 75, p. 55 (hypotheses there: $q_{jj} < 0 < q_{jk}$, $\delta q_j > 0$). Proof: expand.

Proof. **Step S11.11: Expansion.** $-\sum_j \delta q_j x_j^2 = \sum_j q_{jj} x_j^2 + \sum_j \sum_{l \ne j} q_{jl} x_j^2$ and $\sum_{j<k} q_{jk}(x_j - x_k)^2 = \sum_j \sum_{l \ne j} q_{jl} x_j^2 - 2\sum_{j<k} q_{jk} x_j x_k$. The sum of the two with the sign as stated is $\sum_j q_{jj} x_j^2 + 2\sum_{j<k} q_{jk} x_j x_k = x^\top Q x$. □

Lemma 11.5 (Laplace coefficients: $b_s^{(1)} > b_s^{(2)}$; the case of Fej04r Lemma 74 that is needed)**.** *For every $s > 0$ and $\alpha \in (0,1)$, with $A(t) = 1 + \alpha^2 - 2\alpha\cos t$ (Definition 10.6),*

$$b_s^{(1)}(\alpha) - b_s^{(2)}(\alpha) = \frac{2s\alpha}{\pi} \int_0^\pi \sin^2 t \, (1 - \cos t) \, A(t)^{-s-1} \, dt > 0.$$

In particular $b_{3/2}^{(1)}(\alpha) > b_{3/2}^{(2)}(\alpha) > 0$.

Source: Fej04r Lemme 74, pp. 54–55 states $b_s^{(j)} > b_s^{(j+1)}$ for all $s \in \mathbb{R}_+ \setminus \mathbb{N}$, $j \in \mathbb{N}$, by an induction on s using three-term recurrences (displayed with the resulting positive difference formula on p. 55). Only $j = 1$, $s = \frac{3}{2}$ is used; the proof below is ours, independent of the recurrences, and avoids importing the stronger general result.

Proof. **Step S11.12: Reduction to $[0,\pi]$.** $w(t) = A(t)^{-s}$ is even and 2π-periodic, so $b_s^{(k)} = \frac{1}{\pi}\int_0^\pi \cos(kt)\, w(t)\, dt$.

Step S11.13: Integration by parts. Let $H(t) = \sin t\,(1 - \cos t)$. Then $H'(t) = \cos t - \cos 2t$, $H(0) = H(\pi) = 0$, and $H > 0$ on $(0,\pi)$. So $b_s^{(1)} - b_s^{(2)} = \frac{1}{\pi}\int_0^\pi H' w \, dt = -\frac{1}{\pi}\int_0^\pi H w' \, dt$, and $-w'(t) = 2s\alpha \sin t \, A(t)^{-s-1} \ge 0$ on $[0,\pi]$. This is the stated formula ((E5)); the integrand is continuous, ≥ 0 and positive on $(0,\pi)$. For the last claim use Lemma 10.7 (positivity of $b_s^{(k)}$ from the series). □

Proposition 11.6 (Fej04r Proposition 73)**.** *Let $n \ge 2$, $a_1 < \cdots < a_n$ (so $0 < \alpha_{ij} = a_i/a_j < 1$ for $i < j$), all masses positive. With $\mathfrak{a}_{ij}, \mathfrak{b}_{ij}$ of Definition 10.10 put*

$$c_{\hat{q}}(a) := \min_j \sum_{k \ne j} (\mathfrak{a}_{jk} - \mathfrak{b}_{jk}) > 0.$$

Then for all $x \in \mathbb{R}^n$: $x^\top \hat{q} x \le -\sum_j \big(\sum_{k\neq j}(\mathfrak{a}_{jk} - \mathfrak{b}_{jk})\big)x_j^2 \le -c_{\hat{q}}|x|^2$, and

$$x^\top Q_h x \le -\frac{c_{\hat{q}}(a)}{\max_i \Lambda_i}|x|^2.$$

So $\hat{q}$ and Q_h are negative definite, every eigenvalue satisfies $\sigma_j \le -c_{\hat{q}}/\max_i \Lambda_i < 0$, and $\Omega_j = 2\sigma_j < 0$. The bound $c_{\hat{q}}/\max \Lambda$ is continuous, hence locally uniformly positive, on $\{a_1 < \cdots < a_n\}$. (For $n = 1$ there is no pair and $Q_h = 0$.)

Source: Fej04r Prop. 73 and proof, pp. 54–55 (proof via Lemmas 74, 75 and a maximum-component argument for definiteness). The source's remark that Herman omitted the sign of the Newtonian potential is consistent with Part 8, where $V_{ij} = -Gm_im_j/|q^{(i)} - q^{(j)}|$. The quantitative proof is ours.

Proof. **Step S11.14: Strong diagonal dominance.** Definition 10.10: $\hat{q}_{jk} = \mathfrak{b}_{jk} > 0$ $(j \neq k)$, $\hat{q}_{jj} = -\sum_{k\neq j}\mathfrak{a}_{jk}$, hence $\delta\hat{q}_j = \sum_{k\neq j}(\mathfrak{a}_{jk} - \mathfrak{b}_{jk})$. For $j < k$: $\mathfrak{a}_{jk} - \mathfrak{b}_{jk} = \frac{Gm_jm_ka_j}{4a_k^2}\big(b^{(1)}_{3/2} - b^{(2)}_{3/2}\big)(\alpha_{jk}) > 0$ by Lemma 11.5, and the same for $k < j$ by symmetry of $\mathfrak{a}, \mathfrak{b}$. As $n \ge 2$ every j has at least one $k \neq j$, so $\delta\hat{q}_j > 0$.

Step S11.15: Lemma 11.4. gives the first two inequalities (the dropped term $-\sum_{j<k}\hat{q}_{jk}(x_j - x_k)^2 \le 0$).

Step S11.16: Congruence. $x^\top Q_h x = (\Delta^{-1}x)^\top \hat{q}(\Delta^{-1}x) \le -c_{\hat{q}}|\Delta^{-1}x|^2 \le -c_{\hat{q}}|x|^2/\max_i \Lambda_i$ since $\Delta^{-1} = \mathrm{diag}(\Lambda_i^{-1/2})$. An eigenvalue is $x^\top Q_h x$ for a unit eigenvector x. Continuity: $\mathfrak{a}_{jk}, \mathfrak{b}_{jk}$ are continuous in a (Lemma 10.7). □

11.4. **Local analytic diagonalisation; the symplectic rotation.**

Proposition 11.7 (Local analytic spectral decomposition)**.** *Let $\mathcal{W} \subset \mathbb{R}^m$ be open, $M : \mathcal{W} \to \mathrm{Sym}_n$ real analytic, $w^0 \in \mathcal{W}$ such that $M(w^0)$ has n distinct eigenvalues $\sigma_1^0 < \cdots < \sigma_n^0$ (equivalently* $\mathrm{discr}(M(w^0)) \neq 0$*, `Matrix.discr`). Then there are an open ball $\mathcal{W}' \ni w^0$ and real analytic $\sigma_j : \mathcal{W}' \to \mathbb{R}$, $\varrho : \mathcal{W}' \to SO(n)$ with $\sigma_1 < \cdots < \sigma_n$ on $\mathcal{W}'$, $M(w) = \varrho(w)\,\mathrm{diag}(\sigma(w))\varrho(w)^\top$ and $\sigma_j(w^0) = \sigma_j^0$. The spectrum of $M(w)$ is simple on $\mathcal{W}'$. On a connected open set, σ_j is unique, and ϱ is unique up to multiplying columns by ± 1 (with* $\det = 1$*).*

Proof. **Step S11.17: Eigenvalues.** $p(w, s) = \det(sI - M(w))$ is real analytic in (w, s) and $\partial_s p(w^0, \sigma_j^0) = \prod_{k\neq j}(\sigma_j^0 - \sigma_k^0) \neq 0$. By (E2) there are analytic $\sigma_j(w)$ near w^0 with $p(w, \sigma_j(w)) = 0$, $\sigma_j(w^0) = \sigma_j^0$. Shrinking to a ball the $\sigma_j(w)$ stay in disjoint intervals around the σ_j^0, so they are n distinct real roots of the degree-n polynomial $p(w, \cdot)$: they are all of its roots, they are the eigenvalues of $M(w)$ with $\sigma_1 < \cdots < \sigma_n$, and $M(w)$ has simple spectrum.

Step S11.18: Eigenprojections by the adjugate. Fix w, an orthonormal eigenbasis v_k $(M(w)v_k = \sigma_k v_k)$. In this basis $\mathrm{adj}(\sigma_j I - M) = \mathrm{diag}\big(\prod_{l\neq k}(\sigma_j - \sigma_l)\big)_k$ has only its (j, j) entry nonzero, so $\mathrm{adj}(\sigma_j I - M) = p_s(\sigma_j)\, v_j v_j^\top$ with $p_s = \partial_s p(w, \cdot) \neq 0$. Hence $\Pi_j(w) := \mathrm{adj}(\sigma_j(w)I - M(w))/\partial_s p(w, \sigma_j(w)) = v_j v_j^\top$ is the orthogonal projection onto the σ_j-eigenspace, and *real analytic in w* (adjugate: polynomial in the entries).

Step S11.19: Eigenvectors. $\mathrm{tr}\,\Pi_j(w^0) = 1$, so some diagonal entry $\Pi_j(w^0)_{kk} > 0$, with $k = k(j)$; on a smaller ball $\Pi_j(w)_{kk} > 0$ and $v_j(w) := \Pi_j(w)e_k/\sqrt{\Pi_j(w)_{kk}}$ is analytic, of unit norm, an eigenvector for $\sigma_j(w)$ with $\Pi_j v_j = v_j$, hence $v_j = \pm$ the vectors of the eigenbasis. The $v_j(w)$ are orthonormal.

Step S11.20: $SO(n)$ and uniqueness. $\det[v_1|\ldots|v_n]$ is continuous with values ± 1, so constant on the ball; replace v_1 by $-v_1$ if it equals -1. Uniqueness: an eigenvalue branch is a continuous root of p, unique by (E2) and the ordering; eigenvectors for a simple eigenvalue are determined up to sign, locally constant. □

Proposition 11.8 (Symplectic lift of a Λ-dependent rotation)**.** *Let $\mathcal{W} \subset \mathbb{R}^n$ be open (actions Λ) and $\varrho : \mathcal{W} \to SO(n)$ real analytic. On $\mathbb{R}^n \times \mathcal{W} \times \mathbb{R}^n \times \mathbb{R}^n \ni (\lambda, \Lambda, \xi, \eta)$ with the chart form $\Omega_{\mathrm{chart}} = d\Lambda \wedge d\lambda + d\eta \wedge d\xi$ (Theorem 9.12, Part 7) define*

$$\Phi_\varrho(\lambda, \Lambda, \xi, \eta) = (\tilde\lambda, \Lambda, \tilde\xi, \tilde\eta), \quad \tilde\xi = \varrho(\Lambda)^\top \xi,\ \tilde\eta = \varrho(\Lambda)^\top \eta,\ \tilde\lambda_l = \lambda_l - \eta^\top \big(\partial_{\Lambda_l}\varrho\big)\varrho^\top \xi \quad (l = 1, \dots, n).$$

Then Φ_ϱ is a real analytic diffeomorphism of $\mathbb{R}^n \times \mathcal{W} \times \mathbb{R}^n \times \mathbb{R}^n$ onto itself with inverse $\xi = \varrho\tilde\xi$, $\eta = \varrho\tilde\eta$, $\lambda_l = \tilde\lambda_l + \tilde\eta^\top(\varrho^\top\partial_{\Lambda_l}\varrho)\tilde\xi$; it is symplectic, $\Phi_\varrho^*\Omega_{\mathrm{chart}} = \Omega_{\mathrm{chart}}$ (a 1-chart in the sense of Definition 2.3); $\Phi_\varrho(\lambda + c, \Lambda, \xi, \eta) = \Phi_\varrho(\lambda, \Lambda, \xi, \eta) + (c, 0, 0, 0)$ for every $c \in \mathbb{R}^n$ (in particular it commutes with 2π-shifts of the angles); and it preserves Λ and $|\xi|, |\eta|$ (Euclidean norms).
Source: Ours; the lift of a parameter-dependent symplectic map is standard (Maurer–Cartan). Numerically confirmed ($n = 3$, Jacobian by finite differences, $\|J^\top\Omega J - \Omega\|_\infty = 10^{-5}$, scratch script `.drafts/P/bp-scratch-bpP-p9/chk.py`).

Proof. **Step S11.21: Bijectivity.** $\varrho(\Lambda)$ is orthogonal; the displayed inverse is read off, with $H_l(\Lambda, \tilde\xi, \tilde\eta) := \tilde\eta^\top\Omega_l\tilde\xi$, $\Omega_l := \varrho^\top\partial_{\Lambda_l}\varrho$, since $\eta^\top(\partial_l\varrho)\varrho^\top\xi = \tilde\eta^\top\varrho^\top(\partial_l\varrho)\tilde\xi = H_l$. All maps are analytic. The commutation with angle translations holds because $\tilde\lambda - \lambda$ does not depend on λ.

Step S11.22: Antisymmetry. $\varrho^\top\varrho = I$ gives $\Omega_l^\top = -\Omega_l$; $d\varrho = \varrho\sum_l \Omega_l d\Lambda_l$, $d\varrho^\top = -\sum_l \Omega_l\varrho^\top d\Lambda_l$.

Step S11.23: Pull-back computation. Substitute $\xi = \varrho\tilde\xi$, $\eta = \varrho\tilde\eta$ in $d\eta^\top \wedge d\xi = \sum_i d\eta_i \wedge d\xi_i$:

$$d\eta^\top \wedge d\xi = d\tilde\eta^\top \wedge d\tilde\xi + \sum_l \big(d\tilde\eta^\top\Omega_l\tilde\xi + \tilde\eta^\top\Omega_l d\tilde\xi\big) \wedge d\Lambda_l - \sum_{l,m} \tilde\eta^\top\Omega_l\Omega_m\tilde\xi\ d\Lambda_l \wedge d\Lambda_m,$$

using $\varrho^\top d\varrho = \sum \Omega_l d\Lambda_l$ and $d\varrho^\top\varrho = -\sum\Omega_l d\Lambda_l$ (the mixed terms $d\tilde\eta^\top\varrho^\top \wedge d\varrho\tilde\xi$, $\tilde\eta^\top d\varrho^\top \wedge \varrho\, d\tilde\xi$ and $\tilde\eta^\top d\varrho^\top \wedge d\varrho\,\tilde\xi$). The first bracket is $d_{(\tilde\xi,\tilde\eta)}H_l$, the differential of H_l in the variables $(\tilde\xi, \tilde\eta)$ only. The full differential is $dH_l = d_{(\tilde\xi,\tilde\eta)}H_l + \sum_m \tilde\eta^\top(\partial_m\Omega_l)\tilde\xi\, d\Lambda_m$, so

$$d\eta^\top \wedge d\xi = d\tilde\eta^\top \wedge d\tilde\xi + \sum_l dH_l \wedge d\Lambda_l - \sum_{l,m} \tilde\eta^\top\big(\partial_m\Omega_l + \Omega_m\Omega_l\big)\tilde\xi\ d\Lambda_m \wedge d\Lambda_l.$$

Step S11.24: Maurer–Cartan. $\partial_m\Omega_l = \partial_m\varrho^\top\,\partial_l\varrho + \varrho^\top\partial_m\partial_l\varrho = -\Omega_m\Omega_l + \varrho^\top\partial_m\partial_l\varrho$ (as $\partial_m\varrho^\top = -\Omega_m\varrho^\top$), so $\partial_m\Omega_l + \Omega_m\Omega_l = \varrho^\top\partial_m\partial_l\varrho$ is symmetric in (m, l) and the last sum vanishes against $d\Lambda_m \wedge d\Lambda_l$.

Step S11.25: Conclusion. $d\eta^\top \wedge d\xi = d\tilde\eta^\top \wedge d\tilde\xi - \sum_l d\Lambda_l \wedge dH_l$, hence $\sum_l d\Lambda_l \wedge d\lambda_l + d\eta \wedge d\xi = \sum_l d\Lambda_l \wedge d(\lambda_l - H_l) + d\tilde\eta \wedge d\tilde\xi = \sum_l d\Lambda_l \wedge d\tilde\lambda_l + d\tilde\eta \wedge d\tilde\xi$, i.e. $\Phi_\varrho^*\Omega_{\mathrm{chart}} = \Omega_{\mathrm{chart}}$. Euclidean norms are preserved since ϱ is orthogonal. □

Theorem 11.9 (Diagonalisation and the frozen Birkhoff data)**.** *In the situation of Prop. 10.2 and Theorem 10.11 with $n \geq 2$, let $\mathcal{W} \subset \{\Lambda : a(\Lambda)$ increasing$\}$ be open, and suppose that $Q_h(\Lambda)$ has simple spectrum on $\mathcal{W}$ and σ_j, ϱ are the analytic functions of Proposition 11.7 on $\mathcal{W}$ (this holds on a ball around every Λ^0 where $\operatorname{discr} Q_h \neq 0$; if $\mathcal{W}$ is connected the labelling by increasing order is global on $\mathcal{W}$). Then:*

*(a) (*Quadratic part.*) With $(\tilde\xi, \tilde\eta) = \Phi_\varrho$-image of Prop. 11.8,*

$$\xi^\top Q_h\xi + \eta^\top Q_h\eta = \sum_j \sigma_j(\tilde\xi_j^2 + \tilde\eta_j^2) = \sum_j \Omega_j\tilde R_j, \qquad \tilde R_j := \tfrac12(\tilde\eta_j^2 + \tilde\xi_j^2), \quad \Omega_j := 2\sigma_j < 0.$$

*(b) (*Secular function of the rotated problem.*) For the two matched pairs $(S, \hat f) \in \{(S_{\mathrm{fix}}, \hat f_{\mathrm{fix}}), (S_{\mathrm{mov}}, \hat f_{\mathrm{mov}})\}$ (Prop. 10.2; not independently chosen) the angle-average of $\hat f \circ \Phi_\varrho^{-1}$ is*

$$\big\langle \hat f \circ \Phi_\varrho^{-1}\big\rangle(\Lambda, \tilde\xi, \tilde\eta) = S(\Lambda, \varrho\tilde\xi, \varrho\tilde\eta) = C_0 + [\delta] + \sum_j \Omega_j(\Lambda)\tilde R_j + O(|(\tilde\xi, \tilde\eta)|^4),$$

where $[\delta]=0$ *for fix and* $=\delta(\Lambda)$ *for mov (Theorem 10.13), locally uniformly in* $\Lambda\in\mathcal{W}$, *on the set where* $(\xi,\Lambda,\eta)=(\varrho\tilde\xi,\Lambda,\varrho\tilde\eta)\in\mathcal{N}_{\mathbb{R}}(\rho_*)$*; for* $\mu>0$ *the Hamiltonians* $H^{\Psi}_{\mathrm{fix}},H^{\Psi}_{\mathrm{mov}}$ *transformed by* Φ_ϱ^{-1} *are* $h(\Lambda)+\mu\,\hat f\circ\Phi_\varrho^{-1}$ *(*h *depends on* Λ *only, which* Φ_ϱ *preserves).*

(c) *(*Frozen dictionary.*) With* $(I,p,q)=(\Lambda,\tilde\eta,\tilde\xi)$ *as in §10.1,* $P_0=C_0(+\delta)$, $\Omega_j(\mu,I)=2\sigma_j(a(I))$ *(independent of* μ*) and* $R_j=\frac12(p_j^2+q_j^2)$ *of Definition 1.14, the zeroth- and first-order terms of* HasBirkhoff *for* $P=\hat f\circ\Phi_\varrho^{-1}$ *are those of (b).* The matrix β and the integrability clause (i) are not treated here *(Part 11b, Corollary 10.14(c)). The domains: if* $\tilde\xi_j^2+\tilde\eta_j^2<\delta_0^2$ *for all* j *then* $|\xi|,|\eta|<\sqrt{n}\,\delta_0$*, so* $(\xi,\Lambda,\eta)\in\mathcal{N}_{\mathbb{R}}(\rho_*)$ *as soon as* $\sqrt{n}\,\delta_0\le\rho_*$ *and* $|\Lambda-\Lambda^*|_\infty<\rho_*$.

(d) *(*Transport.*)* $\Psi\circ\Phi_\varrho^{-1}$ *is a symplectic chart of Part 1 (Definition 2.3) on the preimage of* $\mathcal{O}_n$, *satisfies the lattice compatibility (Q1)–(Q2) of Theorem 2.9 whenever* Ψ *does (the angle-shift of* Φ_ϱ *is independent of* λ*), and the torus-transport theorem applies to it.*

Source: Fej04r Formule (36)–(37), pp. 50–51 and the diagonalisation around Prop. 73, pp. 54–55; CP09 Thm. 4 (properly degenerate form $\sum\Omega_j(u_j^2+v_j^2)/2$*), CP10 (*$A_2''$*). The factor: Remark 10.12(2) (halving of* Q_h*, E-P-Fej04r-3). The lift (b), (d) is ours; Fej04r and CP11 pass to "Birkhoff variables" without mentioning the* Λ*-dependence of the rotation.*

Source gap: The sources treat the diagonalising rotation as a plain symplectic change of the slow variables; because ϱ *depends on the actions* Λ *it is symplectic on the full phase space only after the angle correction of Prop. 11.8. The correction is a shift of* λ *independent of* λ *and so does not affect any average over* λ *(part (b)).*

Proof. **Step S11.26: (a).** $Q_h=\varrho\,\mathrm{diag}(\sigma)\varrho^\top$ gives $\xi^\top Q_h\xi=\sum_j\sigma_j\tilde\xi_j^2$, similarly for η; and $\sigma_j(\tilde\xi_j^2+\tilde\eta_j^2)=2\sigma_j\tilde R_j$. Negativity: Prop. 11.6.

Step S11.27: (b). $\Phi_\varrho^{-1}(\tilde\lambda,\Lambda,\tilde\xi,\tilde\eta)=(\tilde\lambda+H(\Lambda,\tilde\xi,\tilde\eta),\Lambda,\varrho\tilde\xi,\varrho\tilde\eta)$, with H independent of $\tilde\lambda$. Hence $\hat f\circ\Phi_\varrho^{-1}(\tilde\lambda,\cdot)=\hat f(\tilde\lambda+H,\Lambda,\varrho\tilde\xi,\varrho\tilde\eta)$, which is continuous and 2π-periodic in $\tilde\lambda$ (Prop. 10.2(b)). The average over $\tilde\lambda\in[0,2\pi]^n$ is invariant under the translation by the constant vector H (periodicity, (E2)), so it equals $S(\Lambda,\varrho\tilde\xi,\varrho\tilde\eta)$. Insert Theorem 10.11, Theorem 10.13 and (a): the remainder $O(|(\xi,\eta)|^4)$ becomes $O(|(\tilde\xi,\tilde\eta)|^4)$ since $|\varrho\tilde\xi|=|\tilde\xi|$. The statement on $H^\Psi\circ\Phi_\varrho^{-1}$ is Theorem 9.12(b),(c) and Φ_ϱ^{-1} preserving Λ.

Step S11.28: (c). Read off from (a), (b) and Definition 1.14; the norm comparison is $|\xi|_\infty\le|\xi|=|\tilde\xi|\le\sqrt{n}\max_j|\tilde\xi_j|$ and $\max_j|\tilde\xi_j|<\delta_0$.

Step S11.29: (d). Compositions of symplectic charts are symplectic charts (the pull-back of Ω_{cart} by $\Psi\circ\Phi_\varrho^{-1}$ is $(\Phi_\varrho^{-1})^*\Omega_{\mathrm{chart}}=\Omega_{\mathrm{chart}}$ by Prop. 11.8, Theorem 9.12). (Q1): $\Phi_\varrho^{-1}(z+2\pi e_k)=\Phi_\varrho^{-1}(z)+2\pi e_k$ and Part 7's Ψ has the same property; (Q2): if $\Psi\Phi_\varrho^{-1}(z')=\Psi\Phi_\varrho^{-1}(z)+\sigma_k$ then, by the property of Ψ, $\Phi_\varrho^{-1}z'=\Phi_\varrho^{-1}z+2\pi m$, hence $z'=z+2\pi m$ as Φ_ϱ is injective and commutes with the shifts. □

Remark 11.10 (Dictionary: Fej04r, CP09/CP10, frozen). (1) Frozen Ω_j multiplies $R_j=\frac12(p_j^2+q_j^2)$ (Definition 1.14; CP09 Thm. 4 writes $\Omega_j(u_j^2+v_j^2)/2$). Under the frozen chart (Part 7, $|u|^2=2\Gamma$) $S=C_0+\sum\sigma_j(\tilde\xi_j^2+\tilde\eta_j^2)=C_0+\sum2\sigma_j\tilde R_j$, so $\Omega_j=2\sigma_j$ with σ_j the eigenvalues of the *halved* Q_h of Definition 10.10; the list $\Omega=(2\sigma_j)_j$ is the eigenvalue list of the matrix printed in Fej04r (36)–(37) in the ξ-basis, i.e. of $Q_h^{\mathrm F}=2Q_h$ (suspected erratum E-P-Fej04r-3, Remark 10.12(2)); the σ_j themselves are the eigenvalues of the halved Q_h. Both sets of eigenvalues are negative (Prop. 11.6). (2) Sign: $\Omega_j<0$ for all j. CP10 (A_2''), (8) only requires $|\sum_j\Omega_jk_j|\ge\mathrm{const}>0$ for $0<|k|_1\le4$, which is insensitive to the overall sign. (3) Coordinate permutation: the Lean order of the entries of Q_h is that of increasing axes (Q_h index i = planet i); the eigenvalue labelling is increasing in σ and

independent of it; the frozen $\Omega : \mathbb{R} \times \mathbb{R}^a \to \mathbb{R}^b$ is total, so Part 12 sets it to 0 off the patch (nothing is claimed there). (4) $n = 1$: there is no secular problem, $Q_h = 0$ and $\Omega_1 = 0$ (no pair); statements with $n \geq 2$ above do not apply.

11.5. **Distinct and non-resonant spectrum: the interface for Parts 10–11.**

Definition 11.11 (Non-resonance up to order four; the function Φ_4)**.** Let $K_4 = \{k \in \mathbb{Z}^n : 0 < |k|_1 \leq 4\}$. For $\sigma \in \mathbb{R}^n$ put

$$\mathrm{NR}_4(\sigma) :\Longleftrightarrow \ \sigma_j \text{ pairwise distinct and } k \cdot \sigma \neq 0 \ \ \forall k \in K_4,$$

and

$$\mathcal{R}_k(\sigma) = \prod_{\pi \in S_n} \sum_j k_j \sigma_{\pi(j)} \quad (k \in K_4), \qquad \mathcal{D}(\sigma) = \prod_{j<l} (\sigma_j - \sigma_l)^2, \qquad \Phi_4(\sigma) = \mathcal{D}(\sigma) \prod_{k \in K_4} \mathcal{R}_k(\sigma).$$

For $a \in \{a_1 < \cdots < a_n\}$ write $\Phi_4(a) := \Phi_4(\sigma(a))$ for the eigenvalues $\sigma(a)$ of $Q_h(a)$ in any order. (Since $\sigma \mapsto 2\sigma$ rescales, $\mathrm{NR}_4(\sigma) \Leftrightarrow \mathrm{NR}_4(\Omega)$, $\Omega = 2\sigma$; and the quantitative form $|\Omega \cdot k| \geq \alpha > 0$ on K_4 of CP10 (8) holds on every compact subset of $\{\mathrm{NR}_4\}$ by continuity.)

Source: CP10 Remark 1 and (A_2'), pp. 548–549, formula (8) with 6 replaced by 4; Prop. 2 (27), p. 553.

Proposition 11.12 (The good set)**.** *(a) Φ_4 is a symmetric polynomial in σ; hence $a \mapsto \Phi_4(a)$ is a polynomial in the coefficients of* $\det(sI - Q_h(a))$ *and is* real analytic on $\{a_1 < \cdots < a_n\}$*, and independent of the labelling.*

(b) $\Phi_4(a) \neq 0$ if and only if $\mathrm{NR}_4(\sigma(a))$*; then $Q_h(a)$ has simple spectrum, all $\Omega_j = 2\sigma_j \neq 0$, and Propositions 11.7, Theorem 11.9 apply on a ball around a.*

(c) Let $\mathcal{A} \subset \{a_1 < \cdots < a_n\}$ be open and connected. Either $\Phi_4 \equiv 0$ on $\mathcal{A}$ or $\mathcal{G} := \{a \in \mathcal{A} : \Phi_4(a) \neq 0\}$ is open, dense and of full Lebesgue measure in $\mathcal{A}$. So: one *point $a^0 \in \mathcal{A}$ with* $\mathrm{NR}_4(\sigma(a^0))$ *gives an open dense full-measure subset on every point of which the eigenvalues are analytic and distinct and Ω is non-resonant to order four, uniformly on compact subsets of $\mathcal{G}$.*

Proof. **Step S11.30: (a).** $\mathcal{D}$ is symmetric (squares), and $\mathcal{R}_k$ is invariant under $\sigma \mapsto \sigma \circ \pi'$ because the product runs over all of S_n; so Φ_4 is a symmetric polynomial and, by (E3), a polynomial Π in the elementary symmetric polynomials of σ, which up to sign are the coefficients of $\det(sI - Q_h(a))$ ((E1)); these are polynomials in the entries of $Q_h(a)$, real analytic by Definition 10.10.

Step S11.31: (b). $\mathcal{D} \neq 0$ iff the σ_j are distinct; $\mathcal{R}_k \neq 0$ iff $\sum k_j \sigma_{\pi(j)} \neq 0$ for all π, iff $k' \cdot \sigma \neq 0$ for all rearrangements k' of k. K_4 is invariant under rearrangement of coordinates, so the conjunction over $k \in K_4$ is $k \cdot \sigma \neq 0$ for all $k \in K_4$. Simple spectrum: Prop. 11.7 (with discr $= \mathcal{D} \neq 0$).

Step S11.32: (c). Identity theorem (E4) for the analytic function Φ_4 on connected $\mathcal{A}$: if it is not identically zero, it does not vanish on any open set; its zero set is closed, nowhere dense and null. Uniformity on compacts: continuity of $a \mapsto \sigma(a)$ (continuity of roots; or locally Prop. 11.7) and finiteness of K_4. □

Lemma 11.13 (The top eigenvalue is always simple)**.** *For $n \geq 2$ and increasing axes, the largest eigenvalue $\sigma_{\max}$ of Q_h (the one closest to 0) is simple and has an eigenvector with all entries nonzero of one sign.*

Proof. **Step S11.33: Positivity.** Off the diagonal, $(Q_h)_{jk} = \mathfrak{b}_{jk}/\sqrt{\Lambda_j \Lambda_k} > 0$ for *all* $j \neq k$ (Definition 10.10, including non-adjacent planets). Take c with $M = Q_h + cI$ entrywise > 0. Let x be a unit eigenvector for $\sigma_{\max}$; $|x|$ (entrywise) satisfies $|x|^\top M |x| \geq x^\top M x = \max_{|y|=1} y^\top M y$, so $|x|$ is also a maximiser, hence an eigenvector for $\sigma_{\max}$, and $M|x| = (\sigma_{\max} + c)|x|$ has strictly positive entries, hence $|x| > 0$ entrywise. So every eigenvector for $\sigma_{\max}$ has all entries nonzero. If the eigenspace had dimension ≥ 2 it would contain a nonzero vector with a vanishing first entry, a contradiction. Sign: $|x|$ and x are both eigenvectors of a simple eigenvalue, so $x = \pm|x|$. □

Corollary 11.14 ($n = 2$: explicit)**.** *Let $n = 2$, $\alpha = a_1/a_2$, $\mathfrak{a} = \mathfrak{a}_{12}$, $\mathfrak{b} = \mathfrak{b}_{12}$. Then*

$$Q_h = \begin{pmatrix} -\mathfrak{a}/\Lambda_1 & \mathfrak{b}/\sqrt{\Lambda_1\Lambda_2} \\ \mathfrak{b}/\sqrt{\Lambda_1\Lambda_2} & -\mathfrak{a}/\Lambda_2 \end{pmatrix}, \quad \operatorname{tr} Q_h = -\mathfrak{a}\Big(\frac{1}{\Lambda_1} + \frac{1}{\Lambda_2}\Big), \quad \det Q_h = \frac{\mathfrak{a}^2 - \mathfrak{b}^2}{\Lambda_1\Lambda_2} > 0,$$

$$\operatorname{tr}^2 - 4\det = \mathfrak{a}^2\Big(\frac{1}{\Lambda_1} - \frac{1}{\Lambda_2}\Big)^2 + \frac{4\mathfrak{b}^2}{\Lambda_1\Lambda_2} > 0.$$

Hence for every $a_1 < a_2$ the eigenvalues $\sigma_1 < \sigma_2 < 0$ are distinct and real analytic, with a global analytic rotation $\varrho \in SO(2)$ on all of $\{a_1 < a_2\}$ (no patch needed for $n = 2$; the discriminant never vanishes and $q_{12} = \mathfrak{b}/\sqrt{\Lambda_1\Lambda_2} > 0$ gives a global frame, explicit below)*, and, with $t = \sigma_1/\sigma_2 > 1$,*

$$\mathrm{NR}_4(\sigma) \iff t \notin \{2, 3\} \iff \big(2\operatorname{tr}^2 - 9\det\big)\big(3\operatorname{tr}^2 - 16\det\big) \neq 0.$$

So for $n = 2$ the functions Φ_4 and $\mathcal{R} := (2\operatorname{tr}^2 - 9\det)(3\operatorname{tr}^2 - 16\det)$ have the same zero set *on $\{a_1 < a_2\}$ (the negative simple-spectrum locus). They are* not *equal up to a nonvanishing factor: each ratio-2 and ratio-3 resonance factor occurs in Φ_4 to the fourth power (all signed k and both permutations), so at a resonance Φ_4 vanishes to higher order than $\mathcal{R}$. Only the equivalence $\Phi_4(a) \neq 0 \iff \mathcal{R}(a) \neq 0$ is used; $\mathcal{R}$ is an explicit analytic function of a built from $b^{(1)}_{3/2}, b^{(2)}_{3/2}$, and Part 10 must show that it is nonzero at one point (then $\Phi_4 \not\equiv 0$ by this equivalence).*

Proof. **Step S11.34: Matrix.** Definition 10.10 with $n = 2$: $\hat{q}_{11} = \hat{q}_{22} = -\mathfrak{a}$ (as $\mathfrak{a}_{21} = \mathfrak{a}_{12}$), $\hat{q}_{12} = \mathfrak{b}$; conjugate by Δ^{-1}. $\det = (\mathfrak{a}^2 - \mathfrak{b}^2)/(\Lambda_1\Lambda_2) > 0$ by Lemma 11.5; the discriminant of a 2×2 symmetric matrix is $(q_{11} - q_{22})^2 + 4q_{12}^2$, here > 0 because $\mathfrak{b} > 0$. *Global frame:* writing $Q_h = (q_{jk})$, $q_{12} > 0$, and $\sigma_1 = \frac{1}{2}\big(\operatorname{tr} - \sqrt{\operatorname{tr}^2 - 4\det}\big)$ (analytic since the discriminant is > 0), the vector $w = (q_{12}, \sigma_1 - q_{11})$ satisfies $(Q_h - \sigma_1)w = 0$ and $|w| \geq q_{12} > 0$. Then $v^{(1)} = w/|w|$, $v^{(2)} = Jv^{(1)}$ (J the rotation by $\pi/2$) and $\varrho = (v^{(1)}\, v^{(2)}) \in SO(2)$ are real analytic on $\{a_1 < a_2\}$; $v^{(2)} \perp v^{(1)}$ is an eigenvector for the other eigenvalue σ_2 because Q_h is symmetric with simple spectrum.

Step S11.35: Resonances. $\sigma_j < 0$ (Prop. 11.6). For $k \in K_4$ with a zero entry, $k \cdot \sigma \neq 0$. For $k_1k_2 > 0$ the two entries of k have the same sign, so $k \cdot \sigma \neq 0$ (it is negative if $k_1, k_2 > 0$ and positive if both are negative). For $k_1k_2 < 0$ ($|k|_1 \leq 4$): $k \cdot \sigma = 0 \iff \sigma_1/\sigma_2 = -k_2/k_1 \in \{1, 2, \frac{1}{2}, 3, \frac{1}{3}\}$ (from $(1, -1), (2, -2) \mapsto 1$; $(1, -2), (2, -1) \mapsto 2, \frac{1}{2}$; $(1, -3), (3, -1) \mapsto 3, \frac{1}{3}$). Distinctness excludes 1; with $\sigma_1 < \sigma_2 < 0$, $t = \sigma_1/\sigma_2 > 1$ so $\frac{1}{2}, \frac{1}{3}$ are impossible. Finally $\operatorname{tr}^2/\det = t + 1/t + 2$ equals $\frac{9}{2}$ at $t = 2$ and $\frac{16}{3}$ at $t = 3$, and $t \mapsto t + 1/t$ is injective on $(1, \infty)$. □

11.6. **Sources, deviations, errata, forward references.**
Sources read. Fej04r pp. 52–56 ("7.1 Lemmes sur les valeurs propres", Lemmas 70–75, Prop. 73 with its proof, start of 7.3); Fej07e (`fejoz04_erratum.txt`); CP10 pp. 548–549, 553 (non-resonance of order four, Prop. 2); CP09 Thm. 4 for the form $\sum \Omega_j(u_j^2 + v_j^2)/2$ (via Part 8); Part 8 of this blueprint in full; Mathlib names in (E1)–(E3) were checked in `.lake/packages/mathlib`.
Deviations. (1) Lemma 11.2 is a new statement: no ordering of d, no hypothesis on $\|D\|$ (the printed bound $\propto (1 + \|D\|/\sigma)$ is replaced by an absolute constant), eigenvectors included; the proof is a direct eigenvector elimination, not the conjugation $e^{-\varepsilon U}(D + \varepsilon A)e^{\varepsilon U}$ (whose bound $\|U\| \leq \|A\|/\sigma$ is not elementary for the operator norm, while for the Frobenius norm the product estimates change). (2) Corollary 11.3 adds the boundedness hypothesis on A_ε. (3) Lemma 74 is replaced by Lemma 11.5 (one inequality, complete proof). (4) The lift Prop. 11.8 and the explicit Φ_4 are not in the sources. (5) Eigenvalue labelling by increasing value; Q_h halved against the print (Part 8).
Suspected errata (for the ledger). E-P-Fej04r-4 *withdrawn* (review P9-02): the $\frac{3}{\sigma}$ versus $\frac{2}{\sigma}$ is not an inconsistency (the proof carries an extra $1 + O(\varepsilon)$ absorbed by 3); the only remark is that "ε assez petit" is not quantified, a quantitative enhancement and not a false statement. E-P-Fej04r-5 *withdrawn* (review P9-03): Lemma 74's proof on pp. 54–55 displays the recurrences and completes the induction; we keep our independent $j = 1$ integration-by-parts proof only because it avoids

importing the general result. E-P-Fej04r-6 (minor): Cor. 72 does not say that A_ε is bounded. Already recorded: E-P-Fej04r-3 (factor $\frac{1}{2}$), E-P-Fej04r-2 (Lemma 76).
Doubts about frozen statements. None. The Birkhoff form is stated for the given (p, q) (Definition 1.14): with ϱ depending on the action it is reached from the physical chart only through Φ_ϱ (Prop. 11.8), so the *perturbation* handed to P-G is $\hat{f} \circ \Phi_\varrho^{-1}$, not $\hat{f}$.
External facts used. (E1)–(E5); (E2); Parts 7, 8, 0 and Part 1 (chart, torus transport).
Forward references and consumers. Part 10: Prop. 11.12(c) (must exhibit one point with $\Phi_4 \neq 0$ on the connected domain of axes, or a point of each patch), Corollary 11.14 ($n = 2$: show $(2\,\mathrm{tr}^2 - 9\det)(3\,\mathrm{tr}^2 - 16\det) \not\equiv 0$; asymptotically $\sigma_{1,2} \sim$ explicit, giving ratios $\neq 2, 3$), Cor. 11.3 (induction for $n \geq 3$), Lemma 11.13. Parts 11b/12: Prop. 11.7, Theorem 11.9 (the rotation Φ_ϱ and $\Omega = 2\sigma$), Definition 11.11. Part 13 (fix): same, by Corollary 10.14. No forward references to unwritten parts in this part.
Label index. `lemma70`, `lemma71`, `cor72`, `l75`, `l74`, `definite`, `local-analytic`, `lift`, `diagonalize`, `dict`, `good`, `good-set`, `top-simple`, `n2`.

12. Nondegeneracy of the planetary frequency data

Parts 8–9 produced the real analytic symmetric matrix $Q_h(a)$ of the secular quadratic form, its eigenvalues σ_j (the frozen $\Omega_j = 2\sigma_j$), and the single analytic function Φ_4 of the axes whose non-vanishing means *simple spectrum and non-resonance up to order four* (Definition 11.11). This part proves that these objects are non-degenerate in the two senses the downstream parts need, and states the result in their exact form.

- `good-set` (Theorem 12.8, Corollary 12.9, Lemma 12.1): $\Phi_4 \not\equiv 0$ on the connected domain $\{0 < a_1 < \cdots < a_n\}$, by exhibiting points in the hierarchical regime $a_1 \ll a_2 \ll \cdots \ll a_n$; hence the good set is open, dense and of full measure, and every neighbourhood of every prescribed axis vector contains a ball of good axes. Only the local holomorphy of the Laplace coefficients is used; no analytic continuation of the eigenvalues, no universal cover (Fej04r Lemma 76 is false, Remark 10.8);
- `asymptotics` (Lemma 12.4, Corollary 12.5, Lemma 12.6): Fej04r Formula 69 with explicit remainders, and the exact structure of Q_h when the innermost axis tends to 0;
- `r-nondegenerate` (Theorem 12.14, Theorem 12.15): the frequency map $a \mapsto (\nu(a), 2\sigma(a)) \in \mathbb{R}^{2n}$ is Rüssmann-nondegenerate (CP09 Def. 1) on *every* connected open set of simple spectrum, for every $n \geq 2$. The proof is the symmetric-product argument (a polynomial in the characteristic coefficients, single valued and analytic on the convex domain of axes) plus the peeling induction on n that Fej04r Prop. 77 only sketches (a source gap, Theorem 12.14);
- `index-amount` (Lemma 12.18, Corollary 12.20, Lemma 12.21, Theorem 12.22): the index and amount of nondegeneracy on a compact patch in the exact form of CP09 Lemma 1/(2)–(3), and the interface statement for Parts 11a, 12, 13.

The quartic torsion (the frozen matrix β) is not treated here (Part 11b).

12.1. **Conventions and external facts.** Fix $n \geq 2$, $G, m_0 > 0$, $\kappa = Gm_0$, $m \in (0, \infty)^n$ as in Part 8, and write

$$\mathcal{A}_n = \{a \in \mathbb{R}^n : \ 0 < a_1 < \cdots < a_n\}$$

(planet 1 innermost, the Lean order). $\mathcal{A}_n$ is an *open convex* cone (finitely many strict linear inequalities), hence connected. The data $\Lambda_i(a) = m_i\sqrt{\kappa a_i}$, $a_i = \Lambda_i^2/(m_i^2\kappa)$ define a real analytic diffeomorphism $a \mapsto \Lambda(a)$ of $\mathcal{A}_n$ onto an open set $\mathcal{A}_n^\Lambda \subset \mathbb{R}^n$ (the frozen action variable $I = \Lambda$, Part 8, §10.1), with real analytic inverse. The fast frequencies are

$$\nu_i(a) = \frac{\kappa^2 m_i^3}{\Lambda_i^3} = \kappa^{1/2} a_i^{-3/2} \qquad (i = 1, \ldots, n)$$

(Theorem 9.12(b); they do not depend on the masses when expressed in the axes). $Q_h(a) \in \mathrm{Sym}_n$ is the matrix of Definition 10.10 (real analytic in a, Lemma 10.7); $\sigma_1(a) \le \cdots \le \sigma_n(a)$ are its eigenvalues *sorted increasingly, with multiplicity*. Recall $\sigma_j < 0$ for all j (Proposition 11.6). With $\mathcal{D}$, K_4, NR_4, Φ_4 as in Definition 11.11 put

$$\mathcal{U}_n = \{a \in \mathcal{A}_n : \ \mathcal{D}(\sigma(a)) \neq 0\}, \qquad \mathcal{G}_n = \{a \in \mathcal{A}_n : \ \Phi_4(a) \neq 0\} \subset \mathcal{U}_n.$$

$\mathcal{D}(\sigma(a))$ is continuous (symmetric polynomial in the characteristic coefficients), so $\mathcal{U}_n$ and $\mathcal{G}_n$ are open. For vectors, $|x|$ is the Euclidean norm, $|x|_\infty$ the max norm; $\|M\| \le \|M\|_F$ as in §11.1. Frozen conventions: all statements here are about the axes a or the actions Λ (no angles occur); the passage between CP's 2π-angles and the period-one frequency $w = \omega/2\pi$ is a positive rescaling of the frequency components, which preserves every statement below (Rüssmann nondegeneracy and index/amount up to a constant factor, Remark 12.17).

External facts (those of Part 9 are reused by label).

(F1) *Identity theorem and integral domain.* Identity theorem (E4): a real analytic function on a connected open set $\mathcal{O}$ that vanishes on a nonempty open subset vanishes identically. Mathlib: `AnalyticOnNhd.eqOn_zero_of_preconnected_of_eventuallyEq_zero`. If all derivatives of a real analytic function vanish at a point it vanishes on a ball (the Taylor series converges to the function), hence identically on $\mathcal{O}$.

(F2) Fundamental theorem of symmetric polynomials (E3), here over the coefficient ring $\mathbb{R}[X]$ (it holds over every commutative ring): a symmetric polynomial in $\sigma_1, \dots, \sigma_n$ with coefficients in $\mathbb{R}[X]$ is a polynomial in $e_1(\sigma), \dots, e_n(\sigma)$ with coefficients in $\mathbb{R}[X]$.

(F3) Spectral theorem and charpoly (E1); local analytic eigenvalues (E2), Proposition 11.7.

(F4) *Polarization.* A symmetric ν-linear form T on $\mathbb{R}^N$ is zero iff $a \mapsto T(a, \dots, a)$ is zero iff $\max_{|a|=1} |T(a, \dots, a)| = 0$. (Not found in Mathlib in this form; the classical identity $T(a_1, \dots, a_\nu) = \frac{1}{\nu! 2^\nu} \sum_{\epsilon \in \{\pm 1\}^\nu} \epsilon_1 \cdots \epsilon_\nu T(s_\epsilon, \dots, s_\epsilon)$, $s_\epsilon = \sum \epsilon_i a_i$.)

(F5) Leibniz formula for the symmetric tensor derivative of a product: $\partial^\nu (fg)(y)(a, \dots, a) = \sum_{k=0}^{\nu} \binom{\nu}{k} \partial^k f(y)(a^k)\, \partial^{\nu-k} g(y)(a^{\nu-k})$.

Lemma 12.1 (Finitely many analytic conditions)**.** *Let $\mathcal{O} \subset \mathbb{R}^N$ be open and connected and $g_1, \dots, g_r : \mathcal{O} \to \mathbb{R}$ real analytic and $\not\equiv 0$. Then $g = g_1 \cdots g_r \not\equiv 0$, and $\{g \neq 0\}$ is open, dense in $\mathcal{O}$ and has full Lebesgue measure in $\mathcal{O}$. In particular, if $g_1 = \Phi_4$ and $g_2, \dots, g_r$ are further analytic functions of the axes (for instance the torsion determinant of Part 11b) that do not vanish identically on $\mathcal{A}_n$, the set where all of them are nonzero is open, dense and of full measure in $\mathcal{A}_n$.*

Proof. **Step S12.1: Product.** For $r = 2$: $\{g_1 \neq 0\}$ is open; if $g_1 g_2 \equiv 0$ then $g_2 = 0$ on it. A nonempty open set on which g_2 vanishes forces $g_2 \equiv 0$ ((F1)); so $\{g_1 \neq 0\} = \emptyset$, i.e. $g_1 \equiv 0$, a contradiction ($\{g_1 \neq 0\}$ is nonempty). Induct on r.

Step S12.2: Zero set. By Proposition 11.12(c)/(E4), the zero set of a not identically zero analytic function on a connected open set is closed with empty interior and null. □

Lemma 12.2 (Weyl's inequality)**.** *Let $A, E \in \mathrm{Sym}_n$ and let $\lambda_1(M) \le \cdots \le \lambda_n(M)$ be the sorted eigenvalues of $M \in \mathrm{Sym}_n$. Then $|\lambda_l(A+E) - \lambda_l(A)| \le \|E\|$ for every l. In particular the sorted eigenvalues are continuous (indeed 1-Lipschitz for the operator norm) functions of the matrix.*

Proof. **Step S12.3: Min–max.** $\lambda_l(M) = \min_{\dim V = l} \max_{x \in V, |x|=1} x^\top M x$. ($\le$: take $V = \mathrm{span}(v_1, \dots, v_l)$, v_j an orthonormal eigenbasis ((E1)), $x^\top M x = \sum \lambda_j x_j^2 \le \lambda_l$. $\ge$: any l-dimensional V meets $\mathrm{span}(v_l, \dots, v_n)$ of dimension $n - l + 1$ in a unit vector x, and $x^\top M x \ge \lambda_l$.) Mathlib has the spectral theorem but, to our knowledge, no Courant–Fischer; this is the only place it is needed.

Step S12.4: Perturbation. $|x^\top (A+E) x - x^\top A x| \le \|E\|$ for unit x, so the min–max values differ by at most $\|E\|$. □

Lemma 12.3 (Sorted eigenvalues are analytic where the spectrum is simple)**.** *On $\mathcal{U}_n$ the sorted eigenvalues $\sigma_1 < \cdots < \sigma_n$ are real analytic functions of a. Moreover $\mathcal{U}_2 = \mathcal{A}_2$.*

Proof. **Step S12.5: Local branches.** For $a^0 \in \mathcal{U}_n$ Proposition 11.7 gives a ball and analytic $\tilde\sigma_1 < \cdots < \tilde\sigma_n$ on it which are *all* the eigenvalues of $Q_h(a)$; hence $\tilde\sigma_j = \sigma_j$ (sorted).

Step S12.6: $n = 2$**.** Corollary 11.14: the discriminant is positive on all of $\mathcal{A}_2$. □

12.2. **Laplace coefficients and the secular matrix as the innermost axis tends to** 0**.**

Lemma 12.4 (Laplace coefficients $b^{(1)}_{3/2}, b^{(2)}_{3/2}$ for small argument)**.** *For $k \in \{1,2\}$ and $0 \le \alpha \le \frac12$, with $t_1 = \frac32$, $t_2 = \frac{15}{8}$,*

$$t_k \alpha^k \;\le\; b^{(k)}_{3/2}(\alpha) \;\le\; t_k \alpha^k (1 + 4\alpha^2).$$

Source: Fej04r Formule 69, p. 52 (the asymptotics $C_1 = -\frac{1}{a_j}\big(\frac34 (a_k/a_j)^2 + O((a_k/a_j)^4)\big)$, $C_2 = O((a_k/a_j)^3)/a_j$ as $a_k/a_j \to 0$, asserted without proof; the corresponding statements for $b^{(1)}_{3/2}, b^{(2)}_{3/2}$ are proved here from the series).

Proof. **Step S12.7: Series.** By Lemma 10.7, $b^{(k)}_{3/2}(\alpha) = \sum_{q\ge0} T_q$, $T_q = \frac{(3/2)_{q+k}(3/2)_q}{(q+k)!\,q!}\alpha^{2q+k} \ge 0$. The first term is $T_0 = \frac{(3/2)_k}{k!}\alpha^k = t_k\alpha^k$ ($\frac32$ and $\frac32\cdot\frac52/2$).

Step S12.8: Ratio. $T_{q+1}/T_q = \frac{(q+k+3/2)(q+3/2)}{(q+k+1)(q+1)}\alpha^2$. For $k \ge 1$ the first factor is $1 + \frac{1}{2(q+k+1)} \le \frac54$ and the second $1 + \frac{1}{2(q+1)} \le \frac32$, so $T_{q+1} \le \frac{15}{8}\alpha^2 T_q \le 2\alpha^2 T_q$.

Step S12.9: Tail. $\sum_{q\ge1} T_q \le T_0 \frac{2\alpha^2}{1-2\alpha^2} \le 4\alpha^2 T_0$ since $2\alpha^2 \le \frac12$. □

Corollary 12.5 (Formula 69 with remainders)**.** *For $i < j$ and $\alpha = a_i/a_j \le \frac12$, with $\mathfrak{a}_{ij}, \mathfrak{b}_{ij}$ of Definition 10.10,*

$$\mathfrak{a}_{ij} = \frac38 \frac{G m_i m_j\, \alpha^2}{a_j}(1+\theta), \quad \mathfrak{b}_{ij} = \frac{15}{32}\frac{G m_i m_j\, \alpha^3}{a_j}(1+\theta'), \qquad 0 \le \theta, \theta' \le 4\alpha^2.$$

In particular $\mathfrak{a}_{ij} \le \frac34 G m_i m_j a_i^2 a_j^{-3}$ and $\mathfrak{b}_{ij} \le \frac{15}{16} G m_i m_j a_i^3 a_j^{-4}$.
Source: Fej04r Formule 69, p. 52, in the normalisation of Remark 10.12: Fej04r's C_1, C_2 are $-2\mathfrak{a}/(Gm_im_j)$, $+2\mathfrak{b}/(Gm_im_j)$ (inner/outer exchanged), so $\frac34\alpha^2/a_j$ there is $\frac38\alpha^2/a_j$ here; C_0 is not needed.

Proof. $\mathfrak{a}_{ij} = \frac{G m_i m_j a_i}{4a_j^2} b^{(1)}_{3/2}(\alpha) = \frac{G m_i m_j \alpha}{4a_j} b^{(1)}_{3/2}$ and similarly $\mathfrak{b}_{ij}$; insert Lemma 12.4 and $1+4\alpha^2 \le 2$. □

Lemma 12.6 (Peeling the innermost planet)**.** *Let $n \ge 2$, $\hat a = (a_2, \dots, a_n)$ with $0 < a_2 < \cdots < a_n$, and $a = (a_1, \hat a) \in \mathcal{A}_n$ with $0 < a_1 \le \min\{1, a_2/2\}$. Let $\hat Q(\hat a) \in \mathrm{Sym}_{n-1}$ be the matrix Q_h of the $(n-1)$-planet system with the same G, m_0, masses $(m_2, \dots, m_n)$ and axes $\hat a$ if $n \ge 3$, and $\hat Q := (0)$ if $n = 2$ (Definition 10.10 has no pair for one planet). Then with $D = 0 \oplus \hat Q(\hat a)$ (first row and column the innermost planet)*

$$E := Q_h(a) - D, \qquad \|E\| \le \|E\|_F \le C\, a_1^{3/2},$$

where C depends on $(G, m_0, m, \hat a)$ but not on a_1. More precisely: $E_{jk} = 0$ for $j \ne k$, $j, k \ge 2$;

$$|E_{11}| \le \frac{3G}{4\sqrt\kappa}\, a_1^{3/2} \sum_{k\ge2} m_k a_k^{-3}, \qquad |E_{jj}| \le \frac{3Gm_1}{4\sqrt\kappa}\, a_1^2 a_j^{-7/2}\ (j \ge 2),$$

$$|E_{1k}| \le \frac{15G\sqrt{m_1 m_k}}{16\sqrt\kappa}\, a_1^{11/4} a_k^{-17/4}\ (k \ge 2).$$

Source: Fej04r Prop. 77, proof, p. 58 ("$Q_h = \begin{pmatrix} \hat{Q}_h + O(a_n^{5/2}) & O(a_n^{5/2}) \\ \cdot & b_{nn} \end{pmatrix}$", Fej04r's decreasing order, innermost planet n). In the present normalisation (increasing order, halved Q_h) the exponents are 2 for the perturbation of the $(n-1)$-block and $\frac{11}{4}$ for the coupling, see the notes; numerically confirmed in `.drafts/P/bp-scratch-bpP-p10/chk.py`.

Proof. **Step S12.10: Entries.** By Definition 10.10, $Q_{jk} = \hat{q}_{jk}/\sqrt{\Lambda_j \Lambda_k}$, $\hat{q}_{jk} = \mathfrak{b}_{jk}$ $(j \neq k)$, $\hat{q}_{jj} = -\sum_{k \neq j} \mathfrak{a}_{jk}$, and $\mathfrak{a}_{jk}, \mathfrak{b}_{jk}, \Lambda_j$ depend only on the pair resp. the planet. Hence for $j, k \geq 2$ the entries of $Q_h(a)$ and of $\hat{Q}$ agree, except $Q_{jj} = \hat{Q}_{jj} - \mathfrak{a}_{1j}/\Lambda_j$ (for $n = 2$: $Q_{22} = -\mathfrak{a}_{12}/\Lambda_2 = 0 - \mathfrak{a}_{12}/\Lambda_2$). The first row is $Q_{11} = -\sum_{k \geq 2} \mathfrak{a}_{1k}/\Lambda_1$, $Q_{1k} = \mathfrak{b}_{1k}/\sqrt{\Lambda_1 \Lambda_k}$.

Step S12.11: Bounds. Here $a_1/a_k \leq \frac{1}{2}$ for $k \geq 2$, so Corollary 12.5 applies. With $\Lambda_i = m_i\sqrt{\kappa a_i}$: $\mathfrak{a}_{1k}/\Lambda_1 \leq \frac{3}{4} G m_1 m_k a_1^2 a_k^{-3}/(m_1\sqrt{\kappa a_1})$, which is the bound for E_{11}; $\mathfrak{a}_{1j}/\Lambda_j \leq \frac{3}{4} G m_1 m_j a_1^2 a_j^{-3}/(m_j\sqrt{\kappa a_j})$; and $\mathfrak{b}_{1k}/\sqrt{\Lambda_1\Lambda_k} \leq \frac{15}{16} G m_1 m_k a_1^3 a_k^{-4}/\big((m_1 m_k)^{1/2}\kappa^{1/2}(a_1 a_k)^{1/4}\big)$ (indeed $\sqrt{\Lambda_1\Lambda_k} = \sqrt{m_1 m_k}\sqrt{\kappa}\,(a_1 a_k)^{1/4}$), i.e. $\frac{15 G\sqrt{m_1 m_k}}{16\sqrt{\kappa}} a_1^{11/4} a_k^{-17/4}$. As $a_1 \leq 1$, every power $a_1^2, a_1^{11/4}$ is $\leq a_1^{3/2}$. □

12.3. **A good point: $\Phi_4 \not\equiv 0$ (part A).**

Proposition 12.7 (The case $n = 2$)**.** *Let $n = 2$ and $r(a) := \operatorname{tr}^2 Q_h / \det Q_h$, $t := \sigma_1/\sigma_2 > 1$ (so $r = t + t^{-1} + 2$). Then with $\mathfrak{a} = \mathfrak{a}_{12}$, $\mathfrak{b} = \mathfrak{b}_{12}$,*

$$r(a) = \frac{(\Lambda_1 + \Lambda_2)^2}{\Lambda_1 \Lambda_2} \cdot \frac{1}{1 - (\mathfrak{b}/\mathfrak{a})^2} > \frac{\Lambda_2}{\Lambda_1} + 2, \qquad \frac{\Lambda_2}{\Lambda_1} = \frac{m_2}{m_1}\sqrt{\frac{a_2}{a_1}}, \qquad t \geq r - 3.$$

Consequently, if $a_2/a_1 \geq 16\, m_1^2/m_2^2$ then $r > 6$ and $\mathrm{NR}_4(\sigma(a))$ holds, i.e. $\Phi_4(a) \neq 0$; and as $a_1/a_2 \to 0$ one has $r \to \infty$, $t \to \infty$.

Source: Fej04r Prop. 77, $n = 2$ case, pp. 56–57 ((41): $\sigma_1, \sigma_2 \sim -\frac{3}{8} m_1 m_2 \frac{a_{\mathrm{in}}^2}{a_{\mathrm{out}}^3} \Lambda_{1,2}^{-1}$ in Fej04r's normalisation,); the exact formula for r is ours and needs no asymptotics of Laplace coefficients.

Proof. **Step S12.12: Exact trace–determinant ratio.** Corollary 11.14: $\operatorname{tr} = -\mathfrak{a}(\Lambda_1 + \Lambda_2)/(\Lambda_1\Lambda_2)$, $\det = (\mathfrak{a}^2 - \mathfrak{b}^2)/(\Lambda_1\Lambda_2) > 0$. Hence $r = \mathfrak{a}^2(\Lambda_1 + \Lambda_2)^2/\big(\Lambda_1\Lambda_2(\mathfrak{a}^2 - \mathfrak{b}^2)\big)$, the displayed formula. Since $0 < \mathfrak{b} < \mathfrak{a}$ (Lemma 11.5), $(\mathfrak{b}/\mathfrak{a})^2 \in (0, 1)$ and the second factor exceeds 1; $(\Lambda_1 + \Lambda_2)^2/(\Lambda_1\Lambda_2) = \Lambda_2/\Lambda_1 + 2 + \Lambda_1/\Lambda_2 > \Lambda_2/\Lambda_1 + 2$.

Step S12.13: t versus r. $\sigma_1 < \sigma_2 < 0$ gives $t > 1$ and $r = \sigma_1/\sigma_2 + 2 + \sigma_2/\sigma_1 = t + 1/t + 2$ $(\operatorname{tr} = \sigma_1 + \sigma_2, \det = \sigma_1\sigma_2)$; $1/t < 1$ gives $t > r - 3$.

Step S12.14: Non-resonance. By Corollary 11.14, $\mathrm{NR}_4 \iff t \notin \{2, 3\}$, and $t + 1/t \in \{\frac{5}{2}, \frac{10}{3}\}$ at these values, whereas $a_2/a_1 \geq 16 m_1^2/m_2^2$ gives $\Lambda_2/\Lambda_1 \geq 4$, so $t + 1/t = r - 2 > \Lambda_2/\Lambda_1 \geq 4$. □

Theorem 12.8 (Existence of good axes, hierarchical regime)**.** *Let $n \geq 2$, $G, m_0 > 0$, $m \in (0, \infty)^n$. For every $\epsilon \in (0, 1)$ there is $a \in \mathcal{A}_n$ with $a_j/a_{j+1} < \epsilon$ for $j = 1, \ldots, n-1$ and $\mathrm{NR}_4(\sigma(a))$, i.e. $\Phi_4(a) \neq 0$. In particular $\Phi_4 \not\equiv 0$ on $\mathcal{A}_n$.*

Source: Fej04r Prop. 77, pp. 56–58 (induction on n by letting the innermost semi-major axis tend to 0; the printed induction step is "par un raisonnement analogue à ci-dessus" and Cor. 72); the proof here is ours: Weyl's inequality replaces Cor. 72, because negative definiteness already gives $\sigma_n < 0$.

Source gap: Fej04r's induction (p. 58) shows no scales and no transfer of non-resonance; supplied here.

Proof. Induction on n for all data (G, m_0, m) simultaneously.

Step S12.15: $n = 2$. Take $a_2 = 1$ and $a_1 < \min\{\epsilon, m_2^2/(16 m_1^2)\}$ and apply Proposition 12.7.

Step S12.16: Step $n - 1 \to n$, $n \geq 3$: the chosen point. Apply the hypothesis to the system $(G, m_0, (m_2, \ldots, m_n))$: there is $\hat{a} \in \mathcal{A}_{n-1}$ (axes $a_2 < \cdots < a_n$) with $a_j/a_{j+1} < \epsilon$ $(2 \leq j < n)$ and

$\mathrm{NR}_4(\delta)$ for the sorted eigenvalues $\delta_1 \le \cdots \le \delta_{n-1}$ of $\hat{Q}(\hat{a})$. By Proposition 11.6 (for $n-1 \ge 2$) all $\delta_l < 0$, and

$$\kappa_0 := \min\{|\hat{k}\cdot\delta| :\ \hat{k}\in\mathbb{Z}^{n-1},\ 0<|\hat{k}|_1\le 4\} > 0.$$

Let $a=(a_1,\hat{a})$, with a_1 so small that $a_1 \le \min\{1, a_2/2\}$, $a_1 < \epsilon a_2$ and $4\|E\| < \kappa_0$, where E is the matrix of Lemma 12.6 ($\|E\| \le C a_1^{3/2} \to 0$).

Step S12.17: Location of the eigenvalues. The sorted spectrum of $D = 0 \oplus \hat{Q}$ is $\delta_1 \le \cdots \le \delta_{n-1} < 0$ followed by 0 (all $\delta_l < 0$). By Lemma 12.2, $|\sigma_l(a) - \delta_l| \le \|E\|$ for $l \le n-1$ and $|\sigma_n(a)| \le \|E\|$; moreover $\sigma_n(a) < 0$ by Proposition 11.6 for the n-planet system.

Step S12.18: No resonance. Let $k = (\hat{k}, k_n) \in K_4$, $\hat{k}\in\mathbb{Z}^{n-1}$. If $\hat{k}=0$, $k\cdot\sigma = k_n\sigma_n(a) \ne 0$ ($k_n \ne 0$, $\sigma_n<0$). If $\hat{k}\ne 0$ then $|k_n| \le 4 - |\hat{k}|_1$ and

$$|k\cdot\sigma| \ge |\hat{k}\cdot\delta| - \sum_{l<n}|\hat{k}_l|\|E\| - |k_n|\|E\| \ge \kappa_0 - |k|_1\|E\| \ge \kappa_0 - 4\|E\| > 0.$$

(For $\hat{k}$ with $0<|\hat{k}|_1\le 4$, $\hat{k}\cdot\delta\ne 0$ is the definition of κ_0.) So $k\cdot\sigma\ne 0$ for all $k\in K_4$; this includes the differences $e_j - e_l$, so σ has distinct entries. Hence $\mathrm{NR}_4(\sigma(a))$, and $a_1/a_2 < \epsilon$ by the choice of a_1. □

Corollary 12.9 (The good set; good patches)**.** *For every $n\ge 2$ and all data: $\mathcal{G}_n = \{\Phi_4 \ne 0\}$ is open, dense and of full Lebesgue measure in $\mathcal{A}_n$, and on every point of it the eigenvalues are simple, real analytic (locally, Proposition 11.7) and $\Omega = 2\sigma$ is non-resonant to order 4, uniformly on compact subsets. Consequently for every $a^*\in\mathcal{A}_n$ and $\eta>0$ there are $a_0\in\mathcal{G}_n$ and $r_0>0$ with $\overline{B}_\infty(a_0,2r_0)\subset\mathcal{G}_n\cap B_\infty(a^*,\eta)$.*

Proof. $\mathcal{A}_n$ is connected and Φ_4 is real analytic on it (Proposition 11.12(a)), not identically zero by Theorem 12.8; so Proposition 11.12(c) gives the first sentence. $\mathcal{G}_n\cap B_\infty(a^*,\eta)\cap\mathcal{A}_n$ is nonempty open (density), hence contains a closed cube. □

Remark 12.10 (Connectedness without continuation). Fej04r reaches the same conclusion on the *complexified* domain $A^{\mathbb{C}}$ and the universal cover of the complement of the discriminant, claimed simply connected although the loop $a\mapsto e^{it}a$ projects to winding one (E-P-Fej04r-2, Part 0). None of this is used: $\mathcal{A}_n$ itself is convex, Φ_4 is single valued and analytic on it, and the identity theorem is applied directly. The exp chart $L=\{\mathrm{Re}\,l_1>\cdots>\mathrm{Re}\,l_n,\ |\mathrm{Im}\,l_j|<\pi/4\}$ of `statement.md` item 2 is not needed either.

12.4. **R-nondegeneracy of $(\nu,2\sigma)$ (part B).** For $a\in\mathcal{U}_n$ put $\varpi(a) = (\nu_1(a),\dots,\nu_n(a),2\sigma_1(a),\dots,2\sigma_n(a))\in\mathbb{R}^{2n}$ (real analytic on $\mathcal{U}_n$, Lemma 12.3); its expression in the actions is $\tilde{\varpi}=\varpi\circ a(\Lambda)$ on $\Lambda(\mathcal{U}_n)$.

Definition 12.11 (Rüssmann nondegeneracy)**.** A real analytic $\omega: B\to\mathbb{R}^m$ on a nonempty open connected $B\subset\mathbb{R}^N$ is *R-nondegenerate* if for every $c\in\mathbb{R}^m\setminus\{0\}$ the function $y\mapsto\langle c,\omega(y)\rangle$ is not identically zero on B.

Source: CP09 Def. 1, p. 5 (equivalently, $\omega(B)$ lies in no hyperplane through 0); Rus01 §1.2, p. 124 ("nondegenerate").

Definition 12.12 (Symmetric product of the relation forms)**.** For $c=(c',c'')\in\mathbb{R}^n\times\mathbb{R}^n$ and $\pi\in S_n$ put

$$L_{c,\pi}(a) = \sum_i c'_i\nu_i(a) + \sum_j c''_j\sigma_{\pi(j)}(a), \qquad F_c(a) = \prod_{\pi\in S_n} L_{c,\pi}(a) \qquad (a\in\mathcal{A}_n),$$

with the sorted eigenvalues $\sigma_l(a)$ of $Q_h(a)$. Since the product runs over all of S_n, F_c is unchanged if σ is replaced by any rearrangement.

Proposition 12.13 (F_c is real analytic on all of $\mathcal{A}_n$)**.** *F_c is real analytic on $\mathcal{A}_n$ (not only on $\mathcal{U}_n$); and on $\mathcal{U}_n$, $L_{c,\mathrm{id}} = \langle c, (\nu, \sigma)\rangle$ is one of its factors.*

Proof. **Step S12.19: Polynomial in the characteristic coefficients.** Let $x(a) = \sum c_i' \nu_i(a)$, real analytic on $\mathcal{A}_n$ ($a_i > 0$). $P(X;\sigma) := \prod_{\pi\in S_n}(X + \sum_j c_j'' \sigma_{\pi(j)}) \in \mathbb{R}[X][\sigma_1, \dots, \sigma_n]$ is symmetric in σ (a permutation of σ permutes the factors). By (F2), $P = \Pi(X; e_1(\sigma), \dots, e_n(\sigma))$ for a polynomial Π. Hence $F_c(a) = \Pi(x(a); e_1(\sigma(a)), \dots, e_n(\sigma(a)))$, and $(-1)^k e_k(\sigma(a))$ is the coefficient of s^{n-k} in $\det(sI - Q_h(a))$, a polynomial in the entries of $Q_h(a)$, real analytic by Definition 10.10. Compositions and products of real analytic functions are real analytic ((E3)). □

Theorem 12.14 ($F_c \not\equiv 0$ for $c \neq 0$)**.** *For every $n \geq 2$, all $G, m_0 > 0$, $m \in (0,\infty)^n$ and every $c \in \mathbb{R}^{2n} \setminus \{0\}$, $F_c \not\equiv 0$ on $\mathcal{A}_n$.*

Source: Fej04r Prop. 77, pp. 56–58 (for $n = 2$ via the universal cover of the complement of the discriminant and (41); general n "par un raisonnement analogue"); the symmetric-product device is ours (PLAN.md, Part 10), the peeling induction is ours.

Source gap: Fej04r's general-n induction is not written out; supplied here. For the base case the printed argument needs the universal cover of a domain whose simple connectedness is false as printed; the present argument does not.

Proof. Induction on n, for all data. Throughout $a = (a_1, \hat{a})$, $\hat{a} = (a_2, \dots, a_n)$ fixed, $a_1 \downarrow 0$ (so $a_1 < a_2$ eventually); $\hat{x}(\hat{a}) = \sum_{i\geq 2} c_i' \nu_i(a_i)$ does not depend on a_1. By Lemma 12.6 and Weyl (Lemma 12.2) all $\sigma_l(a)$ stay bounded as $a_1 \to 0$.

Step S12.20: Case A: $c_1' \neq 0$ (every $n \geq 2$). $\nu_1(a_1) = \kappa^{1/2} a_1^{-3/2} \to \infty$ while the other summands of every $L_{c,\pi}$ stay bounded; so $|L_{c,\pi}| \to \infty$ for all π simultaneously, and $F_c(a_1, \hat{a}) \neq 0$ for small a_1, for any fixed $\hat{a}$.

Step S12.21: Case B: $c_1' = 0$, $n = 2$. Write $c = (0, c_2; c_3, c_4)$, $\sigma_1 < \sigma_2 < 0$ (Corollary 11.14), $s = \sigma_2/\sigma_1 \in (0,1)$; $s = 1/t \leq 1/(r-3) \to 0$ as $a_1 \to 0$ (Proposition 12.7). If $c_2 \neq 0$: both $L_{c,\pi} \to c_2\nu_2(a_2) \neq 0$, since $\sigma_l \to 0$ (Lemma 12.6, $\hat{Q} = 0$). If $c_2 = 0$ then $(c_3, c_4) \neq 0$ and $L_{c,\mathrm{id}} = \sigma_1(c_3 + c_4 s)$, $L_{c,\mathrm{swap}} = \sigma_1(c_4 + c_3 s)$. If $c_3 = 0$: $L_{\mathrm{id}} = \sigma_1 c_4 s \neq 0$, $L_{\mathrm{swap}} = \sigma_1 c_4 \neq 0$; if $c_4 = 0$: $L_{\mathrm{id}} = \sigma_1 c_3 \neq 0$, $L_{\mathrm{swap}} = \sigma_1 c_3 s \neq 0$; if both are nonzero, $c_3 + c_4 s \neq 0 \neq c_4 + c_3 s$ for s small. In every case $F_c(a_1, a_2) \neq 0$ for small a_1.

Step S12.22: Case C: $c_1' = 0$, $n \geq 3$. The $(n-1)$-planet system $(G, m_0, (m_2, \dots, m_n))$ has the same ν_i ($i \geq 2$) and the matrix $\hat{Q}(\hat{a})$, with sorted eigenvalues $\delta_1 \leq \cdots \leq \delta_{n-1} < 0$. By Lemma 12.6 and Lemma 12.2, as $a_1 \to 0$:

$$\sigma_l(a) \to \delta_l \ (l \leq n-1), \qquad \sigma_n(a) \to 0.$$

For $d \in \{1, \dots, n\}$ let $\iota_d : \{1, \dots, n-1\} \to \{1, \dots, n\} \setminus \{d\}$ be the increasing bijection and $c^{(d)} := \big((c_2', \dots, c_n'); (c''_{\iota_d(l)})_{l\leq n-1}\big) \in \mathbb{R}^{2(n-1)}$. The map $\pi \mapsto (d, \pi'')$, $d = \pi^{-1}(n)$, $\pi'' = \pi \circ \iota_d \in S_{n-1}$, is a bijection $S_n \to \{1..n\} \times S_{n-1}$, and

$$\lim_{a_1\to 0} L_{c,\pi}(a_1, \hat{a}) = \hat{x}(\hat{a}) + \sum_{j\neq d} c_j'' \delta_{\pi(j)} = L^{(n-1)}_{c^{(d)},\pi''}(\hat{a})$$

(the term $c_d'' \sigma_n$ tends to 0). Hence $\lim_{a_1\to 0} F_c(a_1, \hat{a}) = \prod_{d=1}^n F^{(n-1)}_{c^{(d)}}(\hat{a})$. *Subcase C1: $c^{(d)} \neq 0$ for all d.* By the induction hypothesis each $F^{(n-1)}_{c^{(d)}} \not\equiv 0$ on the connected open set $\mathcal{A}_{n-1}$ (axes $a_2 < \cdots < a_n$) and is real analytic there (Proposition 12.13); by Lemma 12.1 there is $\hat{a}$ at which all n factors are $\neq 0$. Then $\lim F_c(a_1, \hat{a}) \neq 0$, so $F_c(a_1, \hat{a}) \neq 0$ for small a_1. *Subcase C2: $c^{(d)} = 0$ for some d.* Then $c_2', \dots, c_n' = 0$ and $c_j'' = 0$ for $j \neq d$; with $c_1' = 0$ and $c \neq 0$ this means $c = c_d'' e_d$ (in the σ-block), $c_d'' \neq 0$. Then $L_{c,\pi} = c_d'' \sigma_{\pi(d)}(a) \neq 0$, since $c_d'' \neq 0$ and all $\sigma_l < 0$ (Proposition 11.6); so F_c vanishes nowhere on $\mathcal{A}_n$. □

Theorem 12.15 (R-nondegeneracy of the planetary frequency data)**.** *Let $n \geq 2$ and let $\mathcal{P} \subset \mathcal{U}_n$ be nonempty, open and connected (for instance a ball in $\mathcal{G}_n$, Corollary 12.9; for $n = 2$ also $\mathcal{P} = \mathcal{A}_2$). Then:*

- *(a) $\varpi|_{\mathcal{P}} : \mathcal{P} \to \mathbb{R}^{2n}$ is real analytic and R-nondegenerate; equivalently, in the actions, $\tilde{\varpi} = \varpi \circ a(\Lambda)$ is R-nondegenerate on the nonempty open connected set $\Lambda(\mathcal{P}) \subset \mathbb{R}^n$;*
- *(b) the fast frequency map $\nu : \mathcal{P} \to \mathbb{R}^n$ alone is R-nondegenerate, and so is $\Lambda \mapsto \nu(\Lambda) = (\kappa^2 m_i^3 \Lambda_i^{-3})_i$;*
- *(c) the same holds for (ν, Ω) with $\Omega_j = 2\sigma_{\pi(j)}$ in any labelling $\pi \in S_n$ and with any positive rescaling of the components ($\Omega = 2\sigma$, σ, σ/π, $\nu/2\pi, \dots$).*

Proof. **Step S12.23: (a).** Suppose $\langle c, \varpi \rangle \equiv 0$ on $\mathcal{P}$ for some $c = (c', c_\Omega) \neq 0$. Put $\tilde{c} = (c', 2c_\Omega) \neq 0$. Then $\langle c, \varpi \rangle = L_{\tilde{c},\mathrm{id}}$ on $\mathcal{P}$, one factor of $F_{\tilde{c}}$ (Proposition 12.13), so $F_{\tilde{c}} \equiv 0$ on the nonempty open $\mathcal{P}$. $F_{\tilde{c}}$ is real analytic on the connected set $\mathcal{A}_n$, so $F_{\tilde{c}} \equiv 0$ ((F1)), contradicting Theorem 12.14. The statement in the actions: $a \mapsto \Lambda(a)$ is an analytic diffeomorphism, so connectedness and identical vanishing are preserved.

Step S12.24: (b). If $\sum c_i \kappa^{1/2} a_i^{-3/2} \equiv 0$ on an open set, differentiating in a_i gives $-\frac{3}{2} c_i \kappa^{1/2} a_i^{-5/2} = 0$, so $c = 0$.

Step S12.25: (c). A relabelling or positive rescaling of the components of ω replaces c by another nonzero vector. □

Remark 12.16 (The symmetric-product argument for $n = 2$, as recorded in PLAN.md). The plan's device is valid and is the case $n = 2$ of Theorem 12.15. Checked points: (1) $F(a) = \prod_{\pi \in S_2} (c_1\nu_1 + c_2\nu_2 + c_3\sigma_{\pi 1} + c_4\sigma_{\pi 2})$ is a polynomial in $x = c_1\nu_1 + c_2\nu_2$, $\operatorname{tr} Q_h$ and $\det Q_h$ (explicitly $(x + c_3\sigma_1 + c_4\sigma_2)(x + c_3\sigma_2 + c_4\sigma_1) = x^2 + (c_3 + c_4)\, x \operatorname{tr} Q_h + c_3 c_4 \operatorname{tr}^2 Q_h + (c_3 - c_4)^2 \det Q_h$), hence single valued and real analytic on the convex $\mathcal{A}_2$ (the domain of Part 8; no complexification); (2) a relation on a real patch gives $F = 0$ there and hence everywhere (identity theorem); (3) the plan then says "one of the finitely many factors vanishes on an open subset of the asymptotic region": this is true but, for $n = 2$, needs that the labelled σ_1, σ_2 are analytic on the region, which holds because the discriminant is never zero (Corollary 11.14); the present proof avoids this step by evaluating F at a point (Case B), where it equals a product of explicit nonzero terms. (4) No universal cover is used. The contradiction with Fej04r (41) is replaced by $t = \sigma_1/\sigma_2 \to \infty$, Proposition 12.7.

Remark 12.17 (Scalings). The frozen fast frequency is $w = \omega/2\pi$ for the period-one parametrisation (Part 1), the secular frequency enters through $\Omega_j = 2\sigma_j$ with the chart's $R_j = \frac{1}{2}(p_j^2 + q_j^2)$ (Remark 11.10); all of these differ from $\tilde{\varpi}$ by a positive diagonal matrix, which maps c to a nonzero c (Theorem 12.15(c)). The index below is unchanged and the amount changes by a factor between the squares of the smallest and largest diagonal entries.

12.5. **Index and amount of nondegeneracy on a compact patch.** Following CP09: for a C^ν function g on an open set $B \subset \mathbb{R}^N$, $\partial^\nu g(y)$ is the symmetric ν-linear form of the ν-th derivative and $|\partial^\nu g(y)| := \max_{|a|=1} |\partial^\nu g(y)(a, \dots, a)|$ (Euclidean unit sphere, as CP09 p. 5; for $\nu = 0$ this is $|g(y)|$). By (F4), $|\partial^\nu g(y)| = 0$ iff all partial derivatives of order ν vanish at y. For vector-valued maps $|\partial^\nu \omega(y)| := \max_{|a|=1} |\partial^\nu \omega(y)(a^\nu)|_2$.

Lemma 12.18 (Finite type on a compact set; CP09 Lemma 1 = Rus01 Lemma 18.2)**.** *Let $B \subset \mathbb{R}^N$ be open and connected, $K \subset B$ compact* and nonempty*, $\mathcal{C}$ a compact* nonempty *metric space and $g : \mathcal{C} \times B \to \mathbb{R}$ such that for each $c \in \mathcal{C}$ the function $g(c, \cdot)$ is real analytic and $\not\equiv 0$ on B, and every partial derivative $\partial_y^\alpha g$ is jointly continuous on $\mathcal{C} \times B$. Then there are $\mu_0 \in \mathbb{N}$ and $\beta > 0$ with*

$$\max_{0 \leq \nu \leq \mu_0} |\partial_y^\nu g(c, y)| \;\geq\; \beta \qquad \forall (c, y) \in \mathcal{C} \times K.$$

Source: CP09 Lemma 1 and (2), p. 5 (the proof is cited to Rus01 Lemma 18.2, p. 185); the proof below is ours, by compactness.

Proof. **Step S12.26: Pointwise.** Fix (c_0, y_0). If $\partial^\nu g(c_0,\cdot)(y_0) = 0$ for all ν, then $g(c_0,\cdot) \equiv 0$ near y_0 and hence on B ((F1)), a contradiction. So $|\partial^{\nu_0} g(c_0, y_0)| > 0$ for some $\nu_0 = \nu(c_0, y_0)$.

Step S12.27: Open cover. By joint continuity ($\partial^\nu g$ is a continuous tensor, so its norm is continuous) $|\partial^{\nu_0} g| > 0$ on a neighbourhood $N(c_0, y_0)$ in $\mathcal{C} \times B$. Cover the compact nonempty $\mathcal{C} \times K$ by finitely many such neighbourhoods, with orders $\nu_1, \dots, \nu_s$; put $\mu_0 = \max_i \nu_i$.

Step S12.28: Minimum. $M(c, y) = \max_{\nu \le \mu_0} |\partial^\nu g(c, y)|$ is continuous and positive on the nonempty compact $\mathcal{C} \times K$, so the minimum is attained and $\beta := \min M > 0$. (On an empty product the displayed inequality is vacuous and no minimum is defined; the nonemptiness guards are what make β and the next definition meaningful.) □

Definition 12.19 (Index and amount)**.** Let $K \neq \emptyset$ be compact and $\mathcal{C} \neq \emptyset$ compact, and let $g : \mathcal{C} \times B \to \mathbb{R}$ be such that every $\partial_y^\nu g$ ($\nu \in \mathbb{N}$) exists and is jointly continuous (no nondegeneracy assumed). For $\mu \in \mathbb{N}$ let $\beta_\mu(g, K) = \min_{\mathcal{C}\times K} \max_{0 \le \nu \le \mu} |\partial^\nu g| \ge 0$ (a minimum of a continuous function on a nonempty compact set); it is nondecreasing in μ. The *index* $\mu_0(g, K)$ is defined *when some* $\beta_\mu > 0$ (this holds under Lemma 12.18): it is the least positive integer μ with $\beta_\mu > 0$, and the *amount* is $\beta(g, K) = \beta_{\mu_0}$; for every $\mu \ge \mu_0$, $\beta_\mu \ge \beta_{\mu_0} > 0$. For $\mathcal{C} = S^{m-1}$ ($m \ge 1$, nonempty) and $g = \langle c, \omega\rangle^2$ write $\beta_\mu(\omega, K)$.
Source: CP09 Def. 2, p. 6 ("smallest positive integer"), with $g(c, y) = \langle c, \omega(y)\rangle^2$, $\mathcal{C} = S^{m-1}$; CP09 Thm. 3 allows every $\mu \ge \mu_0$ with $\beta = \beta_\mu$.

Corollary 12.20 (CP09 Lemma 1 for an R-nondegenerate map)**.** *If $\omega : B \to \mathbb{R}^m$ is real analytic and R-nondegenerate (Definition 12.11) and $K \subset B$ is compact and nonempty ($m \ge 1$, so $S^{m-1} \neq \emptyset$), then $g(c, y) = \langle c, \omega(y)\rangle^2$ on $S^{m-1} \times B$ satisfies Lemma 12.18: there are $\mu_0 = \mu_0(\omega, K)$ and $\beta = \beta(\omega, K) > 0$ with $\max_{0\le\nu\le\mu_0} |\partial^\nu \langle c, \omega\rangle^2(y)| \ge \beta$ for all $c \in S^{m-1}$, $y \in K$. The same holds for any family of the form $g(c, y) = \prod_{l=1}^r \langle c_l, f_l(y)\rangle$, $c \in (S^{m_l-1})_l$, with f_l real analytic and R-nondegenerate (Lemma 12.1: a product of nonzero analytic functions is nonzero), and for $g = \langle c, \omega\rangle$ without the square.*

Proof. g is polynomial in c and real analytic in y, so all $\partial_y^\alpha g$ are jointly continuous; $g(c,\cdot) \not\equiv 0$ because $\langle c, \omega\rangle \not\equiv 0$ and the square of a nonzero real function is nonzero. Apply Lemma 12.18. □

Lemma 12.21 (Finite-order stability of the amount)**.** *Let $\omega, \tilde\omega : B \to \mathbb{R}^m$ be real analytic ($m \ge 1$), $K \subset B$ compact and nonempty, $\mu \in \mathbb{N}$, $M = \max_{\nu\le\mu} \sup_K |\partial^\nu \omega|$, $\theta = \max_{\nu\le\mu} \sup_K |\partial^\nu(\tilde\omega - \omega)|$. Then for all $c \in S^{m-1}$, $y \in K$, $\nu \le \mu$,*

$$\big|\, |\partial^\nu \langle c, \tilde\omega\rangle^2(y)| - |\partial^\nu \langle c, \omega\rangle^2(y)| \,\big| \le 2^\nu \theta(2M + \theta),$$

hence $\beta_\mu(\tilde\omega, K) \ge \beta_\mu(\omega, K) - 2^\mu\theta(2M + \theta)$ (valid for every μ, with $\beta_\mu \ge 0$ possibly zero). If moreover $\beta_\mu(\omega, K) > 0$ (equivalently, when ω is R-nondegenerate, $\mu \ge \mu_0(\omega, K)$; Theorem 12.22(d) supplies this guard for $\mu \ge \mu_0$) and $2^\mu\theta(2M + \theta) \le \frac12 \beta_\mu(\omega, K)$, then $\beta_\mu(\tilde\omega, K) \ge \frac12 \beta_\mu(\omega, K) > 0$, and $\tilde\omega$ then satisfies the conclusion of Corollary 12.20 with the same μ and amount $\frac12\beta_\mu(\omega, K)$ (no nondegeneracy of $\tilde\omega$ is assumed). Without the premise $\beta_\mu(\omega, K) > 0$ the smallness condition can hold with both sides 0 and no positive amount follows (e.g. $\omega = \tilde\omega = (1, y^2)$, $B = (-1, 1)$, $K = [-\frac12, \frac12]$, $\mu = 1$: at $y = 0$, $c = (0, 1)$ one has $\langle c, \omega\rangle^2 = y^4$, all derivatives through order 1 vanish, so $\beta_1 = 0$).
Source: Ours; this is the "finite-order stability lemma" of the Part 11a statement list (`reviews/p11a-russmann-check.md`, row `p11a:rank-compact`), in CP09's convention.

Proof. **Step S12.29: Product rule.** With $f = \langle c, \omega\rangle$, $\tilde f = \langle c, \tilde\omega\rangle$ and $|c| = 1$: $|\partial^k f(y)(a^k)| \le |\partial^k \omega(y)(a^k)|_2 \le M$ and $|\partial^k(\tilde f - f)(y)(a^k)| \le \theta$ for unit a (Cauchy–Schwarz), so $|\partial^k(\tilde f + f)| \le 2M + \theta$.

Step S12.30: Difference. $\tilde f^2 - f^2 = (\tilde f - f)(\tilde f + f)$, and by the Leibniz formula (F5), $|\partial^\nu(\tilde f^2 - f^2)(y)(a^\nu)| \le \sum_k \binom{\nu}{k}\theta(2M+\theta) = 2^\nu\theta(2M+\theta)$. Taking the maximum over unit a and then over $\nu \le \mu$ and the minimum over (c,y) gives the claim (the maxima of two families that differ termwise by at most ϵ differ by at most ϵ). □

12.6. **The interface for Parts 11a, 12 and 13.**

Theorem 12.22 (Good patch with nondegenerate frequency data)**.** *Let $n \ge 2$, $G, m_0 > 0$, $m \in (0,\infty)^n$, $a^* \in \mathcal{A}_n$ and $\eta > 0$. There are $a_0 \in \mathcal{A}_n$ and $r_0 > 0$ (depending only on these data) such that, with $B = B_\infty(a_0, 2r_0)$, $K_a = \overline{B}_\infty(a_0, r_0)$:*

(a) $\overline{B}_\infty(a_0,2r_0) \subset \mathcal{G}_n \cap B_\infty(a^*,\eta)$; so on B the sorted eigenvalues $\sigma_1 < \cdots < \sigma_n < 0$ of $Q_h(a)$ are real analytic (Lemma 12.3), Q_h has simple spectrum, and the rotation ϱ of Proposition 11.7 exists locally (Theorem 11.9 applies around every point of B);

(b) (non-resonance) $\alpha_4 := \min\{|k\cdot 2\sigma(a)| :\ a \in K_a,\ k \in K_4\} > 0$, and $|k\cdot\Omega| \ge \alpha_4$ with $\Omega = 2\sigma$ for $0 < |k|_1 \le 4$ on K_a;

(c) (R-nondegeneracy) in the frozen action variable $I = \Lambda$: $B^\Lambda = \Lambda(B)$ is open and connected, $K = \Lambda(K_a) \subset B^\Lambda$ is compact with meas $K > 0$*, and*

$$\tilde\varpi = (\nu(\Lambda), \Omega(\Lambda)) :\ B^\Lambda \to \mathbb{R}^n \times \mathbb{R}^n, \qquad \nu_i = \kappa^2 m_i^3 \Lambda_i^{-3}, \quad \Omega_j(\Lambda) = 2\sigma_j(a(\Lambda)),$$

is real analytic and R-nondegenerate (Definition 12.11; so are ν alone and Ω up to relabelling, rescaling as in Theorem 12.15(c));

(d) (index and amount) there are $\mu_0 = \mu_0(\tilde\varpi, K) \in \mathbb{N}$ and $\beta = \beta(\tilde\varpi, K) > 0$, depending only on (n, G, m_0, m, a_0, r_0) and in particular independent of the coupling parameter*, such that for every $c \in S^{2n-1}$ and $\Lambda \in K$*

$$\max_{0\le\nu\le\mu_0} \left|\partial_\Lambda^\nu \langle c, \tilde\varpi(\Lambda)\rangle^2\right| \ \ge\ \beta,$$

and for every $\mu \ge \mu_0$ the same with μ and $\beta_\mu \ge \beta > 0$ (CP09 (2)–(3)); and Lemma 12.21 holds for $\omega = \tilde\varpi$, with its positivity premise $\beta_\mu(\tilde\varpi, K) > 0$ guaranteed for every $\mu \ge \mu_0$ by this item.

The patch can be taken inside every prescribed neighbourhood of a^*. If further analytic functions of the axes that are $\not\equiv 0$ on $\mathcal{A}_n$ (torsion determinants, Part 11b) must also be nonzero, a_0, r_0 can be chosen in the set where all of them and Φ_4 are nonzero (Lemma 12.1).

Proof. **Step S12.31: (a).** Corollary 12.9.

Step S12.32: (b). $a \mapsto \min_{k\in K_4}|k\cdot 2\sigma(a)|$ is continuous (Lemma 12.2), positive on $\mathcal{G}_n$ ($\Phi_4 \ne 0$ iff NR$_4$, Proposition 11.12(b), and NR$_4$ is invariant under $\sigma \mapsto 2\sigma$), and K_a is compact.

Step S12.33: (c). B is a ball in the max norm (a cube), connected, contained in $\mathcal{U}_n$; Λ is an analytic diffeomorphism, so B^Λ is open and connected, K is compact and meas $K > 0$ (it contains the image of an open cube). R-nondegeneracy: Theorem 12.15 with $\mathcal{P} = B$.

Step S12.34: (d). Corollary 12.20 with $\omega = \tilde\varpi$ on B^Λ and the compact K; the dependence only on the data is clear, since $\tilde\varpi$ is a function of Λ alone (Q_h does not involve the coupling parameter). $\beta_\mu \ge \beta_{\mu_0} > 0$ is monotonicity (Definition 12.19); K is nonempty because meas $K > 0$ (part (c)) and $S^{2n-1} \ne \emptyset$, so the guards of Lemma 12.18 hold. □

Remark 12.23 (What the consumers take from this part). *Part 11a* (maximal analytic weak-nondegeneracy KAM, route (a), P-FIXED, general n): the hypothesis "$(\nabla h, \Omega)$ is R-nondegenerate on B^Λ, with index μ_0 and amount β on K" of CP09 Thm. 4 (pp. 8, 5–6) in the form of Theorem 12.22(c),(d); the $\mu \ge \mu_0$ variant is the one used in the preparation (CP09 Lemma 8). The preparation of CP09 §2.3 changes $\tilde\varpi$ by perturbations small in C^μ on shrinking sets; Lemma 12.21 is the form of stability that is available without further hypotheses. *Parts 12, 13*: the good patch

B for P-PLANAR ($n = 2$, route (c)) and P-FIXED; for $n = 2$ Corollary 11.14 and Proposition 12.7 give the explicit exclusion set $\{(2\,\mathrm{tr}^2 - 9\det)(3\,\mathrm{tr}^2 - 16\det) = 0\}$ (two analytic curves in $\mathcal{A}_2$, none in the regime $a_2/a_1 \geq 16m_1^2/m_2^2$); and because $\Phi_4 \neq 0$ is an open dense condition (Lemma 12.1), a patch can be chosen on which $\det\beta \neq 0$ as well, once Part 11b shows $\det\beta \not\equiv 0$.

12.7. **Sources, deviations, errata, forward references.**
Sources read. Fej04r pp. 52–58 (Formule 69 with (38)–(39); Lemmas 70–71, p. 53; Lemma 76, Prop. 77 with proof, pp. 56–58; Lemma 78 ff., p. 58–59 for the induction pattern), `fejoz04_erratum.txt` (no statement on Prop. 77), CP09 pp. 5–8 (Def. 1, Lemma 1, Def. 2, Thm. 3, Thm. 4), `reviews/p11a-russmann-check.md` (row `rank-compact`, Rus01 §§18.2–18.3 conventions), Parts 7–9 of this blueprint. Numerical checks: `.drafts/P/bp-scratch-bpP-p10/chk.py` ($b^{(k)}_{3/2}$ bounds of Lemma 12.4; r, t of Proposition 12.7 for $m = (0.7, 1.9)$ at $a = (1,2), (1,5), (0.1,1), (0.01,1)$, e.g. $t = 7.46, 6.63, 8.76, 27.1$; the scaling $E_{11} \sim -0.868\,a_1^{3/2}$, $E_{22} \sim -0.315\,a_1^2$, $E_{12} \sim 0.650\,a_1^{11/4}$ for $n = 3$; $\min_{k\in K_4}|k\cdot\sigma| = 2.6\cdot 10^{-3}$ at $a = (1, 1.7, 2.9)$; the six functions $(\nu, 2\sigma)$ at 40 random points of a small patch have a matrix of full numerical rank). This is evidence only.
Deviations from the source. (1) No complexification, universal cover or exp chart; the symmetric product and the convexity of $\mathcal{A}_n$ replace them. (2) Prop. 77's conclusion ("gauche", Definition 57, the Arnold–Pyartli condition) is *not* proved; only the weaker Rüssmann nondegeneracy is, which is what CP09 Thm. 4 and the frozen P-FIXED route need. (3) Fej04r's Cor. 72 is replaced by Weyl's inequality plus $\sigma_n < 0$.
Suspected errata (for the ledger). E-P-Fej04r-7 (minor): Prop. 77 proof, p. 58, the block form of Q_h as the innermost axis tends to 0 is printed with errors $O(a_n^{5/2})$. The assertion about the *outer diagonal block* (and the associated eigenvalue error) is false: in the present normalisation that block is perturbed by $O(a^2)$ (the diagonal terms $-\mathfrak{a}_{1j}/\Lambda_j$ with $\mathfrak{a}_{1j} \sim a^2$, nonzero leading term). The printed coupling bound $O(a^{5/2})$ is *not* false: the sharper $O(a^{11/4})$ implies it, and is a sharpening of a valid weaker bound. The three scales are kept distinct: the $(1,1)$ entry is $\sim a^{3/2}$, the outer diagonal perturbation a^2, the coupling $a^{11/4}$ (Lemma 12.6, numerical check above). Immaterial for the argument. E-P-Fej04r-8 (gap, not an error): the general-n step is "par un raisonnement analogue"; the analogy needs both the transfer of the relation and the case in which the dropped σ coefficient is the only nonzero one (Case C2), neither of which is discussed. Already recorded: E-P-Fej04r-2 (Lemma 76, the simple connectedness of $A^{\mathbb{C}}$), E-P-Fej04r-3 (halving).
Doubts about frozen statements. None. (P-FIXED and P-PLANAR assert only existence of tori; the nondegeneracy statements here are proof obligations, not premises, Part 0 §1.31.)
External facts used. (F1)–(F5); (E1)–(E5); (E3). Not in Mathlib as far as we found: Courant–Fischer (proved in Lemma 12.2), polarization for symmetric multilinear maps ((F4)).
Forward references and consumers. Part 11a: Theorem 12.22(c),(d), Lemma 12.21, Corollary 12.20. Part 11b: Lemma 12.1 (to combine $\det\beta \not\equiv 0$ with Φ_4). Parts 12, 13: Theorem 12.22, Corollary 12.9. No forward references to unwritten parts.
Label index. `generic`, `weyl`, `sorted-analytic`, `laplace-bounds`, `formula69`, `peel`, `n2`, `good-point`, `good-set`, `noncont`, `rnd-def`, `Fc-def`, `Fc-analytic`, `F-nonzero`, `rnd`, `n2-check`, `scaling`, `finite-type`, `index-def`, `cp09-lemma1`, `stability`, `interface`, `consumers`.

13. The planar quartic Birkhoff form and the torsion

This part produces, for the planar two-planet secular function, exactly the hypotheses on (P_0, Ω, β) that the frozen theorem P-G (Theorem 1.28) needs: the secular Birkhoff form HasBirkhoff (Definition 1.14) with a symmetric torsion matrix β whose determinant is bounded away from zero on a patch near every prescribed axis vector. Route (c) of the plan: no new KAM theorem, only the finite computation below plus the frozen P-G.

- `quartic` (Lemma 13.1, Lemma 13.2, Proposition 13.3, Theorem 13.4): the six invariant quartic coefficients $c_1, \dots, c_6$ of Proposition 10.16, derived from the Kepler expansion of the Poincaré chart to order four (third- and fourth-order single-planet variations are kept) and reduced to the two Laplace coefficients $b_{1/2}^{(0)}, b_{1/2}^{(1)}$ with explicit rational coefficients; c_1, c_2 agree with Part 8's definition, and none of this is copied from CP11;
- `birkhoff` (Lemmas 13.5–13.7, Proposition 13.8, Proposition 13.9): Birkhoff normalisation to order 4 by a finite, explicit homological equation, realised as a symplectic map of the full phase space that preserves the actions and commutes with angle translations, so that the average of the transformed perturbation is the normalised secular function *exactly*;
- `beta` (Definition 13.10, Proposition 13.11, Theorem 13.12): the torsion $\hat\beta$ is the torus average of the rotated quartic form, in closed form in (c_ν, Π_1); the frozen HasBirkhoff follows;
- `torsion` (Proposition 13.14, Theorem 13.15, Corollary 13.16): $\det\hat\beta = -\frac{117}{16}(G/\kappa)^2 a_1^3 a_2^{-7}(1 + O(a_1^2/a_2^2))$ as $a_1/a_2 \to 0$ (this recovers CP11 (8.4), pp. 33–34, and Arnold's check), so $\det\hat\beta \not\equiv 0$ on the connected domain $\{0 < a_1 < a_2\}$ and, by the identity theorem, is nonzero on an open dense full-measure set; patches with $\Phi_4 \neq 0$ and $\det\hat\beta \neq 0$ exist inside every neighbourhood of every axis vector.

13.1. **Setting and external facts.** Throughout $n = 2$, $G, m_0 > 0$, $\kappa = Gm_0$, $m = (m_1, m_2) \in (0,\infty)^2$, and we use the objects of Parts 7–9: the chart Ψ and $w = (\lambda, \xi, \Lambda, \eta)$, $u_i = (\xi_i, \eta_i) \in \mathbb{R}^2$, $a_i = \Lambda_i^2/(m_i^2\kappa)$, $\mathcal{A}_2 = \{0 < a_1 < a_2\}$ (increasing order), $\alpha = a_1/a_2 \in (0,1)$, the secular functions $S_{\mathrm{fix}}, S_{\mathrm{mov}}$ and the common jet of Corollary 10.14, $C_0, \mathfrak{a} = \mathfrak{a}_{12}, \mathfrak{b} = \mathfrak{b}_{12}, Q_h$ (Definition 10.10), the eigenvalues $\sigma_1 < \sigma_2 < 0$ and the rotation $\varrho \in SO(2)$ (Corollary 11.14, Proposition 11.7), Φ_4 and NR_4 (Definition 11.11), and the Laplace coefficients $b_s^{(k)}$ (Definition 10.6, Lemma 10.7). Abbreviate

$$\beta_0 := b_{1/2}^{(0)}(\alpha), \qquad \beta_1 := b_{1/2}^{(1)}(\alpha), \qquad \Delta := 1 - \alpha^2, \qquad \mathrm{pref} := -\frac{Gm_1m_2}{a_2}.$$

Complex notation for the plane: $\zeta = v_0 + iv_1$ for an eccentricity vector $v = (v_0, v_1)$ (Definition 9.5) and $X = e^{i\lambda}$. For $F = F(\Lambda, z)$ on the slow variables, $z = (z_1, z_2)$, $z_j = (x_j, y_j) \in \mathbb{R}^2$ and $R_j = \frac{1}{2}(x_j^2 + y_j^2)$, with the chart form $d\Lambda \wedge d\lambda + dy \wedge dx$ of Part 9; $\{F, G\} := dF(X_G)$ in the slow variables (Λ a parameter), X_G the Hamiltonian field of the frozen sign convention $\dot x = \partial_y G$, $\dot y = -\partial_x G$ (Corollary 10.14(b)). *Dictionary:* $q = x$, $p = y$ in the sense of Part 8 (position ξ, momentum η).

External facts.

(F1) *Analytic ODE.* The flow of a real-analytic vector field depends real-analytically on time, initial data and parameters on its (open) domain of definition. (Standard; cf. [N:`bpN:p3:local-exist`] for existence, uniqueness and continuation, Part 1 `ode`.)

(F2) *Identity theorem* on a connected open set ((E4), (F1)), and its consequence Lemma 12.1.

(F3) A real-analytic function whose jet of order $k-1$ at $z = 0$ vanishes is $O(|z|^k)$ locally uniformly in the parameters (Taylor, (E3)).

(F4) *Finite exact computation.* A polynomial or rational identity with rational coefficients, in finitely many variables, of bounded degree, is verified by a finite exact computation (Lean: `ring`, `field_simp`, `norm_num`). The displayed expansions of Lemma 13.1 and the identities of Proposition 13.3 and Theorem 13.4 are of this kind; they were generated and checked with a computer algebra system and numerically (§13.6).

13.2. **The quartic jet of the pair interaction.**

Lemma 13.1 (Kepler expansion to order four)**.** *For the Poincaré chart with $a = 1$ (Definition 9.5), in the complex notation $z_{\mathrm{orb}} = q_0 + iq_1$ for the position q, as a formal power series in $(\zeta, \bar\zeta)$ with*

coefficients Laurent polynomials in $X = e^{i\lambda}$,

$$z_{\mathrm{orb}} = X + \left(\tfrac{1}{2}X^2\bar\zeta - \tfrac{3}{2}\zeta\right) + \left(\tfrac{3}{8}X^3\bar\zeta^2 - \tfrac{1}{2}X\zeta\bar\zeta + \tfrac{1}{8}X^{-1}\zeta^2\right) + \left(\tfrac{1}{3}X^4\bar\zeta^3 - \tfrac{3}{8}X^2\zeta\bar\zeta^2 + \tfrac{1}{24}X^{-2}\zeta^3\right) + \left(\tfrac{125}{384}X^5\bar\zeta^4 - \tfrac{3}{8}X^3\zeta\bar\zeta^3 - \tfrac{1}{64}X\zeta^2\bar\zeta^2 + \tfrac{1}{24}X^{-1}\zeta^3\bar\zeta + \tfrac{3}{128}X^{-3}\zeta^4\right) + O(|\zeta|^5),$$

where λ is the mean longitude and the terms are grouped by degree $1,2,3,4$ in $(\zeta,\bar\zeta)$. Moreover z_{orb} at fixed λ is real analytic in v for $|v|<1$, and its degree-d homogeneous part is the d-th Taylor polynomial.

Proof. **Step S13.1: Kepler's equation.** By Proposition 9.4(a), $F - v_0\sin F + v_1\cos F = \lambda$. Since $v_0\sin F - v_1\cos F = \mathrm{Im}(\bar\zeta e^{iF})$, writing $F = \lambda + \phi$, $E = e^{iF} = Xe^{i\phi}$, the equation is $\phi = \mathrm{Im}(\bar\zeta E) = (\bar\zeta E - \zeta\bar E)/(2i)$, a fixed-point equation whose right side is divisible by $\zeta,\bar\zeta$: Picard iteration from $\phi = 0$ determines ϕ modulo degree d after d steps (formal inverse function theorem; the iteration is a finite polynomial computation by (F4)).

Step S13.2: Position. $q = a(kc - v + \frac{v\cdot c}{1+k}v)$ with $a = 1$, $k = \sqrt{1-|\zeta|^2}$, $v\cdot c = \mathrm{Re}(\bar\zeta E)$ and $c \leftrightarrow E$ gives $z_{\mathrm{orb}} = kE - \zeta + \frac{\mathrm{Re}(\bar\zeta E)}{1+k}\zeta$. Insert the series of ϕ, of $k = 1 - \frac{1}{2}\zeta\bar\zeta - \dots$ and of $(1+k)^{-1}$, expand and collect. The result is the displayed table (the degree-≤ 2 part is the classical $x \approx a(\cos\lambda - \frac{3}{2}e\cos\varpi + \frac{1}{2}e\cos(2\lambda - \varpi))$; the degree $3,4$ parts are needed for the bidegrees $(3,1),(1,3)$, cf. Proposition 10.16(c)).

Step S13.3: Analyticity. Proposition 9.4(a) and the formulas above are analytic in v for $|v| < 1$, and the Taylor polynomials of an analytic function are the homogeneous parts of its (formal) series. □

Lemma 13.2 (Laplace coefficients: relations and reduction to two functions)**.** *Let $s \in \mathbb{R}$, $k \ge 0$, $\alpha \in (0,1)$ (the circular denominator $1 + \alpha^2 - 2\alpha\cos t$ is positive, so (R1)–(R3) hold for every real s; the divisors $(s-1)\alpha$, $s\alpha$, $s-1-k$ are guarded where the relations are solved) and $b_s^{(k)} = b_s^{(k)}(\alpha)$ (Definition 10.6), $b_s^{(-k)} = b_s^{(k)}$. Then*

(R1) $b_{s-1}^{(k)} = (1+\alpha^2)b_s^{(k)} - \alpha\big(b_s^{(k+1)} + b_s^{(k-1)}\big)$;

(R2) $k\,b_{s-1}^{(k)} = (s-1)\alpha\big(b_s^{(k-1)} - b_s^{(k+1)}\big)$;

(R3) (three-term recurrence) $(s-1-k)\,\alpha\,b_s^{(k+1)} = (k+s-1)\,\alpha\,b_s^{(k-1)} - k(1+\alpha^2)\,b_s^{(k)}$ *for $k \ge 1$.*

Consequently, for every $s \in \frac{1}{2} + \mathbb{N}$ and $k \in \mathbb{N}$, $b_s^{(k)} = r_{s,k}^{(0)}(\alpha)\,\beta_0 + r_{s,k}^{(1)}(\alpha)\,\beta_1$ with $r_{s,k}^{(0)}, r_{s,k}^{(1)} \in \mathbb{Q}(\alpha)$ whose denominators divide $\alpha^k\Delta^{2(s-1/2)}\cdot\prod(\cdot)$ times a product of linear factors $(s'-1-k')$ not vanishing for half-integer s'. Explicitly, for $s \ge \frac{1}{2}$:

$$b_{s+1}^{(0)} = \frac{(1+\alpha^2)b_s^{(0)} + 2\alpha\big(1-\frac{1}{s}\big)b_s^{(1)}}{\Delta^2}, \qquad b_{s+1}^{(1)} = \frac{(1+\alpha^2)\big(1-\frac{1}{s}\big)b_s^{(1)} + 2\alpha\,b_s^{(0)}}{\Delta^2}, \tag{2}$$

and $b_{s+1}^{(k+1)} = b_{s+1}^{(k-1)} - k\,b_s^{(k)}/(s\alpha)$ for $k \ge 1$.

Proof. **Step S13.4: (R1).** $1 + \alpha^2 - 2\alpha\cos t = \Delta_t$ gives $\Delta_t\cdot\Delta_t^{-s} = \Delta_t^{-(s-1)}$ and $2\cos t\cos kt = \cos(k+1)t + \cos(k-1)t$; integrate against $\frac{1}{2\pi}$.

Step S13.5: (R2). $\int_0^{2\pi}\frac{d}{dt}\big(\sin kt\,\Delta_t^{-(s-1)}\big)dt = 0$, i.e. $k\int\cos kt\,\Delta_t^{-(s-1)} = (s-1)2\alpha\int\sin kt\sin t\,\Delta_t^{-s}$, and $2\sin kt\sin t = \cos(k-1)t - \cos(k+1)t$.

Step S13.6: (R3). Eliminate $b_{s-1}^{(k)}$ between (R1) and (R2).

Step S13.7: Reduction. At level $s = \frac{1}{2}$ the recurrence (R3) (denominator $(s-1-k) = -(k+\frac{1}{2}) \ne 0$) gives all $b_{1/2}^{(k)}$ from $b_{1/2}^{(0)}, b_{1/2}^{(1)}$. For the step $s \to s+1$: (R2) at $s+1$ gives $b_{s+1}^{(k-1)} - b_{s+1}^{(k+1)} = k\,b_s^{(k)}/(s\alpha)$,

i.e. the last formula; (R1) at $s+1$ for $k=0$ and $k=1$ reads $(1+\alpha^2)b^{(0)}_{s+1} - 2\alpha b^{(1)}_{s+1} = b^{(0)}_s$ and $(1+\alpha^2)b^{(1)}_{s+1} - \alpha(b^{(2)}_{s+1}+b^{(0)}_{s+1}) = b^{(1)}_s$ with $b^{(2)}_{s+1} = b^{(0)}_{s+1} - b^{(1)}_s/(s\alpha)$, i.e. $(1+\alpha^2)b^{(1)}_{s+1} - 2\alpha b^{(0)}_{s+1} = (1-\frac{1}{s})b^{(1)}_s$. This 2×2 system has determinant $(1+\alpha^2)^2 - 4\alpha^2 = \Delta^2$; solving gives (2). Induct on s and on k. □

Proposition 13.3 (The jet of $\mathcal{W}$ to order four, by Fourier averaging)**.** *Let $0<\alpha<1$. The function $\mathcal{W}(\alpha; v_1, v_2) = \frac{1}{(2\pi)^2}\int\int |\alpha z_{\rm orb}(\lambda_1;\zeta_1) - z_{\rm orb}(\lambda_2;\zeta_2)|^{-1} d\lambda_1 d\lambda_2$ of Proposition 10.16(a) ($S_{12} = -Gm_1m_2a_2^{-1}\mathcal{W}$; here v_i is the eccentricity vector of planet i and ζ_i its complex form) is real analytic near $v=0$, and its Taylor expansion is $\mathcal{W} = \mathcal{W}_0 + \mathcal{W}_2 + \mathcal{W}_4 + O(|v|^6)$ with $\mathcal{W}_0 = \beta_0$,*

$$\mathcal{W}_2 = \mathsf{A}\,|\zeta_1|^2 + \mathsf{A}\,|\zeta_2|^2 + \mathsf{B}\,\mathrm{Re}(\bar\zeta_1\zeta_2),$$
$$\mathcal{W}_4 = \mathsf{p}_1|\zeta_1|^4 + \mathsf{p}_2|\zeta_2|^4 + \mathsf{p}_3|\zeta_1|^2|\zeta_2|^2 + \mathsf{p}_4\,\mathrm{Re}(\bar\zeta_1\zeta_2)^2 + \mathsf{p}_5|\zeta_1|^2\,\mathrm{Re}(\bar\zeta_1\zeta_2) + \mathsf{p}_6|\zeta_2|^2\,\mathrm{Re}(\bar\zeta_1\zeta_2),$$

where, with $\Delta = 1-\alpha^2$, $\beta_0 = b^{(0)}_{1/2}$, $\beta_1 = b^{(1)}_{1/2}$,

$$\mathsf{A} = \frac{\alpha\,[\,2\alpha\beta_0 - (1+\alpha^2)\beta_1\,]}{4\Delta^2}\ \Big(=\tfrac14\alpha\, b^{(1)}_{3/2}\Big), \qquad \mathsf{B} = \frac{2(1-\alpha^2+\alpha^4)\beta_1 - \alpha(1+\alpha^2)\beta_0}{2\Delta^2}\ \Big(=-\tfrac12\alpha\, b^{(2)}_{3/2}\Big),$$
$$\mathsf{p}_1 = \frac{\alpha\,[(\alpha+14\alpha^3+33\alpha^5)\beta_0 - (2-6\alpha^2+46\alpha^4+6\alpha^6)\beta_1]}{64\,\Delta^4},$$
$$\mathsf{p}_2 = \frac{\alpha\,[(33\alpha+14\alpha^3+\alpha^5)\beta_0 - (6+46\alpha^2-6\alpha^4+2\alpha^6)\beta_1]}{64\,\Delta^4},$$
$$\mathsf{p}_3 = \frac{3\alpha\,[(7\alpha+2\alpha^3+7\alpha^5)\beta_0 - (2+6\alpha^2+6\alpha^4+2\alpha^6)\beta_1]}{32\,\Delta^4},$$
$$\mathsf{p}_4 = \frac{3\alpha\,[(-\alpha+18\alpha^3-\alpha^5)\beta_0 + (2-10\alpha^2-10\alpha^4+2\alpha^6)\beta_1]}{16\,\Delta^4},$$
$$\mathsf{p}_5 = \frac{(4-15\alpha^2+50\alpha^4+9\alpha^6)\beta_1 - (2\alpha+4\alpha^3+42\alpha^5)\beta_0}{16\,\Delta^4},$$
$$\mathsf{p}_6 = \frac{\alpha^2\,[(9+50\alpha^2-15\alpha^4+4\alpha^6)\beta_1 - (42\alpha+4\alpha^3+2\alpha^5)\beta_0]}{16\,\Delta^4}.$$

The same coefficient of $|\zeta_1|^2$ and $|\zeta_2|^2$ and the vanishing of all further monomials (such as Im*-type terms) are part of the statement. As power series in α (convergent for $|\alpha|<1$, Lemma 10.7):*

$$\begin{aligned}
\mathsf{A} &= \tfrac38\alpha^2 + \tfrac{45}{64}\alpha^4 + \tfrac{525}{512}\alpha^6 + \dots, & \mathsf{B} &= -\tfrac{15}{16}\alpha^3 - \tfrac{105}{64}\alpha^5 - \tfrac{4725}{2048}\alpha^7 - \dots,\\
\mathsf{p}_1 &= \tfrac{135}{512}\alpha^4 + \tfrac{2625}{2048}\alpha^6 + \dots, & \mathsf{p}_2 &= \tfrac{15}{32}\alpha^2 + \tfrac{945}{512}\alpha^4 + \tfrac{4725}{1024}\alpha^6 + \dots,\\
\mathsf{p}_3 &= \tfrac{9}{16}\alpha^2 + \tfrac{585}{256}\alpha^4 + \tfrac{1575}{256}\alpha^6 + \dots, & \mathsf{p}_4 &= \tfrac{315}{128}\alpha^4 + \tfrac{4725}{512}\alpha^6 + \dots,\\
\mathsf{p}_5 &= -\tfrac{45}{64}\alpha^3 - \tfrac{525}{128}\alpha^5 - \dots, & \mathsf{p}_6 &= -\tfrac{75}{32}\alpha^3 - \tfrac{2205}{256}\alpha^5 - \dots.
\end{aligned} \tag{3}$$

Source: Ours. The method is that of Theorem 10.11 carried to order four; Fej04r (35)–(37) and CP11 (B.5)–(B.6) give no derivation. Independent checks: §13.6.

Proof. **Step S13.8: Reduction to a formal series.** Put $z_1 = \alpha z_{\rm orb}(\lambda_1;\zeta_1) = \alpha X_1 + \alpha\delta_1$, $z_2 = z_{\rm orb}(\lambda_2;\zeta_2) = X_2 + \delta_2$ with $\delta_i = z_{\rm orb} - X_i$ (degree ≥ 1 in ζ_i), $X_i = e^{i\lambda_i}$. By Proposition 10.2 and (E2) the Taylor coefficients of the average at $v=0$ are the averages of the Taylor coefficients of the integrand, which is analytic in v for λ in the compact torus. So it suffices to expand $|z_1-z_2|^{-1}$ formally in $(\zeta_i, \bar\zeta_i)$ (Lemma 13.1) and average the coefficients.

Step S13.9: Binomial expansion. $|z_1-z_2|^2 = \mathsf{s} + D$, $\mathsf{s} = 1+\alpha^2-2\alpha\cos(\lambda_1-\lambda_2)$ (the circular value, >0), $D = \overline{(\alpha X_1 - X_2)}\,d + (\alpha X_1 - X_2)\,\bar d + d\bar d$ with $d = \alpha\delta_1 - \delta_2$; D has no constant term in $(\zeta, \bar\zeta)$ and is a Laurent polynomial in X_1, X_2 at each degree. Hence, as formal series, $|z_1-z_2|^{-1} = \sum_{\ell=0}^{4}\binom{-1/2}{\ell}\mathsf{s}^{-1/2-\ell}D^\ell$ modulo degree 5, only finitely many terms contribute to each

degree. The degree-d coefficient is therefore $\sum_{\ell\le d}\mathsf{s}^{-1/2-\ell}\,L_{d,\ell}(X_1,X_2;\alpha)$, $L_{d,\ell}$ a Laurent polynomial in X_1,X_2 with coefficients polynomial in α and in the monomials of $\zeta_1,\bar\zeta_1,\zeta_2,\bar\zeta_2$.

Step S13.10: Averaging rule. Write $X_1^pX_2^r=e^{ip(\lambda_1-\lambda_2)}e^{i(p+r)\lambda_2}$. Since s depends on $\lambda_1-\lambda_2$ only, integration over λ_2 (at fixed $\theta=\lambda_1-\lambda_2$) kills the monomial unless $p+r=0$, and $\frac{1}{2\pi}\int e^{ip\theta}\mathsf{s}^{-s}d\theta=b_s^{(|p|)}(\alpha)$ (Definition 10.6). So $\langle X_1^pX_2^r\mathsf{s}^{-s}\rangle=\delta_{p+r,0}\,b_s^{(|p|)}$.

Step S13.11: Reduction and extraction. By Lemma 13.2 every $b_s^{(k)}$ occurring ($s\in\{\frac12,\dots,\frac92\}$, $k\le 8$) is a rational combination of β_0,β_1. The result is a polynomial in $\zeta_1,\bar\zeta_1,\zeta_2,\bar\zeta_2$ of degrees $0,2,4$ (odd degrees vanish by (S1), Proposition 10.3), invariant under simultaneous rotation $\zeta\mapsto e^{i\theta}\zeta$ (S2) and under $\zeta\mapsto\bar\zeta$ (S3): the monomials $\zeta_1^a\bar\zeta_1^b\zeta_2^c\bar\zeta_2^d$ occurring have $a+c=b+d$; the span of the real invariants of degree 4 is spanned by the six listed monomials. Reading off the coefficients of $|\zeta_1|^4$, $|\zeta_2|^4$, $\zeta_1\bar\zeta_1\zeta_2\bar\zeta_2$, $\bar\zeta_1^2\zeta_2^2$, $\zeta_1\bar\zeta_1^2\zeta_2$, $\bar\zeta_1\zeta_2^2\bar\zeta_2$ and using $\mathrm{Re}(\bar\zeta_1\zeta_2)=\frac12(\bar\zeta_1\zeta_2+\zeta_1\bar\zeta_2)$, $\mathrm{Re}(\bar\zeta_1\zeta_2)^2=\frac14(\bar\zeta_1^2\zeta_2^2+\zeta_1^2\bar\zeta_2^2)+\frac12|\zeta_1|^2|\zeta_2|^2$ gives $\mathsf{p}_1,\dots,\mathsf{p}_6$ (and p_3 is the coefficient of $\zeta_1\bar\zeta_1\zeta_2\bar\zeta_2$ minus $\frac12\mathsf{p}_4$); all other monomials cancel. This is the finite exact computation (F4); its output is displayed. The leading series (3) follow by inserting $\beta_0=\sum_q((\frac12)_q/q!)^2\alpha^{2q}$, $\beta_1=\sum_q\frac{(1/2)_{q+1}(1/2)_q}{(q+1)!q!}\alpha^{2q+1}$ (Lemma 10.7); equivalently from the unreduced sum of the terms $\alpha^jb_s^{(k)}$ before Step S13.11.

Step S13.12: Consistency with Part 8. $\mathsf{A}=\frac14\alpha b_{3/2}^{(1)}$ and $\mathsf{B}=-\frac12\alpha b_{3/2}^{(2)}$ reproduce the quadratic part of Theorem 10.11: $S_{12}\ni\mathrm{pref}\,(\mathsf{A}|v_1|^2+\dots)$ with $v_i=u_i/\sqrt{\Lambda_i}$ to leading order gives $-\mathfrak{a}/\Lambda_1\cdot|u_1|^2$, and $\mathrm{pref}\cdot\mathsf{B}/\sqrt{\Lambda_1\Lambda_2}\,u_1\cdot u_2=2\mathfrak{b}\,u_1\cdot u_2/\sqrt{\Lambda_1\Lambda_2}$; the identities hold identically in α by the reduction (finite check (F4)). The multipole expansion gives an independent test of the leading terms: the inner ellipse contributes $\mathcal{W}\ni\alpha^2\frac14\langle r^2\rangle+\alpha^4\frac{9}{64}\langle r^4\rangle$ with $\langle r^2\rangle=1+\frac32e^2$, $\langle r^4\rangle=1+5e^2+\frac{15}{8}e^4$ (inner axis 1), so $\mathsf{A}=\frac38\alpha^2+\dots$ and $\mathsf{p}_1=\frac{9}{64}\cdot\frac{15}{8}\alpha^4+\dots=\frac{135}{512}\alpha^4+\dots$, as in (3). □

Theorem 13.4 (The six quartic coefficients)**.** *For $a\in\mathcal{A}_2$ put $\Lambda_i=m_i\sqrt{\kappa a_i}$, $\alpha=a_1/a_2$, and*

$$c_1=\mathrm{pref}\,\frac{\mathsf{p}_1-\mathsf{A}/4}{\Lambda_1^2},\qquad c_2=\mathrm{pref}\,\frac{\mathsf{p}_2-\mathsf{A}/4}{\Lambda_2^2},\qquad c_3=\mathrm{pref}\,\frac{\mathsf{p}_3}{\Lambda_1\Lambda_2},\qquad c_4=\mathrm{pref}\,\frac{\mathsf{p}_4}{\Lambda_1\Lambda_2},$$
$$c_5=\mathrm{pref}\,\frac{\mathsf{p}_5-\mathsf{B}/8}{\Lambda_1^{3/2}\Lambda_2^{1/2}},\qquad c_6=\mathrm{pref}\,\frac{\mathsf{p}_6-\mathsf{B}/8}{\Lambda_1^{1/2}\Lambda_2^{3/2}},$$

(with $\mathsf{A},\mathsf{B},\mathsf{p}_\nu$ of Proposition 13.3, $\mathrm{pref}=-Gm_1m_2/a_2$). They are real analytic on $\mathcal{A}_2$, and the degree-four part of S_{fix} and of S_{mov} in (ξ,η) is

$$\mathcal{Q}_4(u)=c_1|u_1|^4+c_2|u_2|^4+c_3|u_1|^2|u_2|^2+c_4(u_1\cdot u_2)^2+c_5|u_1|^2(u_1\cdot u_2)+c_6|u_2|^2(u_1\cdot u_2).$$

Thus $c_\nu^{12}=c_\nu$ in Proposition 10.16; c_1,c_2 coincide with the diagonal formula of Proposition 10.16(b) (equivalently $c_1=\frac{m_1m_2}{2\Lambda_1^2}r_1^{\mathrm{ex}}(a_1,a_2)$, $c_2=\frac{m_1m_2}{2\Lambda_2^2}r_1^{\mathrm{ex}}(a_2,a_1)$ for $G=m_0=1$).

Proof. **Step S13.13: Exact conversion.** $v_i=\sqrt{(1-\Gamma_i/2\Lambda_i)/\Lambda_i}\,u_i=\Lambda_i^{-1/2}(1-|u_i|^2/(4\Lambda_i))^{1/2}u_i$ ($\Gamma_i=|u_i|^2/2$). Put $w_i=u_i/\sqrt{\Lambda_i}$; then $v_i=w_i(1-\frac18|w_i|^2+O(|w|^4))$, so $|v_i|^2=|w_i|^2-\frac14|w_i|^4+O(|w|^6)$ and $v_1\cdot v_2=w_1\cdot w_2\,(1-\frac18|w_1|^2-\frac18|w_2|^2)+O(|w|^6)$. Insert in $\mathcal{W}_2+\mathcal{W}_4$ (Proposition 13.3; $|\zeta_i|=|v_i|$, $\mathrm{Re}(\bar\zeta_1\zeta_2)=v_1\cdot v_2$): the degree-four part in w is

$$\mathcal{W}_4(w)-\tfrac{\mathsf{A}}{4}(|w_1|^4+|w_2|^4)-\tfrac{\mathsf{B}}{8}\,(w_1\cdot w_2)(|w_1|^2+|w_2|^2),$$

and multiplying by pref and using $|w_1|^4=|u_1|^4/\Lambda_1^2$, $|w_1|^2(w_1\cdot w_2)=|u_1|^2(u_1\cdot u_2)/(\Lambda_1^{3/2}\Lambda_2^{1/2})$, etc., gives the displayed c_ν.

Step S13.14: Analyticity and the jet. $\beta_0,\beta_1,\Lambda_i$ are real analytic on $\mathcal{A}_2$ (Lemma 10.7) and the denominators $\Delta^4=(1-\alpha^2)^4$ do not vanish. By Proposition 10.16's structure and Corollary 10.14, S_{fix} and S_{mov} have the same degree-four part, equal to that of S_{12} (pair decomposition, Step S10.21).

Step S13.15: c_1, c_2**.** For $\zeta_2 = 0$ the planetary pair average is a function of $E = |v_1|^2$ alone: $\mathcal{W}(\alpha; \zeta_1, 0) = \beta_0 + \mathsf{A}E + \mathsf{p}_1 E^2 + \dots$, while Step S10.24 gives $S_{12}|_{u_2=0} = -Gm_1m_2[h(0) + \frac{1}{4}h''(0)E + \frac{1}{64}h^{(4)}(0)E^2 + O(E^3)]$ (inner eccentric; E is the same e^2). Comparing, $\operatorname{pref}\mathsf{A} = -Gm_1m_2h''(0)/4$ and $\operatorname{pref}\mathsf{p}_1 = -Gm_1m_2h^{(4)}(0)/64$, hence $\operatorname{pref}(\mathsf{p}_1 - \mathsf{A}/4)/\Lambda_1^2 = -Gm_1m_2(h^{(4)}/64 - h''/16)/\Lambda_1^2$, the formula of Proposition 10.16(b). The case $i = 2$ is the same with the outer planet eccentric (Step S10.24, $i > j$). Numerically both sides agree to 10^{-14} (§13.6). □

13.3. **Birkhoff normalisation to order four.** Fix an open set $\mathcal{O} \subset \mathbb{R}^2$ of actions and real-analytic $\Omega = (\Omega_1, \Omega_2) : \mathcal{O} \to \mathbb{R}^2$ with the non-resonance

$$k \cdot \Omega(\Lambda) \neq 0 \qquad \forall k \in \mathbb{Z}^2,\ 0 < |k|_1 \leq 4,\ \Lambda \in \mathcal{O} \qquad \text{(i.e. } \mathrm{NR}_4(\Omega)\text{)}. \tag{4}$$

Put $N_0(\Lambda, z) = \sum_j \Omega_j(\Lambda) R_j(z)$ on $z = (z_1, z_2) \in \mathbb{R}^4$. A *quartic family* is $F(\Lambda, z) = \sum_{|\gamma|=4} f_\gamma(\Lambda) z^\gamma$ with real-analytic f_γ (monomials in x_1, y_1, x_2, y_2) and real values. Identify $z_j \leftrightarrow x_j + iy_j$. The *rotation action* is $T_\theta z = (e^{-i\theta_1} z_1, e^{-i\theta_2} z_2)$, $\theta \in \mathbb{R}^2$; by the frozen sign convention it is the flow of N_0: $\psi^t_{N_0} = T_{t\Omega}$ ($\dot z_j = -i\Omega_j z_j$ from $\dot x = \partial_y N_0$, $\dot y = -\partial_x N_0$).

Lemma 13.5 (Character decomposition)**.** *For a quartic family F and $k \in \mathbb{Z}^2$ let $F_k(\Lambda, z) = (2\pi)^{-2} \int_{[0,2\pi]^2} e^{-ik\cdot\theta} F(\Lambda, T_\theta z)\, d\theta$. Then $F_k(\Lambda, T_\phi z) = e^{ik\cdot\phi} F_k(\Lambda, z)$; $F_k \neq 0$ only if $|k|_1 \leq 4$; $F = \sum_k F_k$ (finite sum); $\overline{F_k} = F_{-k}$ for real F; F_k is a polynomial in $(z, \bar z)$ with coefficients real analytic in Λ, and $\{F_k, N_0\} = i\,(k \cdot \Omega)\, F_k$. In particular $\langle F \rangle := F_0$ is the T^2-average of F, and it is a real quartic family.*

Proof. **Step S13.16: Monomials.** In $z_j, \bar z_j$ the quartic family is a combination of $z^\alpha \bar z^\beta$, $|\alpha| + |\beta| = 4$, and $T_\theta(z^\alpha \bar z^\beta) = e^{ik\cdot\theta} z^\alpha \bar z^\beta$ with $k = \beta - \alpha$, $|k|_1 \leq 4$. Orthogonality of characters gives $F_k = \sum_{\beta-\alpha=k} f_{\alpha\beta} z^\alpha \bar z^\beta$ and $F = \sum F_k$.

Step S13.17: Reality and bracket. For real F, conjugating the definition gives $\overline{F_k} = F_{-k}$. $\{F_k, N_0\} = \frac{d}{dt} F_k(T_{t\Omega} z)|_0 = i(k \cdot \Omega) F_k$ by the covariance just proved. □

Lemma 13.6 (Homological equation)**.** *Let F be a quartic family on $\mathcal{O}$ and assume (4). Then*

$$\chi := \sum_{k \neq 0} \frac{F_k}{i\,(k \cdot \Omega)}$$

is a real quartic family (coefficients real analytic on $\mathcal{O}$) and $\{\chi, N_0\} = F - \langle F \rangle$, equivalently $\{N_0, \chi\} = -(F - \langle F \rangle)$.

Proof. **Step S13.18: Well defined, real.** The sum has finitely many terms with $0 < |k|_1 \leq 4$, so $k \cdot \Omega \neq 0$ and analytic; $\overline{F_k/(ik\cdot\Omega)} = F_{-k}/(i(-k)\cdot\Omega)$ shows χ real.

Step S13.19: Equation. Lemma 13.5: $\{\chi, N_0\} = \sum_{k\neq 0} \frac{i(k\cdot\Omega)F_k}{i(k\cdot\Omega)} = F - F_0$; antisymmetry of $\{\cdot,\cdot\}$ gives the second form. □

Lemma 13.7 (Time-one map of a quartic Hamiltonian)**.** *Let χ be a quartic family on $\mathcal{O}$, $\mathcal{K} \subset \mathcal{O}$ compact. There are $C_1, r_1 > 0$ such that for $\Lambda \in \mathcal{K}$ and $|z|_2 < r_1$ the flow ϕ^t_χ of $X_\chi(\Lambda, \cdot)$ exists for $t \in [0,1]$, $|\phi^t_\chi(z)|_2^2 \leq 2|z|_2^2$, and $Y(\Lambda, z) := \phi^1_\chi(\Lambda, z)$ is real analytic in (Λ, z) on a neighbourhood of $\mathcal{K} \times \{|z| < r_1\}$, $Y(\Lambda, z) = z + O(|z|^3)$ uniformly, and Y is injective with inverse $\phi^1_{-\chi}$ on its image.*

Proof. **Step S13.20: Bound.** X_χ is a polynomial vector field, homogeneous of degree 3 in z with coefficients analytic in Λ: $|X_\chi| \leq C_1|z|^3$ on $\mathcal{K}$. For $u = |z(t)|^2$, $\dot u = 2z \cdot X_\chi \leq 2C_1 u^2$, so $1/u(t) \geq 1/u_0 - 2C_1 t$; if $u_0 \leq r_1^2 := 1/(4C_1)$ then $u(t) \leq 2u_0$ on $[0,1]$ (and the solution cannot blow up, [N:bpN:p3:gen-continuation]).

Step S13.21: Analyticity, expansion. (F1); $\phi^1_\chi(z) - z = \int_0^1 X_\chi(\phi^t)dt = O(|z|^3)$. The inverse is the time-one map of $-\chi$ (uniqueness, [N:bpN:p3:gen-uniqueness]). □

Proposition 13.8 (Birkhoff normal form of order four)**.** *Let $\mathcal{O}$, Ω satisfy (4), let $K(\Lambda, z)$ be real analytic on a neighbourhood of $\mathcal{O} \times \{0\}$ with*

$$K = C + N_0 + F + \mathcal{R},$$

$C = C(\Lambda)$, F a quartic family, and $\mathcal{R}$ analytic with vanishing 5-jet at $z = 0$ (so $\mathcal{R} = O(|z|^6)$ locally uniformly, (F3)). Let χ be given by Lemma 13.6 and $Y = \phi^1_\chi$. Then for every compact $\mathcal{K} \subset \mathcal{O}$ there are $C_2, r_2 > 0$ with

$$\Big|K(\Lambda, Y(\Lambda, z)) - \big(C(\Lambda) + N_0(\Lambda, z) + \langle F\rangle(\Lambda, z)\big)\Big| \le C_2|z|^6 \qquad (\Lambda \in \mathcal{K},\ |z| < r_2).$$

Here $\langle F\rangle$ is the T^2-average, a quartic polynomial in R_1, R_2 only (Lemma 13.5): $\langle F\rangle = \sum_{i\le j} \varphi_{ij}(\Lambda) R_i R_j$.
Source: Standard Birkhoff normalisation (Siegel–Moser; Fej04r/CP10 Prop. 2 step), here in the finite closed form of the homological equation; CP10 Prop. 2, p. 553, and Step 4 p. 559 use track N's iterative Lie transform, not needed for order four.

Proof. **Step S13.22: Taylor along the flow.** For analytic K and $t \in [0,1]$, $\frac{d}{dt} K \circ \phi^t_\chi = \{K, \chi\} \circ \phi^t_\chi$ (definition of $\{\cdot,\cdot\}$; Part N, [N:bpN:p5:lie-derivative], [N:bpN:p5:taylor]), hence $K \circ Y = K + \{K, \chi\} + \int_0^1 (1-t)\{\{K, \chi\}, \chi\} \circ \phi^t_\chi \, dt$.

Step S13.23: Orders. $\{N_0, \chi\} = -(F - \langle F\rangle)$ (Lemma 13.6) is quartic. $\{F, \chi\}$ is a polynomial of degree $4 + 4 - 2 = 6$ ($dF = O(|z|^3)$, $X_\chi = O(|z|^3)$), $\{\mathcal{R}, \chi\} = O(|z|^8)$ and $\{C, \chi\} = 0$ (C does not depend on z). So $\{K, \chi\} = -(F - \langle F\rangle) + O(|z|^6)$. Further $\{K, \chi\} = O(|z|^4)$ analytic, so $\{\{K, \chi\}, \chi\} = O(|z|^3) \cdot O(|z|^3) = O(|z|^6)$ (analytic, (F3)), and on $\phi^t_\chi(z)$ with $|\phi^t_\chi z|^2 \le 2|z|^2$ (Lemma 13.7) it is $\le C|z|^6$.

Step S13.24: Sum. $K \circ Y = C + N_0 + F + \mathcal{R} - (F - \langle F\rangle) + O(|z|^6) = C + N_0 + \langle F\rangle + O(|z|^6)$. $\langle F\rangle$ is invariant under T_θ, hence a polynomial in $|z_1|^2, |z_2|^2$ (invariant polynomials of the torus action in $(z, \bar z)$ are $z^\alpha \bar z^\alpha$), of degree 2 in (R_1, R_2). □

Proposition 13.9 (The Birkhoff map on the full phase space)**.** *In the situation of Lemma 13.7, let $\mathcal{O}' \subset \mathcal{O}$ be open with $\overline{\mathcal{O}'}$ compact and contained in $\mathcal{O}$ (relatively compact, $\mathcal{O}' \Subset \mathcal{O}$), and let $r_1 = r_1(\overline{\mathcal{O}'})$ be the radius of Lemma 13.7 for $\mathcal{K} = \overline{\mathcal{O}'}$. With Λ a variable, define on $\mathbb{R}^2 \times \mathcal{O}' \times \{|z|_2 < r_1\} \ni (\lambda, \Lambda, z)$*

$$\Phi_\chi(\lambda, \Lambda, z) = \big(\lambda + g(\Lambda, z),\ \Lambda,\ Y(\Lambda, z)\big), \qquad g_l(\Lambda, z) = \int_0^1 \partial_{\Lambda_l} \chi\big(\Lambda, \phi^t_\chi(\Lambda, z)\big)\, dt.$$

(A uniform radius r_1 on an arbitrary open $\mathcal{O}$ is false: for $\chi = x_1^3 y_1/\Lambda_1$ on $\mathcal{O} = (0,1)^2$ the solution with $y = 0$ is $x_1(t) = x_1(0)(1 - 2x_1(0)^2 t/\Lambda_1)^{-1/2}$, which blows up before $t = 1$ once $\Lambda_1 < 2x_1(0)^2$; the radius of Lemma 13.7 is uniform only on compact sets, hence the restriction to $\mathcal{O}' \Subset \mathcal{O}$.) Then Φ_χ is the time-one map of the Hamiltonian flow of $\chi(\Lambda, z)$ for the chart form $\Omega_{\text{chart}} = d\Lambda \wedge d\lambda + dy \wedge dx$; hence it is a real analytic symplectic diffeomorphism of $\mathbb{R}^2 \times \mathcal{O}' \times \{|z|_2 < r_1\}$ onto its image, preserves Λ, commutes with angle translations $\Phi_\chi(\lambda + c, \Lambda, z) = \Phi_\chi(\lambda, \Lambda, z) + (c, 0, 0)$, and the composition $T := \Phi_\varrho^{-1} \circ \Phi_\chi$ with the lift of Proposition 11.8 (defined on all of $\mathbb{R}^2 \times \mathcal{O}$, hence on the image) has the same three properties on the same domain and the form

$$T(\lambda, \Lambda, z) = \big(\lambda + g(\Lambda, z) + H(\Lambda, Y(\Lambda, z)),\ \Lambda,\ \varrho(\Lambda) Y(\Lambda, z)\big)$$

with $H_l(\Lambda, \tilde z) = \tilde y^\top (\varrho^\top \partial_{\Lambda_l} \varrho) \tilde x$ (Proposition 11.8); ϱY means ϱ applied to the x- and to the y-vectors.

Proof. **Step S13.25: Hamilton's equations.** The Hamiltonian χ does not depend on λ, so $\dot\Lambda = -\partial_\lambda\chi = 0$, $\dot z = X_\chi(\Lambda, z)$ and $\dot\lambda_l = \partial_{\Lambda_l}\chi(\Lambda, z(t))$; integrating gives $\lambda(1) = \lambda + g$, $z(1) = Y$.

Step S13.26: Properties. Existence of the flow up to time one for every (Λ, z) of the domain is Lemma 13.7 on $\mathcal{K} = \overline{\mathcal{O}'}$; this is where $\mathcal{O}' \Subset \mathcal{O}$ is used. Hamiltonian flows are symplectic ([N:bpN:p5:symplectic], [N:bpN:p5:canonical]); analytic by (F1); Λ is constant; the vector field does not involve λ so λ-translations commute. T: composition of two maps with these properties (Proposition 11.8). □

13.4. **The torsion matrix.** For $a \in \mathcal{A}_2$ let $\sigma_1 < \sigma_2 < 0$ be the eigenvalues of $Q_h(a)$ (Corollary 11.14) and

$$\Pi_1(a) = \frac{Q_h(a) - \sigma_2 I}{\sigma_1 - \sigma_2} = \begin{pmatrix} \pi_{11} & \pi_{12} \\ \pi_{12} & \pi_{22} \end{pmatrix}$$

the orthogonal projection onto the σ_1-eigenspace (rank one: $\pi_{11} + \pi_{22} = 1$, $\pi_{11}\pi_{22} = \pi_{12}^2$). With $\varrho = [e_1|e_2] \in SO(2)$ one has $\Pi_1 = e_1 e_1^\top$.

Definition 13.10 (Torsion $\hat\beta$)**.** With $c_1, \dots, c_6$ of Theorem 13.4 put $\delta_\pi = \pi_{11} - \pi_{22}$ and

$$\begin{aligned} \mathsf{a} &= c_1\pi_{11}^2 + c_2\pi_{22}^2 + (c_3 + c_4)\pi_{11}\pi_{22} + (c_5\pi_{11} + c_6\pi_{22})\pi_{12}, \\ \mathsf{d} &= c_1\pi_{22}^2 + c_2\pi_{11}^2 + (c_3 + c_4)\pi_{11}\pi_{22} - (c_5\pi_{22} + c_6\pi_{11})\pi_{12}, \\ \mathsf{b} &= 4(c_1 + c_2)\pi_{11}\pi_{22} + c_3\delta_\pi^2 + c_4\big(\tfrac12\delta_\pi^2 - 2\pi_{11}\pi_{22}\big) + 2(c_6 - c_5)\pi_{12}\delta_\pi, \end{aligned}$$

and $\hat\beta(a) = \begin{pmatrix} 8\mathsf{a} & 4\mathsf{b} \\ 4\mathsf{b} & 8\mathsf{d} \end{pmatrix}$, symmetric. All entries are real analytic on $\mathcal{A}_2$ ($\sigma_{1,2} = \frac12(\operatorname{tr} \mp\sqrt{\operatorname{tr}^2 - 4\det}$) with positive discriminant, Corollary 11.14), *independent of the choice of* ϱ (only Π_1 enters), of the coupling parameter, and the same for S_{fix} and S_{mov} (Corollary 10.14).

Proposition 13.11 ($\hat\beta$ is the torus average of the rotated quartic)**.** *Let $\varrho = [e_1|e_2] \in SO(2)$ diagonalise $Q_h(a)$ and let $\tilde{\mathcal{Q}}_4(z) = \mathcal{Q}_4(\varrho x, \varrho y)$, i.e. $u_i = \sum_j \varrho_{ij} z_j$ in $\mathcal{Q}_4$ (Theorem 13.4). Its T^2-average (Lemma 13.5) is*

$$\langle \tilde{\mathcal{Q}}_4 \rangle = \tfrac12 \sum_{i,j} \hat\beta_{ij}(a) R_i R_j = 4\mathsf{a}\, R_1^2 + 4\mathsf{b}\, R_1 R_2 + 4\mathsf{d}\, R_2^2.$$

Proof. **Step S13.27: Gram variables.** Put $r_j = |z_j|^2 = 2R_j$ and $g = z_1 \cdot z_2 = x_1x_2 + y_1y_2$. With $\varrho = \begin{pmatrix} \gamma & -s \\ s & \gamma \end{pmatrix}$, $e_1 = (\gamma, s)$, $\pi_{11} = \gamma^2$, $\pi_{22} = s^2$, $\pi_{12} = \gamma s$: $u_1 = \gamma z_1 - s z_2$, $u_2 = s z_1 + \gamma z_2$, so

$$A_1 := |u_1|^2 = \pi_{11} r_1 + \pi_{22} r_2 - 2\pi_{12} g, \quad A_2 := |u_2|^2 = \pi_{22} r_1 + \pi_{11} r_2 + 2\pi_{12} g,$$
$$B_\cdot := u_1 \cdot u_2 = \pi_{12}(r_1 - r_2) + \delta_\pi g,$$

and $\tilde{\mathcal{Q}}_4 = c_1 A_1^2 + c_2 A_2^2 + c_3 A_1 A_2 + c_4 B_\cdot^2 + c_5 A_1 B_\cdot + c_6 A_2 B_\cdot$ is a polynomial in (r_1, r_2, g) of degree ≤ 2 in g.

Step S13.28: Averaging. Under T_θ, r_j is invariant and $g = \operatorname{Re}(\bar z_1 z_2) \mapsto \operatorname{Re}(e^{i(\theta_1 - \theta_2)} \bar z_1 z_2)$, so $\langle g \rangle = 0$ and $\langle g^2 \rangle = \frac12 r_1 r_2$ ($\langle \cos^2 \rangle = \frac12$; $\operatorname{Re}(e^{i\vartheta} \bar z_1 z_2) = \sqrt{r_1 r_2}\cos(\vartheta + \vartheta_0)$). Collecting the coefficients of r_1^2, r_2^2 and $r_1 r_2 + \frac12 g^2$ in the expansion (and using $\pi_{11}\pi_{22} = \pi_{12}^2$, $\pi_{11} + \pi_{22} = 1$ to write $A_\cdot$, $B_\cdot$ products in the π's): coefficient of r_1^2 is a, of r_2^2 is d and of $r_1 r_2$ (after $g^2 \mapsto \frac12 r_1 r_2$) is b; for instance the $r_1 r_2$ coefficient of A_1^2 is $2\pi_{11}\pi_{22}$ and $\frac12 \cdot 4\pi_{12}^2 = 2\pi_{12}^2$ comes from g^2, total $4\pi_{11}\pi_{22}$, which gives the term $4c_1\pi_{11}\pi_{22}$ of b; the others are identical bookkeeping, verified symbolically and numerically (§13.6). Hence $\langle \tilde{\mathcal{Q}}_4 \rangle = \mathsf{a} r_1^2 + \mathsf{b} r_1 r_2 + \mathsf{d} r_2^2 = 4\mathsf{a} R_1^2 + 4\mathsf{b} R_1 R_2 + 4\mathsf{d} R_2^2$, and $\frac12 \sum \hat\beta_{ij} R_i R_j = \frac12(8\mathsf{a} R_1^2 + 8\mathsf{b} R_1 R_2 + 8\mathsf{d} R_2^2)$. □

Theorem 13.12 (Secular Birkhoff form, planar)**.** *Let $n=2$, $G, m_0>0$, $m_1, m_2>0$. Let $K_a=\overline{B}_\infty(a^0,\rho)\subset\{a\in\mathcal{A}_2:\Phi_4(a)\neq 0\}$ be a closed cube, $\mathcal{O}_a\supset K_a$ open and inside $\{\Phi_4\neq 0\}$ and simply connected, $\Lambda(\cdot)$ the analytic diffeomorphism $a\mapsto\Lambda$, $K=\Lambda(K_a)$, $\mathcal{O}=\Lambda(\mathcal{O}_a)$. Then there are an analytic $\varrho:\mathcal{O}\to SO(2)$ diagonalising Q_h, analytic χ (Lemma 13.6 with $F=\tilde{\mathcal{Q}}_4$, $\Omega=2\sigma$), an open $\mathcal{O}'$ with $K\subset\mathcal{O}'\Subset\mathcal{O}$, constants $r_*>0$, $C_*>0$ and the analytic symplectic map $T=\Phi_\varrho^{-1}\circ\Phi_\chi$ (Proposition 13.9, for this $\mathcal{O}'$) on $\mathcal{T}=\{(\lambda,\Lambda,z):\Lambda\in\mathcal{O}', |z|_2<r_*\}$, with $r_*\le\rho_K/\sqrt{2}$ and $r_*\le\min\{r_1(\overline{\mathcal{O}'}),r_2\}$ (ρ_K of Step S13.30, r_1 of Lemma 13.7, r_2 of Proposition 13.8 for $\mathcal{K}=\overline{\mathcal{O}'}$), such that T takes values in the domain of analyticity of $\hat f_{\mathrm{mov}}$ (and of $\hat f_{\mathrm{fix}}$), i.e. in the polydisc domain of Proposition 10.2, and such that for $\hat f\in\{\hat f_{\mathrm{fix}},\hat f_{\mathrm{mov}}\}$ and $P:=\hat f\circ T$:*

(a) P is real analytic on $\mathcal{T}$, 2π-periodic in λ, independent of any coupling parameter, and $H^\Psi\circ T=h(\Lambda)+\mu P$ for $H^\Psi=h+\mu\hat f$ (T preserves Λ);

(b) for $\Lambda\in K$, $|z|_2<r_*$,

$$\langle P\rangle(\Lambda,z)=P_0(\Lambda)+\sum_{j=1,2}\Omega_j(\Lambda)R_j+\tfrac12\sum_{i,j}\hat\beta_{ij}(a(\Lambda))R_iR_j+\mathrm{Rem},\qquad |\mathrm{Rem}|\le C_*|z|^6,$$

with $\Omega_j=2\sigma_j(a(\Lambda))<0$, $P_0=C_0(a)$ for $\hat f_{\mathrm{fix}}$ and $C_0(a)+\delta(\Lambda)$ for $\hat f_{\mathrm{mov}}$, $\langle\cdot\rangle$ the average over $\lambda\in[0,2\pi]^2$;

(c) with the dictionary $(I,\varphi,p,q)=(\Lambda,\lambda,y,x)$, the frozen HasBirkhoff$(P,V,r_0,\mu_0,P_0,\Omega,\beta)$ *of Definition 1.14 holds for $V=\operatorname{int}K$, any $\mu_0>0$, every $r_0\in(0,r_*/\sqrt{2}]$ (the clauses are monotone in r_0; Part 12 uses $r_0=r_*/2$, for which the closed cylinder $\{|z|_2\le\sqrt{2}r_0\}$ lies in the open set $\mathcal{T}$; any $r_0<r_*/\sqrt{2}$ works likewise) and $\Omega(\mu,I)=2\sigma(a(I))$, $\beta(\mu,I)=\hat\beta(a(I))$, $P_0(\mu,I)$ as above (all independent of μ), where P is extended to the rest of phase space as in Part 12; $\beta(\mu,I)$ is symmetric.*

The data ϱ,χ,r_*,C_* depend only on (G,m_0,m,K_a).

Proof. **Step S13.29: Rotation.** Q_h has simple spectrum on $\mathcal{O}_a$, Π_1 is analytic there, and e_1 (a unit vector in the range of Π_1) lifts analytically to the angle ψ on the simply connected $\mathcal{O}_a$ (covering $S^1\to S^1$); put $\varrho=[e_1|e_2]$, $e_2=Je_1$ (Proposition 11.7, Theorem 11.9(a)).

Step S13.30: The function K. Put $K(\Lambda,z)=S_{\mathrm{fix}}(\Lambda,\varrho x,\varrho y)$. By Theorem 10.11, Corollary 10.4 and Theorem 13.4, $S_{\mathrm{fix}}=C_0+\xi^\top Q_h\xi+\eta^\top Q_h\eta+\mathcal{Q}_4+\mathcal{R}_6$ with $\mathcal{R}_6$ analytic with vanishing 5-jet; by Theorem 11.9(a), $K=C_0+N_0+\tilde{\mathcal{Q}}_4+\mathcal{R}$ with $\Omega=2\sigma$, $\mathcal{R}=\mathcal{R}_6\circ(\varrho\cdot)$. K is analytic on a neighbourhood $\mathcal{N}_K$ of $K\times\{0\}$ in the sense of Part 8: choose ρ_K by compactness (cover K by finitely many $\mathcal{N}(\rho_*(\Lambda^*)/2)$) so that the real polydiscs of radius ρ_K around $(0,\Lambda_c,0)$, $\Lambda_c\in K$, lie in the domain of Proposition 10.2.

Step S13.31: Normal form. $F=\tilde{\mathcal{Q}}_4$ is a quartic family (coefficients analytic in Λ through c_ν,ϱ), and $\mathrm{NR}_4(\Omega)$ holds on $\mathcal{O}$ since $\Phi_4\neq0$ (Proposition 11.12(b), $\mathrm{NR}_4(\sigma)\Leftrightarrow\mathrm{NR}_4(2\sigma)$). Proposition 13.8: $K\circ Y=C_0+N_0+\langle\tilde{\mathcal{Q}}_4\rangle+O(|z|^6)$, and $\langle\tilde{\mathcal{Q}}_4\rangle=\frac12\sum\hat\beta_{ij}R_iR_j$ by Proposition 13.11.

Step S13.32: Average of P. $T(\lambda,\Lambda,z)=(\lambda+h_T(\Lambda,z),\Lambda,\varrho Y)$ with h_T independent of λ (Proposition 13.9). $\hat f$ is continuous and 2π-periodic in λ (Proposition 10.2(b)), so the λ-average of $\hat f\circ T$ is the average of $\hat f$ at the point $(\Lambda,\varrho x',\varrho y')$, $z'=Y(\Lambda,z)$ (translation invariance, (E2); the point $(\lambda+h_T,\Lambda,\varrho Y)=T(\lambda,\Lambda,z)$ is defined because $\Lambda\in\mathcal{O}'$ and $|z|_2<r_*\le r_1$, Proposition 13.9), i.e. $S(\Lambda,\varrho x',\varrho y')=K(\Lambda,Y)+[\delta]$; here $|\varrho Y|\le|Y|\le\sqrt{2}|z|$ so the image point lies in the polydisc for $r_*\le\rho_K/\sqrt{2}$ (the radius condition used here, together with $r_*\le r_1$ for the flow); thus T takes values in the analyticity domain of $\hat f_{\mathrm{mov}}$, which gives (a). This is (b); the remainder bound is Proposition 13.8 (applied for $\mathcal{K}=K$, $|z|<r_*\le r_2$) with the $[\delta]$ identification of Theorem 10.13.

Step S13.33: The frozen clauses. Definition 1.14 has two clauses (integrability, and the ϵ–δ estimate), quantified over $\mu\in[0,\mu_0]$, $I\in V$. Let $0<r_0\le r_*/\sqrt{2}$. Antecedents are oscillator discs:

$p_j^2+q_j^2<r_0^2$ $(j=1,2)$, which give $|z|_2^2=\sum_j(x_j^2+y_j^2)<2r_0^2\le r_*^2$, so the point lies in $\mathcal{T}$. (i) P is continuous on $\mathcal{T}$, hence integrable over the torus on such discs (Definition 1.14, first clause); the frozen secularAverage is $\langle P\rangle$ (dictionary, §10.1). (ii) Let $\epsilon>0$ and $\delta\le r_0$ with $\delta^2\le\epsilon/(64C_*)$; for the oscillator discs $p_j^2+q_j^2<\delta^2$ the point lies in $\mathcal{T}$ as just shown, and only for the remainder estimate we use $|z|_2\le 2|(p,q)|_\infty$: $|\mathrm{Rem}|\le C_*|z|^6\le 64C_*|(p,q)|_\infty^6\le\epsilon|(p,q)|_\infty^4$, uniformly in $I\in K$. The data $\hat\beta,\Omega,P_0$ do not depend on μ, so both clauses hold for all $\mu\in[0,\mu_0]$ (the functions are constant in μ, also at $\mu=0$). Symmetry of $\hat\beta$ is built in; it supplies the separate symmetry hypothesis of P-G (it is not a clause of HasBirkhoff). □

Remark 13.13 (Two clauses of P-G that remain for Part 12). (1) htorsion asks $B\le|\det\beta|$ on V: by Corollary 13.16 below, $\det\hat\beta$ is continuous and nonzero on the compact K_a, so $B:=\min_{K_a}|\det\hat\beta|>0$ works. (2) The joint analyticity of P on a neighbourhood of closure(cylinder), the extension of P outside $\mathcal{T}$, hperiod (which holds because T commutes with 2π-shifts and $\hat f$ is 2π-periodic), the fast diffeomorphism and the boundary collar are Part 12's. For $n=2$ and P-FIXED the same P_0, Ω, $\hat\beta$ (up to δ) serve (Corollary 10.14).

13.5. **The torsion determinant.**

Proposition 13.14 (Hierarchical asymptotics of $\hat\beta$)**.** *Fix $G,m_0,m_1,m_2>0$, $g_0:=G/\kappa=1/m_0$, and take $a_2=1$, $a_1=\alpha\in(0,\alpha_0)$ (α_0 depends only on m_1/m_2). Then, with $O(\cdot)$ constants depending only on m_1/m_2,*

$$\hat\beta_{11}=\frac{3g_0m_2}{4m_1}\,\alpha\,\big(1+O(\alpha^2)\big),\qquad \hat\beta_{12}=-\frac{9g_0}{4}\,\alpha^{3/2}\big(1+O(\alpha^2)\big),\quad \hat\beta_{22}=-\frac{3g_0m_1}{m_2}\,\alpha^2\big(1+O(\alpha^2)\big),$$
$$\det\hat\beta=-\frac{117}{16}\,g_0^2\,\alpha^3\,\big(1+O(\alpha^2)\big).$$

For general axes, by homogeneity ($\hat\beta(sa_1,sa_2)=s^{-2}\hat\beta(a_1,a_2)$), $\det\hat\beta=-\frac{117}{16}\,(G/\kappa)^2\,a_1^3a_2^{-7}\,(1+O(a_1^2/a_2^2))$ as $a_1/a_2\to0$ with m fixed.

Source: Ours. Cross-check: CP11 (8.3)–(8.4), p. 34 (planar block $\tau_{\rm pl}=m_1m_2\frac{a_1^2}{a_2^3}\left(\begin{smallmatrix}3/(4\Lambda_1^2) & -9/(4\Lambda_1\Lambda_2)\\ -9/(4\Lambda_1\Lambda_2) & -3/\Lambda_2^2\end{smallmatrix}\right)$ for $G=m_0=1$, $\det\tau_{\rm pl}=-\frac{117}{16\Lambda_1^2\Lambda_2^2}(m_1m_2a_1^2/a_2^3)^2$), Arnold 1963 [2, p. 138, (3.4.31)]; with $\Lambda_1^2=m_1^2a_1$, $\Lambda_2^2=m_2^2a_2$ these are exactly the three limits above and (8.4): the two computations agree. CP11's asymptotics (8.2) for $r_1(a_1,a_2)=\frac{3}{16}\alpha^2$, $r_1(a_2,a_1)=-\frac34\alpha^2$ also agree with c_1,c_2 below.

Proof. **Step S13.34: Inputs.** Here $\Lambda_1=m_1\sqrt{\kappa\alpha}$, $\Lambda_2=m_2\sqrt\kappa$, $\mathrm{pref}=-Gm_1m_2$. From Lemma 10.7, $b^{(1)}_{3/2}=\frac32\alpha(1+O(\alpha^2))$, $b^{(2)}_{3/2}=\frac{15}{8}\alpha^2(1+O(\alpha^2))$, so $\mathfrak{a}=\frac38Gm_1m_2\alpha^2(1+O(\alpha^2))$, $\mathfrak{b}=\frac{15}{32}Gm_1m_2\alpha^3(1+O(\alpha^2))$ and

$$Q_{11}=-\frac{\mathfrak{a}}{\Lambda_1}=-\frac{3Gm_2}{8\sqrt\kappa}\alpha^{3/2}(1+O(\alpha^2)),\quad Q_{22}=-\frac{\mathfrak{a}}{\Lambda_2}=-\frac{3Gm_1}{8\sqrt\kappa}\alpha^2(1+O(\alpha^2)),$$
$$Q_{12}=\frac{\mathfrak{b}}{\sqrt{\Lambda_1\Lambda_2}}=O(\alpha^{11/4}).$$

The difference $\Delta_Q:=Q_{11}-Q_{22}=Q_{11}\big(1+O(\alpha^{1/2})\big)$ is nonzero and negative for small α.

Step S13.35: Projection. With $R=\sqrt{\Delta_Q^2+4Q_{12}^2}\ge|\Delta_Q|$: $\sigma_2-\sigma_1=R$, $\pi_{12}=Q_{12}/(\sigma_1-\sigma_2)=-Q_{12}/R$, so $|\pi_{12}|\le|Q_{12}|/|\Delta_Q|=O(\alpha^{5/4})$; $\pi_{22}=(\sigma_2-Q_{22})/R$ and $\sigma_2-Q_{22}=\frac12(\Delta_Q+R)=\frac12(R-|\Delta_Q|)\in[0,Q_{12}^2/|\Delta_Q|]$, so $0\le\pi_{22}\le Q_{12}^2/\Delta_Q^2=O(\alpha^{5/2})$ and $\pi_{11}=1-\pi_{22}$.

Step S13.36: Sizes of c_ν. By (3) (with $\mathrm{pref}=-Gm_1m_2$): $\mathsf{p}_1-\mathsf{A}/4=-\frac{3}{32}\alpha^2(1+O(\alpha^2))$, $\mathsf{p}_2-\mathsf{A}/4=\frac38\alpha^2(1+O(\alpha^2))$, $\mathsf{p}_3=\frac{9}{16}\alpha^2(1+O(\alpha^2))$, $\mathsf{p}_4=O(\alpha^4)$, $\mathsf{p}_5-\mathsf{B}/8=O(\alpha^3)$, $\mathsf{p}_6-\mathsf{B}/8=O(\alpha^3)$.

Dividing by $\Lambda_1^2 = m_1^2\kappa\alpha$, $\Lambda_2^2 = m_2^2\kappa$, $\Lambda_1\Lambda_2 = m_1m_2\kappa\alpha^{1/2}$, $\Lambda_1^{3/2}\Lambda_2^{1/2} \asymp \alpha^{3/4}$, $\Lambda_1^{1/2}\Lambda_2^{3/2} \asymp \alpha^{1/4}$:

$$c_1 = \frac{3g_0m_2}{32m_1}\alpha(1+O(\alpha^2)), \quad c_2 = -\frac{3g_0m_1}{8m_2}\alpha^2(1+O(\alpha^2)), \quad c_3 = -\frac{9g_0}{16}\alpha^{3/2}(1+O(\alpha^2)),$$
$$c_4 = O(\alpha^{7/2}),\ c_5 = O(\alpha^{9/4}),\ c_6 = O(\alpha^{11/4}).$$

Step S13.37: The entries. Insert into Definition 13.10 with the bounds of Step S13.35 ($\pi_{12} = O(\alpha^{5/4})$, $\pi_{11}\pi_{22} = \pi_{12}^2 = O(\alpha^{5/2})$, $\delta_\pi^2 = 1 - 4\pi_{12}^2$): $\mathsf{a} = c_1 + O(\alpha^{7/2})$ (the corrections are $c_1\pi_{22} = O(\alpha^{7/2})$, $c_3\pi_{12}^2 = O(\alpha^4)$, $c_5\pi_{12} = O(\alpha^{7/2})$); $\mathsf{d} = c_2 + O(\alpha^4)$ ($c_6\pi_{12} = O(\alpha^4)$, $c_3\pi_{12}^2 = O(\alpha^4)$, $c_1\pi_{22}^2$ negligible); $\mathsf{b} = c_3 + O(\alpha^{7/2})$ ($4(c_1+c_2)\pi_{12}^2 = O(\alpha^{7/2})$, $c_4 = O(\alpha^{7/2})$, $(c_6 - c_5)\pi_{12}\delta_\pi = O(\alpha^{7/2})$, $c_3(\delta_\pi^2 - 1) = O(\alpha^4)$). Since $c_1 \asymp \alpha$, $c_2 \asymp \alpha^2$, $c_3 \asymp \alpha^{3/2}$ the relative errors are $O(\alpha^{5/2})$, $O(\alpha^2)$, $O(\alpha^2)$. Then $\hat\beta_{11} = 8\mathsf{a}$, $\hat\beta_{12} = 4\mathsf{b}$, $\hat\beta_{22} = 8\mathsf{d}$ give the three limits.

Step S13.38: Determinant. $\det\hat\beta = 64\,\mathsf{ad} - 16\,\mathsf{b}^2 = g_0^2\alpha^3\big[64\cdot\frac{3}{32}\cdot(-\frac{3}{8}) - 16\cdot\frac{81}{256}\big](1+O(\alpha^2)) = g_0^2\alpha^3\big[-\frac{9}{4} - \frac{81}{16}\big](1+O(\alpha^2)) = -\frac{117}{16}g_0^2\alpha^3(1+O(\alpha^2))$, the two summands having the same sign, so there is no cancellation. Homogeneity: S_{12} is homogeneous of degree -1 in the lengths and the actions scale as $\Lambda \mapsto s^{1/2}\Lambda$, so $\hat\beta = \partial_R^2 S$ scales as $s^{-1}\cdot s^{-1}$. □

Theorem 13.15 ($\det\hat\beta \not\equiv 0$ and the torsion theorem)**.** *$\det\hat\beta$ is real analytic on the connected (convex) open set $\mathcal{A}_2 = \{0 < a_1 < a_2\}$ and not identically zero. Hence $\{\det\hat\beta \neq 0\}$ is open, dense and of full Lebesgue measure in $\mathcal{A}_2$; the same for $\{\Phi_4\det\hat\beta \neq 0\}$.*

Source: Ours. This is the "full torsion" statement of CP11 §8.1, p. 34, with its proof (Remark 7.3(iii), p. 32: Birkhoff invariants are analytic in the axes and the determinant does not vanish identically, so by the identity theorem not on an open dense set); here with the invariants recomputed.

Proof. **Step S13.39: Analyticity.** The entries of $\hat\beta$ are real analytic on $\mathcal{A}_2$ (Definition 13.10).

Step S13.40: A point. By Proposition 13.14 (with $a_2 = 1$, $a_1 = \alpha$ small) $\det\hat\beta < 0$; so $\det\hat\beta \not\equiv 0$.

Step S13.41: Identity theorem. (F2) ((E4)) on connected $\mathcal{A}_2$: the zero set of $\det\hat\beta$ is closed, has empty interior and measure zero. Together with Theorem 12.8 ($\Phi_4 \not\equiv 0$ on $\mathcal{A}_2$) Lemma 12.1 gives the last claim. □

Corollary 13.16 (Patches near every axis vector)**.** *For every $a^* \in \mathcal{A}_2$ and $\eta > 0$ there are $a_0 \in \mathcal{A}_2$ and $r_0 > 0$ with $\overline{B}_\infty(a_0, 2r_0) \subset B_\infty(a^*,\eta) \cap \{\Phi_4 \neq 0\} \cap \{\det\hat\beta \neq 0\}$, and then $B := \min_{\overline{B}_\infty(a_0,r_0)}|\det\hat\beta| > 0$, $\hat\beta$ symmetric and Theorem 13.12 applies with $K_a = \overline{B}_\infty(a_0,r_0)$, $\mathcal{O}_a = B_\infty(a_0,2r_0)$ (a cube: simply connected). Moreover Part 10's frequency data (Theorem 12.22) can be taken on the same patch (the extra analytic condition $\det\hat\beta \neq 0$ is covered by the final sentence of that theorem).*

Proof. Theorem 13.15: $\mathcal{G} = \{\Phi_4\det\hat\beta \neq 0\}$ is open and dense in $\mathcal{A}_2$, so $B_\infty(a^*,\eta)\cap\mathcal{G}\cap\mathcal{A}_2$ is a nonempty open set containing a closed cube $\overline{B}_\infty(a_0,2r_0)$; $|\det\hat\beta|$ is continuous and positive on the compact $\overline{B}_\infty(a_0,r_0)$. □

13.6. **Numerical evidence, comparison with CP11.** All items below are supplementary numerical checks; the proofs above rest on the exact identities and analytic arguments only.

Scratch scripts: `.drafts/P/bp-scratch-bpP-p11b/` (`orbit.py`, `pair.py`, `avg.py`, `reduce.py`, `coeffs.py`, `series.py`, `final.py`, `endtoend.py`, `cmp.py`, `scan.py`, `beta_asym.py`).

(1) *Pair jet* (`num1.py`): at $\alpha = 0.45$, $\zeta_1 = 0.05e^{0.7i}$, $\zeta_2 = 0.04e^{-1.1i}$, the brute-force torus average of $|z_1 - z_2|^{-1}$ (Kepler's equation solved by Newton, 96^2 nodes) minus $\mathcal{W}_0 + \cdots + \mathcal{W}_4$ (before reduction) is $1.4\cdot10^{-8}$; halving ζ gives $2.1\cdot10^{-10}$, ratio $64 = 2^6$: the error is consistent with $O(|v|^6)$ (a supplementary check at one amplitude direction; the six coefficients are established by the exact identities above, not by this test).

(2) *Closed forms vs series* (`final.py`): the reduced rational formulas of Proposition 13.3 agree with the unreduced power series (3) (to $M = 16$) at $\alpha = 0.3$ to the truncation error ($\le 3 \cdot 10^{-7}$).

(3) *End to end* (`endtoend.py`): $G = \kappa = 1$, $m = (0.7, 1.9)$, $a = (1, 3)$. The exact $S_{12}(u_1, u_2) = \operatorname{pref} \mathcal{W}$ computed by brute force from the chart (128^2 nodes, the exact map $u \mapsto v$) minus $C_0 + \xi^\top Q_h \xi + \eta^\top Q_h \eta + \mathcal{Q}_4$ equals $-1.5 \cdot 10^{-10}$ at $|u| \approx 0.06$ and $-2.4 \cdot 10^{-12}$ at half amplitude (ratio 64). Values: $c_1 = 0.0065937$, $c_2 = -0.0028633$ (equal to Remark 10.17), $c_3 = -0.019563$, $c_4 = -0.009108$, $c_5 = 0.018701$, $c_6 = 0.011446$; $\sigma = (-0.034350, -0.005582)$; $\hat\beta = \left(\begin{smallmatrix} 0.0050153 & -0.060765 \\ -0.060765 & -0.010875 \end{smallmatrix}\right)$, $\det \hat\beta = -3.747 \cdot 10^{-3}$.

(4) *Torus average* (`cmp.py`): $\hat\beta$ computed from Definition 13.10 equals the numerical T^2-average of $\tilde{\mathcal{Q}}_4$ (48^2 angles) to 10^{-15}.

(5) *Torsion vs* α (`scan.py`, $m = (0.7, 1.9)$, $a_2 = 1$): $\det \hat\beta / (-\frac{117}{16}\alpha^3) = 1.0011,\ 1.0062,\ 1.0215,\ 1.065$ at $\alpha = 0.02, 0.05, 0.1, 0.2$ (consistent with an error of order α^2), and $\det \hat\beta$ changes sign between $\alpha = 0.6$ (-1.49) and $\alpha = 0.7$ ($+0.879$), which suggests zeros of $\det \hat\beta$ in $\mathcal{A}_2$; this is not used, and only the genericity of Theorem 13.15 is claimed.

(6) *Frequency shifts of the nonlinear flow* (`freq2.py`): for the Hamiltonian $K = \xi^\top Q_h \xi + \eta^\top Q_h \eta + \mathcal{Q}_4$ at the same point, integrate Hamilton's equations (DOP853, 10^{-12}) from the pure mode-j initial datum of amplitude ρ ($R = \rho^2/2$) and fit the phase of the mode coordinates. Frequency shift per unit R: mode 1 in mode 1: 0.00474, 0.00390 for $\rho = 0.05, 0.1$ against $\hat\beta_{11} = 0.00502$; mode 2 in mode 1: -0.0754, -0.0634 against $\hat\beta_{12} = -0.0608$; mode 2 in mode 2: -0.01087 against $\hat\beta_{22} = -0.01087$ (differences $O(R)$ as expected). This is a supplementary numerical check of the Birkhoff step; it is not a proof and gives no certified error order.

Comparison with CP11. (8.3), (8.4) and (8.2) of CP11 (pp. 33–34, rendered page read) agree with Proposition 13.14 (all three entries and the constant $-\frac{117}{16}$), which is an independent derivation (it checks the asymptotic regime of CP11's calculation, not every axis vector). CP11's printed finite-α formulas (r_1 of (B.6), r_2, r_3, r_4) were not used; the repaired r_1 (exponent $37\alpha^3\beta_4^{(9)}$, review P8-01/02) agrees with c_1 in the numerical value above (Remark 10.17).

13.7. **Sources, deviations, errata, forward references.**
Sources read. CP11 pp. 31–34 (page 34 rendered as an image), Part 8–10 of this blueprint in their revised versions; the review P8 and the Part 8–10 consolidation notes.
Deviations. (1) β is derived from the Birkhoff *finite* homological equation and the T^2-average; no Lie-transform machinery of track N beyond the flow/Taylor formula. (2) The Birkhoff map is realised on the full phase space as a λ-independent Hamiltonian flow; its angle shift g does not affect the λ-average. (3) The exceptional set is the zero set of an analytic function: the independent calculation proves that $\det \hat\beta$ is not identically zero (small-α asymptotics), followed by the analytic open-dense argument (as in CP11, Remark 7.3(iii)); it does not prove nonvanishing at every axis vector.
Suspected errata (for the ledger). E-P-CP11-2 (already recorded): (B.6) r_1 exponent $\alpha^4 \to \alpha^3$ in the term $37\alpha^4\beta_4^{(9)}$; the repaired formula reproduces c_1 at the test point, r_2, r_3, r_4 unchecked. None new: the asymptotics (8.2)–(8.4) are correct.
Doubts about frozen statements. None. The frozen HasSecularBirkhoffForm asks for the secular average *of the physical perturbation in the stated coordinates*; the coordinate change T of Proposition 13.9/Part 9 is not mentioned in the sources and must be part of Part 12's definition of P (a symplectic, Λ-preserving, angle-translation commuting change of the cylinder coordinates).
External facts used. (F1)–(F4); (E2), (E3), (E4), (F1).

Forward references and consumers. Part 12: Theorem 13.12, Corollary 13.16, Remark 13.13. Cited from other tracks: [N:bpN:p3:gen-continuation], [N:bpN:p3:gen-uniqueness], [N:bpN:p5:symplectic], [N:bpN:p5:canonical], [N:bpN:p5:lie-derivative], [N:bpN:p5:taylor]. Label index. orbit, laplace-red, pair-jet, quartic-explicit, char, homological, flow, lie, lift-map, beta-def, beta-formula, birkhoff-form, remain, asymptotics, torsion, patch.

14. Part 11: finite preparation and the properly degenerate local wrapper for P-FIXED

This part prepares the fixed-Sun Hamiltonian of Theorem 1.30 (P-FIXED, general $n \geq 2$) for the maximal-case analytic KAM theorem of Part 11a, and transports the resulting tori back to Cartesian coordinates. It follows CP09 §2.3 (finite-order averaging, polar normalisation of the secular oscillators, the amount of non-degeneracy of the prepared frequency map, Lemma 8 and its polynomial budget) for the properly degenerate Hamiltonian $H = h(I) + \varepsilon f$ of CP09 Theorem 4 (here: $d = p = n$ fast angles and secular oscillators, so $N = d+p = 2n$), with every quantitative gap of the source closed (reviews p11-route-study.md, p11a-russmann-check.md: R-CP-1, R-CP-2, A2, A4–A7). The route and its deviations in one paragraph.

Route. (A) Part 7 (product chart), Part 8 (secular function), Part 9 (diagonalising rotation ϱ) put H_{fix} into the form $\tilde{h}(J)+\varepsilon F(\theta, J, x, y)$ with F holomorphic on a strip and a complex action polydisc, $\theta \in \mathbb{T}^n$ the fast angles (period one), (x, y) the rotated secular Cartesian pairs (Prop. 14.4). (B) A base point J_0 with Diophantine joint frequency $(\omega, \tilde{\Omega})(J_0)$ is chosen by Part 11a-1 (Lemma 14.3; this replaces CP09 Lemma 3 and the erroneous set $K \cap D_{\gamma_0,\tau_0}$ of CP09 Lemma 3, errata item E-P-CP09-4 in §14.7). (C) *Fast averaging* (CP09 Lemma 4): Part 2's oscillator averaging removes the θ-dependence up to $(\varepsilon M_F)^3$ on the action polydisc of radius $\asymp (\log \frac{1}{\varepsilon})^{-(\tau_0+1)}$ (Prop. 14.6). (D) *Polar normalisation and slow averaging*: on the shell $|x_j|^2 + |y_j|^2 \approx \varepsilon$ the oscillators are put in action–angle form $(\tilde{\zeta}, \tilde{\rho})$, and a second two-scale averaging, now over all $2n$ angles with divisors α_1 (fast) and $\varepsilon\alpha_2$ (slow), makes the remainder $O(\varepsilon^{\nu_2+3/2})$ for an arbitrary fixed ν_2 (Lemma 14.7, Prop. 14.8). *This replaces CP09 Lemmas 5–7* (equilibrium shift, quadratic diagonalisation, finite Birkhoff normal form): the Birkhoff step is not needed for the Rüssmann gate because only the J-derivatives of the prepared frequency map enter the amount of non-degeneracy (Lemma 14.10), and the exponentially small slow remainder replaces the $O(|(u, v)|^{\nu_2+1})$ remainder. It also bypasses the errata A5 (polar domain) and A6 (Birkhoff formula) of the review; the price is that $\tilde{\Omega}(J_0)$ must be Diophantine, which Lemma 14.3 provides for free (deviation D-11-1, §14.7). (E) The *prepared system* $\hat{H}_\varepsilon = N_\varepsilon + P_\varepsilon$ in rescaled actions $J = J_0 + \varepsilon r$, $\tilde{\rho} = \varepsilon(2\pi + \rho')$ lives on a fixed compact $K_\varepsilon = [-\frac{1}{2}, \frac{1}{2}]^{2n}$ with $\vartheta = \frac{1}{4}$ and $\varepsilon^* = \frac{1}{2}$, so that *every chart constant of Part 11a-5 is independent of* ε *except the amount* $\beta_\varepsilon \asymp \varepsilon^{\bar{\mu}+2}$ (Prop. 14.9, Lemma 14.10); this is the repaired form of CP09 Prop. 2 (R-CP-2) and of CP09 (42)–(47) (A4). (F) Theorem 14.13 applies the KAM input (Definition 14.1) and transports the tori through the total chart to the tube U_{ζ_*}.

14.1. Conventions, external facts, and the KAM input.

Conventions. Frozen conventions of Part 0: angles of period one, characters $\chi_k = e^{2\pi i k\cdot\theta}$, max norms $|\cdot|_\infty$ on $\mathbb{R}^m, \mathbb{C}^m$ and induced operator norms. $D(c, r) = \{v \in \mathbb{C}^m : |v-c|_\infty < r\}$ (complex polydisc), $\mathbb{T}_\sigma = \{\theta \in \mathbb{C}^m : |\mathrm{Im}\,\theta|_\infty < \sigma\}$. For $V \subset \mathbb{C}^m$ open and a function u on $\mathbb{T}_\sigma \times V$, 1-periodic in the angles, the *weighted norm* is that of [N:bpN:p4:norm],

$$|u|_{V,\sigma} := \sup_{x\in V} \sum_k |\hat{u}_k(x)|\, e^{2\pi|k|_1\sigma},$$

(Fourier expansion in the angles only; the remaining variables, including oscillator coordinates, play the rôle of the parameter x); $\sup|u| \leq |u|_{V,\sigma}$ and $|u|_{V,\sigma} \leq \coth(\pi\sigma')^m \sup_{\mathbb{T}_{\sigma+\sigma'}\times V}|u|$ ([N:bpN:p4:sup-weighted]). $c, c_1, \ldots$ denote positive constants; a constant is always listed with

the data it depends on. Period one is used for *all* angles of the prepared problem. The period conversion is the canonical chart $\mathcal{T}(\theta, J, \zeta, \tilde{\rho}) = (2\pi\theta, J/2\pi, 2\pi\zeta, \tilde{\rho}/2\pi)$ of Remark 3.12(3) (a 1-chart, $dJ \wedge d\theta = d\Lambda \wedge d\lambda$): the Hamiltonian is the same function, frequencies are divided by 2π, and the frozen frequency is $w = \omega_{\mathrm{CP}}/2\pi$ (Lemma 2.11).
External facts. (each used by name below): (e1) a real analytic function on an open set of $\mathbb{R}^m$ extends holomorphically to a complex neighbourhood of every compact subset ([K:bp:p1:holomorphic-analytic]); (e2) Cauchy estimates for holomorphic functions on polydiscs, $|\partial^\alpha u| \le \alpha!\, \sup|u|/\delta^{|\alpha|}$ at points whose polydisc of radius δ lies in the domain ([N:bpN:p4:general-cauchy]); (e3) uniform continuity of a continuous function on a compact set; (e4) $\log\frac{1}{\varepsilon} \ge 1$ and $\varepsilon^a(\log\frac{1}{\varepsilon})^b \to 0$ as $\varepsilon \to 0$ for $a > 0$; (e5) the Fourier average over an angle torus of a holomorphic periodic function is holomorphic in the remaining variables ([N:bpN:p4:coefficients]).

Definition 14.1 (Quantitative KAM input, in the form in which it is used)**.** This is a *citation with exact interface*, not a hypothesis of any lemma of this part: it is [P:bpP:p11a:weak-kam-quantitative] together with [P:bpP:p11a:budget] (File 11a-5; forward references, not yet written). It is used only in the proof of Theorem 14.13. Fix $N = 2n$ and the squared convention $\rho = 2$ of Definition 15.1. A *KAM datum* is $\mathfrak{d} = (\mu, \beta, K, \vartheta, s, M, C_1, \tau_D, \varepsilon^*)$ with $\mu \in \mathbb{N}_{\ge 1}$, $\beta, \vartheta, s, M, C_1 > 0$, $K \subset \mathbb{R}^N$ compact, $\varepsilon^* \in (0, \operatorname{meas} K)$, $\tau_D > \max\{N\mu/2 - 1,\ N - 1\}$. A Hamiltonian $H = h(r) + f(\theta, r)$, $(\theta, r) \in \mathbb{T}_s \times (K + 4\vartheta) \subset \mathbb{C}^N \times \mathbb{C}^N$ (symplectic form $dr \wedge d\theta$), is $\mathfrak{d}$-*admissible* if h, f are holomorphic, f is 1-periodic in θ, both are real on real points, and, with $\omega = \nabla h$,

- (A1) $\beta_{2,\mu}(\omega, K) \ge \beta$ (Definition 15.1);
- (A2) $|D^k g_2[\omega](c, y)|_{\mathrm{op}} \le M$ for $c \in S^{N-1}$, $1 \le k \le \mu + 1$, $y \in K_\vartheta$;
- (A3) $|\omega|_\infty \le C_1$ and $|D^2 h| \le C_1$ on $K + 3\vartheta$.

Input. There are $\gamma > 0$ and $\varepsilon_{\mathrm{adm}} > 0$, depending only on $\mathfrak{d}$, such that for every $\mathfrak{d}$-admissible H with $\sup_{\mathbb{T}_s \times (K+4\vartheta)}|f| \le \varepsilon_f \le \varepsilon_{\mathrm{adm}}$:

- (K1) a compact $\mathcal{H} \subset K$ with $\operatorname{meas}\mathcal{H} \ge \operatorname{meas} K - \varepsilon^*$;
- (K2) for $b \in \mathcal{H}$ a real analytic $\mathcal{K}_b : \mathbb{R}^N \to \mathbb{R}^N \times \mathbb{R}^N$ (positions, actions) with $\mathcal{K}_b(\theta + k) = \mathcal{K}_b(\theta) + (k, 0)$ $(k \in \mathbb{Z}^N)$ — equivalently $u_b := \mathcal{K}_b - (\mathrm{id}, b)$ is $\mathbb{Z}^N$-periodic — $D\mathcal{K}_b$ injective, $\mathcal{K}_b^*(dr \wedge d\theta) = 0$, $\mathcal{K}_b(\theta) = \mathcal{K}_b(\theta')$ only if $\theta' - \theta \in \mathbb{Z}^N$, and a frequency $w_b \in \mathbb{R}^N$ that is (γ, τ_D)-Diophantine with $s \mapsto \mathcal{K}_b(\theta + s w_b)$ having derivative $X_H(\mathcal{K}_b(\theta + t w_b))$ at every t;
- (K3) $\sup_{b,\theta}|u_b(\theta)|_\infty \le \Delta$ with $\Delta = \Delta(\varepsilon_f; \mathfrak{d}) \ge 0$;
- (K4) the evaluation map $(b, \theta) \mapsto \mathcal{K}_b(\theta)$ is injective on $\mathcal{H} \times [0,1)^N$, and $\operatorname{meas}\mathcal{K}(\mathcal{H} \times [0,1)^N) \ge 2^{-2N}\operatorname{meas}\mathcal{H}$ (the set is Borel);
- (K5) *(budget)* if $(\mu, K, \vartheta, s, M, C_1, \tau_D, \varepsilon^*)$ are held fixed there are $c_*, c_D > 0$, $B_0 \ge 1$, $\kappa_D > 0$, $a_D \ge 0$ (depending only on these) with $\varepsilon_{\mathrm{adm}} \ge c_* \beta^{B_0}$ and $\Delta \le c_D\, \varepsilon_f^{\kappa_D} \beta^{-a_D}$ for $\varepsilon_f \le \varepsilon_{\mathrm{adm}}$, uniformly in $\beta \in (0, 1]$.

14.2. **Step A: the Hamiltonian in working coordinates.**
The fixed data and the patch. $n \ge 2$, $G, m_0 > 0$, $\kappa = G m_0$, $m \in (0,\infty)^n$, $a \in \mathbb{R}^n$ with $0 < a_1 < \cdots < a_n$, $\eta > 0$ (Theorem 1.30; Lean index shift as in Part 12). ζ_*, d, D are those of Definition 8.15; $\Lambda_i^* = m_i\sqrt{\kappa a_i}$, $\rho_* = \rho_*(m, \kappa, a)$ of Proposition 10.2. Let $\delta_* > 0$ be the constant of Lemma 18.1 for $\zeta = \zeta_*$ (the proof of that lemma uses $n = 2$ only through the number of planets; Lemma 14.12 restates it). Choose, by Theorem 12.22 (with the prescribed neighbourhood of a so small that $|\Lambda(a') - \Lambda^*|_\infty < \min\{\rho_*, \delta_*\}/8$), the patch a_0, r_0, K_a and put $K_\Lambda = \Lambda(K_a)$, the index $\bar{\mu} = \mu_0 \ge 1$ and amount $\bar{\beta} > 0$ of $\tilde{\varpi} = (\nu, \Omega)$ in the squared convention (Theorem 12.22(d)), depending only on the data. In the period-one working variable $J = 2\pi\Lambda$ put

$$\tilde{h}(J) = h(J/2\pi), \quad \omega(J) = \nabla\tilde{h}(J) = \tfrac{1}{2\pi}\nu(J/2\pi), \quad \tilde{\Omega}(J) = \tfrac{1}{2\pi}\Omega(J/2\pi), \quad \varpi_J = (\omega, \tilde{\Omega}) :\ 2\pi B^\Lambda \to \mathbb{R}^{2n}, \quad K_J = 2\pi K$$

with $\Omega_j = 2\sigma_j$ (Remark 11.10) and $\tilde\Omega_j < 0$.

Lemma 14.2 (Index and amount in the working variable)**.** *ϖ_J is real analytic and R-nondegenerate on $2\pi B^\Lambda$, and $\beta_{2,\bar\mu}(\varpi_J, K_J) \ge (2\pi)^{-2-\bar\mu}\bar\beta =: \beta_J > 0$.*

Proof. **Step S14.1: Constant factor and dilation.** $\varpi_J(J) = (2\pi)^{-1}\tilde\varpi(J/2\pi)$. Hence $g_2[\varpi_J](c,J) = (2\pi)^{-2} g_2[\tilde\varpi](c, J/2\pi)$ (Definition 15.1) and $D^\nu_J = (2\pi)^{-\nu} D^\nu_\Lambda$, so $\max_{\nu\le\bar\mu}|D^\nu_J g_2[\varpi_J](c,J)| \ge (2\pi)^{-2-\bar\mu}\max_{\nu\le\bar\mu}|D^\nu_\Lambda g_2[\tilde\varpi](c,\Lambda)| \ge (2\pi)^{-2-\bar\mu}\bar\beta$ for $c \in S^{2n-1}$, $J \in K_J$ (Theorem 12.22(d)). R-nondegeneracy is invariant under a positive diagonal rescaling and an affine change of the variable (Theorem 12.15(c), Remark 12.17). □

Lemma 14.3 (Base point)**.** *Put $\tau_0 = \max\{n\bar\mu, 2n\}$ and $\varepsilon_0^* = \frac12 \operatorname{meas} K_J$. There are $\gamma_0 \in (0,1]$ and $J_0 \in \operatorname{int} K_J$, depending only on the data, with $|k\cdot\varpi_J(J_0)| \ge \gamma_0|k|_1^{-\tau_0}$ for all $k \in \mathbb{Z}^{2n}\setminus\{0\}$. In particular $\omega(J_0)$ and $\tilde\Omega(J_0)$ are (γ_0,τ_0)-Diophantine in $\mathbb{Z}^n$ (Definition 1.1).*

Proof. **Step S14.2: Hypotheses of the measure estimate.** Apply Corollary 15.11 with $N = n$, $m = 2n$, $\rho = 2$, $\mu = \bar\mu$, $\omega = \varpi_J$, $\tilde\omega_k = \varpi_J$ for all k, the compact set K_J (nonempty, positive measure, $d = \operatorname{diam}_\infty K_J > 0$), $\vartheta > 0$ so small that $(K_J)_\vartheta \subset 2\pi B^\Lambda$, $\beta = \beta_J$ (Lemma 14.2) and $M = \max\{\beta_J, M_1\}$ of Lemma 15.4. (D1) is Lemma 14.2, (D2) is Lemma 15.4, (D3) is trivial because $\tilde\omega_k = \omega$. The exponent condition is $\tau_0 > m\mu/\rho - 1 = n\bar\mu - 1$.

Step S14.3: Choice of γ_0. By the Corollary $\operatorname{meas}(K_J \setminus H_\gamma) \le C_{\mathrm{Dio}}\gamma^{2/\bar\mu}$ for $\gamma \le \gamma_*(\varepsilon_0^*)$; take $\gamma_0 = \min\{1, \gamma_*(\varepsilon_0^*)\}$. Then $\operatorname{meas} H_{\gamma_0} \ge \operatorname{meas} K_J - \varepsilon_0^* > 0$. Here $H_{\gamma_0} = \{b \in K_J : |k\cdot\varpi_J(b)| \ge \gamma_0|k|_1^{-\tau_0}\ \forall k \ne 0\}$ since the map does not depend on k.

Step S14.4: Interior point. $K_J \setminus \operatorname{int} K_J$ is the boundary of a cube (the map $a \mapsto 2\pi\Lambda(a)$ acts coordinatewise and is increasing, so it maps the cube K_a to a cube), a null set. Hence $H_{\gamma_0} \cap \operatorname{int} K_J \ne \emptyset$; take J_0 in it. The restriction to $k = (k_1, 0)$ and $k = (0, k_2)$ is the last sentence. □

This replaces CP09 Lemma 3 (`p11a-russmann-check`, A7): the object is the set of *parameters* b with $\varpi_J(b) \in \mathcal{D}_{\gamma_0,\tau_0}$, not $K \cap \mathcal{D}$; the exponent of CP09 ($\tau_0 \ge d\mu_0$) is replaced by the weaker $\tau_0 > n\bar\mu - 1$ proved in File 11a-1 (deviation D5 there).

Proposition 14.4 (Holomorphic secular form near the base point)**.** *Let J_0, γ_0, τ_0 be as in Lemma 14.3. There are $r_C \in (0,1]$, $\sigma_F \in (0, \frac14]$, $r_z \in (0,1]$, $M_F \ge 1$, $L \ge 1$ and a function F, all depending only on the data (and J_0), such that, with $\mathcal{J} = D(J_0, 2r_C) \subset \mathbb{C}^n$ and $z = (x,y) \in \mathbb{C}^{2n}$ (position x, momentum y):*

- *(a) F is holomorphic on $\mathbb{T}_{2\sigma_F} \times \mathcal{J} \times B_{r_z}$ ($B_{r_z} = D(0,r_z) \subset \mathbb{C}^{2n}$), 1-periodic in θ, real on real points, and $|F| \le M_F/\coth(\pi\sigma_F)^n$ there; hence $|F|_{\mathcal{J}\times B_{r_z},\sigma_F} \le M_F$ in the sense of §14.1 with z as a parameter. Moreover $\omega, \tilde\Omega$ are holomorphic on $\mathcal{J}$ and $|D^j\varpi_J|_{\mathrm{op}} \le L$ for $0 \le j \le \bar\mu + 1$ there (so $|D\omega|, |D\tilde\Omega|, |\varpi_J| \le L$).*
- *(b) (identity) $\Psi_c := \Psi^{(n)} \circ \Phi_\varrho^{-1} \circ \mathcal{T}_1$, with $\mathcal{T}_1(\theta,J,x,y) = (2\pi\theta, J/2\pi, x, y)$ and $\varrho : D(\Lambda_0, 2r_C/2\pi) \to SO(n)$, $\Lambda_0 = J_0/2\pi$, the analytic rotation of Proposition 11.7 (diagonalising Q_h), is a symplectic 1-chart on $\mathcal{O}_c = \{(\theta,J,z) \in \mathbb{R}^{2n}\times\mathbb{R}^{2n} : J \in \mathcal{J}\cap\mathbb{R}^n,\ |z|_\infty < r_z\}$ with values in the collision-free domain, and for all real ε and $(\theta,J,z) \in \mathcal{O}_c$: $H_{\mathrm{fix}}\big(\Psi_c(\theta,J,z)\big) = \tilde h(J) + \varepsilon F(\theta,J,z)$.*
- *(c) (lattice) Ψ_c satisfies (Q1)–(Q2) of Theorem 2.9 for the lattice $\iota_k = ((k,0),0)$ shifting θ and $\sigma_k = 0$ (the secular positions x are not shifted; the secular lattice enters after the polar chart, Lemma 14.5).*
- *(d) (mean) $F_0 := \hat F_0$ (the Fourier average over θ) is holomorphic on $\mathcal{J}\times B_{r_z}$ and*

$$F_0(J,z) = C(J) + \sum_{j=1}^n \Omega_j(J)\tfrac12(x_j^2+y_j^2) + R_4(J,z), \qquad |R_4(J,z)| \le M_F\big(|z|_\infty/r_z\big)^4,$$

with $C(J) = C_0(a(J/2\pi))$, $\Omega_j(J) = 2\pi\tilde{\Omega}_j(J)$ and real analytic C. Both models would give this form (Corollary 10.14); only F_0 of the fixed-Sun model is used.

Proof. **Step S14.5: Rotation and its extension.** Q_h has simple spectrum on the ball $\mathcal{W} \ni \Lambda_0$ (Theorem 12.22(a), $J_0 \in K_J$), so σ_j, ϱ are real analytic on a ball $D_{\mathbb{R}}(\Lambda_0, t_1)$ (Proposition 11.7); extend holomorphically to a complex polydisc (e1), shrinking so that $\varrho^\top \varrho = 1$ (identity theorem), ϱ and $\partial\varrho$ are bounded by 2, and ϱ is real at real points. This fixes r_C (with $2r_C \le \pi t_1$).

Step S14.6: The map Φ_ϱ^{-1} on the complex domain. By Proposition 11.8, $\Phi_\varrho^{-1}(\tilde{\lambda}, \Lambda, \tilde{\xi}, \tilde{\eta}) = (\tilde{\lambda} + H_\varrho, \Lambda, \varrho\tilde{\xi}, \varrho\tilde{\eta})$ with $H_{\varrho,l} = \tilde{\eta}^\top(\varrho^\top \partial_{\Lambda_l}\varrho)\tilde{\xi}$, a holomorphic function with $|H_\varrho|_\infty \le c n^2 |z|_\infty^2$ for $|z|_\infty < 1$ (bilinear in z, entries bounded by 4). For $|z|_\infty < r_z$ with $r_z \le \rho_*/(4\sqrt{n})$ one has $|\varrho\tilde{\xi}|_\infty \le |\tilde{\xi}|_2 \le \sqrt{n}|\tilde{\xi}|_\infty < \rho_*$.

Step S14.7: Strip extension of $\hat{f}_{\mathrm{fix}}$. Part 8 proves the holomorphy of $\hat{f}_{\mathrm{fix}}$ in $w' = (\xi, \Lambda, \eta) \in \mathcal{N}(\rho_*)$ only for real λ (Proposition 10.2(b)). Here the angular strip is needed. By Proposition 9.8 $s_{ij}(\lambda, w')$ is holomorphic on $\{|\mathrm{Im}\,\lambda| < \frac{1}{4}\} \times \mathcal{N}(\rho_0)$ and 2π-periodic; by Proposition 10.2(a) $\mathrm{Re}\, s_{ij} \ge \frac{1}{2}(a_j - a_i)^2$ at real λ and $w' \in \mathcal{N}(\rho_*)$ (the a_i here are the ordered axes of the data, a_i^* of Part 8). By (e3) applied to $[0, 2\pi]^n \times \overline{\mathcal{N}(\rho_*/2)}$ (and the 2π-periodicity in $\mathrm{Re}\,\lambda$) there is $\sigma' \in (0, \frac{1}{4}]$ with $\mathrm{Re}\, s_{ij} \ge \frac{1}{4}(a_j - a_i)^2$ for $|\mathrm{Im}\,\lambda|_\infty \le 2\pi\sigma'$ (a strip of period-one width σ' in $\theta = \lambda/2\pi$) and $w' \in \mathcal{N}(\rho_*/2)$. On this set $\hat{f}_{\mathrm{fix}} = -G\sum_{i<j} m_i m_j s_{ij}^{-1/2}$ (principal root) is holomorphic and bounded by $c(m, G, a)$.

Step S14.8: Definition of F and (a). $F := \hat{f}_{\mathrm{fix}} \circ \Phi_\varrho^{-1} \circ \mathcal{T}_1$ is holomorphic where the composition is defined: $\theta \in \mathbb{T}_{2\sigma_F}$ with $2\sigma_F \le \sigma'/2$, the shift $|\mathrm{Im}\, H_\varrho| \le c n^2 r_z^2 < \pi\sigma'/2$ after shrinking r_z, $\mathcal{J} \subset D(J_0, 2r_C)$ so $\Lambda \in D(\Lambda_0, r_C/\pi) \subset \mathcal{N}$-range. F is 1-periodic in θ ($\hat{f}_{\mathrm{fix}}$ is 2π-periodic in λ and H_ϱ does not depend on $\tilde{\lambda}$), real at real points (ϱ, $\hat{f}$ real). Its sup is some M_∞; put $M_F = \max\{1, \coth(\pi\sigma_F)^n M_\infty\}$ and use [N:bpN:p4:sup-weighted](b) with the extra strip σ_F. L bounds ϖ_J and its derivatives up to order $\bar{\mu}+1$ on $\mathcal{J}$ (holomorphic on a neighbourhood of $\overline{\mathcal{J}}$ after shrinking r_C; Cauchy, (e2)).

Step S14.9: (b) Identity and chart. For real points the composition $\Psi^{(n)} \circ \Phi_\varrho^{-1} \circ \mathcal{T}_1$ lies in $\Psi^{(n)}(\mathcal{N}_{\mathbb{R}}(\rho_*) \times \mathbb{R}^n) \subset W$ (Proposition 10.2(a)); Theorem 9.12(c) gives $H_{\mathrm{fix}} \circ \Psi^{(n)} = h(\Lambda) + \varepsilon\hat{f}_{\mathrm{fix}}$ and Φ_ϱ preserves Λ (Proposition 11.8), $\mathcal{T}_1$ maps $J \mapsto \Lambda = J/2\pi$: $h(\Lambda) = \tilde{h}(J)$. Symplecticity: $\Psi^{(n)}$ (Theorem 9.12(a)), Φ_ϱ^{-1} (Proposition 11.8), $\mathcal{T}_1$ ($d\Lambda \wedge d\lambda = dJ \wedge d\theta$), Remark 2.10(3); the chart is real analytic because F and the maps are (Lemma 2.7).

Step S14.10: (c) Lattice. $\mathcal{T}_1(\theta + k, \cdot) = \mathcal{T}_1(\theta, \cdot) + (2\pi k, 0, 0, 0)$, Φ_ϱ^{-1} commutes with $\tilde{\lambda}$-shifts, $\Psi^{(n)}$ is invariant under $2\pi\mathbb{Z}^n$-shifts of λ with injectivity modulo them (Theorem 9.12(a)); $\mathcal{T}_1$ and Φ_ϱ are injective. This is Theorem 11.9(d) with $\mathcal{T}_1$.

Step S14.11: (d) Mean. The θ-average of F is, by (e5) and Theorem 11.9(b), $S_{\mathrm{fix}}(\Lambda, \varrho x, \varrho y) = C_0 + \sum_j \Omega_j \tilde{R}_j + O(|(x, y)|^4)$ with $\tilde{R}_j = \frac{1}{2}(x_j^2 + y_j^2)$; the bound with explicit constant follows from Cauchy on B_{r_z} after enlarging M_F: the function $R_4 = F_0 - C - \sum \Omega_j \tilde{R}_j$ is holomorphic with vanishing jet to order 3 at $z = 0$ (Corollary 10.4: even in (ξ, η), no linear term, quadratic part as stated), hence $|R_4| \le \sup|R_4|(|z|_\infty/r_z)^4$ (Schwarz lemma on polydiscs applied on the lines $z = tz_0$). □

Lemma 14.5 (Total chart and lattice compatibility)**.** *Let $\Theta^{(1)}(\zeta, \tilde{\rho}) = \sqrt{\tilde{\rho}/\pi}\,(\sin 2\pi\zeta, \cos 2\pi\zeta)$ componentwise for $\tilde{\rho}_j > 0$ (so $x_j = \sqrt{\tilde{\rho}_j/\pi}\sin 2\pi\zeta_j$, $y_j = \sqrt{\tilde{\rho}_j/\pi}\cos 2\pi\zeta_j$, $\frac{1}{2}(x_j^2 + y_j^2) = \tilde{\rho}_j/2\pi$). Then $\hat{\Theta}(\theta, J, \zeta, \tilde{\rho}) = (\theta, J, \Theta^{(1)}(\zeta, \tilde{\rho}))$ is a symplectic 1-chart for $dJ \wedge d\theta + d\tilde{\rho} \wedge d\zeta$ (the $\mathcal{T}$-conjugate of the Part 7 polar chart Θ_n), real analytic on $\{\tilde{\rho}_j > 0\}$, $\hat{\Theta}(\theta, J, \zeta + k, \tilde{\rho}) = \hat{\Theta}(\theta, J, \zeta, \tilde{\rho})$, and if $\hat{\Theta}(\cdot) = \hat{\Theta}(\cdot')$ then $\zeta' - \zeta \in \mathbb{Z}^n$ and the other coordinates agree. Consequently, for every open*

$\mathcal{O} \subset \{(\theta, J, \zeta, \tilde\rho) : \tilde\rho_j > 0\}$ with $\hat\Theta(\mathcal{O}) \subset \mathcal{O}_c$ and $\mathcal{O} + \iota_k = \mathcal{O}$ ($\iota_k = ((k,0),0)$ shifting (θ,ζ) by $k \in \mathbb{Z}^{2n}$), the chart $\Xi_0 := \Psi_c \circ \hat\Theta$ satisfies (Q1)–(Q2) of Theorem 2.9 for $d = 2n$ with $\sigma_k = 0$.

Proof. **Step S14.12: Polar chart.** Lemma 9.15(a),(b),(c) (with $\psi = 2\pi\zeta$, $R = \tilde\rho/2\pi$) and Remark 2.10(1),(3).

Step S14.13: Lattice. $\Xi_0(w + \iota_k) = \Psi_c(\hat\Theta w)$ since $\hat\Theta$ is $\mathbb{Z}^n$-periodic in ζ and $\Psi_c(\cdot + (k_\theta)) = \Psi_c(\cdot)$ (Proposition 14.4(c)), (Q1) with $\sigma_k = 0$. (Q2): $\Xi_0(w') = \Xi_0(w)$ gives $\hat\Theta w' = \hat\Theta w + \iota_{(m,0)}$ by (c) of the proposition, then $\zeta' - \zeta \in \mathbb{Z}^n$ by the polar statement, $\tilde\rho' = \tilde\rho$, $J' = J$, so $w' = w + \iota_m$. □

14.3. **Step C: fast averaging (CP09 Lemma 4).** Fix $\nu_1 := 3$. For $0 < \varepsilon < 1$ with $\varepsilon M_F < 1$ put, with $J_0, \gamma_0, \tau_0, r_C, \sigma_F, r_z, M_F, L$ of Proposition 14.4,

(5)
$$K_1 = \frac{3(\nu_1 - 1)}{\pi\sigma_F} \log\frac{1}{\varepsilon M_F}, \quad \alpha_1 = \tfrac{1}{2}\gamma_0 K_1^{-\tau_0}, \quad s_1 = \min\Big\{r_C,\ \frac{\gamma_0}{4L\,K_1^{\tau_0+1}}\Big\}, \quad d_1 = \min\{s_1\sigma_F,\ r_z^2\}.$$

(So $e^{-2\pi K_1\sigma_F/6} = (\varepsilon M_F)^{\nu_1 - 1}$, $\kappa_1 := 2\pi K_1\sigma_F \geq 6$ for small ε, and $s_1 \asymp (\log\frac{1}{\varepsilon})^{-(\tau_0+1)}$, $\alpha_1 \asymp (\log\frac{1}{\varepsilon})^{-\tau_0}$, $d_1 \asymp (\log\frac{1}{\varepsilon})^{-(\tau_0+1)}$.) Let $\mathcal{D}_1 = \mathbb{T}_{\sigma_F} \times D(J_0, s_1) \times B_{r_z}$ and $\mathcal{D}_1' = \mathbb{T}_{\sigma_F/6} \times D(J_0, s_1/2) \times B_{r_z/2}$.

Proposition 14.6 (Fast averaging with the secular pairs as spectators)**.** *There is $\varepsilon_1 \in (0,1)$ depending only on the data such that for $0 < \varepsilon \leq \varepsilon_1$ the inequality (S1): $2^{12}(1+n)K_1\,\varepsilon M_F\,\sigma_F \leq \alpha_1 d_1$ holds, and there is a holomorphic, injective, real, symplectic map $\Phi_\varepsilon^1 : \mathcal{D}_1' \to \mathcal{D}_1$ for $dJ\wedge d\theta + dx\wedge dy$ ($\Phi_\varepsilon^1 - \mathrm{id}$ $\mathbb{Z}^n$-periodic in θ) with*

$$H \circ \Phi_\varepsilon^1 = \tilde h(J) + \varepsilon G(J,z) + f_1(\theta, J, z) \quad \text{on } \mathcal{D}_1', \qquad H := \tilde h + \varepsilon F, \quad G := F_0 + \varepsilon\bar g,$$

where $G = G(J,z)$ is holomorphic and real, f_1 is real, and

$$\sup_{D(J_0,s_1/2)\times B_{r_z/2}} |\varepsilon\bar g| \leq \eta_1(\varepsilon) := \frac{400(1+n)\,\varepsilon M_F^2}{2\pi\alpha_1 d_1}, \qquad |f_1|_{D(J_0,s_1/2)\times B_{r_z/2},\sigma_F/6} \leq (\varepsilon M_F)^{\nu_1},$$

and the displacement bounds $2\pi\alpha_1\sigma_F|J - J'| \leq 3\varepsilon M_F$, $2\pi\alpha_1 s_1|\theta - \theta'| \leq 9\varepsilon M_F$, $2\pi\alpha_1 r_z|z - z'| \leq 9\varepsilon M_F$ for $(\theta', J', z') \in \mathcal{D}_1'$, $(\theta, J, z) = \Phi_\varepsilon^1(\theta', J', z')$. Note $\eta_1(\varepsilon) \leq c\,\varepsilon(\log\frac{1}{\varepsilon})^{2\tau_0+1}$ with $c = c(\mathrm{data})$ (the exponent $2\tau_0 + 1$ is that of CP09 (25), $\Omega - \hat\Omega = O(\varepsilon\log^{2\tau_0+1}\varepsilon^{-1})$).

Proof. **Step S14.14: Application of Part 2.** Proposition 3.10 with $n \to n$ (angles), $m \to n$ (oscillator pairs; (p,q) there is (y,x) here), $D = \{J_0\}$, $r = s_1$, $r_p = r_q = r_z$, $\bar s = 0$, $s = \sigma_F$, $K = K_1$, $\alpha = \alpha_1$, $h = \tilde h$ (independent of z), $f = \varepsilon F$, $E = \varepsilon M_F$ (Proposition 14.4(a)), $b = 1 + n$, $d = d_1$; $\kappa = \kappa_1 \geq 6$.

Step S14.15: (O1). For $J \in D(J_0, s_1)$ and $0 < |k|_1 \leq K_1$: $|k\cdot\omega(J)| \geq |k\cdot\omega(J_0)| - |k|_1|\omega(J) - \omega(J_0)|_\infty \geq \gamma_0 K_1^{-\tau_0} - K_1 L s_1 \geq \frac{3}{4}\gamma_0 K_1^{-\tau_0} \geq \alpha_1$ (Lemma 14.3, [K:bp:p1:ext-mvt], definition of s_1).

Step S14.16: (O2) is (S1). $K_1, \alpha_1^{-1}, d_1^{-1}$ are polynomials in $\log\frac{1}{\varepsilon}$, so the ratio $K_1\varepsilon M_F\sigma_F/(\alpha_1 d_1) \to 0$ (e4); ε_1 is the largest value with (S1) and $\kappa_1 \geq 6$ for all smaller ε.

Step S14.17: Conclusion. Proposition 3.10 gives Φ_ε^1 and $H \circ \Phi_\varepsilon^1 = \tilde h + g + f_*$, $g = g(J,z)$ independent of θ, with $\|g - \hat f_0\| \leq 400bE^2/(\bar\alpha d)$, $\bar\alpha = 2\pi\alpha_1$, $\hat f_0 = \varepsilon F_0$; put $\varepsilon^2\bar g = g - \varepsilon F_0$ and $f_1 = f_*$, $\|f_*\| \leq e^{-\kappa_1/6}E = (\varepsilon M_F)^{\nu_1 - 1}\varepsilon M_F$; the displacement bounds are those of that proposition (with $\bar\alpha s_1|\theta - \theta'|$ replaced by $\bar\alpha r|\theta - \theta'|$, $r = s_1$). The sup bound for $\varepsilon\bar g$ is [N:bpN:p4:sup-weighted](a). □

14.4. **Step D: polar normalisation and slow averaging.** CP09 now passes to normal form coordinates and normalises the quadratic and higher Taylor terms of G in z up to a finite order. Here the dependence on z is kept and removed in action–angle variables on the shell $\tilde\rho_j \approx 2\pi\varepsilon$, by averaging over the polar angles. That needs one extension of Part 2's two-scale averaging.

Lemma 14.7 (Two-scale averaging with two action radii)**.** *Let $n = n_1 + n_2$, $I^0 \in \mathbb{R}^n$, $r_1, r_2 > 0$, $r_* = \min\{r_1, r_2\}$, $V = \{I \in \mathbb{C}^n : |I_i - I_i^0| < r_1 \ (i \le n_1),\ |I_i - I_i^0| < r_2 \ (i > n_1)\}$, $s > 0$, $K > 0$ with $\kappa = 2\pi K s \ge 6$, $0 < \alpha_2 \le \alpha_1$, $\bar\alpha_j = 2\pi\alpha_j$. Let h be holomorphic on V and real on real points, $\omega = \partial_I h$, with*

(B1) $|k\cdot\omega(I)| \ge \alpha_1$ for $k_1 \ne 0$ and $\ge \alpha_2$ for $k_1 = 0$, $k \ne 0$, $I \in V$, $|k|_1 \le K$ ($k = (k_1, k_2) \in \mathbb{Z}^{n_1} \times \mathbb{Z}^{n_2}$);

(B2) f holomorphic on $\mathbb{T}_s \times V$, 1-periodic, real on real points, $E := |f|_{V,s} \le \alpha_2 r_*/(128K)$.

Then the conclusions (a)–(d) of Proposition 3.4 hold with D_r replaced by V, $\mathcal{D}_{r/2,\bar s+s/6}$ by $\mathbb{T}_{s/6} \times \frac12 V$ ($\frac12 V$: radii $r_1/2, r_2/2$), $\bar s = 0$, with $\bar\alpha_2 r$ in (d) replaced by $\bar\alpha_2 r_*$, and in addition

$$|g - \hat f_0|_{\frac12 V, s/6} \le \frac{130\,E^2}{\bar\alpha_2 r_* s}, \qquad g = g(I') \text{ real}, \quad H\circ\Psi = h + g + f_*, \quad |f_*|_{\frac12 V, s/6} \le e^{-\kappa/6}E.$$

Proof. **Step S14.18: The proof of Part 2 goes through.** Proposition 3.4 is proved from Lemma 3.2, Lemma 3.3 and N Parts 4–5. The action radius enters only through (i) the bracket and field bounds, in the form $1/(\rho\sigma)$ with ρ the loss in the action variables, and (ii) the weights $\rho/(r_j - r + \rho)$ of $|g|_*$ ([N:bpN:p6:setting]). Take the losses block by block, $\rho_{(j)} = r_j/(8N')$ in block j (N' the number of steps, so that the domains V_i are the products with radii $\frac34 r_j - 2(i-1)\rho_{(j)}$). All ratios in (ii) are then identical to those of the isotropic proof. In (i) every estimate of N Part 4 bounds $\sum_i |\partial_{\theta_i} u|\cdot\max_j|\partial_{I_j} w|$, so only $\max_j 1/\rho_{(j)} = 1/\rho_{(*)}$, $\rho_{(*)} = r_*/(8N')$, occurs (this is the same device as Lemma 3.8, with $\delta_\nu = \rho_{(*)}\sigma$); the scaled norm $\nu(a, b_1, b_2) = \max\{|a|/\sigma, |b_1|/\rho_{(1)}, |b_2|/\rho_{(2)}\}$ has balls of relative size $\le \|\phi\|/(\rho_{(*)}\sigma)$ in every block, the quantity bounded in Step S3.7. Hence the proof of Proposition 3.4 applies verbatim with r replaced by r_* wherever a product $\bar\alpha\rho\sigma$ or $\bar\alpha r s$ occurs, and with the blocks' own radii in the domains V_i. The displacements of I_j ($\le \varepsilon/(e\bar\alpha_j\sigma)$) do not involve r.

Step S14.19: The quadratic bound for g. In Step S3.5 (Case A) one has $\varepsilon_1 := \|f_1\| \le \frac87(1 + \frac2e) y_0 E$, $y_0 = 48E/(\bar\alpha_2 r_* s)$, so $\varepsilon_1 \le 1.984 \cdot 48E^2/(\bar\alpha_2 r_* s) \le 96E^2/(\bar\alpha_2 r_* s)$; Step S3.8 gives $\|g - g_0\| \le \frac43\varepsilon_1 \le 128E^2/(\bar\alpha_2 r_* s)$. In Case B, $g = \hat f_0$. (Part 2 records only the consequence $\le E/\kappa$ of this bound.) □

Source gap: Lemma 14.7 is a variant of Proposition 3.4 (two action radii), obtained by the same list of changes as Lemma 3.9; it is the statement that Part 11 needs and is not in CP10/CP09 (CP09 Lemma 4 cites [3, Prop. A.1] with the single radius s). It may be absorbed into Part 2.

For $0 < \varepsilon \le \varepsilon_1$ let $\Pi(\zeta, \tilde\rho) := \Theta^{(1)}(\zeta, \tilde\rho)$ (Lemma 14.5) on the *polar shell*

$$\mathcal{P}_\varepsilon = \{(\zeta, \tilde\rho) \in \mathbb{C}^n \times \mathbb{C}^n :\ |\operatorname{Im}\zeta|_\infty < \sigma_F/6,\ |\tilde\rho_j - 2\pi\varepsilon| < \pi\varepsilon \ (\forall j)\}.$$

Π is holomorphic on $\mathcal{P}_\varepsilon$ ($\operatorname{Re}\tilde\rho_j > \pi\varepsilon > 0$, principal square root) and $|\Pi(\zeta,\tilde\rho)|_\infty \le \sqrt{3\varepsilon}\, e^{\pi\sigma_F/3}$; hence $\Pi(\mathcal{P}_\varepsilon) \subset B_{r_z/2}$ as soon as $\sqrt{3\varepsilon}e^{\pi\sigma_F/3} \le r_z/2$, which we assume ($\varepsilon \le \varepsilon_2$). Put $\sigma_3 = \sigma_F/12$ and, for $\varepsilon \le \varepsilon_2$,

(6)

$$\hat G(J,\zeta,\tilde\rho) = G(J, \Pi(\zeta,\tilde\rho)), \quad \hat G_0 = \textstyle\int_{\mathbb{T}^n} \hat G\, d\zeta, \quad h_0(J,\tilde\rho) = \tilde h(J) + \varepsilon\hat G_0(J,\tilde\rho), \quad f_2 = \varepsilon(\hat G - \hat G_0) + f_1 \circ (\mathrm{id}, \Pi),$$

$$K_2 = \frac{3\nu_2}{\pi\sigma_3}\log\frac1\varepsilon, \quad \alpha_{(1)} = \tfrac14\gamma_0 K_2^{-\tau_0}, \quad \alpha_{(2)} = \tfrac14\varepsilon\gamma_0 K_2^{-\tau_0}, \quad s_2 = \min\Big\{\frac{s_1}{2},\ \frac{\gamma_0}{8L\,K_2^{\tau_0+1}}\Big\}, \quad r_* = \pi\varepsilon.$$

Here $\nu_2 \geq 1$ is a real number fixed in Theorem 14.13; $e^{-2\pi K_2\sigma_3/6} = \varepsilon^{\nu_2}$. The slow frequency of $\tilde\Omega$ is the second block: $\partial_{\tilde\rho} h_0 = \varepsilon(\tilde\Omega(J) + \cdots)$ with $\frac12(x_j^2+y_j^2) = \tilde\rho_j/2\pi$ and $\Omega_j/2\pi = \tilde\Omega_j$ (Proposition 14.4(d)).

Proposition 14.8 (Polar normalisation and slow averaging)**.** *Let $\nu_2 \geq 1$. There are $\varepsilon_2 \in (0,\varepsilon_1]$ and $c_2 \geq 1$ depending only on the data and ν_2 such that for $0<\varepsilon\leq\varepsilon_2$, with $V_2 = D(J_0,s_2)\times D_{\tilde\rho}$ ($D_{\tilde\rho} = \{|\tilde\rho_j - 2\pi\varepsilon| < \pi\varepsilon\}$, radii s_2 for J and $r_2 = \pi\varepsilon$ for $\tilde\rho$), $n_1 = n_2 = n$, angles (θ,ζ), $s = \sigma_3$:*

- *(a) (*the perturbation*) $|f_2|_{V_2,\sigma_3} \leq E_2 := c_2\,\varepsilon^{5/2}(\log\frac1\varepsilon)^{2\tau_0+1}$, and $\hat G_0 = C(J) + \sum_j \tilde\Omega_j(J)\tilde\rho_j + \hat Z(J,\tilde\rho)$ with $\hat Z$ holomorphic on $D(J_0,s_1/2)\times D(0,\pi r_z^2/8)$ and $|\partial_J^\alpha\partial_{\tilde\rho}\hat Z| \leq c_2(\varepsilon+\eta_1(\varepsilon))\alpha!\,(4/s_1)^{|\alpha|}$ on $D(J_0,s_1/4)\times D_{\tilde\rho}$;*
- *(b) (*(B1)–(B2) for h_0 and f_2*) (B1) holds on V_2 with $\alpha_1\to\alpha_{(1)}$, $\alpha_2\to\alpha_{(2)}$ and $K = K_2$, and (B2) holds: $E_2 \leq \alpha_{(2)}\pi\varepsilon/(128K_2)$;*
- *(c) (*averaging*) Lemma 14.7 applies to (h_0,f_2) and gives $\Psi_2 : \mathbb{T}_{\sigma_3/6}\times\frac12 V_2 \to \mathbb{T}_{\sigma_3}\times V_2$ with $(H\circ\Phi_\varepsilon^1\circ\hat\Theta)\circ\Psi_2 = h_0 + g_2 + f_3$, where $\hat\Theta = (\mathrm{id}_{\theta,J},\Pi)$, $g_2 = g_2(J,\tilde\rho)$ is real and*
$$|g_2| \leq c_2\,\varepsilon^3(\log\tfrac1\varepsilon)^{5\tau_0+2} \quad \text{on } \tfrac12 V_2, \qquad |f_3|_{\frac12 V_2,\sigma_3/6} \leq \varepsilon^{\nu_2}E_2;$$
- *(d) (*displacements*) for $(\theta,\zeta,J,\tilde\rho) = \Psi_2(\theta',\zeta',J',\tilde\rho')$: $|\tilde\rho-\tilde\rho'| \leq c_2\varepsilon^{3/2}(\log\frac1\varepsilon)^{3\tau_0+1}$, $|J-J'| \leq c_2\varepsilon^{5/2}(\log\frac1\varepsilon)^{3\tau_0+1}$, $|\theta-\theta'|, |\zeta-\zeta'| \leq c_2\varepsilon^{1/2}(\log\frac1\varepsilon)^{3\tau_0+1}$.*

Proof. **Step S14.20: (a) Size of $\hat G - \hat G_0$ and $\hat Z$.** $G = C + \sum\Omega_j\tilde R_j + R_4 + \varepsilon\bar g$ with $\tilde R_j = \tilde\rho_j/2\pi$ independent of ζ. So $\hat G - \hat G_0 = (R_4+\varepsilon\bar g)\circ\Pi - \langle(R_4+\varepsilon\bar g)\circ\Pi\rangle_\zeta$. On $\mathcal{P}_\varepsilon$, $|\Pi|_\infty^4 \leq 9\varepsilon^2 e^{4\pi\sigma_F/3}$ so $|R_4\circ\Pi| \leq 9e^{4\pi\sigma_F/3}M_F\varepsilon^2/r_z^4$; and $|\varepsilon\bar g(J,z) - \varepsilon\bar g(J,0)| \leq 2\eta_1|z|_\infty/r_z$ by Cauchy (e2) and Proposition 14.6, so the ζ-oscillation of $\varepsilon\bar g\circ\Pi$ is $\leq 4\eta_1\sqrt{3\varepsilon}e^{\pi\sigma_F/3}/r_z$. Hence $\sup|\hat G - \hat G_0| \leq c(\varepsilon^2 + \eta_1\sqrt{\varepsilon})$, and $\varepsilon\cdot\eta_1\sqrt{\varepsilon} \leq c\varepsilon^{5/2}(\log\frac1\varepsilon)^{2\tau_0+1}$ is the dominant term (the contribution $\varepsilon\cdot\varepsilon^2$ is smaller). The part $f_1\circ(\mathrm{id},\Pi)$ is bounded by $(\varepsilon M_F)^3\coth(\pi\sigma_3)^{2n}$ on the strip, by Proposition 14.6 and [N:`bpN:p4:sup-weighted`](b) (sup on $\mathbb{T}_{\sigma_F/6}$, weighted norm at $\sigma_3 = \sigma_F/12$). Passing from sup to $|\cdot|_{V_2,\sigma_3}$ costs $\coth(\pi\sigma_3)^{2n}$; the oscillator coordinates are parameters. The mean: the ζ-average of an analytic function of $z = \Pi(\zeta,\tilde\rho)$ keeps only even Taylor degrees, hence is a power series in $\tilde\rho$ convergent for $|\tilde\rho_j| < \pi r_z^2/8$ (so $|\Pi| \leq r_z/2$ after rotation, uniformly); the Cauchy estimates (e2) on $D(J_0,s_1/2)\times D(0,\pi r_z^2/8)$ bound $\partial_J^\alpha\partial_{\tilde\rho}$ of $\hat Z = \langle(R_4+\varepsilon\bar g)\circ\Pi\rangle_\zeta$ by $\sup|R_4+\varepsilon\bar g|/\ldots$; at $\tilde\rho \approx 2\pi\varepsilon$ the R_4-part is $\leq c\varepsilon$ (it is $O(\tilde\rho^2)$, so $\partial_{\tilde\rho}R_4 = O(\tilde\rho)$) and the $\bar g$-part is $\leq c\eta_1$.

Step S14.21: (b) Non-resonance on V_2. $\partial_J h_0 = \omega(J) + \varepsilon\partial_J\hat G_0$ and $\partial_{\tilde\rho}h_0 = \varepsilon(\tilde\Omega(J) + \partial_{\tilde\rho}\hat Z)$. For $k = (k_1,k_2)$, $|k|_1 \leq K_2$, $J\in D(J_0,s_2)$: $k\cdot\partial h_0 = k_1\cdot\omega(J) + k_2\cdot\varepsilon\tilde\Omega(J) + \varepsilon\,k\cdot(\partial_J\hat G_0,\partial_{\tilde\rho}\hat Z)$. By Lemma 14.3, $|k_1\cdot\omega(J_0)| \geq \gamma_0 K_2^{-\tau_0}$, $|k_2\cdot\tilde\Omega(J_0)| \geq \gamma_0K_2^{-\tau_0}$ and $|k_i\cdot(\varpi(J)-\varpi(J_0))| \leq K_2Ls_2 \leq \frac18\gamma_0K_2^{-\tau_0}$; and $\varepsilon K_2|(\partial_J\hat G_0,\partial_{\tilde\rho}\hat Z)| \leq \varepsilon K_2 c(1+M_F/s_1) \leq \frac18\gamma_0K_2^{-\tau_0}$ for $\varepsilon\leq\varepsilon_2$ (polynomial in $\log\frac1\varepsilon$ against ε; here $|\partial_J\hat G_0| \leq cM_F/s_1$ by (e2)). Case $k_1\neq0$: $|k\cdot\partial h_0| \geq \gamma_0K_2^{-\tau_0} - |k_2|\varepsilon|\tilde\Omega| - \ldots \geq \frac12\gamma_0K_2^{-\tau_0} \geq \alpha_{(1)}$ (the term $\varepsilon|k_2\cdot\tilde\Omega| \leq \varepsilon K_2L$ is absorbed as above). Case $k_1 = 0$, $k_2\neq0$: $|k\cdot\partial h_0| \geq \varepsilon\big(\gamma_0K_2^{-\tau_0} - \frac14\gamma_0K_2^{-\tau_0}\big) \geq \alpha_{(2)}$. $\alpha_{(2)} \leq \alpha_{(1)}$.

Step S14.22: (B2). $E_2 \leq c_2\varepsilon^{5/2}(\log\frac1\varepsilon)^{2\tau_0+1}$ and $\alpha_{(2)}\pi\varepsilon/(128K_2) = \frac{\pi\gamma_0}{512}\varepsilon^2K_2^{-\tau_0-1}$; the ratio of the former to the latter is $\varepsilon^{1/2}$ times a power of $\log\frac1\varepsilon$, so (B2) holds for $\varepsilon\leq\varepsilon_2$ (e4). $\kappa = 2\pi K_2\sigma_3 = 6\nu_2\log\frac1\varepsilon \geq 6$.

Step S14.23: (c),(d). Lemma 14.7 with $s=\sigma_3$, $r_* = \pi\varepsilon \leq s_2$, $\bar\alpha_2 = 2\pi\alpha_{(2)}$: $g_2 := g$, $f_3 := f_*$, $e^{-\kappa/6} = \varepsilon^{\nu_2}$; $\hat f_0 = \int f_2 = \int_{\mathbb{T}^{2n}} f_1\circ(\mathrm{id},\Pi)$ (the term $\varepsilon(\hat G - \hat G_0)$ has zero ζ-mean and does not depend on θ), of size $c\varepsilon^3$; and $|g_2| \leq |\hat f_0| + 130E_2^2/(\bar\alpha_2 r_*\sigma_3) \leq c\varepsilon^3 + c\,\varepsilon^5(\log\frac1\varepsilon)^{4\tau_0+2}/(\varepsilon\cdot\varepsilon\,K_2^{-\tau_0})$, which is the stated bound ($K_2 \asymp \log\frac1\varepsilon$). The displacements: $|\tilde\rho-\tilde\rho'| \leq 3E_2/(\bar\alpha_2\sigma_3)$, $|J-J'| \leq 3E_2/(\bar\alpha_1\sigma_3)$, $|\theta-\theta'| \leq 9E_2/(\bar\alpha_2 r_*)$, with $\bar\alpha_2 \asymp \varepsilon K_2^{-\tau_0}$. □

14.5. **Step E: the prepared system and its frequency map.** Put $\tilde\rho_0 = (2\pi\varepsilon,\dots,2\pi\varepsilon)$ and let $A_\varepsilon(\theta,r,\zeta,\rho') = (\theta,\ J_0+\varepsilon r,\ \zeta,\ \varepsilon(2\pi\mathbf{1}+\rho'))$ (here $\mathbf{1}=(1,\dots,1)$). A_ε is an ε-chart ($A_\varepsilon^*(dJ\wedge d\theta + d\tilde\rho\wedge d\zeta) = \varepsilon(dr\wedge d\theta + d\rho'\wedge d\zeta)$) commuting with all angle shifts. It maps the polydisc $\{|r|_\infty<2,\ |\rho'|_\infty<\frac{\pi}{2}\}$ into $\frac12 V_2$ (radii $s_2/2$ for J, $\pi\varepsilon/2$ for $\tilde\rho$ around $\tilde\rho_0$) as soon as $2\varepsilon<s_2/2$. Define, with $\mathcal{E}=\{(\theta,r,\zeta,\rho') : |\mathrm{Im}(\theta,\zeta)|<\sigma_3/6,\ |r|_\infty<2,\ |\rho'|_\infty<\frac{\pi}{2}\}$,

$$N_\varepsilon(r,\rho') = \tfrac{1}{\varepsilon}\big(h_0+g_2\big)(J_0+\varepsilon r,\ \varepsilon(2\pi\mathbf{1}+\rho')),\qquad P_\varepsilon = \tfrac{1}{\varepsilon}f_3\circ A_\varepsilon,\qquad \hat H_\varepsilon = \tfrac{1}{\varepsilon}H_2\circ A_\varepsilon = N_\varepsilon + P_\varepsilon,$$

where $H_2 := H\circ\Phi^1_\varepsilon\circ\hat\Theta\circ\Psi_2$ is the Hamiltonian of Proposition 14.8(c). The compact set and the data of Definition 14.1 are

$$K_\varepsilon=[-\tfrac12,\tfrac12]^{2n},\quad \vartheta=\tfrac14,\quad \varepsilon^*=\tfrac12,\quad s=s_*:=\sigma_3/6=\sigma_F/72,\quad N=2n,\quad \mu=\bar\mu,\quad \tau_D=\max\{n\bar\mu,2n\},$$

so $\mathrm{meas}\,K_\varepsilon=1$, $d=1$, $K_\varepsilon+4\vartheta\subset D(0,\frac32)^{2n}\subset\{|r|<2,\ |\rho'|<\frac{\pi}{2}\}$. Frequency map of N_ε: $\hat\omega_\varepsilon = \nabla_{(r,\rho')}N_\varepsilon = (\partial_J h,\ \partial_{\tilde\rho}h)(J_0+\varepsilon r,\varepsilon(2\pi\mathbf{1}+\rho'))$ with $h=h_0+g_2$. It is convenient to write $\hat\omega_\varepsilon=(\hat\omega^{(1)},\varepsilon\hat\omega^{(2)})$ with

$$\hat\omega^{(1)} = \omega+\varepsilon\partial_J\hat G_0+\partial_J g_2,\qquad \hat\omega^{(2)} = \tilde\Omega+\partial_{\tilde\rho}\hat Z+\varepsilon^{-1}\partial_{\tilde\rho}g_2\qquad\text{(arguments as above)}.$$

Proposition 14.9 (The prepared system)**.** *For $0<\varepsilon\le\varepsilon_2$ the functions $N_\varepsilon,P_\varepsilon$ are holomorphic on $\mathcal{E}$, 1-periodic in the angles, real on real points, N_ε depends on (r,ρ') only, and $H_2\circ A_\varepsilon=\varepsilon\hat H_\varepsilon$. Moreover, with $c_3=c_3(data,\nu_2)$:*

- *(a) $\sup_{\mathcal{E}}|P_\varepsilon|\le c_3\,\varepsilon^{\nu_2+3/2}(\log\frac1\varepsilon)^{2\tau_0+1}=:\varepsilon_f(\varepsilon)$;*
- *(b) $|\hat\omega_\varepsilon|_\infty\le C_1:=2L+1$ and $|D^2N_\varepsilon|_\infty\le C_1$ on $K_\varepsilon+3\vartheta$;*
- *(c) for $|\alpha|\le\bar\mu+1$ with α a derivative in J only: $\sup_{J\in D(J_0,s_1/4)}|\partial_J^\alpha(\hat\omega^{(i)}-\varpi^{(i)})|\le\theta_\varepsilon := c_3\,\varepsilon(\log\frac1\varepsilon)^{5\tau_0+2}$ ($i=1,2$, derivative norm as in Definition 15.1 up to a factor $c(n,\bar\mu)$; $\varpi^{(1)}=\omega$, $\varpi^{(2)}=\tilde\Omega$), uniformly in $\tilde\rho\in\varepsilon(2\pi\mathbf{1}+D(0,\frac{\pi}{2}))$;*
- *(d) for the full derivatives of order $k\le\bar\mu+1$ in (r,ρ') on $K_\varepsilon+\vartheta$: $|D^kg_2[\hat\omega_\varepsilon](c,y)|_{\mathrm{op}}\le M_*$ ($c\in S^{2n-1}$), with $M_*=2^{\bar\mu+1}(2L+1)^2$ independent of ε.*

Proof. **Step S14.24: Holomorphy, reality, periodicity.** Φ^1_ε, $\hat\Theta$, Ψ_2 are real holomorphic with the lattice properties, $H_2\circ A_\varepsilon = H\circ(\Phi^1\circ\hat\Theta\circ\Psi_2\circ A_\varepsilon)$; the domain inclusions are those of Propositions 14.6, 14.8: Ψ_2 is defined on $\mathbb{T}_{\sigma_3/6}\times\frac12V_2\supset A_\varepsilon(\mathcal{E})$ and maps into $\mathbb{T}_{\sigma_3}\times V_2$, on which $\hat\Theta$ is defined and takes values in $\mathcal{D}'_1$ ($|\Pi|\le r_z/2$, $J\in D(J_0,s_1)$, strip $\sigma_3\le\sigma_F/6$). N_ε is $\frac1\varepsilon(h_0+g_2)\circ A_\varepsilon$, a function of (r,ρ'); $P_\varepsilon=\varepsilon^{-1}f_3\circ A_\varepsilon$.

Step S14.25: (a). $\sup|f_3|\le\|f_3\|\le\varepsilon^{\nu_2}E_2$ ([N:bpN:p4:sup-weighted](a)), divide by ε.

Step S14.26: (b). Write $h=h_0+g_2$. $\partial_rN_\varepsilon=\partial_Jh=\omega+\varepsilon\partial_J\hat G_0+\partial_Jg_2$ and $\partial_{\rho'}N_\varepsilon=\partial_{\tilde\rho}h=\varepsilon(\tilde\Omega+\partial_{\tilde\rho}\hat Z)+\partial_{\tilde\rho}g_2$; by (e2) with radii $s_1/4$ (in J) and $\pi\varepsilon/4$ resp. $\pi r_z^2/16$ (in $\tilde\rho$), Proposition 14.8(a),(c) and $|\hat G_0|\le c$ these are $\le L+o(1)$ and $o(1)$. The Hessian: $\partial_r^2N_\varepsilon=\varepsilon\partial_J^2h$, $\partial_r\partial_{\rho'}N_\varepsilon=\varepsilon\partial_J\partial_{\tilde\rho}h$, $\partial_{\rho'}^2N_\varepsilon=\varepsilon\partial_{\tilde\rho}^2h$, with $\partial_{\tilde\rho}^2h=\varepsilon\partial^2\hat Z+\partial_{\tilde\rho}^2g_2$, $\partial_{\tilde\rho}^2g_2=O(\varepsilon^3\varepsilon^{-2}\log^c)$; all entries are $o(1)$ as $\varepsilon\to0$ (polynomial in $\log\frac1\varepsilon$ against a positive power of ε). So $|\hat\omega_\varepsilon|_\infty\le C_1=2L+1$ and $|D^2N_\varepsilon|\le C_1$ for $\varepsilon\le\varepsilon_3$.

Step S14.27: (c). $\hat\omega^{(1)}-\omega=\varepsilon\partial_J\hat G_0+\partial_Jg_2$: ∂_J^α of the first term is $\le\varepsilon\alpha!\,M_F(2/s_1)^{|\alpha|+1}$ by (e2) (the sup of $\hat G_0$ is $\le cM_F$ on $D(J_0,s_1/2)$) and of the second is $\le|\alpha|!\sup|g_2|(4/s_2)^{|\alpha|+1}\le c\varepsilon^3\log^c$. $\hat\omega^{(2)}-\tilde\Omega=\partial_{\tilde\rho}\hat Z+\varepsilon^{-1}\partial_{\tilde\rho}g_2$: the first is $\le c_2(\varepsilon+\eta_1)$ by Proposition 14.8(a) (all J-derivatives included), the second is bounded by Cauchy in $\tilde\rho$ at radius $\pi\varepsilon/4$ and in J at radius $s_2/4$: $\varepsilon^{-1}\cdot\frac{4}{\pi\varepsilon}\sup|g_2|(4/s_2)^{|\alpha|}\alpha!\le c\varepsilon\log^c\frac1\varepsilon$. *Only J-derivatives enter.*

Step S14.28: (d). By Lemma 15.4, $|D^kg_2[\hat\omega_\varepsilon]|_{\mathrm{op}}\le2^kM_1^2$ with $M_1=\|\hat\omega_\varepsilon\|^{\bar\mu+1}_{\mathrm{op},K_\varepsilon+\vartheta}$. The main terms of $\hat\omega_\varepsilon$ are $(\omega(J_0+\varepsilon r),\ \varepsilon\tilde\Omega(J_0+\varepsilon r))$: independent of ρ', and each r-derivative of order $j\ge1$

carries a factor ε^j, so their norms in $C^{\bar\mu+1}$ are $\le L$. The remaining terms are $\varepsilon\partial_J\hat G_0+\partial_J g_2$ and $\varepsilon\partial_{\tilde\rho}\hat Z+\partial_{\tilde\rho}g_2$: a derivative of order $j=j_J+j_\rho$ in (r,ρ') is ε^j times a derivative in $(J,\tilde\rho)$, bounded by (e2) with radii $s_1/4$, resp. $\pi r_z^2/16$ for $\hat Z$ and $\pi\varepsilon/4$ for g_2 (where $\sup|g_2|\le c\varepsilon^3\log^c$ against the factor $\varepsilon^{j_\rho}\varepsilon^{-j_\rho-1}$); every such term is $o(1)$. Hence $M_1\le 2L+1$ for $\varepsilon\le\varepsilon_3$ and $M_*:=2^{\bar\mu+1}(2L+1)^2$ works. □

Lemma 14.10 (Directional stability and target scaling; R-CP-2)**.** *Let $\varepsilon\le\varepsilon_3\le\varepsilon_2$ be such that $2^{\bar\mu}\theta_\varepsilon(2L+\theta_\varepsilon)\le\frac12\beta_J$ with θ_ε of Proposition 14.9(c) and β_J of Lemma 14.2. Then, in the sense of Definition 15.1 with $\rho=2$, $m=N=2n$,*

$$\beta_{2,\bar\mu}(\hat\omega_\varepsilon,K_\varepsilon)\ \ge\ \beta_\varepsilon:=\tfrac12\,\varepsilon^{\bar\mu+2}\beta_J>0.$$

Proof. **Step S14.29: Normalisation of the second block.** Put $\hat\omega_\varepsilon=\operatorname{diag}(\mathbf{1}_n,\varepsilon\mathbf{1}_n)\,\check\omega_\varepsilon$, $\check\omega_\varepsilon=(\hat\omega^{(1)},\hat\omega^{(2)})$ (Prop. 14.9(c) notation). For $D=\operatorname{diag}(1,\varepsilon)$ and $c\in S^{2n-1}$: $\langle c,D\check\omega\rangle=\langle Dc,\check\omega\rangle=|Dc|\langle\hat c,\check\omega\rangle$ with $\hat c=Dc/|Dc|\in S^{2n-1}$, $\varepsilon\le|Dc|\le1$. So $g_2[D\check\omega](c,\cdot)=|Dc|^2g_2[\check\omega](\hat c,\cdot)$ and $\beta_{2,\mu}(D\check\omega,K)\ge\varepsilon^2\beta_{2,\mu}(\check\omega,K)$.

Step S14.30: Restriction to r-directions. Fix $\rho_1'\in[-\frac12,\frac12]^n$, $\tilde\rho_1=\varepsilon(2\pi\mathbf{1}+\rho_1')$, and on $V=D_{\mathbb R}(J_0,s_1/4)\subset\mathbb R^n$ consider $\psi_\varepsilon(J)=\check\omega_\varepsilon$ evaluated at $(J,\tilde\rho_1)$ and $\psi(J)=\varpi_J(J)$, both maps $V\to\mathbb R^{2n}$ in $C^{\bar\mu}$. By Proposition 14.9(c), $\theta:=\|\psi_\varepsilon-\psi\|^{\bar\mu}_{\mathrm{dg},A}\le\theta_\varepsilon$ ($A\supset$ the compact $\{J_0+\varepsilon r:|r|\le\frac12\}$ for $\varepsilon\le s_1/2$), and $M_0:=\|\psi\|^{\bar\mu}_{\mathrm{dg},A}\le c(n,\bar\mu)L=:L_1$ (Proposition 14.4(a)); we read L in the hypothesis of the lemma as L_1. Lemma 15.3 (with $N=n$, $m=2n$, $\rho=2$) gives, since $\beta_{2,\bar\mu}(\psi,A)\ge\beta_J$ (Lemma 14.2, $A\subset K_J$ for small ε because $J_0\in\operatorname{int}K_J$) and $2^{\bar\mu}\theta(2M_0+\theta)\le\frac12\beta_J$: $\beta_{2,\bar\mu}(\psi_\varepsilon,A)\ge\frac12\beta_J$, where the n-dimensional amount is the minimum over (c,J) of $\max_\nu\max_{a\in S^{n-1}}|D_J^\nu g_2[\psi_\varepsilon](c,J)(a^\nu)|$.

Step S14.31: Rescaling of the actions. The (r,ρ')-derivative of $g_2[\check\omega_\varepsilon$ in rescaled variables$](c,\cdot)$ in a direction $(a,0)$, $a\in S^{n-1}$, at (r,ρ_1') equals $\varepsilon^\nu D_J^\nu g_2[\psi_\varepsilon](c,J)(a^\nu)$ with $J=J_0+\varepsilon r$ (chain rule for the dilation, Lemma 15.13); the full tensor norm (Definition 15.1, maximum over all unit directions $a\in S^{2n-1}$) is at least its value on directions $(a,0)$ (note $|(a,0)|_2=1$). Therefore $\beta_{2,\bar\mu}(\check\omega_\varepsilon,K_\varepsilon)\ge\min\{1,\varepsilon^{\bar\mu}\}\frac12\beta_J=\varepsilon^{\bar\mu}\frac12\beta_J$ (as $\max_\nu\varepsilon^\nu x_\nu\ge\varepsilon^{\bar\mu}\max_\nu x_\nu$, $\varepsilon<1$); the dependence on ρ' is only through the frozen parameter, and the minimum over ρ_1' is trivial because the bound is uniform. Combine with Step S14.29: $\beta_{2,\bar\mu}(\hat\omega_\varepsilon,K_\varepsilon)\ge\varepsilon^2\cdot\varepsilon^{\bar\mu}\frac12\beta_J=\beta_\varepsilon$. Euclidean norms on spheres as in Definition 15.1. □

Remark 14.11 (R-CP-2 and what is not claimed). CP09 p. 17 asserts $\beta_\varepsilon\ge\varepsilon^{\bar\mu}\bar\beta/2$ from $\Psi_\varepsilon=\Psi_0+O(\varepsilon)$ in C^0 of the *rescaled* variable; that is insufficient (the amount is $\asymp\varepsilon^{\bar\mu}$, a C^0-perturbation of size ε is larger). Here the perturbation is measured in $C^{\bar\mu}$ in the *unscaled* action J (Proposition 14.9(c)), where it is $O(\varepsilon\log^{5\tau_0+2}\frac1\varepsilon)$ against the unscaled amount β_J; the rescaling then multiplies both by the same ε^ν (Lemma 14.10). The ρ'-dependence of N_ε is not controlled in $C^{\bar\mu}$ at all (only through Proposition 14.9(d), a bound), and is not needed: the amount in the full sense is bounded below by the amount in the r-directions. Also corrected: CP09 (17), (28) and Prop. 2 carry a spurious factor $\frac12$ in front of $\hat\Omega\cdot(\rho_0+\varepsilon\rho)$ (with $z_j=\sqrt{2(\rho_{0j}+\rho_j)}e^{-i\zeta_j}$ one has $\frac12(u_j^2+v_j^2)=\rho_{0j}+\rho_j$), which changes only the constant in the amount.

14.6. Step F: the local wrapper and the transport of the tori.

Lemma 14.12 (Forward tube for n planets)**.** *Let $a_i>0$, $\Lambda_i^*=m_i\sqrt{\kappa a_i}$, $\zeta>0$. There is $\delta_\zeta\in(0,\frac14\min_i\Lambda_i^*]$, depending only on (G,m_0,m,a,ζ), such that every $w=((\lambda,\xi),(\Lambda,\eta))\in\mathbb R^{4n}$ with $|\Lambda_i-\Lambda_i^*|\le\delta_\zeta$ and $\Gamma_i(w)\le\delta_\zeta$ for all i lies in $\mathcal O_n$ and satisfies $|\Psi^{(n)}(w)-\operatorname{circ}(\lambda/2\pi)|_\infty<\zeta$; in particular $\Psi^{(n)}(w)\in U_\zeta$.*

Proof. Steps S18.1–S18.3 of Lemma 18.1 with 2 replaced by n in the index ranges and $\mathbb R^8$ by $\mathbb R^{4n}$; nothing else changes (compact set Q_0, uniform continuity (e3) of $\Psi^{(n)}$, $\Psi^{(n)}(w_0)=\operatorname{circ}(\lambda/2\pi)$ by Lemma 9.14). □

Theorem 14.13 (Local wrapper for P-FIXED: tori in the tube)**.** *Let $n \geq 2$, $G, m_0 > 0$, $m_i > 0$, $0 < a_1 < \cdots < a_n$, $\eta > 0$, and ζ_*, d, D as in Definition 8.15. Then there is $\varepsilon_0 > 0$ such that for every $\varepsilon \in (0, \varepsilon_0)$ there is a Borel set $S_\varepsilon \subset \mathrm{Ph}(2n)$ with $0 < \operatorname{meas} S_\varepsilon < \infty$, $S_\varepsilon \subset U_{\zeta_*}$, and every $z \in S_\varepsilon$ lies on some $\mathcal{K} : \mathbb{R}^{2n} \to \mathrm{Ph}(2n)$ with $\mathcal{K}(\mathbb{R}^{2n}) \subset S_\varepsilon$ and* $\mathrm{IsPhysTorus}(H_{\mathrm{fix}}, w, \mathcal{K})$*, for $H_{\mathrm{fix}} =$ `fixedHamiltonian` $G\ m_0\ m\ \varepsilon$. (Part 13 applies Proposition 8.22 to obtain* $\mathrm{FixKol}(G, m_0, m, a, \varepsilon, \eta, d, D, S_\varepsilon)$ *and the frozen P-FIXED.) The constants depend only on the data; the order of choice is: data $\to$ patch, $(\bar\mu, \bar\beta)$, $J_0, \gamma_0 \to r_C, \sigma_F, r_z, M_F, L \to B_0, \kappa_D, a_D$ (Definition 14.1(K5), for the fixed data $\mathfrak{d}_*$ of Remark 14.14) $\to \nu_2 \to \varepsilon_0 \to \varepsilon$.*

Remark 14.14 (The data of the KAM input). For $\varepsilon \leq \varepsilon_3$ the Hamiltonian $\hat H_\varepsilon = N_\varepsilon + P_\varepsilon$ on $\mathbb{T}_{s_*} \times (K_\varepsilon + 4\vartheta)$ is $\mathfrak{d}_\varepsilon$-admissible with $\mathfrak{d}_\varepsilon = (\bar\mu, \beta_\varepsilon, K_\varepsilon, \frac14, s_*, M_*, C_1, \tau_D, \frac12)$ (periodicity, reality and holomorphy: Proposition 14.9; (A1): Lemma 14.10; (A2): Proposition 14.9(d); (A3): (b)). All entries but $\beta_\varepsilon = \frac12 \beta_J\, \varepsilon^{\bar\mu+2}$ are independent of ε ($K_\varepsilon = [-\frac12, \frac12]^{2n}$, $d_\varepsilon = \operatorname{diam}_\infty K_\varepsilon = 1$, $\vartheta_\varepsilon = \frac14$); $\mathfrak{d}_* = (\bar\mu, \cdot, K_\varepsilon, \frac14, s_*, M_*, C_1, \tau_D, \frac12)$ denotes the common part to which (K5) is applied; $\tau_D > \max\{n\bar\mu - 1, 2n-1\}$.

Proof of Theorem 14.13. **Step S14.32: Choice of ν_2 and ε_0.** Let $B_0, \kappa_D, a_D, c_*, c_D$ be the constants of (K5) for $\mathfrak{d}_*$. Choose $\nu_2 := \max\big\{(\bar\mu+2)B_0 + 2,\ \frac{(\bar\mu+2)a_D+1}{\kappa_D} + 2\big\}$. By Proposition 14.9(a) $\varepsilon_f(\varepsilon) \leq c_3 \varepsilon^{\nu_2 + 3/2} \log^c \leq \varepsilon^{\nu_2}$ for small ε, while $\varepsilon_{\mathrm{adm}}(\mathfrak{d}_\varepsilon) \geq c_*(\frac12\beta_J)^{B_0} \varepsilon^{(\bar\mu+2)B_0} \geq \varepsilon^{\nu_2 - 2} \cdot c'$; and $\Delta \leq c_D \varepsilon_f^{\kappa_D} (\frac12 \beta_J \varepsilon^{\bar\mu+2})^{-a_D} \leq c'' \varepsilon^{\kappa_D \nu_2 - (\bar\mu+2)a_D} \leq c'' \varepsilon \leq \frac12$. Hence there is $\varepsilon_0 \leq \varepsilon_3$ (depending only on the data, ν_2 and the constants above) with: $\varepsilon_f(\varepsilon) \leq \varepsilon_{\mathrm{adm}}(\mathfrak{d}_\varepsilon)$, $\Delta(\varepsilon_f; \mathfrak{d}_\varepsilon) \leq \frac12$ and all smallness conditions of Propositions 14.6, 14.8, 14.9, 14.10, for every $\varepsilon \in (0, \varepsilon_0)$. This is the quantifier order demanded by R-CP-1 and A4: the admissible threshold is a *lower* bound $c\varepsilon^B$, and ν_2 is fixed *before* ε.

Step S14.33: The KAM input. Apply Definition 14.1 to $\hat H_\varepsilon$ (Remark 14.14) and $\varepsilon_f = \varepsilon_f(\varepsilon)$: compact $\mathcal{H}_\varepsilon \subset K_\varepsilon$, $\operatorname{meas} \mathcal{H}_\varepsilon \geq \frac12$, maps $\mathcal{K}_b = (\theta + u_b^\theta,\ \zeta + u_b^\zeta;\ r_b, \rho_b')$ for $b = (b_r, b_{\rho'}) \in \mathcal{H}_\varepsilon$ with defect $|u_b| \leq \frac12$, so the real points (r, ρ') of the tori satisfy $|r|_\infty, |\rho'|_\infty \leq 1 < \frac{\pi}{2}$ (the positions θ, ζ are unrestricted by periodicity).

Step S14.34: Pull-back through the chain of charts. Let $\Xi_\varepsilon := \Psi_c \circ \Phi_\varepsilon^1 \circ \hat\Theta \circ \Psi_2 \circ A_\varepsilon$, defined on $\mathcal{R}_\varepsilon = \{(\theta, r, \zeta, \rho') \in \mathbb{R}^{4n} : |r|_\infty < 2,\ |\rho'|_\infty < \frac{\pi}{2}\}$ (each factor is defined and real analytic there by the domain chains of Propositions 14.6, 14.8, 14.4, Lemma 14.5), a c-chart with $c = \varepsilon$ (product of the conformal factors, all other factors 1). Its lattice properties (Q1)–(Q2) with $\sigma_k = 0$, $\iota_k = ((k,0),0)$ shifting (θ, ζ) by $k \in \mathbb{Z}^{2n}$, hold: A_ε (angles unchanged), Ψ_2 (equivariant by Lemma 14.7(a), injective), $\hat\Theta$ and Ψ_c (Lemma 14.5), Φ_ε^1 ($\Phi^1 - \mathrm{id}$ $\mathbb{Z}^n$-periodic in θ, injective). By construction $H_{\mathrm{fix}} \circ \Xi_\varepsilon = H_2 \circ A_\varepsilon = \varepsilon \hat H_\varepsilon$ on $\mathcal{R}_\varepsilon$ (Proposition 14.4(b), Propositions 14.6, 14.8(c)), i.e. $\hat H_\varepsilon = H_{\mathrm{fix}}^{\Xi_\varepsilon}$ in the notation of Definition 2.3.

Step S14.35: Physical tori. For $b \in \mathcal{H}_\varepsilon$ the map $\theta \mapsto \mathcal{K}_b(\theta) \in \mathcal{R}_\varepsilon$ (Step S14.33) satisfies (L1)–(L6) of Definition 2.8 with $G = \hat H_\varepsilon$, $O = \mathcal{R}_\varepsilon$, $w = w_b$ and $\iota_k = ((k,0),0)$ (here the $2n$ positions are (θ, ζ) and $\mathcal{K}_b(\theta + k) = \mathcal{K}_b(\theta) + (k, 0)$: (K2)); differentiability of $\hat H_\varepsilon$ is holomorphy. w_b is (γ, τ_D)-Diophantine with $\tau_D \geq 2n > d - 1$ for $d = 2n$. Theorem 2.9(b) gives $\mathrm{IsPhysTorus}(H_{\mathrm{fix}}, w_b, \Xi_\varepsilon \circ \mathcal{K}_b)$, and $(\Xi_\varepsilon \circ \mathcal{K}_b)(\mathbb{R}^{2n}) = \Xi_\varepsilon(\mathcal{K}_b(\mathbb{R}^{2n}))$.

Step S14.36: The set S_ε, measure. $A := \mathcal{K}(\mathcal{H}_\varepsilon \times \mathbb{R}^{2n})$ is Borel, $\mathbb{Z}^{2n}$-invariant, and its unit cell $\mathcal{K}(\mathcal{H}_\varepsilon \times [0,1)^{2n})$ has meas $\geq 2^{-4n} \operatorname{meas} \mathcal{H}_\varepsilon > 0$ (K4); put $S_\varepsilon = \Xi_\varepsilon(A)$. By Lemma 8.19(b) (a c-chart with (Q1)–(Q2), $c = \varepsilon$; $A \subset \mathcal{R}_\varepsilon$ open invariant) S_ε is Borel and $\operatorname{meas} S_\varepsilon = \varepsilon^{2n} \operatorname{meas}(A \cap Q) > 0$.

Step S14.37: Inclusion in the tube. Let $(\theta, r, \zeta, \rho') \in A$: $|r|_\infty \leq 1$, $|\rho'|_\infty \leq 1$ (Step S14.33). Then The real action coordinates are $J = J_0 + \varepsilon r$ and $\tilde\rho_j = \varepsilon(2\pi + \rho_j') \in \varepsilon[2\pi - 1, 2\pi + 1]$. Apply the displacements: Ψ_2 changes $\tilde\rho$ by $\leq c_2 \varepsilon^{3/2} \log^{\tau_0+1}$ and J by $\leq c_2 \varepsilon^{5/2} \log^{2\tau_0+1}$ (Proposition 14.8(d)),

so the image has $\tilde\rho_j \in \varepsilon[2\pi-2, 2\pi+2]$, $|J-J_0|_\infty \le 2\varepsilon$ for small ε; Π gives $|z|_\infty^2 \le \tilde\rho_j/\pi \le (2\pi+2)\varepsilon/\pi < 3\varepsilon$; Φ_ε^1 moves J by $\le 3\varepsilon M_F/(2\pi\alpha_1\sigma_F) \le c\varepsilon\log^{\tau_0}\frac{1}{\varepsilon}$ and z by $\le 9\varepsilon M_F/(2\pi\alpha_1 r_z) \le c\varepsilon\log^{\tau_0}\frac{1}{\varepsilon}$ (Proposition 14.6); finally $(\xi,\eta) = \varrho(x,y)$ with $\sum_i(\xi_i^2+\eta_i^2) = \sum_j(x_j^2+y_j^2) < 7n\varepsilon$ (ϱ orthogonal, Proposition 11.8), so $\Gamma_i = \frac12(\xi_i^2+\eta_i^2) \le 4n\varepsilon$, and $|\Lambda-\Lambda^*|_\infty \le |\Lambda-\Lambda_0|+|\Lambda_0-\Lambda^*| < \delta_*/8 + c\varepsilon\log^{\tau_0}\frac{1}{\varepsilon}$ (patch choice). For ε small the hypotheses of Lemma 14.12 (with $\zeta=\zeta_*$, $\delta_{\zeta_*}=\delta_*$) hold, so $\Xi_\varepsilon(\theta,r,\zeta,\rho') = \Psi^{(n)}(\dots) \in U_{\zeta_*}$, i.e. $S_\varepsilon \subset U_{\zeta_*} \subset U_\eta$ (Lemma 8.16(a)). The decreasing choice of ε_0 so that these inequalities hold is added to Step S14.32. □

Remark 14.15 (Displacement in the polar and Birkhoff charts). In the polar chart the oscillator actions $\tilde\rho_j$ of every torus lie in $\varepsilon[2\pi-2,2\pi+2]$ (Step S14.37), in particular they are positive, uniformly over the family; this is the quantitative form of CP09 Theorem 5 ($\bar w$ of norm 2ε). There is no Birkhoff chart in this route (it is replaced by Step D); the rotation chart Φ_ϱ preserves Λ and the Euclidean norms of (ξ,η), which is what the tube needs (Step S14.37).

14.7. **Sources, deviations, errata, forward references.**

Sources read. CP09 (key CP09 in `docs/sources.json`; arXiv v1 numbering, 26 pages) pp. 5–8 (Def. 1, Lemma 1, Def. 2, Thm. 3, Thm. 4 and Remark 4), pp. 9–17 (§2.3.1: Prop. 1, Lemmas 2–7, (11)–(30); §2.3.2: Prop. 2), pp. 17–21 (§2.3.3: Lemma 8, Remark 5, (34)–(47)); `reviews/p11-route-study.md`, `reviews/p11a-russmann-check.md` (R-CP-1, R-CP-2, A2–A7); Parts 1, 2, 6–10, 11a-1, 11a-2 of this blueprint; `PLAN-11a.md` §4; OVERNIGHT.md entries of 21:15 (Tracks I, N, P).

Deviations. **D-11-1** Steps C–D replace CP09 Lemmas 4–7: fast averaging as in CP09 Lemma 4 (derived from Part 2), then *no* equilibrium shift, quadratic diagonalisation or finite Birkhoff form; instead a polar normalisation on the shell $\tilde\rho \approx 2\pi\varepsilon$ and a second two-scale averaging over all angles (needs $\tilde\Omega(J_0)$ Diophantine, Lemma 14.3). **D-11-2** $\nu_1 = 3$ is fixed; the arbitrary order is carried by ν_2 (exponential smallness in the slow averaging), not by ν_1. **D-11-3** The rescaled data are $\vartheta = \frac14$, $\varepsilon^* = \frac12$, $K_\varepsilon = [-\frac12,\frac12]^{2n}$, independent of ε, instead of CP09's $\vartheta = \varepsilon/16$, $\varepsilon^* = \varepsilon^{p+1}$ (their ρ-domain is $D^p(0,\varepsilon)$ in the variable already divided by ε). **D-11-4** Max norms and period one throughout; the Euclidean balls of CP09 (11), (40) are replaced by polydiscs (no $\sqrt{n}$ is hidden: Steps S14.6, S14.37).

Suspected errata (for the ledger). E-P-CP09-1: (17), (28), Prop. 2: factor $\frac12$ in front of $\hat\Omega\cdot(\rho_0+\varepsilon\rho)$ is inconsistent with (8), (24), (27) (see Remark 14.11). E-P-CP09-2 (= A4): p. 21 after (47), "$\epsilon_0 \le \bar c\epsilon^{\nu_0}$" gives only an upper bound for the admissible size; the claim needs a lower bound (Step S14.32). E-P-CP09-3 (= A5): (27) with $\rho_0 \in (\mathbb{R}^+)^p$, $|\rho_0| = 2\epsilon$ and polydisc $D^p(0,|\rho_0|/2)$ may contain $\rho_j = -\rho_{0j}$ where $\sqrt{\rho_{0j}+\rho_j}$ vanishes; here all components of $\tilde\rho_0$ are equal and the polydisc has radius $\frac12\tilde\rho_{0j}$. E-P-CP09-4 (= A7): Lemma 3 for $K\cap D_{\gamma_0,\tau_0}$ (should be $K\cap\omega^{-1}(D)$), proof uses "Theorem 2" for Lemma 2. E-P-CP09-5 (= A6): Lemma 7 states $(\theta,r) = (\tilde x,\tilde y)$ and puts $\langle\hat\Omega, I\rangle$ twice in $Q^\star$; not used here. E-P-CP09-6 (= R-CP-2): see Remark 14.11. E-P-CP09-7: Prop. 1 (p. 10) states $\Phi_\epsilon : \mathbb{T}^d\times B^d(0,s/5)\times\cdots\times B^p(0,\epsilon)\to\cdots$ and N_ϵ in (17) in the same variables; (28) divides by ϵ after A_ϵ, so the domains in (16)–(17) and in the Rüssmann application (40) are not those of (28); the definition of Step E fixes them.

Doubts about frozen statements. None. The frozen P-FIXED asserts only existence; no secular non-degeneracy is a premise (Theorem 12.22 is proved in Part 10).

Forward references and what is received. *Part 11a-5 must supply* (Definition 14.1): the exact statement of [P:`bpP:p11a:weak-kam-quantitative`] for $N=2n$, $\rho=2$, angles of period one, *primary* (equivariant) $\mathcal{K}_b$, with the data $(\mu,\beta,K,\vartheta,s,M,C_1,\tau_D,\varepsilon^*)$ and admissibility (A1)–(A3), the domain $\mathbb{T}_s\times(K+4\vartheta)$ (if 11a-5 needs only $K+2\vartheta$ nothing changes), and the polynomial budget (K5): $\varepsilon_{\rm adm} \ge c_*\beta^{B_0}$ and $\Delta \le c_D\varepsilon_f^{\kappa_D}\beta^{-a_D}$ for fixed $(K,\vartheta,s,M,C_1,\tau_D,\varepsilon^*)$, $\beta\to0$. Part 11a-5's `reference-constants` is *not* used (Lemma 14.10 proves the amount directly). *Part 11 supplies to 11a-5 (and 13)*: for each small ε the data $\mathfrak{d}_\varepsilon$ of Remark 14.14, with $K_\varepsilon = [-\frac12,\frac12]^{2n}$, $\vartheta=\frac14$, $s=\sigma_F/72$, $\varepsilon^*=\frac12$, M_*, C_1 independent

of ε and $\beta_\varepsilon = \frac{1}{2}\beta_J\varepsilon^{\bar\mu+2}$ (the only ε-dependent entry), the size $\varepsilon_f(\varepsilon) = O(\varepsilon^{\nu_2+3/2}\log^{2\tau_0+1})$, and, to Part 13, Theorem 14.13. *Other forward references*: Part 13 (assembly via Proposition 8.22); Part 11a-4 for (K2)–(K4) ([P:`bpP:p11a:limit-torus`], [P:`bpP:p11a:displacement`], [P:`bpP:p11a:evaluation-bilipschitz`]). Track-N citations: [N:`bpN:p4:norm`], [N:`bpN:p4:sup-weighted`], [N:`bpN:p4:coefficients`], [N:`bpN:p4:general-cauchy`]; track K: [K:`bp:p1:ext-mvt`], [K:`bp:p1:holomorphic-analytic`].
Label index. `kam-input`, `norm-dict`, `select`, `holo-chart`, `total-chart`, `avg-fast`, `aniso-avg`, `avg-slow`, `prepared`, `amount`, `r-cp-2`, `forward-tube`, `wrapper`, `data-star`, `displacement`.

15. Part 11a-1: finite-order nondegeneracy and the measure of the small-divisor set

This is the first of the five files of Part 11a (the maximal-case analytic weak-nondegeneracy KAM theorem; split plan in `PLAN-11a.md`). It proves, from scratch and in the frozen conventions, the *measure* half of Rüssmann's argument: a finite-order index and amount of nondegeneracy on a compact patch of parameters, its stability under C^μ perturbations, a scalar sublevel estimate, and the measure of the set of parameters whose (perturbed) frequency vector violates a power-law small-divisor condition. No KAM iteration occurs here; Files 11a-2 to 11a-5 consume the results below as black boxes.

- `rank-compact` (Definition 15.1, Lemma 15.2, Lemma 15.3, Lemma 15.4): index and amount of an R-nondegenerate map on a compact set, in the two scalar conventions $\langle c,\omega\rangle$ and $\langle c,\omega\rangle^2$; stability of the amount under a C^μ-small change of the frequency map (Rus01 §§18.2–18.3, pp. 185–186; CP09 Lemma 1, p. 5).
- `scalar-sublevel` (Lemmas 15.5, 15.6, 15.7, Theorem 15.8): the sublevel estimate $\mathrm{meas}\{|\tilde g|\le\varepsilon\}\le C(\varepsilon/\beta)^{1/\mu}$ for a function with a finite-order amount β, stable under C^μ perturbation (Rus01 Thm 17.1 and Lemmas 17.2, 17.3, pp. 180–183). The one-variable estimate is proved by an elementary divided-difference argument; Bakhtin's theorem, which Rus01 imports, is *not* used.
- `diophantine-measure` (Theorem 15.10, Corollary 15.11, Lemma 15.13): the measure loss of the set $\{b\in K:\ |k\cdot\tilde\omega_k(b)|\ge\Psi(k)\ \forall k\ne0\}$ for *k-dependent* perturbed frequency maps $\tilde\omega_k$ and for the power divisor $\Psi(k)=\gamma|k|_1^{-\tau}$, $\tau>m\mu/\rho-1$, with every constant explicit in the index, the amount, the chart constants (d,ϑ,M) (Rus01 §18.5, pp. 189–193, with $p=q=0$).

15.1. **Conventions and external facts.**

Dimensions and norms. $N\ge1$ is the dimension of the parameter space (in the application the action space, $N=m$ in the maximal case), $m\ge1$ the dimension of the frequency vector and of the lattice $\mathbb{Z}^m$ of Fourier modes. On $\mathbb{R}^N$ we write $|y|_2$ for the Euclidean and $|y|_\infty$ for the max norm, $|y|_\infty\le|y|_2\le\sqrt{N}|y|_\infty$; likewise on $\mathbb{R}^m$. *The frozen max norm is used for diameters and boxes* (d below), and for the lattice $|k|_1$ in the Diophantine condition of Definition 1.1; *the Euclidean norm is used for balls, directions and derivative tensors* (an orthogonal change of variables is needed in Lemma 15.7). Every conversion carries an explicit factor $\sqrt{N}$ or $\sqrt{m}$ (Rus01 uses $|\cdot|_2$ throughout, p. 123 and §16.2, p. 178). For $K\subset\mathbb{R}^N$ nonempty compact,

$$d=d_\infty(K)=\sup_{y,y'\in K}|y-y'|_\infty,\qquad K_\vartheta:=\{y\in\mathbb{R}^N:\ \mathrm{dist}_2(y,K)\le\vartheta\}\ \ (\vartheta>0),$$

a compact set (Rus01's real thickening $K\oplus\vartheta$, defined in §10.1.7 and used on p. 180, in the Euclidean norm; $d_\infty\le d_2\le\sqrt{N}d_\infty$).

Derivative tensors. For $V \subset \mathbb{R}^N$ open, $g \in C^k(V, \mathbb{R}^r)$ ($r \geq 1$, Euclidean target), $0 \leq \nu \leq k$, $y \in V$, the ν-th derivative $D^\nu g(y)$ is a symmetric ν-linear map $(\mathbb{R}^N)^\nu \to \mathbb{R}^r$. Put $D^\nu g(y)(a^\nu) = D^\nu g(y)(a, \dots, a)$ and

$$|D^\nu g(y)|_{\mathrm{dg}} = \max_{|a|_2=1} |D^\nu g(y)(a^\nu)|_2, \qquad |D^\nu g(y)|_{\mathrm{op}} = \max_{|a_1|_2=\cdots=|a_\nu|_2=1} |D^\nu g(y)(a_1, \dots, a_\nu)|_2,$$

($\nu = 0$: both equal $|g(y)|_2$). Then $|\cdot|_{\mathrm{dg}} \leq |\cdot|_{\mathrm{op}}$. *The two norms are used for different purposes:* the *diagonal* norm $|\cdot|_{\mathrm{dg}}$ measures nondegeneracy and the perturbation (it is CP09's and Part 10's $|\partial^\nu g(y)|$, p. 5; Rus01 §16.2.1, p. 178); the *operator* norm $|\cdot|_{\mathrm{op}}$ measures the size of the top derivative in the Taylor step of Theorem 15.8. Rus01 has one norm (the diagonal one) and pays the classical polarization constant $\sigma_\nu = \nu^\nu/\nu!$ (p. 178, p. 182); using $|\cdot|_{\mathrm{op}}$ for the top derivative avoids polarization altogether (and a computable bound for $|D^k g|_{\mathrm{op}}$ is Lemma 15.4). For $A \subset V$ and $0 \leq \mu \leq k$,

$$\|g\|^\mu_{\mathrm{dg},A} = \max_{0\leq\nu\leq\mu} \sup_{y\in A} |D^\nu g(y)|_{\mathrm{dg}}, \qquad \|g\|^\mu_{\mathrm{op},A} = \max_{0\leq\nu\leq\mu} \sup_{y\in A} |D^\nu g(y)|_{\mathrm{op}}.$$

By polarization $|D^\nu g|_{\mathrm{op}} \leq \frac{\nu^\nu}{\nu!} |D^\nu g|_{\mathrm{dg}}$ (classical; not needed below).

Lean totality. Every division below is by a positive number (β, ε, $|k|_2$ with $k \neq 0$, ϑ, s); real powers $x^{1/\mu}$ are applied to $x \geq 0$ and $|k|_1^{-\tau}$ to $|k|_1 \geq 1$ (as in Remark 1.27: the power in Definition 1.1 is the usual one for $k \neq 0$). meas is Lebesgue (outer) measure on $\mathbb{R}^N$; the sublevel sets on compact domains and the good sets H are closed (hence measurable); differences such as $K \setminus H$ and countable unions are Borel, and where measurability is not needed the statements concern outer measure.

External facts (standard; the Mathlib names are indicative).

(F1) *Rolle and Lagrange interpolation.* If $u \in C^q([a,b])$ has $q+1$ distinct zeros, then $u^{(q)}$ has a zero in (a,b) (apply Rolle's theorem q times; `exists_deriv_eq_zero`). For distinct nodes $t_0, \dots, t_q$ and values y_i the unique polynomial of degree $\leq q$ with $P(t_i) = y_i$ is $P = \sum_i y_i \prod_{j\neq i} \frac{t-t_j}{t_i-t_j}$ (`Lagrange.interpolate`); its coefficient of t^q is $\sum_i y_i / \prod_{j\neq i}(t_i - t_j)$ and $P^{(q)} = q!$ times it.

(F2) *Lebesgue measure.* Countable subadditivity and monotonicity; $t \mapsto \mathrm{meas}(E \cap [a,t])$ is continuous and nondecreasing for measurable $E \subset [a,b]$ (Lebesgue measure has no atoms); a linear isometry Q of $(\mathbb{R}^N, |\cdot|_2)$ (an orthogonal matrix on $\mathrm{Fin}\, N \to \mathbb{R}$) satisfies $\mathrm{meas}(Q(E)) = \mathrm{meas}\, E$ ($|\det Q| = 1$; `MeasureTheory.Measure.addHaar_image_linearMap`); Tonelli on $\mathbb{R}^{N-1} \times \mathbb{R}$, [K:`bp:p1:ext-measure`].

(F3) *Mean value inequality*, [K:`bp:p1:ext-mvt`], and the chain rule for $t \mapsto g(x+ta)$ ($g \in C^q$): $\frac{d^q}{dt^q} g(x+ta) = D^q g(x+ta)(a^q)$.

(F4) *Orthonormal completion.* Every unit vector $a \in \mathbb{R}^N$ is the last column of an orthogonal matrix (for the Euclidean inner product: `Orthonormal.exists_orthonormalBasis_extension`; directly on $\mathrm{Fin}\, N \to \mathbb{R}$: $Q = I$ if $a = e_N$, otherwise the Householder matrix $Q = I - 2vv^\top/(v^\top v)$, $v = e_N - a$, which is orthogonal, $|\det Q| = 1$, $Qe_N = a$; the ambient norm stays the max norm and Euclidean lengths are used explicitly).

(F5) The lattice sum $\sum_{k\in\mathbb{Z}^m\setminus\{0\}} |k|_1^{-\tau'-1} \leq S(m,\tau') = 2 \cdot 3^{m-1} \frac{\tau'+2-m}{\tau'+1-m}$ for $\tau' > m-1$, [K:`bp:p4:lattice-sum`] (with [K:`bp:p4:lattice-count`]).

(F6) *Leibniz formula* for ℓ-linear forms: for $f, h \in C^k(V, \mathbb{R})$ and $a_1, \dots, a_k$, $D^k(fh)(y)(a_1, \dots, a_k) = \sum_{S\subset\{1..k\}} D^{|S|} f(y)(a_S)\, D^{k-|S|} h(y)(a_{S^c})$; for equal arguments it is $\sum_j \binom{k}{j} D^j f(a^j) D^{k-j} h(a^{k-j})$ (also Part 10, (F5)).

15.2. **Index and amount of nondegeneracy; stability.**

Definition 15.1 (Scalar families, index and amount)**.** Fix $N, m \geq 1$, an open set $V \subset \mathbb{R}^N$, a regularity $L \in \mathbb{N} \cup \{\infty, \omega\}$, $\omega \in C^L(V, \mathbb{R}^m)$, and $\rho \in \{1, 2\}$. All orders j below are $\leq L$ (for

unbounded-order statements take ω real analytic or C^∞). For $c \in S^{m-1} = \{c \in \mathbb{R}^m : |c|_2 = 1\}$ put

$$g_\rho[\omega](c,y) := \langle c, \omega(y)\rangle^\rho \qquad (y \in V).$$

For a nonempty compact $K \subset V$ and an integer $0 \le \mu \le L$ set

$$\beta_{\rho,\mu}(\omega,K) := \min_{c\in S^{m-1},\, y\in K} \max_{0\le\nu\le\mu} |D^\nu g_\rho[\omega](c,y)|_{\mathrm{dg}} \quad \ge 0.$$

(A minimum of a continuous function on a nonempty compact set: $(c,y,a) \mapsto |D^\nu g_\rho(c,y)(a^\nu)|$ is continuous, hence so is its maximum over the compact sphere $a \in S^{N-1}$ and then over $\nu \le \mu$, Rus01 18.3.1, p. 186.) $\mu \mapsto \beta_{\rho,\mu}$ is nondecreasing; $\beta_{\rho,0}$ is the natural minimum of $|g_\rho|$ (not Rus01's artificial convention $\beta(f,0,K) := 0$, irrelevant for the positive-index definition). When some $\beta_{\rho,\mu}(\omega,K) > 0$ with $1 \le \mu \le L$, the *index* $\mu_0 = \mu_0(\rho,\omega,K)$ is the least positive integer with $\beta_{\rho,\mu_0} > 0$ and the *amount* is $\beta_\rho(\omega,K) = \beta_{\rho,\mu_0}$; for $\mu \ge \mu_0$, $\beta_{\rho,\mu} \ge \beta_\rho(\omega,K)$.
Source: CP09 Def. 2, p. 6, and Part 10 Definition 12.19 for $\rho = 2$ (the same object with $\mathcal{C} = S^{m-1}$); Rus01 §18.3.1, p. 186, with $\beta(f,\mu,K)$ for $f = \langle c,\omega\rangle$ ($\rho = 1$) and $\beta(f,0,K) := 0$. The three numbers (Rus01's product family $\langle c,\omega\rangle\langle \hat c,\omega\rangle$, the squared family $\rho = 2$, and the unsquared family $\rho = 1$) are *different* in general; the theorems below hold for each of $\rho = 1, 2$ with its own index and amount (deviation D4). Rus01 uses the product family $\langle c,\omega\rangle\langle \hat c,\omega\rangle$ with independent $c, \hat c$ for the squared divisors (p. 187); CP09's displayed data use $\langle c,\omega\rangle^2$.

Lemma 15.2 (Finite type on a compact set: $\rho = 1, 2$)**.** *Let $B \subset \mathbb{R}^N$ be open and connected, $\omega : B \to \mathbb{R}^m$ real analytic and R-nondegenerate (Definition 12.11), $K \subset B$ nonempty and compact. Then for $\rho \in \{1,2\}$ there are $\mu_0 \in \mathbb{N}$, $\mu_0 \ge 1$, and $\beta > 0$ such that $\max_{0\le\nu\le\mu_0} |D^\nu g_\rho[\omega](c,y)|_{\mathrm{dg}} \ge \beta$ for all $c \in S^{m-1}$, $y \in K$; i.e. $\mu_0(\rho,\omega,K)$ and $\beta_\rho(\omega,K) > 0$ exist, and $\beta_{\rho,\mu}(\omega,K) \ge \beta_\rho(\omega,K)$ for every $\mu \ge \mu_0$.*
Source: Rus01 Lemma 18.2 and §18.3.1, pp. 185–186; CP09 Lemma 1, p. 5. Proved in Part 10, Corollary 12.20 ("and for $g = \langle c,\omega\rangle$ without the square").

Proof. **Step S15.1: Reduction to Part 10.** S^{m-1} is compact and nonempty ($m \ge 1$) and $K \ne \emptyset$. For $\rho = 1, 2$, $g_\rho[\omega](c,\cdot)$ is real analytic on B and $\not\equiv 0$ for each $c \in S^{m-1}$ by R-nondegeneracy (a nonzero real analytic function on a connected open set; its square is not identically zero either), and every y-derivative is jointly continuous in (c,y) (polynomial in c). Lemma 12.18 (finite type on a compact set) gives μ_0, β. The witness order may be 0 (e.g. $m = 1$, $\omega \equiv 1$); replace it by $\max\{1,\mu_0\}$ (monotonicity of $\beta_{\rho,\mu}$ in μ; ω is analytic so every order is available). The index is the least positive μ with $\beta_{\rho,\mu} > 0$, at most $\max\{1,\mu_0\}$. □

Lemma 15.3 (Finite-order stability of the amount)**.** *Let $V \subset \mathbb{R}^N$ be open, $\omega, \tilde\omega \in C^\mu(V,\mathbb{R}^m)$, $\mu \ge 0$, $A \subset V$ nonempty, and*

$$M_0 := \|\omega\|^\mu_{\mathrm{dg},A}, \qquad \theta := \|\tilde\omega - \omega\|^\mu_{\mathrm{dg},A}.$$

For $c \in S^{m-1}$ put $e_c = g_\rho[\tilde\omega](c,\cdot) - g_\rho[\omega](c,\cdot)$. Then for $0 \le \nu \le \mu$ and $y \in A$,

$$|D^\nu e_c(y)|_{\mathrm{dg}} \le \theta \quad (\rho = 1), \qquad |D^\nu e_c(y)|_{\mathrm{dg}} \le 2^\nu\theta(2M_0 + \theta) \quad (\rho = 2).$$

Write $\eta_\rho(\theta) = \theta$ for $\rho = 1$ and $\eta_2(\theta) = 2^\mu\theta(2M_0 + \theta)$. Consequently, if $K \subset A$ is compact, $\beta_{\rho,\mu}(\tilde\omega,K) \ge \beta_{\rho,\mu}(\omega,K) - \eta_\rho(\theta)$, and if $\beta_{\rho,\mu}(\omega,K) \ge \beta > 0$ and $\eta_\rho(\theta) \le \beta/2$, then $\beta_{\rho,\mu}(\tilde\omega,K) \ge \beta/2$ and $|D^\nu e_c(y)|_{\mathrm{dg}} \le \beta/2$ on A for all c, $\nu \le \mu$ (hypothesis (S3) of Theorem 15.8). No analyticity of $\tilde\omega$ and no nondegeneracy of $\tilde\omega$ is assumed. The premise $\beta_{\rho,\mu}(\omega,K) > 0$ cannot be dropped (Part 10, Lemma 12.21, final example).
Source: Ours; Part 10 Lemma 12.21 for $\rho = 2$ and $\tilde\omega$ analytic (its proof uses only C^μ and is repeated here for C^μ and for both ρ); the finite-order stability row `p11a:rank-compact` of `reviews/p11a-russmann-check.md`.

Proof. Fix c, $y \in A$, a unit a, and put $f = \langle c, \omega\rangle$, $\tilde f = \langle c, \tilde\omega\rangle$. By Cauchy–Schwarz ($|c|_2 = 1$), $|D^k f(y)(a^k)| \le |D^k\omega(y)(a^k)|_2 \le M_0$ and $|D^k(\tilde f - f)(y)(a^k)| \le \theta$ for $k \le \mu$.

Step S15.2: $\rho = 1$. $D^\nu e_c(y)(a^\nu) = D^\nu(\tilde f - f)(y)(a^\nu)$, of modulus $\le \theta$; maximise over a.

Step S15.3: $\rho = 2$. $e_c = (\tilde f - f)(\tilde f + f)$ and $|D^k(\tilde f + f)(y)(a^k)| \le 2M_0 + \theta$ ($\tilde f + f = 2f + (\tilde f - f)$). By the Leibniz formula (F6) with equal arguments, $|D^\nu e_c(y)(a^\nu)| \le \sum_k \binom{\nu}{k}\theta(2M_0+\theta) = 2^\nu\theta(2M_0+\theta)$.

Step S15.4: Amount. For each $(c, y) \in S^{m-1} \times K$ and $\nu \le \mu$, $|D^\nu g_\rho[\tilde\omega](c,y)|_{\mathrm{dg}} \ge |D^\nu g_\rho[\omega](c,y)|_{\mathrm{dg}} - |D^\nu e_c(y)|_{\mathrm{dg}} \ge |D^\nu g_\rho[\omega](c,y)|_{\mathrm{dg}} - \eta_\rho(\theta)$ (triangle inequality for the seminorm $|\cdot|_{\mathrm{dg}}$). Maximise over ν, minimise over (c, y) (maxima of two families that differ termwise by at most ϵ differ by at most ϵ). □

Lemma 15.4 (A computable bound for the top derivative)**.** *Let $\omega \in C^{\mu+1}(V, \mathbb{R}^m)$, $A \subset V$, $M_1 := \|\omega\|^{\mu+1}_{\mathrm{op},A}$. Then for every $c \in S^{m-1}$, $1 \le k \le \mu+1$ and $y \in A$: $|D^k g_1[\omega](c,y)|_{\mathrm{op}} \le M_1$ and $|D^k g_2[\omega](c,y)|_{\mathrm{op}} \le 2^k M_1^2 \le 2^{\mu+1} M_1^2$.*

Proof. $\rho = 1$: $D^k\langle c, \omega\rangle(a_1..a_k) = \langle c, D^k\omega(a_1..a_k)\rangle$, Cauchy–Schwarz. $\rho = 2$: apply (F6) to $f = h = \langle c, \omega\rangle$; there are 2^k subsets S, and each term is bounded by $M_1 \cdot M_1$ (orders $\le k \le \mu+1$, including order 0: $|f| \le |\omega|_2 \le M_1$). □

15.3. **The scalar sublevel estimate.** *Choice.* The route study offers two proofs of the one-variable estimate: Fej04r Théorème 50 (Pyartli, `sources/incoming-P/fejoz04_herman_arnold_revised.*`, §5.2) and the Bakhtin-based Lemma 17.2 of Rus01 (p. 180, which cites [2] and proves only the reduction to it). Théorème 50 is stated for a C^∞ *curve* $\nu : I \to \mathbb{R}^p$ that is (a,b)-"gauche", with an unspecified constant $C_{\tau,p}$ and for γ "assez petit"; it is not stated for a perturbed function of finitely many derivatives and does not display the dependence on (μ, β). We therefore *prove* the one-variable estimate directly (Lemma 15.6); the proof is a few lines, uses nothing beyond Rolle's theorem, and has the same constant as Rus01 (17.4) for $q = 1, 2$ (for $q = 3$: 6 against $4 \cdot 3^{1/3} \approx 5.77$). Neither Bakhtin's theorem nor Théorème 50 is used.

Lemma 15.5 (Divided differences)**.** *Let $q \ge 1$, $a < b$, $f \in C^q([a,b], \mathbb{R})$ and $a \le t_0 < t_1 < \cdots < t_q \le b$. Then there is $\xi \in (t_0, t_q)$ with*

$$\sum_{i=0}^{q} \frac{f(t_i)}{\prod_{j\ne i}(t_i - t_j)} = \frac{f^{(q)}(\xi)}{q!}.$$

Proof. **Step S15.5: Interpolation.** Let P be the Lagrange interpolation polynomial of degree $\le q$ with $P(t_i) = f(t_i)$ ((F1)); its t^q-coefficient is $c = \sum_i f(t_i)/\prod_{j\ne i}(t_i - t_j)$, so $P^{(q)} \equiv q!\, c$.

Step S15.6: Rolle. $u = f - P \in C^q([t_0, t_q])$ has $q+1$ distinct zeros t_i; by Rolle (F1) there is $\xi \in (t_0, t_q)$ with $u^{(q)}(\xi) = 0$, i.e. $f^{(q)}(\xi) = q!\, c$. □

Lemma 15.6 (One-variable sublevel estimate; Rus01 Lemma 17.2)**.** *Let $q \ge 1$ and $\beta > 0$, $a < b$, and $f \in C^q([a,b], \mathbb{R})$ with $|f^{(q)}(t)| \ge \beta$ for all $t \in [a,b]$. Then for every $\varepsilon > 0$*

$$\mathrm{meas}\{t \in [a,b] :\ |f(t)| \le \varepsilon\} \ \le\ 2q\Big(\frac{\varepsilon}{\beta}\Big)^{1/q}.$$

Source: Rus01 Lemma 17.2, p. 180: "$\mathrm{meas}_1\{t \in [a,b] \mid |f(t)| \le \varepsilon\} \le 4\,(q!\,\varepsilon/2\beta)^{1/q}$*" (17.4). The printed proof constructs a C^q extension of f to $\mathbb{R}$ (its Taylor polynomial at a plus the integral of the constantly extended q-th derivative) and then* cites *Bakhtin [2] (an extremal property of Chebyshev polynomials) for (17.4). The proof below is ours and avoids the Bakhtin dependency (deviation D1); it does not correct the extension step, and no erratum is claimed.*

Proof. Let $E = \{t \in [a,b] : |f(t)| \le \varepsilon\}$ (closed, f continuous), $L = \operatorname{meas} E$. If $L = 0$ there is nothing to prove; let $L > 0$.

Step S15.7: Distribution function. $\phi(t) = \operatorname{meas}(E \cap [a,t])$ is continuous and nondecreasing from $\phi(a) = 0$ to $\phi(b) = L$ ((F2)).

Step S15.8: Equidistributed nodes in E. Put $t_0 = \min E$ (nonempty compact) and, for $1 \le i \le q$, $t_i = \min\{t \in [a,b] : \phi(t) \ge iL/q\}$ (nonempty closed set since $\phi(b) = L$). Then $\phi(t_0) = 0$ (as $E \cap [a,t_0) = \emptyset$ and points have measure 0) and $\phi(t_i) = iL/q$ for $i \ge 1$ (continuity; $\phi(t) < iL/q$ for $t < t_i$; $t_i > a$ since $\phi(a) = 0 < iL/q$). Moreover $t_i \in E$ for $i \ge 1$: otherwise, E being closed, $E \cap (t', t_i] = \emptyset$ for some $t' < t_i$, so $\phi(t') = \phi(t_i) = iL/q$, contradicting minimality. For $0 \le i < j \le q$,

$$(j-i)\tfrac{L}{q} = \phi(t_j) - \phi(t_i) = \operatorname{meas}(E \cap (t_i, t_j]) \le t_j - t_i,$$

so $t_0 < \cdots < t_q$ and $t_j - t_i \ge (j-i)L/q$.

Step S15.9: Product of gaps. For each i, $\prod_{j\ne i}|t_i - t_j| \ge \prod_{j<i}(i-j)\frac{L}{q}\prod_{j>i}(j-i)\frac{L}{q} = i!\,(q-i)!\,(L/q)^q$.

Step S15.10: Divided difference. By Lemma 15.5, $\beta/q! \le |f^{(q)}(\xi)|/q! = \big|\sum_i f(t_i)/\prod_{j\ne i}(t_i - t_j)\big| \le \varepsilon \sum_i \frac{1}{i!(q-i)!}(q/L)^q = \varepsilon\frac{2^q}{q!}(q/L)^q$, using $|f(t_i)| \le \varepsilon$ ($t_i \in E$) and $\sum_i \frac{1}{i!(q-i)!} = \frac{2^q}{q!}$. Hence $L^q \le (2q)^q\varepsilon/\beta$. □

Lemma 15.7 (Sublevel set in a ball; Rus01 Lemma 17.3)**.** *Let $V \subset \mathbb{R}^N$ be open, $q \ge 1$, $g \in C^q(V,\mathbb{R})$, $U = \{y : |y - y_0|_2 \le r\} \subset V$ a closed ball ($r > 0$), $a \in \mathbb{R}^N$ with $|a|_2 = 1$, $\beta > 0$ and $|D^q g(y)(a^q)| \ge \beta$ for all $y \in U$. Then for every $\varepsilon > 0$*

$$\operatorname{meas}\{y \in U : |g(y)| \le \varepsilon\} \;\le\; 2q\,(2r)^{N-1}\Big(\frac{\varepsilon}{\beta}\Big)^{1/q}.$$

Source: Rus01 Lemma 17.3, p. 180–181 (17.6): the bound there is $(2\pi e/n)^{n/2}\,(q!\varepsilon/2\beta)^{1/q}\,r^{n-1}$ after the Gamma-function estimate (17.9) (Whittaker–Watson, p. 253). Here the projection of the ball is bounded by the circumscribed cube of side $2r$, which removes the Stirling estimate (deviation D2).

Proof. **Step S15.11: Rotate.** Choose an orthogonal Q with last column a ((F4)); the map $(x',t) \mapsto y_0 + Q(x',t)$ preserves Lebesgue measure ((F2)) and maps $\{(x',t) : |x'|_2^2 + t^2 \le r^2\}$ onto U. For $N = 1$ there is no x' and $Q = \pm 1$.

Step S15.12: Sections. Fix $x' \in \mathbb{R}^{N-1}$ with $|x'|_2 \le r$; the section is $t \in I_{x'} = [-\sqrt{r^2 - |x'|^2}, \sqrt{r^2 - |x'|^2}]$ and $f_{x'}(t) = g(y_0 + Q(x',0) + ta)$ (the base point is $y_0 + Q(x',0)$) is C^q on $I_{x'}$ with $f_{x'}^{(q)}(t) = D^q g(\cdot)(a^q)$ ((F3)), $|f_{x'}^{(q)}| \ge \beta$. Lemma 15.6 gives section measure $\le 2q(\varepsilon/\beta)^{1/q}$ (zero if $I_{x'}$ is a point).

Step S15.13: Tonelli. The set $\{|x'|_2 \le r\} \subset [-r,r]^{N-1}$ has measure $\le (2r)^{N-1}$; integrate the section bound ((F2)). □

Theorem 15.8 (Sublevel estimate for a C^μ-perturbed function of finite type; Rus01 Thm 17.1)**.** *Let $N, \mu \ge 1$ be integers, $K \subset \mathbb{R}^N$ nonempty compact with $d = d_\infty(K) > 0$, $\vartheta > 0$, $V \subset \mathbb{R}^N$ open with $K_\vartheta \subset V$, $g \in C^{\mu+1}(V,\mathbb{R})$, $\tilde g \in C^\mu(V,\mathbb{R})$, and $\beta > 0$, $M \ge \beta$ such that*

(S1) (finite type) for every $y \in K$ there is $\nu \in \{0,\dots,\mu\}$ with $|D^\nu g(y)|_{\mathrm{dg}} \ge \beta$;
(S2) (size) $|D^k g(y)|_{\mathrm{op}} \le M$ for $1 \le k \le \mu+1$, $y \in K_\vartheta$;
(S3) (closeness) $|D^\nu(\tilde g - g)(y)|_{\mathrm{dg}} \le \beta/2$ for $0 \le \nu \le \mu$, $y \in K_\vartheta$.

Then for every $0 < \varepsilon < \beta/4$,

$$\operatorname{meas}\{y \in K :\ |\tilde g(y)| \le \varepsilon\} \;\le\; C_{\mathrm{sub}}\Big(\frac{\varepsilon}{\beta}\Big)^{1/\mu}, \qquad C_{\mathrm{sub}} = C_{\mathrm{sub}}(N,\mu,d,\vartheta,M/\beta) := 2^{N+2}\mu\,(\sqrt{N}d)^{N-1}\frac{M}{\beta}\Big(1+\big(4+\vartheta^{-1}\big)$$

The constant depends only on N, μ, the max-diameter d of K, the thickening ϑ and the ratio M/β of the size of the top derivatives to the amount.

Source: Rus01 Thm 17.1, p. 180 (17.2), proof §17.4, pp. 181–183, where the constant is $\mathcal{B}\, d^{n-1}(n^{-1/2} + 2d + \vartheta^{-1}d)(\dots)$ with $\mathcal{B} = 3(2\pi e)^{n/2}(\mu_0+1)^{\mu_0+2}[(\mu_0+1)!]^{-1}$ and the range $\varepsilon \le \beta/(2\mu_0+2)$. The covering argument (cubes of side d/m, balls U_j around points $b^{(j)} \in K$, Taylor step (17.12)–(17.14)) is followed; deviations D1–D3 (constants, norms, range of ε).

Proof. Put $\kappa := \lceil \frac{M}{\beta}(4+\vartheta^{-1})\sqrt{N}d \rceil \ge 1$ (the number of subdivisions; Rus01's m), $r := \sqrt{N}d/\kappa$. Since $M/\beta \ge 1$,

$$\kappa \ge \tfrac{4M\sqrt{N}d}{\beta}, \qquad \kappa \ge \tfrac{\sqrt{N}d}{\vartheta}, \qquad \kappa \le 1 + \tfrac{M}{\beta}(4+\vartheta^{-1})\sqrt{N}d \le \tfrac{M}{\beta}\big(1 + (4+\vartheta^{-1})\sqrt{N}d\big),$$

so $Mr \le \beta/4$ and $r \le \vartheta$.

Step S15.14: Cover by balls. $K \subset Q = \prod_i [l_i, l_i + d]$ with $l_i = \min_{y\in K} y_i$ (each coordinate range of K is $\le d$). Divide Q into κ^N closed subcubes Q_j of side d/κ; for the at most κ^N indices with $Q_j \cap K \ne \emptyset$ choose $b_j \in Q_j \cap K$, and let $U_j = \{y : |y - b_j|_2 \le r\}$. Every $y \in Q_j$ has $|y - b_j|_\infty \le d/\kappa$, so $|y - b_j|_2 \le r$; thus $K \subset \bigcup_j U_j$, and $U_j \subset K_\vartheta \subset V$ ($r \le \vartheta$, $b_j \in K$); U_j is convex.

Step S15.15: A lower bound for the top-order derivative of $\tilde{g}$ on U_j. Fix j and by (S1) choose $\nu = \nu(j) \le \mu$ with $|D^\nu g(b_j)|_{\mathrm{dg}} \ge \beta$; if $\nu \ge 1$ choose a unit $a = a_j$ with $|D^\nu g(b_j)(a^\nu)| \ge \beta$. For $y \in U_j$, the function $\varphi(s) = D^\nu g(b_j + s(y - b_j))(a^\nu)$ ($s \in [0,1]$) is C^1 ($\nu \le \mu$, $g \in C^{\mu+1}$) with $\varphi'(s) = D^{\nu+1}g(\cdot)(a, \dots, a, y - b_j)$, hence $|\varphi'| \le M|y - b_j|_2 \le Mr$ by (S2) (the segment lies in $U_j \subset K_\vartheta$; $\nu+1 \in [1, \mu+1]$), and by the mean value inequality (F3) $|D^\nu g(y)(a^\nu) - D^\nu g(b_j)(a^\nu)| \le Mr$. With (S3),

$$|D^\nu \tilde{g}(y)(a^\nu)| \;\ge\; \beta - Mr - \beta/2 \;\ge\; \beta/4 \qquad (y \in U_j,\ \nu = \nu(j) \ge 1),$$

and for $\nu(j) = 0$ the same computation gives $|\tilde{g}(y)| \ge \beta/4 > \varepsilon$ on U_j.

Step S15.16: One ball. If $\nu(j) = 0$, $\{y \in U_j : |\tilde{g}| \le \varepsilon\} = \emptyset$. If $q = \nu(j) \ge 1$, Lemma 15.7 for $\tilde{g} \in C^q(V)$, $\beta/4$ in place of β, gives meas $\le 2q(2r)^{N-1}(4\varepsilon/\beta)^{1/q} \le 2\mu(2r)^{N-1} \cdot 4\,(\varepsilon/\beta)^{1/\mu}$, because $\varepsilon/\beta < 1$ makes $(\cdot)^{1/q} \le (\cdot)^{1/\mu}$ and $q \le \mu$, $4^{1/\mu} \le 4$.

Step S15.17: Sum. Summing over $\le \kappa^N$ balls ((F2)), meas $\le \kappa^N \cdot 8\mu(2r)^{N-1}(\varepsilon/\beta)^{1/\mu} = 2^{N+2}\mu(\sqrt{N}d)^{N-1}\kappa\,(\varepsilon/\beta)^{1/\mu}$, since $\kappa^N r^{N-1} = \kappa(\sqrt{N}d)^{N-1}$; insert the bound for κ. □

Remark 15.9 (Comparison with Rus01 17.1, deviations). (D1) The constant C_{sub} is not Rus01's $\mathcal{B}\cdots$ (no $(\mu+1)!$, no Gamma function); it is explicit and polynomial in μ except for the unavoidable M/β. (D2) Two tensor norms (no polarization constant σ_ν). (D3) The range is $\varepsilon < \beta/4$ rather than $\varepsilon \le \beta/(2\mu_0+2)$; for $\mu \ge 1$, $2\mu + 2 \ge 4$ so ours is the larger range up to the strict inequality. The order of the arguments in Rus01 (choice of m minimising (17.17), (17.15)) is replaced by the explicit κ. The case $\mu = 0$ (nonvanishing g) needs no estimate: by (S1)–(S3) $|\tilde{g}| > \varepsilon$ on K for $\varepsilon < \beta/4$.

15.4. **The measure of the small-divisor set.**

Theorem 15.10 (Measure of the set where the divisors are large; Rus01 Thm 18.5 with $p = q = 0$)**.** *Let $N, m, \mu \ge 1$, $\rho \in \{1, 2\}$, $K \subset \mathbb{R}^N$ nonempty compact with $d = d_\infty(K) > 0$, $\vartheta > 0$, $V \subset \mathbb{R}^N$ open with $K_\vartheta \subset V$, $\omega \in C^{\mu+1}(V, \mathbb{R}^m)$, and $\beta > 0$, $M \ge \beta$ with*

- *(D1) (index and amount) $\beta_{\rho,\mu}(\omega, K) \ge \beta$ (Definition 15.1; e.g. $\mu \ge \mu_0(\rho, \omega, K)$ and $\beta = \beta_\rho(\omega, K)$, Lemma 15.2);*
- *(D2) (size) $|D^k g_\rho[\omega](c, y)|_{\mathrm{op}} \le M$ for all $c \in S^{m-1}$, $1 \le k \le \mu + 1$, $y \in K_\vartheta$ (by Lemma 15.4, $M = \max\{\beta, M_1^{(\rho)}\}$ with $M_1^{(1)} = \|\omega\|^{\mu+1}_{\mathrm{op},K_\vartheta}$, $M_1^{(2)} = 2^{\mu+1}(\|\omega\|^{\mu+1}_{\mathrm{op},K_\vartheta})^2$ works).*

Let $\Psi : \mathbb{Z}^m \setminus \{0\} \to (0, \infty)$ (a divisor profile) and, for each $k \in \mathbb{Z}^m \setminus \{0\}$, let $\tilde{\omega}_k \in C^\mu(V, \mathbb{R}^m)$ satisfy

- *(D3) (closeness) $|D^\nu\big(g_\rho[\tilde{\omega}_k](c, \cdot) - g_\rho[\omega](c, \cdot)\big)(y)|_{\mathrm{dg}} \le \beta/2$ for all $c \in S^{m-1}$, $0 \le \nu \le \mu$, $y \in K_\vartheta$ (by Lemma 15.3: it suffices that $\eta_\rho(\theta_k) \le \beta/2$ with $\theta_k = \|\tilde{\omega}_k - \omega\|^\mu_{\mathrm{dg},K_\vartheta}$);*

(D4) (smallness of the divisors) $\big(\Psi(k)/|k|_2\big)^\rho < \beta/4$ *for all* $k \neq 0$.
Put $H := \{b \in K : \ |k \cdot \tilde\omega_k(b)| \geq \Psi(k) \text{ for all } k \in \mathbb{Z}^m \setminus \{0\}\}$. *Then* H *is compact and*

$$\operatorname{meas}(K \setminus H) \ \leq \ C_{\mathrm{sub}}\, \beta^{-1/\mu} \sum_{k \in \mathbb{Z}^m \setminus \{0\}} \Big(\frac{\Psi(k)}{|k|_2}\Big)^{\rho/\mu},$$

with $C_{\mathrm{sub}} = C_{\mathrm{sub}}(N, \mu, d, \vartheta, M/\beta)$ *of Theorem 15.8 (the right side may be* $+\infty$*, in which case there is no claim).*
Source: Rus01 Thm 18.5, p. 189, proof pp. 190–193, specialised to $p = q = 0$ *(no elliptic or hyperbolic determinants, amount of degeneracy* $\alpha = 1$*), the maximal case. The* k*-dependence of* $\tilde\omega_k$ *(Rus01 uses one* $\tilde\chi$ *per application) is added because the iteration tests a different frequency map per Fourier mode. The good set* H *tests one map per mode; it is* not *the statement that one vector is Diophantine. The bridge to the limiting frequency is Lemma 15.12 and Remark 15.14; deviations D3–D5.*

Proof. **Step S15.18: Compactness.** For each k, $b \mapsto |k \cdot \tilde\omega_k(b)|$ is continuous ($\mu \geq 1$), so $\{b \in K : |k \cdot \tilde\omega_k(b)| \geq \Psi(k)\}$ is closed; an intersection of closed subsets of the compact K is compact.

Step S15.19: Union of sublevel sets. If $b \in K \setminus H$ there is $k \neq 0$ with $|k \cdot \tilde\omega_k(b)| < \Psi(k)$. With $c = k/|k|_2 \in S^{m-1}$ this reads $|\langle c, \tilde\omega_k(b)| < \Psi(k)/|k|_2$, hence $|g_\rho[\tilde\omega_k](c, b)| < \varepsilon_k := (\Psi(k)/|k|_2)^\rho$. Therefore $K \setminus H \subset \bigcup_{k \neq 0} \{b \in K : |g_\rho[\tilde\omega_k](c_k, b)| \leq \varepsilon_k\}$, $c_k = k/|k|_2$.

Step S15.20: One k**.** Apply Theorem 15.8 to $g = g_\rho[\omega](c_k, \cdot) \in C^{\mu+1}(V)$ (as $\omega \in C^{\mu+1}$), $\tilde g = g_\rho[\tilde\omega_k](c_k, \cdot) \in C^\mu(V)$ and $\varepsilon = \varepsilon_k \in (0, \beta/4)$ ((D4)): (S1) is (D1) at c_k, (S2) is (D2), (S3) is (D3). This gives meas $\leq C_{\mathrm{sub}}(\varepsilon_k/\beta)^{1/\mu} = C_{\mathrm{sub}}\beta^{-1/\mu}(\Psi(k)/|k|_2)^{\rho/\mu}$.

Step S15.21: Sum. Countable subadditivity ((F2)). □

Corollary 15.11 (Power-law divisors; the measure loss)**.** *In the setting of Theorem 15.10 let* $\tau > \frac{m\mu}{\rho} - 1$ *and* $s := \frac{\rho(\tau+1)}{\mu} - m > 0$*, and let* $\gamma > 0$ *be such that* $(\sqrt{m}\,\gamma)^\rho < \beta/4$*. Suppose (D1)–(D3) hold (for all* $k \neq 0$*) and put*

$H_\gamma := \{b \in K : \ |k \cdot \tilde\omega_k(b)| \geq \gamma |k|_1^{-\tau} \ \ \forall k \in \mathbb{Z}^m \setminus \{0\}\}$ (*one map* $\tilde\omega_k$ *per mode; not yet* IsDiophantine *of a single* ω

Then H_γ *is compact and*

$$\operatorname{meas}(K \setminus H_\gamma) \leq C_{\mathrm{Dio}}\, \gamma^{\rho/\mu}, \qquad C_{\mathrm{Dio}} := C_{\mathrm{sub}}\, \beta^{-1/\mu}\, m^{\rho/(2\mu)} \cdot 2 \cdot 3^{m-1}\Big(1 + \frac{1}{s}\Big).$$

Consequently, for every $\varepsilon^* \in (0, \operatorname{meas} K)$ *and every* γ *with*

$$0 < \gamma \leq \gamma_*(\varepsilon^*) := \min\Big\{\tfrac{1}{2}\, m^{-1/2}(\beta/4)^{1/\rho},\ \big(\varepsilon^*/C_{\mathrm{Dio}}\big)^{\mu/\rho}\Big\}$$

one has $\operatorname{meas} H_\gamma \geq \operatorname{meas} K - \varepsilon^*$*. The data on which* γ_* *depends:* (N, m, μ, ρ, τ)*, the index and amount* (μ, β)*, the chart constants* (d, ϑ, M) *and* ε^**; it does* not *depend on* ω *or* $\tilde\omega_k$ *otherwise.*
Source: Rus01 §18.5, pp. 191–193: the lattice summation with $\Phi_{m\mu_0} = \sup_{T \geq 1} T^m \Phi(T)^{1/\mu_0}$ *(p. 193) is replaced by the explicit lattice sum of Track K ((F5),* [K:bp:p4:lattice-sum]*); CP09 Lemma 8, p. 18 (CP09 (36): the threshold is* $\gamma \propto (\varepsilon^*)^{\mu/2}$*, i.e. measure loss* $\propto \gamma^{2/\mu}$*, the case* $\rho = 2$ *of the loss exponent* ρ/μ *above, equivalently the threshold exponent* μ/ρ *in* γ_**). Deviation D5: for the power* $\Phi(T) = T^{-\tau_D}$ *the summability condition is* $\rho(\tau+1)/\mu > m$*, i.e.* $\tau > m\mu/\rho - 1$*, weaker than Rus01/CP09's* $\tau_D > m\mu$*.*

Proof. **Step S15.22: Profile.** Put $\Psi(k) = \gamma|k|_1^{-\tau}$. For $k \neq 0$, $|k|_1 \geq 1$ and $|k|_2 \geq |k|_1/\sqrt{m}$, so $\Psi(k)/|k|_2 \leq \sqrt{m}\,\gamma |k|_1^{-\tau-1} \leq \sqrt{m}\,\gamma$ and (D4) follows from $(\sqrt{m}\gamma)^\rho < \beta/4$.

Step S15.23: Lattice sum. $\sum_{k \neq 0} (\Psi(k)/|k|_2)^{\rho/\mu} \leq (\sqrt{m}\gamma)^{\rho/\mu} \sum_{k \neq 0} |k|_1^{-\rho(\tau+1)/\mu}$. Put $\tau' = \rho(\tau+1)/\mu - 1$, so that $\tau' + 1 - m = s > 0$, i.e. $\tau' > m - 1$; by (F5) the sum is $\leq S(m, \tau') = 2 \cdot 3^{m-1}(s+1)/s$.

Step S15.24: Conclusion. Theorem 15.10 gives $\operatorname{meas}(K\setminus H_\gamma) \le C_{\mathrm{sub}}\beta^{-1/\mu}m^{\rho/2\mu}\gamma^{\rho/\mu}\cdot 2\cdot 3^{m-1}(1+1/s) = C_{\mathrm{Dio}}\gamma^{\rho/\mu}$. If $\gamma\le\gamma_*$, then $(\sqrt{m}\gamma)^\rho \le 2^{-\rho}\beta/4 < \beta/4$ and $C_{\mathrm{Dio}}\gamma^{\rho/\mu}\le\varepsilon^*$. □

Lemma 15.12 (From the mode-wise good set to a single Diophantine vector)**.** *Let $\gamma>0$, $\tau\in\mathbb{R}$, $k\in\mathbb{Z}^m\setminus\{0\}$ and $u,v\in\mathbb{R}^m$ with*

$$|k\cdot u| \;\ge\; 2\gamma|k|_1^{-\tau} \qquad \text{and} \qquad |u-v|_\infty \;\le\; \gamma|k|_1^{-\tau-1}.$$

Then $|k\cdot v|\ge\gamma|k|_1^{-\tau}$. Consequently, if $b\in H_{2\gamma}$ (Corollary 15.11 with 2γ in place of γ; mode k tested against $\tilde\omega_k$) and a limiting frequency $\omega_*(b)$ satisfies the tail condition $|\omega_*(b)-\tilde\omega_k(b)|_\infty\le\gamma|k|_1^{-\tau-1}$ *for every $k\neq 0$, then $\omega_*(b)$ is (γ,τ)-Diophantine in the frozen sense (Definition 1.1).*

Proof. $|k\cdot(u-v)|\le|k|_1|u-v|_\infty\le\gamma|k|_1^{-\tau}$ (Hölder, ℓ^1–ℓ^∞), so $|k\cdot v|\ge|k\cdot u|-\gamma|k|_1^{-\tau}\ge\gamma|k|_1^{-\tau}$. □

Lemma 15.13 (Change of chart: affine similarities)**.** *Let $y=y_0+\lambda y'$ with $\lambda>0$, $K'=\lambda^{-1}(K-y_0)$, $V'=\lambda^{-1}(V-y_0)$, $\omega'(y')=\omega(y_0+\lambda y')$ (and likewise $\tilde\omega_k'$). Then $g_\rho[\omega'](c,y')=g_\rho[\omega](c,y_0+\lambda y')$ and $D^kg_\rho[\omega'](c,y')=\lambda^kD^kg_\rho[\omega](c,y_0+\lambda y')$ for both tensor norms; $d'=d/\lambda$, $\vartheta'=\vartheta/\lambda$, $(K')_{\vartheta'}=\lambda^{-1}(K_\vartheta-y_0)$, and $\operatorname{meas}E'=\lambda^{-N}\operatorname{meas}E$ for $E'=\lambda^{-1}(E-y_0)$. Consequently:*

- *(a) $\beta_{\rho,\mu}(\omega',K')\ge\min\{1,\lambda^\mu\}\,\beta_{\rho,\mu}(\omega,K)$; thus (D1) holds in the chart y' with $\beta'=\min\{1,\lambda^\mu\}\beta$;*
- *(b) (D2) holds in the chart y' with $M'=\max\{1,\lambda^{\mu+1}\}M\ge\beta'$;*
- *(c) (D3) is* not *scale invariant: its left side is multiplied by λ^ν, so in the chart y' it must be (re)checked with the bound $\beta'/2$ (it holds if the original left side is $\le\min\{\lambda^{-\mu},1\}\,\beta'/2$, in particular if it is $\le\min\{1,\lambda^\mu\}\min\{1,\lambda^{-\mu}\}\beta/2$);*
- *(d) the Diophantine sets H, H_γ of Theorem 15.10 and Corollary 15.11 are defined by the same inequalities $|k\cdot\tilde\omega_k|\ge\Psi(k)$, so $H=y_0+\lambda H'$ and $\operatorname{meas}(K\setminus H)=\lambda^N\operatorname{meas}(K'\setminus H')$.*

Source: Ours (elementary); the chart changes that CP09's polar and action rescalings produce enter here, `reviews/p11a-russmann-check.md` ("chart constants").

Proof. Chain rule for the affine map: $D^k(g\circ A)(y')(a_1,\dots,a_k)=\lambda^kD^kg(Ay')(a_1,\dots,a_k)$; and $\max_{0\le\nu\le\mu}\lambda^\nu x_\nu\ge\min\{1,\lambda^\mu\}\max_\nu x_\nu$ for $x_\nu\ge0$; for (b), $\lambda^k\le\max\{1,\lambda^{\mu+1}\}$ for $1\le k\le\mu+1$. Diameters and thickenings scale by λ^{-1}, Lebesgue measure by λ^{-N} ((F2)). □

Remark 15.14 (Use of the chart lemma, and what consumers must supply). *(a) What the finite preparation (Part 11) and the measure statement need.* Part 10 (Theorem 12.22(c),(d), Corollary 12.20) provides, on a good patch $K=\Lambda(K_a)\subset B^\Lambda$, the index μ_0 and amount β of the planetary frequency data $\tilde\varpi=(\nu,\Omega)$ in the squared convention ($\rho=2$, $m=2n$, $N=n$), independent of the coupling. Part 11 (CP09 finite preparation, Lemmas 4–7 and Proposition 2) must then supply, for the *prepared maximal frequency map* ω_ε (defined on a neighbourhood of a rescaled patch, after the polar change of variables), *(i)* a patch K_ε with $d_\varepsilon,\vartheta_\varepsilon$ and an index/amount (μ,β_ε) of ω_ε, obtained by Lemma 15.3 from Part 10's β and a *C^μ-estimate* for $\omega_\varepsilon-\omega_0$ on $K_{\varepsilon,\vartheta}$ (this is CP09 Prop. 2 with derivative control, defect R-CP-2 of the review: a C^0 bound does not suffice); *(ii)* the size M_ε of (D2). With these, Corollary 15.11 gives the measure of the set of Diophantine frequencies in the prepared chart. *(b) What File 11a-3 must supply per Fourier mode, and how the excluded measure accumulates.* The iteration tests a mode k at one stage $\nu(k)=\min\{\nu:|k|_1\le T_\nu\}$, defined with *the lattice norm of the cutoff* ($T_{\nu-1}<|k|_1\le T_\nu$; File 11a-2 truncates in $|k|_1$, never in $|k|_2$). It must supply, for every $k\neq0$:

- (I1) *the map:* $\tilde\omega_k:=\tilde\omega_{\nu(k)}\in C^\mu(V,\mathbb{R}^m)$, the chain extension of the stage-$\nu(k)$ frequency $\omega_{\nu(k)}$ (equal to $\omega_{\nu(k)}$ on the retained set $K_{\nu(k)}$), with closeness (D3) on K_ϑ uniformly in k (accumulated C^μ-size of the increments, via Lemma 15.3);
- (I2) *the test:* the retained set at stage ν is defined by the mode-wise conditions $|k\cdot\tilde\omega_k(b)|\ge2\gamma|k|_1^{-\tau}$ for the modes of the shell $T_{\nu-1}<|k|_1\le T_\nu$ (not by the cutoff-wide quantity $\Phi(T_\nu)$), so that $K_\infty\supset H_{2\gamma}\cap\{\text{tail condition}\}$;

(I3) *the tail (corrected form, Remark 17.16):* with $\ell_j \geq \sup|\Delta\omega_j|_\infty$ the sizes of the frequency increments, for *every* $m \geq 0$ (including $m = 0$) $\sum_{j\geq m} \ell_j \leq \gamma\, T_m^{-\tau-1}$. Then for $b \in K_\infty$ and $k \neq 0$ with $m = \nu(k)$ (i.e. $T_{m-1} < |k|_1 \leq T_m$), $|\omega_*(b) - \omega_m(b)|_\infty \leq \sum_{j\geq m} \ell_j \leq \gamma T_m^{-\tau-1} \leq \gamma|k|_1^{-\tau-1}$. The endpoint is T_m, the *upper* end of the shell, where $|k|_1^{-\tau-1}$ is smallest (an earlier version used T_{m-1}, which controls only the smallest $|k|_1$ of the shell; review P11A2-01). The finite-stage version needed for the homological equation is not derived from the limit (two bounds to a limit would give 2γ): it is proved on the finite retained set, for $\omega_\nu - \omega_{\nu(k)}$, in Lemma 17.15.

Then: (i) $\omega_*(b)$ is (γ,τ)-Diophantine on $K_\infty \cap H_{2\gamma}$ (Lemma 15.12); (ii) at stage ν, for b in the finite retained set $K_{\nu+1}$ (not only on K_∞) and $0 < |k|_1 \leq T_\nu$: $|k\cdot\omega_\nu(b)| \geq 2\gamma|k|_1^{-\tau} - \gamma|k|_1^{-\tau} \geq \gamma T_\nu^{-\tau}$, where the frequency difference is bounded by the finite telescoping sum $|\omega_\nu(b) - \omega_{\nu(k)}(b)|_\infty \leq \sum_{\nu(k)\leq j<\nu} \ell_j \leq \gamma T_{\nu(k)}^{-\tau-1} \leq \gamma|k|_1^{-\tau-1}$, so that the subtracted divisor error $|k\cdot(\omega_\nu - \omega_{\nu(k)})(b)| \leq |k|_1|\omega_\nu(b) - \omega_{\nu(k)}(b)|_\infty$ is at most $\gamma|k|_1^{-\tau}$ (Lemma 17.15); this is the cutoff-wide divisor of the homological equation (the passage $\Phi(|k|_1) \to \Phi(T_\nu)$ uses only $|k|_1 \leq T_\nu$); (iii) *accumulation:* each mode lies in exactly one shell, so the measure removed at stage ν is at most $C_{\rm sub}\beta^{-1/\mu}\sum_{T_{\nu-1}<|k|_1\leq T_\nu}(\Psi(k)/|k|_2)^{\rho/\mu}$, $\Psi(k) = 2\gamma|k|_1^{-\tau}$, and the total over all stages is the right side of Theorem 15.10, $\leq C_{\rm Dio}(2\gamma)^{\rho/\mu} \leq \varepsilon^*$ (Corollary 15.11, γ replaced by 2γ in γ_*). The shrinking sets K_ν cause no loss because every mode is measured over the original K through the extension (I1). Diophantinity of ω_* is the output of (i), not an identification with H_γ. This is a requirement list for 11a-3, not an assumption of any frozen statement. *(c)* All of Part 11a is formulated with period-one angles (characters $e^{2\pi i k\cdot\theta}$), so the divisors are the frozen ones, $|k\cdot w| \geq \gamma|k|_1^{-\tau}$ (Definition 1.1), and only the comparison $|k|_2 \geq |k|_1/\sqrt{m}$ enters here. Data given with CP's 2π-angles are converted once by Part 11 (`bpP:p1:period-convert`); Rus01's own 2π-periodic constants are not reused. For ω real analytic the constant M of (D2) is obtained from the general Cauchy estimate of arbitrary order ([N:`bpN:p4:general-cauchy`]; [K:`bp:p1:cauchy-open`] covers only orders $j \leq 3$) and Lemma 15.4; this is carried out in File 11a-5.

15.5. **Sources, deviations, errata, forward references.**
Sources read. Rus01 §§17.1–17.4 (pp. 180–183), 17.5 (p. 183, determinant derivatives; not needed for $p = q = 0$), 18.1–18.5 (pp. 185–193), §16.2 (p. 178), Thm 16.7 (p. 179) for the consumer interface, §§7.2–7.5 (pp. 134–136) for the neighbouring cutoff; the Rus01 pages were read from the pinned PDF (print page = pdf page +118). Fej04r §5.2, Théorème 50 (statement only). CP09 pp. 5–6 (Lemma 1, Def. 2, via Part 10). `reviews/p11a-russmann-check.md` (rows `rank-compact`, `scalar-sublevel`, `diophantine-measure`, errata A3, R-CP-2), Part 10.
Deviations from the source (all recorded). (D1) The one-variable estimate (17.4) is proved (Lemma 15.6) instead of imported from Bakhtin; constants are ours, explicit and polynomial in μ. (D2) Two tensor norms; no polarization constant; no Gamma-function estimates (ball volume bounded by the circumscribed cube). (D3) Range $\varepsilon < \beta/4$ in place of $\beta/(2\mu_0+2)$; the constants $C_{\rm sub}, C_{\rm Dio}$ are recomputed (not Rus01's (18.12), (18.13), (18.30)). (D4) Index and amount are those of the scalar family $\langle c,\omega\rangle^\rho$, $\rho \in \{1,2\}$, rather than of the product family of Rus01 §18.3.4 ($\langle c,\omega\rangle\langle\hat c,\omega\rangle$); each theorem holds for each ρ with its own (μ,β), and the consumer picks one convention and keeps it. No comparison between the numbers of different conventions is claimed (review, §18.3 note). (D5) The summability condition is $\tau > m\mu/\rho - 1$ via the explicit lattice sum of Track K; it covers CP09's $\tau_D > m\mu$ (with CP09's squared index) and Rus01's $\Phi_{m\mu_0} < \infty$ for the power $\Phi(T) = T^{-\tau_D}$ (finite exactly when $\tau_D \geq m\mu_0$ in the normalised case, and $\Phi \equiv 1$ on $[0,1]$). (D6) The Diophantine condition is stated with $|k|_1$ (frozen Definition 1.1); Rus01 uses $|k|_2$ with a general Φ; the comparison costs the factor $\sqrt{m}$ (in γ_*). (D7) k-dependent perturbed frequency maps (new but a trivial generalisation of the proof). (D8) Elliptic/hyperbolic divisors $(p,q>0)$, α (amount

of degeneracy, Rus01 §18.1) and §17.5 (derivatives of determinants) are not needed in the maximal case and are dropped.
Suspected errata. None found in the pages used. Rus01 Lemma 17.2 depends on an unproved citation (recorded in the source note of Lemma 15.6); this is an external input, not an error.
Doubts about frozen statements. None (Part 11a-1 touches no frozen declaration; the conclusion of Part 11a is assembled in File 11a-5).
External facts used. (F1)–(F6); (F4), (F5), (F1) through Part 10. Not in Mathlib in this form as far as we know: iterated Rolle for q nodes in the form of Lemma 15.5 (a few lines from `exists_deriv_eq_zero` and `Lagrange`); the divided-difference estimate is proved in the text.
Numerical check. `.drafts/P/bp-scratch-bpP-p11a-1/chk2.py` samples 4000 random polynomials f with $|f^{(q)}| \geq \beta$ on random intervals, $q \leq 5$, and compares the grid-measured sublevel measure with $2q(\varepsilon/\beta)^{1/q}$: the maximal ratio is $0.991 \leq 1$ (sharp for $q = 1$). Evidence only.
Forward references and consumers. File 11a-3 (parameter selection): Theorem 15.10, Corollary 15.11, Lemma 15.3. File 11a-5 (quantitative theorem and budget): Corollary 15.11 (constants $C_{\mathrm{sub}}, C_{\mathrm{Dio}}, \gamma_*$), Lemma 15.13, Lemma 15.2. Part 11 (finite preparation): Lemma 15.3, Remark 15.14(a). No forward reference to an unwritten label other than those named in `PLAN-11a.md`.
Label index. `index-def`, `rank-compact`, `stability`, `M-bound`, `divided-difference`, `scalar-sublevel`, `ball-sublevel`, `sublevel`, `sublevel-dev`, `diophantine-measure`, `power-law`, `limit-bridge`, `chart-scaling`, `consumers`.

16. Part 11a-2: Fourier and Taylor cutoffs, the maximal homological equation, the canonical step

This is the second of the five files of Part 11a (split plan `PLAN-11a.md`; the file is independent of File 11a-1 except for the divisor interface of Lemma 15.12). It proves, in the frozen conventions (angles of period 1, characters $e^{2\pi i k\cdot\theta}$, max norms), the *analytic* half of the maximal-case KAM step of Rüssmann, replacing every external or defective ingredient:

- `fourier-cutoff`, `loss-choice`, `loss-sum` (Lemmas 16.2, 16.3, 16.4): an elementary Fourier cutoff in $|k|_1$ with an explicit constant c_N, the loss d_T with error $\leq \alpha$, and the sum of the losses over a geometric schedule. This replaces Rus01 Thm 7.3 (via Akhiezer) and the repaired Rus05 estimate (deviation D-4).
- `taylor-cutoff`, `cutoff` (Lemma 16.5, Proposition 16.6): the Cauchy–Taylor cutoff of order 1 in the action variable and the composed approximation $R \mapsto R_*$ (replacing Rus01 Thms 7.2, 7.4; deviation D-5).
- `param-cauchy`, `divisor-wide`, `divisor-perturb`, `homological-maximal` (Lemmas 16.7, 16.8, 16.9, Proposition 16.10): parameter Cauchy estimates of every order, the cutoff-wide divisor from the per-mode interface of File 11a-1, its extension to a complex neighbourhood of the parameter set, and the maximal homological equation with η-linear normal form (Rus01 §11, pp. 148–153, case $p = q = 0$).
- `affine-flow`, `real-lipschitz`, `canonical-step` (Lemma 16.11, Lemma 16.12, Proposition 16.13): the time-one flow of $\varphi_0(\theta) + \varphi_1(\theta)\cdot\eta$ as a simple canonical map (Rus01 §12, pp. 154–155), through track N's flow package, and the one-step conjugation $H\circ\Phi = N_+ + R_+$ with every domain loss and constant explicit.

16.1. Conventions, domains, external facts.

Variables. $N \geq 1$ is the number of angles and of actions. The point of the extended phase space is $(\theta, \eta, b) \in \mathbb{C}^N \times \mathbb{C}^N \times \mathbb{C}^N$; θ is an angle (period one), η an action *translated to* 0, b the parameter of the family (Rus01's b, §14). All norms on $\mathbb{R}^N$, $\mathbb{C}^N$ are max norms $|\cdot|_\infty$, and $|k|_1 = \sum |k_i|$; for a matrix $|M|_\infty$ is the induced norm (maximal row sum); for a symmetric j-linear map $\|D^j u\| = \sup\{|D^j u(a_1,\dots,a_j)|_\infty : |a_i|_\infty \leq 1\}$. For $E \subset \mathbb{R}^N$ and $t > 0$ put $E + t = \{b \in \mathbb{C}^N : \exists b' \in E,\ |b - b'|_\infty < t\}$, an open set, and for $b' \in E$ the polydisc $P(b', t) \subset E + t$.

Definition 16.1 (Domains and sup norms)**.** For $r, s > 0$ and $B \subset \mathbb{C}^N$ put $T_r = \{\theta \in \mathbb{C}^N : |\operatorname{Im}\theta|_\infty < r\}$ ([K:bp:p0:domains]) and

$$\mathcal{D}(r, s; B) = \{(\theta, \eta, b) : \ \theta \in T_r, \ |\eta|_\infty < s, \ b \in B\}.$$

For F on $\mathcal{D}(r, s; B)$, or on $T_r \times B$ ($s = 0$ is not written; the norm is then $\|F\|_{r;B}$), $\|F\|_{r,s;B} = \sup|F|_\infty$ over the domain. A function is *admissible* if it is holomorphic in all its variables (B open) and 1-periodic in each θ_j. For $T \geq 0$ the *truncation* of a bounded admissible (or merely continuous periodic) F is $\mathcal{T}_T F = \sum_{|k|_1 \leq T} \hat{F}_k \chi_k$, with $\hat{F}_k = \int_{[0,1]^N} F(x)\chi_{-k}(x)\,dx$ (coefficients taken for each fixed (η, b)) and $\chi_k = e^{2\pi i k\cdot\theta}$; its average is $\langle F\rangle = \hat{F}_0$. A function F is a *trigonometric polynomial of degree* $\leq T$ if $F = \mathcal{T}_T F$.

Convention for the bracket and the field (that of track N, [N:bpN:p5:setting]): $X_\phi = (\partial_\eta\phi, -\partial_\theta\phi)$ on $\mathbb{C}^N \times \mathbb{C}^N$ (angles first), $\{u, v\} = Du \cdot X_v = \partial_\theta u \cdot \partial_\eta v - \partial_\eta u \cdot \partial_\theta v$, and the generating function acts by the time-one flow $\Phi = X_\phi^1$. This is the same scalar Hamiltonian field and bracket as in Rus01 after the conversion of rows to columns (Rus01 uses row vectors and $J_{\mathrm{src}} = \left(\begin{smallmatrix} 0 & -\mathbf{1} \\ \mathbf{1} & 0 \end{smallmatrix}\right)$, so that its row field $S_x J_{\mathrm{src}} = (S_y, -S_x)$ is the transpose of X_S above); the homological equation below has the same positive sign as Rus01's $\{S, N\} = \partial_\omega S$, (11.12), up to the factor 2π of the period-one convention. The sign was fixed once to match [N:bpN:p5:taylor]; it does not affect any estimate. External facts. Besides [K:bp:p1:ext-cauchy], [K:bp:p1:ext-mvt], [K:bp:p1:ext-integral] and [N:bpN:p4:ext-param]:

(X1) *Cauchy coefficient estimate in one variable.* If g is holomorphic on $|\zeta| < R$ with values in $\mathbb{C}^q$ (max norm) and $|g| \leq M$, then $g(\zeta) = \sum_{j\geq 0} g_j \zeta^j$ on $|\zeta| < R$ with $|g_j| \leq MR^{-j}$. (Cauchy integral formula for the coefficients; Mathlib: `Complex.hasSum_taylorSeries_on_ball`, `norm_cauchyPowerSeries_le`, names not machine-checked here.) Applied successively in several variables, if G is holomorphic and bounded by M on a polydisc of radii $R_1, \dots, R_j$ then $|\partial_{\zeta_1} \cdots \partial_{\zeta_j} G(0)| \leq M \prod_i R_i^{-1}$.

(X2) $|k \cdot x| \leq |k|_1 |x|_\infty$ for $k \in \mathbb{Z}^N$, $x \in \mathbb{C}^N$, and $|\chi_k(\theta)| \leq e^{2\pi|k|_1|\operatorname{Im}\theta|_\infty}$.

(X3) A linear inhomogeneous ODE $y' = P(t)y + q(t)$ with continuous coefficients on $[0, 1]$ has a unique global solution, affine in $y(0)$; Gronwall: if $y' = P(t)y$, $|P| \leq \Lambda$, then $|y(t)| \leq e^{\Lambda t}|y(0)|$. (Mathlib: `ODE_solution_unique`, Gronwall.)

Lean totality. Nothing here divides by zero: the only division is by $2\pi i\, k \cdot \omega(b)$ for $0 < |k|_1 \leq T$, where the hypotheses give $|k \cdot \omega(b)| \geq D/2 > 0$ (Lemma 16.9); $\mathcal{T}_T$ is a finite sum; no fderiv of a non-differentiable function occurs, because every derivative taken is of an admissible function on its open domain.

16.2. The elementary Fourier cutoff and the choice of the loss.

Lemma 16.2 (Fourier cutoff in $|k|_1$)**.** *Let $N \geq 1$, $r > 0$ and put*

$$c_N = 2\Big(1 + \frac{1}{\pi}\Big)\Big(\frac{3N}{e\pi}\Big)^{N-1}, \qquad \varepsilon_N(T, d) = c_N\,(1 + d^{-N})\,e^{-\pi T d} \qquad (T \geq 0, \ d > 0).$$

Then $c_N \geq 1$. Let $F : T_r \to \mathbb{C}^q$ (max norm) be holomorphic, 1-periodic and bounded, $T \geq 0$ and $0 < d < r$. Then $\mathcal{T}_T F$ is a trigonometric polynomial of degree $\leq T$, entire in θ, and

(a) $\|F - \mathcal{T}_T F\|_{r-d} \leq \varepsilon_N(T, d)\,\|F\|_r$;
(b) $\|\mathcal{T}_T F\|_{r-d} \leq (1 + \varepsilon_N(T, d))\,\|F\|_r$;
(c) if F is real on $\mathbb{R}^N$, then so is $\mathcal{T}_T F$.

The dimension N is that of the angle; there is no rescaling of T or d.

Source: Ours. Replaces Rus01 Thm 7.3 p. 134 (approximation by trigonometric polynomials with multipliers, from [21] via Akhiezer's theorem) and its repair Rus05 §§9–11 pp. 26–31 (deviation D-4). Idea: Cauchy decay of coefficients ([K:bp:p1:decay]*) and a lattice count (*[K:bp:p4:lattice-count]*); no multipliers, no smoothing.*

Proof. **Step S16.1: $c_N \ge 1$.** $c_1 = 2(1+1/\pi) > 2$. For $N \ge 2$, $c_N = 2(1+1/\pi)(3N/(e\pi))^{N-1}$ and $3N/(e\pi) \ge 6/(e\pi) > 0.70$; for $N = 2$, $c_2 > 2 \cdot 1.31 \cdot 0.70 > 1.8$; for $N \ge 3$, $3N/(e\pi) \ge 9/(e\pi) > 1$, so $c_N \ge 2$.

Step S16.2: Decay and Fourier series. By [K:bp:p1:decay] (with $\rho' \uparrow r$), $|\hat F_k| \le e^{-2\pi|k|_1 r}\|F\|_r$ (componentwise, hence in the max norm). Hence for every $\rho < r$, $\sum_k |\hat F_k| e^{2\pi|k|_1\rho} < \infty$, and by [K:bp:p1:summation] $F = \sum_k \hat F_k \chi_k$ on T_ρ; $\mathcal{T}_T F$ is a finite sum, entire. On T_{r-d}, $F - \mathcal{T}_T F = \sum_{|k|_1 > T} \hat F_k \chi_k$. For (c): if F is real on $\mathbb{R}^N$ then $\hat F_{-k} = \overline{\hat F_k}$ and $\mathcal{T}_T F$ is a real sum.

Step S16.3: Tail. By (X2), for $\theta \in T_{r-d}$, $|F(\theta) - \mathcal{T}_T F(\theta)| \le \sum_{|k|_1>T} \|F\|_r\, e^{-2\pi|k|_1 r} e^{2\pi|k|_1(r-d)} = \|F\|_r \sum_{|k|_1>T} e^{-2\pi|k|_1 d}$.

Step S16.4: Count. Group by $j = |k|_1$ ($j \ge 1$ integer, $j > T$) and use $\#\{|k|_1 = j\} \le 2\cdot 3^{N-1} j^{N-1}$ ([K:bp:p4:lattice-count]): $\sum_{|k|_1>T} e^{-2\pi|k|_1 d} \le 2\cdot 3^{N-1} \sum_{j>T} j^{N-1} e^{-\pi j d}\, e^{-\pi j d}$.

Step S16.5: Polynomial against exponential. By [K:bp:p1:exp-sums](a) with $\tau = N-1$ and $a = \pi d$: $j^{N-1} e^{-\pi j d} \le \big(\frac{N-1}{e\pi d}\big)^{N-1}$ (read $0^0 = 1$). The integers $j > T$ are $j_0, j_0+1, \dots$ with $j_0 > T$, so $\sum_{j>T} e^{-\pi j d} \le e^{-\pi j_0 d}/(1-e^{-\pi d}) \le e^{-\pi T d}(1 + \frac{1}{\pi d})$, since $1/(1-e^{-x}) = 1 + 1/(e^x - 1) \le 1 + 1/x$.

Step S16.6: Assembly. The tail sum is at most $2\cdot 3^{N-1}\big(\frac{N-1}{e\pi}\big)^{N-1}\big(d^{-(N-1)} + \frac{d^{-N}}{\pi}\big)e^{-\pi T d}$. Now $d^{-(N-1)} \le 1 + d^{-N}$ (if $d \ge 1$ the left side is ≤ 1, if $d < 1$ it is $\le d^{-N}$), so the bracket is $\le (1+\frac{1}{\pi})(1+d^{-N})$; and $N - 1 \le N$ gives the factor $(3N/(e\pi))^{N-1}$. This is (a); (b) is (a) and the triangle inequality ($\|F\|_{r-d} \le \|F\|_r$). □

Lemma 16.3 (Choice of the loss d_T)**.** *Let $N \ge 1$, $0 < \alpha \le 1$, $T \ge 1$ and*

$$d_T = d_T(\alpha) = \frac{\log(2c_N/\alpha) + N\log(\pi T)}{\pi T} \quad (> 0).$$

If $d_T \le 1$, then $\varepsilon_N(T, d_T) \le \alpha$. Moreover $T \mapsto d_T(\alpha)$ is decreasing on $[1,\infty)$ for fixed α, and $\alpha \mapsto d_T(\alpha)$ is decreasing in α.

Source: Ours; corrects the first draft of `PLAN-11a.md`, where $d = (N+1)\log(1/\alpha)/(2\pi T)$ does not give error $\le \alpha$ because the factor d^{-N} grows like T^N (review item P11A1-P02).

Proof. **Step S16.7: The exponent.** Put $A = \pi T d_T = \log(2c_N/\alpha) + N\log(\pi T)$. Since $c_N \ge 1$, $\alpha \le 1$, $T \ge 1$: $A \ge \log 2 + N\log\pi > 1$.

Step S16.8: The estimate. For $d = d_T \le 1$: $1 + d^{-N} \le 2d^{-N}$ and $d^{-N} = (\pi T/A)^N \le (\pi T)^N$ ($A \ge 1$). And $e^{-\pi T d} = e^{-A} = \frac{\alpha}{2c_N}(\pi T)^{-N}$. Hence $\varepsilon_N(T, d_T) \le c_N \cdot 2(\pi T)^N \cdot \frac{\alpha}{2c_N}(\pi T)^{-N} = \alpha$.

Step S16.9: Monotonicity. With $a = \log(2c_N/\alpha) \ge \log 2 > 0$: $\frac{d}{dT}\frac{a + N\log(\pi T)}{\pi T} = \frac{\pi(N - a - N\log(\pi T))}{(\pi T)^2} < 0$ because $\pi T \ge \pi > e$. The α-monotonicity is clear. □

Lemma 16.4 (Sum of the losses on a geometric schedule)**.** *Let $\Theta > 1$, $T_0 \ge 1$, $T_\nu = T_0\Theta^\nu$, and let $\alpha_\nu \in (0,1]$ satisfy $\log(2c_N/\alpha_\nu) \le a + b\nu$ ($a, b \ge 0$). Then $d_\nu := d_{T_\nu}(\alpha_\nu) \le (a + b\nu + N\log(\pi T_0) + N\nu\log\Theta)/(\pi T_0\Theta^\nu)$ and*

$$\sum_{\nu\ge 0} d_\nu \le \frac{1}{\pi T_0}\Big[\big(a + N\log(\pi T_0)\big)\frac{\Theta}{\Theta - 1} + (b + N\log\Theta)\frac{\Theta}{(\Theta-1)^2}\Big].$$

In particular for $\alpha_\nu = \alpha_0\delta^{\kappa\nu}$ ($0 < \delta < 1$): $a = \log(2c_N/\alpha_0)$, $b = \kappa\log(1/\delta)$. The hypothesis $d_\nu \le 1$ needed in Lemma 16.3 is not *implied and must be checked by the consumer (it holds for all ν as soon as the displayed bound is ≤ 1).*

Source: Ours.

Proof. $\log(\pi T_\nu) = \log(\pi T_0) + \nu \log \Theta$ and the definition of d_T give the bound on d_ν; sum with $\sum_{\nu\ge 0} \Theta^{-\nu} = \Theta/(\Theta-1)$ and $\sum_{\nu\ge 0} \nu\Theta^{-\nu} = \Theta/(\Theta-1)^2$. □

16.3. **The Taylor cutoff in the action and the composed cutoff.**

Lemma 16.5 (Taylor cutoff of order 1 in η)**.** *Let $r, s > 0$, $B \subset \mathbb{C}^N$ open, $F : \mathcal{D}(r, s; B) \to \mathbb{C}^q$ admissible, $M = \|F\|_{r,s;B} < \infty$. Put $F^{(\le 1)}(\theta, \eta, b) = F(\theta, 0, b) + \partial_\eta F(\theta, 0, b)\,\eta$ ($\partial_\eta F$ the Jacobian in η). Then $F^{(\le 1)}$ is admissible (entire in η) and, for $\theta \in T_r$, $b \in B$:*

(a) $|\partial_{\eta_j} F(\theta, 0, b)|_\infty \le M/s$ for each j; $|F^{(\le 1)}(\theta, \eta, b)|_\infty \le M(1+|\eta|_\infty/s)$, so $\|F^{(\le 1)}\|_{r,s;B} \le 2M$;

(b) for $0 < q < 1$: $\|F - F^{(\le 1)}\|_{r,qs;B} \le M\,\dfrac{q^2}{1-q}$.

Source: Ours; replaces Rus01 Thm 7.4 p. 135 (algebraic polynomials, from Lemma 6 of [18], external) in the case needed here, polynomials of degree ≤ 1 (deviation D-5). The q plays the role of δ^β of Rus01 (7.2).

Proof. **Step S16.10: One variable.** Fix (θ, b) and $\eta \ne 0$ with $a = |\eta|_\infty < s$ ($\eta = 0$ is trivial). $g(\zeta) = F(\theta, \zeta\eta/a, b)$ is holomorphic on $|\zeta| < s$ (since $|\zeta\eta/a|_\infty = |\zeta|$) and bounded by M. By (X1), $g(\zeta) = \sum g_j \zeta^j$ with $|g_j| \le M s^{-j}$, and $g_0 = F(\theta, 0, b)$, $g_1 = \partial_\eta F(\theta, 0, b)\eta/a$.

Step S16.11: (a). $|F^{(\le 1)}| = |g_0 + g_1 a| \le M + Ma/s$. For the column bound apply the same argument along e_j ($a = s^-$): $|\partial_{\eta_j} F(\theta, 0, b)| \le M/s$ ([K:bp:p1:ext-cauchy]).

Step S16.12: (b). At $\zeta = a$: $|F(\theta, \eta, b) - F^{(\le 1)}(\theta, \eta, b)| = |\sum_{j\ge 2} g_j a^j| \le M \sum_{j\ge 2} (a/s)^j = M\frac{(a/s)^2}{1-a/s}$, increasing in a; for $a < qs$ it is $\le Mq^2/(1-q)$. □

Proposition 16.6 (The composed cutoff $R \mapsto R_*$)**.** *Let $r > d > 0$, $s > 0$, $B \subset \mathbb{C}^N$ open, $R : \mathcal{D}(r, s; B) \to \mathbb{C}$ admissible with $\varepsilon = \|R\|_{r,s;B} < \infty$; $T \ge 0$, $0 < q < 1$, and $E_T := \varepsilon_N(T, d)$ (Lemma 16.2). Put*

$$R_* = \mathcal{T}_T\big(R^{(\le 1)}\big) = R_{*0}(\theta, b) + R_{*1}(\theta, b) \cdot \eta, \quad R_{*0} = \mathcal{T}_T[R(\cdot, 0, \cdot)], \quad R_{*1,j} = \mathcal{T}_T[\partial_{\eta_j} R(\cdot, 0, \cdot)].$$

Then R_{*0}, $R_{*1,j}$ are trigonometric polynomials of degree $\le T$ in θ, admissible on $\mathbb{C}^N \times B$ (holomorphic in (θ, b), 1-periodic), and

(a) $\|R_{*0}\|_{r-d;B} \le (1+E_T)\varepsilon$, $\|R_{*1,j}\|_{r-d;B} \le (1+E_T)\varepsilon/s$ for each j, $\|R_*\|_{r-d,s;B} \le 2(1+E_T)\varepsilon$;

(b) $\|R - R_*\|_{r-d,qs;B} \le \varepsilon\big(\frac{q^2}{1-q} + 2E_T\big)$;

(c) if R is real for real (θ, η, b), so is R_*.

With $T = T_\nu$, $d = d_\nu$, $\alpha = \alpha_\nu$ of Lemma 16.3: $E_T \le \alpha$ and (b) reads $\le \varepsilon(q^2/(1-q) + 2\alpha)$.

Source: Rus01 Thm 7.2 p. 134 and (13.10) p. 159 (the composed estimate $|F - F_{T\beta\delta}| \le 2\delta|F|$); the statement and the constants are ours (D-4, D-5).

Proof. **Step S16.13: Coefficients.** Holomorphy of $R_{*0}, R_{*1,j}$ in (θ, b): the Fourier coefficients $b \mapsto \int_{[0,1]^N} R(x, 0, b)\chi_{-k}(x)\,dx$ are holomorphic by [N:bpN:p4:ext-param] (the integrand is jointly continuous with continuous b-derivative, R being holomorphic on the open set), and R_* is a finite sum of them times characters.

Step S16.14: (a). Apply Lemma 16.2(b) for each fixed (η, b) to $F = R^{(\le 1)}(\cdot, \eta, b)$ on T_r with $\|F\|_r \le 2\varepsilon$ (Lemma 16.5(a)): $\|R_*(\cdot, \eta, b)\|_{r-d} \le 2(1 + E_T)\varepsilon$. For R_{*0} take $F = R(\cdot, 0, b)$ ($\|F\|_r \le \varepsilon$) and for $R_{*1,j}$ take $F = \partial_{\eta_j} R(\cdot, 0, b)$ ($\|F\|_r \le \varepsilon/s$, Lemma 16.5(a)).

Step S16.15: (b). $R - R_* = (R - R^{(\le 1)}) + (R^{(\le 1)} - \mathcal{T}_T R^{(\le 1)})$. On $\mathcal{D}(r - d, qs; B) \subset \mathcal{D}(r, qs; B)$ the first term is $\le \varepsilon q^2/(1-q)$ (Lemma 16.5(b)); the second is $\le E_T \cdot 2\varepsilon$ by Lemma 16.2(a) applied for each (η, b).

Step S16.16: (c). $R^{(\le 1)}$ is real at real points ($\partial_\eta R$ is real there); then Lemma 16.2(c). □

16.4. **Parameter estimates and divisors.**

Lemma 16.7 (Cauchy estimates in the parameter, every order)**.** *Let $E \subset \mathbb{R}^N$, $0 < t' < t$, and $u : E + t \to \mathbb{C}^q$ holomorphic with $|u|_\infty \le M$. Then for every $b \in E + t'$ and $j \ge 1$: $\|D^j u(b)\| \le j^j M\,(t-t')^{-j}$ (norm as in the conventions). The same holds for a function of (θ, η, b) on $\mathcal{D}(r, s; E+t)$ bounded by M, for the b-derivatives, at every (θ, η).*
Source: Ours (the general-order version; [K:bp:p1:cauchy-open] *covers only orders ≤ 3,* [N:bpN:p4:general-cauchy] *only the first derivative).*

Proof. **Step S16.17: Polydisc.** Put $\varrho = t - t'$, fix $b \in E + t'$ and $a_1, \dots, a_j$ with $|a_i|_\infty \le 1$. There is $b' \in E$ with $|b - b'| < t'$. For $\zeta \in \mathbb{C}^j$ with $|\zeta_i| < \varrho/j$ we have $|\sum \zeta_i a_i|_\infty < \varrho$, so $b + \sum \zeta_i a_i \in P(b', t) \subset E + t$ and $G(\zeta) = u(b + \sum \zeta_i a_i)$ is holomorphic and bounded by M on this polydisc.

Step S16.18: Iterated Cauchy. $D^j u(b)(a_1, \dots, a_j) = \partial_{\zeta_1} \cdots \partial_{\zeta_j} G(0)$, so by (X1) it is bounded by $M(j/\varrho)^j$. For the vector-valued case take each component and the max norm. □

Lemma 16.8 (Cutoff-wide divisor from the per-mode interface)**.** *Let $\gamma > 0$, $\tau \ge 0$, $T > 0$, $v \in \mathbb{R}^N$, and for each $k \in \mathbb{Z}^N$ with $0 < |k|_1 \le T$ let $u_k \in \mathbb{R}^N$ satisfy $|k \cdot u_k| \ge 2\gamma |k|_1^{-\tau}$ and $|u_k - v|_\infty \le \gamma |k|_1^{-\tau-1}$. Then for all $0 < |k|_1 \le T$,*

$$|k \cdot v| \;\ge\; \gamma |k|_1^{-\tau} \;\ge\; \gamma\, T^{-\tau}.$$

Source: Ours; Lemma 15.12 of File 11a-1 and the monotonicity of $x \mapsto x^{-\tau}$.

Proof. Lemma 15.12 gives the first inequality for each k with $u = u_k$; the second is $|k|_1 \le T$, $\tau \ge 0$. In the consumer, $u_k = \omega_{\nu(k)}(b') = \tilde{\omega}_k(b')$ and $v = \omega_\nu(b')$ at stage ν (File 11a-3, requirements (I1)–(I3) of Remark 15.14, with (I3) in the corrected form of Remark 17.16); the hypotheses here are those requirements, evaluated at one parameter point. The *finite-stage* version that the iteration actually proves, for every b in the retained set $K_{\nu+1}$ and without any limit frequency, is Lemma 17.15 (review P11A2-01): its telescoping $|\omega_\nu(b) - \omega_{\nu(k)}(b)|_\infty \le \gamma |k|_1^{-\tau-1}$ is exactly the hypothesis $|u_k - v| \le \gamma |k|_1^{-\tau-1}$ of this lemma. □

Lemma 16.9 (Divisors on a complex neighbourhood of the retained set)**.** *Let $E \subset \mathbb{R}^N$, $t > 0$, $T > 0$, $D > 0$, $\omega : E + t \to \mathbb{C}^N$ holomorphic with $\|D\omega(b)\| \le L$ for all $b \in E + t$ (operator norm for $|\cdot|_\infty$), and*

$$tTL \le D/2, \qquad |k \cdot \omega(b')| \ge D \quad \text{for all } b' \in E,\ 0 < |k|_1 \le T.$$

Then $|k \cdot \omega(b)| \ge D/2$ for all $b \in E + t$ and $0 < |k|_1 \le T$.
Source: Rus01 §11.4 p. 150, step 3 (extension of 11.3.7 from K_+ to B_+); ours in the maximal case with the power-law D and the max norm.

Proof. Let $b \in E + t$, $b' \in E$ with $|b - b'|_\infty < t$. The polydisc $P(b', t) \subset E + t$ is convex, so the mean value inequality ([K:bp:p1:ext-mvt], complex normed spaces) gives $|\omega(b) - \omega(b')|_\infty \le Lt$. By (X2), $|k \cdot (\omega(b) - \omega(b'))| \le TLt \le D/2$, whence $|k \cdot \omega(b)| \ge D - D/2$. □

16.5. **The maximal homological equation.** For $\delta > 0$ put $\Lambda_\delta = \coth(\pi\delta)^N - 1 = \sum_{k \ne 0} e^{-2\pi\delta|k|_1}$ ([K:bp:p1:exp-sums](b) with $x = 2\pi\delta$); $\Lambda_\delta \le (1 + \frac{1}{\pi\delta})^N - 1$ and $\Lambda_\delta \le (4N/(\pi\delta))^N$ ([K:bp:p1:exp-sums](b),(c)).

Proposition 16.10 (Homological equation, maximal case with η-linear normal form)**.** *Let $E \subset \mathbb{R}^N$, $t > 0$, $B = E + t$, $T > 0$, $D > 0$, $\omega : B \to \mathbb{C}^N$ holomorphic, real on $B \cap \mathbb{R}^N$, with $\|D\omega\| \le L$ on B and the hypotheses of Lemma 16.9 ($tTL \le D/2$ and $|k \cdot \omega| \ge D$ on E for $0 < |k|_1 \le T$). Let $r_* > \delta > 0$ and let $P_0 : \mathbb{C}^N \times B \to \mathbb{C}$, $P_1 = (P_{1,j}) : \mathbb{C}^N \times B \to \mathbb{C}^N$ be admissible trigonometric polynomials of degree $\le T$ in θ with $\|P_0\|_{r_*;B} \le M_0$, $\|P_{1,j}\|_{r_*;B} \le M_1$. Then there are unique*

admissible trigonometric polynomials $\varphi_0, \varphi_{1,j}$ of degree $\le T$ with zero average, holomorphic on $\mathbb{C}^N \times B$, such that for every $b \in B$

$$\omega(b)\cdot\partial_\theta\varphi_0 = P_0 - \langle P_0\rangle, \qquad \omega(b)\cdot\partial_\theta\varphi_{1,j} = P_{1,j} - \langle P_{1,j}\rangle \quad (1\le j\le N),$$

and, with Λ_δ as above:

- *(a)* $\|\varphi_0\|_{r_*-\delta;B} \le \dfrac{\Lambda_\delta}{\pi D}M_0$, $\|\varphi_{1,j}\|_{r_*-\delta;B} \le \dfrac{\Lambda_\delta}{\pi D}M_1$;
- *(b) the averages $\langle P_0\rangle$, $\langle P_{1,j}\rangle$ are holomorphic on B and bounded by M_0, M_1 (no divisor is needed for them);*
- *(c) for $0<t'<t$, $j\ge 1$, $b\in E+t'$ and $\theta\in T_{r_*-\delta}$: $\|D_b^j\varphi_0\| \le j^j\frac{\Lambda_\delta M_0}{\pi D}(t-t')^{-j}$, likewise for $\varphi_{1,j}$ with M_1, and for the averages with M_0, M_1;*
- *(d) if P_0, P_1 are real for real (θ, b), then φ_0, φ_1 and the averages are real for real (θ, b), $b\in B\cap\mathbb{R}^N$;*
- *(e) $|\hat\varphi_{1,j,k}(b)| \le M_1 e^{-2\pi|k|_1 r_*}/(\pi D)$ for $0<|k|_1\le T$, $b\in B$ (used by Lemma 16.12).*

In the conventions of Section 16.1, for $\varphi=\varphi_0+\varphi_1\cdot\eta$ and $N_b = h+\omega\cdot\eta$: $\{N_b,\varphi\} = -(P-\langle P\rangle)$ with $P = P_0+P_1\cdot\eta$, $\langle P\rangle = \langle P_0\rangle + \langle P_1\rangle\cdot\eta$.

Source: Rus01 §11.4 pp. 149–151 (equation (11.12), $U_k = G_k/(i\langle k,\omega\rangle)$, and the estimate (11.18), (11.20)–(11.21) with $p=q=0$); sup-norm Fourier estimate (11.2) p. 148 replaced by [K:bp:p1:decay]. *The cutoff-wide divisor, the loss δ through Λ_δ instead of the count $(2T+1)^n$ of (11.21), and the parameter Cauchy estimates are ours.*

Proof. **Step S16.19: Divisors.** By Lemma 16.9, $|k\cdot\omega(b)| \ge D/2$ for all $b\in B$, $0<|k|_1\le T$.

Step S16.20: Coefficients. Write $P_0 = \sum_{|k|_1\le T} p_k(b)\chi_k$ with p_k holomorphic on B (Step S16.13). Since $\partial_\omega\chi_k = 2\pi i(k\cdot\omega)\chi_k$ ([K:bp:p1:fourier-basic](a)), the equation holds iff $\hat\varphi_{0,k} = p_k/(2\pi i\, k\cdot\omega(b))$ for $0<|k|_1\le T$ and $\hat\varphi_{0,0}=0$; this defines φ_0 uniquely, holomorphic on B (finite sum of quotients of holomorphic functions with non-vanishing denominator); likewise $\varphi_{1,j}$. Holomorphy in θ and periodicity are clear.

Step S16.21: (a),(e). By [K:bp:p1:decay], $|p_k|\le M_0 e^{-2\pi|k|_1 r_*}$, so $|\hat\varphi_{0,k}|\le M_0 e^{-2\pi|k|_1 r_*}/(\pi D)$, and for $\theta\in T_{r_*-\delta}$, $|\varphi_0| \le \sum_{k\ne0}|\hat\varphi_{0,k}|e^{2\pi|k|_1(r_*-\delta)} \le \frac{M_0}{\pi D}\sum_{k\ne0} e^{-2\pi\delta|k|_1} = \frac{\Lambda_\delta M_0}{\pi D}$. Same for $\varphi_{1,j}$; (e) is the coefficient bound just used.

Step S16.22: (b),(c). $\langle P_0\rangle = p_0(b)$ is holomorphic, $|p_0|\le M_0$ by [K:bp:p1:decay] ($k=0$). (c) is Lemma 16.7 applied, for fixed θ, to $b\mapsto\varphi_0(\theta,b)$ on $B = E+t$ with $M = \Lambda_\delta M_0/(\pi D)$.

Step S16.23: (d). At real b, ω is real and $\overline{p_k} = p_{-k}$ (P_0 real), so $\overline{\hat\varphi_{0,k}} = \overline{p_k}/\overline{2\pi i\, k\cdot\omega} = p_{-k}/(2\pi i(-k)\cdot\omega) = \hat\varphi_{0,-k}$.

Step S16.24: The bracket. $\partial_\theta N = 0$, $\partial_\eta N = \omega$, so $\{N,\varphi\} = -\omega\cdot\partial_\theta\varphi = -(\omega\cdot\partial_\theta\varphi_0) - (\omega\cdot\partial_\theta\varphi_1)\cdot\eta = -(P-\langle P\rangle)$. □

16.6. **The time-one flow as a simple canonical map.**

Lemma 16.11 (Flow of $\varphi_0(\theta)+\varphi_1(\theta)\cdot\eta$)**.** *Let $r_1>2\sigma>0$, $s>0$, $A_1\ge 0$ with $4(N+1)A_1\le\sigma$ ($A_1=0$ is allowed: then $\varphi=0$, $\Phi=\mathrm{id}$ and every statement is trivial; the proof below divides by A_1 nowhere), and let $\varphi_0: T_{r_1}\to\mathbb{C}$, $\varphi_1: T_{r_1}\to\mathbb{C}^N$ be holomorphic, 1-periodic, with $\|\varphi_0\|_{r_1}\le sA_1$ and $|\varphi_{1,j}|\le A_1$ on T_{r_1}. Let $\varphi(\theta,\eta) = \varphi_0(\theta)+\varphi_1(\theta)\cdot\eta$, $X_\varphi = (\varphi_1(\theta), -\partial_\theta\varphi_0(\theta) - (D\varphi_1(\theta))^\top\eta)$, $\Omega = T_{r_1-\sigma}\times B_s$ ($B_s=\{|\eta|_\infty<s\}$), and*

$$\kappa = \frac{2(N+1)A_1}{\sigma} \le \frac12, \qquad A_\eta = \frac{(N+1)sA_1}{\sigma} \le \frac{s}{4}, \qquad \nu = \nu_{s/2,\sigma},$$

where $\nu(a,c) = \max(|a|/\sigma,\ |c|/(s/2))$ ([N:bpN:p5:setting]). Put $\Omega_{\mathrm h} = T_{r_1-3\sigma/2}\times B_{3s/4}$ and $\Omega_{\mathrm f} = T_{r_1-2\sigma}\times B_{s/2}$. Then:

- *(a) (field) on Ω: $|\partial_\eta\varphi|_\infty\le A_1$, $|\partial_\theta\varphi|_\infty\le A_\eta$, and $\nu(X_\varphi)\le\kappa\le\frac12$.*

(b) (existence, displacement) $\Phi = X_\varphi^1$ *is defined and holomorphic on* Ω_{h} *(*$\Omega_{\mathrm{h}} \subset \Omega^{1/2}$ *in the notation of* [N:bpN:p5:flow]*), the trajectories* $X_\varphi^t(z)$*,* $|t| \le 1$*, stay in* Ω*, and* $|\Phi_\theta - \theta|_\infty \le A_1$*,* $|\Phi_\eta - \eta|_\infty \le A_\eta$*.*

(c) (simple form) $\Phi(\theta, \eta) = (\Theta(\theta),\ \mathbf{M}(\theta)\eta + m(\theta))$ *on* Ω_{h}*, where* Θ *is the time-one map of the autonomous system* $\dot\theta = \varphi_1(\theta)$*, and* $\mathbf{M}(\theta) = (D\Theta(\theta))^{-\top}$*.*

(d) (derivative) on $\Omega_{\mathrm{f}} \subset \Omega^1$*:* $|D\Phi - \mathbf{1}|_\nu \le 2\kappa$*; in particular* $|D\Theta - \mathbf{1}|_\infty \le 2\kappa$ *and* $|\mathbf{M} - \mathbf{1}|_\infty \le 2\kappa$*, and* $|\partial_\theta(\mathbf{M}\eta + m)|_\infty \le \kappa s/\sigma$*.*

(e) (canonical, periodic, real, inverse) $(D\Phi)^\top J\, D\Phi = J$*;* $\Phi(\theta + k, \eta) = \Phi(\theta, \eta) + (k, 0)$ *(*$k \in \mathbb{Z}^N$*); if* φ_0, φ_1 *are real on* $\mathbb{R}^N$*,* Φ *maps real points of* Ω_{h} *to real points;* $\Psi = X_\varphi^{-1}$ *is defined on* Ω_{h} *and* $\Psi \circ \Phi = \mathrm{id} = \Phi \circ \Psi$ *on* Ω_{f}*.*

(f) (parameter) If φ_0, φ_1 *depend on* $b \in B$ *(*$B \subset \mathbb{C}^N$ *open) so that* $(\theta, b) \mapsto \varphi_i(\theta, b)$ *are holomorphic with the same bounds for every* b*, then* $(b, \theta, \eta) \mapsto \Phi_b(\theta, \eta)$ *is holomorphic on* $B \times \Omega_{\mathrm{h}}$*, and the same bounds hold for every* b*.*

Source: Rus01 §12 pp. 154–155 (Lemma 12.1, Definition 12.2: simple canonical transformations) for the form; the existence, displacement, derivative, symplectic and reality statements are [N:bpN:p5:flow]*,* [N:bpN:p5:symplectic] *with* $\varepsilon = \kappa$ *(Pos93 pp. 211–212 through track N); new here: the sup-norm field bounds without weighted norms, the simple form, the parameter.*

Proof. **Step S16.25: (a) Field.** By [K:bp:p1:cauchy-strip] (with $\rho = r_1$, loss σ): on $T_{r_1-\sigma}$, $|\partial_{\theta_i}\varphi_0| \le sA_1/\sigma$ and $|\partial_{\theta_i}\varphi_{1,j}| \le A_1/\sigma$. Hence for $|\eta|_\infty < s$: $|\partial_{\theta_i}\varphi| \le sA_1/\sigma + N \cdot A_1 s/\sigma = A_\eta$ (N terms $|\partial_{\theta_i}\varphi_{1,j}||\eta_j|$), and $\partial_\eta\varphi = \varphi_1$, $|\varphi_1| \le A_1$. Then $\nu(X_\varphi) \le \max(A_1/\sigma,\ A_\eta/(s/2)) = \max(A_1/\sigma, 2(N+1)A_1/\sigma) = \kappa \le \frac{1}{2}$; $A_\eta \le s/4$ is $4(N+1)A_1 \le \sigma$.

Step S16.26: (b) Flow. Ω is open and X_φ is holomorphic on Ω with $\nu(X_\varphi) \le \kappa \le \frac{1}{2}$. [N:bpN:p5:flow](a),(b) give Φ on $\Omega^{1/2} = \{z : \bar B_\nu(z, \frac{1}{2}) \subset \Omega\}$, with trajectories in Ω. The ball $\bar B_\nu(z, c)$ is the polydisc $\bar P(z; c\sigma, cs/2)$ ([N:bpN:p5:setting]), so $\Omega_{\mathrm{h}} \subset \Omega^{1/2}$ and $\Omega_{\mathrm{f}} \subset \Omega^1$. Component bounds: along a trajectory $\dot\theta = \varphi_1$, $\dot\eta = -\partial_\theta\varphi$, bounded by A_1, A_η ([K:bp:p1:ext-mvt]), so $|\Phi_\theta - \theta| \le A_1$, $|\Phi_\eta - \eta| \le A_\eta$.

Step S16.27: (c) Simple form. The θ-equation $\dot\theta = \varphi_1(\theta)$ does not involve η, so by uniqueness ([N:bpN:p5:flow](a)) the θ-component of the trajectory through (θ_0, η_0) is the solution of this autonomous system through θ_0: $\Phi_\theta(\theta_0, \eta_0) = \Theta(\theta_0)$. Given $\theta(t)$ the η-equation is linear inhomogeneous, $\dot\eta = -\partial_\theta\varphi_0(\theta(t)) - (D\varphi_1(\theta(t)))^\top\eta$, so by (X3) $\Phi_\eta = \mathbf{M}(\theta_0)\eta_0 + m(\theta_0)$. By [N:bpN:p5:symplectic], $Y = D\Phi = \begin{pmatrix} D\Theta & 0 \\ * & \mathbf{M} \end{pmatrix}$ satisfies $Y^\top JY = J$; the $(1,2)$-block of $Y^\top JY$ is $(D\Theta)^\top\mathbf{M}$ and that of J is $\mathbf{1}$, so $\mathbf{M} = (D\Theta)^{-\top}$.

Step S16.28: (d). [N:bpN:p5:flow](c): $|D\Phi - \mathbf{1}|_\nu \le 2\kappa$ on Ω^1, i.e. $|W(D\Phi - \mathbf{1})W^{-1}|_\infty \le 2\kappa$ with $W = \mathrm{diag}(\sigma^{-1}\mathbf{1}, (s/2)^{-1}\mathbf{1})$. The diagonal blocks of $W(\cdot)W^{-1}$ equal the diagonal blocks of the matrix, and a block has norm at most the row norm of the whole matrix: $|D\Theta - \mathbf{1}|, |\mathbf{M} - \mathbf{1}| \le 2\kappa$. The lower-left block of $W(D\Phi - \mathbf{1})W^{-1}$ is $(2/s)\sigma\,\partial_\theta(\mathbf{M}\eta + m)$, whence the last bound $(2\kappa \cdot s/(2\sigma))$.

Step S16.29: (e),(f). Symplecticity: [N:bpN:p5:symplectic]; periodicity and reality: [N:bpN:p5:flow](d),(e) (φ_i are $\mathbb{Z}^N$-periodic, so X_φ is invariant under $(\theta, \eta) \mapsto (\theta + k, \eta)$); inverse: [N:bpN:p5:flow](c) and $\Psi = X_{-\varphi}^1$. For (f) consider the field $\tilde F(b, z) = (0, X_{\varphi_b}(z))$ on $B \times \Omega$ with the complex norm $\tilde\nu(\beta, z) = \max(|\beta|_\infty/\varsigma, \nu(z))$ for a $\varsigma > 0$; it is holomorphic with $\tilde\nu(\tilde F) = \nu(X_{\varphi_b}) \le \frac{1}{2}$, and its flow is $(b, \Phi_b(z))$. By [N:bpN:p5:flow](b) the flow is holomorphic on $\tilde\Omega^{1/2} \supset \{b : P(b, \varsigma/2) \subset B\} \times \Omega^{1/2}$; as B is open every b is covered for ς small. □

Lemma 16.12 (Lipschitz bound for the angle map on the reals)**.** *Let* $\varphi_1 : \mathbb{R}^N \to \mathbb{R}^N$ *be a real trigonometric polynomial of degree* $\le T$*, 1-periodic, and* Θ^t *the flow of* $\dot\theta = \varphi_1(\theta)$ *on* $\mathbb{R}^N$ *(global).*

Put $\Lambda = \sup_{x\in\mathbb{R}^N} |D\varphi_1(x)|_\infty$. *Then for all* $|t| \le 1$ *and* $\theta \in \mathbb{R}^N$: $|D\Theta^t(\theta) - \mathbf{1}|_\infty \le e^\Lambda - 1$, $\Theta^t(\theta+k) = \Theta^t(\theta) + k$, $|\Theta^t(\theta) - \theta|_\infty \le \sup|\varphi_1|_\infty$, *and*

$$\Lambda \;\le\; 2\pi T \max_i \sum_{0<|k|_1\le T} |\hat\varphi_{1,i,k}|.$$

If $|\hat\varphi_{1,i,k}| \le M_1 e^{-2\pi|k|_1 r_*}/(\pi D)$ *(Proposition 16.10(e)), then* $\Lambda \le \Lambda_{\mathbb{R}} := 2TM_1\Lambda_{r_*}/D$.
Source: Ours; this is the source of the factor $e^{cC_\nu T_\nu}$ *of Rus01 (13.x), made explicit for the only place where it is needed, the real variables (no domain loss).*

Proof. **Step S16.30: Global flow.** φ_1 is smooth, bounded and Lipschitz, so solutions are global; periodicity of φ_1 gives $\Theta^t(\theta+k) = \Theta^t(\theta) + k$ (uniqueness); $|\Theta^t - \theta| \le |t|\sup|\varphi_1|$.

Step S16.31: Derivative. $Y(t) = D\Theta^t(\theta)$ solves $Y' = D\varphi_1(\Theta^t\theta)Y$, $Y(0) = \mathbf{1}$ ((X3)); $|Y(t)| \le e^{\Lambda|t|}$ and $|Y(t) - \mathbf{1}| \le \int_0^{|t|} \Lambda e^{\Lambda\tau}\,d\tau = e^{\Lambda|t|} - 1$.

Step S16.32: The constant. The i-th row sum of $D\varphi_1(x)$ is $\sum_j|\partial_j\varphi_{1,i}(x)| \le \sum_k|\hat\varphi_{1,i,k}|\,2\pi\sum_j|k_j| \le 2\pi T\sum_{0<|k|_1\le T}|\hat\varphi_{1,i,k}|$ ($|\chi_k| = 1$ on $\mathbb{R}^N$, $k = 0$ drops out). With the decay bound, the sum is $\le M_1\Lambda_{r_*}/(\pi D)$. □

16.7. **The canonical step.**

Proposition 16.13 (Canonical step in the maximal case)**.** ***Data.*** $N \ge 1$; $E \subset \mathbb{R}^N$, $t > 0$, $B = E + t$; $r, s > 0$; $h : B \to \mathbb{C}$ *and* $\omega : B \to \mathbb{C}^N$ *holomorphic, real on real points;* $R : \mathcal{D}(r, s; B) \to \mathbb{C}$ *admissible with* $\varepsilon = \|R\|_{r,s;B} < \infty$ *(real on real points, if reality is wanted);* $\varepsilon = 0$ *is allowed, and then* $A_1 = 0$ *below,* $R_* = 0$, $\varphi = 0$, $\Phi = \mathrm{id}$, $R_+ = 0$ *(the argument never divides by* ε *or* A_1*; this is Lemma 16.11 with* $A_1 = 0$*; review P11A2-03);* $N_b(\theta,\eta) = h(b) + \omega(b)\cdot\eta$ *and* $H = N_b + R$.

Step parameters. $T > 0$; $d, \delta, \sigma > 0$ *with* $r_+ := r - d - \delta - 2\sigma > 0$; $q \in (0, \frac12]$; $D > 0$. *Put* $E_T = \varepsilon_N(T, d)$ *and* $A_1 = \Lambda_\delta(1 + E_T)\varepsilon/(\pi Ds)$.

Hypotheses.

(C1): *(cutoff error)* $E_T \le 1$ *(for example* $E_T \le \alpha \le 1$ *by Lemma 16.3 with* $d = d_T(\alpha) \le 1$*);*
(C2): *(divisor)* $\|D\omega\| \le L$ *on* B, $tTL \le D/2$, *and* $|k\cdot\omega(b')| \ge D$ *for* $b' \in E$, $0 < |k|_1 \le T$ *(for* $E = K_\nu$, $D = \gamma T^{-\tau_D}$ *this is Lemma 16.8);*
(C3): *(smallness)* $4(N+1)A_1 \le \sigma$, *i.e.* $4(N+1)\Lambda_\delta(1+E_T)\varepsilon \le \pi Ds\sigma$.

Conclusions. *Let* $R_* = R_{*0} + R_{*1}\cdot\eta$ *be the cutoff of Proposition 16.6 (with* T, d, q*),* $\Delta h = \langle R_{*0}\rangle$, $\Delta\omega = \langle R_{*1}\rangle$, $N_+ = h + \Delta h + (\omega + \Delta\omega)\cdot\eta$, $D_+ = \mathcal{D}(r_+, qs; B)$, *and* $\varphi = \varphi_0 + \varphi_1\cdot\eta$ *the solution of Proposition 16.10 with* $P = R_*$, $r_* = r - d$, $M_0 = (1 + E_T)\varepsilon$, $M_1 = M_0/s$. *Then:*

(i) (normal form) $\Delta h, \Delta\omega$ *are holomorphic on* B, $|\Delta h| \le (1 + E_T)\varepsilon \le 2\varepsilon$, $|\Delta\omega|_\infty \le (1 + E_T)\varepsilon/s \le 2\varepsilon/s$; N_+ *has the form of* N.
(ii) (the map) $\Phi_b = X^1_{\varphi_b}$ *is defined and holomorphic on* D_+ *jointly in* (θ,η,b)*, with values in* $\mathcal{D}(r_+ + \sigma, s; B)$*; precisely* $|\Phi_\theta - \theta|_\infty \le A_1 \le \sigma/(4(N+1))$, $|\Phi_\eta - \eta|_\infty \le A_\eta := (N+1)sA_1/\sigma \le s/4$; Φ *is simple (Lemma 16.11(c)), canonical,* $\Phi(\theta + k, \eta) = \Phi(\theta,\eta) + (k, 0)$*, real at real points if the data are real; on* D_+: $|D\Phi - \mathbf{1}|_\nu \le 2\kappa$ *with* $\kappa = 2(N+1)A_1/\sigma \le \frac12$ *and* $\nu = \nu_{s/2,\sigma}$; Φ *is invertible with inverse* X_φ^{-1} *(Lemma 16.11(e)).*
(iii) (conjugation) $H_b \circ \Phi_b = N_+ + R_+$ *on* D_+*, where* R_+ *is holomorphic on* D_+*, 1-periodic in* θ*, real at real points if the data are real, and*

$$\|R_+\|_{D_+} \;\le\; \varepsilon\Big(\frac{q^2}{1-q} + 2E_T\Big) + \frac{N(4N+5)(5+4E_T)(1+E_T)\,\Lambda_\delta}{\pi\, D\, s\,\sigma}\,\varepsilon^2$$
$$\le\; \varepsilon\Big(2q^2 + 2E_T\Big) + \frac{18\,N(4N+5)\Lambda_\delta}{\pi Ds\sigma}\varepsilon^2.$$

(iv) (real Lipschitz) for real $b \in E$ and real θ: $|D\Theta_b - \mathbf{1}|_\infty \le e^{\Lambda_{\mathbb{R}}} - 1$, with $\Lambda_{\mathbb{R}} = 2T(1 + E_T)\varepsilon\,\Lambda_{r-d}/(Ds)$, and $\Theta_b(\theta + k) = \Theta_b(\theta) + k$.

(v) (parameter derivatives) for $0 < t' < t$, $j \ge 1$, on $E + t'$ and at every $(\theta, \eta) \in D_+$: $\|D_b^j \Delta h\| \le j^j \frac{(1+E_T)\varepsilon}{(t-t')^j}$, $\|D_b^j \Delta\omega\| \le j^j \frac{(1+E_T)\varepsilon}{s(t-t')^j}$, $\|D_b^j(\Phi_\theta - \theta)\| \le \frac{j^j A_1}{(t-t')^j}$, $\|D_b^j(\Phi_\eta - \eta)\| \le \frac{j^j A_\eta}{(t-t')^j}$, $\|D_b^j R_+\| \le \frac{j^j \|R_+\|_{D_+}}{(t-t')^j}$.

Source: Rus01 §§13.6–13.12 pp. 159–164 (the estimate of the remainder after one step; not read: only the structure is used, the constants are recomputed here), Lemma 11.1 p. 148 (Poisson bracket estimate) replaced by Cauchy estimates, §12; Pos93 pp. 211–212 through [N:bpN:p5:taylor]. *Constants are ours (D-9).*

Proof. **Step S16.33: Cutoff.** Proposition 16.6 with the given T, d, q (here $d < r$ since $r_+ > 0$): $R_* \in$ trigonometric polynomials of degree $\le T$, with $\|R_{*0}\|_{r-d;B} \le M_0$, $\|R_{*1,j}\|_{r-d;B} \le M_1$, $\|R_*\|_{r-d,s;B} \le 2M_0$, $\|R - R_*\|_{r-d,qs;B} \le \varepsilon(\frac{q^2}{1-q} + 2E_T)$.

Step S16.34: Homological equation. Proposition 16.10 with $r_* = r - d$ (and $\delta < r - d$), $P_0 = R_{*0}$, $P_1 = R_{*1}$: hypothesis (C2). We get φ on T_{r_1}, $r_1 = r - d - \delta$, with $\|\varphi_{1,j}\|_{r_1;B} \le \Lambda_\delta M_1/(\pi D) = A_1$ and $\|\varphi_0\|_{r_1;B} \le \Lambda_\delta M_0/(\pi D) = sA_1$, and (i) from part (b) of that proposition ($\Delta h = \langle R_{*0}\rangle$, $|\Delta h| \le M_0$, etc.). Moreover $\{N_b, \varphi\} = -(R_* - \langle R_*\rangle) =: w$.

Step S16.35: Flow. Lemma 16.11 with $r_1 = r - d - \delta = r_+ + 2\sigma$ (so $r_1 - 2\sigma = r_+$), A_1 as above, hypothesis (C3) $= 4(N+1)A_1 \le \sigma$, fibrewise in $b \in B$ and with the parameter statement (f). The domain $\Omega_{\mathrm{f}} = T_{r_1 - 2\sigma} \times B_{s/2} = T_{r_+} \times B_{s/2} \supset D_+$ fibrewise ($q \le \frac12$), so (ii) follows from Lemma 16.11(b)–(f); the displacements are the stated A_1, A_η.

Step S16.36: Taylor identity. Fix $b \in B$ and $z \in D_+ \cap \{b \text{ fixed}\}$. Put $\Omega = T_{r_1 - \sigma} \times B_s$; N_b is entire and R is holomorphic on $T_r \times B_s \supset \Omega$ ($r_1 \le r$); the trajectory $X_\varphi^t(z)$ stays in Ω (Lemma 16.11(b)) and $z \in \Omega^{1/2}$. [N:bpN:p5:taylor] (the displayed formula for $(h + f_0) \circ X^1$) with $h = N_b$, $f_0 = R$, gives

$$H \circ \Phi(z) = N_b(z) + \{N_b, \varphi\}(z) + R(z) + \mathcal{R}(z), \qquad \mathcal{R}(z) = \int_0^1 \{u_t, \varphi\}(X_\varphi^t z)\, dt, \quad u_t = (1-t)w + R.$$

With $\{N_b, \varphi\} = w = -(R_* - \langle R_*\rangle)$: $N_b + w + R = N_b + \langle R_*\rangle + (R - R_*) = N_+ + (R - R_*)$. Hence $H \circ \Phi = N_+ + R_+$ with $R_+ = (R - R_*) + \mathcal{R}$ on D_+. Holomorphy of R_+ in (θ, η, b) is that of H and of Φ (joint, Lemma 16.11(f)); periodicity and reality follow from those of H, Φ and N_+.

Step S16.37: Size of u_t. u_t is holomorphic on $T_{r-d} \times B_s$. There $|w| \le |R_*| + |\langle R_*\rangle| \le 4(1 + E_T)\varepsilon$ ($\langle R_*\rangle(\eta)$ is an average of values of R_* at real θ, hence bounded by $\|R_*\|_{r-d,s;B} \le 2(1 + E_T)\varepsilon$), so $|u_t| \le U := \varepsilon(5 + 4E_T)$ ($\le 9\varepsilon$) for $t \in [0, 1]$.

Step S16.38: Cauchy estimates for the bracket. Let $z \in D_+$ and $z_t = X_\varphi^t(z) = (\theta_t, \eta_t)$. By the displacement bounds, $\theta_t \in T_{r_+ + A_1}$, $|\eta_t| < qs + A_\eta \le qs + s/4$. The polydisc around z_t of radii (σ, ρ_u) with $\rho_u = (1 - q)s/2$ ($\ge s/4$) lies in $T_{r-d} \times B_s$: $r_+ + A_1 + \sigma \le r_+ + \frac98\sigma \le r - d$, and $qs + \frac{s}{4} + \frac{(1-q)s}{2} \le s$ for $q \le \frac12$. By [K:bp:p1:ext-cauchy] (one variable at a time), $|\partial_{\theta_i} u_t(z_t)| \le U/\sigma$ and $|\partial_{\eta_i} u_t(z_t)| \le U/\rho_u \le 4U/s$. Also $z_t \in \Omega$ so $|\partial_{\eta_i}\varphi| \le A_1$, $|\partial_{\theta_i}\varphi| \le A_\eta$ at z_t (Lemma 16.11(a)). Hence

$$|\{u_t, \varphi\}(z_t)| \le \sum_{i=1}^{N} \big(|\partial_{\theta_i} u_t||\partial_{\eta_i}\varphi| + |\partial_{\eta_i} u_t||\partial_{\theta_i}\varphi|\big) \le N\Big(\frac{UA_1}{\sigma} + \frac{4U}{s} \cdot \frac{(N+1)sA_1}{\sigma}\Big) = \frac{N(4N+5)UA_1}{\sigma}.$$

Step S16.39: Assembly. Integrate over $t \in [0, 1]$: $|\mathcal{R}| \le N(4N + 5)UA_1/\sigma$ on D_+; insert $U = \varepsilon(5 + 4E_T)$, $A_1 = \Lambda_\delta(1 + E_T)\varepsilon/(\pi Ds)$ and the bound of $R - R_*$ (Step S16.33; $D_+ \subset T_{r-d} \times B_{qs}$). The second form uses $q^2/(1 - q) \le 2q^2$, $E_T \le 1$, $(5 + 4E_T)(1 + E_T) \le 18$.

Step S16.40: Real Lipschitz bound. Lemma 16.12 with $r_* = r - d$, $M_1 = (1 + E_T)\varepsilon/s$ and Proposition 16.10(e); for real $b \in E$, φ_1 is real (Prop. 16.10(d)) and the real flow of $\dot\theta = \varphi_1$ is the restriction of Θ_b (uniqueness).

Step S16.41: Parameter derivatives. Lemma 16.7 applied, for fixed $(\theta, \eta) \in D_+$, to $b \mapsto \Delta h, \Delta\omega, \Phi_\theta - \theta, \Phi_\eta - \eta, R_+$ on $B = E + t$, using their bounds from (i), (ii), (iii). □

Remark 16.14 (Reading the statement; the next step). (a) *Domain bookkeeping.* The angular strip loses $d + \delta + 2\sigma$ ($d = d_T$ Fourier, δ homological, 2σ Cauchy/flow); the action radius becomes qs ($q \le \frac{1}{2}$); the parameter domain $B = E + t$ is *not* reduced by the step itself (the parameter Cauchy estimates (v) cost $t - t'$ only if derivatives are wanted). The output (N_+, R_+) on D_+ has the input form with $(r, s, \varepsilon, h, \omega) \to (r_+, qs, \varepsilon_+, h+\Delta h, \omega+\Delta\omega)$, $\varepsilon_+ \le$ the bound of (iii). *(b) What the next step needs on the frequency:* holomorphy of $\omega+\Delta\omega$ on B and the Lipschitz bound $L_+ \le L + \sup\|D\Delta\omega\|$, with $\|D\Delta\omega\| \le (1 + E_T)\varepsilon/(s(t - t'))$ on $E + t'$ (v) ($j = 1$); so the next step runs on $E_+ \subset E$ with $t_+ \le t'$: File 11a-3 sets $t_{\nu+1} = t'$ and checks $t_{\nu+1}T_{\nu+1}L_{\nu+1} \le D_{\nu+1}/2$. *(c)* The only place where the *small divisor* enters is D in (C2)–(C3), (iii), (iv); the cutoff T enters through E_T, d and (C2). *(d)* The first term of (iii) is linear in ε (cutoff errors q^2, E_T); only the second is quadratic. In Rus01's scheme $\varepsilon_{\nu+1} = \varepsilon_\nu\delta^\kappa$ is only geometric, so one needs $2q_\nu^2 + 2\alpha_\nu \lesssim \delta^\kappa/2$ and the quadratic term $\lesssim \varepsilon_\nu \cdot \varepsilon_\nu/(D_\nu s_\nu \sigma_\nu) \ll \delta^\kappa\varepsilon_\nu$ (File 11a-3).

16.8. **Sources, deviations, errata, what File 11a-3 takes.**
Sources read. Rus01 §§7.1–7.5 (pp. 134–136), §8.1–8.3 (p. 137), §§11.1–11.4 (pp. 148–153), §12 (pp. 154–155), §13.1–13.2 (p. 156; schedule only); Rus05 §§9–11 (pp. 26–31) as comparison (equation (46)); the review `reviews/p11a-russmann-check.md` (rows `analytic-cutoff`, `homological-maximal`, `canonical-step`; errata R05-1, R05-2) and `reviews/p11a1-review-1.md` (P11A1-P02); track N Parts 4 and 5. Not read: Rus01 §10 (pp. 143–147, the flow of the full vector field with elliptic/hyperbolic variables), §§13.3–13.12 (pp. 157–164): the maximal case needs neither, the one-step remainder is recomputed above.
Deviations (all recorded; D-numbers as in `PLAN-11a.md`).

- **D-4** (Fourier cutoff): Lemma 16.2 replaces Rus01 Thm 7.3 and Rus05 by a short proof from Cauchy decay; no multipliers; truncation in $|k|_1$; the constant c_N is explicit; the loss is d_T of Lemma 16.3, not Rus01's $\rho = \frac{n}{T}\log\frac{4n}{\alpha}$ with error $\frac{4}{5}\alpha$. The cube/Euclid rescaling that causes R05-2 does not occur: T and d are used in the same ℓ^1 lattice norm throughout.
- **D-5** (algebraic cutoff): Lemma 16.5, Taylor remainder of order 2 with ratio q (Rus01's δ^β); no appeal to [18] Lemma 6; only degree ≤ 1 is needed.
- **D-10** (sup norms): all estimates are in sup norms on $\mathcal{D}(r, s; B)$ (as Rus01 §7–§13), not in the weighted norm of track N; the passage between them is not needed, since only $\mathcal{T}_T$ and the homological equation use Fourier coefficients (Cauchy decay, [K:`bp:p1:decay`]).
- **D-11** (homological equation): solved coefficientwise, Rus01 (11.12) with $p = q = 0$; the loss δ enters through $\Lambda_\delta = \coth(\pi\delta)^N - 1$, not through $(2T + 1)^n$ of (11.21); the divisor is $D/2$ on all of B (extension from E, Lemma 16.9).
- **D-12** (sign): $X_\phi = (\partial_\eta\phi, -\partial_\theta\phi)$ and $\omega \cdot \partial_\theta\varphi = +(R_* - \langle R_*\rangle)$ (track N's convention). This is the same convention as Rus01's $\{S, N\} = F - [F]$, (11.12) after row/column conversion (Rus01 §10.1.8 uses $J_{\rm src} = \left(\begin{smallmatrix} 0 & -\mathbf{1} \\ \mathbf{1} & 0 \end{smallmatrix}\right)$ acting on row vectors, so $S_x J_{\rm src} = (S_y, -S_x)$; the displayed matrices differ only because of row versus column multiplication); the positive homological sign corrects the negative sign $\partial_\omega\varphi = -(R - \langle R\rangle)$ of `PLAN-11a.md`.
- **D-13** (higher parameter Cauchy): by polydisc iteration, Lemma 16.7; no polarization constant.
- **D-14** (flow): Rus01 Thm 8.1 (flow in $\mathbb{C}^m$ with $|f(x)| \le Me^{|x|}$ and the Poisson bracket estimate 11.1) is replaced by [N:`bpN:p5:flow`] and Cauchy estimates of $\{u, \varphi\}$, which use the explicit polydisc geometry of the strips.

- **D-15** (strip bookkeeping): the angular loss is $d+\delta+2\sigma$ per step; it is a datum of File 11a-3 (condition (I0), $\sum_\nu(d_\nu+\delta_\nu+2\sigma_\nu)\le s/2$), with $\sum_\nu d_\nu$ given by Lemma 16.4.

Suspected errata. (1) Rus05 p. 30, equation (46) and p. 31 (R05-2, from the review): the dimension rescaling $T\to T/\sqrt{n}$ produces $E(T\rho/n)$, not $E(T\rho)$; not used here, recorded in the ledger by the orchestrator. (2) No new defect found in Rus01 pp. 134–136, 148–155 beyond typographical notation (the convention of rows and columns in §12). The statement of Rus01 Thm 7.3 is correct as quoted, but its proof imports [21] (a theorem of Akhiezer proved elsewhere); see D-4.
Doubts about frozen statements. None: this file touches no frozen declaration.
What File 11a-3 must take from this file.

(T1) *Per step, Proposition 16.13* with a stage index: parameters $(T_\nu, d_\nu, \delta_\nu, \sigma_\nu, q_\nu, D_\nu)$; the hypotheses (C1)–(C3) become the schedule inequalities: $d_\nu = d_{T_\nu}(\alpha_\nu)\le 1$ with $\alpha_\nu\le 1$ (Lemma 16.3); $D_\nu=\gamma T_\nu^{-\tau_D}$ with the divisor hypothesis from Lemma 16.8 (here $E=K_{\nu+1}$, the retained set, with the divisor bound of Lemma 17.15; $T=T_\nu$, $\tau=\tau_D\ge 0$); $t_\nu T_\nu L_\nu\le D_\nu/2$; and $4(N+1)\Lambda_{\delta_\nu}(1+E_{T_\nu})\varepsilon_\nu\le \pi D_\nu s_\nu\sigma_\nu$.

(T2) *The recursions:* $r_{\nu+1}=r_\nu-d_\nu-\delta_\nu-2\sigma_\nu$; $s_{\nu+1}=q_\nu s_\nu$; $\varepsilon_{\nu+1}\le \varepsilon_\nu(2q_\nu^2+2\alpha_\nu)+\frac{18N(4N+5)\Lambda_{\delta_\nu}}{\pi D_\nu s_\nu\sigma_\nu}\varepsilon_\nu^2$; $\omega_{\nu+1}=\omega_\nu+\Delta\omega_\nu$ with $|\Delta\omega_\nu|_\infty\le 2\varepsilon_\nu/s_\nu$ (Rus01 $3\varepsilon_\nu/(2s_\nu)$ when $E_T\le\frac12$); $h_{\nu+1}=h_\nu+\Delta h_\nu$, $|\Delta h_\nu|\le 2\varepsilon_\nu$.

(T3) *Displacement and Lipschitz data for File 11a-4:* $|X_{\nu,\theta}-\theta|_\infty\le A_{1,\nu}$, $|X_{\nu,\eta}-\eta|_\infty\le A_{\eta,\nu}$ on $D_{+,\nu}$; $|DX_\nu-\mathbf{1}|_\nu\le 2\kappa_\nu$ on the complex domain in the stage-dependent weighted norm $\nu=\nu_{s_\nu/2,\sigma_\nu}$ (the summable quantity $\sum\kappa_\nu$ is $2(N+1)\sum A_{1,\nu}/\sigma_\nu$), *and* the unweighted bound $|DX_\nu-\mathbf{1}|_\infty\le 2\kappa_\nu+\kappa_\nu s_\nu/\sigma_\nu$ in the coordinates (θ,η) (diagonal blocks $\le 2\kappa_\nu$, lower-left block $\le\kappa_\nu s_\nu/\sigma_\nu$ by Lemma 16.11(d); maximal row sum), whose sum over ν the consumer must control (File 11a-3, Lemma 17.20(E2): $\le 2^{-12}$); the weighted bound alone is not a bound for the composition in a fixed norm (review P11A2-04); and on the reals $|D\Theta_\nu-\mathbf{1}|_\infty\le e^{\Lambda_{\mathbb{R},\nu}}-1$ with $\Lambda_{\mathbb{R},\nu}=2T_\nu(1+E_{T_\nu})\varepsilon_\nu\Lambda_{r_\nu-d_\nu}/(D_\nu s_\nu)$ (the analogue of Rus01's $e^{cC_\nu T_\nu}$). The factor $\Lambda_{r_\nu-d_\nu}$ is large when $r_\nu-d_\nu$ is small; File 11a-3 keeps $r_\nu\ge s/2$-type lower bounds on the strip.

(T4) *Parameter regularity (for the chain extension, deviation D-7):* (v) of the proposition gives $\|D_b^j\Delta\omega_\nu\|\le j^j(1+E_T)\varepsilon_\nu/(s_\nu(t_\nu-t_{\nu+1})^j)$ on $E+t_{\nu+1}$, for $j\le\mu+1$: this is the *holomorphic* size of the increments, from which the mollified cut-offs of File 11a-3 produce the C^μ-size.

(T5) *The interface to File 11a-1:* Lemma 16.8 is the only point where $\tilde\omega_k$, $\nu(k)$, (I1)–(I3) enter this file; it turns them into the single divisor $D_\nu=\gamma T_\nu^{-\tau_D}$ used by the homological equation, on E only; the finite-stage invariant that pays its hypothesis on the retained set is Lemma 17.15 (File 11a-3), which supersedes the limit-based form of (I3) (Remark 17.16; review P11A2-01); the passage to $E+t$ is Lemma 16.9 (requires $t_\nu T_\nu L_\nu\le D_\nu/2$, another schedule inequality).

What is reused and what is new. Track N: [N:bpN:p5:flow], [N:bpN:p5:symplectic], [N:bpN:p5:taylor], [N:bpN:p5:setting], [N:bpN:p4:ext-param] (as cited). Not used: [N:bpN:p4:projections] (weighted-norm form of the cutoff, $e^{-2\pi K\sigma}$, which has no polynomial prefactor but requires a weighted norm of R, i.e. a $\coth^N$ conversion at every step, [N:bpN:p4:sup-weighted]; the sup-norm version above is cheaper here because the pairing of $|k|_1\le T$ with a width $d\sim T^{-1}\log T$ is the only loss), [N:bpN:p5:composition], [N:bpN:p5:transform], [N:bpN:p5:canonical-map] (weighted norms; Part 6's Iterative Lemma is not usable, see PLAN-11a.md). Track K: [K:bp:p1:decay], [K:bp:p1:summation], [K:bp:p1:exp-sums], [K:bp:p1:cauchy-strip], [K:bp:p1:fourier-basic], [K:bp:p1:ext-cauchy], [K:bp:p1:ext-mvt], [K:bp:p4:lattice-count]. New in this file: all of the statements above; in particular the simple-form and real-Lipschitz statements, the loss choice and its sum, and the explicit constants of the one-step remainder.

External facts used. (X1)–(X3); [K:bp:p1:ext-cauchy], [K:bp:p1:ext-mvt], [K:bp:p1:ext-integral]; [N:bpN:p4:ext-param]. The multivariable Cauchy coefficient estimate is stated in the form used; the Mathlib names are likely, not verified.
Numerical checks. `.drafts/P/bp-scratch-bpP-p11a-2/chk1.py`: for $N \le 6$, $d \in [0.01, 5]$, $T \le 100$, the exact lattice tail $\sum_{|k|_1 > T} e^{-2\pi|k|_1 d}$ is at most $0.12\, c_N(1 + d^{-N})e^{-\pi T d}$ and $\varepsilon_N(T, d_T) \le 0.34\,\alpha$ whenever $d_T \le 1$ (evidence only). `chk2.py`: $N = 1$, $R = \varepsilon_0 \cos(2\pi\theta + \eta)$, $\omega = \sqrt{2}$: the real remainder $H \circ \Phi - N_+$ computed by numerical flow is $3 \cdot 10^{-5}$ ($\approx \varepsilon_0 q^2/2$, the Taylor part; a wrong sign of φ would give $\approx \varepsilon_0$), below the bound of (iii) ($3.8 \cdot 10^{-2}$).
Forward references and label index. No forward reference to an unwritten label. Labels: `c2-conv`, `domain-def`, `fourier-cutoff`, `loss-choice`, `loss-sum`, `taylor-cutoff`, `cutoff`, `param-cauchy`, `divisor-wide`, `divisor-perturb`, `homological-maximal`, `affine-flow`, `real-lipschitz`, `canonical-step`, `step-reading`.

17. Part 11a-3: the iteration schedule, the inductive lemma, initialisation and parameter selection

This is the third of the five files of Part 11a (split plan `PLAN-11a.md`). Files 11a-1 and 11a-2 prove the two halves of one KAM step: the *measure* of the parameters whose frequency violates a power-law small-divisor condition, and the *analytic* step (Fourier and Taylor cutoffs, homological equation, simple canonical map) on a complex neighbourhood of the retained parameter set. This file runs the step infinitely often. It fixes the schedule of every quantity (cutoff T_ν, size ε_ν, strip width r_ν, action radius s_ν, parameter collar t_ν), proves that the hypotheses of File 11a-2 and File 11a-1 hold at every stage, chooses the free parameters explicitly, and extracts the set of parameters on which the frequencies converge to a (γ, τ)-Diophantine vector.
The mechanism. Rüssmann's iteration is only *geometric*: the remainder shrinks by a fixed factor $\varpi < 1$ per step (it is not superlinear), because the cutoff error and the loss of action radius contribute terms linear in ε_ν (Remark 16.14(d)). Geometric decay is enough because the three competing growth rates are all *fixed geometric ratios*: the inverse divisor $D_\nu^{-1} = \gamma^{-1} T_\nu^{\tau}$ grows by 2^τ per step, the inverse action radius by $1/q$, the inverse collar t_ν^{-1} by $2^{\tau+1}$; the decay ratio ϖ and the shrink factor q of the action radius are chosen *once*, depending only on (N, μ, τ), so small that ϖ beats every one of these growth factors at every stage (Lemma 17.4). What remains is a finite list of inequalities at stage $\nu = 0$, polynomial in the data (Definition 17.12); this is where the *shrinking-domain budget* is spent (the term $\frac{N}{2} C_1 s_0^2$ of the initial remainder, Lemma 17.11).
What is proved.

- `geom-recursion`, `schedule-consts`, `schedule-ratios`, `schedule`, `strip-budget`, `schedule-props` (Lemma 17.2, Definitions 17.3, 17.5, Lemmas 17.4, 17.6, 17.7): the geometric-recursion version of track K's iteration lemma and the schedule, with every exponent inequality of Rus01 (13.1.2), (13.1.6) replaced by an explicit one.
- `smooth-cutoff`, `finite-extension` (Lemmas 17.8, 17.9): the chain extension $\tilde{\omega}_\nu$ of the stage-ν frequency to a C^μ map on the real neighbourhood of K by explicit mollified cutoffs, with the closeness to ω that File 11a-1 needs (deviation D-7: it *works*; Rus01 §§19–20 are not needed).
- `initial`, `adm-params`, `adm-choice` (Lemma 17.11, Definition 17.12, Proposition 17.13): the translation of Rus01 §14, the initial remainder, the explicit admissible parameters $(T_0, t_0, s_0, \varepsilon_{\mathrm{adm}})$ and the proof that they satisfy the inequalities (A1)–(A7).
- `finite-divisor`, `finite-measure`, `inductive-lemma` (Lemmas 17.15, 17.17, Proposition 17.18): the *finite-stage divisor invariant* (it replaces the limit-based tail (I3) of File 11a-1, see Remark 17.16), the nonemptiness of the retained set by the measure estimate of File 11a-1, and the inductive lemma.

- `parameter-selection`, `exported-sums` (Theorem 17.19, Lemma 17.20): the retained sets, the limit frequency, its Diophantine property and the measure loss $\le \varepsilon^*$; the displacement, derivative and Lipschitz budgets that File 11a-4 consumes.

Verdicts. (R1) The admissible size $\varepsilon_{\mathrm{adm}}$ is an explicit minimum of three expressions (Remark 17.14); it is *not* a monomial, and its dependence on the data is controlled only when *all* the data are controlled: Proposition 17.21 gives a lower bound $\varepsilon_{\mathrm{adm}} \ge c\,u^B$ along any family of data with polynomially controlled $\vartheta, s, \beta, \varepsilon^*$ from below and M, M_0, C_1, d from above, with an explicit exponent B (independent of the iteration index), and Corollary 17.23 is the special case needed by Part 11 (everything fixed except the amount β). Without a bound on M or on C_1 no such lower bound holds (review P11A3-01; see the Remark after the Proposition). Numerics: Section 17.10. (D-7) The chain extension by mollified cutoffs is valid under the exponent condition $g_3 = \varrho\, 2^{(\tau+1)\mu} \le 2^{-5}$, which the choice of ϖ enforces (this is the analogue of Rus01's $\tau\mu < \tau_0$, Lemma 17.4); no fall-back to Rus01 §§19–20 is needed.

17.1. **Standing data, conventions, external facts.** All conventions are those of Files 11a-1 and 11a-2: angles of period 1, characters $e^{2\pi i k\cdot\theta}$, max norms $|\cdot|_\infty$ on $\mathbb{R}^N, \mathbb{C}^N$ with induced operator norms, $|k|_1 = \sum|k_i|$, domains T_r, $\mathcal{D}(r,s;B) = \{(\theta,\eta,b) : \theta \in T_r,\ |\eta|_\infty < s,\ b \in B\}$ (Definition 16.1, written $\mathcal{D}(r,s;B)$ here), $E + t = \{b \in \mathbb{C}^N : \exists b' \in E,\ |b - b'|_\infty < t\}$, and $\|F\|_{r,s;B}$ the supremum. Euclidean norms and the Euclidean real thickening $K_\vartheta = \{y \in \mathbb{R}^N : \mathrm{dist}_2(y,K) \le \vartheta\}$ appear only where File 11a-1 uses them (derivative tensors $|\cdot|_{\mathrm{dg}}$, $\|\cdot\|^\mu_{\mathrm{dg},A}$), with the factors $\sqrt{N}$ stated. N is the number of actions (equal to the number of angles and to the dimension of the frequency vector: $m = N$ in File 11a-1).

Standing assumption 17.1 (Data of the selection theorem)**.** (a) $N, \mu \ge 1$ integers, $\rho \in \{1,2\}$, $\tau \ge 0$ real with $\tau > \frac{N\mu}{\rho} - 1$ (the Diophantine exponent, called τ_D in the plan).

(b) $K \subset \mathbb{R}^N$ compact with $\mathrm{meas}\, K > 0$; $\vartheta \in (0,1]$; $s \in (0,1]$ (angular strip width); $C_1 \ge 1$; $\varepsilon^* \in (0, \mathrm{meas}\, K)$. (Normalisation: C_1 occurs only as an upper bound for $\|D\omega\|$, so it may always be enlarged to $\max\{1, C_1\}$; this is done so that $\varepsilon_{\mathrm{adm}}$ is not made small by a small Hessian bound, review P11A3-01.)

(c) (Hamiltonian) $U := K + 2\vartheta \subset \mathbb{C}^N$. $h : U \to \mathbb{C}$ holomorphic, real on $U \cap \mathbb{R}^N$; $\omega := \nabla h$ and $\|D\omega(b)\| = |D^2 h(b)|_{\infty\to\infty} \le C_1$ for all $b \in U$. $f : T_s \times U \to \mathbb{C}$ holomorphic, 1-periodic in θ, real on $\mathbb{R}^N \times (U \cap \mathbb{R}^N)$, with $|f| \le \varepsilon$ on $T_s \times U$.

(d) (finite type, File 11a-1) $V := U \cap \mathbb{R}^N \supset K_\vartheta$ is open in $\mathbb{R}^N$, $\omega \in C^\infty(V, \mathbb{R}^N)$ and the hypotheses (D1), (D2) of Theorem 15.10 hold with $m = N$: $\beta_{\rho,\mu}(\omega, K) \ge \beta > 0$ and $|D^k g_\rho[\omega](c,y)|_{\mathrm{op}} \le M$ $(1 \le k \le \mu+1,\ y \in K_\vartheta)$, $M \ge \beta$. Put $M_0 := \|\omega\|^\mu_{\mathrm{dg},K_\vartheta}$.

The loss γ: $\gamma := \gamma_*(\varepsilon^*)/2$ with $\gamma_*(\varepsilon^*)$ of Corollary 15.11 (with $m = N$ and this τ); thus $2\gamma \le \gamma_*(\varepsilon^*)$.

Only $\|D\omega\| \le C_1$ (with $C_1 \ge 1$) is used from the plan's "$C_1 \ge |\omega|, |D^2 h|$". The theorem has *no* hypothesis on ε beyond $\varepsilon \le \varepsilon_{\mathrm{adm}}$ (Definition 17.12), and no hypothesis that any conclusion holds.
External facts. Besides the facts of Files 11a-1, 11a-2 and [K:`bp:p1:ext-mvt`], [K:`bp:p1:ext-cauchy`]:

(Y1) *Tangent line of the logarithm.* $\log x \le \log y + x/y - 1$ for $x, y > 0$ (concavity; Mathlib: `Real.log_le_sub_one_of_pos` applied to x/y).

(Y2) *Convolution with a compactly supported C^k kernel.* If $\psi \in C^k_c(\mathbb{R}^N)$ and g is bounded measurable with compact support, then $g * \psi \in C^k$ and $\partial^\alpha(g*\psi) = g * \partial^\alpha\psi$, $|\partial^\alpha(g*\psi)| \le \|g\|_\infty \|\partial^\alpha\psi\|_{L^1}$ (differentiation under the integral; Mathlib: `HasCompactSupport.hasFDerivAt_convolution_right`).

(Y3) *Continuity from above.* For a decreasing sequence of Lebesgue measurable sets of finite measure, $\mathrm{meas} \bigcap_\nu A_\nu = \lim_\nu \mathrm{meas}\, A_\nu$ (`MeasureTheory.tendsto_measure_iInter_atTop`).

(Y4) Geometric series $\sum_{j\ge m} x^j = x^m/(1-x)$, $0 \le x < 1$, and $1/(1-x) \le 2$ for $x \le \frac{1}{2}$.

Lean totality. Divisions are by $\gamma, T_\nu, s, s_\nu, \sigma_\nu, t_\nu, D_\nu, C_1, \varpi, q$, all positive by construction; log is applied to positive numbers ($16c_N/\varpi$, $32N/s$, πT_ν); powers x^τ have $x \ge 1$ or $x > 0$ with real exponent as in Definition 1.1.

17.2. **Geometric recursion and the constants of the schedule.** Track K's [K:bp:p1:iteration] is a *quadratic* recursion $\varepsilon_{\nu+1} \le c\,\varepsilon_\nu^2/(\dots)$ and does not apply: here the one-step bound of Proposition 16.13(iii) has a term linear in ε_ν (cutoff errors $2q^2 + 2E_T$) beside the quadratic one. The version needed is the following.

Lemma 17.2 (Geometric recursion)**.** *Let $\varpi \in (0,1)$, $c_0 \in [0, \frac{3}{8}\varpi]$, $p > 0$ with $p\varpi \le 1$, $P_0 > 0$, $x_0 \ge 0$ with $P_0 x_0 \le \frac{1}{2}\varpi$, and let $x_\nu \ge 0$ satisfy*

$$x_{\nu+1} \;\le\; x_\nu\big(c_0 + P_0 p^\nu x_\nu\big) \qquad (\nu \ge 0).$$

Then for all $\nu \ge 0$: $x_\nu \le x_0\varpi^\nu$ and $P_0 p^\nu x_\nu \le \frac{1}{2}\varpi$. In particular $x_{\nu+1} \le \frac{7}{8}\varpi x_\nu$.
Source: Ours; replaces [K:bp:p1:iteration] *for the geometric scheme (Rus01 (13.1.7): $\varepsilon_\nu = \varepsilon_0\delta^{\kappa\nu}$, p. 156).*

Proof. **Step S17.1: Induction.** Suppose $x_\nu \le x_0\varpi^\nu$. Then $P_0 p^\nu x_\nu \le P_0 x_0 (p\varpi)^\nu \le \frac{1}{2}\varpi$ because $p\varpi \le 1$. Hence $x_{\nu+1} \le x_\nu(\frac{3}{8}\varpi + \frac{1}{2}\varpi) = \frac{7}{8}\varpi x_\nu \le \varpi x_\nu \le x_0\varpi^{\nu+1}$. The case $\nu = 0$ is the hypothesis $P_0 x_0 \le \frac{1}{2}\varpi$. □

Definition 17.3 (Constants of the schedule; depend only on (N, μ, τ))**.**

$$E_* = 2(\tau+1)(\mu+1) + N + 4, \quad \varpi = 2^{-2E_*}, \quad q = \tfrac{1}{4}\sqrt{\varpi} = 2^{-E_*-2},$$

$$\alpha = \tfrac{1}{8}\varpi, \qquad \varrho = \varpi/q = 4\sqrt{\varpi} = 2^{2-E_*},$$

$$\Theta = 2, \qquad \zeta = \tfrac{1}{2}, \qquad C_Q = 18N(4N+5), \qquad A_0 = \log(16c_N/\varpi)$$

(c_N of Lemma 16.2). Further

$$g_1 = \varrho\,\Theta^\tau\zeta^{-(N+1)}, \qquad g_2 = \varrho\,\Theta^{\tau+1}, \qquad g_3 = \varrho\,\Theta^{(\tau+1)\mu}, \qquad g_4 = g_1\,\zeta\,\Theta^{\tau+1}.$$

Source: Ours. ϖ replaces Rus01's δ^κ ($\kappa = 24/25$, $\tau_0 = 1/9$ in (13.1), p. 156), $\Theta = 2$ replaces the coupling $T_\nu = \Psi^{-1}(\Psi(T_0)\delta^{\tau\nu})$ of 13.1.4; deviation D-16 (Section 17.11).

Lemma 17.4 (The growth factors are beaten)**.** *With the constants of Definition 17.3: $q \le 2^{-5}$, $2q^2 = \frac{1}{8}\varpi$, $2\alpha = \frac{1}{4}\varpi$, $\varrho \le 2^{-5}2^{-(\tau+1)}$ and*

$$g_1,\; g_2,\; g_3,\; g_4 \;\le\; 2^{-5}.$$

Consequently $q \le \zeta$, $\varrho \le \frac{1}{2}$ and $\varrho\,\Theta^{\tau+1} \le \frac{1}{2}$.
Source: Ours; this is where the exponent conditions of Rus01 13.1.2 ($\kappa - \sigma - \tau_0 > \dots$) and the summability $\tau\mu < \tau_0$ of (20.33) are paid.

Proof. **Step S17.2: Exponents.** $\varrho = 2^{2-E_*}$, so $g_1 = 2^{2-E_*+\tau+N+1}$ and $E_* - \tau - N - 3 = 2(\tau+1)(\mu+1) - \tau + 1 \ge 4(\tau+1) - \tau + 1 = 3\tau + 5 \ge 5$ ($\mu \ge 1$). $g_2 = 2^{2-E_*+\tau+1}$ and $E_* - \tau - 3 \ge 3\tau + N + 5 \ge 5$. $g_3 = 2^{2-E_*+(\tau+1)\mu}$ and $E_* - 2 - (\tau+1)\mu = (\tau+1)\mu + 2(\tau+1) + N + 2 \ge 6$. $g_4 = 2^{2\tau+N+3-E_*}$ and $E_* - 2\tau - N - 3 = 2(\tau+1)(\mu+1) + 1 - 2\tau \ge 2\tau + 5 \ge 5$. Finally $E_* \ge 9$ (equality for $(N,\mu,\tau) = (1,1,0)$, which is permitted), so $q = 2^{-E_*-2} \le 2^{-11} \le 2^{-5}$, $2q^2 = \varpi/8$ and $2\alpha = \varpi/4$ by definition; $\varrho\,2^{\tau+1} = g_2 \le 2^{-5}$. □

17.3. **The schedule.**

Definition 17.5 (The schedule)**.** Let the constants of Definition 17.3 and the data of Assumption 17.1 be given, and let $T_0 \ge 1$, $t_0 \in (0,1]$, $s_0 > 0$, $\varepsilon_0 > 0$ be numbers (chosen in Definition 17.12); put $\varsigma_0 = s/24$ and $\lambda_0 = \varepsilon_0/s_0$. For $\nu \ge 0$:

$$T_\nu = T_0 2^\nu, \qquad D_\nu = \gamma T_\nu^{-\tau}, \qquad t_\nu = t_0 2^{-(\tau+1)\nu}, \qquad \varepsilon_\nu = \varepsilon_0 \varpi^\nu, \qquad s_\nu = s_0 q^\nu,$$

$$\delta_\nu = \sigma_\nu = \varsigma_0 2^{-\nu}, \qquad d_\nu = d_{T_\nu}(\alpha) = \frac{\log(2c_N/\alpha) + N\log(\pi T_\nu)}{\pi T_\nu},$$

$$r_0 = s, \qquad r_{\nu+1} = r_\nu - d_\nu - \delta_\nu - 2\sigma_\nu,$$

$$\ell_\nu = 2\varepsilon_\nu/s_\nu, \qquad S_m = \sum_{j \ge m} \ell_j, \qquad E_\nu = \varepsilon_N(T_\nu, d_\nu) \quad \text{(Lemma 16.2)}, \qquad Q_\nu^\sharp = \frac{C_Q (2/\delta_\nu)^N \varepsilon_\nu}{\pi D_\nu s_\nu \sigma_\nu}.$$

The *shells* are $\mathcal{S}_\nu = \{k \in \mathbb{Z}^N : T_{\nu-1} < |k|_1 \le T_\nu\}$ with $T_{-1} := 0$ (so $\mathcal{S}_0 = \{0 < |k|_1 \le T_0\}$), and $\nu(k)$ is the unique ν with $k \in \mathcal{S}_\nu$ ($k \ne 0$); $|k|_1 \le T_{\nu(k)}$. The ℓ^1 norm is the one of the cutoff (File 11a-2 truncates in $|k|_1$).

Lemma 17.6 (The strip budget)**.** *Let $T_0 \ge 1$ and $T_0 \ge \frac{16}{\pi s}\big(A_0 + N\log\frac{32N}{s}\big)$ (assumption (A6) below). Then*

$$\sum_{\nu\ge0} d_\nu \le \frac{s}{4}, \quad d_\nu \le \frac{s}{4} \le \frac{1}{4}, \quad \sum_{\nu\ge0}(\delta_\nu + 2\sigma_\nu) = \frac{s}{4}, \quad r_\nu \ge \frac{s}{2}, \quad r_{\nu+1} = r_\nu - d_\nu - \delta_\nu - 2\sigma_\nu \ge \frac{s}{2},$$

and $E_\nu \le \alpha$ for all ν (so hypothesis (C1) of Proposition 16.13 holds at every stage).
Source: Ours; replaces Rus01 13.1.5–6 and Lemma 13.2.2 (the Bruno-type integral condition $\tau\log\frac{1}{\delta}\int_{T_0}^\infty \log\frac{1}{\Phi}\frac{dT}{T^2} \le \dots$ and $\sum 1/T_\mu$), review A3 items 2, 3; deviation D-15.

Proof. **Step S17.3: Sum of the cutoff losses.** Lemma 16.4 with $\Theta = 2$, $\alpha_\nu = \alpha$ ($a = \log(2c_N/\alpha) = A_0$, $b = 0$): $\sum_\nu d_\nu \le \frac{1}{\pi T_0}[(A_0 + N\log(\pi T_0))\cdot 2 + N\log 2 \cdot 2] = \frac{4}{X}(A_0 + N\log X)$ with $X = 2\pi T_0$.

Step S17.4: Solving for T_0. By (Y1) with $y = 32N/s$: $N\log X \le \frac{sX}{32} + N\log\frac{32N}{s}$, so $\frac{4}{X}(A_0 + N\log X) \le \frac{s}{8} + \frac{4}{X}(A_0 + N\log\frac{32N}{s})$. The assumption on T_0 reads $X \ge \frac{32}{s}(A_0 + N\log\frac{32N}{s})$, i.e. $\frac{4}{X}(A_0 + N\log\frac{32N}{s}) \le \frac{s}{8}$. Hence $\sum d_\nu \le \frac{s}{4}$; each $d_\nu \le \sum d_j \le \frac{s}{4} \le \frac{1}{4}$.

Step S17.5: Losses of δ_ν, σ_ν and the strip. $\sum_\nu(\delta_\nu + 2\sigma_\nu) = 3\varsigma_0\sum 2^{-\nu} = 6\varsigma_0 = \frac{s}{4}$. Therefore $r_\nu = s - \sum_{j<\nu}(d_j + \delta_j + 2\sigma_j) \ge s - \frac{s}{4} - \frac{s}{4} = \frac{s}{2}$, for every ν (including $\nu+1$ in the last identity).

Step S17.6: Cutoff error. $d_\nu = d_{T_\nu}(\alpha) \le 1$ and $T_\nu \ge 1$, $\alpha \le 1$: Lemma 16.3 gives $E_\nu = \varepsilon_N(T_\nu, d_{T_\nu}) \le \alpha$. □

Lemma 17.7 (Elementary properties of the schedule)**.** *Let $\lambda_0 = \varepsilon_0/s_0$, $a = \gamma T_0^{-\tau-1}$ and suppose $4\lambda_0 \le a$, $8\lambda_0 \le C_1 t_0$ and $4C_1 t_0 \le a$ (these are (A1), (A2) below). For all $\nu, m \ge 0$:*

- *(S1) $T_{\nu+1} = 2T_\nu$; $t_{\nu+1} = 2^{-(\tau+1)}t_\nu \le \frac12 t_\nu$; $t_\nu \le \gamma T_\nu^{-\tau-1}/(4C_1)$; $s_\nu = s_0 q^\nu$ and $\varepsilon_\nu/s_\nu = \lambda_0\varrho^\nu$, $\ell_\nu = 2\lambda_0\varrho^\nu$.*
- *(S2) (tail) $S_m \le 4\lambda_0\varrho^m \le \gamma T_m^{-\tau-1}$.*
- *(S3) (derivatives of the increments) $\sum_{j<\nu}\frac{4\varepsilon_j}{s_j t_j} \le \frac{8\lambda_0}{t_0} \le C_1$, and $\sum_{j<\nu}\ell_j t_j^{-\mu} \le 4\lambda_0 t_0^{-\mu}$.*
- *(S4) (quadratic coefficient) $Q_\nu^\sharp = Q_0 g_1^\nu$ with $Q_0 := \frac{C_Q 2^N T_0^\tau \lambda_0}{\pi\gamma\varsigma_0^{N+1}}$.*
- *(S5) (divisor and collar) $t_{\nu+1}T_\nu \cdot 2C_1 \le D_\nu/2$.*

Source: Ours.

Proof. **Step S17.7: (S1).** Definition 17.5; $2^{-(\tau+1)} \le \frac{1}{2}$ as $\tau \ge 0$; $t_\nu = t_0 2^{-(\tau+1)\nu} \le \frac{a}{4C_1} 2^{-(\tau+1)\nu} = \gamma T_\nu^{-\tau-1}/(4C_1)$ by $4C_1 t_0 \le a$. $\varepsilon_\nu/s_\nu = \frac{\varepsilon_0 \varpi^\nu}{s_0 q^\nu} = \lambda_0 \varrho^\nu$.

Step S17.8: (S2). $\varrho \le \frac{1}{2}$, so $S_m = 2\lambda_0 \varrho^m/(1-\varrho) \le 4\lambda_0 \varrho^m$ ((Y4)); $\varrho \le 2^{-(\tau+1)}$ (Lemma 17.4) and $4\lambda_0 \le a$ give $S_m \le a 2^{-(\tau+1)m} = \gamma T_m^{-\tau-1}$.

Step S17.9: (S3). $\frac{\varepsilon_j}{s_j t_j} = \frac{\lambda_0}{t_0} g_2^j$ with $g_2 \le \frac{1}{2}$, so $\sum_j \frac{4\varepsilon_j}{s_j t_j} \le \frac{8\lambda_0}{t_0} \le C_1$. And $\ell_j t_j^{-\mu} = 2\lambda_0 t_0^{-\mu} g_3^j$, $g_3 \le \frac{1}{2}$.

Step S17.10: (S4). $Q_\nu^\sharp = \frac{C_Q 2^N \varsigma_0^{-N} 2^{N\nu} \varepsilon_0 \varpi^\nu T_0^\tau 2^{\tau\nu}}{\pi\gamma\, s_0 q^\nu\, \varsigma_0 2^{-\nu}} = \frac{C_Q 2^N T_0^\tau \lambda_0}{\pi\gamma \varsigma_0^{N+1}} \cdot \big(\varrho 2^{N+1+\tau}\big)^\nu$, and $\varrho\, 2^{N+1+\tau} = g_1$ ($\Theta^\tau \zeta^{-(N+1)} = 2^{\tau+N+1}$).

Step S17.11: (S5). By (S1), $t_{\nu+1} T_\nu 2C_1 \le 2^{-(\tau+1)} \frac{\gamma T_\nu^{-\tau-1}}{4C_1} T_\nu 2C_1 = 2^{-(\tau+2)} \gamma T_\nu^{-\tau} \le D_\nu/2$. □

17.4. **Mollified cutoffs and the chain extension of the frequency.** The measure estimate of File 11a-1 tests Fourier mode k against a map $\tilde\omega_k \in C^\mu$ defined on a *fixed* real neighbourhood of K, whereas ω_ν lives only on the shrinking complex collar B_ν of K_ν. The two are connected by extending ω_ν from K_ν (Rus01 §20: "chains", Lemma 20.3–20.5, pp. 201–203; §§19–20.4 not read, and not followed: deviation D-7).

Lemma 17.8 (Cutoff function with t-scaled derivatives)**.** *Let $\mu \ge 1$, $\psi_1(x) = \kappa_\mu (1-x^2)_+^{\mu+1}$ with $\kappa_\mu > 0$ such that $\int \psi_1 = 1$ (so $\psi_1 \in C^\mu(\mathbb{R})$, $\operatorname{supp}\psi_1 = [-1,1]$), $m_j = \|\psi_1^{(j)}\|_{L^1}$ ($0 \le j \le \mu$, $m_0 = 1$), and*

$$c_k = N^k \max_{\alpha\in\mathbb{N}^N,\ |\alpha|=k} \prod_{i=1}^N m_{\alpha_i} \quad (0 \le k \le \mu),$$

$$c'_{N,\mu} = \sqrt{N} \max_{0\le k\le\mu} 4^k \sum_{j=0}^k \binom{k}{j} c_j\, (k-j)^{k-j} \quad (0^0 = 1).$$

For every nonempty compact $E \subset \mathbb{R}^N$ and $0 < t \le 1$ there is $\chi = \chi_{E,t} \in C^\mu(\mathbb{R}^N, [0,1])$ with $\chi = 1$ on E, $\operatorname{supp}\chi \subset \{y : \operatorname{dist}_\infty(y,E) \le t/2\}$, and for $k \le \mu$, $|a_i|_\infty \le 1$: $|D^k\chi(y)(a_1,\dots,a_k)| \le c_k (4/t)^k$. Source: *Ours; replaces the smoothing of Rus01 §19 (not read).*

Proof. **Step S17.12: Construction.** $a = t/4$, $E_a = \{y : \operatorname{dist}_\infty(y,E) \le a\}$ (compact), $\psi(x) = \prod_i \psi_1(x_i) \in C_c^\mu$, $\psi_a(x) = a^{-N}\psi(x/a)$ (support in $\{|x|_\infty \le a\}$, $\int \psi_a = 1$), $\chi = \mathbf{1}_{E_a} * \psi_a$. Then $0 \le \chi \le 1$. If $y \in E$ then $y - x \in E_a$ for all $|x|_\infty \le a$, so $\chi(y) = 1$. If $\operatorname{dist}_\infty(y,E) > 2a$ then $y - x \notin E_a$ for $|x|_\infty \le a$, so $\chi(y) = 0$.

Step S17.13: Derivatives. By (Y2), $\chi \in C^\mu$ and $|\partial^\alpha \chi| \le \|\partial^\alpha \psi_a\|_{L^1} = a^{-|\alpha|} \prod_i m_{\alpha_i}$ (scaling and Fubini). For $|a_i|_\infty \le 1$, $|D^k\chi(y)(a_1,\dots,a_k)| \le \sum_{i_1,\dots,i_k} |\partial_{i_1}\cdots\partial_{i_k}\chi| \le N^k \max_{|\alpha|=k} |\partial^\alpha\chi| \le c_k a^{-k} = c_k (4/t)^k$. □

Lemma 17.9 (Chain extension and closeness)**.** *Let Assumption 17.1 and a schedule (Definition 17.5) with $t_0 \le \vartheta/\sqrt{N}$ be given, $\nu \ge 0$, $K = K_0 \supset K_1 \supset \cdots \supset K_\nu$ nonempty compact, $B_j = K_j + t_j$, and for $j < \nu$ let $\Delta\omega_j : B_j \to \mathbb{C}^N$ be holomorphic, real on $B_j \cap \mathbb{R}^N$, with $|\Delta\omega_j|_\infty \le \ell_j$. Put $\omega_\nu = \omega + \sum_{j<\nu} \Delta\omega_j$ on B_ν and, with $\chi_j = \chi_{K_j,t_j}$ of Lemma 17.8,*

$$\tilde\omega_\nu \ :=\ \omega + \sum_{j<\nu} \chi_j\, \Delta\omega_j \quad \text{on } V \qquad (\chi_j \Delta\omega_j := 0 \text{ outside } \operatorname{supp}\chi_j).$$

Then:

(a) $\tilde\omega_\nu \in C^\mu(V, \mathbb{R}^N)$ and $\tilde\omega_\nu = \omega_\nu$ on $K_\nu \subset K_j$ (all $j \le \nu$); $\tilde\omega_0 = \omega$.

(b) $\|\tilde\omega_\nu - \omega\|^\mu_{\mathrm{dg},K_\vartheta} \le c'_{N,\mu} \sum_{j<\nu} \ell_j t_j^{-\mu}$.

Source: Ours (deviation D-7); the role of Rus01 (20.7), (20.25), (20.26), pp. 201–203, where the closeness $\|\tilde\chi_\nu - \chi\|^{\mu_0} \le \frac12\alpha^{4p^2}\beta/M^*$ *is the hypothesis of Thm 18.5.*

Proof. **Step S17.14: Supports.** $\operatorname{supp}\chi_j \subset \{\operatorname{dist}_\infty(\cdot,K_j) \le t_j/2\} \subset K + \frac{\vartheta}{2\sqrt N}$ (closed), whose Euclidean distance to K is $\le \vartheta/2$: inside $K_\vartheta \subset V$ and inside $B_j \cap \mathbb{R}^N$ with room $(K_j + t_j/2 \subset B_j)$.

Step S17.15: Regularity and equality. On $B_j\cap\mathbb{R}^N$, $\Delta\omega_j$ is real-analytic; $\chi_j \in C^\mu$ is supported where $\Delta\omega_j$ is defined, so $\chi_j\Delta\omega_j \in C^\mu(V,\mathbb{R}^N)$. On $K_\nu \subset K_j$ $(j<\nu)$ $\chi_j = 1$, so $\tilde\omega_\nu = \omega + \sum_{j<\nu}\Delta\omega_j = \omega_\nu$.

Step S17.16: Derivatives of the increments. For $y\in\mathbb{R}^N$ with $\operatorname{dist}_\infty(y,K_j)\le t_j/2$ we have $y \in K_j + \frac34 t_j$. Lemma 16.7 with $E=K_j$, $t=t_j$, $t'=\frac34 t_j$ gives $\|D^u\Delta\omega_j(y)\| \le u^u \ell_j (t_j/4)^{-u} = u^u 4^u \ell_j t_j^{-u}$ for $u\ge 1$ (max norms on the arguments).

Step S17.17: Leibniz. For a unit vector $a\in\mathbb{R}^N$ ($|a|_2=1$, so $|a|_\infty\le 1$) and $k\le\mu$: $D^k(\chi_j\Delta\omega_j)(a^k) = \sum_{u=0}^k \binom{k}{u} D^{k-u}\chi_j(a^{k-u})\, D^u\Delta\omega_j(a^u)$, hence in the max norm $\le \sum_u \binom{k}{u} c_{k-u}(4/t_j)^{k-u} u^u 4^u \ell_j t_j^{-u}$ (for $u=0$ the bound is ℓ_j, $0^0=1$), and the Euclidean norm of the value is at most $\sqrt N$ times the max norm. Reindexing $u\leftrightarrow k-u$ gives $\le c'_{N,\mu}\ell_j t_j^{-k} \le c'_{N,\mu}\ell_j t_j^{-\mu}$ (as $t_j\le 1$). Sum over $j<\nu$. □

Remark 17.10 (Verdict on D-7). The claim of the plan is true, with the following precise form. The increments $\Delta\omega_j$ are *holomorphic on the whole collar* B_j, so a C^μ cutoff at scale t_j suffices and Whitney-type extension is not needed. The C^μ-size of the j-th increment is $\lesssim c'\ell_j t_j^{-\mu} = 2c'\lambda_0 t_0^{-\mu} g_3^j$ and is summable exactly because $g_3 = \varrho\, 2^{(\tau+1)\mu} < 1$ (it is $\le 2^{-5}$ by Lemma 17.4); this is Rus01's $\tau\mu < \tau_0$ (Rus01 16.3–16.4, (20.33)). The cost is the factor $t_0^{-\mu}$ in the closeness inequality (A5), where $t_0 \lesssim \gamma T_0^{-\tau-1}/C_1$: this is the origin of the large degree in the budget (Section 17.10).

17.5. The initial remainder and the admissible parameters.

Lemma 17.11 (Initialisation; Rus01 §14)**.** *Under Assumption 17.1(c) let* $0<t_0\le\vartheta$, $0<s_0\le\vartheta$, $B_0 = K + t_0$ *and*

$$H_0(\theta,\eta,b) = h(b+\eta) + f(\theta,b+\eta) - h(b), \qquad N_0(\eta,b) = \omega(b)\cdot\eta, \qquad R_0 = H_0 - N_0.$$

Then R_0 *is holomorphic and* 1*-periodic on* $\mathcal{D}(s,s_0;B_0)$*, real at real points, and*

$$\|R_0\|_{s,s_0;B_0} \;\le\; \varepsilon + \tfrac{N}{2}C_1 s_0^2 \;=:\; \varepsilon_0.$$

(The translation $(\theta,\eta)\mapsto(\theta,b+\eta)$ *is symplectic for each* b*; the constant* $h(b)$ *is removed;* $h_0 := 0$*.)*
Source: Rus01 §14.1, (14.2)–(14.5) pp. 165, and Lemma 14.5 p. 168 (there $s_0^2 \approx \vartheta\varepsilon_0/C_\bullet$*, (14.13)); constants ours, in the max norm; the factor* N *is* $|\eta^\top M\eta| \le N\|M\||\eta|_\infty^2$*.*

Proof. **Step S17.18: Domain.** For $b\in B_0$ pick $b'\in K$ with $|b-b'|<t_0\le\vartheta$; for $|\eta|_\infty<s_0\le\vartheta$ and $\lambda\in[0,1]$, $b+\lambda\eta \in K+2\vartheta = U$. So $h(b+\eta)$, $f(\theta,b+\eta)$ are defined and holomorphic; periodicity and reality are inherited.

Step S17.19: Taylor. $h(b+\eta)-h(b)-\omega(b)\cdot\eta = \int_0^1(1-\lambda)\,\eta^\top D\omega(b+\lambda\eta)\eta\,d\lambda$ and $|\eta^\top M\eta| \le |\eta|_\infty|M\eta|_1 \le N|\eta|_\infty|M\eta|_\infty \le N\|M\||\eta|_\infty^2$. So the quadratic part is $\le \frac N2 C_1 s_0^2$ and $|f|\le\varepsilon$. Thus $R_0 = [h(b+\eta)-h(b)-\omega\cdot\eta] + f(\theta,b+\eta)$. □

Definition 17.12 (Admissible parameters)**.** Under Assumption 17.1, with $\varsigma_0 = s/24$, $a = \gamma T_0^{-\tau-1}$ and the constants of Definitions 17.3 and Lemma 17.8, put

$$A_0 = \log\frac{16c_N}{\varpi}, \qquad T_0 = \max\Big\{1,\ \frac{16}{\pi s}\Big(A_0 + N\log\frac{32N}{s}\Big)\Big\}, \qquad t_0 = \min\Big\{\frac{\vartheta}{\sqrt N},\ \frac{a}{4C_1}\Big\},$$

$$\Xi = 2^{\tau+1}\Big(1+\frac{t_0}{\varsigma_0}+\frac{\varsigma_0}{t_0}+T_0\varsigma_0\Big), \qquad \theta_{\max} = \begin{cases} \beta/2 & \rho = 1, \\ \min\big\{\beta,\ \beta/(2^{\mu+1}(2M_0+\beta))\big\} & \rho = 2, \end{cases}$$

$$\lambda_{\max} = \min\Big\{\frac{C_1 t_0}{8},\ \ \min\Big\{\frac{\varpi}{2}, \frac{2^{-10}}{\Xi}\Big\}\frac{\pi\gamma\varsigma_0^{N+1}}{C_Q 2^N T_0^\tau},\ \ \frac{\theta_{\max} t_0^\mu}{4c'_{N,\mu}}\Big\}, \qquad s_0 = \frac{\lambda_{\max}}{NC_1}, \qquad \varepsilon_{\mathrm{adm}} = \frac{\lambda_{\max}^2}{2NC_1}.$$

For $\varepsilon \le \varepsilon_{\mathrm{adm}}$ let $\varepsilon_0 = \varepsilon + \frac{N}{2}C_1 s_0^2$ (Lemma 17.11) and $\lambda_0 = \varepsilon_0/s_0$. All quantities depend on the data $(N, \mu, \rho, \tau, \vartheta, s, C_1, \beta, M_0, \varepsilon^*)$ and on $d = \operatorname{diam}_\infty K$ and M through γ only.
Source: Ours (deviation D-9: explicit $\varepsilon_{\mathrm{adm}}$, replacing Rus01 (13.4)–(13.7), (14.19)–(14.22), pp. 158, 168–170, where ε_0/s_0 is fixed by a list of inequalities with undetermined constants c_1, c_2, c_3, C^*).

Proposition 17.13 (The initial inequalities)**.** *For $\varepsilon \le \varepsilon_{\mathrm{adm}}$ the schedule of Definition 17.5 (with $T_0, t_0, s_0, \varepsilon_0$ of Definition 17.12) satisfies, with $a = \gamma T_0^{-\tau-1}$, $\lambda_0 = \varepsilon_0/s_0$ and $Q_0 = C_Q 2^N T_0^\tau \lambda_0/(\pi\gamma\varsigma_0^{N+1})$:*

- *(A1) $0 < t_0 \le \vartheta/\sqrt{N} \le 1$ and $4C_1 t_0 \le a$;*
- *(A2) $8\lambda_0 \le C_1 t_0$ (hence $4\lambda_0 \le a/8 \le a$);*
- *(A3) $s_0 \le t_0/(8N)$ $(\le \vartheta)$;*
- *(A4) $Q_0 \le \varpi/2$ and $Q_0\,\Xi \le 2^{-10}$;*
- *(A5) $4c'_{N,\mu}\lambda_0 t_0^{-\mu} \le \theta_{\max}$;*
- *(A6) $T_0 \ge 1$ and $T_0 \ge \frac{16}{\pi s}(A_0 + N\log\frac{32N}{s})$ (Lemma 17.6);*
- *(A7) $\varepsilon_0 = \varepsilon + \frac{N}{2}C_1 s_0^2$, $|R_0| \le \varepsilon_0$ on $\mathcal{D}(s, s_0; B_0)$ (Lemma 17.11).*

In particular $\varepsilon_{\mathrm{adm}} > 0$ and $s_0 \le \vartheta$. (Monotonicity in the data is not claimed here; the polynomial lower bound is Proposition 17.21 in Section 17.10, and the monotone minorant is File 11a-5's business.)

Proof. **Step S17.20: $\lambda_0 \le \lambda_{\max}$.** $\lambda_0 = \varepsilon/s_0 + \frac{N}{2}C_1 s_0 = \varepsilon NC_1/\lambda_{\max} + \frac{1}{2}\lambda_{\max} \le \lambda_{\max}$ because $\varepsilon \le \lambda_{\max}^2/(2NC_1)$.

Step S17.21: (A1)–(A5). (A1) is the definition of t_0 ($\vartheta \le 1$). Since $\lambda_0 \le \lambda_{\max}$: (A2) from the first term of the minimum; $C_1 t_0 \le a/4$ gives $4\lambda_0 \le a/8$. (A3): $s_0 = \lambda_{\max}/(NC_1) \le t_0/(8N)$. (A4): $Q_0 = \frac{C_Q 2^N T_0^\tau \lambda_0}{\pi\gamma\varsigma_0^{N+1}} \le \min\{\varpi/2, 2^{-10}/\Xi\}$ by the second term. (A5): third term. (A6): definition of T_0. (A7): Lemma 17.11 (its hypotheses $t_0 \le \vartheta$, $s_0 \le \vartheta$ are (A1), (A3)). □

Remark 17.14 (The three branches of $\varepsilon_{\mathrm{adm}}$). Write $\lambda_{\max} = \min\{\lambda^{(1)}, \lambda^{(2)}, \lambda^{(3)}\}$ for the three terms of Definition 17.12. Then $\varepsilon_{\mathrm{adm}} = \min\{\varepsilon^{(1)}, \varepsilon^{(2)}, \varepsilon^{(3)}\}$ with exactly

$$\varepsilon^{(1)} = \frac{C_1 t_0^2}{128N}, \qquad \varepsilon^{(2)} = \frac{\pi^2\gamma^2\varsigma_0^{2N+2}}{2NC_1 C_Q^2\, 4^N\, T_0^{2\tau}}\ \min\Big\{\frac{\varpi}{2}, \frac{2^{-10}}{\Xi}\Big\}^2, \qquad \varepsilon^{(3)} = \frac{\theta_{\max}^2\, t_0^{2\mu}}{32\, N\, c'^2_{N,\mu}\, C_1}.$$

All three can bind. For instance, if $t_0 = \gamma T_0^{-\tau-1}/(4C_1)$ and $\Xi \simeq 2^{\tau+1}\varsigma_0/t_0$ (the case ς_0/t_0 dominates Ξ), then $\varepsilon^{(2)} \asymp \gamma^4\varsigma_0^{2N} C_1^{-3} T_0^{-4\tau-2}$, of higher degree in γ than $\varepsilon^{(3)}$ when $\mu = 1$; $\varepsilon^{(3)}$ is a multiple of $\theta_{\max}^2 t_0^{2\mu}$, and for $\rho = 2$, $\theta_{\max}$ is $\beta/(2^{\mu+1}(2M_0+\beta))$ or β, not universally β^2/M_0^2 after squaring. No statement about the size of $\varepsilon_{\mathrm{adm}}$ below is made from one branch alone.

17.6. **The finite-stage divisor invariant and the retained sets.** *Stage-ν data.* A *chain of length ν* consists of nonempty compact sets $K = K_0 \supset K_1 \supset \cdots \supset K_\nu$ and holomorphic maps $\Delta\omega_j : B_j = K_j + t_j \to \mathbb{C}^N$ $(j < \nu)$, real on $B_j \cap \mathbb{R}^N$, with $|\Delta\omega_j|_\infty \le \ell_j$ on B_j, such that for $j < \nu$

$$K_{j+1} = \big\{b \in K_j :\ |k\cdot\omega_j(b)| \ge 2\gamma|k|_1^{-\tau} \ \text{ for all } k \in \mathcal{S}_j\big\}, \qquad \omega_j = \omega + \sum_{i<j}\Delta\omega_i \ \text{ on } B_j. \tag{7}$$

($K_{\nu+1}$ is defined by (7) with $j = \nu$ once ω_ν is known; it is compact, as ω_ν is continuous on K_ν.)

Lemma 17.15 (Finite-stage divisor invariant)**.** *Let a chain of length ν be given, let $K_{\nu+1}$ be defined by (7), and assume $S_m \le \gamma T_m^{-\tau-1}$ for all $m \le \nu$ (this is (S2)). Then for every $b \in K_{\nu+1}$ and every $k \in \mathbb{Z}^N$ with $0 < |k|_1 \le T_\nu$:*

$$|\omega_\nu(b) - \omega_{\nu(k)}(b)|_\infty \le \gamma T_{\nu(k)}^{-\tau-1} \le \gamma |k|_1^{-\tau-1}, \qquad |k \cdot \omega_{\nu(k)}(b)| \ge 2\gamma |k|_1^{-\tau},$$

$$\boxed{|k \cdot \omega_\nu(b)| \ \ge \ \gamma |k|_1^{-\tau} \ \ge \ \gamma T_\nu^{-\tau} = D_\nu.}$$

Thus the hypotheses of Lemma 16.8 hold with $u_k = \omega_{\nu(k)}(b)$, $v = \omega_\nu(b)$, $T = T_\nu$, and the divisor hypothesis of (C2) in Proposition 16.13 holds on $E = K_{\nu+1}$ with $D = D_\nu$.
Source: Ours; this replaces *the limit-based tail requirement (I3) of Remark 15.14, which bounded $|\omega_* - \omega_{\nu(k)}|$ on K_∞ only and used the endpoint $T_{\nu(k)-1}$ (review P11A2-01); see Remark 17.16.*

Proof. **Step S17.22: Telescoping.** Let $m = \nu(k) \le \nu$ (as $|k|_1 \le T_\nu$) and $b \in K_{\nu+1} \subset K_j \subset B_j$ for all $j \le \nu$. On B_ν, $\omega_\nu - \omega_m = \sum_{j=m}^{\nu-1} \Delta\omega_j$ (both defined at b, since $B_\nu \subset B_m$), so $|\omega_\nu(b) - \omega_m(b)|_\infty \le \sum_{j=m}^{\nu-1} \ell_j \le S_m \le \gamma T_m^{-\tau-1}$; and $T_m^{-\tau-1} \le |k|_1^{-\tau-1}$ because $|k|_1 \le T_m$, $\tau + 1 > 0$.

Step S17.23: Mode k was tested at stage m. $b \in K_{\nu+1} \subset K_{m+1}$ and $k \in \mathcal{S}_m$, so by (7) $|k \cdot \omega_m(b)| \ge 2\gamma |k|_1^{-\tau}$.

Step S17.24: Conclusion. By $|k \cdot x| \le |k|_1 |x|_\infty$: $|k \cdot (\omega_\nu - \omega_m)(b)| \le |k|_1 \gamma |k|_1^{-\tau-1} = \gamma |k|_1^{-\tau}$, so $|k \cdot \omega_\nu(b)| \ge 2\gamma |k|_1^{-\tau} - \gamma |k|_1^{-\tau}$. Finally $|k|_1 \le T_\nu$, $\tau \ge 0$. (This is also the conclusion of Lemma 16.8, now with all its hypotheses verified; for $m = \nu$ the telescoping sum is empty.) □

Remark 17.16 (The correct form of (I3))*.* Requirement (I3) of Remark 15.14 should read: *for every $m \ge 0$ (including $m = 0$)* $\sum_{j \ge m} \ell_j \le \gamma T_m^{-\tau-1}$, where $\ell_j \ge \sup_{B_j} |\Delta\omega_j|_\infty$; and then, for k with $T_{m-1} < |k|_1 \le T_m$ (i.e. $m = \nu(k)$), $|\omega_*(b) - \omega_m(b)|_\infty \le \sum_{j \ge m} \ell_j \le \gamma T_m^{-\tau-1} \le \gamma |k|_1^{-\tau-1}$. The printed endpoint T_{m-1} is on the wrong side: it would demand the bound $\gamma |k|_1^{-\tau-1}$ only for the smallest $|k|_1$ of the shell, where it is weakest, and gives no control of the largest. In addition the finite-stage version (Lemma 17.15) must be proved for $\omega_\nu - \omega_m$ on the finite retained set, *not* derived from the limit (two bounds to the limit would give 2γ instead of γ). The schedule satisfies this by (S2).

Lemma 17.17 (The retained set is nonempty: finite-stage measure bound)**.** *Let Assumption 17.1, the schedule with (A1)–(A6), a chain of length ν and $K_{\nu+1}$ as in (7) be given. Put*

$$H^{(\nu)} = \Big\{ b \in K : \ |k \cdot \tilde\omega_{\nu(k)}(b)| \ge 2\gamma |k|_1^{-\tau} \ \text{ for all } k \in \mathbb{Z}^N,\ 0 < |k|_1 \le T_\nu \Big\}$$

with $\tilde\omega_j$ the chain extension of Lemma 17.9 (it uses $K_i, \Delta\omega_i$ for $i < j \le \nu$ only). Then $H^{(\nu)}$ is compact, $K_{\nu+1} \supset H^{(\nu)}$, $H^{(\nu)} \supset H^{(\nu+1)}$ whenever both are defined, and

$$\operatorname{meas} H^{(\nu)} \ \ge \ \operatorname{meas} K - \varepsilon^* \ > \ 0.$$

In particular $K_{\nu+1} \neq \emptyset$.
Source: Ours; Rus01 Thm 20.5.1 (any maximal chain is infinite), p. 201–202, whose proof is this argument with Thm 18.5 (here Corollary 15.11); the circularity (the sets K_j depend on the construction) is resolved as there: only finitely many stages, already built, enter $H^{(\nu)}$.

Proof. **Step S17.25: Closeness for every stage.** For $j \le \nu$ the chain of length j is an initial segment, so Lemma 17.9(b) and (S3), (A5) give $\theta_j := \|\tilde\omega_j - \omega\|^\mu_{\mathrm{dg},K_\vartheta} \le c'_{N,\mu} \cdot 4\lambda_0 t_0^{-\mu} \le \theta_{\max}$. By Lemma 15.3, $\eta_\rho(\theta_j) \le \beta/2$ (for $\rho = 2$: $\theta_j \le \beta$ and $2^\mu \theta_j (2M_0 + \theta_j) \le 2^\mu \theta_j (2M_0 + \beta) \le \beta/2$ by definition of $\theta_{\max}$), so (D3) holds for $\tilde\omega_j$.

Step S17.26: A family indexed by all modes. For $k \neq 0$ put $\tilde\omega_k := \tilde\omega_{\min\{\nu(k),\nu\}}$. Then (D1)–(D3) hold for the family $(\tilde\omega_k)$. Corollary 15.11 with $2\gamma \le \gamma_*(\varepsilon^*)$ in place of γ gives that

$H_{2\gamma} := \{b \in K : |k \cdot \tilde\omega_k(b)| \geq 2\gamma|k|_1^{-\tau}\ \forall k \neq 0\}$ is compact with $\operatorname{meas} H_{2\gamma} \geq \operatorname{meas} K - \varepsilon^*$. For $|k|_1 \leq T_\nu$, $\tilde\omega_k = \tilde\omega_{\nu(k)}$, so $H_{2\gamma} \subset H^{(\nu)}$.

Step S17.27: Inclusion in the retained set. Let $b \in H^{(\nu)}$. By induction $b \in K_j$ for $j \leq \nu+1$: $b \in K_0$; if $b \in K_j$, $j \leq \nu$, then $\omega_j(b) = \tilde\omega_j(b)$ (Lemma 17.9(a)), and each $k \in \mathcal{S}_j$ has $\nu(k) = j$, $|k|_1 \leq T_j \leq T_\nu$, so $|k \cdot \omega_j(b)| = |k \cdot \tilde\omega_{\nu(k)}(b)| \geq 2\gamma|k|_1^{-\tau}$, whence $b \in K_{j+1}$ by (7). Compactness of $H^{(\nu)}$: an intersection of finitely many closed sets of K (the $\tilde\omega_{\nu(k)}$ are continuous). □

17.7. **The inductive lemma.**

Proposition 17.18 (Inductive lemma, maximal case; Rus01 Lemma 13.4 p. 158)**.** *Let Assumption 17.1 hold, $\varepsilon \leq \varepsilon_{\mathrm{adm}}$, and take the schedule of Definition 17.5 with the parameters of Definition 17.12. Say that* (P_ν) holds *if there are a chain of length ν (so K_ν, $B_\nu = K_\nu + t_\nu$, $\omega_\nu = \omega + \sum_{j<\nu}\Delta\omega_j$ and Δh_j, $j < \nu$) and functions $h_\nu = \sum_{j<\nu}\Delta h_j$ (with $h_0 = 0$), R_ν such that*

- *(a) h_ν, ω_ν are holomorphic on B_ν and real on $B_\nu \cap \mathbb{R}^N$; $|\Delta h_j| \leq 2\varepsilon_j$ on B_j and $|\Delta\omega_j|_\infty \leq \ell_j$ on B_j ($j < \nu$); $\|D\omega_\nu\| \leq 2C_1$ on B_ν;*
- *(b) R_ν is holomorphic and 1-periodic on $\mathcal{D}(r_\nu, s_\nu; B_\nu)$, real at real points, $\|R_\nu\|_{r_\nu,s_\nu;B_\nu} \leq \varepsilon_\nu$;*
- *(c) $H_\nu := N_\nu + R_\nu$ with $N_\nu(\eta, b) = h_\nu(b) + \omega_\nu(b)\cdot\eta$ satisfies $H_\nu = H_0 \circ X_0 \circ \cdots \circ X_{\nu-1}$ on $\mathcal{D}(r_\nu, s_\nu; B_\nu)$, where H_0 is that of Lemma 17.11 and X_j are the maps of the next statement.*

Then (P_0) *holds, and for every $\nu \geq 0$,* (P_ν) *implies the existence of $K_{\nu+1} \neq \emptyset$ compact (by (7)), $\Delta h_\nu, \Delta\omega_\nu$ and X_ν such that* ($\mathrm{P}_{\nu+1}$) *holds with $B_{\nu+1} = K_{\nu+1} + t_{\nu+1}$ and:*

- *(O1) (finite-stage divisors) for $b \in K_{\nu+1}$, $0 < |k|_1 \leq T_\nu$: $|k\cdot\omega_\nu(b)| \geq \gamma|k|_1^{-\tau} \geq D_\nu$ (Lemma 17.15);*
- *(O2) (normal form) $\Delta h_\nu, \Delta\omega_\nu$ are holomorphic and real on B_ν with $|\Delta h_\nu| \leq 2\varepsilon_\nu$, $|\Delta\omega_\nu|_\infty \leq \ell_\nu = 2\varepsilon_\nu/s_\nu$; $\omega_{\nu+1} = \omega_\nu + \Delta\omega_\nu$, $h_{\nu+1} = h_\nu + \Delta h_\nu$ on $B_{\nu+1}$;*
- *(O3) (the map) for each $b \in B_{\nu+1}$, $X_{\nu,b} = \Phi_b$ of Proposition 16.13 is a simple canonical map on $\mathcal{D}(r_{\nu+1}, s_{\nu+1}; \{b\})$, holomorphic jointly in (θ, η, b) on $\mathcal{D}(r_{\nu+1}, s_{\nu+1}; B_{\nu+1})$, with values in $\mathcal{D}(r_{\nu+1} + \sigma_\nu, s_\nu; B_{\nu+1}) \subset \mathcal{D}(r_\nu, s_\nu; B_{\nu+1})$, $X_\nu(\theta + k, \eta) = X_\nu(\theta, \eta) + (k, 0)$, real at real points; with $A_{1,\nu} = \Lambda_{\delta_\nu}(1+E_\nu)\varepsilon_\nu/(\pi D_\nu s_\nu)$, $A_{\eta,\nu} = (N+1)s_\nu A_{1,\nu}/\sigma_\nu$, $\kappa_\nu = 2(N+1)A_{1,\nu}/\sigma_\nu$: $|X_{\nu,\theta} - \theta|_\infty \leq A_{1,\nu}$, $|X_{\nu,\eta} - \eta|_\infty \leq A_{\eta,\nu}$, $\kappa_\nu \leq \frac12$, and the unscaled derivative bound $|DX_\nu - \mathbf{1}|_\infty \leq 2\kappa_\nu + \kappa_\nu s_\nu/\sigma_\nu$ in (θ, η); on the real part $|D\Theta_{\nu,b} - \mathbf{1}|_\infty \leq e^{\Lambda_{\mathbb{R},\nu}} - 1$ with $\Lambda_{\mathbb{R},\nu} = 2T_\nu(1+E_\nu)\varepsilon_\nu\Lambda_{r_\nu - d_\nu}/(D_\nu s_\nu)$;*
- *(O4) (parameter derivatives) for $0 < t' < t_{\nu+1}$ and all (θ, η) in the domain: $\|D_b^j(X_\nu - \mathrm{id})_\theta\| \leq j^j A_{1,\nu}(t_{\nu+1} - t')^{-j}$ and likewise for the η-component with $A_{\eta,\nu}$ on $K_{\nu+1} + t'$;*
- *(O5) (conjugation) $H_\nu \circ X_\nu = N_{\nu+1} + R_{\nu+1}$ on $\mathcal{D}(r_{\nu+1}, s_{\nu+1}; B_{\nu+1})$.*

Source: Rus01 Lemma 13.4, p. 158 (conditions (13.4)–(13.7) become (A1)–(A7); $|\Delta\omega_\nu| \leq \frac32\varepsilon_\nu/s_\nu$ becomes $\leq 2\varepsilon_\nu/s_\nu$; the Lipschitz factor $e^{24C_\nu T_\nu}$ of 13.4.5.4 becomes $e^{\Lambda_{\mathbb{R},\nu}}$, Lemma 16.12); deviations D-7, D-9, D-15, D-16. Proof by Proposition 16.13.

Proof. **Step S17.28:** (P_0)**.** $K_0 = K$, $B_0 = K + t_0$, $\Delta\omega_j$ empty, $\omega_0 = \omega$, $h_0 = 0$, R_0 of Lemma 17.11: $|R_0| \leq \varepsilon_0$ on $\mathcal{D}(s, s_0; B_0) = \mathcal{D}(r_0, s_0; B_0)$ ((A7), $r_0 = s$). $\|D\omega\| \leq C_1 \leq 2C_1$ on $B_0 \subset U$ since $t_0 \leq \vartheta$. H_0 is the composition of the empty list.

Step S17.29: Retained set, nonempty. Given (P_ν) with its chain, (S2) holds by Lemma 17.7 (hypotheses (A1), (A2)). $K_{\nu+1}$ of (7) is compact and by Lemma 17.17 contains $H^{(\nu)}$, which has positive measure; so $K_{\nu+1} \neq \emptyset$. (O1) is Lemma 17.15.

Step S17.30: Hypothesis (C2). Apply Proposition 16.13 to the stage ν with $E = K_{\nu+1}$, $t = t_{\nu+1}$, $B = E + t = B_{\nu+1}$ ($\subset B_\nu$; the restrictions of h_ν, ω_ν, R_ν to $B_{\nu+1}$), $r = r_\nu$, $s = s_\nu$, $T = T_\nu$, $d = d_\nu$, $\delta = \delta_\nu$, $\sigma = \sigma_\nu$, q, $D = D_\nu$, and with $\varepsilon := \varepsilon_\nu$ (the proposition holds verbatim for any upper bound $\varepsilon \geq \|R\|$: its proof uses $\|R\| \leq \varepsilon$ only, through Proposition 16.6; here $\varepsilon_\nu > 0$, so $A_1 > 0$ as

Lemma 16.11 requires). The divisor part of (C2) is (O1). The bound on $D\omega_\nu$ on $B_{\nu+1}$: $L = 2C_1$ by (P_ν)(a), and $t_{\nu+1}T_\nu L \le D_\nu/2$ is (S5).

Step S17.31: Hypotheses (C1) and the loss. $r_{\nu+1} = r_\nu - d_\nu - \delta_\nu - 2\sigma_\nu \ge s/2 > 0$ and $d_\nu \le \frac14$, $E_\nu \le \alpha \le 1$ (Lemma 17.6); $q \le \frac12$ (Lemma 17.4); $\delta_\nu \le s/24 \le 1$.

Step S17.32: Hypothesis (C3) and the recursion. By [K:bp:p1:exp-sums](b) (as in Section 16.5), $\Lambda_\delta \le (1+\frac{1}{\pi\delta})^N - 1 \le (2/\delta)^N$ for $0<\delta\le 1$ (as $1+\frac1\pi < 2$). Hence the relative coefficient $Q_\nu := C_Q\Lambda_{\delta_\nu}\varepsilon_\nu/(\pi D_\nu s_\nu\sigma_\nu) \le Q_\nu^\sharp = Q_0 g_1^\nu$ ((S4)). Since $\varepsilon_\nu = \varepsilon_0\varpi^\nu$ is a prescribed sequence (the induction below is that of Lemma 17.2, with $c_0 = 2q^2+2\alpha = \frac38\varpi$, $P_0 = Q_0/\varepsilon_0$, $p = g_1/\varpi$, $p\varpi = g_1 \le 1$), only the inequality $Q_\nu \le \varpi/2$ is needed: it is $Q_\nu \le Q_0 g_1^\nu \le Q_0 \le \varpi/2$ ((A4)). Since $C_Q \ge 8(N+1) \ge 4(N+1)(1+E_\nu)$, $Q_\nu \le \varpi/2 < 1$ gives $4(N+1)\Lambda_{\delta_\nu}(1+E_\nu)\varepsilon_\nu \le \pi D_\nu s_\nu\sigma_\nu$: (C3).

Step S17.33: The new remainder. Proposition 16.13(iii), second form: $\|R_+\| \le \varepsilon_\nu(2q^2 + 2E_\nu) + \frac{18N(4N+5)\Lambda_{\delta_\nu}}{\pi D_\nu s_\nu\sigma_\nu}\varepsilon_\nu^2 = \varepsilon_\nu(2q^2+2E_\nu+Q_\nu) \le \varepsilon_\nu(\frac18\varpi + \frac14\varpi + \frac12\varpi) \le \varepsilon_\nu\varpi = \varepsilon_{\nu+1}$, by $2q^2 = \varpi/8$, $2E_\nu \le 2\alpha = \varpi/4$ (Lemma 17.4). The domain of R_+ is $\mathcal{D}(r_{\nu+1}, qs_\nu; B_{\nu+1}) = \mathcal{D}(r_{\nu+1}, s_{\nu+1}; B_{\nu+1})$; periodicity and reality are in the proposition. This is ($\mathrm{P}_{\nu+1}$)(b), and (O5) with (c) by composition.

Step S17.34: The increments on the whole collar. Let $R_* = \mathcal{T}_{T_\nu}(R_\nu^{(\le1)})$ be formed on $\mathcal{D}(r_\nu, s_\nu; B_\nu)$ (Proposition 16.6 with $B = B_\nu$; it does not involve the divisor), $\Delta h_\nu = \langle R_{*0}\rangle$, $\Delta\omega_\nu = \langle R_{*1}\rangle$. These are holomorphic on B_ν (Step S16.13), real at real points (Proposition 16.6(c)), $|\Delta h_\nu| \le (1+E_\nu)\varepsilon_\nu \le 2\varepsilon_\nu$ and $|\Delta\omega_\nu|_\infty \le (1+E_\nu)\varepsilon_\nu/s_\nu \le \ell_\nu$ (Proposition 16.6(a) and $k=0$ of [K:bp:p1:decay]); on $B_{\nu+1}$ they are the $\Delta h, \Delta\omega$ of Proposition 16.13(i), which gives (O2).

Step S17.35: $D\omega_{\nu+1}$. For $b \in B_{\nu+1}$ and $j \le \nu$ pick $b' \in K_{\nu+1} \subset K_j$ with $|b-b'| < t_{\nu+1} \le t_j/2$: $b \in K_j + \frac{t_j}{2}$. Lemma 16.7 ($E = K_j$, $t = t_j$, $t' = t_j/2$, $u = 1$): $\|D\Delta\omega_j(b)\| \le \ell_j(t_j/2)^{-1} = 4\varepsilon_j/(s_j t_j)$. So $\|D\omega_{\nu+1}\| \le C_1 + \sum_{j\le\nu} 4\varepsilon_j/(s_j t_j) \le 2C_1$ by (S3): ($\mathrm{P}_{\nu+1}$)(a). (Δh_ν is holomorphic on $B_\nu \supset B_{\nu+1}$ and real.)

Step S17.36: The map, (O3), (O4). Proposition 16.13(ii),(iv),(v) with the displayed $A_1 = A_{1,\nu}$, $A_\eta = A_{\eta,\nu}$, $\kappa = \kappa_\nu \le \frac12$ ((C3) gives $4(N+1)A_1 \le \sigma$, i.e. $\kappa \le \frac12$). The unscaled derivative bound: in the block form of Lemma 16.11(c),(d) the first row has norm $|D\Theta - \mathbf{1}| \le 2\kappa$ and the second row $|\partial_\theta(\mathbf{M}\eta + m)| + |\mathbf{M} - \mathbf{1}| \le \kappa s_\nu/\sigma_\nu + 2\kappa$, maximal row sum (review P11A2-04). □

17.8. **Parameter selection.**

Theorem 17.19 (Parameter selection; Rus01 Thm 16.7 p. 179, proof §20.7 pp. 201–203)**.** *Let Assumption 17.1 hold and $0 \le \varepsilon \le \varepsilon_{\mathrm{adm}}$ (Definition 17.12). Then the iteration of Proposition 17.18 can be carried out for all $\nu \ge 0$, and:*

- *(a) $K = K_0 \supset K_1 \supset \cdots$, $K_\infty := \bigcap_\nu K_\nu$ is compact; for $b \in K_\infty$ the limits $\omega_*(b) = \omega(b) + \sum_{j\ge0}\Delta\omega_j(b)$ and $h_*(b) = \sum_{j\ge0}\Delta h_j(b)$ exist, $|\omega_*(b) - \omega_m(b)|_\infty \le S_m \le \gamma T_m^{-\tau-1}$, $|h_*(b) - h_m(b)| \le 4\varepsilon_m$, and $\omega_* : K_\infty \to \mathbb{R}^N$, $h_* : K_\infty \to \mathbb{R}$ are continuous;*
- *(b) the set $H_\infty := \bigcap_{\nu\ge0} H^{(\nu)} = \{b \in K : |k\cdot\tilde\omega_{\nu(k)}(b)| \ge 2\gamma|k|_1^{-\tau}\ \forall k \ne 0\}$ is compact, $H_\infty \subset K_\infty$, and $\operatorname{meas} H_\infty \ge \operatorname{meas} K - \varepsilon^*$;*
- *(c) for every $b \in H_\infty$ the vector $\omega_*(b)$ is (γ,τ)-Diophantine in the frozen sense: $|k\cdot\omega_*(b)| \ge \gamma|k|_1^{-\tau}$ for all $k \in \mathbb{Z}^N \setminus \{0\}$.*

In particular $\operatorname{meas} K_\infty \ge \operatorname{meas} H_\infty \ge \operatorname{meas} K - \varepsilon^*$, and the quantifier order is data $\to \varepsilon^* \to (\gamma, \varepsilon_{\mathrm{adm}}) \to f$: $\gamma, \varepsilon_{\mathrm{adm}}$ depend on the data and ε^* only. No premise equals a conclusion: the Diophantine property is an output.

Source: Rus01 Thm 16.7 (p. 179: $K_\infty \ne \emptyset$, $\operatorname{meas} K_\infty \ge \operatorname{meas} H \ge \operatorname{meas} K - \varepsilon^*$) and Thm 20.5 (pp. 201–203); the explicit constants (20.33)–(20.35) are replaced by Definition 17.12 (deviation D-9); mode-wise selection with the finite-stage invariant (Remark 17.16), after Remark 15.14.

Proof. **Step S17.37: Infinite iteration.** Proposition 17.18 gives (P_0) and $(\mathrm{P}_\nu) \Rightarrow (\mathrm{P}_{\nu+1})$, hence (P_ν) for all ν with nonempty compact K_ν decreasing. K_∞ is a decreasing intersection of nonempty compact sets (so nonempty), closed and bounded.

Step S17.38: Limits. For $b \in K_\infty$: $b \in K_j \subset B_j$ for all j and $\sum_j |\Delta\omega_j(b)|_\infty \le \sum \ell_j = S_0 < \infty$ ((S2)); $\sum_j |\Delta h_j| \le 2\sum \varepsilon_j < \infty$. The partial sums ω_ν converge uniformly on K_∞ (each $\Delta\omega_j$ is continuous on $K_\infty \subset B_j$, uniform Weierstrass test, [K:`bp:p1:uniform-limits`]) so ω_* is continuous; the tail bound is $\sum_{j\ge m} \ell_j = S_m$, and $\sum_{j\ge m} 2\varepsilon_j \le 4\varepsilon_m$ ($\varpi \le \frac12$).

Step S17.39: The set H_∞. $H^{(\nu)}$ is compact and $H^{(\nu)} \supset H^{(\nu+1)}$ (more modes are tested; $\tilde\omega_{\nu(k)}$ is the same map for a given k whichever ν is used, because the chain extension $\tilde\omega_j$ depends only on the first j stages). So $H_\infty = \bigcap H^{(\nu)}$ is compact; by (Y3) and Lemma 17.17, $\operatorname{meas} H_\infty = \lim \operatorname{meas} H^{(\nu)} \ge \operatorname{meas} K - \varepsilon^*$. Also $H_\infty \subset K_\nu$ for every ν (Lemma 17.17, $H^{(\nu)} \subset K_{\nu+1}$), so $H_\infty \subset K_\infty$.

Step S17.40: Diophantine property. Let $b \in H_\infty$, $k \ne 0$, $m = \nu(k)$. Then $u := \omega_m(b) = \tilde\omega_m(b)$ ($b \in K_m$) satisfies $|k\cdot u| \ge 2\gamma |k|_1^{-\tau}$ ($b \in H^{(m)}$), and $v := \omega_*(b)$ satisfies $|u - v|_\infty \le S_m \le \gamma T_m^{-\tau-1} \le \gamma |k|_1^{-\tau-1}$ (part (a), $|k|_1 \le T_m$). Lemma 15.12 gives $|k\cdot\omega_*(b)| \ge \gamma|k|_1^{-\tau}$. □

17.9. What File 11a-4 takes: summable budgets.

Lemma 17.20 (Exported sums)**.** *In the situation of Theorem 17.19 (so (A1)–(A7), with Q_0 as in (S4)) let $A_{1,\nu}, A_{\eta,\nu}, \kappa_\nu, \Lambda_{\mathbb{R},\nu}$ be as in Proposition 17.18(O3). Then for all $\nu \ge 0$:*

- *(E1)* $A_{1,\nu} \le 2\sigma_\nu Q_\nu / C_Q$, $\kappa_\nu \le Q_\nu/16$, $A_{\eta,\nu} = \frac12 \kappa_\nu s_\nu$, $\Lambda_{\mathbb{R},\nu} \le \dfrac{4\pi}{C_Q} T_0 \varsigma_0 Q_\nu$, *where* $Q_\nu \le Q_0 g_1^\nu \le Q_0 2^{-5\nu}$;
- *(E2) (angle, action and Lipschitz budgets, unscaled)* $\sum_\nu (2\kappa_\nu + \kappa_\nu s_\nu/\sigma_\nu) \le 2^{-12}$, $\sum_\nu \Lambda_{\mathbb{R},\nu} \le 2^{-12}$, $\sum_\nu A_{1,\nu} \le 3\varsigma_0 Q_0/C_Q$, $\sum_\nu A_{\eta,\nu} \le s_0 Q_0/16$;
- *(E3) (tails against the collars)* $\sup_{\nu\ge0} t_{\nu+1}^{-1} \sum_{j\ge\nu} (A_{1,j} + A_{\eta,j}) \le 2^{-10}$, *and* $\sum_{j\ge\nu} \Lambda_{\mathbb{R},j} \le 2^{-10}$;
- *(E4) (parameter Cauchy) on* $K_{\nu+1} + t_{\nu+1}/2$: $\|D_b(X_\nu - \mathrm{id})\| \le 2(A_{1,\nu} + A_{\eta,\nu})/t_{\nu+1}$.

Source: Ours; replaces Rus01 15.1–15.2 (the summability $\sum C_\nu T_\nu < \infty$, (14.20)) and (A3) item 3 of the review; the unscaled form is the correction P11A2-04 of the review of File 11a-2.

Proof. **Step S17.41: (E1).** $A_{1,\nu} = (1+E_\nu)\sigma_\nu Q_\nu/C_Q \le 2\sigma_\nu Q_\nu/C_Q$ by definition of Q_ν. $\kappa_\nu = 2(N+1)A_{1,\nu}/\sigma_\nu \le 4(N+1)Q_\nu/C_Q \le Q_\nu/16$ since $C_Q = 18N(4N+5) \ge 64(N+1)$. $A_{\eta,\nu} = (N+1)s_\nu A_{1,\nu}/\sigma_\nu = \frac12 \kappa_\nu s_\nu$. $\Lambda_{\mathbb{R},\nu} \le 4T_\nu \varepsilon_\nu \Lambda_{\delta_\nu}/(D_\nu s_\nu) = \frac{4\pi}{C_Q} T_\nu \sigma_\nu Q_\nu$, because $r_\nu - d_\nu \ge \delta_\nu + s/2 > \delta_\nu$ (Lemma 17.6) and $\Lambda_\cdot$ decreases; and $T_\nu\sigma_\nu = T_0\varsigma_0(\Theta\zeta)^\nu = T_0\varsigma_0$. $Q_\nu \le Q_\nu^\sharp = Q_0 g_1^\nu$ (Step S17.32).

Step S17.42: (E2). $\sum_\nu Q_\nu \le \frac{32}{31} Q_0$. $s_\nu/\sigma_\nu = (s_0/\varsigma_0)(2q)^\nu \le s_0/\varsigma_0 \le t_0/\varsigma_0$ ((A3)), so $\sum(2\kappa_\nu + \kappa_\nu s_\nu/\sigma_\nu) \le \frac{1}{16}\cdot\frac{32}{31} Q_0(2 + t_0/\varsigma_0) \le Q_0\Xi/7 \le 2^{-12}$ ((A4), $2 + t_0/\varsigma_0 \le 2\Xi$). $\sum \Lambda_{\mathbb{R},\nu} \le \frac{4\pi}{C_Q} T_0\varsigma_0 \frac{32}{31} Q_0 \le \frac{13}{162} Q_0 \Xi \le 2^{-12}$. $\sum A_{1,\nu} \le \frac{2\varsigma_0 Q_0}{C_Q} \sum(\zeta g_1)^\nu \le 3\varsigma_0 Q_0/C_Q$; $\sum A_{\eta,\nu} \le \frac{1}{32} s_0 Q_0 \sum (q g_1)^\nu \le s_0 Q_0/16$.

Step S17.43: (E3). $\sum_{j\ge\nu} A_{1,j} \le \frac{2\varsigma_0 Q_0}{C_Q} \sum_{j\ge\nu} (\zeta g_1)^j \le \frac{3\varsigma_0 Q_0}{C_Q} (\zeta g_1)^\nu$ and $t_{\nu+1} = t_0 2^{-(\tau+1)} (2^{-(\tau+1)})^\nu$, so the quotient is $\le \frac{3\varsigma_0 Q_0 \Theta^{\tau+1}}{C_Q t_0} g_4^\nu \le \frac{3}{C_Q} Q_0 \Xi \le 2^{-10}$ ($g_4 \le 1$). Similarly $\sum_{j\ge\nu} A_{\eta,j} \le \frac{s_0 Q_0}{16} (q g_1)^\nu$ and $s_0 \le t_0$, $q g_1 \le g_4$: quotient $\le Q_0\Xi/16$. Sum of the two is $\le 2^{-10}$. $\sum_{j\ge\nu} \Lambda_{\mathbb{R},j} \le 2^{-12}$ by (E2).

Step S17.44: (E4). Proposition 16.13(v) with $t' = \frac12 t_{\nu+1}$, $j = 1$ gives $\le A/(t_{\nu+1}/2)$ for each component, with $A = A_{1,\nu}, A_{\eta,\nu}$; the max norm of the pair is at most the sum. □

What File 11a-4 must take from this file.

(T1) The sets $K_\nu \supset H_\infty$ and the maps X_ν, holomorphic on $\mathcal{D}(r_{\nu+1}, s_{\nu+1}; B_{\nu+1})$, with $r_{\nu+1} \geq s/2$, $s_{\nu+1} = s_0 q^{\nu+1}$, $B_{\nu+1} = K_{\nu+1} + t_{\nu+1}$; the composition $X_0 \circ \cdots \circ X_\nu$ is defined on $\mathcal{D}(r_{\nu+1}, s_{\nu+1}; B_{\nu+1})$ with values in $\mathcal{D}(s, s_0; B_0)$ (domain bookkeeping of Proposition 17.18(O3)). The remainders satisfy $|R_\nu| \leq \varepsilon_0 \varpi^\nu$ on $\mathcal{D}(r_\nu, s_\nu; B_\nu)$ and $H_\nu = h_\nu + \omega_\nu \cdot \eta + R_\nu$. The shrinking-domain estimates give exactly two *zero-section limits*: for $b \in K_\infty$ and every θ in the strip, $|R_\nu(\theta, 0, b)| \leq \varepsilon_\nu \to 0$, and, by Cauchy in η on $|\eta| < s_\nu/2$, $|\partial_\eta R_\nu(\theta, 0, b)| \leq 2\varepsilon_\nu/s_\nu = \ell_\nu \to 0$. Hence the limit torus (the image of the zero section under $X_0 \circ \cdots \circ X_\nu$), on which $H_\nu = h_\nu + \omega_\nu \cdot \eta + R_\nu$ has vanishing remainder and vanishing first η-jet of the remainder in the limit, is what this file delivers. It does *not* deliver convergence of $\partial_\eta^2 R_\nu$, nor a limiting normal form $h_* + \omega_* \cdot \eta + O(\eta^2)$ on a fixed action neighbourhood: $s_\nu \to 0$ and $\varepsilon_\nu / s_\nu^2 = (\varepsilon_0/s_0^2)\, 16^\nu \to \infty$. If File 11a-4 wants an off-section normal form, it must prove the convergence and extension of the affine action coefficients $\Delta h_\nu, \Delta\omega_\nu$ (and of the maps) on a common domain; the physical consumer needs only the torus and its first-jet identity.

(T2) Displacement of the defect: $\sum_\nu A_{1,\nu} \leq 3\varsigma_0 Q_0 / C_Q$ (angles), $\sum_\nu A_{\eta,\nu} \leq s_0 Q_0/16$ (actions), both explicit in the data (E2); the actions $I = b + \eta$ differ from b by the action defect, whose total size is at most s_0 (the initial translation) plus $\sum A_{\eta,\nu}$ on $|\eta| < s_0$; for $\eta = 0$ it is $\leq \sum A_{\eta,\nu}$.

(T3) The *unscaled* derivative budgets, not the stage-norm bound $2\kappa_\nu$: $|DX_\nu - \mathbf{1}|_\infty \leq 2\kappa_\nu + \kappa_\nu s_\nu/\sigma_\nu$ in (θ, η) with $\sum \leq 2^{-12}$ (E2), and $|D\Theta_\nu - \mathbf{1}|_\infty \leq e^{\Lambda_{\mathbb{R},\nu}} - 1$ on the reals with $\sum \Lambda_{\mathbb{R},\nu} \leq 2^{-12}$; these are the inputs for the λ-Lipschitz bound in θ (the compositions of maps close to the identity in the fixed sup-norm). The parameter derivatives (E4) and tails (E3): $\sum_{j \geq \nu}(A_{1,j} + A_{\eta,j}) \leq 2^{-10} t_{\nu+1}$ (the plan's "κ-versus-τ" condition: it holds because $g_4 \leq 2^{-5}$ and $Q_0 \Xi \leq 2^{-10}$). The local-to-global argument takes $b, b' \in K_\infty$ with $|b - b'|_\infty < t_{\nu+1}/2$: the segment $[b, b']$ lies in $K_{\nu+1} + t_{\nu+1}/2$ (within $t_{\nu+1}/2$ of $b \in K_{\nu+1}$), where (E4) applies; no convexity of K_∞ is used.

(T4) The frequency: ω_* on K_∞ and $w_b = \omega_*(b)$ is (γ, τ)-Diophantine exactly on a set containing H_∞ (Theorem 17.19(c)); $|\omega_* - \omega_\nu|_\infty \leq S_\nu$ on K_∞.

(T5) The inequalities (A1)–(A7), (E2)–(E3) are *part of the admissibility* $\varepsilon \leq \varepsilon_{\mathrm{adm}}$ (the factor Ξ in $\lambda_{\max}$ is what makes (E2), (E3) hold); 11a-4 adds no further smallness unless its λ-Lipschitz constant must be $< 2^{-10}$, in which case it multiplies $\lambda_{\max}$ by a constant.

17.10. **The polynomial budget and its numerical check (risk R1).** Part 11 chooses the order of its finite preparation after the exponent of the budget, so the budget must be a proved lower bound in the exact form in which Part 11 uses it: every datum except one is held fixed and the *amount* β tends to zero. The only data on which $\varepsilon_{\mathrm{adm}}$ depends are (N, μ, ρ, τ) (fixed integers and exponents) and $(\vartheta, s, \beta, \varepsilon^*, M, M_0, C_1, d)$, where $d = \operatorname{diam}_\infty K$ and d, M enter only through γ (β enters through γ and directly through $\theta_{\max}$). The next proposition bounds $\varepsilon_{\mathrm{adm}}$ from below along an arbitrary one-parameter family of such data, in terms of explicit polynomial rates.

Proposition 17.21 (Polynomial lower bound for $\varepsilon_{\mathrm{adm}}$)**.** *Fix $N, \mu \geq 1$, $\rho \in \{1, 2\}$ and $\tau > \frac{N\mu}{\rho} - 1$. Let $s_\tau := \frac{\rho(\tau+1)}{\mu} - N > 0$ (the lattice gap of Corollary 15.11; it is a constant, not the strip width s). Let $c \in (0, 1]$ and exponents $a_\vartheta, a_s, a_\beta, a_*, a_M, a_0, a_C, a_d \geq 0$ be given, and let, for every $u \in (0, 1]$, data $(\vartheta, s, \beta, \varepsilon^*, M, M_0, C_1, d, K, \ldots)$ as in Assumption 17.1 (with $C_1 \geq 1$) be given such that*

$$\vartheta \geq c\, u^{a_\vartheta}, \quad s \geq c\, u^{a_s}, \quad \beta \geq c\, u^{a_\beta}, \quad \varepsilon^* \geq c\, u^{a_*},$$
$$M \leq c^{-1} u^{-a_M}, \quad M_0 \leq c^{-1} u^{-a_0}, \quad C_1 \leq c^{-1} u^{-a_C}, \quad d \leq c^{-1} u^{-a_d}.$$

Put

$$D = Na_d + a_M + a_\vartheta + \big(1+\tfrac{1}{\mu}\big)a_\beta, \qquad G = \max\Big\{\frac{a_\beta}{\rho},\ \frac{\mu}{\rho}\,(a_* + D)\Big\}, \qquad L = \begin{cases} a_s + 1 & a_s > 0, \\ 0 & a_s = 0, \end{cases}$$

$$J = \max\{a_\vartheta,\ G + (\tau+1)L + a_C\}, \qquad X = \max\{J, L\},$$
$$H = \max\big\{J,\ G + (N+1)a_s + \tau L + X,\ a_\beta + a_0 + \mu J\big\}, \qquad B = 2H + a_C.$$

Then there is $c' > 0$, depending only on (N, μ, ρ, τ, c) and on the strip rate a_s (for $a_s = 0$: on (N, μ, ρ, τ, c) alone; in no case on u, on the family once the rates are fixed, or on the iteration index), with

$$\varepsilon_{\mathrm{adm}} \ \geq\ c'\, u^B \qquad (u \in (0, 1]).$$

Source: Ours; this replaces the monomial asymptotics asserted in the first version of this section (reviews P11A3-01, -02). The exponent formula is the review's, with one more entry checked: the factor $1 + 1/s_\tau$ in C_{Dio} is a constant, not the strip width.

Proof. Below $C, c_i > 0$ denote constants depending only on (N, μ, ρ, τ, c) and a_s (this is needed only in Step S17.46; all other steps are free of the rates); every exponent is ≥ 0 and $u \leq 1$, so enlarging an exponent weakens a bound of the form u^{-e} or u^e in the right direction.

Step S17.45: The loss γ. Since $M/\beta \leq c^{-2}u^{-a_M - a_\beta}$, $d \leq c^{-1}u^{-a_d}$ and $\vartheta \geq cu^{a_\vartheta}$: $C_{\mathrm{sub}} = 2^{N+2}\mu(\sqrt{N}d)^{N-1}\frac{M}{\beta}(1 + (4 + \vartheta^{-1})\sqrt{N}d) \leq Cu^{-(N-1)a_d - a_M - a_\beta - a_d - a_\vartheta}$ (the last bracket is $\leq C(1 + u^{-a_\vartheta})(1 + u^{-a_d}) \leq Cu^{-a_\vartheta - a_d}$). Hence $C_{\mathrm{Dio}} = C_{\mathrm{sub}}\beta^{-1/\mu}N^{\rho/2\mu}2 \cdot 3^{N-1}(1 + 1/s_\tau) \leq Cu^{-D}$, because $\beta^{-1/\mu} \leq c^{-1/\mu}u^{-a_\beta/\mu}$. So both entries of $\gamma_*(\varepsilon^*) = \min\{\frac{1}{2}N^{-1/2}(\beta/4)^{1/\rho}, (\varepsilon^*/C_{\mathrm{Dio}})^{\mu/\rho}\}$ are $\geq c_1 u^{a_\beta/\rho}$, respectively $\geq c_1 u^{(\mu/\rho)(a_* + D)}$, and $\gamma = \gamma_*/2 \geq c_2 u^G$.

Step S17.46: The cutoff. $T_0 = \max\{1, \frac{16}{\pi s}(A_0 + N\log\frac{32N}{s})\}$ with A_0 depending on (N, μ, τ) only. If $a_s = 0$, $s \geq c$ and $T_0 \leq C$. If $a_s > 0$, $\log\frac{32N}{s} \leq \log\frac{32N}{c} + a_s\log\frac{1}{u} \leq C_0(1 + a_s)u^{-1}$ (as $\log\frac{1}{u} \leq \frac{1}{u}$) and $\frac{1}{s} \leq c^{-1}u^{-a_s}$, so $T_0 \leq C_0(1 + a_s)u^{-a_s - 1}$; the factor $1 + a_s$ is absorbed in C, which is why C and hence c' may depend on a_s. In both cases $T_0 \leq Cu^{-L}$, and $a = \gamma T_0^{-\tau-1} \geq c_3 u^{G + (\tau+1)L}$.

Step S17.47: The collar and Ξ. $t_0 = \min\{\vartheta/\sqrt{N}, a/(4C_1)\} \geq c_4 u^J$ (use $C_1 \leq c^{-1}u^{-a_C}$). Since $\varsigma_0 = s/24 \leq 1$, $t_0 \leq 1$ and $\varsigma_0 \geq c_5 u^{a_s}$: $\Xi = 2^{\tau+1}(1 + \frac{t_0}{\varsigma_0} + \frac{\varsigma_0}{t_0} + T_0\varsigma_0) \leq C(1 + u^{-a_s} + u^{-J} + u^{-L}) \leq Cu^{-X}$ ($a_s \leq L$).

Step S17.48: The three terms. (1) $C_1 t_0/8 \geq c_4 u^J/8$ ($C_1 \geq 1$). (2) $\min\{\varpi/2, 2^{-10}/\Xi\} \geq c_6 u^X$ and $\frac{\pi\gamma\varsigma_0^{N+1}}{C_Q 2^N T_0^\tau} \geq c_7 u^{G + (N+1)a_s + \tau L}$, so the second term is $\geq c_8 u^{G + (N+1)a_s + \tau L + X}$. (3) For $\rho = 1$, $\theta_{\max} = \beta/2 \geq \frac{c}{2}u^{a_\beta}$. For $\rho = 2$: if $\beta \leq 2M_0$ then $2M_0 + \beta \leq 4M_0$ and $\beta/(2^{\mu+1}(2M_0 + \beta)) \geq 2^{-\mu-3}\beta/M_0$; otherwise $2M_0 + \beta < 2\beta$ and it is $\geq 2^{-\mu-2}$. Hence $\theta_{\max} \geq 2^{-\mu-3}\min\{\beta, \beta/M_0, 1\} \geq c_9 u^{a_\beta + a_0}$. In both cases the third term is $\geq c_{10}u^{a_\beta + a_0 + \mu J}$ by $t_0 \geq c_4 u^J$. Altogether $\lambda_{\max} \geq c_{11}u^H$.

Step S17.49: Conclusion. $\varepsilon_{\mathrm{adm}} = \lambda_{\max}^2/(2NC_1) \geq c_{11}^2 u^{2H}\, c\, u^{a_C}/(2N)$. □

Remark 17.22 (Why the hypotheses are all needed). The dependence of c' on a_s cannot be removed: take fixed (N, μ, ρ, τ, c) and the family with only $s = c\,u^A$ varying ($a_s = A$, all other rates 0). Then $B = 2\mu(\tau + 1)(A + 1)$, and at the same $u = e^{-1}$ one has $T_0 \geq c_0 e^A(A + b)$ for fixed $c_0, b > 0$, so $\varepsilon_{\mathrm{adm}} \leq \varepsilon^{(3)} \leq CT_0^{-2\mu(\tau+1)}$ gives $\varepsilon_{\mathrm{adm}}/u^B \leq C'(A + b)^{-2\mu(\tau+1)} \to 0$ as $A \to \infty$ (review P11A3R2-01). The corollary below has $a_s = 0$. Dropping the upper bound on M destroys the bound: take $N = 2$, $h(I) = \frac{1}{2}|I|_2^2$ on a fixed box with $\mu = 3$, $\rho = 2$, fixed $\vartheta, s, \beta, \varepsilon^*$, and let the admissible upper bound $M = M(u)$ for the derivatives of $g_\rho[\omega]$ be $e^{1/u}$ (any larger bound is again a bound); then $\gamma \propto M^{-3/2}$ and, as the closeness branch eventually binds, $\varepsilon_{\mathrm{adm}} \propto M^{-9}$, which is smaller than every cu^B. Dropping the normalisation $C_1 \geq 1$ does the same: for $h(I) = I$ ($N = 1$) every $C_1 > 0$ is a

valid Hessian bound, and $\varepsilon^{(1)} = C_1 t_0^2/(128N)$ is proportional to C_1 for small C_1. The diameter d and the size M_0 of ω enter likewise and are therefore data of the family.

Corollary 17.23 (The budget in the form used by Part 11)**.** *Fix $N, \mu \geq 1$, $\rho \in \{1, 2\}$, $\tau > \frac{N\mu}{\rho} - 1$, and fix* all *data of Assumption 17.1 — K, ϑ, s, ε^*, M, M_0, C_1 — except the amount β, which is free in $(0, \beta_{\max}]$, $\beta_{\max} = \min\{1, M\}$. Put*

$$G_0 = \frac{\mu + 1}{\rho}, \qquad B_0 = 2\max\{2G_0,\ 1 + \mu G_0\}.$$

Then there is $c_* > 0$, depending only on the fixed data, with $\varepsilon_{\mathrm{adm}} \geq c_*\, \beta^{B_0}$ for all $\beta \in (0, \beta_{\max}]$. For $\rho = 2$, $B_0 = \mu(\mu + 1) + 2$.

Source: Ours; the form of the polynomial budget (K5) of Definition 14.1: the fixed-data case $a_\beta = 1$, all other exponents 0*, of Proposition 17.21.*

Proof. **Step S17.50: Specialise.** Take $u = \beta \in (0, 1]$, $a_\beta = 1$, $a_\vartheta = a_s = a_* = a_M = a_0 = a_C = a_d = 0$ and c the minimum of 1, ϑ, s, ε^*, M^{-1}, M_0^{-1}, C_1^{-1}, d^{-1} (all fixed). Then $D = 1 + \frac{1}{\mu}$, $G = \max\{\frac{1}{\rho}, \frac{\mu}{\rho}(1 + \frac{1}{\mu})\} = \frac{\mu+1}{\rho} = G_0$, $L = 0$, $J = \max\{0, G_0\} = G_0$, $X = G_0$, $H = \max\{G_0,\ 2G_0,\ 1 + \mu G_0\}$, $B = 2H$. Since $2G_0 \geq G_0$ this is B_0. For $\rho = 2$, $G_0 = \frac{\mu+1}{2}$ and $1 + \mu G_0 \geq 2G_0$ iff $\mu \geq 1$, so $B_0 = 2 + \mu(\mu + 1)$. □

Reading of the exponent. Each of the three branches of Remark 17.14 contributes one entry of H: the first and third through J (the collar $t_0 \gtrsim u^J$), the second through $G + (N+1)a_s + \tau L + X$ (the quadratic branch, including Ξ). The schedule contributes only constants, powers of 2 ($\varpi = 2^{-2E_*}$, $E_* = 2(\tau+1)(\mu+1) + N + 4$), so the dependence on the iteration index is nil; the constant c' carries $2^{-O(\mu\tau)}$. The exponent is conservative in $L = a_s + 1$ (which absorbs the logarithm in T_0) and in the maximum of three exponents.

The numerical check. The scripts are in `.drafts/P/bp-scratch-pfix/` (`sched.py`, `rand.py`, `budget.py`; mpmath, 60 digits; the bump-function L^1 norms behind $c'_{N,\mu}$ are *approximations* by a trapezoidal rule, not certified constants; the rest uses exact $\Lambda_\delta = \coth(\pi\delta)^N - 1$, d_ν, E_ν, not the bounds of the proofs). `sched.py` builds the schedule of Definition 17.5 with $C_1 \to \max\{1, C_1\}$, with M and M_0 as separate inputs, $\gamma = \gamma_*/2$, $\varepsilon = \varepsilon_{\mathrm{adm}}$, and checks, for $\nu < 90$: (C1)–(C3); $\sum d_\nu \leq s/4$, $r_\nu \geq s/2$; $\|D\omega_\nu\| \leq 2C_1$; the recursion; the finite-stage tails $S_m \leq \gamma T_m^{-\tau-1}$ for *all* m, the infinite parts of the sums being bounded by the exact geometric envelope $\ell_{n-1}\varrho/(1 - \varrho)$; the closeness sum (envelope ratio $g_3 \leq 2^{-5}$); $Q_0 \leq \varpi/2$, $Q_0\Xi \leq 2^{-10}$; the budgets (E2) *against the stated* 2^{-12} (both sums), and (E3) against 2^{-10} (tail ratios checked $\leq 2^{-4}$). Four cases $(N, \mu, \rho, \tau; C_1, \vartheta, s, \beta, M{=}M_0) = (1, 2, 1, 2; 2, .5, .5, .3, 3)$, $(2, 3, 2, 3; 2, .3, .3, .2, 4)$, $(4, 4, 2, 8; 1, .2, .2, .1, 2)$, $(6, 5, 2, 15; 1, .1, .1, .05, 2)$: all inequalities hold; $\varepsilon_{\mathrm{adm}} \approx 2.4 \cdot 10^{-86}$, $1.1 \cdot 10^{-158}$, $4.2 \cdot 10^{-442}$, $1.9 \cdot 10^{-1036}$, with binding branch (Remark 17.14) the quadratic one in the first case and the closeness one in the other three. `rand.py`: 40 random data sets with independent M, M_0, C_1 below and above 1: no failure; the quadratic branch binds in 17 and the closeness branch in 23 runs, the first never. `budget.py`: along families with a single rate equal to 1 and along the family with all eight rates equal to 1, for $(N, \mu, \rho, \tau) = (1, 2, 1, 2)$ and $(2, 3, 2, 3)$, the measured log–log slopes of $\varepsilon_{\mathrm{adm}}$ never exceed the exponent B of Proposition 17.21; they equal B for $\beta, M, C_1, \varepsilon^*$ (and approach it for ϑ, d), e.g. 14 for β in both cases (= B_0 of Corollary 17.23), 8 and 9 for M; for s the measured slope (12.2, 24.6) is below B (24, 48), reflecting the conservative $L = a_s + 1$; for all rates 1: $63.2 \leq 77$ and $92.6 \leq 116$. These are tests of the formulas; the proofs are Proposition 17.13, Lemma 17.20 and Proposition 17.21.

17.11. Sources, deviations, errata, what is reused.

Sources read. Rus01 §13.1–13.4 (pp. 156–158; the printed formulas (13.1.2), (13.1) are partly garbled in the text layer of the pinned PDF: only $\kappa = 24/25$, $\tau_0 = 1/9$ (read from the rendered page; the text layer drops the denominator) and the structure $\varepsilon_\nu = \varepsilon_0\delta^{\kappa\nu}$, $t_\nu = t_0\delta^{\tau\nu}$, $T_\nu = \Psi^{-1}(\Psi(T_0)\delta^{\tau\nu})$ are used), §14.1–14.8 (pp. 165–170), §16.1–16.7 (pp. 178–179), §20.5–20.7 (pp. 201–203). Not read: §§13.5–13.12 (one-step remainder; recomputed in File 11a-2), §15 (convergence; File 11a-4), §§19–20.4 (smoothing, chains; replaced). Files 11a-1, 11a-2; `reviews/p11a2-review-1.md` (P11A2-01, -03, -04); `PLAN-11a.md` (deviation table, R1, R2, R5).
Deviations (D-numbers as in PLAN-11a.md; D-16 to D-18 new).

- **D-3** (power divisor): $D_\nu = \gamma T_\nu^{-\tau}$ (not $\gamma\Phi(T_\nu)$ with $\Psi = T^{-d}\Phi$); no coupling of T_ν to ε_ν through Ψ^{-1}, so Rus01's exponent bookkeeping $\kappa - \sigma - \tau_0 > \dots$ and the integral condition 13.1.6 do not occur (Lemma 17.6).
- **D-7** (chain extension): mollified cutoffs; verified, Remark 17.10. Condition $g_3 \le 2^{-5}$ instead of $\tau\mu < \tau_0$.
- **D-9** (explicit $\varepsilon_{\mathrm{adm}}$): Definition 17.12.
- **D-15** (strip bookkeeping): $\sum(d_\nu + \delta_\nu + 2\sigma_\nu) \le s/2$, condition (A6).
- **D-16** (decay ratio and growth ratios): $\varpi = 2^{-2E_*}$ instead of δ^κ with $\kappa = \frac{24}{25}$; $\Theta = 2$, $\zeta = \frac{1}{2}$ free; the price is the constants $2^{-O(\mu\tau)}$ in $\varepsilon_{\mathrm{adm}}$.
- **D-17** (collar): $t_\nu = t_0 2^{-(\tau+1)\nu}$ with $t_0 \le \gamma T_0^{-\tau-1}/(4C_1)$, so that $t_{\nu+1}T_\nu\|D\omega_\nu\| \le D_\nu/2$ (Rus01 (13.4): $t_0 \le \gamma\Psi(T_0)\vartheta/(8(2p^2+p)(3C_\bullet+3)^{2p^2+p})$, p. 158).
- **D-18** (mode-wise test): $K_{\nu+1}$ tests the shell $\mathcal{S}_\nu$ with margin $2\gamma|k|_1^{-\tau}$ against ω_ν itself (not the cutoff-wide $\frac{1}{2}\gamma\Phi(T_\nu)$ of Rus01 (16.1)); the finite-stage invariant (Lemma 17.15) gives the cutoff-wide divisor.

Suspected errata. None new in the pages read. (The text layer of Rus01 p. 156 is garbled at (13.1.2) and (13.1); not a defect of the source.) Defect inherited from File 11a-1: the endpoint and the limit-based form of (I3) (Remark 17.16), already noted by the review of File 11a-2.
Doubts about frozen statements. None: no frozen declaration is touched. Note for the consumers: the geometric scheme gives $\varepsilon_{\mathrm{adm}}$ polynomial but with a large degree; Part 11's finite preparation order must exceed the exponent B_0 of Corollary 17.23; its interface item (K5) must list M_0, d and $C_1 \ge 1$ among the fixed data.
External facts used. (Y1)–(Y4); Files 11a-1, 11a-2's lists; [K:`bp:p1:uniform-limits`], [K:`bp:p1:exp-sums`], [K:`bp:p1:decay`].
Forward references. Part 11a-4 (limit, tori, displacement, family): (O3), (O4), Lemma 17.20; Part 11a-5 (budget): Definition 17.12, Theorem 17.19, Proposition 17.21, Corollary 17.23. No unwritten label is cited except those.
Label index. `sel-data`, `geom-recursion`, `schedule-consts`, `schedule-ratios`, `schedule`, `strip-budget`, `schedule-props`, `smooth-cutoff`, `finite-extension`, `d7-verdict`, `initial`, `adm-params`, `adm-choice`, `finite-divisor`, `i3-corrected`, `finite-measure`, `inductive-lemma`, `parameter-selection`, `exported-sums`, `adm-branches`, `adm-budget`, `adm-budget-necessary`, `adm-budget-beta`.

18. P-PLANAR: assembly through the frozen P-G

This part proves the frozen theorem P-PLANAR (Theorem 1.29) by route (c) of `PLAN.md`: the *frozen* P-G (Theorem 1.28, an input, proved in Part 5) is applied to the planar secular Birkhoff form of Part 11b in the Poincaré product chart of Part 7, and its cylinder tori are transported to physical coordinates by Part 1 and Part 6. Nothing about KAM theory is proved here: the part is the bookkeeping of data, thresholds and clauses.

- `forward-tube`, `collar`, `fast-diffeo` (Lemmas 18.1, 18.2, 18.3): the elementary facts about the chart image of a small cylinder and the box V that P-G needs;

- `data`, `chart`, `perturbation` (Definition 18.4, Proposition 18.6, Definition 18.5): the data of P-G in the order in which they are chosen; the total chart $\Psi_{\mathrm{tot}} = \Psi^{(2)} \circ T$ (T of Part 11b) and the perturbation $P = \hat{f}_{\mathrm{mov}} \circ T$;
- `pg-hypotheses` (Proposition 18.7): every hypothesis (H1)–(H6) and every datum of P-G, including joint analyticity, periodicity, the fast diffeomorphism, the boundary collar and $B = \min|\det\hat{\beta}|$;
- `transport`, `measure` (Propositions 18.8, 18.9): cylinder tori $\to$ physical tori, measurability and $0 < \mathrm{meas} < \infty$;
- `planar` (Theorem 18.10) and `quantifiers` (Remark 18.11): the complete proof of the frozen statement and the order in which every constant is chosen.

18.1. **Setting, dictionary, external facts.** Fix $G, m_0 > 0$, $\kappa = Gm_0$, $m, a : \mathtt{Fin}\,2 \to \mathbb{R}$ with $m_i > 0$, $0 < a_0 < a_1$, and $\eta > 0$ (the data of Theorem 1.29). *Index dictionary.* The Lean planets $0, 1$ are the planets $1, 2$ of Parts 7–11b (inner planet first, *increasing* axes, §10.1); $\mathcal{A}_2 = \{0 < a_1 < a_2\}$, $\Lambda_i^* = m_i\sqrt{\kappa a_i}$, $a_i(\Lambda) = \Lambda_i^2/(m_i^2\kappa)$ in those parts. I write the axes as $a = (a_1, a_2)$ in this part. A point of $\mathrm{Ph}(4) = \mathrm{Ph}(2+2) = \mathrm{Ph}(2\cdot 2)$ is $w = ((\lambda,\xi),(\Lambda,\eta))$ with $\lambda, \xi, \Lambda, \eta \in \mathbb{R}^2$; for the frozen P-G ($a = b = 2$) this is $z = \mathrm{assemble}(I, \varphi, p, q) = ((\varphi, q), (I, p))$ with

$$(I, \varphi, p, q) = (\Lambda, \lambda, \eta, \xi) \qquad \text{(dictionary of §10.1)},$$

and the chart $\Psi = \Psi^{(2)}$ of Theorem 9.12. The numerals $2\cdot 2$ and $2+2$ both reduce to 4 (the Lean types `Fin (2*2)` and `Fin (2+2)` agree by `rfl`); the first two coordinates of each block are the fast ones. Throughout $\Gamma_i(w) = \frac{1}{2}(\xi_i^2 + \eta_i^2)$ and $z = (x, y) \in \mathbb{R}^2 \times \mathbb{R}^2$ denotes the oscillator variable of Part 11b, with $(x, y) = (q, p)$ in the P-G dictionary, $R_j = \frac{1}{2}(x_j^2 + y_j^2)$, and $|z|_2^2 = |x|^2 + |y|^2$ (Euclidean). A point w of the cylinder $\mathcal{C}(V, r)$ (Definition 1.10, $b = 2$) satisfies $x_j^2 + y_j^2 < r^2$ for $j = 1, 2$, hence

$$|z|_2^2 < 2r^2 \qquad \text{on } \mathcal{C}(V, r). \tag{8}$$

External facts.

(F1) Composition of real analytic maps is real analytic; a function that agrees on an open set with a real analytic function is real analytic (and differentiable) there ((X1)).
(F2) A continuous map on a compact metric space is uniformly continuous (Heine–Cantor).
(F3) An injective continuous image of a Borel set is Borel (`MeasurableSet.image_of_continuousOn_injOn`, (X4)).
(F4) Elementary: on $(0, \infty)$ the map $s \mapsto s^{1/3}$ is real analytic; $\log(1/r) > 0$ and $r^6/(\log(1/r))^c > 0$ for $0 < r < 1$, $c > 0$.

18.2. **Elementary lemmas.**

Lemma 18.1 (Forward tube: small chart cylinders lie in the circular tube)**.** *Let $a_i > 0$, $\Lambda_i^* = m_i\sqrt{\kappa a_i}$ and $\zeta > 0$. There is $\delta_\zeta \in \big(0, \frac{1}{4}\min_i \Lambda_i^*\big]$, depending only on (G, m_0, m, a, ζ), such that every $w = ((\lambda,\xi),(\Lambda,\eta)) \in \mathbb{R}^8$ with $|\Lambda_i - \Lambda_i^*| \le \delta_\zeta$ and $\Gamma_i(w) \le \delta_\zeta$ for $i = 1, 2$ lies in $\mathcal{O}_2$ and satisfies $|\Psi^{(2)}(w) - \mathrm{circ}(\lambda/2\pi)|_\infty < \zeta$ (max norm on $\mathrm{Ph}(4)$). In particular $\Psi^{(2)}(w) \in U_\zeta$ (Definition 1.17).*
Source: Ours (uniform continuity of the chart); the converse inclusion is Lemma 9.14.

Proof. **Step S18.1: A compact set.** Put $\delta_0 = \frac{1}{4}\min_i \Lambda_i^*$ and $Q_0 = \{w : \lambda \in [0, 2\pi]^2,\ |\Lambda_i - \Lambda_i^*| \le \delta_0,\ \xi_i^2 + \eta_i^2 \le 2\delta_0\}$, compact. On Q_0: $\Lambda_i \ge \frac{3}{4}\Lambda_i^* > 0$ and $\Gamma_i \le \delta_0 < \Lambda_i$, so $Q_0 \subset \mathcal{O}_2$ (Definition 9.11).

Step S18.2: Uniform continuity. $\Psi^{(2)}$ is continuous on $\mathcal{O}_2$ (Theorem 9.12(a)), hence uniformly continuous on Q_0 (F2): there is $\delta_1 > 0$ such that $w, w' \in Q_0$, $|w - w'|_\infty < \delta_1$ imply $|\Psi^{(2)}(w) - \Psi^{(2)}(w')|_\infty < \zeta$. Set $\delta_\zeta = \min\{\delta_0, \delta_1/2, \delta_1^2/4\}$.

Step S18.3: Comparison with the circular torus. Let w be as in the statement; replacing λ by its residue in $[0,2\pi)^2$ changes neither $\Psi^{(2)}(w)$ nor $\mathrm{circ}(\lambda/2\pi)$ (period 2π, Theorem 9.12(a); circ has period one in ϕ), so we may assume $\lambda \in [0,2\pi]^2$. Then $w \in Q_0$ and $w_0 = ((\lambda,0),(\Lambda^*,0)) \in Q_0$ with $|w-w_0|_\infty \le \max\{\delta_\zeta, \sqrt{2\delta_\zeta}\} < \delta_1$ (since $|\xi_i|, |\eta_i| \le \sqrt{2\Gamma_i}$). By Step S18.2 and $\Psi^{(2)}(w_0) = \mathrm{circ}(\lambda/2\pi)$ (Lemma 9.14, last sentence), $|\Psi^{(2)}(w) - \mathrm{circ}(\lambda/2\pi)|_\infty < \zeta$. □

Lemma 18.2 (Boxes have an $O(t)$ collar)**.** *Let $V = \prod_{i=1}^2 (\ell_i, u_i) \subset \mathbb{R}^2$ with $L_i = u_i - \ell_i > 0$ (an open, connected, bounded set). Put $A_c = 4(L_1 + L_2)$ and $d_c = \frac{1}{2}\min_i L_i$. Then for all $0 < t < d_c$,*

$$\mathrm{meas}\{I \in \mathbb{R}^2 : \ \exists J \in \partial V, \ |I - J|_\infty < t\} \le A_c\, t.$$

This is exactly the boundary collar datum of Theorem 1.28 with $A = A_c$, $d = d_c$.
Source: Ours, elementary; the collar is a corrected-contract premise (audit P2, CP10 p. 561).

Proof. **Step S18.4: Inclusion.** Let N_t be the set in question. If $I \in N_t$ and $J \in \partial V$ with $|I-J|_\infty < t$, then $I \in \prod(\ell_i - t, u_i + t) =: E_t$ ($J \in \overline{V}$). Moreover $I \notin \prod[\ell_i + t, u_i - t] =: E_t'$: a point $J \in \partial V$ has some coordinate $J_i \in \{\ell_i, u_i\}$, and every $I \in E_t'$ has $|I_i - J_i| \ge t$, so $|I-J|_\infty \ge t$. Hence $N_t \subset E_t \setminus E_t'$.

Step S18.5: Measure. For $t < d_c$ all $L_i - 2t > 0$, $E_t' \subset E_t$ and $\mathrm{meas}(E_t \setminus E_t') = \prod(L_i + 2t) - \prod(L_i - 2t) = 4t(L_1 + L_2)$. □

Lemma 18.3 (The fast diffeomorphism for the Kepler part)**.** *Let $h(\Lambda) = -\sum_{i=1}^2 \kappa^2 m_i^3/(2\Lambda_i^2)$ (Theorem 9.12(b); Lean's total division gives the summand 0 at $\Lambda_i = 0$, never used) and let $W = \prod_{i=1}^2 (\ell_i', u_i')$ with $0 < \ell_i' < u_i'$. Then* HasFastDiffeo(h, W) *(Definition 1.13) holds, with $\omega_h(\Lambda)_i = \nu_i(\Lambda_i) = \kappa^2 m_i^3 \Lambda_i^{-3}$ and inverse $g(v)_i = (\kappa^2 m_i^3/v_i)^{1/3}$.*

Proof. **Step S18.6: Analyticity, frequencies.** h is real analytic on $\{\Lambda_1\Lambda_2 \neq 0\} \supset W$, and $\omega_h(\Lambda)_i = Dh(\Lambda)(e_i) = \kappa^2 m_i^3 \Lambda_i^{-3}$ (Theorem 9.12(b)); this is clause (i) of Definition 1.13.

Step S18.7: Open image. Each ν_i is a strictly decreasing homeomorphism of (ℓ_i', u_i') onto the open interval $(\kappa^2 m_i^3/u_i'^3, \ \kappa^2 m_i^3/\ell_i'^3) \subset (0,\infty)$, so $\omega_h(W)$ is the product of these intervals, open (clause (ii)).

Step S18.8: The inverse. Put $g(v)_i = (\kappa^2 m_i^3/v_i)^{1/3}$ (`Real.rpow`). On $\omega_h(W) \subset (0,\infty)^2$ it is real analytic (F4), hence C^1; $g(\omega_h(\Lambda)) = \Lambda$ for $\Lambda \in W$, and for $v \in \omega_h(W)$: $g(v) \in W$ and $\omega_h(g(v)) = v$ (each ν_i is a bijection of the interval onto its image with inverse $v_i \mapsto (\kappa^2 m_i^3/v_i)^{1/3}$). This is clause (iii). □

18.3. The data of P-G, in the order of choice.

Definition 18.4 (Data of the application of P-G)**.** Given (G, m_0, m, a, η) as in §18.1, choose successively:

(D1) (physical constants) $\zeta_* = \min\{\eta, \gamma_a/6\}$, $d = \gamma_a/2$, $D = 2a_{\max} + \gamma_a$ with $a_{\max} = \max a_i$, $\gamma_a = \min\{a_1, a_2, a_2 - a_1\}$ (Definition 8.15, $n = 2$, indices shifted); they depend on a (and η for ζ_*) only. Put $\delta = \delta_{\zeta_*}$ of Lemma 18.1.

(D2) (axis neighbourhood) $\eta' \in (0, \frac{1}{2}\min\{a_1, a_2 - a_1\})$ such that $a' \in \mathbb{R}^2$, $|a' - a|_\infty < \eta'$ imply $a' \in \mathcal{A}_2$ and $|\Lambda(a') - \Lambda^*|_\infty < \delta$, where $\Lambda(a')_i = m_i\sqrt{\kappa a_i'}$ (continuity of Λ at a).

(D3) (good patch) $a^\circ \in \mathcal{A}_2$, $\rho_a > 0$ with $\overline{B}_\infty(a^\circ, 2\rho_a) \subset B_\infty(a, \eta') \cap \{\Phi_4 \neq 0\} \cap \{\det \hat\beta \neq 0\}$ (Corollary 13.16); $K_a = \overline{B}_\infty(a^\circ, \rho_a)$, $O_a = B_\infty(a^\circ, 2\rho_a)$ and $B = \min_{K_a} |\det \hat\beta| > 0$.

(D4) (action boxes) $K = \Lambda(K_a) = \prod[\ell_i, u_i]$, $\mathcal{O} = \Lambda(O_a) = \prod(\ell_i', u_i')$ (the map Λ acts coordinatewise and is increasing, so images of cubes are boxes, with $0 < \ell_i' < \ell_i < u_i < u_i'$), $V = \mathrm{int}\, K = \prod(\ell_i, u_i)$ and $W = \mathcal{O}$.

(D5) (Birkhoff data) the rotation ϱ, the generating function χ, the constants $r_*, C_* > 0$, an open $\mathcal{O}'$ with $K \subset \mathcal{O}' \Subset \mathcal{O}$, the domain $\mathcal{T} = \{(\lambda, \Lambda, z) : \Lambda \in \mathcal{O}',\ |z|_2 < r_*\}$ and the map $T = \Phi_\varrho^{-1} \circ \Phi_\chi : \mathcal{T} \to \mathcal{O}_2$ of Theorem 13.12 (applied to the cube K_a of (D3), $\hat{f} = \hat{f}_{\mathrm{mov}}$); they depend on (G, m_0, m, K_a) only.
(D6) (P-G data) $a = b = 2$, V, W as in (D4); collar constants $A = A_c$, d_{c} (Lemma 18.2); $r_0 = r_*/2$, $\mu_0^{\mathrm{G}} = 1$, B of (D3); and $H_0, P, P_0, \Omega, \beta$ of Definition 18.5.
(D7) (P-G constants) $r_{\mathrm{G}}, C_{\mathrm{G}}, c_{\mathrm{G}}$ delivered by Theorem 1.28 for the data (D6).
(D8) (secular radius) $r = \frac{1}{2}\min\{r_{\mathrm{G}}, \sqrt{\delta/2}\}$. Then $0 < r < r_{\mathrm{G}} < r_0$ and $r < 1$.
(D9) (coupling threshold) $\mu_0 = \min\{1,\ r^6/(\log(1/r))^{c_{\mathrm{G}}}\} > 0$.

No item depends on the coupling μ; items (D1)–(D3) depend on (G, m_0, m, a, η) only, and (D4)–(D9) on those and the choices before them.

Definition 18.5 (The perturbation and the Birkhoff data as total functions)**.** With the notation of Definition 18.4 put (all functions total, with arbitrary values where the formula is not defined, which are never used because every use is guarded by $I \in V$ or $w \in \mathcal{T}$):

$$H_0(I) = h(I), \qquad \Psi_{\mathrm{tot}} := \Psi^{(2)} \circ T \ \text{ on } \mathcal{T} \ (\text{and } := 0 \text{ elsewhere}),$$
$$P(\mu, w) = \hat{f}_{\mathrm{mov}}\big(T(w)\big) \ \text{ for } w \in \mathcal{T}, \qquad P(\mu, w) = 0 \ \text{ for } w \notin \mathcal{T},$$
$$P_0(\mu, I) = C_0(a(I)) + \delta_{\mathrm{S}}(I), \qquad \Omega(\mu, I) = 2\sigma(a(I)), \qquad \beta(\mu, I) = \hat{\beta}(a(I)) \qquad (I \in V),$$

and $P_0 = 0$, $\Omega = 0$, $\beta = 0$ for $I \notin V$. Here $\hat{f}_{\mathrm{mov}} = f_{\mathrm{mov}} \circ \Psi^{(2)}$ (Theorem 9.12(c)), δ_{S} is the function δ of Theorem 10.13 (renamed to avoid the clash with the constant δ of (D1)), C_0, $\sigma = (\sigma_1, \sigma_2)$ (eigenvalues of Q_h) and $\hat{\beta}$ are those of Definitions 10.10, 13.10, and $a(I) \in K_a$ for $I \in K$. P does not depend on μ.

Proposition 18.6 (The total chart)**.** *In the situation of Definition 18.4:*

(a) $\mathcal{T}$ is open, $\mathcal{T} + \iota_k' = \mathcal{T}$ for $\iota_k' = ((2\pi k, 0), 0)$, $k \in \mathbb{Z}^2$ (angle translations of the fast block), and $T(\mathcal{T}) \subset \mathcal{O}_2 \cap {\Psi^{(2)}}^{-1}(W_{\mathrm{col}})$ (the collision-free domain of Lemma 8.2, on which $\hat{f}$ is analytic).
(b) Ψ_{tot} is a 1-chart on $\mathcal{T}$ (Definition 2.3); in particular $D\Psi_{\mathrm{tot}}$ is invertible on $\mathcal{T}$.
(c) $\Psi_{\mathrm{tot}}(w + \iota_k') = \Psi_{\mathrm{tot}}(w)$ for $w \in \mathcal{T}$, $k \in \mathbb{Z}^2$, and if $w, w' \in \mathcal{T}$, $\Psi_{\mathrm{tot}}(w') = \Psi_{\mathrm{tot}}(w)$ then $w' = w + \iota_k'$ for some $k \in \mathbb{Z}^2$.
(d) For every real μ and every $w \in \mathcal{T}$: $H_{\mathrm{mov}}(\Psi_{\mathrm{tot}}(w)) = H_0(\Lambda) + \mu P(\mu, w)$, i.e. $H_{\mathrm{mov}}^{\Psi_{\mathrm{tot}}} = H_\mu$ on $\mathcal{T}$, with $H_\mu = H_0 + \mu P(\mu, \cdot)$ of Definition 1.11.
(e) For all $r \in (0, r_0]$ with $2r^2 \le \delta$ and every $w \in \mathcal{C}(V, r)$: $\mathcal{C}(V, r) \subset \mathcal{T}$ and $\Psi_{\mathrm{tot}}(w) \in U_{\zeta_*} \subset U_\eta$. (For the r of (D8): $2r^2 \le \delta/2 < \delta$.)

Proof. **Step S18.9: (a) Domain.** $\mathcal{T}$ is defined by conditions on (Λ, z) only, hence open and invariant under translations of λ. The inclusion of the image is the statement "with values in the domain of $\hat{f}$" of Theorem 13.12, whose proof (Step S13.32) takes $r_* \le \rho_K/\sqrt{2}$ so that $|\varrho Y| \le |Y| \le \sqrt{2}|z|$ lies in the polydisc of Proposition 10.2, where $\Psi^{(2)} \in W_{\mathrm{col}}$ (part (a) there); also $r_* \le r_1$ of Lemma 13.7.

Step S18.10: (b) Chart. $T = \Phi_\varrho^{-1} \circ \Phi_\chi$ is a real analytic symplectic diffeomorphism onto its image for $\Omega_{\mathrm{chart}} = d\Lambda \wedge d\lambda + dy \wedge dx$, the frozen Ω in the order $((\lambda, x), (\Lambda, y))$ (Proposition 13.9, Proposition 11.8); $\Psi^{(2)}$ is a 1-chart on $\mathcal{O}_2$ (Theorem 9.12(a)). By Remark 2.10(3) the composition is a 1-chart on $\mathcal{T}$; invertibility of the derivative is Lemma 2.1(b).

Step S18.11: (c) Lattice. T commutes with angle translations: $T(w + \iota_k') = T(w) + \iota_k'$ (Proposition 13.9), and $\Psi^{(2)}(w + \iota_k') = \Psi^{(2)}(w)$ (Theorem 9.12(a)); so $\Psi_{\mathrm{tot}}(w + \iota_k') = \Psi_{\mathrm{tot}}(w)$. If

$\Psi_{\mathrm{tot}}(w') = \Psi_{\mathrm{tot}}(w)$, then $T(w') = T(w) + \iota'_k$ for some $k \in \mathbb{Z}^2$ (injectivity modulo $2\pi\mathbb{Z}^2$ in Theorem 9.12(a)), i.e. $T(w') = T(w+\iota'_k)$ with $w+\iota'_k \in \mathcal{T}$; T is injective (diffeomorphism onto its image), so $w' = w + \iota'_k$.

Step S18.12: (d) The Hamiltonian. By Theorem 9.12(c) (the Lean total functions, $c = 1$), $H_{\mathrm{mov}}(\Psi^{(2)}(v)) = h(\Lambda(v)) + \mu\hat{f}_{\mathrm{mov}}(v)$ for all $v \in \mathcal{O}_2$ and all μ. Put $v = T(w)$: T preserves Λ (Proposition 13.9), so $h(\Lambda(v)) = H_0(\Lambda(w))$ and $\hat{f}_{\mathrm{mov}}(v) = P(\mu, w)$.

Step S18.13: (e) The tube. Let $w \in \mathcal{C}(V, r)$, $r \le r_0$. Then $\Lambda \in V \subset K \subset \mathcal{O}'$ and $|z|_2 < \sqrt{2}r \le \sqrt{2}r_0 < r_*$ by (8), so $w \in \mathcal{T}$. Write $T(w) = (\lambda', \Lambda, \xi, \eta)$, $(\xi, \eta) = \varrho(\Lambda)Y$ with $Y = (x', y')$, $|Y|_2^2 \le 2|z|_2^2 < 4r^2$ (Lemma 13.7), and $\varrho \in SO(2)$ acts on the planet index so $|\xi|^2 + |\eta|^2 = |Y|_2^2$. Hence $\Gamma_i(T(w)) \le \frac{1}{2}(|\xi|^2 + |\eta|^2) < 2r^2 \le \delta$. Moreover $\Lambda \in K$ gives $|\Lambda - \Lambda^*|_\infty < \delta$ by (D2), (D3). Lemma 18.1 with $\zeta = \zeta_*$ yields $\Psi_{\mathrm{tot}}(w) \in U_{\zeta_*}$, and $U_{\zeta_*} \subset U_\eta$ by Lemma 8.16(a). □

18.4. **The hypotheses of P-G.**

Proposition 18.7 (All data and hypotheses of P-G hold)**.** *With the data of Definition 18.4 and the functions of Definition 18.5, every datum and every hypothesis (H1)–(H6) of Theorem 1.28 (with $a = b = 2$) holds: $V \subset \mathbb{R}^2$ is open, connected and bounded; $W \supset \overline{V}$ is open; the collar holds with $A = A_c$, $d = d_c$; $r_0, \mu_0^{\mathrm{G}}, B > 0$; and*

(H1) $(\mu, w) \mapsto P(\mu, w)$ *is real analytic at every point of* $[0,1] \times \overline{\mathcal{C}(V, r_0)}$*;*
(H2) $P(\mu, w + \iota'_k) = P(\mu, w)$ *for* $\mu \in [0,1]$, $w \in \mathcal{C}(V, r_0)$, $k \in \mathbb{Z}^2$*;*
(H3) $\mathrm{HasFastDiffeo}(H_0, W)$*;*
(H4) $\mathrm{HasBirkhoff}(P, V, r_0, 1, P_0, \Omega, \beta)$*;*
(H5) $|\det \beta(\mu, I)| \ge B$ *for* $\mu \in [0,1]$, $I \in V$*;*
(H6) $\beta(\mu, I)$ *is symmetric for* $\mu \in [0,1]$, $I \in V$*.*

Proof. **Step S18.14: The sets.** $V = \prod(\ell_i, u_i)$ is open, connected, bounded; $W = \mathcal{O} = \prod(\ell'_i, u'_i)$ is open and $\overline{V} = K \subset \mathcal{O}$ by (D4). The collar is Lemma 18.2. $r_0 = r_*/2 > 0$, $\mu_0^{\mathrm{G}} = 1$, $B > 0$ by Corollary 13.16.

Step S18.15: (H1) Joint analyticity. Let $w \in \overline{\mathcal{C}(V, r_0)}$: $\Lambda \in \overline{V} = K \subset \mathcal{O}'$ and $x_j^2 + y_j^2 \le r_0^2$, so $|z|_2 \le \sqrt{2}r_0 = r_*/\sqrt{2} < r_*$ and $w \in \mathcal{T}$, which is open (Proposition 18.6(a)). On $\mathbb{R} \times \mathcal{T}$ the function $(\mu, w) \mapsto \hat{f}_{\mathrm{mov}}(T(w))$ is real analytic: T is analytic on $\mathcal{T}$ and $\hat{f}_{\mathrm{mov}}$ is analytic on its domain, which contains $T(\mathcal{T})$ (Theorem 13.12(a), Proposition 10.2(a)); (F1). The independence of μ makes the joint analyticity trivial.

Step S18.16: (H2) Periodicity. For $w \in \mathcal{C}(V, r_0) \subset \mathcal{T}$: $w + \iota'_k \in \mathcal{T}$, $T(w + \iota'_k) = T(w) + \iota'_k$ (Proposition 13.9) and $\hat{f}_{\mathrm{mov}} = f_{\mathrm{mov}} \circ \Psi^{(2)}$ is 2π-periodic in each λ_i (Theorem 9.12(a)); so $P(\mu, w + \iota'_k) = P(\mu, w)$. For $k \in \mathbb{Z}^2$ the frozen $\mathrm{shift}(k)$ is exactly ι'_k (Definition 1.8).

Step S18.17: (H3). Lemma 18.3 with $W = \mathcal{O}$ ($0 < \ell'_i$).

Step S18.18: (H4) The Birkhoff form. Theorem 13.12(c) gives $\mathrm{HasBirkhoff}(P, V, r_*/\sqrt{2}, \mu_0, P_0, \Omega, \beta)$ for every $\mu_0 > 0$, in the dictionary of §18.1. The clauses are monotone in the radius: (i) is a statement for $p_j^2 + q_j^2 < r_0^2$, which shrinks with r_0; in (ii) the number δ with $0 < \delta \le r_*/\sqrt{2}$ may be replaced by $\min\{\delta, r_0\}$ for $r_0 = r_*/2$, since the estimate is asked for $p_j^2 + q_j^2 < \delta^2$ only. Hence $\mathrm{HasBirkhoff}(P, V, r_0, 1, P_0, \Omega, \beta)$. (Here the zero extension of P outside $\mathcal{T}$ is invisible: the points $\mathrm{assemble}(I, \varphi, p, q)$ with $I \in V$, $p_j^2 + q_j^2 < r_0^2$ lie in $\mathcal{C}(V, r_0) \subset \mathcal{T}$.)

Step S18.19: (H5), (H6). For $I \in V$, $a(I) \in \operatorname{int} K_a$ and $\beta(\mu, I) = \hat{\beta}(a(I))$ is symmetric (Definition 13.10); $|\det \beta(\mu, I)| \ge \min_{K_a} |\det \hat{\beta}| = B$ by (D3) and Remark 13.13(1). □

18.5. **Transport of the tori.**

Proposition 18.8 (Cylinder tori of H_μ are physical tori of H_{mov})**.** *Let $\mu \in \mathbb{R}$, $0 < r \le r_0$, $w \in \mathbb{R}^4$ (a frequency) and $K : \mathbb{R}^4 \to \mathrm{Ph}(4)$ with* IsCylTorus$(H_\mu, \mathcal{C}(V,r), w, K)$ *(Definition 1.15), and assume $\gamma > 0$, $\tau > 4$ with w (γ,τ)-Diophantine. Then $\mathcal{K}_{\mathrm{ph}} := \Psi_{\mathrm{tot}} \circ K$ satisfies* IsPhysTorus$(H_{\mathrm{mov}}, w, \mathcal{K}_{\mathrm{ph}})$ *(Definition 1.24), $d = 4$, $H_{\mathrm{mov}} =$ `movingHamiltonian` $G\, m_0\, m\, \mu$), and $\mathcal{K}_{\mathrm{ph}}(\mathbb{R}^4) = \Psi_{\mathrm{tot}}(K(\mathbb{R}^4))$.*

Proof. Write $\mathcal{C} = \mathcal{C}(V,r) \subset \mathcal{T}$ (Proposition 18.6(e)). Part 1's Theorem 2.9 is not used: its lattice clause (L2) shifts *all* four position coordinates, whereas a cylinder torus has periodic slow coordinates; the four steps below redo it for the frozen cylinder predicate.

Step S18.20: (i) Analyticity, differentiability. K is analytic with values in $\mathcal{C} \subset \mathcal{T}$, where Ψ_{tot} is analytic (Proposition 18.6(b)); (F1). By Proposition 18.6(d), $H_{\mathrm{mov}} \circ \Psi_{\mathrm{tot}} = H_\mu$ on the open set $\mathcal{T}$, and H_μ is differentiable at $K(\theta)$ (IsCylTorus (ii)); Lemma 2.4 (with $c = 1$, $O = \mathcal{T}$) gives that H_{mov} is differentiable at $\Psi_{\mathrm{tot}}(K(\theta))$.

Step S18.21: (ii) Periodicity. For $k \in \mathbb{Z}^4$, IsCylTorus (iii) gives $K(\theta + k) = K(\theta) + \iota'_{k_{\mathrm{f}}}$, $k_{\mathrm{f}} = (k_1, k_2)$, so $\mathcal{K}_{\mathrm{ph}}(\theta + k) = \Psi_{\mathrm{tot}}(K(\theta) + \iota'_{k_{\mathrm{f}}}) = \mathcal{K}_{\mathrm{ph}}(\theta)$ by Proposition 18.6(c).

Step S18.22: (iii) Injectivity modulo $\mathbb{Z}^4$. If $\mathcal{K}_{\mathrm{ph}}(\theta') = \mathcal{K}_{\mathrm{ph}}(\theta)$ then Proposition 18.6(c) gives $K(\theta') = K(\theta) + \iota'_k$ for some $k \in \mathbb{Z}^2$, i.e. $K(\theta') = K(\theta) + \mathrm{shift}(k)$, and IsCylTorus (iv) gives $\theta' - \theta \in \mathbb{Z}^4$.

Step S18.23: (iv) Derivative, Lagrangian. $D\mathcal{K}_{\mathrm{ph}}(\theta) = D\Psi_{\mathrm{tot}}(K\theta) DK(\theta)$ with $D\Psi_{\mathrm{tot}}$ invertible, so injective by IsCylTorus (v); $\Omega(D\mathcal{K}_{\mathrm{ph}} u, D\mathcal{K}_{\mathrm{ph}} v) = \Omega(DKu, DKv) = 0$ (Proposition 18.6(b), IsCylTorus (vi)).

Step S18.24: (v) Flow. For θ, t the curve $y(s) = K(\theta + sw) \in \mathcal{T}$ has derivative $X_{H_\mu}(y(t)) = X_{H_{\mathrm{mov}}^{\Psi_{\mathrm{tot}}}}(y(t))$ at $s = t$ (IsCylTorus (vii) and $H_{\mathrm{mov}}^{\Psi_{\mathrm{tot}}} = H_\mu$ on $\mathcal{T}$), and $H_{\mathrm{mov}}^{\Psi_{\mathrm{tot}}}$ is differentiable there; Corollary 2.5 gives that $s \mapsto \mathcal{K}_{\mathrm{ph}}(\theta + sw)$ has derivative $X_{H_{\mathrm{mov}}}(\mathcal{K}_{\mathrm{ph}}(\theta + tw))$ at t.

Step S18.25: (vi) Diophantine. $\tau > 4 > d - 1 = 3$ (Lemma 2.15(a) if one wishes to lower τ); this is clause (vii) of Definition 1.24. The range statement is immediate. □

Proposition 18.9 (Measurability, positivity, finiteness)**.** *Let $0 < r < r_0$ with $2r^2 \le \delta$, let $\mu \in \mathbb{R}$ and let $S \subset \mathrm{Ph}(4)$ satisfy* IsCylKol(H_μ, V, r, μ, C, S) *(Definition 1.16) and $0 < \mathrm{meas}(S \cap \mathcal{F}(V,r))$. Then $S_{\mathrm{ph}} := \Psi_{\mathrm{tot}}(S)$ is Borel (the frozen* `MeasurableSet`*), $S_{\mathrm{ph}} \subset U_{\zeta_*}$, and $0 < \mathrm{meas}\, S_{\mathrm{ph}} < \infty$.*

Proof. **Step S18.26: Invariance.** S is a union of whole tori $K(\mathbb{R}^4)$, and $K(\mathbb{R}^4) + \iota'_k = K(\mathbb{R}^4)$ for $k \in \mathbb{Z}^2$ (IsCylTorus (iii) with $K(\theta) + \iota'_k = K(\theta + (k, 0))$). So $S + \iota'_k = S$, and $S \subset \mathcal{C}(V,r) \subset \mathcal{T}$.

Step S18.27: Reduction to the cell. Let $\mathcal{F} = \mathcal{F}(V,r)$ (Definition 1.10). Every $s \in S$ is $s' + \iota'_k$ with $s' \in S \cap \mathcal{F}$ (shift the fast angles into $[0, 2\pi)^2$), so $\Psi_{\mathrm{tot}}(S) = \Psi_{\mathrm{tot}}(S \cap \mathcal{F})$ by Proposition 18.6(c). Moreover Ψ_{tot} is injective on $\mathcal{F}$: $\Psi_{\mathrm{tot}}(w') = \Psi_{\mathrm{tot}}(w)$ with $w, w' \in \mathcal{F}$ forces $w' = w + \iota'_k$ and $0 \le \lambda_i, \lambda_i + 2\pi k_i < 2\pi$, so $k = 0$.

Step S18.28: Borel, measure. S is Borel (frozen `MeasurableSet`), $\mathcal{F}$ is Borel, so $S \cap \mathcal{F}$ is Borel, and Ψ_{tot} is continuous and injective on it: $S_{\mathrm{ph}} = \Psi_{\mathrm{tot}}(S \cap \mathcal{F})$ is Borel (F3). Lemma 2.18 ($c = 1$, Ψ_{tot} injective on $A = S \cap \mathcal{F}$) gives $\mathrm{meas}\, S_{\mathrm{ph}} = \mathrm{meas}(S \cap \mathcal{F}) > 0$.

Step S18.29: Tube, finiteness. $S_{\mathrm{ph}} \subset \Psi_{\mathrm{tot}}(\mathcal{C}(V,r)) \subset U_{\zeta_*}$ (Proposition 18.6(e)), hence $\mathrm{meas}\, S_{\mathrm{ph}} < \infty$ by Lemma 8.19(a) (and also $\mathrm{meas}\, S_{\mathrm{ph}} \le \mathrm{meas}\, \mathcal{F}(V,r) < \infty$). □

18.6. **The complete proof of P-PLANAR.**

Theorem 18.10 (P-PLANAR: two planar planets with a moving Sun)**.** *Theorem 1.29 holds: for $G, m_0 > 0$, $m, a : \mathit{Fin}\,2 \to \mathbb{R}$ with $m_i, a_i > 0$, $a_0 < a_1$ and $\eta > 0$ there are $\mu_0 > 0$ and $0 < d < D$ such that for every $0 < \mu < \mu_0$ there is $S \subset \mathrm{Ph}(2 \cdot 2)$ with $0 < \operatorname{meas} S < \infty$ and* $\mathrm{MovKol}(G, m_0, m, a, \mu, \eta, d, D, S)$.

Frozen Lean declaration: `KAM.Planetary.planarTwoPlanetKAM`

Source: CP09 Thm. 4, p. 8 and Fej04r Prop. 77, in the route (c) form of `PLAN.md`: the frozen P-G (CP11 Thm. 11.1, CP10 Thm. 1.3) applied to the planar secular Birkhoff form of Part 11b.

Proof. **Step S18.30: Constants.** Choose d, D and all constants of Definition 18.4, (D1)–(D9): $0 < d < D$ (Definition 8.15); $\mu_0 = \min\{1, r^6/(\log(1/r))^{c_{\mathrm{G}}}\} > 0$ since $0 < r < 1$ (F4). These depend on (G, m_0, m, a, η) only (Remark 18.11), and *not* on μ.

Step S18.31: P-G applies. Proposition 18.7 gives all data and (H1)–(H6) of Theorem 1.28; hence $r_{\mathrm{G}}, C_{\mathrm{G}}, c_{\mathrm{G}}$ with $0 < r_{\mathrm{G}} < r_0$, $r_{\mathrm{G}} < 1$ exist ((D7)); they do not depend on the r, μ below.

Step S18.32: The coupling. Let $0 < \mu < \mu_0$. The pair (r, μ) of (D8) satisfies the premise of the conclusion of P-G: $0 < r < r_{\mathrm{G}}$ ($r \le r_{\mathrm{G}}/2$), $0 < \mu < 1 = \mu_0^{\mathrm{G}}$, and $\mu < \mu_0 \le r^6/(\log(1/r))^{c_{\mathrm{G}}}$. So there is $S_{\mathrm{cyl}} \subset \mathrm{Ph}(4)$ with $\mathrm{IsCylKol}(H_\mu, V, r, \mu, C_{\mathrm{G}}, S_{\mathrm{cyl}})$ and $0 < \operatorname{meas}(S_{\mathrm{cyl}} \cap \mathcal{F}(V, r))$ (and the further measure bounds of P-G, not needed here).

Step S18.33: The physical set. Put $S = \Psi_{\mathrm{tot}}(S_{\mathrm{cyl}})$. By Proposition 18.9 (with $2r^2 \le \delta/2 < \delta$, $r < r_0$): S is Borel, $S \subset U_{\zeta_*}$ and $0 < \operatorname{meas} S < \infty$. These are the first two conjuncts of the conclusion, and clauses "`MeasurableSet` S" and "$S \subset$ `circularTube` η" of MovKol ($U_{\zeta_*} \subset U_\eta$, Lemma 8.16(a)).

Step S18.34: Tori. Let $z \in S$, $z = \Psi_{\mathrm{tot}}(s)$ with $s \in S_{\mathrm{cyl}}$. IsCylKol gives $w \in \mathbb{R}^4$, K, θ with $K(\theta) = s$, $K(\mathbb{R}^4) \subset S_{\mathrm{cyl}}$, $\mathrm{IsCylTorus}(H_\mu, \mathcal{C}(V, r), w, K)$, and $\gamma > 0$, $\tau > 2 + 2 = 4$ with w (γ, τ)-Diophantine. Proposition 18.8 gives $\mathrm{IsPhysTorus}(H_{\mathrm{mov}}, w, \mathcal{K}_{\mathrm{ph}})$ for $\mathcal{K}_{\mathrm{ph}} = \Psi_{\mathrm{tot}} \circ K$; $\mathcal{K}_{\mathrm{ph}}(\theta) = z$ and $\mathcal{K}_{\mathrm{ph}}(\mathbb{R}^4) \subset \Psi_{\mathrm{tot}}(S_{\mathrm{cyl}}) = S$. This is hypothesis (S2) of Proposition 8.21; (S1) is Step S18.33.

Step S18.35: The physical clauses. Proposition 8.21 (with ζ_*, d, D of (D1), indices shifted) turns (S1), (S2) into $\mathrm{MovKol}(G, m_0, m, a, \mu, \eta, d, D, S)$: for every $\phi \in \mathbb{R}^4$, $\mathrm{NoColl}(\mathcal{K}_{\mathrm{ph}}(\phi)_1)$ (Lemma 8.16(d)), the barycentric reconstruction of $t \mapsto \mathcal{K}_{\mathrm{ph}}(\phi + tw)$ is an all-time Newton solution (Proposition 8.11 via Corollary 8.14(a), or Corollary 8.18: $\mathcal{K}_{\mathrm{ph}}(\phi + tw) \in S \subset U_{\zeta_*}$ for all t and $\mathcal{K}_{\mathrm{ph}}(\cdot + tw)$ has derivative $X_{H_{\mathrm{mov}}}$ by the flow clause), and $\mathrm{PairBounds}(d, D;$ `barycentricPositions` m_0 m μ $\mathcal{K}_{\mathrm{ph}}(\phi)_1)$ with the same d, D for every μ (Lemma 8.16(e), exact identities $x_i - x_j = q_i - q_j$, Lemma 8.10(a)). □

Remark 18.11 (Order of the constants against the frozen quantifiers). The frozen statement is $\forall (G, m_0, m, a, \eta)\ \exists (\mu_0, d, D)\ \forall \mu \in (0, \mu_0)\ \exists S$. The constants of Definition 18.4 are chosen in this order, and each depends *only* on the quantities listed.

Constant	Depends on	Where
γ_a, ζ_*, d, D	a (ζ_* also η)	(D1)
$\delta = \delta_{\zeta_*}$	(G, m_0, m, a, η)	Lemma 18.1
η'	(G, m_0, m, a, δ)	(D2)
a°, ρ_a, K_a, B	(G, m_0, m, a, η')	(D3)
K, V, W, A_c, d_c	$(G, m_0, m, a^\circ, \rho_a)$	(D4), Lemma 18.2
$\varrho, \chi, r_*, C_*, \mathcal{O}', T$	(G, m_0, m, K_a)	(D5)
$r_0 = r_*/2$, $\mu_0^{\mathrm{G}} = 1$	r_*	(D6)
$r_{\mathrm{G}}, C_{\mathrm{G}}, c_{\mathrm{G}}$	all of the above (P-G data)	(D7)
r	r_{G}, δ	(D8)
μ_0	r, c_{G}	(D9)

Only after (μ_0, d, D) are fixed does $\mu \in (0, \mu_0)$ enter: it selects S_{cyl} in P-G (Step S18.32) and then $S = \Psi_{\mathrm{tot}}(S_{\mathrm{cyl}})$. In particular the secular radius r is a *fixed* number, chosen before μ so that the whole chart image $\Psi_{\mathrm{tot}}(\mathcal{C}(V, r))$ lies in the ζ_*-tube (Proposition 18.6(e)) and so that the coupling window of P-G, $\mu < r^6/(\log(1/r))^{c_G}$, contains $(0, \mu_0)$. Thus d, D do not depend on η through anything but $\zeta_* \le \eta$, and $d < D$ always; μ_0 depends on η through δ and the patch, as it must (smaller tubes force smaller couplings).

Remark 18.12 (Uniform tube and pair bounds: the route-study contract, checked). The assembly contract of `reviews/p11-route-study.md` ("Uniform tube and pair bounds") is met as follows. (1) ζ is chosen from η and the axes ($\zeta_* = \min\{\eta, \gamma_a/6\}$), with the factor $\sqrt{2}$ between the phase max norm and the Euclidean length built into Lemma 8.16(b), (c) ($2\sqrt{2}\zeta_* \le \frac{\sqrt{2}}{3}\gamma_a < \frac{1}{2}\gamma_a$). (2) The axis patch and the secular radius are chosen before μ so that the whole chart image lies in U_{ζ_*} (Lemma 18.1, Proposition 18.6(e)); the pair bounds are exact and μ-independent. (3) No $O(\mu)$ estimate of the barycentre is used.

18.7. **Notes: deviations, doubts, errata, forward references.**
Replaced plan items. `PLAN.md`'s `hamiltonian-form` (the pair (h, f_0) verifying CP09 Thm. 4 (7)–(8) with $d = p = 2$) belongs to route (a) and is not used; the labels of the plan map to this part as `choose-patch` → Definition 18.4 (D2)–(D3); `transport` → Proposition 18.8 (+ Proposition 18.9); `physical` → Theorem 18.10 Steps S18.33–S18.35; `quantifiers` → Remark 18.11.
Deviations. (1) *Radius.* Theorem 13.12(c) states HasBirkhoff for $r_0 = r_*/\sqrt{2}$; the closure of $\mathcal{C}(V, r_*/\sqrt{2})$ reaches $|z|_2 = r_*$, outside the open $\mathcal{T}$, which P-G's (H1) (analyticity on the *closed* cylinder) cannot allow. I use $r_0 = r_*/2$ and the monotonicity argument of Step S18.18. (2) *Torus transport.* Part 1's Theorem 2.9(a) and Lemma 2.19 are stated with (L2): $K(\theta + k) = K(\theta) + \iota_k$ for *all* $k \in \mathbb{Z}^{a+b}$ and (Q1) for the whole lattice ι_k, which shifts the slow position coordinates q as well. A P-G cylinder torus has *periodic* slow coordinates (IsCylTorus (iii)), so those statements do not apply to the output of P-G as written (they do apply in the polar oscillator coordinates, Lemma 9.15, if the torus is known to wind once around the origin of each slow disc, which the frozen conclusion does not assert). Proposition 18.8 and Proposition 18.9 are the direct versions for the Cartesian cylinder; they need only the fast-angle lattice, the injectivity of the chart modulo it (Proposition 18.6(c)) and Lemma 2.18. Part 6's Lemma 8.19(b) is likewise not needed (only part (a)). The statement of Lemma 2.18 that $\Psi(A)$ is Borel for Borel A is used here only on a set where Ψ_{tot} is injective ((F3)); it needs that hypothesis. (3) No Part 10 R-nondegeneracy (Theorem 12.22) is used: P-G needs only $\Phi_4 \neq 0$ through Theorem 13.12 and $\det \hat{\beta} \neq 0$.
Doubts about frozen statements. None. The index set `Phase` $(2 \cdot 2)$ and `Phase` $(2 + 2)$ coincide as types. The conclusion does not assert that the physical tori have positive distance to a collision beyond the tube bounds (these are provided by d).
Suspected source errata. None new; E-P-Fej04r-2, E-P-Fej04r-3, E-P-CP11-1, E-P-CP11-2 (Parts 0, 8, 11b) are used only through Parts 8–11b.
Open points that this part does not settle. The part is conditional on the statements of Parts 1, 5–11b it cites, in particular on the proof of P-G (Part 5) and on the Birkhoff map T of Part 11b taking values in the analyticity domain of $\hat{f}_{\mathrm{mov}}$.
External facts used. (F1)–(F4).
Label index. `forward-tube`, `collar`, `fast-diffeo`, `data`, `perturbation`, `chart`, `pg-hypotheses`, `transport`, `measure`, `planar`, `quantifiers`, `contract`.

19. D25: P-AMD-CONFINEMENT and P-EXP-SAFE

This part proves the two D25 theorems of Part 0, §1.7: the deterministic stopped-input helper P-AMD-CONFINEMENT (Theorem 1.44) and P-EXP-SAFE (Theorem 1.45). Throughout $G = m_0 = 1$, $n \ge 2$ is the number of planets, $\mathcal{B}$ is an admissible box (Definition 1.37, fields

$m, \ell, \ell^0, u^0, u, e_*, A_0, A_1, d, R$) and $H_{\mathrm{mov}} = H^{(\mu)}_{\mathrm{mov}}$ is the moving-Sun Hamiltonian of Definition 1.22 with $G = m_0 = 1$.

- `elements`, `ecc-bound`, `geometry`, `closure` (Lemmas 19.1–19.4): the algebra of prograde elliptic points (eccentricity, radial and momentum bounds) and the elementary consequences of low AMD and separated annuli: eccentricity bound, radial separation, pair bounds, a compact collision-free set, and closedness of the sub-region of the outer region cut out by the stopped bound (round 2 §2, Lemmas 2 and 5);
- `moving-ode`, `ang-mom` (Lemma 19.5, Proposition 19.6): the frozen moving-Sun solution predicate is the solution predicate of the C^1 field $X_{H_{\mathrm{mov}}}$ on the collision-free set; exact conservation of $J = \sum_i \mathrm{cr}(q_i, p_i)$ for the literal system (round 2 Lemma 1);
- `amd` (Theorem 19.7): the complete proof of the frozen P-AMD-CONFINEMENT: first exit, bootstrap with the stopped bound, compact continuation (round 2 Lemma 6, with the stopped-source correction of round 3);
- `enlarge`, `real-domain`, `h-data`, `perturbation` (§§19.5–19.6): the larger domains $G' \supset G$, the slow ball D, the complex width ρ, the Hessian data of the Kepler Hamiltonian and the analytic perturbation P with its bound B_0 (the data demanded by finding F-NADD-4 of the N audit);
- `lift`, `stopped` (Proposition 19.15, 19.17): lifting of an arbitrary finite solution segment to the Poincaré chart, and the discharge of StoppedBound from the frozen N-PARTIAL-CONVEX ([N:`bpN:p0:partial`], an input), for H_{mov} and, by time reversal, $-H_{\mathrm{mov}}$;
- `expsafe` (Theorem 19.19) and `constants` (Remark 19.20): the complete proof of the frozen P-EXP-SAFE and the order of every constant against the frozen quantifiers.

19.1. **Conventions and external facts.** *Notation.* For $z = (q, p) \in \mathrm{Ph}(2n)$ write $q_i, p_i \in \mathbb{R}^2$ for the body vectors, $r_i = |q_i|_2$, and, where defined, $E_i, a_i, \Lambda_i, \mathbf{e}_i$ as in Definition 1.33, $C_i(z) = \mathrm{cr}(q_i, p_i)$, $e_i(z) = |\mathbf{e}_i(z)|_2$, $J(z) = \sum_i C_i(z)$, $\mathcal{A}(z) = \sum_i \Lambda_i(z) - J(z)$. The *outer* and *inner* regions are

$$\mathcal{R}_{\mathrm{out}} = \mathrm{LowAMD}(m, \ell, u, A_1), \qquad \mathcal{R}_{\mathrm{in}} = \mathrm{LowAMD}(m, \ell^0, u^0, A_0) \subset \mathcal{R}_{\mathrm{out}}$$

(the inclusion is $\ell_i < \ell_i^0$, $u_i^0 < u_i$, $A_0 < A_1$). $W = \{z : \mathrm{NoColl}(q)\}$ is the collision-free set (Lemma 8.2). $a = \alpha_n$, $b = \beta_n = \frac{3}{2}a$ (Definition 1.42). All norms on phase space are the max norms of Part 0; physical lengths are $|\cdot|_2$.

External facts (Mathlib level; used by label).

(X1) Planar algebra: Cauchy–Schwarz $|u \cdot v| \le |u|_2 |v|_2$; the Lagrange identity $(u \cdot v)^2 + \mathrm{cr}(u, v)^2 = |u|_2^2 |v|_2^2$; bilinearity of cr and $\mathrm{cr}(u, u) = 0$, $\mathrm{cr}(u, v) = -\mathrm{cr}(v, u)$.

(X2) Continuity and compactness: preimages of open sets under continuous maps are open; closed bounded sets of $\mathbb{R}^N$ are compact; a continuous function on a nonempty compact set attains its extrema; the neighbourhood form ((E7)): if $K \subset V \subset \mathbb{R}^N$, K compact, V open, some $\delta > 0$ has the closed δ-neighbourhood of K inside V.

(X3) Calculus on $\mathbb{R}$: a function continuous on an interval with derivative 0 at every point is constant; chain rule for `HasDerivAt`; if y has derivative $v(t)$ then $s \mapsto y(-s)$ has derivative $-v(-s)$.

(X4) *Complexification of real analytic functions.* If f is real analytic at $x \in \mathbb{R}^N$, there is $r > 0$ and a holomorphic g on the polydisc $D(x, r) = \{z \in \mathbb{C}^N : |z - x|_\infty < r\}$ with $g = f$ on $D(x, r) \cap \mathbb{R}^N$ (the power series of f at x converges on a complex polydisc of the same radii).

(X5) *Identity theorem.* If g is holomorphic on a connected open $\mathcal{O} \subset \mathbb{C}^N$ and vanishes on a nonempty open subset of $\mathcal{O} \cap \mathbb{R}^N$, then $g \equiv 0$ (the complex Taylor coefficients at a real point are the real ones; then [K:`bp:p1:holomorphic-analytic`] and the identity theorem for analytic functions on connected sets). Convex sets, in particular polydiscs and their intersections, are connected.

(X6) Holomorphic calculus: finite sums, products, quotients by non-vanishing functions of holomorphic functions are holomorphic; the monomial z^{-k} is holomorphic on $z \neq 0$ with derivative $-kz^{-k-1}$; holomorphic functions are continuous ([K:bp:p1:holomorphic-analytic]).

Cited results. Track N, Part 3 (the ODE layer): [N:bpN:p3:gen-sol] (`IsSolIn`), [N:bpN:p3:gen-symmetry], [N:bpN:p3:gen-uniqueness], [N:bpN:p3:gen-maximal], [N:bpN:p3:gen-continuation], [N:bpN:p3:gen-continuation-compact], [N:bpN:p3:gen-uniform-compact], [N:bpN:p3:gen-exit], [N:bpN:p3:ham-smooth]; the frozen theorem N-PARTIAL-CONVEX [N:bpN:p0:partial] with its definitions [N:bpN:p0:partial-domain]–[N:bpN:p0:partial-exponents]. Track P: Lemma 8.2, Proposition 8.6, Lemma 8.10, Proposition 8.11; Corollary 2.5; Theorem 9.12, Proposition 9.16. *Totality:* Remark 1.43 (every Kepler quantity is used under ProgradeElliptic).

19.2. **Prograde elliptic points: elements, eccentricity, geometry.**

Lemma 19.1 (Elements of one planet)**.** *Let $m > 0$, $(q,p) \in \mathbb{R}^2 \times \mathbb{R}^2$ with $r = |q|_2 > 0$, $E = \frac{|p|_2^2}{2m} - \frac{m}{r} < 0$ and $C = \mathrm{cr}(q,p) > 0$. Put $a = -m/(2E)$, $\Lambda = m\sqrt{a}$, $\mathbf{e} = \big(\frac{|p|_2^2}{m^2} - \frac{1}{r}\big)q - \frac{q\cdot p}{m^2}p$ and $e = |\mathbf{e}|_2$ (Definition 1.33). Then:*

(a) $a > 0$, $\Lambda > 0$, $\Lambda^2 = m^2 a$ and $E = -m^3/(2\Lambda^2)$;
(b) $e^2 = 1 - C^2/\Lambda^2$; hence $0 < C \le \Lambda$, $0 \le e < 1$ and $C = \Lambda\sqrt{1-e^2}$;
(c) $q \cdot \mathbf{e} = C^2/m^2 - r = a(1-e^2) - r$;
(d) $a(1-e) \le r \le a(1+e)$;
(e) $|p|_2^2 < 2m^2/r$;
(f) $\Lambda - C = \Lambda\big(1 - \sqrt{1-e^2}\big) \ge 0$, and $\Lambda - C \ge \Lambda(1 - \sqrt{1-e'^2})$ whenever $e \ge e' \in [0,1)$.

Proof. **Step S19.1: (a).** $E < 0$ and $m > 0$ give $a = m/(2|E|) > 0$, so $\Lambda = m\sqrt{a} > 0$ and $\Lambda^2 = m^2 a = -m^3/(2E)$, i.e. $E = -m^3/(2\Lambda^2)$.

Step S19.2: (b). Put $s = |p|_2^2/m^2$. Expanding $|\mathbf{e}|_2^2$ ((X1)):

$$e^2 = (s - \tfrac{1}{r})^2 r^2 - 2(s - \tfrac{1}{r})\tfrac{(q\cdot p)^2}{m^2} + \tfrac{(q\cdot p)^2}{m^4}|p|_2^2 = (sr-1)^2 + \tfrac{(q\cdot p)^2}{m^2}\big(\tfrac{2}{r} - s\big).$$

By the Lagrange identity $(q\cdot p)^2 = r^2 m^2 s - C^2$, so $e^2 = (sr-1)^2 + (r^2 s - \frac{C^2}{m^2})(\frac{2}{r} - s) = 1 - \frac{C^2}{m^2}(\frac{2}{r} - s)$. Since $E = \frac{m}{2}s - \frac{m}{r}$ we have $\frac{2}{r} - s = -\frac{2E}{m}$, hence $e^2 = 1 + \frac{2EC^2}{m^3} = 1 - \frac{C^2}{\Lambda^2}$ by (a). As $e^2 \ge 0$ and $C > 0$: $0 < C \le \Lambda$, $e < 1$.

Step S19.3: (c). $q\cdot\mathbf{e} = (s - \frac{1}{r})r^2 - \frac{(q\cdot p)^2}{m^2} = sr^2 - r - r^2 s + \frac{C^2}{m^2} = \frac{C^2}{m^2} - r$, and $C^2 = \Lambda^2(1-e^2) = m^2 a(1-e^2)$ by (a), (b).

Step S19.4: (d). By (c) and Cauchy–Schwarz, $|a(1-e^2) - r| \le re$. The inequality $r \ge a(1-e^2) - re$ gives $r(1+e) \ge a(1-e)(1+e)$, i.e. $r \ge a(1-e)$; the inequality $r - a(1-e^2) \le re$ gives $r(1-e) \le a(1-e)(1+e)$, i.e. $r \le a(1+e)$ as $e < 1$.

Step S19.5: (e). $E < 0$ means $|p|_2^2/(2m) < m/r$.

Step S19.6: (f). (b) gives $\Lambda - C = \Lambda(1 - \sqrt{1-e^2})$; $e \mapsto 1 - \sqrt{1-e^2}$ is nonnegative and increasing on $[0,1)$. □

Lemma 19.2 (Low AMD bounds every eccentricity)**.** *Let $z \in \mathrm{Ph}(2n)$ with $\mathrm{ProgradeElliptic}(m,z)$ and $\ell \in (0,\infty)^n$, $A \in \mathbb{R}$, $e_* \in (0,1)$ with $\Lambda_i(z) > \ell_i$ for all i, $\mathcal{A}(z) < A$ and $A \le \ell_i(1 - \sqrt{1-e_*^2})$ for all i. Then for every i*

$$e_i(z) < e_*, \qquad C_i(z) > \Lambda_i(z)\sqrt{1-e_*^2}.$$

Proof. **Step S19.7: Each summand is at most the total.** $\mathcal{A}(z) = \sum_j(\Lambda_j - C_j)$ and each summand is ≥ 0 (Lemma 19.1(f)); so $\Lambda_i - C_i \leq \mathcal{A}(z) < A$.

Step S19.8: Eccentricity. $A \leq \ell_i(1-\sqrt{1-e_*^2}) < \Lambda_i(1-\sqrt{1-e_*^2})$ (as $\Lambda_i > \ell_i$ and $1-\sqrt{1-e_*^2} > 0$). Hence $C_i > \Lambda_i\sqrt{1-e_*^2}$, i.e. $1-e_i^2 = C_i^2/\Lambda_i^2 > 1-e_*^2$ (Lemma 19.1(b)): $e_i < e_*$. □

Lemma 19.3 (Geometry of the outer region)**.** *Let $\mathcal{B}$ be admissible and $z \in \mathcal{R}_{\rm out}$. Then for all i and all $j \neq i$:*

(a) $\ell_i^2/m_i^2 < a_i < u_i^2/m_i^2$ and $e_i < e_*$;
(b) $a_i(1-e_i) \leq |q_i|_2 \leq a_i(1+e_i)$ and $d \leq |q_i|_2 \leq R$;
(c) for $i<j$: $|q_j|_2 - |q_i|_2 \geq d$; hence $|q_i - q_j|_2 \geq d$ and $|q_i-q_j|_2 \leq |q_i|_2 + |q_j|_2 \leq 2R$ for all $i \neq j$; in particular $z \in W$;
(d) $|p_i|_2 < m_i\sqrt{2/d}$.

Consequently $\mathcal{R}_{\rm out} \subset K_1 := \{(q,p) :\ d \leq |q_i|_2 \leq R,\ |q_i-q_j|_2 \geq d\ (i\neq j),\ |p_i|_2 \leq m_i\sqrt{2/d}\ \forall i\}$, a compact subset of W.

Proof. **Step S19.9: (a).** $\Lambda_i = m_i\sqrt{a_i} \in (\ell_i, u_i)$ gives the axis bounds; $e_i < e_*$ is Lemma 19.2 with $A = A_1$ (hypothesis (B2)).

Step S19.10: (b). By Lemma 19.1(d), $|q_i|_2 \geq a_i(1-e_i) > (\ell_i/m_i)^2(1-e_*) \geq d$ by (a) and (B3); and $|q_i|_2 \leq a_i(1+e_i) < (u_i/m_i)^2(1+e_*) \leq R$.

Step S19.11: (c). For $i<j$: $|q_j|_2 - |q_i|_2 \geq a_j(1-e_j) - a_i(1+e_i) > (\ell_j/m_j)^2(1-e_*) - (u_i/m_i)^2(1+e_*) \geq d$ by (a) and (B4). Then $|q_i-q_j|_2 \geq ||q_j|_2 - |q_i|_2| \geq d > 0$ (reverse triangle inequality, (E5)) and the upper bound is the triangle inequality with (b). So NoColl(q).

Step S19.12: (d). Lemma 19.1(e) and $|q_i|_2 \geq d$.

Step S19.13: The compact set. K_1 is closed (finitely many non-strict inequalities between continuous functions) and bounded in $\mathbb{R}^{4n}$ (each $|q_i|_2 \leq R$, $|p_i|_2 \leq m_i\sqrt{2/d}$), hence compact ((X2)); $K_1 \subset W$ because $d > 0$. □

Lemma 19.4 (Openness and closure)**.** *Let $\mathcal{B}$ be admissible and $\delta > 0$ with $\ell_i + 2\delta < \ell_i^0$, $u_i^0 + 2\delta < u_i$, $A_0 + 2n\delta < A_1$.*

(a) $\mathcal{R}_{\rm out}$ is open in Ph($2n$)*, and Λ_i, $\mathcal{A}$ are continuous on the open set $\{z :\ |q_i|_2 > 0,\ E_i(z) < 0\ \forall i\}$.*
(b) Let $\mathcal{R}_\delta = \{z \in \mathcal{R}_{\rm out} :\ \ell_i + \delta \leq \Lambda_i(z) \leq u_i - \delta\ \forall i,\ \mathcal{A}(z) \leq A_0 + n\delta\}$. If $z^{(k)} \in \mathcal{R}_\delta$ and $z^{(k)} \to z^*$ with $|q_i^*|_2 > 0$ for all i, then $z^* \in \mathcal{R}_\delta$.

Proof. **Step S19.14: (a) Continuity.** E_i is continuous on $\{|q_i|_2 > 0\}$ (open). On $\{E_i < 0\}$, $a_i = -m_i/(2E_i)$, $\Lambda_i = m_i\sqrt{a_i}$ are continuous; C_i is a polynomial, so $\mathcal{A} = \sum\Lambda_i - \sum C_i$ is continuous. $\mathcal{R}_{\rm out}$ is a finite intersection of preimages of open sets ((X2)).

Step S19.15: (b) Uniform margins along the sequence. For each k, $\ell_i + \delta \leq \Lambda_i(z^{(k)}) \leq u_i - \delta$, hence $E_i(z^{(k)}) = -m_i^3/(2\Lambda_i^2) \leq -m_i^3/(2(u_i-\delta)^2) =: -\epsilon_i < 0$ (Lemma 19.1(a)), and by Lemma 19.2 (with $A = A_1$) $C_i(z^{(k)}) > \Lambda_i\sqrt{1-e_*^2} \geq (\ell_i+\delta)\sqrt{1-e_*^2} =: c_i > 0$.

Step S19.16: Passage to the limit. E_i and C_i are continuous at z^* ($|q_i^*|_2 > 0$), so $E_i(z^*) \leq -\epsilon_i < 0$ and $C_i(z^*) \geq c_i > 0$: Prograde**Elliptic**(m, z^*) (the masses are positive). By (a) $\Lambda_i, \mathcal{A}$ are continuous at z^*, so $\ell_i + \delta \leq \Lambda_i(z^*) \leq u_i - \delta$ and $\mathcal{A}(z^*) \leq A_0 + n\delta < A_1$. As $\ell_i < \ell_i + \delta$ and $u_i - \delta < u_i$, $z^* \in \mathcal{R}_{\rm out}$, hence $z^* \in \mathcal{R}_\delta$. □

19.3. **The moving-Sun ODE and the conservation of angular momentum.**

Lemma 19.5 (Moving-Sun solutions are solutions of a C^1 field on W)**.** *Let $\mu \in \mathbb{R}$, $H = H_{\mathrm{mov}}$, $F = X_H$ (Definition 1.2). Then W is open, F is C^1 on W and, for every $J \subset \mathbb{R}$ and $z : \mathbb{R} \to \mathrm{Ph}(2n)$,*

$$\mathrm{IsMovSolOn}(m, \mu, J, z) \iff \mathit{IsSolIn}\, F\, W\, J\, z \quad ([\mathrm{N{:}}\texttt{bpN:p3:gen-sol}]).$$

In particular [N:`bpN:p3:gen-uniqueness`], [N:`bpN:p3:gen-maximal`], [N:`bpN:p3:gen-continuation-compact`], [N:`bpN:p3:gen-exit`] *apply to* IsMovSolOn. *The same holds for $-H$ and $-F$.*

Proof. **Step S19.17: Regularity.** W is open and H is C^∞ on W (Lemma 8.2); F is C^1 on W ([N:`bpN:p3:ham-smooth`], $k = 1$).

Step S19.18: Equivalence. ($\Rightarrow$) $\mathrm{NoColl}(z(t)_1)$ is $z(t) \in W$, and the derivative clause is the same. ($\Leftarrow$) $z(t) \in W$ is NoColl; H is differentiable at $z(t)$ by the regularity step; the momentum-domain clause $z(t)_2 \in \texttt{univ}$ is trivial. The statement for $-H$ is [N:`bpN:p3:gen-symmetry`](b) and $X_{-H} = -X_H$. □

Proposition 19.6 (Conservation of the total angular momentum)**.** *Let $\mu \in \mathbb{R}$, $J_0 \subset \mathbb{R}$ an interval and $z : \mathbb{R} \to \mathrm{Ph}(2n)$ with* IsMovSolOn(m, μ, J_0, z). *Then $t \mapsto J(z(t)) = \sum_i \mathrm{cr}(q_i(t), p_i(t))$ is constant on J_0.*

Proof. **Step S19.19: The equations.** Fix $t \in J_0$ and write $P = \sum_i p_i$. By Lemma 19.5 and Proposition 8.6 ($G = m_0 = 1$): $q_i' = p_i/m_i + \mu P$ and $p_i' = -m_i\, q_i/r_i^3 + \mu m_i f_i(q)$, with $f_i(q) = \sum_{j \neq i} m_j \frac{q_j - q_i}{|q_j - q_i|_2^3}$.

Step S19.20: Derivative. cr is bilinear, so $J(z(\cdot))$ has derivative $\sum_i [\mathrm{cr}(q_i', p_i) + \mathrm{cr}(q_i, p_i')]$ at t ((X3)). First sum: $\sum_i \mathrm{cr}(p_i/m_i, p_i) + \mu \sum_i \mathrm{cr}(P, p_i) = 0 + \mu\, \mathrm{cr}(P, P) = 0$. Second sum: $\mathrm{cr}(q_i, q_i) = 0$ kills the central term, leaving $\mu \sum_i m_i\, \mathrm{cr}(q_i, f_i) = \mu \sum_{i \neq j} m_i m_j \frac{\mathrm{cr}(q_i, q_j - q_i)}{|q_j - q_i|_2^3} = \mu \sum_{i \neq j} m_i m_j \frac{\mathrm{cr}(q_i, q_j)}{|q_j - q_i|_2^3}$. The summand s_{ij} satisfies $s_{ji} = -s_{ij}$ (antisymmetry of cr, evenness of $|\cdot|_2$), so the sum over ordered pairs vanishes ((E4)).

Step S19.21: Constancy. $J(z(\cdot))$ is continuous on J_0 (differentiable at each point) with derivative 0: constant ((X3)). □

19.4. **Proof of P-AMD-CONFINEMENT.**

Theorem 19.7 (P-AMD-CONFINEMENT, Theorem 1.44)**.** *Under the hypotheses of Theorem 1.44, the conclusions (a)–(d) hold.*
Frozen Lean declaration: `KAM.Planetary.amdConfinement`
Source: Round 2 §2, Lemmas 1, 2, 5, 6; round 3 (the stopped form of the input); LP17 (9)–(11), p. 2.

Proof. Let $\mathcal{B}, \mu, T, \delta, z_0 \in \mathcal{R}_{\mathrm{in}}$ and StoppedBound$(\mathcal{B}, \mu, T, \delta, z_0)$ be given, with $\ell_i + 2\delta < \ell_i^0$, $u_i^0 + 2\delta < u_i$, $A_0 + 2n\delta < A_1$. Write $\mathcal{R}_\delta$ as in Lemma 19.4.

Step S19.22: The maximal solution. $z_0 \in \mathcal{R}_{\mathrm{in}} \subset \mathcal{R}_{\mathrm{out}}$, so $z_0 \in W$ (Lemma 19.3(c)). By Lemma 19.5 and [N:`bpN:p3:gen-maximal`] there are $T_- < 0 < T_+$ ($T_\pm$ possibly infinite) and $z : \mathbb{R} \to \mathrm{Ph}(2n)$ with $z(0) = z_0$ and IsMovSolOn$(m, \mu, (T_-, T_+), z)$, such that every IsMovSolOn-solution w on an interval $J' \ni 0$ with $w(0) = z_0$ satisfies $J' \subset (T_-, T_+)$ and $w = z$ on J'.

Step S19.23: Forward: no exit before $\min(T, T_+)$**.** *Claim:* $z(t) \in \mathcal{R}_{\mathrm{out}}$ for all $t \in [0, T_+) \cap [0, T]$. Suppose not. $\mathcal{R}_{\mathrm{out}}$ is open (Lemma 19.4(a)) and $z(0) \in \mathcal{R}_{\mathrm{out}}$, so by [N:`bpN:p3:gen-exit`] there is a first exit time $t^* \in (0, T_+)$, with $z([0, t^*)) \subset \mathcal{R}_{\mathrm{out}}$ and $z(t^*) \notin \mathcal{R}_{\mathrm{out}}$; the supposition means $t^* \leq T$. For every $s \in (0, t^*)$ the restriction of z satisfies the hypotheses of the stopped bound with $(l, r) = (0, s)$: $l \leq 0 \leq r$, $l < r$, $z(0) = z_0$, IsMovSolOn$(m, \mu, [0, s], z)$ (restriction), and $z(t) \in \mathcal{R}_{\mathrm{out}}$

on $[0,s]$. Since $[0,s]\subset[-T,T]$ it gives $|\Lambda_i(z(t))-\Lambda_i(z_0)|\le\delta$ for $t\in[0,s]$, hence for all $t\in[0,t^*)$ and all i.

Step S19.24: The margins at times before t^*. Let $t\in[0,t^*)$. As $\ell_i^0<\Lambda_i(z_0)<u_i^0$: $\Lambda_i(z(t))\ge\Lambda_i(z_0)-\delta>\ell_i^0-\delta>\ell_i+\delta$ and $\Lambda_i(z(t))\le\Lambda_i(z_0)+\delta<u_i^0+\delta<u_i-\delta$. By Proposition 19.6 (on $[0,t]$), $J(z(t))=J(z_0)$, so by Definition 1.34

$$\mathcal{A}(z(t))-\mathcal{A}(z_0)=\sum_i\big(\Lambda_i(z(t))-\Lambda_i(z_0)\big)\le n\delta, \tag{9}$$

so $\mathcal{A}(z(t))\le\mathcal{A}(z_0)+n\delta<A_0+n\delta$. Thus $z(t)\in\mathcal{R}_\delta$ for all $t\in[0,t^*)$.

Step S19.25: Closure at t^*. $z(t^*)\in W$, so $|q_i(t^*)|_2>0$; $z(t)\to z(t^*)$ as $t\uparrow t^*$ (continuity of the solution). Lemma 19.4(b) gives $z(t^*)\in\mathcal{R}_\delta\subset\mathcal{R}_{\text{out}}$, contradicting the choice of t^*. This proves the claim; moreover Step S19.24 applies to *every* $t\in[0,T_+)\cap[0,T]$ (for $t>0$ apply the stopped bound on $[0,t]$; $t=0$ is trivial), so $z(t)\in\mathcal{R}_\delta$ there.

Step S19.26: Backward. The same argument with $t\mapsto z(-t)$ in [N:`bpN:p3:gen-exit`] ([N:`bpN:p3:gen-symmetry`](b)) and the stopped bound on $[-s,0]$, $(l,r)=(-s,0)$: $z(t)\in\mathcal{R}_\delta$ for all $t\in(T_-,0]\cap[-T,0]$.

Step S19.27: The solution reaches $\pm T$ and a bit more. Suppose $T_+\le T$. By Step S19.23, $z(t)\in\mathcal{R}_{\text{out}}\subset K_1$ (Lemma 19.3) for all $t\in[0,T_+)$, and $K_1\subset W$ is compact: [N:`bpN:p3:gen-continuation-compact`] gives $T_+=\infty$, a contradiction. So $T_+>T$; likewise $T_->-T$. Hence $[-T,T]\subset(T_-,T_+)$ and $z(t)\in\mathcal{R}_\delta\subset\mathcal{R}_{\text{out}}$ for all $t\in[-T,T]$.

Step S19.28: (a) Existence and uniqueness. IsMovSolOn$(m,\mu,[-T,T],z)$ by restriction; the derivatives at $\pm T$ are two-sided because z is a solution on the open interval $(T_-,T_+)\ni\pm T$. Let $U\subset\mathbb{R}$ be open and convex (an interval) with $0\in U$ and w with $w(0)=z_0$, IsMovSolOn(m,μ,U,w). Then $w=z$ on the interval $U\cap(T_-,T_+)\ni0$ by [N:`bpN:p3:gen-uniqueness`] (Lemma 19.5), hence on $U\cap[-T,T]$: MovUniqueOn$(m,\mu,[-T,T],z)$.

Step S19.29: (b) Newton. Proposition 8.11 with $G=m_0=1$ and $J=[-T,T]$ gives IsNewtonOn$(\text{mass}_\mu,[-T,T],Q,V)$ for $Q=\texttt{barycentricPositions}\,1\,m\,\mu\,q$, $V=\texttt{barycentricVelocities}\,1\,m\,\mu\,p$ (Remark 8.12).

Step S19.30: (c) Region, momentum, geometry. For $t\in[-T,T]$: $z(t)\in\mathcal{R}_{\text{out}}$ (Step S19.27); $J(z(t))=J(z(0))$ by Proposition 19.6 on $[-T,T]$; $e_i<e_*$ and $d\le|q_i|_2\le R$ by Lemma 19.3(a),(b). For PairBounds$(d,2R;Q)$ use Lemma 8.10(a): for planets $i\ne j$, $|Q_j-Q_i|_2=|q_j-q_i|_2\in[d,2R]$ (Lemma 19.3(c)), and for the Sun and planet i, $|Q_i-Q_0|_2=|q_i|_2\in[d,R]\subset[d,2R]$; $|Q_0-Q_i|_2=|Q_i-Q_0|_2$.

Step S19.31: (d) AMD and actions. For $t\in[-T,T]$, Step S19.24 (and its backward analogue, for $t<0$) gives $|\Lambda_i(z(t))-\Lambda_i(z_0)|\le\delta$, and (9) gives $\mathcal{A}(z(t))\le\mathcal{A}(z_0)+n\delta\le A_0+n\delta$ (with $\mathcal{A}(z_0)<A_0$). □

Remark 19.8 (Where each hypothesis is used). The margins $\ell_i+2\delta<\ell_i^0$, $u_i^0+2\delta<u_i$ and $A_0+2n\delta<A_1$ give the strict inequality $z(t)\in\mathcal{R}_\delta\subset\mathcal{R}_{\text{out}}$ with room δ, which is what makes the limit point $z(t^*)$ lie in the *open* outer region (no circularity at first exit); $\mathcal{A}\ge\Lambda_i-C_i$ and J conserved make the AMD bound exact (no assumption of conservation of $\mathcal{A}$: it is *not* conserved); admissibility (B2)–(B4) give eccentricity, radial and pair bounds for every point of $\mathcal{R}_{\text{out}}$, independent of μ; (B1) $m_i>0$ guards all divisions. The stopped bound is used only on solutions that already stay in $\mathcal{R}_{\text{out}}$ on the tested segment.

19.5. **The enlarged analytic domain.** From here on $\Psi = \Psi^{(n)}$ is the product Poincaré chart of Theorem 9.12 with $\kappa = Gm_0 = 1$ and weights m; a chart point is $w = ((\lambda,\xi),(\Lambda,\eta)) \in \mathrm{Ph}(2n)$ with $\lambda,\xi,\Lambda,\eta \in \mathbb{R}^n$ (fast block first: this is the phase-space order $(\varphi, \mathrm{I}) = ((\varphi_{\mathrm{f}}, q),(I,p))$ of [N:bpN:p0:partial] with n fast and $k = n$ slow degrees of freedom), $\Gamma_i(w) = \frac{1}{2}(\xi_i^2 + \eta_i^2)$, $\Gamma(w) = \sum_i \Gamma_i(w)$, $\mathcal{O}_n = \{\Lambda_i > 0,\ \Gamma_i < \Lambda_i\}$, $\tau_k(w) = w + ((2\pi k, 0), 0)$ for $k \in \mathbb{Z}^n$, and $H^\Psi = H_{\mathrm{mov}} \circ \Psi$, $\hat f = f_{\mathrm{mov}} \circ \Psi$ with $f_{\mathrm{mov}} = \frac{1}{2}|\sum_i p_i|_2^2 - \sum_{i<j} \frac{m_i m_j}{|q_i - q_j|_2}$ (Lemma 8.3, $G = m_0 = 1$). Put $\mathcal{W} = \{w \in \mathcal{O}_n : \Psi(w) \in W\}$, open and τ_k-invariant. We recall from Theorem 9.12 and Proposition 9.16(d):

(Ch1) Ψ is a real analytic 1-chart on $\mathcal{O}_n$, $\Psi \circ \tau_k = \Psi$, and $\Psi(\mathcal{O}_n) = \{z : \mathrm{ProgradeElliptic}(m, z)\}$ (onto).

(Ch2) For $z = \Psi(w)$: $\Lambda_i(z) = \Lambda_i$ (the chart action), $C_i(z) = \Lambda_i - \Gamma_i(w)$, $\mathcal{A}(z) = \Gamma(w)$, $a_i(z) = \Lambda_i^2/m_i^2$, $E_i(z) = -m_i^3/(2\Lambda_i^2)$, $\mathbf{e}_i(z) = v_i$ with $|v_i|_2^2 = 1 - (1 - \Gamma_i/\Lambda_i)^2$.

(Ch3) For all $w \in \mathcal{O}_n$ and μ: $H^\Psi(w) = h(\Lambda) + \mu \hat f(w)$, with $h(\Lambda) = -\sum_i m_i^3/(2\Lambda_i^2)$; $\hat f$ is real analytic on $\mathcal{W}$, $\hat f \circ \tau_k = \hat f$, and H^Ψ is C^∞ on $\mathcal{W}$ with $D\Psi\, X_{H^\Psi} = X_{H_{\mathrm{mov}}} \circ \Psi$ there (Lemma 2.4).

(Part 7 proves (Ch2) for $n = 2$ and states that it holds componentwise for every n; the proof is the one-planet identity of Proposition 9.7(d),(e) in each block.)

Lemma 19.9 (Enlarged data)**.** *Let $\mathcal{B}$ be admissible. There are $\kappa > 0$, $e_1 \in (e_*, 1)$ and $A^+ > A_1$ such that, with $r_i^- = \big(\frac{\ell_i - 2\kappa}{m_i}\big)^2(1 - e_1)$ and $r_i^+ = \big(\frac{u_i + 2\kappa}{m_i}\big)^2(1 + e_1)$,*

(E1) $\kappa \le \ell_i/8$ for all i;

(E2) $A^+ < (\ell_i - 2\kappa)\big(1 - \sqrt{1 - e_1^2}\big)$ for all i;

(E3) $r_i^- \ge d/2$ for all i, and $r_j^- - r_i^+ \ge d/2$ for all $i < j$.

They depend only on $(n, \mathcal{B})$. Put $G = \prod_i (\ell_i - \kappa, u_i + \kappa)$, $G' = \prod_i (\ell_i - 2\kappa, u_i + 2\kappa)$ (open convex boxes in $\mathbb{R}^n$), $D = \{(\xi,\eta) \in \mathrm{Ph}(n) : \Gamma(\xi,\eta) < A^+\}$ (an open Euclidean ball of radius $\sqrt{2A^+}$ in $\mathbb{R}^{2n}$), and $\varepsilon_D = (\sqrt{2A^+} - \sqrt{2A_1})/\sqrt{2n} > 0$.

Proof. **Step S19.32: Strict inequalities at the limit point.** The three families are inequalities between continuous functions of $(\kappa, e_1, A^+) \in \mathbb{R}^3$ with $e_1 < 1$, finitely many in number. At $(\kappa, e_1, A^+) = (0, e_*, A_1)$: (E1) holds strictly; (E2) holds strictly by (B2); (E3) holds in the stronger form $r_i^- \ge d$, $r_j^- - r_i^+ \ge d$ by (B3), (B4) (with $\ge d > d/2$ since $d > 0$).

Step S19.33: Choice. By continuity there is a neighbourhood of this point on which all of them hold; pick in it $\kappa > 0$, $e_1 > e_*$, $A^+ > A_1$ ($e_1 < 1$). The positivity of ε_D is $A^+ > A_1$. □

Lemma 19.10 (The real domain is collision free)**.** *With the data of Lemma 19.9 put $\mathcal{K} = \{w \in \mathrm{Ph}(2n) :\ \Lambda_i \in [\ell_i - 2\kappa, u_i + 2\kappa]\ \forall i,\ \Gamma(w) \le A^+\}$ (all λ, ξ, η subject only to the last condition). Then $\mathcal{K} \subset \mathcal{W}$, and for $w \in \mathcal{K}$, $z = \Psi(w)$: $e_i(z) < e_1$, $|q_i|_2 \ge d/2$ and $|q_i - q_j|_2 \ge d/2$ ($i \ne j$). Moreover for $w \in \mathrm{Ph}(2n)$ with $\Lambda \in \prod_i [\ell_i, u_i]$ and $\Gamma(w) \le A_1$ one has $w \in \mathcal{K}$.*

Proof. **Step S19.34: Chart domain.** $\Lambda_i \ge \ell_i - 2\kappa \ge \frac{3}{4}\ell_i > 0$ by (E1). $\Gamma_i \le \Gamma \le A^+ < (\ell_i - 2\kappa)(1 - \sqrt{1 - e_1^2}) \le \Lambda_i(1 - \sqrt{1 - e_1^2}) < \Lambda_i$ by (E2). So $w \in \mathcal{O}_n$ and $k_i := 1 - \Gamma_i/\Lambda_i > \sqrt{1 - e_1^2}$.

Step S19.35: Eccentricity and radii. By (Ch2), $e_i(z)^2 = 1 - k_i^2 < e_1^2$ and $a_i = \Lambda_i^2/m_i^2$; z is prograde elliptic (Ch1). Lemma 19.1(d): $r_i^- \le a_i(1 - e_i) \le |q_i|_2 \le a_i(1 + e_i) \le r_i^+$ since $\Lambda_i \in [\ell_i - 2\kappa, u_i + 2\kappa]$ and $e_i < e_1$.

Step S19.36: Separation. By (E3), $|q_i|_2 \ge d/2 > 0$; for $i < j$: $|q_j - q_i|_2 \ge |q_j|_2 - |q_i|_2 \ge r_j^- - r_i^+ \ge d/2 > 0$. So NoColl($q$), $z \in W$, $w \in \mathcal{W}$.

Step S19.37: Last assertion. Obvious: $[\ell_i, u_i] \subset [\ell_i - 2\kappa, u_i + 2\kappa]$ and $A_1 < A^+$. □

Lemma 19.11 (Kepler Hamiltonian: data for N-PARTIAL-CONVEX)**.** *Let κ, G, G' be as in Lemma 19.9 and $\rho \in (0,\kappa]$. Define $h : \mathbb{C}^n \to \mathbb{C}$ by $h(\Lambda) = -\sum_i m_i^3/(2\Lambda_i^2)$ (total; the value at a zero coordinate plays no role). Put $\bar u_i = u_i + 3\kappa$,*

$$g_h = \min_i \frac{m_i^3}{\bar u_i^3}, \qquad \varsigma_h = \min_i \frac{3m_i^3}{\bar u_i^4}, \qquad M_{\mathbb{C}} = \max_i \frac{3m_i^3}{(5\ell_i/8)^4} \qquad (\textit{all positive}).$$

*Then: (i) $\overline{B}(I,\rho) \subset G'$ for every $I \in G$ (closed max-norm ball); (ii) h is complex differentiable and complex C^2 on G'_ρ (Definition **??**) with $\operatorname{Re}\Lambda_i \ge \ell_i - 3\kappa \ge \frac58\ell_i > 0$ there; (iii) $\operatorname{Im} h(I) = 0$ for real $I \in G'_\rho$; (iv) $\|D^2h(z)\| \le M_{\mathbb{C}}$ for $z \in G'_\rho$ (operator norm, max norms); (v) $\texttt{HasDefiniteFastPart}\, G'\, \rho\, g_h\, \varsigma_h\, h$ holds with $\sigma = -1$, and $\texttt{HasDefiniteFastPart}\, G'\, \rho\, g_h\, \varsigma_h\, (-h)$ holds with $\sigma = +1$; (iv) and (ii), (iii) hold for $-h$ as well.*

Proof. **Step S19.38: (i).** For $I_i \in (\ell_i - \kappa, u_i + \kappa)$ (an *open* interval) and $|v_i - I_i| \le \rho \le \kappa$: $v_i \ge I_i - \kappa > \ell_i - 2\kappa$ and $v_i \le I_i + \kappa < u_i + 2\kappa$, i.e. $v \in G'$.

Step S19.39: (ii),(iii). If $z \in G'_\rho$ there is $J \in G'$ with $|z - J|_\infty < \rho$; so $\operatorname{Re} z_i > \ell_i - 2\kappa - \rho \ge \ell_i - 3\kappa \ge \frac58\ell_i$ by (E1). Hence $z_i \ne 0$ and h is a finite sum of monomials z_i^{-2}, holomorphic ((X6)); a holomorphic function is complex C^2 ($h_i' = m_i^3 z_i^{-3}$, $h_{ii}'' = -3m_i^3 z_i^{-4}$). For real I the value is real.

Step S19.40: (iv). $D^2h(z) = \operatorname{diag}(-3m_i^3 z_i^{-4})$, whose operator norm in the max norm is $\max_i 3m_i^3|z_i|^{-4} \le \max_i 3m_i^3(5\ell_i/8)^{-4} = M_{\mathbb{C}}$ since $|z_i| \ge \operatorname{Re} z_i \ge \frac58\ell_i$.

Step S19.41: (v). The set U of real points of G'_ρ is contained in $\prod[\ell_i - 3\kappa, u_i + 3\kappa]$ and every $I \in U$ has $0 < I_i \le \bar u_i$. There $h_{\mathbb{R}} = h|_{\mathbb{R}^n}$ is C^∞, $\omega_h(I)_i = m_i^3 I_i^{-3} \ge m_i^3 \bar u_i^{-3} \ge g_h$, so $|\omega_h(I)|_\infty \ge g_h > 0$, and $D^2h_{\mathbb{R}}(I)[v,v] = -\sum_i 3m_i^3 I_i^{-4} v_i^2 \le -\varsigma_h \sum_i v_i^2$: $\varsigma_h \sum v_i^2 \le (-1)D^2h_{\mathbb{R}}(I)[v,v]$ with $\sigma = -1$. For $-h$ the frequency has the same modulus and $D^2(-h)_{\mathbb{R}}[v,v] \ge \varsigma_h \sum v_i^2$, $\sigma = +1$. The statements on Im and D^2 for $-h$ are clear. □

19.6. **The analytic perturbation.**

Lemma 19.12 (Uniform complex tube)**.** *Let $\mathcal{W} \subset \mathrm{Ph}(2n) = \mathbb{R}^{4n}$ be open and τ_k-invariant ($k \in \mathbb{Z}^n$), $f : \mathcal{W} \to \mathbb{R}$ real analytic with $f \circ \tau_k = f$, $K_* \subset \mathcal{W}$ compact and $K_0 = \bigcup_k \tau_k(K_*)$. There are $r_0 > 0$ and a holomorphic function $f^{\mathbb{C}}$ on the open set $\mathcal{T} = \{z \in \mathbb{C}^{4n} : \exists x \in K_0,\ |z - x|_\infty < r_0\}$ with $f^{\mathbb{C}} = f$ on $\mathcal{T} \cap \mathbb{R}^{4n}$ (which is contained in $\mathcal{W}$) and $f^{\mathbb{C}} \circ \tau_k = f^{\mathbb{C}}$ on $\mathcal{T}$ (τ_k acting on $\mathbb{C}^{4n}$ by the same formula; $\mathcal{T}$ is τ_k-invariant).*

Proof. **Step S19.42: Local extensions.** For $x \in K_*$, (X4) gives $r_x > 0$ and g_x holomorphic on $D(x, r_x)$ with $g_x = f$ on $D(x, r_x) \cap \mathbb{R}^{4n}$; shrinking r_x we may assume $D(x, r_x) \cap \mathbb{R}^{4n} \subset \mathcal{W}$ ($\mathcal{W}$ is open).

Step S19.43: A Lebesgue number. The polydiscs $D(x, r_x/2)$, $x \in K_*$, cover the compact K_*: choose $x_1, \dots, x_s$ with $K_* \subset \bigcup_j D(x_j, r_j/2)$ ($r_j = r_{x_j}$) and put $r_0 = \min_j r_j/2$. If $x \in K_*$, $x \in D(x_j, r_j/2)$ then $D(x, r_0) \subset D(x_j, r_j)$.

Step S19.44: Gluing. On $U_* = \bigcup_j D(x_j, r_j)$ put $f_* = g_{x_j}$ on $D(x_j, r_j)$. This is well defined: $D(x_j, r_j) \cap D(x_k, r_k)$ is convex, hence connected, both functions are holomorphic on it and equal f on its real points; its real points form a nonempty open set whenever it is nonempty (the polydiscs have real centres, so $\operatorname{Re} z$ lies in both if z does, and the real trace of a polydisc is open). By (X5), $g_{x_j} = g_{x_k}$ there. So f_* is holomorphic on U_* with $f_* = f$ on $U_* \cap \mathbb{R}^{4n}$.

Step S19.45: Periodisation. Let $z \in \mathcal{T}$: $|z - x|_\infty < r_0$ with $x = \tau_k(x_*)$, $x_* \in K_*$. Then $\tau_{-k} z \in D(x_*, r_0) \subset U_*$ (Step S19.43); put $f^{\mathbb{C}}(z) = f_*(\tau_{-k} z)$. If also $z \in D(x', r_0)$, $x' = \tau_{k'}(x'_*)$, the two functions $f_* \circ \tau_{-k}$ and $f_* \circ \tau_{-k'}$ are holomorphic on the convex set $D(x, r_0) \cap D(x', r_0) \ni z$ and agree with f on its real points, a nonempty open set since it contains $\operatorname{Re} z$ (they equal $f \circ \tau_{-k} = f$

by periodicity); by (X5) they coincide, so $f^{\mathbb{C}}$ is well defined and locally equal to $f_* \circ \tau_{-k}$, hence holomorphic. The identities $f^{\mathbb{C}} = f$ on real points and $f^{\mathbb{C}} \circ \tau_k = f^{\mathbb{C}}$ follow from the construction (replace k by $k + k'$). □

Proposition 19.13 (The perturbation P and its bound B_0)**.** *Let $\kappa, e_1, A^+, G, G', D$ be as in Lemma 19.9. There are $\rho \in (0, \kappa]$, $B_0 \geq 1$ and $P : \mathbb{C}^{2n} \times \mathbb{C}^{2n} \to \mathbb{C}$, depending only on $(n, \mathcal{B})$ and the choices of Lemma 19.9, such that*

- *(i) `IsPartialPerturbation` $G'\, D\, \rho\, P$ (Definition **??**, with n fast and $k = n$ slow variables) and likewise for $-P$;*
- *(ii) for every real w in $\mathcal{V} = \{w \in \mathrm{Ph}(2n) : \ \Lambda \in G,\ (\xi, \eta) \in D\}$ (an open set, contained in $\mathcal{K} \subset \mathcal{W}$): $P(\mathrm{cx}(w)) = \hat{f}(w)/B_0 \in \mathbb{R}$.*

Proof. **Step S19.46: Analytic input.** $\hat{f}$ is real analytic on the open set $\mathcal{W}$ and τ_k-invariant (Ch3). Let $K_* = \{w : \lambda \in [0, 2\pi]^n,\ \Lambda \in \overline{G'},\ \Gamma(w) \leq A^+\}$, compact (the conditions bound λ, Λ and $\xi_i^2 + \eta_i^2 \leq 2A^+$) and $K_* \subset \mathcal{K} \subset \mathcal{W}$ (Lemma 19.10); $K_0 = \bigcup_k \tau_k(K_*) = \{w : \Lambda \in \overline{G'}, \Gamma \leq A^+\}$. Lemma 19.12 gives r_0 and $f^{\mathbb{C}} = \hat{f}^{\mathbb{C}}$ on $\mathcal{T}$.

Step S19.47: Choice of ρ, B_0. $\rho = \min(\kappa, r_0/2)$. The tube $F_\rho = \mathcal{F}(G', D, \rho)$ of Definition **??** consists of z with $\mathrm{Re}\, z \in \mathbb{R}^n \times G' \times D \subset K_0$ (in chart order: $\mathrm{Re}\,\Lambda \in G'$, $\mathrm{Re}(\xi, \eta) \in D$, all $|\mathrm{Im}\, z_i| < \rho$); thus $|z - \mathrm{Re}\, z|_\infty < \rho < r_0$ and $F_\rho \subset \mathcal{T}$, F_ρ is open and τ_k-invariant. For $z \in F_\rho$ choose k with $\mathrm{Re}\,\lambda(\tau_{-k} z) \in [0, 2\pi)^n$; then $\tau_{-k} z$ lies in the compact set $\{z : \mathrm{dist}_\infty(z, K_*) \leq \rho\} \subset \mathcal{T}$ on which the continuous $f^{\mathbb{C}}$ is bounded by some B'; $f^{\mathbb{C}}(z) = f^{\mathbb{C}}(\tau_{-k} z)$. Put $B_0 = \max(1, B')$: $|f^{\mathbb{C}}| \leq B_0$ on F_ρ.

Step S19.48: Definition of P. $P = f^{\mathbb{C}}/B_0$ on $\mathcal{T}$ and $P = 0$ elsewhere.

Step S19.49: (P1),(P2). P is holomorphic, hence complex differentiable, on $F_\rho \subset \mathcal{T}$; $|P| \leq 1$ there.

Step S19.50: (P3). A real w with $\mathrm{cx}(w) \in F_\rho$ has $w \in \mathbb{R}^n \times G' \times D \subset K_0 \subset \mathcal{T} \cap \mathbb{R}^{4n}$, so $P(\mathrm{cx}\, w) = \hat{f}(w)/B_0 \in \mathbb{R}$ (Lemma 19.12). This also proves (ii), because $\mathcal{V} \subset \mathbb{R}^n \times G' \times D$ and $\mathcal{V} \subset \mathcal{K}$ (Lemma 19.10, $G \subset G'$, $\Gamma < A^+$); $\mathcal{V}$ is open as the preimage of open sets under coordinate projections.

Step S19.51: (P4). $\mathrm{cx}(\texttt{fastShift}\, \ell) = \tau_\ell$ on the fast angles ($2\pi\ell$ added to λ); for $z \in F_\rho$, $P(\tau_\ell z) = f^{\mathbb{C}}(\tau_\ell z)/B_0 = P(z)$ (Lemma 19.12).

Step S19.52: $-P$. (P1)–(P4) are invariant under $P \mapsto -P$. □

19.7. Lifting finite solution segments to the Poincaré chart.

Lemma 19.14 (Uniform data over compact slow sets)**.** *Let $\mathcal{W} \subset \mathbb{R}^{4n}$ be open and τ_k-invariant, $F : \mathbb{R}^{4n} \to \mathbb{R}^{4n}$ C^1 on $\mathcal{W}$ with $F \circ \tau_k = F$ on $\mathcal{W}$, and $Q \subset \mathbb{R}^{3n}$ compact (the coordinates (ξ, Λ, η)) with $\mathbb{R}^n \times Q \subset \mathcal{W}$ (λ-coordinates free). Then $\Phi = \mathbb{R}^n \times Q$ has uniform data (δ, M, L) for F on $\mathcal{W}$ (Definition **??**), some $\delta \in (0, \pi]$.*

Proof. **Step S19.53: A compact cell.** $[-\pi, 3\pi]^n \times Q \subset \mathcal{W}$ is compact; by (X2) there is $\delta \in (0, \pi]$ with its closed δ-neighbourhood $C \subset \mathcal{W}$; C is compact and [N:`bpN:p3:gen-lip`] gives M, L with $\|F\| \leq M$ and F L-Lipschitz on C.

Step S19.54: Translation. For $x = (\lambda, y) \in \Phi$ choose k with $\lambda - 2\pi k \in [0, 2\pi)^n$ and $x' = \tau_{-k} x$; since $\delta \leq \pi$, $\overline{B}(x', \delta) \subset C$ (max norm). Then $\overline{B}(x, \delta) = \tau_k \overline{B}(x', \delta) \subset \mathcal{W}$, and $F \circ \tau_k = F$ transfers the bounds. □

Proposition 19.15 (Lifting a solution segment)**.** *Let $\mathcal{B}$ be admissible, $\mu \in \mathbb{R}$, $l \le 0 \le r$, and $z : \mathbb{R} \to \mathrm{Ph}(2n)$ with* $\mathrm{IsMovSolOn}(m, \mu, [l, r], z)$ *and $z(t) \in \mathcal{R}_{\mathrm{out}}$ for all $t \in [l, r]$. Then there is $y : \mathbb{R} \to \mathrm{Ph}(2n)$ such that for all $t \in [l, r]$: $y(t) \in \mathcal{W}$, $\Psi(y(t)) = z(t)$, and y has derivative $X_{H^\Psi}(y(t))$ at t (H^Ψ is differentiable at $y(t)$). Moreover $\Lambda_i(y(t)) = \Lambda_i(z(t)) \in (\ell_i, u_i)$ and $\Gamma(y(t)) = \mathcal{A}(z(t)) < A_1$, so $y(t) \in \mathcal{K}$.*

Proof. **Step S19.55: Initial point.** $z(0) \in \mathcal{R}_{\mathrm{out}}$ is prograde elliptic, so $z(0) = \Psi(w_0)$ for some $w_0 \in \mathcal{O}_n$ (Ch1); $z(0) \in W$ gives $w_0 \in \mathcal{W}$. By (Ch2) for every $t \in [l, r]$ and every $w \in \mathcal{O}_n$ with $\Psi(w) = z(t)$: $\Lambda_i(w) = \Lambda_i(z(t)) \in (\ell_i, u_i)$ and $\Gamma(w) = \mathcal{A}(z(t)) < A_1$; by Lemma 19.10 (last assertion) such w lie in $\mathcal{K} \subset \mathcal{W}$. This proves the last sentence once y exists.

Step S19.56: The maximal chart solution. $F = X_{H^\Psi}$ is C^1 on the open $\mathcal{W}$ (Ch3) and τ_k-periodic (τ_k is a translation and $H^\Psi \circ \tau_k = H^\Psi$ on the invariant open set $\mathcal{W}$; differentiating, $DH^\Psi \circ \tau_k = DH^\Psi$ and so $X_{H^\Psi} \circ \tau_k = X_{H^\Psi}$: the proof of [N:`bpN:p3:ham-periodic`], which is stated for angle blocks, uses only that the shift is a translation). Let $y_{\max}$ be the maximal solution through w_0, with maximal interval $(S_-, S_+) \ni 0$ ([N:`bpN:p3:gen-maximal`]).

Step S19.57: It projects to the physical solution. By Corollary 2.5, $x = \Psi \circ y_{\max}$ has derivative $X_{H_{\mathrm{mov}}}(x(t))$ at every $t \in (S_-, S_+)$, with $x(t) \in W$ (as $y_{\max}(t) \in \mathcal{W}$); i.e. $\mathtt{IsSolIn}\, X_{H_{\mathrm{mov}}}\, W\, (S_-, S_+)\, x$. Also z is a solution in W on $[l, r]$ (Lemma 19.5), and $x(0) = \Psi(w_0) = z(0)$. By [N:`bpN:p3:gen-uniqueness`] on the interval $(S_-, S_+) \cap [l, r]$, $x = z$ there.

Step S19.58: The chart solution lives on all of $[l, r]$. Suppose $S_+ \le r$. For $t \in [0, S_+)$: $z(t) = \Psi(y_{\max}(t))$ with $z(t) \in \mathcal{R}_{\mathrm{out}}$, so by Step S19.55 $y_{\max}(t) \in \mathbb{R}^n \times Q_0$ with $Q_0 = \{(\xi, \Lambda, \eta) : \Lambda_i \in [\ell_i, u_i],\ \Gamma \le A_1\}$ compact and $\mathbb{R}^n \times Q_0 \subset \mathcal{K} \subset \mathcal{W}$ (Lemma 19.10). Lemma 19.14 gives uniform data for $\Phi = \mathbb{R}^n \times Q_0$ and [N:`bpN:p3:gen-continuation`] gives $S_+ = \infty$: contradiction. So $S_+ > r$, and likewise $S_- < l$.

Step S19.59: Conclusion. $y = y_{\max}$ has all the properties on $[l, r] \subset (S_-, S_+)$; H^Ψ is differentiable on $\mathcal{W}$. □

19.8. **The stopped action bound from N-PARTIAL-CONVEX.** Throughout this subsection the data of Lemma 19.9 $(\kappa, e_1, A^+, G, G', D, \varepsilon_D)$, Lemma 19.11 $(\rho, g_h, \varsigma_h, M_{\mathbb{C}}, h)$ and Proposition 19.13 (B_0, P) are fixed, with ρ and B_0 of Proposition 19.13 (so $\rho \le \kappa$ and Lemma 19.11 applies). For $\nu > 0$ let, on $\mathrm{Ph}(2n)$ (chart coordinates, N's phase order),

$$H_\nu^\pm(\varphi, \mathsf{I}) = \pm\Big[\mathrm{Re}\, h\big(\mathrm{fast}(\mathsf{I})\big) + \nu\, \mathrm{Re}\, P\big(\mathrm{cx}(\varphi, \mathsf{I})\big)\Big], \qquad H_\nu = H_\nu^+.$$

H_ν^+ is the Hamiltonian H_ν of N-PARTIAL-CONVEX for the data (h, P), and H_ν^- the one for $(-h, -P)$ (both satisfy its hypotheses by Lemma 19.11, Proposition 19.13); $H_\nu^- = -H_\nu^+$.

Lemma 19.16 (The N-Hamiltonian is the chart Hamiltonian on $\mathcal{V}$)**.** *For $\mu > 0$ and $\nu = B_0\mu$: $H_\nu^+ = H^\Psi$ on the open set $\mathcal{V} = \{w : \Lambda \in G,\ (\xi, \eta) \in D\}$ (with $H^\Psi = H_{\mathrm{mov}}^{(\mu)} \circ \Psi$). Consequently, if $w \in \mathcal{V}$, H_ν^+ is differentiable at w and $X_{H_\nu^+}(w) = X_{H^\Psi}(w)$; and $X_{H_\nu^-} = -X_{H_\nu^+}$ there.*

Proof. **Step S19.60: Equality.** For $w \in \mathcal{V}$ and real: $\mathrm{cx}(w)$ has real part w and zero imaginary part, and $\mathrm{Re}\,\Lambda \in G \subset G'$, $\mathrm{Re}(\xi, \eta) \in D$, so $\mathrm{cx}(w) \in F_\rho$ (Definition **??**). By Proposition 19.13(ii), $\nu\, \mathrm{Re}\, P(\mathrm{cx}\, w) = B_0\mu\hat{f}(w)/B_0 = \mu\hat{f}(w)$; $\mathrm{Re}\, h(\Lambda) = h(\Lambda)$ is real. So $H_\nu^+(w) = h(\Lambda) + \mu\hat{f}(w) = H^\Psi(w)$ by (Ch3) ($w \in \mathcal{V} \subset \mathcal{K} \subset \mathcal{O}_n$).

Step S19.61: Fields. Two functions equal on an open set have the same `fderiv` there; H^Ψ is differentiable on $\mathcal{W} \supset \mathcal{V}$ (Ch3). The last claim is linearity of X. □

Proposition 19.17 (Stopped action bound)**.** *Let $n \geq 2$ and $\mathcal{B}$ be admissible. There is $\nu_0 \in (0,1]$, depending only on $(n,\mathcal{B})$ and the choices above, such that for every $\mu > 0$ with $\nu := B_0\mu < \nu_0$, every $z_0 \in \mathrm{Ph}(2n)$, and all $T > 0$, $\delta > 0$ with*

$$T \leq \nu^{-1}\exp(\nu^{-a}), \qquad \delta \geq \tfrac{1}{2}\nu^b \qquad (a = \alpha_n,\ b = \beta_n),$$

the predicate StoppedBound$(\mathcal{B},\mu,T,\delta,z_0)$ *holds.*

Proof. **Step S19.62: The threshold.** Apply [N:`bpN:p0:partial`] with n fast actions, $k = n$ slow pairs ($n \geq 2$, $k \geq 1$), the sets G, G' (open, convex), D (open), the numbers $\rho, g_h, \varsigma_h, M_{\mathbb{C}}$ (positive, $\overline{B}(I,\rho) \subset G'$ for $I \in G$) and the function h: its hypotheses are exactly Lemma 19.11 (i)–(v) (with $\sigma = -1$). It yields $\mu_+ > 0$. Apply it again to $-h$ (Lemma 19.11, $\sigma = +1$): $\mu_- > 0$. Both thresholds depend on $(n,k,G,G',D,\rho,g_h,\varsigma_h,M_{\mathbb{C}},h)$ only, not on P. Put

$$\nu_0 = \min\{1,\ \mu_+,\ \mu_-,\ \kappa^{1/b},\ \varepsilon_D^{1/b}\}.$$

For $\nu < \nu_0$: $\nu^b < \min(\kappa,\varepsilon_D)$ (as $b > 0$, $\nu < 1$). The exponents of [N:`bpN:p0:partial-exponents`] are α_n, β_n for the same n (Definition 1.42; the number of fast actions equals the number of planets).

Step S19.63: The segment. Let $l \leq 0 \leq r$, $l < r$, z with $z(0) = z_0$, IsMovSolOn$(m,\mu,[l,r],z)$ and $z(t) \in \mathcal{R}_{\mathrm{out}}$ for all $t \in [l,r]$; let $t \in [l,r] \cap [-T,T]$ and $i < n$. Proposition 19.15 gives y. For $s \in [l,r]$: $\Lambda(y(s)) \in \prod(\ell_i,u_i) \subset G$ and $(\xi,\eta)(y(s)) \in D$, because $\Gamma(y(s)) = \mathcal{A}(z(s)) < A_1 < A^+$; so $y(s) \in \mathcal{V}$, and $\Lambda_i(y(s)) = \Lambda_i(z(s)) =$ `keplerAction` (Ch2). We must show $|\Lambda_i(y(t)) - \Lambda_i(y(0))| \leq \delta$ (note $y(0)$ projects to z_0, so $\Lambda_i(y(0)) = \Lambda_i(z_0)$). For $t = 0$ this is trivial.

Step S19.64: y solves the N-system on $[0,r]$ and on the reversed segment. By Lemma 19.16, on $[l,r]$: H_ν^+ is differentiable at $y(s)$ and y has derivative $X_{H^\Psi}(y(s)) = X_{H_\nu^+}(y(s))$. Also $\mathrm{fast}(y(s)_2) = \Lambda(y(s)) \in G$. Hence `IsSolutionOn` H_ν^+ $\{\mathsf{I} : \mathrm{fast}\,\mathsf{I} \in G\}$ $[0,r]$ y (Definition **??**). For the reversed curve $\tilde{y}(s) = y(-s)$, $s \in [0,-l]$: by (X3) it has derivative $-X_{H_\nu^+}(y(-s)) = X_{H_\nu^-}(\tilde{y}(s))$ and H_ν^- is differentiable at $\tilde{y}(s)$ (Lemma 19.16): `IsSolutionOn` H_ν^- $\{\mathrm{fast} \in G\}$ $[0,-l]$ $\tilde{y}$.

Step S19.65: Erosions. (Z2) $\Lambda(z_0) \in \prod(\ell_i,u_i)$, and for $v \in \mathbb{R}^n$ with $|v - \Lambda(z_0)|_\infty \leq \nu^b < \kappa$: $v_i > \ell_i - \kappa$, $v_i < u_i + \kappa$; thus $\Lambda(y(0)) = \Lambda(z_0) \in \mathrm{core}(G,\nu^b)$ (Definition **??**). Slow: for $s \in [l,r]$ put $x = (\xi,\eta)(y(s))$, $|x|_2^2 = 2\Gamma(y(s)) < 2A_1$. If $|v - x|_\infty \leq \nu^b < \varepsilon_D$ then $|v - x|_2 \leq \sqrt{2n}\nu^b < \sqrt{2A^+} - \sqrt{2A_1}$, hence $|v|_2 < \sqrt{2A^+}$, i.e. $v \in D$. So $x \in \mathrm{core}(D,\nu^b)$ for every $s \in [l,r]$.

Step S19.66: Positive times. Let $t \in (0,r] \cap [0,T]$. Then $C := r > 0$, $t \leq \min(C,T) \leq \min(C,\nu^{-1}\exp(\nu^{-a}))$ and, for all $u \in [0,t]$, $(\mathrm{slow}(y(u)_1),\mathrm{slow}(y(u)_2)) = (\xi,\eta)(y(u)) \in \mathrm{core}(D,\nu^b)$ by Step S19.65. [N:`bpN:p0:partial`] (with $\mu := \nu \in (0,\mu_+)$, the perturbation P of Proposition 19.13(i), the constant $C = r$, the solution y) gives $|\Lambda(y(t)) - \Lambda(y(0))|_\infty \leq \frac{1}{2}\nu^b \leq \delta$.

Step S19.67: Negative times. Let $t \in [l,0) \cap [-T,T]$, $s_0 = -t \in (0,-l] \cap [0,T]$. Apply [N:`bpN:p0:partial`] to $(-h,-P)$ (threshold μ_-, $\nu < \mu_-$), $C = -l > 0$ and the solution $\tilde{y}$: its conditions (Z1), (Z2) and the slow-erosion condition on $[0,s_0]$ are Steps S19.64, S19.65 (the slow curve of $\tilde{y}$ is that of y reversed); $s_0 \leq \min(-l,\nu^{-1}\exp(\nu^{-a}))$. Hence $|\Lambda(\tilde{y}(s_0)) - \Lambda(\tilde{y}(0))|_\infty = |\Lambda(y(t)) - \Lambda(y(0))|_\infty \leq \frac{1}{2}\nu^b \leq \delta$.

Step S19.68: Translation to physical actions. In both cases $|\Lambda_i(z(t)) - \Lambda_i(z_0)| = |\Lambda_i(y(t)) - \Lambda_i(y(0))| \leq \delta$ by Step S19.63. □

Remark 19.18 (What N provides and what it does not). Proposition 19.17 gives StoppedBound only for $\nu = B_0\mu < \nu_0$ and the specific T,δ; it is not the unconditional hypothesis of Theorem 1.44, which must never be taken as a proof of it. N supplies no slow confinement: the slow erosion is verified on the tested segment from membership of $z(s)$ in $\mathcal{R}_{\mathrm{out}}$ ($\mathcal{A} < A_1 < A^+$), exactly the hypothesis of StoppedBound. N's initial erosion $\mathrm{core}(G,\nu^b)$ is used through $\Lambda(z_0) \in (\ell,u)$ only.

19.9. **Proof of P-EXP-SAFE.**

Theorem 19.19 (P-EXP-SAFE, Theorem 1.45)**.** *The statement of Theorem 1.45 holds, with the constants of Remark 19.20.*

Frozen Lean declaration: `KAM.Planetary.planetaryExponentialSafety`

Source: Nek77 Thm. 12.3, pp. 55–56 and §12.4 (Poincaré variables, Lemma 12.4.B, pp. 56–57: analyticity of the perturbed Hamiltonian on a complex neighbourhood of a compact domain), via Nek77 Thm. 4.4 (N-PARTIAL-CONVEX) and Prop. 19.17; round 2 §2.

Proof. **Step S19.69: Fixed data and constants.** Let $n \geq 2$, $\mathcal{B}$ admissible, and $\kappa, e_1, A^+, G, G', D, \varepsilon_D, \rho, B_0, P, \nu_0$ as in the preceding subsections. Put

$$\theta = \min\Big\{\kappa,\ \varepsilon_D,\ \min_i(\ell_i^0 - \ell_i),\ \min_i(u_i - u_i^0),\ \tfrac{A_1 - A_0}{n}\Big\} > 0,$$

$$\mu_* = \frac{1}{B_0}\min\Big\{\frac{\nu_0}{2},\ \big(\tfrac{\theta}{2}\big)^{1/b}\Big\}, \qquad c = B_0^{-a}, \qquad C = B_0^b \max_i \frac{u_i}{m_i^2}.$$

All are positive and depend only on $(n, \mathcal{B})$ (the Lean quantifier order: μ_*, c, C before μ and z_0).

Step S19.70: Parameters. Let $0 < \mu < \mu_*$, $z_0 \in \mathcal{R}_{\mathrm{in}}$, $\nu = B_0\mu$, $T = \exp(c\mu^{-a})$, $\delta = \frac{1}{2}\nu^b$. Then $\nu < \nu_0/2 < \nu_0 \leq 1$, $\nu^b \leq \theta/2$ and $\nu^{-a} = B_0^{-a}\mu^{-a} = c\mu^{-a}$, so $T = \exp(\nu^{-a}) \leq \nu^{-1}\exp(\nu^{-a})$ (as $\nu \leq 1$). Moreover $2\delta = \nu^b \leq \theta/2 < \min_i(\ell_i^0 - \ell_i)$ and $< \min_i(u_i - u_i^0)$, and $2n\delta = n\nu^b \leq n\theta/2 < A_1 - A_0$.

Step S19.71: The stopped bound. Proposition 19.17 applies ($\nu < \nu_0$, $T \leq \nu^{-1}\exp(\nu^{-a})$, $\delta = \frac{1}{2}\nu^b$): StoppedBound$(\mathcal{B}, \mu, T, \delta, z_0)$.

Step S19.72: The helper. The hypotheses of Theorem 19.7 hold: $\mu, T, \delta > 0$; the margins $\ell_i + 2\delta < \ell_i^0$, $u_i^0 + 2\delta < u_i$, $A_0 + 2n\delta < A_1$ (Step S19.70); $z_0 \in \mathcal{R}_{\mathrm{in}}$; the stopped bound. It gives z with $z(0) = z_0$ and clauses (a)–(d), which contain all the clauses of Theorem 1.45(a), and $|\Lambda_i(z(t)) - \Lambda_i(z_0)| \leq \delta$ for $|t| \leq T$.

Step S19.73: Axis drift. For $|t| \leq T$ and each i both $z(t)$ and z_0 are prograde elliptic with $\Lambda_i(z(t)), \Lambda_i(z_0) \in (0, u_i)$, and $a_i = \Lambda_i^2/m_i^2$ (Lemma 19.1(a)); thus

$$|a_i(z(t)) - a_i(z_0)| = \frac{|\Lambda_i(z(t)) - \Lambda_i(z_0)|\,\big(\Lambda_i(z(t)) + \Lambda_i(z_0)\big)}{m_i^2} \leq \frac{2u_i}{m_i^2}\delta = \frac{u_i}{m_i^2}B_0^b\mu^b \leq C\mu^b.$$

Step S19.74: Assembly. The function z and the horizon $T = \exp(c\mu^{-a})$ are those demanded by Theorem 1.45; the first four clause groups are those of Theorem 1.44(a)–(c). □

Remark 19.20 (Order of every constant against the frozen quantifiers). The chain of choices, with what each depends on:

Constant	Depends on (and is chosen after)
κ, e_1, A^+; G, G', D, ε_D (Lemma 19.9)	$n, \mathcal{B}$ only (strict admissibility inequalities and continuity)
$\rho \leq \kappa$, $B_0 \geq 1$, P (Prop. 19.13)	the above; the compact real set $\overline{G'} \times \overline{D} \times \mathbb{T}^n$ (via r_0 of Lemma 19.12); *not* on μ
$g_h, \varsigma_h, M_{\mathbb{C}}$ (Lemma 19.11)	m, ℓ, u, κ
$\mu_\pm$ (N-PARTIAL-CONVEX, applied to $\pm h$)	$n, k = n, G, G', D, \rho, g_h, \varsigma_h, M_{\mathbb{C}}, h$; *before* P, μ, C, z (N's order), hence independent of B_0 and of P
ν_0 (Prop. 19.17)	$\mu_\pm, \kappa, \varepsilon_D, b$
θ, μ_*, c, C (Thm. 19.19)	$\nu_0, B_0, \kappa, \varepsilon_D, \ell^0 - \ell, u - u^0, (A_1 - A_0)/n, a, b, u, m$: i.e. on $(n, \mathcal{B})$ only — **before** μ and z_0, as in the frozen $\exists\,\mu_*\,c\,C\,\forall\mu\,\forall z_0$
$\nu = B_0\mu$, $T = \exp(c\mu^{-a})$, $\delta = \frac{1}{2}\nu^b$	μ (and constants); independent of z_0
z	z_0, μ (maximal solution)

In Theorem 19.7 (P-AMD-CONFINEMENT) the frozen order is $\mathcal{B}, \mu, T, \delta$, then z_0 with the stopped bound, and no constant is produced; the proof only uses the hypotheses. Exponents: $a = \alpha_n$ and $b = \beta_n = \frac{3}{2}a$ are the frozen ones, *not* an optimisation: b enters only through $\nu^b \le \theta/2$ and through C.

Remark 19.21 (Index bookkeeping between N and P). N-PARTIAL-CONVEX has n fast actions and k slow pairs and exponents $a_n = 2/(6n^2 - 3n + 14)$, $b_n = \frac{3}{2}a_n$ ([N:`bpN:p0:partial-exponents`]); P-EXP-SAFE has n planets and the same formulas α_n, β_n (Nek77 (12.4)–(12.5), p. 56: s = number of planets, page image read). We take N's n = number of planets and $k = n$: each planet contributes one fast action Λ_i and one slow pair (ξ_i, η_i). No theorem with $2n$ fast actions is used, and the exponents are those with the planet count. The N phase space $\mathrm{Ph}(n + k) = \mathrm{Ph}(2n)$ and P's physical phase space $\mathrm{Ph}(n \cdot 2)$ are both $\mathbb{R}^{2n} \times \mathbb{R}^{2n}$; the Poincaré chart is the dictionary (fast block $(\lambda; \Lambda)$ first, then slow block $(\xi; \eta)$), see Theorem 9.12 and Part 0's block convention. The N time unit is that of $H_\nu = h + \mu\hat{f}$, i.e. of H_{mov} (the Hamiltonians are not rescaled; B_0 only rescales the coupling: $\mu\hat{f} = \nu P$ on $\mathcal{V}$).

19.10. **Sources, deviations, doubts, forward references.**
Sources read. Nek77 §12.1–12.4, printed pp. 54–57 (statement of Thm. 12.3 with $a = 2/(6s^2 - 3s + 14)$, $b = \frac{3}{2}a$, p. 56, from the page image; the text layer loses the formulas); Lemma 12.4.B p. 57 (analytic extension to a complex neighbourhood of a compact domain: our Lemma 19.12, Prop. 19.13); LP17 p. 2, (7)–(11) (Hamiltonian, total momentum, AMD); the frozen Lean files and the audits `docs/tracks/P/AUDIT-additions-v0.3.md` §6, `docs/tracks/N/AUDIT-additions-v0.3.md` §6, `AUDIT-additions-v0.4-delta.md` §6; round 2 §2 and round 3 item 1c.
Deviations. (1) Nek77's proof of Thm. 12.3 (domains $B(\alpha, \beta, \gamma)$, barycentric elements, $N(v)$ in a domain of planetary motions) is *not* followed: the frozen statement is the low-AMD specialisation of D25 and its proof here is the one prescribed by the audits (chart, collars, N-PARTIAL-CONVEX, deterministic helper). The AMD argument replaces Nek77's use of $N(v)$ by exact conservation of J. (2) The N input has a Euclidean-erosion/closed-imaginary-bound form in the source and a max-norm/open-width form in the frozen N statement ([N:`bpN:p0:partial`]); we use only the frozen form. (3) The Poincaré chart uses the unscaled heliocentric variables with the perturbation μf_{mov} containing the indirect term (Remark 9.13); no reduced-mass correction is needed and none is made, so no $O(\mu)$ element-comparison error is added to δ (the audit's alternative route).
Doubts about frozen statements. None found. Checked: (i) MovUniqueOn quantifies over open convex $U \ni 0$ and requires w to be an IsMovSolOn solution on U, which forces $U \subset$ the maximal interval; proved by [N:`bpN:p3:gen-uniqueness`]. (ii) The derivatives at $t = \pm T$ in IsMovSolOn are two-sided, so one needs $T_\pm > \pm T$ strictly: shown in Step S19.27. (iii) StoppedBound allows $l = 0$ or $r = 0$ but not $l = r = 0$; used with $(0, s)$, $(-s, 0)$. (iv) The theorems assert Λ_i-drift $\le \delta$ for *all* $t \in [-T, T]$, which for P-EXP-SAFE needs the stopped bound on both sides of 0 (done through $-H_\nu$). (v) B.radius $\ge |q_i|_2$ and PairBounds$(d, 2R)$ use $|q_i - q_j|_2 \le |q_i|_2 + |q_j|_2 \le 2R$; true.
Suspected errata. None in a pinned source. (The N audit's slow-ball collar $(\sqrt{2A^+} - \sqrt{2A_1})/(2\sqrt{2n})$, AUDIT v0.4-delta §6 item 3, is conservative; our ε_D is the sharp value for max-norm balls of $\mathbb{R}^{2n}$.)
Forward references and consumers. None to unwritten parts: this part cites only Parts 0, 1, 6, 7 and tracks N (Parts 0 and 3, the frozen [N:`bpN:p0:partial`]). Track N's Parts 11a–c prove [N:`bpN:p0:partial`]; if it slips, Theorem 19.19 is conditional on it as an input (not a premise of the frozen statement).
External facts used. (X1)–(X6) of §19.1; (E4), (E5), (E7).
Label index. `elements`, `ecc-bound`, `geometry`, `closure`, `moving-ode`, `ang-mom`, `amd`, `amd-uses`, `enlarge`, `real-domain`, `h-data`, `tube`, `perturbation`, `uniform-data`, `lift`, `hnu`, `stopped`, `st-limits`, `expsafe`, `constants`, `index`.

20. D25: P-SECULAR-BOX

This part proves the two frozen D25 theorems about the exactly averaged Sun–Jupiter–Saturn model: `KAM.Planetary.secularBoxSafety` (Theorem 20.4) and its specialisation to the cited decimal masses, `KAM.Planetary.secularBoxAtCitedMasses` (Theorem 20.5). Part 0 does not transcribe the objects of P-SECULAR-BOX; §20.2 does. Throughout $G = m_0 = 1$, there are two planets, indexed $i \in \{0,1\}$ (Jupiter, Saturn), the weights are $m = (1,\sigma)$ with $\sigma = \texttt{saturn}$, and $\mu > 0$.

Idea. The averaged Hamiltonian H_{sec} does not depend on the mean longitudes, so the actions Λ_i are constant. It is invariant under the simultaneous rotation of both eccentricity vectors, so $\mathcal{A} = \Gamma_0 + \Gamma_1$ is constant as well. For a point of the input box, $\mathcal{A}$ is smaller than $\frac{16373}{2500000}$, while a single eccentricity as large as $\frac{3}{20}$ would already force $\Gamma_i > \frac{87}{100} \cdot \frac{11}{1000} > \mathcal{A}$. Hence the slow variables (ξ,η) stay in a compact subset of the region where H_{sec} is smooth, for every $\mu > 0$ (no smallness: μ enters only the speeds). The mean longitudes are recovered by one quadrature; the radial bounds are the identity $|q_i|_2 = a_i(1 - v_i \cdot u(F_i))$ and rational arithmetic. Nothing here concerns the Newtonian problem.

- §20.2: transcription of `SecularInputBox`, `SecularRegular`, and the two frozen theorems;
- §20.3: every rational inequality (Lemma 20.7) with the exact check;
- §20.4: algebra of the explicit chart (eccentricity, radius, Jacobian, rotation covariance);
- §20.5: the open region $\mathcal{S}$ and smoothness of H_{sec} there; λ-independence and rotation invariance;
- §20.6: the reduced slow flow, global existence, lifting of λ, uniqueness;
- §20.7: proof of the two theorems; §20.8: Lean plan, totality, deviations, doubts.

20.1. **Conventions and external facts.** *Coordinates.* $w = ((\lambda,\xi),(\Lambda,\eta)) \in \texttt{Phase}(2+2)$, with $\texttt{fast}\, w_1 = \lambda$, $\texttt{slow}\, w_1 = \xi$, $\texttt{fast}\, w_2 = \Lambda$, $\texttt{slow}\, w_2 = \eta$ (Section 9.6). Write $u_i = (\xi_i,\eta_i) \in \mathbb{R}^2$, $\Gamma_i = \frac{1}{2}(\xi_i^2+\eta_i^2) = \frac{1}{2}|u_i|_2^2$ (`secularGamma`), $\mathcal{A} = \Gamma_0+\Gamma_1$ (`secularAMD`), $k_i = 1-\Gamma_i/\Lambda_i$ (`secularRoot`), $v_i = \rho_i u_i$ with $\rho_i = \sqrt{(1-\Gamma_i/(2\Lambda_i))/\Lambda_i}$ (`secularEccVector`), $e_i = |v_i|_2$, $a_i = (\Lambda_i/m_i)^2$ (`secularAxis`), $c(F) = (\cos F, \sin F)$ (`orbitalUnit`), $x_i(F) = v_i \cdot c(F)$, $\mathcal{J}_i(F) = 1 - x_i(F)$ (`longitudeJacobian`), and $q_i(w;F)$, $p_i(w;F)$ the values of `ellipsePosition`, `ellipseMomentum`; so

$$q_i = a_i\big(k_i c(F) - v_i + \tfrac{x_i}{1+k_i} v_i\big), \qquad p_i = \tfrac{m_i^2}{\Lambda_i \mathcal{J}_i}\big(J_0 c(F) - \tfrac{x_i}{1+k_i} J_0 v_i\big), \quad J_0(a,b) = (-b,a)\ (\texttt{quarterTurn}).$$

`secularAxis` and `secularRoot` are total Lean functions; they are only used where $\Lambda_i > 0$, $\Gamma_i < \Lambda_i$ (Remark 20.6). R_θ is the rotation of $\mathbb{R}^2$ by θ. $\Delta = [0,2\pi]^2$. The Hamiltonian field is `hamiltonianVectorField`: for $z = (q,p)$, $X_H(z) = ((\partial_{p_k} H)_k, (-\partial_{q_k} H)_k)$ (Definition 1.2); with $q = (\lambda,\xi)$, $p = (\Lambda,\eta)$: $\dot\lambda = \partial_\Lambda H$, $\dot\xi = \partial_\eta H$, $\dot\Lambda = -\partial_\lambda H$, $\dot\eta = -\partial_\xi H$.

External facts (Mathlib level; used by label).

(X1) Real algebra: $\sqrt{\cdot}$ is increasing, $\sqrt{y}^2 = y$ for $y \ge 0$; $|c(F)|_2 = 1$, $c(F+2\pi) = c(F)$, $R_\theta c(F) = c(F+\theta)$; rotations preserve $\cdot$, $|\cdot|_2$ and commute with J_0; `norm_num`/`linear_combination` for rational identities and polynomial identities modulo given relations.

(X2) Topology: open sets are preserved by finite intersections and preimages under continuous maps; closed bounded sets of $\mathbb{R}^N$ are compact; a continuous function on a compact set is bounded; a continuous function on an open set that is positive on a compact K is bounded below on K; an interval is connected, and a subset of a connected set that is nonempty, open and closed in it is all of it.

(X3) Integration: continuous functions are integrable on compact boxes; Fubini on Δ; for a continuous $g : \mathbb{R} \to \mathbb{R}$ of period 2π, $\int_0^{2\pi} g(s-\theta)\,ds = \int_0^{2\pi} g(s)\,ds$ (`Function.Periodic.intervalIntegral_add_eq`); fundamental theorem of calculus: $t \mapsto \int_0^t f$ has derivative $f(t)$ at t if f is continuous.

(X4) Differentiation under the integral: Part 8 (E2), used for $k = 2$.

(X5) Calculus: chain rule; a C^k function composed with an affine map is C^k; a function of a real variable which is continuous on an interval and has derivative 0 at each point is constant; `HasDerivAt` for a pair, resp. a function with values in $\mathbb{R}^N$, holds iff it holds for the coordinates; if $H(z + s\,b) = H(z)$ for all s and H is differentiable at z, then $DH(z)b = 0$.

Cited results. Track N, Part 3: [N:`bpN:p3:gen-sol`], [N:`bpN:p3:gen-uniqueness`], [N:`bpN:p3:gen-maximal`], [N:`bpN:p3:gen-continuation-compact`], [N:`bpN:p3:ham-smooth`], [N:`bpN:p3:ham-reduction`] (pattern only; see Lemma 20.16). Track P: Proposition 9.16(a) (the chart quantities and $|v_i|_2^2 = 1 - k_i^2$), Corollary 9.17 and Proposition 10.3(S2) are consistency checks (Remark 20.15); the proofs below do not depend on them.

20.2. **The frozen objects, transcribed.**

Definition 20.1 (Explicit secular objects)**.** For $m \in \mathbb{R}^2$, $\mu \in \mathbb{R}$, $w \in$ `Phase`$(2+2)$ and $F \in \mathbb{R}^2$ let $\mathbf{z}(w;F) =$ `ellipsePhase` $m\,w\,F = \big((q_i(w;F_i))_i, (p_i(w;F_i))_i\big) \in$ `Phase`$(2\cdot 2)$ (Lean: coordinate $k \mapsto$ `finProdFinEquiv.symm` k). Then

$$\mathcal{I}_\mu(w;F) = H_{\mathrm{mov}}^{(\mu,m)}\big(\mathbf{z}(w;F)\big)\prod_{i<2}\mathcal{J}_i(w;F_i) \qquad (\texttt{secularIntegrand}\ m\ \mu\ (\texttt{assemble}\,\Lambda\,F\,\eta\,\xi)),$$

$$H_{\mathrm{sec}}^{(m,\mu)}(w) = (2\pi)^{-2}\int_\Delta \mathcal{I}_\mu\big((F,\xi;\Lambda,\eta);F\big)\,dF \quad (\texttt{exactSecularHamiltonian}\ m\ \mu),$$

$H_{\mathrm{mov}}^{(\mu,m)} =$ `movingHamiltonian` $1\,1\,m\,\mu$ (Definition 1.22).

Frozen Lean declaration: `KAM.Planetary.secularIntegrand`, `KAM.Planetary.exactSecularHamiltonian`, `KAM.Planetary.ellipsePhase`, `KAM.Planetary.ellipsePosition`, `KAM.Planetary.ellipseMomentum`, `KAM.Planetary.longitudeJacobian`

Source: LP17 (B.5), p. 16: $\langle G\rangle_\lambda = (2\pi)^{-m}\int G\,d^m\lambda$; the substitution $\lambda_i \leftrightarrow F_i$ is the new elementary specialisation (Part 7, Corollary 9.17). The Lean definition is the integral in F, not a mean-longitude average, so no change of variable is needed to prove the theorems.

Definition 20.2 (`SecularInputBox`)**.** For $\sigma \in \mathbb{R}$, $w \in$ `Phase`$(2+2)$ (weights $m = (1,\sigma)$, `jupiterSaturnWeights`): $\mathrm{Box}_\sigma(w)$ holds iff (i) for $i = 0,1$: $0 < \Lambda_i$, $0 \le \Gamma_i < \Lambda_i$; (ii) $\frac{103}{20} \le a_0 \le \frac{21}{4}$ and $\frac{189}{20} \le a_1 \le \frac{193}{20}$; (iii) $|v_0|_2 \le \frac{3}{50}$, $|v_1|_2 \le \frac{7}{100}$. All phases λ and all perihelia are allowed.

Frozen Lean declaration: `KAM.Planetary.SecularInputBox`, `KAM.Planetary.jupiterSaturnWeights`

Source: Round 2 §3 (`r2-secular-box-arithmetic.json`); an illustrative canonical-element box, not an ephemeris enclosure.

Definition 20.3 (`SecularRegular`)**.** $\mathrm{Reg}_{\sigma,\mu}(w)$ holds iff: (i) $m_i > 0$, $\Lambda_i > 0$, $0 \le \Gamma_i < \Lambda_i$ for $i = 0,1$; (ii) for every $F \in \mathbb{R}^2$: $\mathcal{J}_i(w;F_i) > 0$ $(i = 0,1)$ and $\mathrm{NoColl}(\mathbf{z}(w;F)_1)$ (Definition 1.18: all $|q_i|_2 > 0$ and $|q_0 - q_1|_2 > 0$ in both orders); (iii) $F \mapsto \mathcal{I}_\mu((F,\xi;\Lambda,\eta);F)$ is integrable on Δ; (iv) H_{sec} is (Fréchet) differentiable at w.

Frozen Lean declaration: `KAM.Planetary.SecularRegular`

Theorem 20.4 (P-SECULAR-BOX, `secularBoxSafety`)**.** *Let $\sigma \in [\frac{29}{100}, \frac{31}{100}]$, $\mu > 0$, $w_0 \in$ `Phase`$(2+2)$ with $\mathrm{Box}_\sigma(w_0)$, $m = (1,\sigma)$, $H = H_{\mathrm{sec}}^{(m,\mu)}$. There is $w : \mathbb{R} \to$ `Phase`$(2+2)$ with*

- *(a) $w(0) = w_0$ and `IsHamiltonianSolutionOn` H `univ univ` w (for every $t \in \mathbb{R}$: H is differentiable at $w(t)$ and w has derivative $X_H(w(t))$ at t);*
- *(b) for every open convex $U \subset \mathbb{R}$ with $0 \in U$ and every $v : \mathbb{R} \to$ `Phase`$(2+2)$ with $v(0) = w_0$, `IsHamiltonianSolutionOn` H `univ` U v and $\mathrm{Reg}_{\sigma,\mu}(v(t))$ for all $t \in U$: $v(t) = w(t)$ for all $t \in U$;*
- *(c) for every $t \in \mathbb{R}$: $\mathrm{Reg}_{\sigma,\mu}(w(t))$; $\Lambda(w(t)) = \Lambda(w_0)$; $\mathcal{A}(w(t)) = \mathcal{A}(w_0) < \frac{16373}{2500000}$; $|v_i(w(t))|_2 < \frac{3}{20}$ $(i = 0,1)$; and for every $F \in \mathbb{R}^2$:*
  - *$\frac{1751}{400} \le |q_i(w(t);F_i)|_2 \le \frac{4439}{400}$ $(i = 0,1)$;*

- $\frac{399}{200} \le |q_1(w(t);F_1)|_2 - |q_0(w(t);F_0)|_2$*;*
- $\mathrm{PairBounds}\big(\frac{399}{200}, \frac{4439}{200};$ `barycentricPositions` $1\, m\, \mu\, (\mathbf{z}(w(t);F)_1)\big)$*, i.e. all* $|Q_j - Q_i|_2$*,* $i \neq j \in \{0,1,2\}$*, lie in* $[\frac{399}{200}, \frac{4439}{200}]$.

Frozen Lean declaration: `KAM.Planetary.secularBoxSafety`

Source: LP17 Appendix B, p. 16 ((B.5), (B.8)) for the averaging and the conservation mechanism; the box and the geometry are the elementary specialisation of round 2 §3, proved here.

Theorem 20.5 (`secularBoxAtCitedMasses`)**.** *The conclusion of Theorem 20.4, literally, with* $\sigma = \frac{476}{1591}$ *and* $\mu = \frac{4773}{5000000}$ *substituted everywhere, holds for every* w_0 *with* $\mathrm{Box}_{476/1591}(w_0)$.

Frozen Lean declaration: `KAM.Planetary.secularBoxAtCitedMasses`

Source: FH24v1 §4, p. 6 (decimal model masses 0.0009546 *and* 0.0002856*); a model-parameter specialisation, not a claim about measured masses.*

Remark 20.6 (Lean totality). Division by Λ_i, $1 + k_i$, $\Lambda_i \mathcal{J}_i$, $|q_i|_2$, $|q_0 - q_1|_2$, $\sqrt{\cdot}$ in ρ_i and in `planeLength`, and the Bochner integral and `fderiv` in H_{sec} are all guarded: at every point where they are used (the region $\mathcal{S}$ below), $\Lambda_i > 0$, $k_i > \frac{49}{50}$, $\mathcal{J}_i \ge 1 - e_i > \frac{4}{5}$, the radicand of ρ_i exceeds $\frac{99}{100\Lambda_i}$, and $|q_i|_2 > 0$, $|q_0 - q_1|_2 > 0$ for every F (Lemma 20.12); the integrand is continuous on the compact Δ, hence integrable, so the integral is the genuine one (Proposition 20.13); H_{sec} is C^∞ on $\mathcal{S}$, so `fderiv` is the derivative. The theorem asserts regularity as a conclusion; the hypotheses give only $\Lambda_i > 0$, $0 \le \Gamma_i < \Lambda_i$ and the box.

20.3. **The rational arithmetic.**

Lemma 20.7 (Numerical certificate)**.** *Each of the following is an equality or inequality between explicit rational numbers (no square root appears). Let* $\Lambda_{\mathrm{J}} = \sqrt{a_0}$, $\Lambda_{\mathrm{S}} = \sigma\sqrt{a_1}$ *be the actions of a box point.*

label	*statement*	*value*
N1	$\frac{29}{100} \le \frac{476}{1591} \le \frac{31}{100}$*;* $3 \cdot 1591 = 4773$*;* $\frac{4773}{5000000} \cdot \frac{476}{1591} = \frac{357}{1250000}$	0.29918*;* 0.0002856
N2	$(\frac{87}{100})^2 < \frac{103}{20}$ *and* $\frac{21}{4} < (\frac{23}{10})^2$	$0.7569 < 5.15$*;* $5.25 < 5.29$
N3	$(\frac{87}{100})^2 < (\frac{29}{100})^2 \frac{189}{20}$ *and* $(\frac{31}{100})^2 \frac{193}{20} < (\frac{97}{100})^2$	$0.7569 < 0.794745$*;* $0.927365 < 0.9409$
N4a	$(\frac{99819}{100000})^2 < 1 - (\frac{3}{50})^2$	$0.99638328 < 0.9964$
N4b	$(\frac{49877}{50000})^2 < 1 - (\frac{7}{100})^2$	*(see script)* < 0.9951
N4c	$\frac{23}{10} \cdot \frac{181}{100000} + \frac{97}{100} \cdot \frac{123}{50000} = \frac{16373}{2500000}$	0.0065492
N5a	$\frac{16373}{2500000} \Big/ \frac{87}{100} < \frac{19}{2500}$	$0.0075278 < 0.0076$
N5b	$1 - (\frac{2481}{2500})^2 = \frac{94639}{6250000} < (\frac{3}{20})^2$	$0.015142 < 0.0225$
N6a	$\frac{51}{10} < \frac{103}{20}$*,* $\frac{21}{4} < \frac{53}{10}$*,* $\frac{47}{5} < \frac{189}{20}$*,* $\frac{193}{20} < \frac{97}{10}$	
N6b	$50 \cdot \frac{16373}{2500000} < \frac{87}{100}$	$0.32746 < 0.87$
N6c	$1 - (\frac{49}{50})^2 = \frac{99}{2500} < (\frac{1}{5})^2$	$0.0396 < 0.04$
N6d	$\frac{53}{10} \cdot \frac{6}{5} < \frac{47}{5} \cdot \frac{4}{5}$	$6.36 < 7.52$
N7a	$\frac{103}{20} \cdot \frac{17}{20} = \frac{1751}{400}$*;* $\frac{193}{20} \cdot \frac{23}{20} = \frac{4439}{400}$	4.3775*;* 11.0975
N7b	$\frac{189}{20} \cdot \frac{17}{20} - \frac{21}{4} \cdot \frac{23}{20} = \frac{399}{200}$	$8.0325 - 6.0375 = 1.995$
N7c	$\frac{21}{4} \cdot \frac{23}{20} = \frac{483}{80} \le \frac{4439}{400}$*;* $\frac{189}{20} \cdot \frac{17}{20} \ge \frac{1751}{400}$*;* $\frac{399}{200} \le \frac{1751}{400}$	
N7d	$\frac{4439}{400} + \frac{483}{80} \le \frac{4439}{200}$	$17.135 \le 22.195$

Source's own chain *(checked, not used):* $(\frac{989}{1000})^2 > 1 - (\frac{3}{20})^2$ *(so* $1 - \sqrt{1 - (3/20)^2} > \frac{11}{1000}$*) and* $\frac{87}{100} \cdot \frac{11}{1000} = \frac{957}{100000} > \frac{16373}{2500000}$*. All were verified in exact rational arithmetic (Python* `fractions`*,* `.drafts/P/bp-scratch-bpP-p14b/check.py`*; every line prints* `OK`*). In Lean each is a* `norm_num` *goal.*

Source: Ours; new elementary arithmetic (the box and its rational certificate), checked in exact rational arithmetic.

Proof. Arithmetic; the script evaluates both sides exactly. The decimal values are for orientation. □

Remark 20.8 (Route against the source's chain). The source bounds $e_i < \frac{3}{20}$ through $1-\sqrt{1-e^2} > \frac{11}{1000}$ and $\mathcal{A} < \frac{957}{100000}$. The proof below avoids square roots of rationals: it works with $k_i = \sqrt{1-e_i^2}$ as a real number, uses $k^2 = 1-e^2$ and compares squares (N4, N5), which is the same inequality in a form that `nlinarith`-free `norm_num` reaches directly. The conserved bound $\frac{16373}{2500000}$ is the one in the frozen statement.

20.4. **Algebra of the explicit chart.**

Lemma 20.9 (Eccentricity, radius, Jacobian). *Let $m_i > 0$, $\Lambda_i > 0$, $0 \le \Gamma_i < \Lambda_i$ (a point with these properties is called* elliptic*), $F \in \mathbb{R}$. Then for $i \in \{0,1\}$:*

(a) $k_i \in (0,1]$, the radicand of ρ_i is positive, $|v_i|_2^2 = 1-k_i^2$, $e_i < 1$, $k_i = \sqrt{1-e_i^2}$ and $\Gamma_i = \Lambda_i(1-k_i)$;
(b) $\mathcal{J}_i(F) \ge 1 - e_i > 0$;
(c) $|q_i(w;F)|_2 = a_i(1-x_i(F))$; hence $a_i(1-e_i) \le |q_i|_2 \le a_i(1+e_i)$ and $|q_i|_2 > 0$.

Source: Ours; direct calculation from the frozen definitions (the explicit chart formulas are not taken verbatim from LP17).

Proof. **Step S20.1: (a).** $0 \le \Gamma_i/\Lambda_i < 1$ gives $k_i \in (0,1]$ and $1-\Gamma_i/(2\Lambda_i) > \frac{1}{2}$, so $\rho_i^2 = (1-\Gamma_i/(2\Lambda_i))/\Lambda_i > 0$. With $|u_i|_2^2 = 2\Gamma_i$: $|v_i|_2^2 = \rho_i^2 \cdot 2\Gamma_i = \frac{2\Gamma_i}{\Lambda_i} - \frac{\Gamma_i^2}{\Lambda_i^2} = 1-k_i^2$ (this is Proposition 9.16(a)). As $k_i > 0$: $e_i^2 < 1$ and $k_i = \sqrt{1-e_i^2}$; $\Gamma_i = \Lambda_i(1-k_i)$ is the definition of k_i.

Step S20.2: (b). Cauchy–Schwarz: $x_i = v_i \cdot c \le |v_i|_2\,|c|_2 = e_i$, so $\mathcal{J}_i = 1-x_i \ge 1-e_i > 0$.

Step S20.3: (c) the identity. Drop the index, put $k = k_i$, $x = x_i(F)$, $c = c(F)$, $\kappa = 1 - \frac{x}{1+k}$, so that $q/a = kc - \kappa v$ and $|q/a|_2^2 = k^2 - 2k\kappa x + \kappa^2 e^2$ ($|c|_2 = 1$, $v\cdot c = x$). Use $e^2 = 1-k^2 = (1-k)(1+k)$ and $\kappa = \frac{1+k-x}{1+k}$; multiplying by $1+k$,

$$(1+k)|q/a|_2^2 = k^2(1+k) - 2kx(1+k-x) + (1+k-x)^2(1-k).$$

Collect with $A = 1+k$: the coefficient of x^2 is $2k+(1-k) = A$, that of x is $-2kA - 2A(1-k) = -2A$, the constant term is $k^2A + A^2(1-k) = A(k^2+1-k^2) = A$. Hence $(1+k)|q/a|_2^2 = A(1-x)^2$ and $|q/a|_2^2 = (1-x)^2$; as $1-x > 0$ (b), $|q|_2 = a(1-x)$.

Step S20.4: Radial bounds. $-e \le x \le e$ (Cauchy–Schwarz) gives $a(1-e) \le |q|_2 \le a(1+e)$, and $|q|_2 \ge a(1-e) > 0$ ($a > 0$). □

Lemma 20.10 (Rotation covariance). *Let w be elliptic (both planets), $\theta \in \mathbb{R}$, and let w^θ have the same λ and Λ, and $(\xi_i', \eta_i') = R_\theta(\xi_i, \eta_i)$ for $i = 0, 1$. Then Γ_i, k_i, a_i are unchanged, $v_i(w^\theta) = R_\theta v_i(w)$, and for all $F \in \mathbb{R}$*

$$x_i(w^\theta; F) = x_i(w; F-\theta), \qquad \mathcal{J}_i(w^\theta; F) = \mathcal{J}_i(w; F-\theta),$$
$$q_i(w^\theta; F) = R_\theta q_i(w; F-\theta), \qquad p_i(w^\theta; F) = R_\theta p_i(w; F-\theta).$$

Source: Ours; direct calculation from the frozen definitions (the explicit chart formulas are not taken verbatim from LP17).

Proof. **Step S20.5: Slow data.** $|R_\theta u_i|_2 = |u_i|_2$, so $\Gamma_i, k_i, \rho_i, a_i$ are unchanged and $v_i(w^\theta) = \rho_i R_\theta u_i = R_\theta v_i$.

Step S20.6: Scalars. $c(F) = R_\theta c(F-\theta)$, hence $x_i(w^\theta; F) = R_\theta v_i \cdot R_\theta c(F-\theta) = v_i \cdot c(F-\theta)$ ((X1)); likewise $\mathcal{J}_i$.

Step S20.7: Vectors. Insert $c(F) = R_\theta c(F-\theta)$, $v' = R_\theta v$, the unchanged scalars $k, a, \Lambda, x/(1+k)$ into the formulas for q, p: $q' = a(kR_\theta c - R_\theta v + \frac{x}{1+k} R_\theta v) = R_\theta q(w; F-\theta)$; and $J_0 R_\theta = R_\theta J_0$ gives $p' = R_\theta p(w; F-\theta)$. □

20.5. **The region $\mathcal{S}$ and smoothness of H_{sec}.** Fix $\sigma \in [\frac{29}{100}, \frac{31}{100}]$ (so $m = (1, \sigma)$, $m_i > 0$) and $\mu > 0$ (the proofs of this subsection work for every real μ).

Definition 20.11 (The region $\mathcal{S}$)**.** $\mathcal{S} = \mathcal{S}_\sigma$ is the set of $w \in \texttt{Phase}(2+2)$ with, for $i = 0, 1$: $\Lambda_i > 0$ and $50\,\Gamma_i < \Lambda_i$; and $\frac{51}{10} < a_0 < \frac{53}{10}$, $\frac{47}{5} < a_1 < \frac{97}{10}$. $\mathcal{S}$ does not constrain λ, and is invariant under the maps $w \mapsto w^\theta$. It is open: it is a finite intersection of strict inequalities between continuous functions of w ((X2); a_i is continuous, $m_i \neq 0$).
Source: Ours; direct calculation from the frozen definitions (the explicit chart formulas are not taken verbatim from LP17).

Lemma 20.12 (Geometry on $\mathcal{S}$)**.** *For $w \in \mathcal{S}$, $F \in \mathbb{R}^2$: $k_i > \frac{49}{50}$, $e_i < \frac{1}{5}$, $\mathcal{J}_i(F_i) > \frac{4}{5}$; $|q_0(w; F_0)|_2 < \frac{53}{10} \cdot \frac{6}{5}$ and $|q_1(w; F_1)|_2 > \frac{47}{5} \cdot \frac{4}{5}$; so $|q_0|_2 < |q_1|_2$, all of $|q_i|_2$, $|q_0 - q_1|_2$ are positive, and the radicand of ρ_i exceeds $\frac{99}{100\,\Lambda_i}$. In particular every $w \in \mathcal{S}$ is elliptic,* Reg*(i), (ii) hold at w, and $\mathbf{z}(w; F)_1$ has* NoColl*.*
Source: Ours; direct calculation from the frozen definitions (the explicit chart formulas are not taken verbatim from LP17).

Proof. **Step S20.8: Elliptic.** $\Gamma_i \geq 0$, $50\Gamma_i < \Lambda_i$ gives $\Gamma_i < \Lambda_i$, $k_i = 1 - \Gamma_i/\Lambda_i > \frac{49}{50}$ and $1 - \Gamma_i/(2\Lambda_i) > \frac{99}{100}$.

Step S20.9: Eccentricity. Lemma 20.9(a): $e_i^2 = 1 - k_i^2 < 1 - (\frac{49}{50})^2 = \frac{99}{2500} < \frac{1}{25}$ (N6c). (b): $\mathcal{J}_i \geq 1 - e_i > \frac{4}{5}$.

Step S20.10: Radii. Lemma 20.9(c): $|q_0|_2 \leq a_0(1 + e_0) < \frac{53}{10} \cdot \frac{6}{5} = 6.36$ and $|q_1|_2 \geq a_1(1 - e_1) > \frac{47}{5} \cdot \frac{4}{5} = 7.52$ (N6d). The reverse triangle inequality gives $|q_1 - q_0|_2 \geq |q_1|_2 - |q_0|_2 > 0$, and $|q_0 - q_1|_2 = |q_1 - q_0|_2$. □

Proposition 20.13 (Smoothness of H_{sec})**.** *Let $m = (1, \sigma)$, $\mu \in \mathbb{R}$. (a) For $w \in \mathcal{S}$: $H_{\mathrm{sec}}(w) = (2\pi)^{-2} \int_\Delta \widehat{\mathcal{I}}(w; F)\, dF$ with $\widehat{\mathcal{I}}(w; F) = H_{\mathrm{mov}}^{(\mu,m)}(\mathbf{z}(w; F)) \prod_i \mathcal{J}_i(w; F_i)$, and $H_{\mathrm{sec}}(w)$ does not depend on λ (indeed on $\mathcal{S}$, $H_{\mathrm{sec}}(\lambda + s, \xi; \Lambda, \eta) = H_{\mathrm{sec}}(\lambda, \xi; \Lambda, \eta)$ for all $s \in \mathbb{R}^2$). (b) $\widehat{\mathcal{I}}$ is C^∞ on $\mathcal{S} \times \mathbb{R}^2$ and 2π-periodic in each F_i. (c) H_{sec} is C^∞ on $\mathcal{S}$; in particular C^2, differentiable at every point of $\mathcal{S}$, and* $\mathrm{Reg}_{\sigma,\mu}(w)$ *(iii), (iv) hold for $w \in \mathcal{S}$.*
Source: Ours; direct calculation from the frozen definitions, with [N:bpN:p3:ham-smooth] *used downstream (differentiation under the integral: (E2)).*

Proof. **Step S20.11: λ-independence.** $\texttt{fast}\,(\texttt{assemble}\,\Lambda\,F\,\eta\,\xi)_1 = F$ and $\texttt{slow}(\cdot)_1 = \xi$, $\texttt{fast}(\cdot)_2 = \Lambda$, $\texttt{slow}(\cdot)_2 = \eta$ (`Fin.append` lemmas). The definitions of $q_i, p_i, v_i, k_i, a_i, \mathcal{J}_i$ use only ξ, η, Λ and the argument F (they never read $\texttt{fast}\,w_1$ except through `ellipsePhase`'s argument), hence $\mathcal{I}_\mu((F, \xi; \Lambda, \eta); F) = \widehat{\mathcal{I}}(w; F)$ for every w with these slow data. This is (a).

Step S20.12: Smoothness of the chart. On $\mathcal{S} \times \mathbb{R}^2$: $\Lambda_i > 0$, $\Gamma_i = \frac{1}{2}(\xi_i^2 + \eta_i^2)$ is polynomial; $\rho_i = \sqrt{\cdot}$ of a function $> \frac{99}{100\Lambda_i} > 0$; k_i, $1 + k_i > \frac{99}{50}$, $\mathcal{J}_i > \frac{4}{5}$ are bounded away from 0; so v_i, x_i, q_i, p_i (quotients by these and products with $c(F_i)$) are C^∞ ((X5)). By Lemma 20.12, $|q_i|_2^2 > 0$ and $|q_0 - q_1|_2^2 > 0$, so $|q_i|_2 = \sqrt{|q_i|_2^2}$, $1/|q_i|_2$ and $1/|q_0 - q_1|_2$ are C^∞. Every term of H_{mov} (Definition 1.22: $\frac{1}{m_i}$, $\frac{\mu}{m_0}$, $|p_i|_2^2$, $p_i \cdot p_j$, $1/|q_i|_2$, $1/|q_0 - q_1|_2$) is then C^∞, and so is $\widehat{\mathcal{I}}$. Periodicity: F_i enters only through $c(F_i)$.

Step S20.13: Parameter integral. Fix a point $w_* \in \mathcal{S}$ and a compact neighbourhood $N \subset \mathcal{S}$ of it. All partial derivatives of $\widehat{\mathcal{I}}$ of order ≤ 2 in w are continuous on $N \times \Delta$ ((X2)); (X4) ($k = 2$) gives that $w \mapsto \int_\Delta \widehat{\mathcal{I}}\, dF$ is C^2 on the interior of N with derivatives under the integral; the same holds for every order, so H_{sec} is C^∞ on $\mathcal{S}$.

Step S20.14: Integrability. $F \mapsto \widehat{\mathcal{I}}(w;F)$ is continuous on the compact Δ ((X3)), hence integrable: Reg(iii). (iv) is differentiability of a C^∞ function on an open set. □

Proposition 20.14 (Rotation invariance)**.** *For $w \in \mathcal{S}$ and $\theta \in \mathbb{R}$: $w^\theta \in \mathcal{S}$ and $H_{\mathrm{sec}}(w^\theta) = H_{\mathrm{sec}}(w)$.*
Source: Ours; direct calculation from the frozen definitions (the explicit chart formulas are not taken verbatim from LP17).

Proof. **Step S20.15: Membership.** Lemma 20.10: Γ_i, Λ_i, a_i are unchanged.

Step S20.16: The Hamiltonian. Let R_θ act on $\mathtt{Phase}(2\cdot 2)$ by R_θ on each body vector of q and of p. Then $\mathtt{bodyVector}(\mathsf{R}_\theta z) = R_\theta\,\mathtt{bodyVector}(z)$ and $|R_\theta a|_2 = |a|_2$, $R_\theta a \cdot R_\theta b = a \cdot b$, $|R_\theta a - R_\theta b|_2 = |a-b|_2$, so $H_{\mathrm{mov}}(\mathsf{R}_\theta z) = H_{\mathrm{mov}}(z)$ term by term.

Step S20.17: The integrand. Lemma 20.10: $\mathbf{z}(w^\theta;F) = \mathsf{R}_\theta \mathbf{z}(w;F-\theta\mathbb{1})$ and $\mathcal{J}_i(w^\theta;F_i) = \mathcal{J}_i(w;F_i-\theta)$, so $\widehat{\mathcal{I}}(w^\theta;F) = \widehat{\mathcal{I}}(w;F-\theta\mathbb{1})$.

Step S20.18: The integral. $F \mapsto \widehat{\mathcal{I}}(w;F)$ is continuous and 2π-periodic in each variable (Proposition 20.13(b)); by Fubini and the periodic translation invariance ((X3)), applied to each variable in turn, $\int_\Delta \widehat{\mathcal{I}}(w;F-\theta\mathbb{1})\,dF = \int_\Delta \widehat{\mathcal{I}}(w;F)\,dF$. Proposition 20.13(a) gives the claim. □

Remark 20.15 (Consistency with Parts 7 and 8). Proposition 20.14 is the F-level form of Proposition 10.3(S2) (there F is shifted by θ together with λ), and by Corollary 9.17, H_{sec} is the double mean-longitude average of $H_{\mathrm{mov}} \circ \Psi$ (LP17 (B.5)). Neither fact is needed for the proofs.

20.6. **The slow flow, its global existence, and the lifting of λ.** For $w \in \mathcal{S}$ put $\omega(w) = \partial_\Lambda H_{\mathrm{sec}}(w) \in \mathbb{R}^2$ (the partial gradient in the Λ-variables): ω is continuous on $\mathcal{S}$ and depends only on (ξ,Λ,η) (Proposition 20.13).

Lemma 20.16 (Predicate and field on $\mathcal{S}$)**.** *$H = H_{\mathrm{sec}}$ is C^2 on the open set $\mathcal{S} \subset$ $\mathtt{Phase}(2+2)$, X_H is C^1 on $\mathcal{S}$, and for $J \subset \mathbb{R}$, $z : \mathbb{R} \to$ $\mathtt{Phase}(2+2)$ with $z(t) \in \mathcal{S}$ for $t \in J$: $\mathtt{IsHamiltonianSolutionOn}\,H\,\mathtt{univ}\,J\,z \iff \mathtt{IsSolIn}\,X_H\,\mathcal{S}\,J\,z$ (*[N:bpN:p3:gen-sol]*).*
Source: Ours; unfolding of the frozen predicates, with [N:bpN:p3:gen-sol] *and* [N:bpN:p3:ham-smooth].

Proof. **Step S20.19: Regularity.** [N:bpN:p3:ham-smooth] ($n=4$, $k=1$) with Proposition 20.13(c).

Step S20.20: Equivalence. The clause $z(t)_2 \in \mathtt{univ}$ is trivial; differentiability of H at $z(t) \in \mathcal{S}$ holds; the derivative clause is the same in both predicates (the pattern of [N:bpN:p3:ham-reduction], with $W = \mathcal{S}$). □

Lemma 20.17 (Structure of X_H)**.** *For $w \in \mathcal{S}$ and $H = H_{\mathrm{sec}}$: $X_H(w) = \big((\omega(w), \partial_\eta H(w)), (0, -\partial_\xi H(w))\big)$, in the block order $((\lambda,\xi),(\Lambda,\eta))$; and for $y = (\xi,\eta)$ fixed, Λ fixed, $\partial_\xi H, \partial_\eta H$ are the partial derivatives of $h_\Lambda(y) = H(0,\xi;\Lambda,\eta)$. Moreover, with $A(y) = \Gamma_0 + \Gamma_1$, $\sum_i \big(\xi_i \partial_{\eta_i} H - \eta_i \partial_{\xi_i} H\big)(w) = 0$.*
Source: Ours; direct calculation from the frozen $\mathtt{hamiltonianVectorField}$ *and Proposition 20.13, with* [N:bpN:p3:ham-reduction].

Proof. **Step S20.21: Components.** Read off $\mathtt{hamiltonianVectorField}$: λ_i-component $\partial_{\Lambda_i} H$, ξ_i-component $\partial_{\eta_i} H$, Λ_i-component $-\partial_{\lambda_i} H$, η_i-component $-\partial_{\xi_i} H$. Proposition 20.13(a): $s \mapsto H(w + s\,e_{\lambda_i})$ is constant, so $\partial_{\lambda_i} H(w) = 0$ ((X5)). The statement on h_Λ is the chain rule for the affine map $y \mapsto (0,\xi;\Lambda,\eta)$.

Step S20.22: Rotation identity. Fix $w \in \mathcal{S}$ and put $g(\theta) = H(w^\theta)$, $\theta \in \mathbb{R}$. By Proposition 20.14, g is constant. The map $\theta \mapsto w^\theta$ is smooth with derivative at $\theta = 0$ the vector with ξ_i-slot $-\eta_i$, η_i-slot ξ_i and zero elsewhere; the chain rule gives $0 = g'(0) = \sum_i (-\eta_i \partial_{\xi_i} H + \xi_i \partial_{\eta_i} H)(w)$. □

Lemma 20.18 (Global slow flow)**.** *Let $w_0 \in \mathcal{S}$ with $\mathcal{A}(w_0) < \frac{16373}{2500000}$ and with $\Lambda_i(w_0) > \frac{87}{100}$ $(i = 0, 1)$. Put $\Lambda^0 = \mathtt{fast}(w_0)_2$ and $\mathcal{S}' = \{y = (\xi, \eta) \in \mathbb{R}^2 \times \mathbb{R}^2 : \ (0, \xi; \Lambda^0, \eta) \in \mathcal{S}\}$ (open). There is a continuous $y = (\xi, \eta) : \mathbb{R} \to \mathcal{S}'$ with $y(0) = (\xi(w_0), \eta(w_0))$ such that y has derivative $\big(\partial_\eta h(y(t)), -\partial_\xi h(y(t))\big)$ at every $t \in \mathbb{R}$ (with $h = h_{\Lambda^0}$), $A(y(t)) = A(y(0))$ for all t, and $y(t) \in \mathcal{S}'$ for all t.*
Source: Ours; application of [N:bpN:p3:gen-continuation-compact] *on the compact set of the proof, with the conservation of Λ and of $\mathcal{A}$ proved directly.*

Proof. **Step S20.23: The field.** h is C^∞ on $\mathcal{S}'$ (composition with an affine map), $\mathcal{S}'$ is open, and $X_h = (\partial_\eta h, -\partial_\xi h)$ is C^1 on $\mathcal{S}'$ ([N:bpN:p3:ham-smooth], with the phase space $\mathbb{R}^2 \times \mathbb{R}^2$, positions ξ, momenta η). $y(0) \in \mathcal{S}'$ since $w_0 \in \mathcal{S}$.

Step S20.24: Maximal solution. [N:bpN:p3:gen-maximal] gives a maximal interval $(T_-, T_+) \ni 0$ and a solution y of X_h in $\mathcal{S}'$ through $y(0)$.

Step S20.25: Conservation of A. On (T_-, T_+): $\frac{d}{dt} A(y(t)) = \sum_i (\xi_i \dot{\xi}_i + \eta_i \dot{\eta}_i) = \sum_i (\xi_i \partial_{\eta_i} h - \eta_i \partial_{\xi_i} h)(y(t)) = 0$ by Lemma 20.17 (at $w = (0, \xi; \Lambda^0, \eta) \in \mathcal{S}$), so $A(y(t)) = A_0 := A(y(0))$ ((X5)).

Step S20.26: A compact set. Let $K = \{y : \ \Gamma_0 + \Gamma_1 = A_0\}$: closed and bounded ($|\xi_i|, |\eta_i| \le \sqrt{2A_0}$), hence compact. If $y \in K$ then $0 \le \Gamma_i \le A_0 < \frac{16373}{2500000}$ and $50\,\Gamma_i < 50 \cdot \frac{16373}{2500000} < \frac{87}{100} < \Lambda_i^0$ (N6b); the axis conditions of $\mathcal{S}$ depend on Λ^0 only, hold at $w_0 \in \mathcal{S}$. So $K \subset \mathcal{S}'$ and $y(t) \in K$ for all $t \in (T_-, T_+)$.

Step S20.27: Continuation. [N:bpN:p3:gen-continuation-compact] (right and left ends) gives $T_\pm = \pm\infty$. □

Proposition 20.19 (Global solution of the full system)**.** *Let w_0 be as in Lemma 20.18. Then there is $w : \mathbb{R} \to \mathtt{Phase}(2+2)$ with $w(0) = w_0$, $w(t) \in \mathcal{S}$, $\Lambda(w(t)) = \Lambda^0$, $\mathcal{A}(w(t)) = \mathcal{A}(w_0)$ for all t, and $\mathtt{IsHamiltonianSolutionOn}\, H_{\mathrm{sec}}\, \mathtt{univ}\, \mathtt{univ}\, w$.*
Source: Ours; application of [N:bpN:p3:gen-sol], [N:bpN:p3:gen-maximal] *and* [N:bpN:p3:gen-continuation-compact]*, with the conserved quantities of Lemma 20.18.*

Proof. **Step S20.28: Quadrature.** Put $\omega_*(t) = \omega\big((0, \xi(t); \Lambda^0, \eta(t))\big)$, continuous in t (ω is continuous on $\mathcal{S}$, y is continuous). Let $\lambda(t) = \lambda(w_0) + \int_0^t \omega_*(s)\, ds$ ((X3)), so $\dot{\lambda} = \omega_*$.

Step S20.29: The curve. $w(t) = ((\lambda(t), \xi(t)), (\Lambda^0, \eta(t)))$. $w(0) = w_0$, since a point of $\mathtt{Phase}(2+2)$ is determined by $\mathtt{fast}$ and $\mathtt{slow}$ of each block ($\mathtt{Fin.append_castAdd_natAdd}$). $w(t) \in \mathcal{S}$: Γ_i, Λ_i, a_i depend on (y, Λ^0) only, and $y(t) \in \mathcal{S}'$. $\mathcal{A}(w(t)) = A(y(t)) = A(y(0)) = \mathcal{A}(w_0)$ (Lemma 20.18).

Step S20.30: The equation. By Lemma 20.17, $X_H(w(t)) = ((\omega_*(t), \partial_\eta h(y(t))), (0, -\partial_\xi h(y(t))))$ (the partials of H in ξ, η at $w(t)$ equal those of h at $y(t)$); each component of w has the corresponding derivative at every t ((X5)). So w has derivative $X_H(w(t))$ and H is differentiable at $w(t) \in \mathcal{S}$: Lemma 20.16. □

Proposition 20.20 (Uniqueness)**.** *Let w be as in Proposition 20.19. Let $U \subset \mathbb{R}$ be open and convex, $0 \in U$, $v : \mathbb{R} \to \mathtt{Phase}(2+2)$ with $v(0) = w_0$ and $\mathtt{IsHamiltonianSolutionOn}\, H_{\mathrm{sec}}\, \mathtt{univ} U\, v$. Then $v = w$ on U. (No regularity of $v(t)$ is used beyond the predicate.)*
Source: Ours; application of [N:bpN:p3:gen-uniqueness] *(and* [N:bpN:p3:gen-maximal]*) to the field of Lemma 20.17.*

Proof. **Step S20.31: The agreement set.** $D = \{t \in U : \ v(t) = w(t)\} \ni 0$. v and w are continuous on U (differentiable at each point), so D is closed in U.

Step S20.32: Open. Let $t_0 \in D$. $w(t_0) \in \mathcal{S}$ (open), v continuous at t_0: there is an open interval $J_0 \ni t_0$, $J_0 \subset U$ with $v(t) \in \mathcal{S}$ on J_0. By Lemma 20.16, v and w are both $\mathtt{IsSolIn}\, X_H\, \mathcal{S}\, J_0$ and agree at t_0; [N:bpN:p3:gen-uniqueness] gives $v = w$ on J_0. So $J_0 \subset D$.

Step S20.33: Conclusion. U is an interval, hence connected; D is nonempty, open, closed in U ((X2)): $D = U$. □

20.7. **Proof of the theorems.**

Proof of Theorem 20.4. Fix σ, μ, w_0 with $\mathrm{Box}_\sigma(w_0)$; $m = (1, \sigma)$; $H = H_{\mathrm{sec}}^{(m,\mu)}$.

Step S20.34: Actions and axes. $\Lambda_0^2 = a_0 \in [\frac{103}{20}, \frac{21}{4}]$, so $\frac{87}{100} < \Lambda_0 < \frac{23}{10}$ (N2; $\Lambda_0 > 0$ by (i)). $\Lambda_1 = \sigma\sqrt{a_1}$ ($a_1 = (\Lambda_1/\sigma)^2$), $\Lambda_1^2 = \sigma^2 a_1 \in [(\frac{29}{100})^2\frac{189}{20}, (\frac{31}{100})^2\frac{193}{20}]$, so $\frac{87}{100} < \Lambda_1 < \frac{97}{100}$ (N3).

Step S20.35: Initial AMD. By Lemma 20.9(a), $k_0 > 0$, $k_0^2 = 1 - e_0^2 \geq 1 - (\frac{3}{50})^2 > (\frac{99819}{100000})^2$ (N4a), so $k_0 > \frac{99819}{100000}$, $1 - k_0 < \frac{181}{100000}$ and $\Gamma_0 = \Lambda_0(1 - k_0) < \frac{23}{10} \cdot \frac{181}{100000}$ (using $\Lambda_0 < \frac{23}{10}$, $1 - k_0 \geq 0$; if $\Gamma_0 = 0$ the bound is trivial). Likewise $k_1 > \frac{49877}{50000}$ (N4b), $\Gamma_1 < \frac{97}{100} \cdot \frac{123}{50000}$. Hence $\mathcal{A}(w_0) < \frac{16373}{2500000}$ (N4c).

Step S20.36: $w_0 \in \mathcal{S}$. $\Gamma_i \leq \mathcal{A}(w_0)$ and $50\Gamma_i < \frac{87}{100} < \Lambda_i$ (N6b); the axis ranges are inside $\mathcal{S}$'s (N6a); $\Lambda_i > 0$. So $w_0 \in \mathcal{S}$ and the hypotheses of Lemma 20.18 hold.

Step S20.37: (a). Proposition 20.19 gives w with $w(0) = w_0$, $w(t) \in \mathcal{S}$, $\Lambda(w(t)) = \Lambda(w_0)$, $\mathcal{A}(w(t)) = \mathcal{A}(w_0) < \frac{16373}{2500000}$ for all t, solving `IsHamiltonianSolutionOn` H `univ univ`. This is (a) and the first conserved clauses of (c) (`fast` $(w(t))_2 =$ `fast` $(w_0)_2$, `secularAMD`).

Step S20.38: (b). Proposition 20.20. (The hypothesis $\mathrm{Reg}(v(t))$ of (b) is not needed.)

Step S20.39: Eccentricities. Fix t, write Γ_i, k_i, e_i at $w(t)$. $\Lambda_i(w(t)) = \Lambda_i > \frac{87}{100}$, $0 \leq \Gamma_i \leq \mathcal{A}(w(t)) < \frac{16373}{2500000}$; so $1 - k_i = \Gamma_i/\Lambda_i < \frac{16373}{2500000}/\frac{87}{100} < \frac{19}{2500}$ (N5a), $k_i > \frac{2481}{2500}$, and $e_i^2 = 1 - k_i^2 < 1 - (\frac{2481}{2500})^2 < (\frac{3}{20})^2$ (N5b). So $|v_i(w(t))|_2 < \frac{3}{20}$.

Step S20.40: Regularity. $w(t) \in \mathcal{S}$: Lemma 20.12 gives Reg(i), (ii) ($\mathcal{J}_i > 0$ for every F, NoColl; the bridge from $\mathbf{z}(w; F)_1$ to $q_i(w; F_i)$ is Step S20.41); Proposition 20.13(c) gives (iii), (iv).

Step S20.41: Bridge. For $F \in \mathbb{R}^2$ and $i < 2$: `bodyVector`$(\mathbf{z}(w; F)_1)\, i = q_i(w; F_i)$. Indeed coordinate k of $\mathbf{z}_1$ is $q_j(w; F_j)$ at the component l, where $(j, l) =$ `finProdFinEquiv.symm` k; for $k =$ `finProdFinEquiv`(i, l) this is (i, l) because `finProdFinEquiv.symm` $\circ$ `finProdFinEquiv` $=$ id.

Step S20.42: Radii. With $e_i < \frac{3}{20}$ and Lemma 20.9(c), $a_i(1 - \frac{3}{20}) \leq |q_i|_2 \leq a_i(1 + \frac{3}{20})$. For $i = 0$: $|q_0|_2 \geq \frac{103}{20} \cdot \frac{17}{20} = \frac{1751}{400}$ and $\leq \frac{21}{4} \cdot \frac{23}{20} = \frac{483}{80} \leq \frac{4439}{400}$. For $i = 1$: $|q_1|_2 \geq \frac{189}{20} \cdot \frac{17}{20} \geq \frac{1751}{400}$ and $\leq \frac{193}{20} \cdot \frac{23}{20} = \frac{4439}{400}$ (N7a, N7c).

Step S20.43: Gap. $|q_1|_2 - |q_0|_2 \geq \frac{189}{20} \cdot \frac{17}{20} - \frac{21}{4} \cdot \frac{23}{20} = \frac{399}{200}$ (N7b).

Step S20.44: Pair bounds. `barycentricPositions` $1\, m\, \mu\, q =$ `Fin.cons` $x_0\, (i \mapsto x_0 +$ `bodyVector` $q\, i)$ with a common x_0, so $Q_{j+1} - Q_0 = b_j$, $Q_{j+1} - Q_{i+1} = b_j - b_i$, $b_j = q_j(w(t); F_j)$ (Step S20.41); $|-u|_2 = |u|_2$ covers both orders. Sun–planet: $|b_j|_2 \in [\frac{1751}{400}, \frac{4439}{400}] \subset [\frac{399}{200}, \frac{4439}{200}]$ (N7c, Step S20.42). Planet–planet: $|b_1 - b_0|_2 \geq |b_1|_2 - |b_0|_2 \geq \frac{399}{200}$ (Step S20.43) and $|b_1 - b_0|_2 \leq |b_1|_2 + |b_0|_2 \leq \frac{4439}{400} + \frac{483}{80} \leq \frac{4439}{200}$ (N7d).

All clauses of the theorem are proved. □

Proof of Theorem 20.5. **Step S20.45: Instantiation.** Apply Theorem 20.4 with $\sigma = \frac{476}{1591}$ (N1: $\frac{29}{100} \leq \frac{476}{1591} \leq \frac{31}{100}$, i.e. $29 \cdot 1591 = 46139 \leq 47600$ and $47600 \leq 31 \cdot 1591 = 49321$), $\mu = \frac{4773}{5000000} > 0$ and the given w_0 (hypothesis $\mathrm{Box}_{476/1591}(w_0)$). The conclusion is literally the claim: the frozen statement is the safety statement with these two literals substituted for `saturn` and `mu`. In Lean: `exact secularBoxSafety _ _ ⟨by norm_num, by norm_num⟩ (by norm_num) w0 hw0`.

Step S20.46: The cited masses. $\mu = 0.0009546$ and $\mu\sigma = \frac{4773}{5000000} \cdot \frac{476}{1591} = \frac{357}{1250000} = 0.0002856$ (N1, since $4773 = 3 \cdot 1591$) are the decimal model masses of FH24v1 §4, p. 6; they are not used in the proof, only to identify the instance. □

20.8. **Lean plan, doubts, deviations.**
Module plan (all under the hard rules). The proof is parameter-uniform, so both providers should be one-line wrappers around ordinary (non-provider) library modules; `secularBoxAtCitedMasses` cannot import the `secularBoxSafety` provider and instead uses the library theorem. Suggested split, each file well under 1500 lines: (1) `KAM/Planetary/SecularBox/Numerics` (Lemma 20.7, `norm_num` only; the scratch script is the oracle); (2) `.../Chart` (Lemmas 20.9, 20.10; the polynomial identity of Step S20.3 is a `linear_combination` with the relations $|c|_2^2 = 1$, $|v|_2^2 = (1-k)(1+k)$, after `field_simp` by $1+k>0$); (3) `.../Region` (Definition 20.11, Lemma 20.12); (4) `.../Smooth` (Proposition 20.13; the one analytic input is (E2); if Mathlib has no ready C^k-parametric-integral lemma over a compact box, it is proved by induction from the first-derivative dominated-convergence theorem); (5) `.../Rotation` (Proposition 20.14, Lemma 20.17); (6) `.../Flow` (§20.6, using [N:bpN:p3:gen-maximal], [N:bpN:p3:gen-continuation-compact], [N:bpN:p3:gen-uniqueness]); (7) `.../Safety` and `.../CitedMasses` (the assembly above). If (4) exceeds the heartbeat default, split the chart smoothness (Step S20.12) per quantity (ρ_i, q_i, p_i, $\mathcal{J}_i$, then H_{mov}); never raise heartbeats.
Frozen statements. No defect found. In particular: the uniqueness clause is true as stated because only $v(0) = w_0$, continuity of v and the predicate are used; the regularity premise on v is then a harmless extra hypothesis. The integrand in `SecularRegular`(iii) is the one of Definition 20.1. The cited-masses theorem is the exact instance.
Deviations. (1) Part 0 has no transcription of the P-SECULAR-BOX objects; §20.2 supplies it (to be moved into Part 0 by the sub-orchestrator if wanted). (2) The proof of $e_i < \frac{3}{20}$ compares squares of rationals (N5) instead of the source's chain through $\frac{11}{1000}$ and $\frac{957}{100000}$; both chains are checked. (3) Rather than quoting Appendix B, conservation of Λ and $\mathcal{A}$ is proved directly from λ-independence and rotation invariance of the explicit F-integral (LP17 (B.8) is the bracket form of the same mechanism).
External facts used. (X1)–(X5), and (E2) (differentiation under the integral).
Forward references. None.